

ANSWER-KEY**EXERCISE # I**

(ONLY ONE OPTION IS CORRECT)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	B	A	B	B	D	C	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	B	B	A	B	A	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	B	D	A	A	B	B	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	D	A	B	C	C	B	C	B	D
Que.	41	42	43	44	45	46	47			
Ans.	A	A	B	C	A	C	A			

EXERCISE # II

(ONE OR MORE THAN ONE OPTION MAY BE CORRECT)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A, B	A, B, C	A, C, D	A, B, C	A, B, D	A, B	A, B, C	A, B, C	A, B, C, D	A, C, D
Que.	11	12	13	14	15	16	17			
Ans.	A, B, C, D	C, D	C, D	A, B, C, D	D	B, C, D	A, B, C, D			

EXERCISE # III

PARAGRAPH & MATRIX MATCH

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	D	A	C	D	A	B	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	B	C	B	B	C	A	D	C
Que.	21	22	23	24						
Ans.	B, C, D	B	A	B						

EXERCISE # J- MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	2	3	4	2	2	2	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	3	1	4	3	2	3	3	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	4	2	1	3	4	2	2	3	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	3	1	1	3	4	4	2	2	3	2
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	2	2	4	4	2	1	3	3	3	3
Que.	51	52	53	54						
Ans.	4	1	2	4						

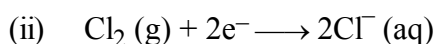
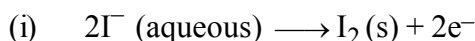
EXERCISE # J-ADVANCED

FILL IN THE BLANKS :

1. In nitrogen, d-orbitals are not present, so in it the possibility of intramolecular multiplicity exists which leads to the completion of octet through π -bond between two nitrogen atoms.

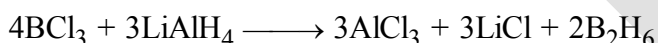
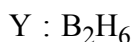
In phosphorus, d-orbitals are present, so in it due to large size of P, the P-P bonds are longer and hence intramolecular multiplicity is ruled out. So, for the completion of octet, it forms the bonds with three other 'P' atoms. Hence due to this reason it shows molecular formula as P_4 .

2. $2I^-$ (aqueous) + $Cl_2 \longrightarrow I_2 + 2Cl^-$ (aqueous)

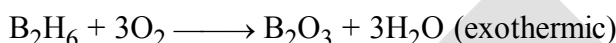


Thus, I^- is oxidised into I_2 by Cl_2 due to higher oxidised potential of Cl_2 than I_2

3. X : BCl_3

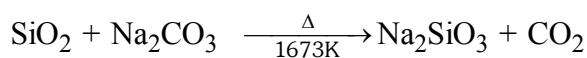
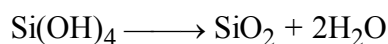
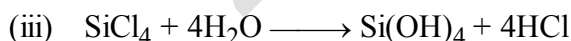
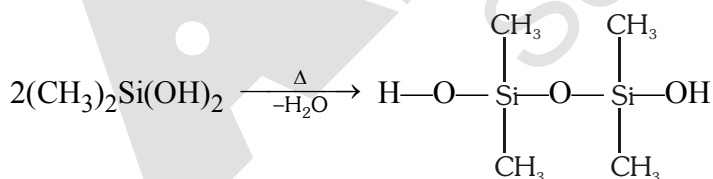
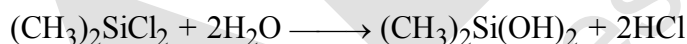
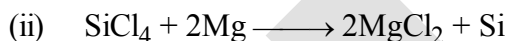


X

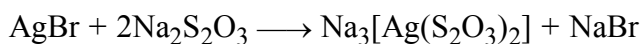


Y

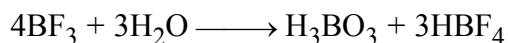
4. (i) $3SiCl_4 + 4Al \longrightarrow 3Si + 4AlCl_3$ (in one step)



5. Unreacted AgBr is removed by hypo ($Na_2S_2O_3$)



6. $^{14}\text{CO}_2$
7. (i) $\text{Al}_4\text{Cl}_3 + 12\text{H}_2\text{O} \longrightarrow 4\text{Al}(\text{OH})_3 + 3\text{CH}_4 \uparrow$
 (ii) $\text{CaNCN} + 3\text{H}_2\text{O} \longrightarrow \text{CaCO}_3 \downarrow + 2\text{NH}_3$
 (iii) $4\text{BF}_3 + 3\text{H}_2\text{O} \longrightarrow \underset{\text{boric acid}}{\text{H}_3\text{BO}_3} + \underset{\text{fluoboric acid}}{3\text{HBF}_4}$
 (iv) $\text{NCl}_3 + 3\text{H}_2\text{O} \longrightarrow \text{NH}_3 + \underset{\text{hypochlorous acid}}{3\text{HOCl}}$
 (v) $2\text{XeF}_4 + 3\text{H}_2\text{O} \longrightarrow \text{Xe} + \underset{\text{xenon trioxide}}{\text{XeO}_3} + \text{F}_2 + 6\text{HCl}$
8. (i) $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \longrightarrow 4\text{Al}(\text{OH})_3 + 3\text{CH}_4$
 (ii) $\text{CaNCN} + 3\text{H}_2\text{O} \longrightarrow \text{CaCO}_3 + 2\text{NH}_3$
 (iii) $\text{BF}_3 + 3\text{H}_2\text{O} \longrightarrow \text{H}_3\text{BO}_3 + 3\text{HF}$
 $3\text{HF} + 3\text{BF}_3 \longrightarrow 3\text{HBF}_4$



9. Oxidation state

- (A) : NaHSO_3 + 4
 (B) : Na_2SO_3 + 4
 (C) : $\text{Na}_2\text{S}_2\text{O}_3$ + 2
 (D) : $\text{Na}_2\text{S}_4\text{O}_6$ + 2.5

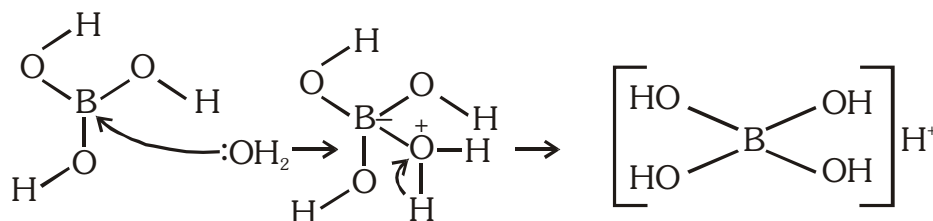
10. $\text{Cl}_2\text{O}_7 < \text{SO}_3 < \text{CO}_2 < \text{B}_2\text{O}_3 < \text{BaO}$

OBJECTIVE TYPE :

- | | | | | | |
|--|-------|-------|-------|-----------|-------|
| 11. D | 12. A | 13. A | 14. C | 15. A,B,D | 16. A |
| 17. C | 18. D | 19. C | 20. C | 21. D | 22. A |
| 23. A | 24. A | | | | |
| 25. (A) $\rightarrow$ Q, S ; (B) $\rightarrow$ R, S ; (C) $\rightarrow$ P ; (D) $\rightarrow$ Q, R | | | | | |
| 26. C | 27. C | 28. B | 29. C | 30. B | 31. A |
| 32. D | 33. A | | | | |

Boron always forms covalent bond because boron requires very high energy of form B^{3+} and again B^{3+} due to its very small size having high polarising power thus cause greater polarisation and eventually significant covalent characteristics-Fajans rule.

34. C



Comprehension # 1 (Q. 35 to 37)

35. A 36. C 37. A or B

Comprehension # 2 (Q.38 to 40)

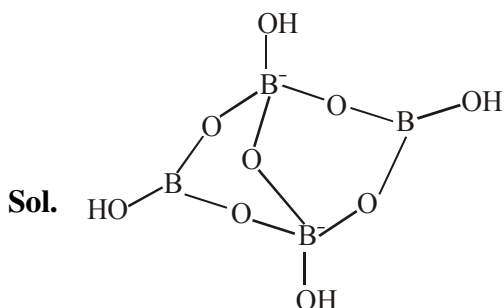
38. C 39. C 40. B 41. B 42. C 43. A
 44. B,D 45. B 46. A,C,D 47. A 48. A 49. D
 50. C 51. A 52. B,D 53. B,C 54. B,C,D 55. B
 56. 6

57. (B)

Sol. The order of radius of 13th group elements is $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$.

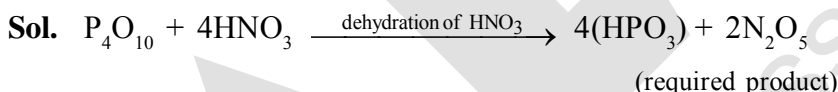
Reason $\Rightarrow$ Due to poor shielding effect of d-orbital, radius of Ga is smaller than Al.

58. (A,C,D)

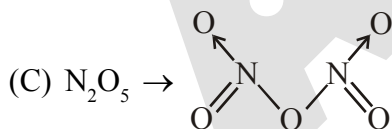


- (A) Having $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ tetranuclear (boron) unit
 (B) All boron atoms not in same plane
 (C) Two boron are sp^2 hybridised and two boron are sp^3 hybridised
 (D) One terminal hydroxide per boron atom is present.

59. (B,D)



- (A) $\text{P}_4 + 20\text{HNO}_3 \rightarrow 4\text{H}_3\text{PO}_4 + 20\text{NO}_2\uparrow + 4\text{H}_2\text{O}$
 (B) N_2O_5 is diamagnetic in nature



N_2O_5 contains one N-O-N bond not N-N bond.

- (D) $\text{Na} + \text{N}_2\text{O}_5 \rightarrow \text{NaNO}_3 + \text{NO}_2\uparrow$
 (Brown gas)

60. A,C,D 61. B,C 62. A 63. C,D
 64. A,C,D 65. B 66. C 67. B,C
 68. 5 or 6 69. (1,4) 70. (19.00) 71. (288.00)
 72. (2,4) 73. (4.00)

Important Notes

SURFACE CHEMISTRY

1. INTRODUCTION

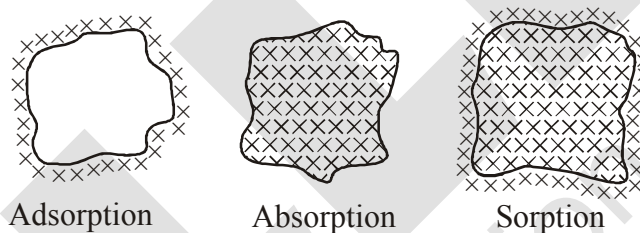
Surface chemistry deals with phenomena that occur at the surfaces or interfaces. The interface or surface is represented by separating the bulk phases by a hyphen or a slash. For example, the interface between a solid and a gas may be represented by solid-gas or solid/gas. Due to complete miscibility, there is no interface between the gases. The bulk phases that we come across in surface chemistry may be pure compounds or solutions. The interface is normally a few molecules thick but its area depends on the size of the particles of bulk phases. Many important phenomena, noticeable amongst these being corrosion, electrode processes, heterogeneous catalysis, dissolution and crystallisation occur at interfaces.

2. ADSORPTION

The tendency of accumulation of molecular species at the surface than in the bulk of a solid (or liquid) is termed **adsorption**. The molecular species or substance which concentrates or accumulates at the surface is termed **adsorbate** and the material on whose surface the adsorption has taken place is called **adsorbent**.

2.1 DISTINCTION BETWEEN ADSORPTION AND ABSORPTION :

In adsorption the concentration of the adsorbate increases only at the surface of the adsorbent, while in absorption the concentration is uniform throughout the bulk of the solid.



There are several examples, which reveal that the surface of a solid has the tendency to attract and retain the molecules of the phase with which it comes into contact. These molecules remain only at the surface and do not go deeper into the bulk. Adsorption is essentially a surface phenomenon. Solids, particularly in finely divided state, have large surface area and therefore, charcoal, silica gel, alumina gel, clay, colloids, metals in finely divided state, etc. act as good adsorbents.

- (i) In adsorption, the composition of final sample at surface & bulk will differ largely, while in absorption, the composition will be nearly same.
- (ii) The rate of adsorption decreases with time but absorption occurs at nearly constant rate.
- (iii) Adsorption is always exothermic but absorption may be endothermic or exothermic.

Adsorption in action :

- (i) If a gas like O_2 , H_2 , CO , Cl_2 , NH_3 or SO_2 is taken in a closed vessel containing powdered charcoal, it is observed that the pressure of the gas in the enclosed vessel decreases. The gas molecules concentrate at the surface of the charcoal, i.e., gases are adsorbed at the surface.
- (ii) In a solution of an organic dye, say methylene blue, when animal charcoal is added and the solution is well shaken, it is observed that the filtrate turns colourless. The molecules of the dye, thus, accumulate on the surface of charcoal, i.e., are adsorbed.

- (iii) Aqueous solution of raw sugar, when passed over beds of animal charcoal, becomes colourless as the colouring substances are adsorbed by the charcoal.
- (iv) The air becomes dry in the presence of silica gel because the water molecules get adsorbed on the surface of the gel.

In absorption as well as adsorption, the concentration or partial pressure of absorbate or adsorbate, decreases.

Ex. 1. A vessel of capacity 8.21L contains NH_3 gas at 1.5 atm and 27°C . Now, 5 gm charcoal is added in the vessel and left for sufficient time. After sufficient time, the pressure of gas decreased to 1.2 atm. Calculate the mass of NH_3 gas adsorbed per gm of charcoal. Neglect the volume of charcoal.

Sol. $\text{Mass of } \text{NH}_3 \text{ gas adsorbed} = \frac{\Delta P \cdot V \cdot M}{RT} = \frac{0.3 \times 8.21 \times 17}{0.0821 \times 300} = 1.7 \text{ gm}$

$\therefore \text{Mass of } \text{NH}_3 \text{ gas adsorbed per gm of charcoal} = \frac{1.7}{5} = 0.34 \text{ gm}$

2.2 MECHANISM OF ADSORPTION :

Adsorption is due to the fact that the surface particles of the adsorbent are in different state than the particles inside the bulk.

- (i) Inside the adsorbent all the force acting between the particles are mutually balanced but on the surface the particles are not surrounded by atoms or molecules of their kind on all sides and hence they possess **Unbalanced or residual attractive force**. These forces of the adsorbent are responsible for attracting the adsorbate particle on its surface.
- (ii) Another important factor featuring adsorption is the heat of adsorption. During adsorption, there is always a decrease in residual forces of the surface, i.e., there is decrease in surface energy which appears as heat. Adsorption, therefore, is invariably an exothermic process. In other words, ΔH of adsorption is always negative. When a gas is adsorbed, the freedom of movement of its molecules become restricted. This amounts to decrease in the entropy of the gas after adsorption, i.e., ΔS is negative. Adsorption is thus accompanied by decrease in enthalpy as well as decrease in entropy of the system. For a process to be spontaneous, the thermodynamic requirement is that, at constant temperature and pressure, ΔG must be negative, i.e., there is a decrease in Gibbs energy. On the basis of equation, $\Delta G = \Delta H - T\Delta S$, ΔG can be negative if ΔH has sufficiently high negative value as $-T\Delta S$ is positive. Thus, in an adsorption process, which is spontaneous, a combination of these two factors makes ΔG negative. As the adsorption proceeds, ΔH becomes less and less negative ultimately ΔH becomes equal to $T\Delta S$ and ΔG becomes zero. At this state equilibrium is attained.

The adsorbate particles may undergo the following further activity :

- (i) If nature & condition will favour, they will diffuse in bulk resulting in adsorption.
- (ii) They may move at the surface.
- (iii) They come out of the surface resulting desorption.

Normally, desorption follows I order kinetics & obeys Arrhenius equation.

Ex.2. Calculate the time upto which a hydrogen molecule may remain absorbed at pt surface at 500 K. ($A = 14 \times 10^{15} \text{ sec}^{-1}$, $E_a = 10 \text{ kcal/mole}$)

Sol.
$$T_{av} = \frac{1}{1.4 \times 10^{15} e^{-10}} = \frac{e^{10}}{1.4 \times 10^{15}}$$

2.3 TYPES OF ADSORPTION :

There are two main types of adsorption of gases on solids.

(i) Physical adsorption or physisorption :

In this adsorption, accumulation of gas on the surface of a solid occurs on account of weak VanderWaals' force.

(ii) Chemical adsorption or chemisorption.

The gas molecules or atoms are held to the solid surface by chemical bonds (covalent or ionic) in nature. Chemisorption has high energy of activation and is, therefore, often referred as **activated adsorption**.

Sometimes these two processes occur simultaneously and it is not easy to ascertain the type of adsorption.

Note :

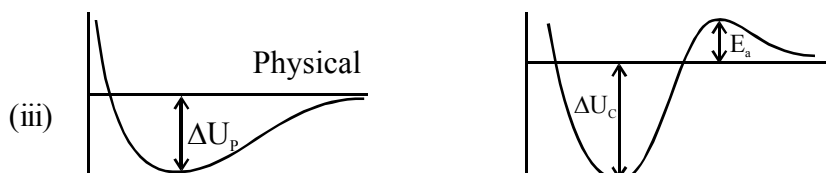
A physical adsorption at low temperature may pass into chemisorption as the temperature is increased. For example, hydrogen is first adsorbed on nickel by van der Waals' force. Molecules of hydrogen then dissociate and hydrogen atoms are held on the surface by chemisorption.

COMPARISON OF CHARACTERISTICS OF PHYSISORPTION AND CHEMISORPTION

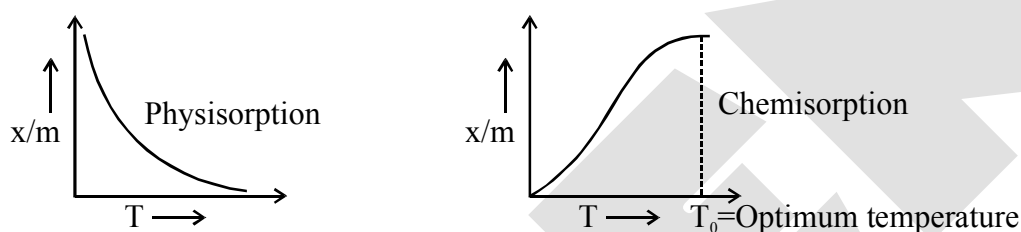
	PHYSICAL ADSORPTION	CHEMICAL ADSORPTION
1	It is caused by weak forces.	It is caused by chemical bond formation
2	It is not specific.	It is highly specific (Bond formation is necessary)
3	It is reversible.	It is irreversible.
4	It depends on the nature of gas. More easily liquefiable gases are adsorbed readily.	It depends on the nature of gas. Gases which form compounds with the adsorbent exhibit chemi-sorption.
5	Enthalpy of adsorption is low ($20-40 \text{ kJ mol}^{-1}$)	Enthalpy of adsorption is low ($80-240 \text{ kJ mol}^{-1}$)
6	Low temperature is favourable. It decreases with increase of temperature.	High temperature is favourable. It increases with the increase in temperature upto certain limit.
7	No appreciable activation energy is involved.	High activation energy is involved.
8	High pressure is favourable. Decrease of pressure causes desorption.	High pressure is favourable. Decrease of pressure does not cause desorption.
9	It forms multilayers on adsorbent surface under high pressure.	It forms unimolecular layer.

Note :

- (i) Chemisorption is irreversible. Either desorption does not occur or molecules other than adsorbate leave the surface.
- (ii) Chemisorption is more exothermic & hence its E_a should be negligible. But in reality it is high due to requirement of bond cleavage in the adsorbate molecule before adsorption. Due to higher E_a , it is called **activated adsorption**.



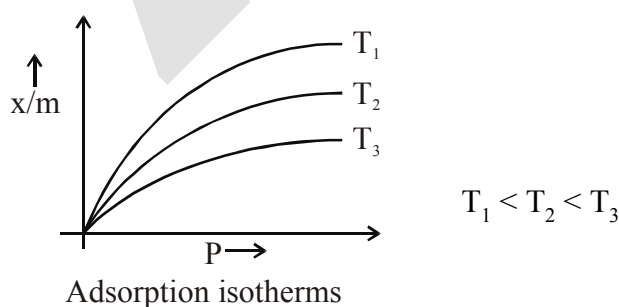
- (iv) On increasing temperature, the extent of physisorption decreases but chemisorption increases.



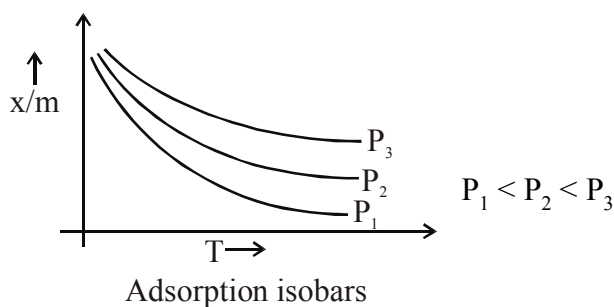
x : mass of adsorbate
and m : mass of adsorbent

2.4 FACTORS AFFECTING ADSORPTION OF GASES ON SOLIDS :

- (i) **Nature of gas :** Easily liquifiable gases (having higher critical temperature) adsorb to the greater extent.
- (ii) **Nature of adsorbent :** Specific surface area represent the surface area of adsorbent per unit mass. Greater the specific surface area, more will be the adsorption. Charcoal, silica gel, alumina gel, colloids are good adsorbents.
- (iii) **Surface area :** Adsorption increases on increasing the surface area.
- (iv) **Pressure :** Adsorption of gas at solid surface (increases due to) decrease in volume of system & hence extent of adsorption increases on increasing pressure.



(v) **Temperature :** $T \uparrow$ Physisorption $\downarrow$ Chemisorption $\uparrow$



2.4.1 Freundlich adsorption isotherm :

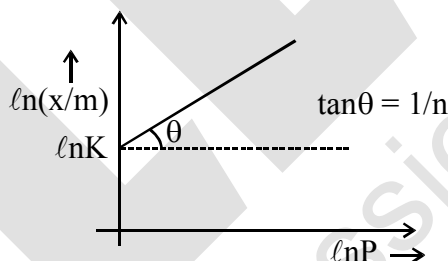
Freundlich, in 1909, gave an empirical relationship between the quantity of gas adsorbed (x) by unit mass of solid adsorbent (m) and pressure at a particular temperature. The relationship can be expressed by the following equation.

$$\frac{x}{m} = K \cdot P^{1/n} \quad (n \geq 1)$$

where K & n are constant, depending on nature of gas, solid & temperature.

For any combination of gas, solid & temperature, the values of k & n may be determined graphically.

$$\ln\left(\frac{x}{m}\right) = \ln K + \frac{1}{n} \ln P$$

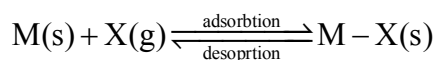


Freundlich isotherm is straight line. Freundlich isotherm fails at high pressure.

2.4.2 Langmuir's adsorption isotherm :

He derived the condition of equilibrium theoretically assuming that adsorption is

- monolayer
- uniform at solid surface
- adsorbate particles do not interact each other at the surface.



At any instant :

Rate of adsorption, $r_a \propto P$

$$\propto (1 - \theta)$$

$$\therefore r_a = K_a P(1 - \theta)$$

where, θ = Fraction of surface area already occupied by adsorbate particles.

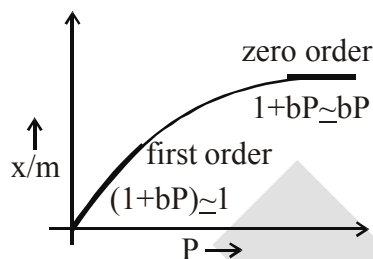
and Rate of desorption, $r_d \propto \theta \Rightarrow r_d = K_d \cdot \theta$

At eqb. $r_a = r_d$

$$\theta = \frac{K_a P}{K_d + K_a P} = \frac{\cancel{K_a} P}{1 + \cancel{K_a} / \cancel{K_d} P} = \frac{K_{eq} P}{1 + K_{eq} P}$$

Now, $\frac{x}{m} \propto \theta$

$$\frac{x}{m} = K\theta = K \frac{K_{eq} P}{1 + K_{eq} P} = \frac{aP}{1 + bP} \quad a, b = \text{constant}$$



2.5 ADSORPTION FROM SOLUTION PHASE :

Solids can adsorb solutes from solutions also. When a solution of acetic acid in water is shaken with charcoal, a part of the acid is adsorbed by the charcoal and the concentration of the acid decreases in the solution. Similarly, the litmus solution when shaken with charcoal becomes colourless. The precipitate of $Mg(OH)_2$ attains blue colour when precipitated in presence of magneson reagent. The colour is due to adsorption of magneson. The following observations have been made in the case of adsorption from solution phase:

- (i) The extent of adsorption decreases with an increase in temperature.
- (ii) The extent of adsorption increases with an increase of surface area of the adsorbent.
- (iii) The extent of adsorption depends on the concentration of the solute in solution.
- (iv) The extent of adsorption depends on the nature of the adsorbent and the adsorbate.

The precise mechanism of adsorption from solution is not known. Freundlich's equation approximately describes the behaviour of adsorption from solution with a difference that instead of pressure, concentration of the solution is taken into account, i.e.,

$$\frac{x}{m} = kC^{1/n}$$

(C is the equilibrium concentration, i.e., when adsorption is complete).

On taking logarithm of the above equation, we have

$$\log \frac{x}{m} = \log k + \frac{1}{n} \log C$$

Plotting $\log \frac{x}{m}$ against $\log C$ a straight line is obtained which shows the validity of Freundlich isotherm.

This can be tested experimentally by taking solutions of different concentrations of acetic acid. Equal volumes of solutions are added to equal amounts of charcoal in different flasks. The final concentration is determined in each flask after adsorption. The difference in the initial and final concentrations give the value of x. Using the above equation, validity of Freundlich isotherm can be established.

2.6 APPLICATIONS OF ADSORPTION :

The phenomenon of adsorption finds a number of applications. Important ones are listed here:

(i) **Production of high vacuum :**

The remaining traces of air can be adsorbed by charcoal from a vessel evacuated by a vacuum pump to give a very high vacuum.

(ii) **Gas masks:**

Gas mask (a device which consists of activated charcoal or mixture of adsorbents) is usually used for breathing in coal mines to adsorb poisonous gases.

(iii) **Control of humidity:**

Silica and aluminium gels are used as adsorbents for removing moisture and controlling humidity.

(iv) **Removal of colouring matter from solutions:**

Animal charcoal removes colours of solutions by adsorbing coloured impurities.

(v) **Heterogeneous catalysis:**

Adsorption of reactants on the solid surface of the catalysts increases the rate of reaction. There are many gaseous reactions of industrial importance involving solid catalysts. Manufacture of ammonia using iron as a catalyst, manufacture of H_2SO_4 by contact process and use of finely divided nickel in the hydrogenation of oils are excellent examples of heterogeneous catalysis.

(vi) **Separation of inert gases:**

Due to the difference in degree of adsorption of gases by charcoal, a mixture of noble gases can be separated by adsorption on coconut charcoal at different temperatures.

(vii) **In curing diseases:** A number of drugs are used to kill germs by getting adsorbed on them.

(viii) **Froth floatation process:**

A low grade sulphide ore is concentrated by separating it from silica and other earthy matter by this method using pine oil and frothing agent

(ix) **Adsorption indicators:**

Surfaces of certain precipitates such as silver halides have the property of adsorbing some dyes like eosin, fluorescein, etc. and thereby producing a characteristic colour at the end point.

(x) **Chromatographic analysis:**

Chromatographic analysis based on the phenomenon of adsorption finds a number of applications in analytical and industrial fields.

3. CATALYSIS

The systematic study of the effect of various foreign substances on the rates of chemical reactions was first made by Berzelius, in 1835. He suggested the term catalyst for such substances. Substances, which accelerate the rate of a chemical reaction and themselves remain chemically and quantitatively unchanged after the reaction, are known as catalysts, and the phenomenon is known as catalysis.

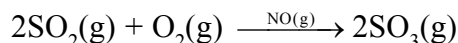
3.1 HOMOGENEOUS AND HETEROGENEOUS CATALYSIS :

Catalysis can be broadly divided into two groups:

(a) Homogeneous catalysis :

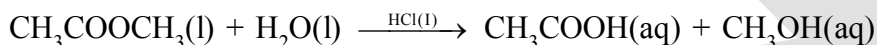
When the reactants and the catalyst are in the same phase (i.e., liquid or gas), the process is said to be homogeneous catalysis. The following are some of the examples of homogeneous catalysis:

- (i) Oxidation of sulphur dioxide into sulphur trioxide with dioxygen in the presence of oxides of nitrogen as the catalyst in the lead chamber process.



The reactants, sulphur dioxide and oxygen, and the catalyst, nitric oxide, are all in the same phase.

- (ii) Hydrolysis of methyl acetate is catalysed by H^+ ions furnished by hydrochloric acid.

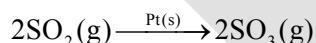


Both the reactants and the catalyst are in the same phase.

3.2 HETEROGENEOUS CATALYSIS :

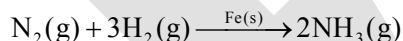
The catalytic process in which the reactants and the catalyst are in different phases is known as heterogeneous catalysis. Some of the examples of heterogeneous catalysis are given below:

- (i) Oxidation of sulphur dioxide into sulphur trioxide in the presence of Pt.



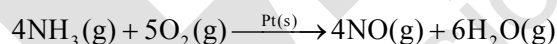
The reactant is in gaseous state while the catalyst is in the solid state.

- (ii) Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron in Haber's process.



The reactants are in gaseous state while the catalyst is in the solid state.

- (iii) Oxidation of ammonia into nitric oxide in the presence of platinum gauze in Ostwald's process.



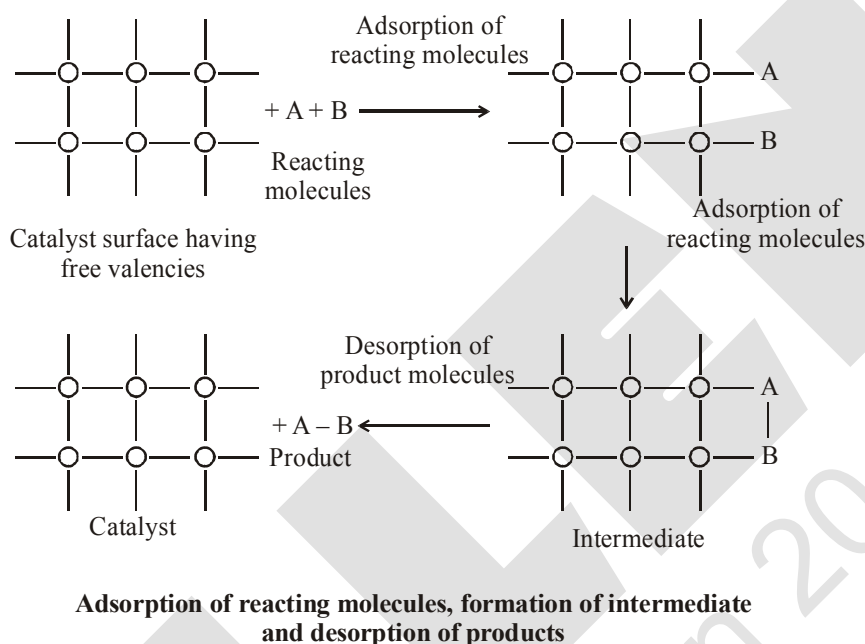
The reactants are in gaseous state while the catalyst is in the solid state

Adsorption Theory of Heterogeneous Catalysis :

This theory explains the mechanism of heterogeneous catalysis. The old theory, known as adsorption theory of catalysis, was that the reactants in gaseous state or in solutions, are adsorbed on the surface of the solid catalyst. The increase in concentration of the reactants on the surface increases the rate of reaction. Adsorption being an exothermic process, the heat of adsorption is utilised in enhancing the rate of the reaction. The modern adsorption theory is the combination of intermediate compound formation theory and the old adsorption theory. The catalytic activity is localised on the surface of the catalyst. The mechanism involves five steps:

- Diffusion of reactants to the surface of the catalyst.
- Adsorption of reactant molecules on the surface of the catalyst.
- Occurrence of chemical reaction on the catalyst's surface through formation of an intermediate
- Desorption of reaction products from the catalyst surface, and thereby, making the surface available again for more reaction to occur.

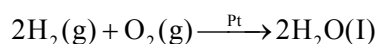
- (v) Diffusion of reaction products away from the catalyst's surface. The surface of the catalyst unlike the inner part of the bulk, has free valencies which provide the seat for chemical forces of attraction. When a gas comes in contact with such a surface, its molecules are held up there due to loose chemical combination. If different molecules are adsorbed side by side, they may react with each other resulting in the formation of new molecules. Thus, formed molecules may evaporate leaving the surface for the fresh reactant molecules. This theory explains why the catalyst remains unchanged in mass and chemical composition at the end of the reaction and is effective even in small quantities. It however, does not explain the action of catalytic promoters and catalytic poisons.



3.3 IMPORTANT FEATURES OF SOLID CATALYSTS :

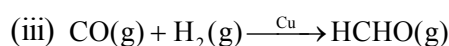
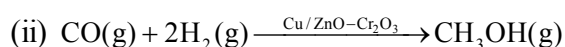
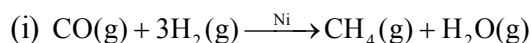
(a) Activity :

The activity of a catalyst depends upon the strength of chemisorption to a large extent. The reactants must get adsorbed reasonably strongly on to the catalyst to become active. However, they must not get adsorbed so strongly that they are immobilised and other reactants are left with no space on the catalyst's surface for adsorption. It has been found that for hydrogenation reaction, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity being shown by groups 7-9 elements of the periodic table



(b) Selectivity :

The selectivity of a catalyst is its ability to direct a reaction to yield a particular product. For example, starting with H_2 and CO , and using different catalysts, we get different products.



Thus, it can be inferred that the action of a catalyst is highly selective in nature, i.e., a given substance can act as a catalyst only in a particular reaction and not for all the reactions.

3.4 SHAPE-SELECTIVE CATALYSIS BY ZEOLITES :

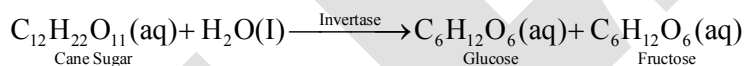
The catalytic reaction that depends upon the pore structure of the catalyst and the size of the reactant and product molecules is called shape-selective catalysis. Zeolites are good shape-selective catalysts because of their honeycomb-like structures. They are microporous aluminosilicates with three dimensional network of silicates in which some silicon atoms are replaced by aluminium atoms giving Al–O–Si framework. The reactions taking place in zeolites depend upon the size and shape of reactant and product molecules as well as upon the pores and cavities of the zeolites.

They are found in nature as well as synthesised for catalytic selectivity. Zeolites are being very widely used as catalysts in petrochemical industries for cracking of hydrocarbons and isomerisation. An important zeolite catalyst used in the petroleum industry is ZSM-5. It converts alcohols directly into gasoline (petrol) by dehydrating them to give a mixture of hydrocarbons.

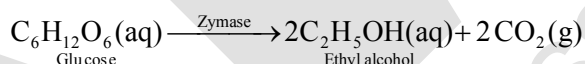
3.5 ENZYME CATALYSIS :

Enzymes are complex nitrogenous organic compounds which are produced by living plants and animals. They are actually protein molecules of high molecular mass and form colloidal solutions in water. They are very effective catalysts, numerous reactions that occur in the bodies of animals and plants to maintain the life process are catalysed by enzymes. The enzymes are, thus, termed as **biochemical catalysts** and the phenomenon is known as **biochemical catalysis**. Many enzymes have been obtained in pure crystalline state from living cells. However, the first enzyme was synthesised in the laboratory in 1969. The following are some of the examples of enzyme-catalysed reactions:

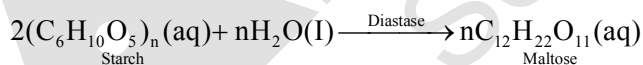
- (i) Inversion of cane sugar: The invertase enzyme converts cane sugar into glucose and fructose.



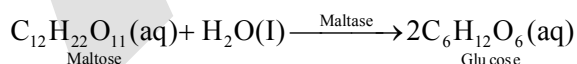
- (ii) Conversion of glucose into ethyl alcohol: The zymase enzyme converts glucose into ethyl alcohol and carbon dioxide.



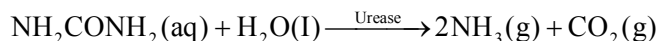
- (iii) Conversion of starch into maltose: The diastase enzyme converts starch into maltose.



- (iv) Conversion of maltose into glucose: The maltase enzyme converts maltose into glucose.



- (v) Decomposition of urea into ammonia and carbon dioxide: The enzyme urease catalyses this decomposition.



- (vi) In stomach, the pepsin enzyme converts proteins into peptides while in intestine, the pancreatic trypsin converts proteins into amino acids by hydrolysis.

- (vii) Conversion of milk into curd: It is an enzymatic reaction brought about by lacto bacilli enzyme present in curd.

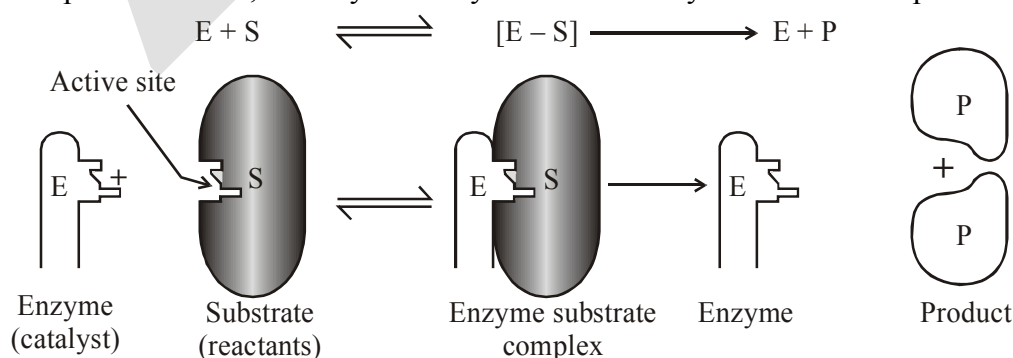
3.5.1 Characteristics of enzyme catalysis :

Enzyme catalysis is unique in its efficiency and high degree of specificity. The following characteristics are exhibited by enzyme catalysts:

- (i) Most highly efficient: One molecule of an enzyme may transform one million molecules of the reactant per minute.
- (ii) Highly specific nature: Each enzyme is specific for a given reaction, i.e., one catalyst cannot catalyse more than one reaction. For example, the enzyme urease catalyses the hydrolysis of urea only. It does not catalyse hydrolysis of any other amide.
- (iii) Highly active under optimum temperature: The rate of an enzyme reaction becomes maximum at a definite temperature, called the optimum temperature. On either side of the optimum temperature, the enzyme activity decreases. The optimum temperature range for enzymatic activity is 298-310K. Human body temperature being 310 K is suited for enzyme-catalysed reactions.
- (iv) Highly active under optimum pH: The rate of an enzyme-catalysed reaction is maximum at a particular pH called optimum pH, which is between pH values 5-7.
- (v) Increasing activity in presence of activators and co-enzymes: The enzymatic activity is increased in the presence of certain substances, known as co-enzymes. It has been observed that when a small non-protein (vitamin) is present along with an enzyme, the catalytic activity is enhanced considerably. Activators are generally metal ions such as Na^+ , Mn^{2+} , CO^{2+} , Cu^{2+} , etc. These metal ions, when weakly bonded to enzyme molecules, increase their catalytic activity. Amylase in presence of sodium chloride i.e., Na^+ ions are catalytically very active.
- (vi) Influence of inhibitors and poisons: Like ordinary catalysts, enzymes are also inhibited or poisoned by the presence of certain substances. The inhibitors or poisons interact with the active functional groups on the enzyme surface and often reduce or completely destroy the catalytic activity of the enzymes. The use of many drugs is related to their action as enzyme inhibitors in the body.

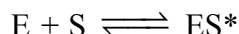
3.5.2 Mechanism of enzyme catalysis :

There are a number of cavities present on the surface of colloidal particles of enzymes. These cavities are of characteristic shape and possess active groups such as $-\text{NH}_2$, $-\text{COOH}$, $-\text{SH}$, $-\text{OH}$, etc. These are actually the active centres on the surface of enzyme particles. The molecules of the reactant (substrate), which have complementary shape, fit into these cavities just like a key fits into a lock. On account of the presence of active groups, an activated complex is formed which then decomposes to yield the products. Thus, the enzyme-catalysed reactions may be considered to proceed in two steps.

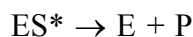


3.5.3 Mechanism of enzyme catalysed reaction :

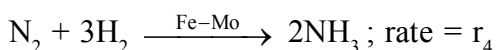
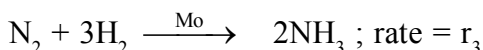
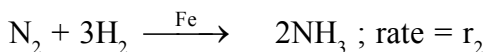
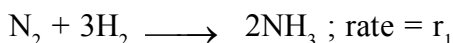
Step 1: Binding of enzyme to substrate to form an activated complex.



Step 2: Decomposition of the activated complex to form product.

**3.6 CATALYTIC PROMOTORS :**

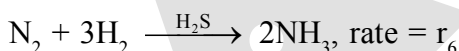
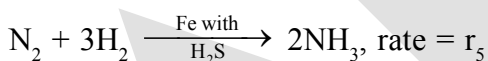
There are the substances, which are not the catalyst of that reaction but increases the activity of catalyst.



Experimentally, $r_1 = r_3 < r_2 < r_4$. It suggests that Mo is not catalyst of this reaction. But when it is mixed with iron, it increased the catalytic activity of iron. Hence, Mo is catalytic promotor in this reaction.

3.7 CATALYTIC POISONS :

These are the substances, which are not the catalyst of that reaction but decreases the activity of catalyst.

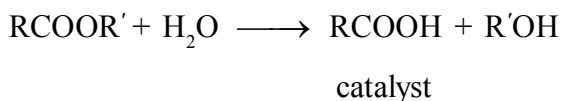


Experimentally, $r_1 = r_6 < r_5 < r_2$. It suggests that H_2S is not catalyst of reaction but it decreases the activity of Fe catalyst. Hence, H_2S is catalytic poison in this reaction.

3.8 AUTO CATALYSIS :

These are the reactions in which some of the product of reaction, catalyses the reaction.

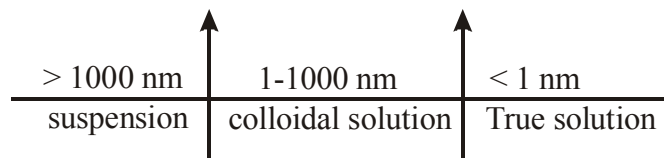
For example



4. COLLOIDS

A colloid is a heterogeneous system in which one substance is dispersed (dispersed phase) as very fine particles in another substance called dispersion medium.

4.1 PARTICLE SIZE :



4.2 PHASE OF COLLOIDS :

A colloidal system is heterogeneous in character. It consists of two phases, namely a dispersed phase and a dispersion medium.

- (a) **Dispersed Phase :** It is the component present in small proportion and is just like a solute in a true solution. For example, in the colloidal state of sulphur in water, the former acts as a dispersed phase.
- (b) **Dispersion Medium :** It is normally the component present in excess and is just like a solvent in a solution.

The particles of the dispersed phase are scattered in the dispersion medium in a colloidal system.

4.3 CLASSIFICATION OF COLLOIDS :

Colloids can be classified in a number of ways based upon some of their important characteristics.

4.3.1 Physical state of Dispersed Phase & Dispersion Medium:

Depending upon whether the dispersed phase and the dispersion medium are solids, liquids or gaseous, eight types of colloidal system are possible.

A gas mixed with another gas forms a homogeneous mixture and not a colloidal system. Typical examples of various type alongwith their characteristic names are given in table.

Types of Colloidal Systems

Dispersed phase	Dispersion medium	Type of colloid	Examples
Solid	Solid	Solid sol	Some coloured glasses and gem stones
Solid	Liquid	Sol	Paints, cell fluids
Solid	Gas	Aerosol	Smoke, dust
Liquid	Solid	Gel	Cheese, butter, jellies
Liquid	Liquid	Emulsion	Milk, hair cream
Liquid	Gas	Aerosol	Fog, mist, cloud, insecticide sprays
Gas	Solid	Solid sol	Pumice stone, foam rubber.
Gas	Liquid	Foam	Froth, whipped cream, soap lather

- * A colloidal system in which the dispersion medium is a liquid or gas are called sols. They are called hydrosols or aqua sols, if the dispersion medium is water. When the dispersion medium is alcohol or benzene, they are accordingly called alcosols or benzosol.

- * Colloidal systems in which the dispersion medium is a gas are called aerosols.
- * Colloids in which the dispersion medium is a solid are called gels, e.g. cheese etc. They have a more rigid structure. Some colloids, such as gelatin, can behave both as a sol and a gel. At high temperature and low concentration of gelatin, the colloid is a hydrosol. But at low temperature and high gelatin concentration, the hydrosol can change into a gel.

4.3.2 Based on interaction or affinity of phases : On the basis of the affinity or interaction between the dispersed phase and the dispersion medium, the colloids may be classified into two types :

- (i) **Lyophilic colloids :** The word 'lyophilic' means liquid-loving. Colloidal sols directly formed by mixing substances like gum, gelatine, starch, rubber, etc., with a suitable liquid (the dispersion medium) are called lyophilic sols. An important characteristic of these sols is that if the dispersion medium is separated from the dispersed phase (say by evaporation), the sol can be reconstituted by simply remixing with the dispersion medium. That is why these sols are also called reversible sols. Which are quite stable.
- (ii) **Lyophobic colloids :** The word 'lyophobic' means liquid-hating. Substances like metals, their sulphides, etc., when simply mixed with the dispersion medium do not form the colloidal sol. Their colloidal sols can be prepared only by special methods. Such sols are called lyophobic sols. These sols are readily precipitated (or coagulated) on the addition of small amounts of electrolytes, by heating or by shaking and hence, are not stable. Further, once precipitated, they do not give back the colloidal sol by simple addition of the dispersion medium. Hence, these sols are also called irreversible sols. Lyophobic sols need stabilising agents for their preservation.

Difference between Lyophilic and Lyophobic sols

Property	Lyophilic sols	Lyophobic sols
1. Nature	Reversible	Irreversible
2. Preparation	They are prepared very easily by shaking or warming the substance with dispersion medium. They do not required any electrolyte for stabilization.	They are difficult to prepare, Special methods are used. Addition of stabiliser is essential for their stability.
3. Stability	They are very stable and are not easily coagulated by electrolytes.	They are generally unstable and get easily coagulated on addition of electrolytes.
4. Charge	Particles carry no or very little charge depending upon the pH of the medium.	Colloidal particles have characteristic charge (positive or negative)
5. Viscosity	Viscosity is much higher than that of the medium.	Viscosity is nearly the same as that of the medium
6. Migration in electric field	The particles may or may not migrate in an electric field.	The colloidal particles migrate either towards cathode or anode in an electric field.
7. Solvation	Particles are heavily solvated.	Particles are not solvated.
8. Action of electrolyte	Large amount of electrolyte is required to cause coagulation.	Small amount of electrolyte is sufficient to cause cogulation.
9. Examples	Mostly organic substances e.g. starch, gums, proteins, gelatin etc.	Generally inorganic substance e.g., metal sols, sulphides and oxides sols.

4.3.3 Classification Based on Type of Particles of the Dispersed Phase :

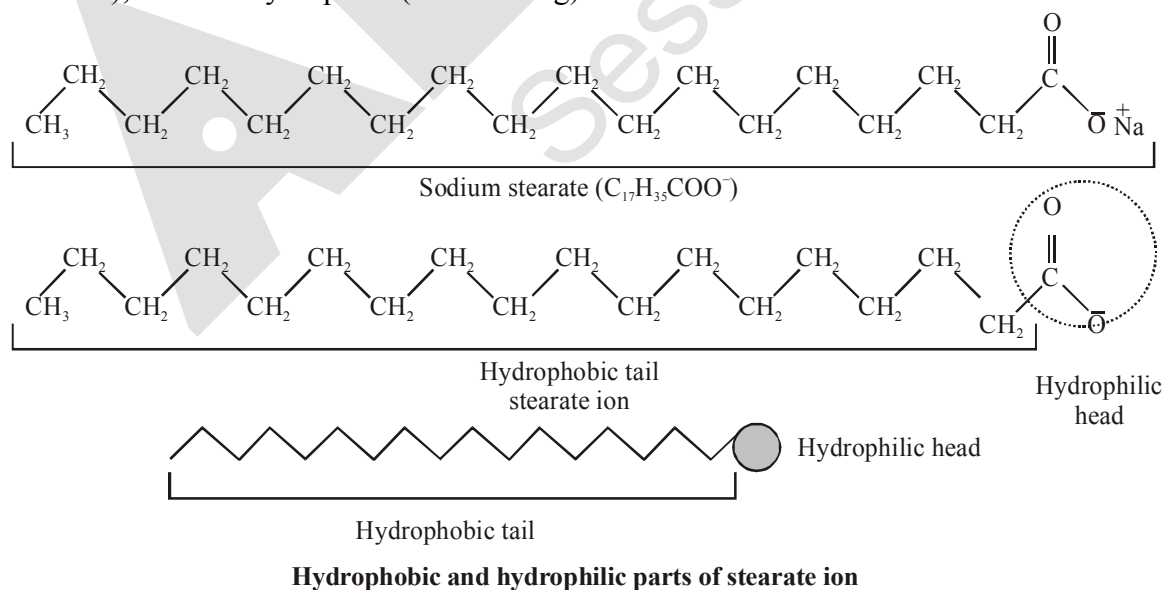
Depending upon the type of the particles of the dispersed phase, colloids are classified as: multimolecular, macromolecular and associated colloids.

- (i) **Multimolecular colloids:** On dissolution, a large number of atoms or smaller molecules of a substance aggregate together to form species having size in the colloidal range (diameter $< 1\text{ nm}$). The species thus formed are called multimolecular colloids. For example, a gold sol may contain particles of various sizes having many atoms. Sulphur sol consists of particles containing a thousand or more of S_8 sulphur molecules.
- (ii) **Macromolecular colloids:** Macromolecules (Unit 15) in suitable solvents form solutions in which the size of the macromolecules may be in the colloidal range. Such systems are called macromolecular colloids. These colloids are quite stable and resemble true solutions in many respects. Examples of naturally occurring macromolecules are starch, cellulose, proteins and enzymes; and those of man-made macromolecules are polythene, nylon, polystyrene, synthetic rubber, etc.
- (iii) **Associated colloids (Micelles):** There are some substances which at low concentrations behave as normal strong electrolytes, but at higher concentrations exhibit colloidal behaviour due to the formation of aggregates. The aggregated particles thus formed are called micelles. These are also known as associated colloids.

The formation of micelles takes place only above a particular temperature called **Kraft temperature** (T_k) and above a particular concentration called **critical micelle concentration** (CMC). On dilution, these colloids revert back to individual ions. Surface active agents such as soaps and synthetic detergents belong to this class. These colloids have both lyophobic and lyophilic parts. Micelles may contain as many as 100 molecules or more.

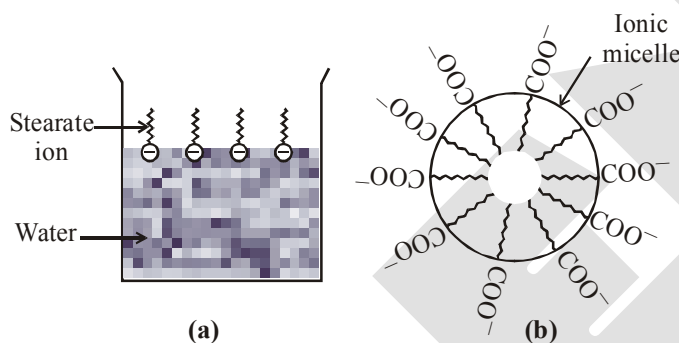
Mechanism of micelle formation :

Let us take the example of soap solutions. Soap is sodium or potassium salt of a higher fatty acid and may be represented as RCOO^-Na^+ (e.g., sodium stearate $\text{CH}_3(\text{CH}_2)_{16}\text{COO}^-\text{Na}^+$, which is a major component of many bar soaps). When dissolved in water, it dissociates into RCOO^- and Na^+ ions. The RCOO^- ions, however, consist of two parts — a long hydrocarbon chain R (also called non-polar 'tail') which is hydrophobic (water repelling), and a polar group COO^- (also called polar-ionic 'head'), which is hydrophilic (water loving).



The RCOO^- ions are, therefore, present on the surface with their COO^- groups in water and the hydrocarbon chains R staying away from it and remain at the surface. But at critical micelle concentration, the anions are pulled into the bulk of the solution and aggregate to form a spherical shape with their hydrocarbon chains pointing towards the centre of the sphere with COO^- part remaining outward on the surface of the sphere. An aggregate thus formed is known as 'ionic micelle'. These micelles may contain as many as 100 such ions.

Similarly, in case of detergents, e.g., sodium laurylsulphate, $\text{CH}_3(\text{CH}_2)_{11}\text{SO}_4^- \text{Na}^+$, the polar group is $-\text{SO}_4^-$ along with the long hydrocarbon chain. Hence, the mechanism of micelle formation here also is same as that of soaps.

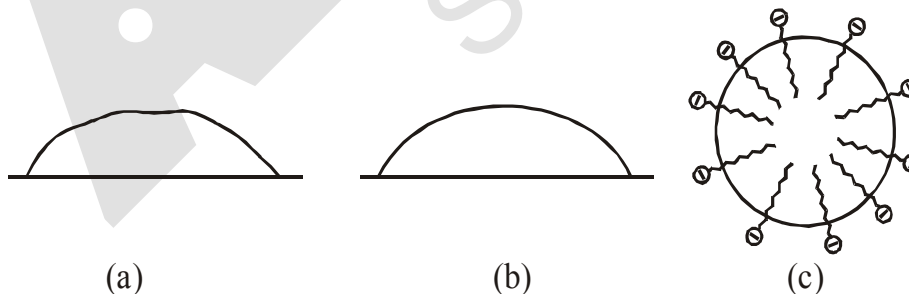


(a) Arrangement of stearate ions on the surface of water at low concentration of soap

(b) Arrangement of stearate ions inside the bulk of water (ionic micelle) at critical micelle concentration of soap

Cleansing action of soaps :

It has been mentioned earlier that a micelle consists of a hydrophobic hydrocarbon – like central core. The cleansing action of soap is due to the fact that soap molecules form micelle around the oil droplet in such a way that hydrophobic part of the stearate ions is in the oil droplet and hydrophilic part projects out of the grease droplet like the bristles. Since the polar groups can interact with water, the oil droplet surrounded by stearate ions is now pulled in water and removed from the dirty surface. The negatively charged sheath around the globules prevents them from coming together and forming aggregates.



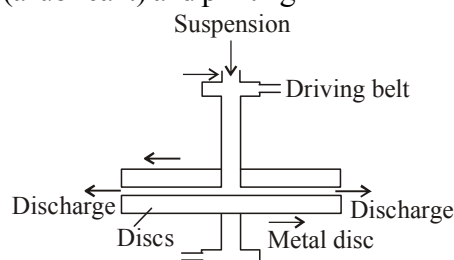
(a) Grease on cloth (b) Stearate ions arranging around the grease droplet and
(c) Grease droplet surrounded by stearate ions (micelle formed)

LIST OF SURFACTANTS AND THEIR CRITICAL MICELLE CONCENTRATION (CMC)

Surfactant	CMC (g/l)	Temp.(°C)
Anionic		
$\text{CH}_3(\text{CH}_2)_6\text{COONa}$	6.5×10^1	20
$\text{CH}_3(\text{CH}_2)_{10}\text{COONa}$	5.6	20–70
$\text{CH}_3(\text{CH}_2)_7\text{OSO}_3\text{Na}$	3.0×10^1	25–50
$\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3\text{Na}$	2.6	25–60
$\text{CH}_3(\text{CH}_2)_5\text{C}_6\text{H}_4\text{SO}_3\text{Na}$	9.8	75
$\text{CH}_3(\text{CH}_2)_{11}\text{C}_6\text{H}_4\text{SO}_3\text{Na}$	4.0×10^{-1}	50–75
Cationic		
$\text{CH}_3(\text{CH}_2)_9\text{NH}_2\cdot\text{HCl}$	8.5	25
$\text{CH}_3(\text{CH}_2)_{11}\text{NH}_2\text{HCl}$	2.7	30–50
$\text{CH}_3(\text{CH}_2)_7\text{N}(\text{CH}_3)_3\text{Br}$	7.8×10^1	25
$\text{CH}_3(\text{CH}_2)_{11}\text{N}(\text{CH}_3)_3\text{Br}$	5.4	25
Non-ionic		
$\text{CH}_3(\text{CH}_2)_7\text{C}_6\text{H}_{11}\text{O}_6$	7.3	25
$\text{C}_{12}\text{H}_{20}\text{O}_9(\text{C}_{16}\text{H}_{31}\text{O}_2)_2$	1.1×10^{-2}	20
$\text{CH}_3(\text{CH}_2)_{10}\text{COOC}_{12}\text{H}_{21}\text{O}_{10}$	7.1×10^{-3}	50

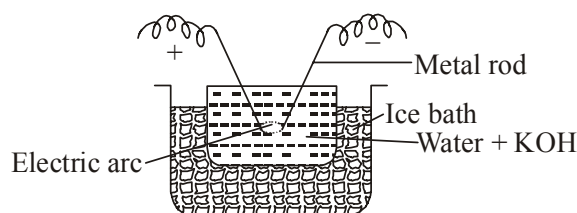
4.4 PREPARATION OF COLLOIDAL SOLUTIONS :

- (1) **Preparation of lyophilic sols :** The colloidal solutions of lyophilic colloids like starch, glue, gelatin etc., can be readily prepared by dissolving these substances in water either by cooling or warming. Solutions of colloidal electrolytes such as soaps and dye stuffs can also be prepared similarly.
 - (2) **Preparation of lyophobic sols :** To get a substance in colloidal form either the substance in bulk is broken down into fine particles of colloidal dimension (1Å to 10^3Å) or increasing the size of molecular particles as to form larger aggregates. In some cases, a third substance is added to increase the stability of the sol. These substances are called stabilizers. Thus, there are two ways by which the lyophobic sols can be prepared :
 - (i) **Dispersion methods :** By splitting coarse aggregates of a substance into colloidal size.
 - (ii) **Condensation methods :** By aggregating very small particles (atoms, ions or molecules) into colloidal size.
- (i) Dispersion Methods :**
- (A) **Mechanical dispersion :** Solid material is first finely ground by usual methods. It is then mixed with dispersion medium which gives a coarse suspension. The suspension is now introduced into the colloid mill. The simplest form of colloid mill consists of two metal discs held at a small distance apart from one another and capable of revolving at a very high speed (about 7000 revolutions per minute) in opposite directions. The particles are ground down to colloidal size and are then dispersed in the liquid. A stabilizer is often added to stabilize the colloidal solution. Colloidal graphite (a lubricant) and printing ink



are made by this method. Tannin is used as a stabilizer in the preparation of colloidal graphite and gum arabic in lampblack colloidal solution (Indian ink).

- (B) **Electro-dispersion (Bredig's arc method)** : This method is suitable for the preparation of colloidal solutions of metals like gold, silver, platinum, etc. An arc is struck between the metal electrodes under the surface of water



container in an ice bath. The intense heat of the arc vaporises some of the metal which condenses under cold water.

- (C) **Ultrasonic dispersion** : The sound waves of high frequency are usually called ultrasonic waves. These waves can be produced when quartz crystal discs are connected with a high frequency generator. The application of ultrasonic waves for the preparation of colloidal solutions was first introduced by wood and Loomis, in 1927. Various substances like oils, mercury, sulphur, sulphides and oxides of metals can be dispersed into colloidal state very easily with the help of ultrasonic waves.

- (D) **Peptization** : Peptization may be defined as the process of converting a precipitate into colloidal sol by shaking it with dispersion medium in the presence of a small amount of electrolyte. The electrolyte used for this purpose is called peptizing agent. This method is applied, generally, to convert a freshly prepared precipitate into a colloidal sol. During peptization, the precipitate adsorbs one of the ions of the electrolyte on its surface. This causes the development of positive or negative charge on precipitates, which ultimately break up into smaller particles of the size of a colloid.

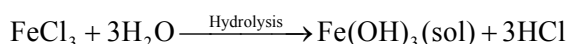
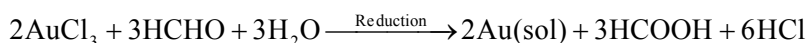
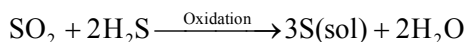
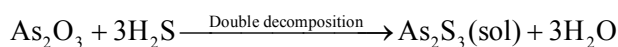
A few examples of sols obtained by peptization are :

- (i) Freshly prepared ferric hydroxide on treatment with a small amount of ferric chloride solution at once forms a dark reddish brown solution. Ferric chloride acts as a peptizing agent.
- (ii) Freshly precipitated silver chloride can be converted into a colloidal solution by a small amount of hydrochloric acid.
- (iii) Cadmium sulphide can be peptized with the help of hydrogen sulphide.

(ii) **Condensation Methods :**

- (A) **By exchange of solvents** : If a solution of sulphur or phosphorus prepared in alcohol is poured into water, a colloidal solution of sulphur or phosphorus is obtained due to low solubility of sulphur or phosphorus in water. Thus, there are a number of substances whose colloidal solutions can be prepared by taking a solution of the substance in one solvent and pouring it into another solvent in which the substance is relatively less soluble.
- (B) **By change of physical state** : Colloidal solutions of certain elements such as mercury and sulphur are obtained by passing their vapour through cold water containing a stabilizer (an ammonium salt or a citrate)

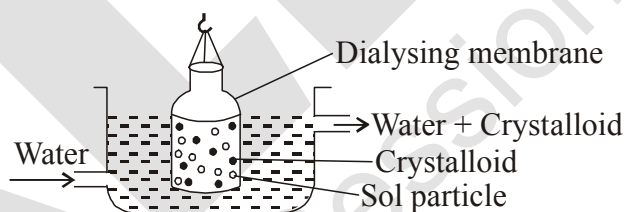
- (C) **Chemical methods** : Colloidal solutions can be prepared by chemical reactions leading to formation of molecules by double decomposition, oxidation, reduction or hydrolysis. These molecules then aggregate leading to formation of sols.



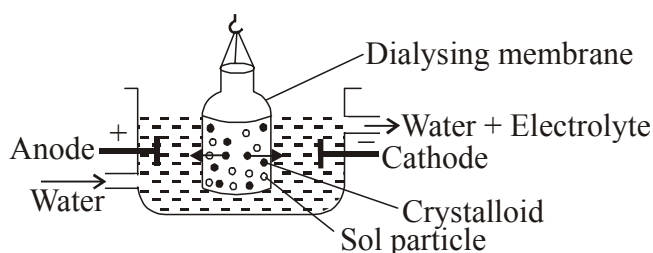
4.5 PURIFICATION OF COLLOIDAL SOLUTIONS :

Colloidal solutions prepared by above methods generally contain excessive amount of electrolytes and some other soluble impurities. The presence of traces of electrolyte is essential for the stability of the colloidal solution but larger quantities coagulate it. It is, therefore, necessary to reduce the concentration of these soluble impurities to a requisite minimum. The process used for reducing of these impurities to a requisite minimum is known as purification of colloidal solution. The purification of colloidal solution is carried out by the following methods.

- (i) **Dialysis** : It is a process of removing a dissolved substance from a colloidal solution by means of diffusion through suitable membrane. Since, particles in true solution (ions or smaller molecules) can pass through animal membranes (bladder) or parchment paper or cellophane sheet but colloidal particles do not, the above can be used for dialysis. The apparatus used for this purpose is called **dialyser**. A bag of suitable membrane containing the colloidal solution is suspended in a vessel through which fresh water is continuously flown figure. The dissolved substances and ions diffuse through the membrane into the outer water and pure colloidal solution is left behind.



- (ii) **Electro-dialysis** : Ordinarily, the process of dialysis is quite slow. It can be made faster by applying an electric field if the dissolved substance in the impure colloidal solution is only electrolyte. The process is then named electro-dialysis. The colloidal solution is placed between two electrodes while pure water is taken in a compartment on each side. Electrodes are fitted in the compartment as shown in figure the ions present in the colloidal solution migrate out to the oppositely charged electrodes.



- (iii) **Ultrafiltration :** Ultrafiltration is the process of separating the colloidal particles from the solvent and soluble solutes present in the colloidal solution by especially prepared filters, which are permeable to all substances except the colloidal particles.

Colloidal particles can pass through ordinary filter paper because the pores are too large. However, the pores of filter paper can be reduced in size by impregnating with collodion solution and subsequently hardened by soaking in formaldehyde. The usual collodion is a 4% solution of nitro-cellulose in a mixture of alcohol and ether. An ultrafilter paper may be prepared by soaking the filter paper in a collodion solution and hardened by formaldehyde and finally drying it. Thus, by using ultrafilter paper, the colloidal particles are separated from rest of the materials. Ultrafiltration is a slow process. To speed up the process, pressure or suction is used.

The colloidal particles left on the ultrafilter paper are then stirred with fresh dispersion medium (solvent) to get a pure colloidal solution.

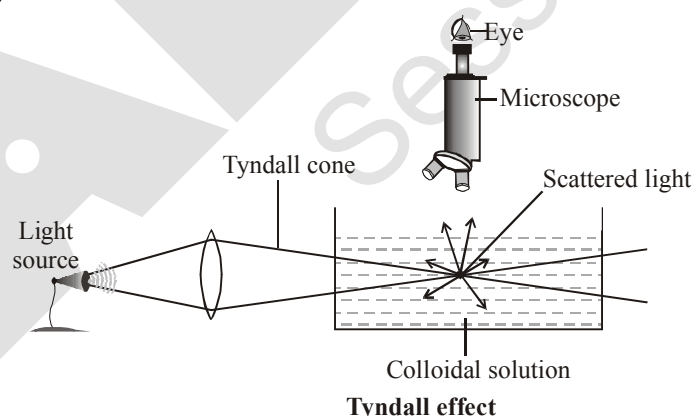
4.6 PROPERTIES OF COLLOIDAL SOLUTIONS :

4.6.1 Colligative properties:

Colloidal particles being bigger aggregates, the number of particles in a colloidal solution is comparatively small as compared to a true solution. Hence, the values of colligative properties (osmotic pressure, lowering in vapour pressure, depression in freezing point and elevation in boiling point) are of small order as compared to values shown by true solutions at same concentrations.

4.6.2 Optical Properties :

- (I) **Tyndal Effect :** When a strong and converging beam of light is passed through a colloidal solution, its path becomes visible (bluish light) when viewed at right angles to the beam of light. This effect is called **Tyndall effect**. The light is observed as a bluish cone which is called **Tyndall cone**. The Tyndall effect is due to scattering of light by the colloidal particles. The scattering of light cannot be due to simple reflection, because the size of the particles is smaller than the wave, length of the visible light and they are unable to reflect light waves. In fact, colloidal particles first absorb light and then a part of the absorbed light is scattered from the surface of the colloidal particles as a light of shorter wavelength. Since maximum scattering of light takes place at right angles to the place of incident light, it becomes visible when seen from that direction.



Tyndall effect

The Tyndall effect is observed under the following conditions :

- The diameter of the dispersed particles must not be much smaller than the wavelength of light employed.
 - The refractive indices of the dispersed phase and the dispersion medium must differ widely. This condition is fulfilled by lyophobic colloids.
- It is important to note that Tyndall effect is not shown by true solutions as their particles are too small to cause scattering.

Note :

Tyndall effect is used to distinguish between a colloidal and true solution. Zsigmondy, in 1903, used Tyndall effect to set up an apparatus known as ultramicroscope. An intense beam of light is focussed on the colloidal solution contained in a glass vessel. The focus of the light is then observed with a microscope at right angles to the beam. Individual colloidal particles appear as bright stars against a dark background. Ultramicroscope does not render the actual colloidal particles visible but only observe the light scattered by them. Thus, ultramicroscope does not provide any information about the size and shape of colloidal particles.

(II) **Colour :** The colour of the colloidal solution is not always the same as the colour of the substances in the bulk. The colour of the colloidal solution depends upon the following factors :

- Size and shape of colloidal particles.
- Wavelength of the light scattered by dispersed particles.
- The way an observer receives the light, i.e., whether by reflection or by transmission.

Examples :

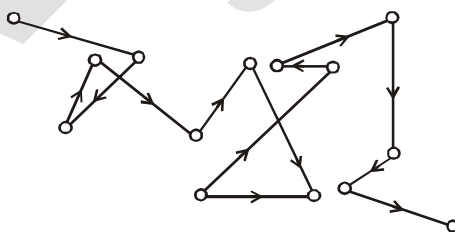
- Finest gold is red in colour. As the size of particles increases, it appears purple, then blue and finally golden.
- Dilute milk gives a bluish tinge in reflected light whereas reddish tinge in transmitted light.

4.6.3 Mechanical Properties :

(a) **Brownian movement :** Colloids particles exhibit a ceaseless random and swarming motion. This kinetic activity of particles suspended in the liquid is called Brownina movement.

Robert Brown a British Botanist first observed this motion with pollen grains suspended in water.

Cause of movement : The Brownian movement has been explained to be due to the unbalanced bombardment of the particles by the molecules of the dispersion medium. The Brownian movement has a stirring effect which does not permit the particles to settle and thus, is responsible for the stability of sols.



Brownian movement

Importance :

Brownian movement does not allow the colloidal particles to settle down due to gravity and thus is responsible for their stability.

4.6.4 Electrical Properties :

(I) Charge of colloidal particles :

Origin of charge : Various reasons have been given regarding the original of charge on the colloidal particles. These are given below :

- (i) **Frictional electrification :** It is believed to be frictional due to the rubbing of the dispersed phase particles with medium molecules.
- (ii) **Dissociation of the surface molecules :** It leads to electric charge on colloidal particles. For example, an aqueous solution of a soap (sodium palmitate) dissociates into ions.

$$\text{C}_{15}\text{H}_{31}\text{COONa} \rightleftharpoons \text{C}_{15}\text{H}_{31}\text{COO}^- + \text{Na}^+$$
 sod. palmitate
 The Na^+ ions pass into the solution while $\text{C}_{15}\text{H}_{31}\text{COO}^-$ ions have a tendency to form aggregates due to weak attractive forces present in the hydrocarbon chains. Thus, the anions which are of colloidal size bear negative charge.
- (iii) **Preferential adsorption of ions from solution :** The charge on the colloidal particles is generally acquired by preferentially adsorbing positive or negative ions from the electrolyte. Thus AgCl particles can adsorb Cl^- ions from chloride solutions and Ag^+ ions from excess Ag^+ ions solutions; the sol. will be negatively charged in the first case and positively charged in the second case.
- (iv) **Capture of electron :** It is from air during preparation of sol. by Bredig's arc method.
- (v) **Dissociation of molecular electrolytes on the surface of particles :** H_2S molecules get adsorbed on sulphides during precipitation. By dissociation of H_2S , H^+ ions are lost and colloidal particles become negatively charged.

Colloidal particles always carry an electric charge. The nature of this charge is the same on all the particles in a given colloidal solution and may be either positive or negative. A list of some common sols with the nature of charge on their particles is given below:

Positively charged sols	Negatively charged sols
Hydrated metallic oxides, e.g. $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, $\text{CrO}_3 \cdot x\text{H}_2\text{O}$ and $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, etc.	Metals, e.g. copper, silver, gold sols.
Basic dye stuffs, e.g., methylene blue sol.	Metallic sulphide, e.g., As_2S_3 , Sb_2S_3 , CdS sols.
Haemoglobin (blood)	Acid dye stuffs, e.g. eosin congo red sols.
Oxides, e.g. TiO_2 sol.	Sols of starch, gum, gelatin, clay, charcoal, etc.

The charge on the sol particles is due to one or more reasons, viz., due to electron capture by sol particles during electrodispersion of metals, due to preferential adsorption of ions from solution and/or due to formulation of electrical double layer. Preferential adsorption of ions is the most accepted reason. The sol particles acquire positive or negative charge by preferential adsorption of positive or negative ions. When two or more ions are present in the dispersion medium, preferential adsorption of the ion common to the colloidal particle usually takes place. This can be explained by taking the following examples:

- (a) When silver nitrate solution is added to potassium iodide solution, the precipitated silver iodide adsorbs iodide ions from the dispersion medium and negatively charged colloidal solution results. However, when KI solution is added to AgNO_3 solution, positively charged sol results due to adsorption of Ag^+ ions from dispersion medium.

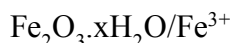


Negatively charged



Positively Charged

- (b) If FeCl_3 is added to excess of hot water, a positively charged sol of hydrated ferric oxide is formed due to adsorption of Fe^{3+} ions. However, when ferric chloride is added to NaOH a negatively charged sol is obtained with adsorption of OH^- ions.

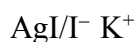


Positively charged

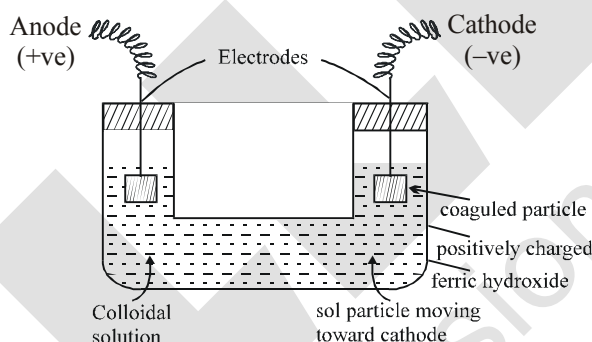


Negatively charged

Having acquired a positive or a negative charge by selective adsorption on the surface of a colloidal particle as stated above, this layer attracts counter ions from the medium forming a second layer, as shown below.



- (II) **Electrophoresis** : In a colloidal solution, the colloidal particles are electrically charged and the dispersion medium has equal but opposite charge. Thus colloidal solution on the whole is electrically neutral. When an electric current is passed through a colloidal solution, the charged particles move towards the oppositely charged electrode where coagulate due to loss of charge.

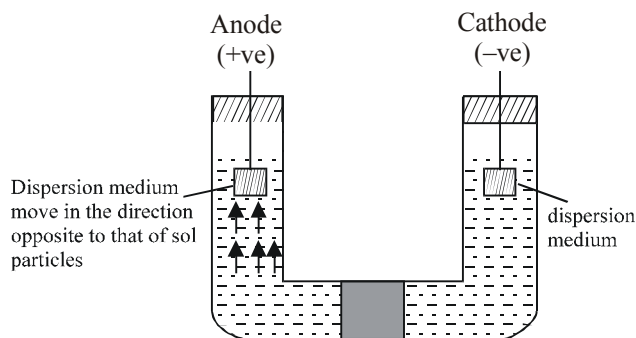


The phenomenon involving the migration of colloidal particles under the influence of electric field towards the oppositely charged electrode, is called electrophoresis or cataphoresis.

This phenomenon is used to determine the charge on the colloidal particles. For example, when a sol. of ferric hydroxide is taken in a U-tube and subjected to electric field, the ferric hydroxide (sol.) particles get accumulated near the cathode. This shows that ferric hydroxide sol. particles are positively charged.

- (III) **Electro-osmosis** : The phenomenon involving the migration of the dispersion medium and not the colloidal particles under the influence of an electric field is electro-osmosis.

Take the pure solvent (dispersion medium) in two limbs of U-tube. In the lower middle portion of U-tube, a porous diaphragm containing the colloidal system is present which divides the U-tube in two sections. In each section of U-tube, an electrode is present, as shown in figure. When the electrode potential is applied to the electrodes, the solid phase of sol. (colloidal system) cannot move but the solvent (dispersion medium) moves through the porous diaphragm towards one of the electrodes. The direction of migration of dispersion medium due to electro-osmosis determines the charge on sol. particles e.g., if the dispersion medium moves towards the cathode (negative electrode), the sol. particles are also negatively charged because the dispersion medium is positively charged as on the whole colloidal solution is neutral.



(IV) **Zeta potential, ξ** : The combination of the two layers of opposite charges around the colloidal particle is called **Helmholtz electrical double layer**. According to modern views, the first layer of ions is firmly held and is termed **fixed layer** while the second layer is mobile which is termed **diffused layer**. Since separation of charge is a seat of potential, the charges of opposite signs on the fixed and diffused parts of the double layer results in a difference in potential between these layers. This potential difference between the fixed layer and the diffused layer of opposite charges is called the **electrokinetic potential or zeta potential**. The presence of equal and similar charges on colloidal particles is largely responsible in providing stability to the colloidal solution, because the repulsive forces between charged particles having same charge prevent them from coalescing or aggregating when they come closer to one another.

(V) **Isoelectric point** : It is the pH at which the colloidal particles do not carry any charge.

4.7 COAGULATION :

The colloidal sols are stable due to the presence of electric charges on the colloidal particles. Because of the electrical repulsion, the particles do not come close to one another to form precipitates. The removal of charge by any means will lead to the aggregation of particles and hence precipitation will occur immediately.

This process by means of which the particles of the dispersed phase in a sol. are precipitated is known as coagulation.

If the coagulated particles instead of settling at the bottom of the container, float on the surface of the dispersion medium, the coagulation is called flocculation.

Most of the sols are coagulated by adding an electrolyte of opposite sign. This is due to the fact that the colloidal particles take up the ions of electrolyte whose charges are opposite to that on colloidal particles with the result that charge on the colloidal particles is neutralized. Thus coagulation takes place. For example, arsenious sulphide sol. (negatively charged) precipitated by adding barium chloride solution. It is due to the fact that the negatively charged particles of the sol. take up barium ions and get neutralized which lower the stability. As a result coagulation takes place.

It is observed that different amounts of different electrolytes is required to bring coagulation of a particular solution.

The minimum concentration of an electrolyte in millimoles per litre required to cause precipitation of one litre of a colloidal solution in two hours is called (**coagulation in value or flocculation value**) of the electrolyte for the sol. The smaller the quantity needed the higher will be the coagulating power of an ion. The reciprocal of coagulation value is regarded as the coagulating power the

For example, the coagulation values of NaCl, BaCl₂ and AlCl₃ for arsenic sulphide sol. are 51, 0.69 and 0.093 millimoles/litre respectively. Thus their coagulating powers are $\frac{1}{51}$, $\frac{1}{0.69}$ and $\frac{1}{0.093}$ i.e., 0.0196, 1.449 and 10.75 respectively.

The coagulation values of a few electrolytes for negatively charged arsenic sulphide and positively charged ferric hydroxide sol. are given in table given below. The valency of the coagulation ion (the ion whose charge is opposite to that of the colloidal particles) is also given.

Coagulation values of different electrolytes

Arsenic sulphide sol.			Ferric hydroxide sol.		
Electrolyte coagulating cation	Valency of cation	coagulation value	Electrolyte coagulating	Valency of anion	coagulation value
K ₂ SO ₄	1	63	KBr	1	138
NaCl	1	51	KNO ₃	1	132
KNO ₃	1	50	KCl	1	103
MgSO ₄	2	0.81	K ₂ CrO ₄	2	0.320
BaCl ₂	2	0.69	K ₂ SO ₄	2	0.215
AlCl ₃	3	0.093	K ₃ Fe(CN) ₆	3	0.096

From the above table, it is clear that the coagulating power of Al³⁺ ions in precipitating the arsenic sulphide sol. is approximately 550 times more than that of sodium (Na⁺) or potassium (K⁺) ions. Again, it is observed that the negatively charged arsenic sulphide sol. is coagulated by cations while positively charged ferric hydroxide sol. is coagulated by anions.

Hardy-Schulz rules : H. Schulze (1882) and W.B. Hardy (1900) suggested the following rules to discuss the effect of electrolytes of the coagulation of the sol.

- (1) Only the ions carrying charge opposite to the one present on the sol. particles are effective to cause coagulation, e.g., the negative charged sol. is best coagulated by cations and a positive sol. is coagulated by anions.
- (2) The charge on coagulating ion influences the coagulation of sol.

In general, the coagulating power of the active ion increases with the valency of the active ion. After observing the regularities concerning the valency of the active ion, a law was proposed by Hardy and Schulz which is termed as Hardy-Schulze law which is stated as follows: "Higher is the valency of the active ion, greater will be its power to precipitate the sol."

Thus, coagulating power of cations is in the order of Al³⁺ > Ba²⁺ or Mg²⁺ > Na⁺ or K⁺.

Similarly, to coagulating the positively charged sol. the coagulating power of anion is in the order of [Fe(CN)₆]⁴⁻ > PO₄³⁻ > SO₄²⁻ > Cl⁻

Ex3. 11.7 gm NaCl is needed for complete coagulation on 50 l of a colloidal solution. What is the coagulation value of electrolyte ?

$$\text{Sol. Coagulation value} = \frac{\text{millimole of electrolyte}}{\text{L of colloidal solution}} = \frac{\frac{11.7}{58.5} \times 1000}{50} = 4$$

Some other methods of coagulation :

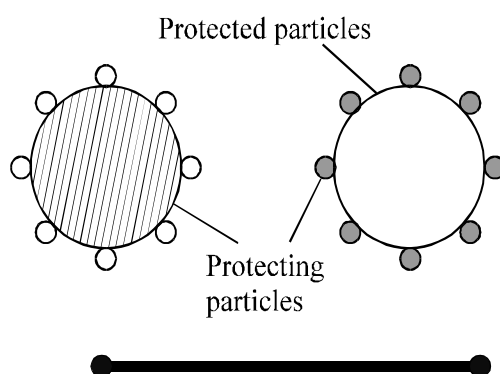
Apart from the addition of electrolyte, coagulation can also be carried out by following methods:

- (i) **By persistent dialysis :** It has been observed that traces of electrolytes are associated with the solution due to which it is stable. If the solution is subjected to prolonged dialysis, the traces of electrolytes are removed and coagulation takes place.
- (ii) **By mutual coagulation of colloids :** When two sols of oppositely charges are mixed together in a suitable proportion, the coagulation takes place. The charge of one is neutralized by the other. For example, when negatively charged arsenic sulphide sol. is added to positively charged ferric hydroxide sol., the precipitation of both occurs simultaneously.
- (iii) **By electrical method :** If the electrical charge of lyophobic sol. is removed by applying any electric field such as in electrophoresis, they also precipitate out.
- (iv) **By boiling:** When a sol is boiled, the adsorbed layer is disturbed due to increased collisions with the molecules of dispersion medium. This reduces the charge on the particles and ultimately lead to settling down in the form of a precipitate.
- (v) **By addition of electrolytes:** When excess of an electrolyte is added, the colloidal particles are precipitated. The reason is that colloids interact with ions carrying charge opposite to that present on themselves. This causes neutralisation leading to their coagulation. The ion responsible for neutralisation of charge on the particles is called the coagulating ion. A negative ion causes the precipitation of positively charged sol and vice versa.

4.8 PROTECTIVE COLLOIDS :

Lyophilic sols are more stable than the lyophobic sols. This is because, lyophilic colloids are extensively hydrated and these hydrated particles do not combine to form large aggregates.

Lyophobic sols are more easily coagulated by the addition of suitable electrolyte. To avoid the precipitation of lyophobic sol, by the addition of electrolyte, some lyophilic colloid is added to it. Such lyophilic colloid is called protective colloid and the protection of lyophobic colloid from the electrolytes is known as **protection**. The substances commonly used as protective colloids are gelatin, albumin, gum arabic, casein, starch, glue etc. A gold sol. containing a little gelatin as protective colloid needs a very large amount of sodium chloride to coagulate the sol.



Explanation : The particles of the protective colloid get adsorbed on the particles of the lyophobic colloid, thereby forming a protective layer around it (figure). The protective layer prevents the precipitating ions from coming in contact with the colloidal particles.

According to a recent view, the increase in stability of the lyophobic colloid is due to the mutual adsorption of the lyophilic and lyophobic colloids. It is immaterial which is adsorbed on which.

Gold number of a protective colloid is a minimum weight of it in milligrams which must be added to 10 ml of a standard red gold sol so that no coagulation of the gold sol. (i.e. change of colour from red to blue) takes place when 1 ml of 10 % sodium chloride solution is rapidly added to it. Obviously, smaller the gold number of a protective colloid, the greater is the protective action.

Ex.4 0.02 gm of lyophilic colloid just present coagulation in 10 ml of a lyophobic colloid on adding 1 ml of 10 % NaCl solution. What is the gold number of lyophilic colloid ?

Sol. Mg of lyophilic colloid = $0.02 \times 1000 = 20$

4.9 EMULSIONS :

An emulsion is a colloidal solution of a liquid. It may be defined as a heterogeneous system consisting of more than one immiscible liquids dispersed in one another in the form of droplets

For example, milk is an emulsion in which small drops of liquid fat are dispersed in aqueous medium.

Cod liver oil is an emulsion in which the water drops are dispersed in the oil. This means in most of the emulsions one of the liquid is water and the other liquid in oil.

The emulsion are classified as :

- (1) **Oil in water type emulsion (O/W):** In this emulsion, oil is the dispersed phase and water is the dispersion medium. It is denoted by O/W or O in W. For example, milk (liquid fat dispersed in water), vanishing cream.
- (2) **Water in oil type :** In this emulsion, water is the dispersed phase and oil is the dispersion medium. It is denoted by W/O or W in O. For example, butter.

The type of emulsion obtained by agitating two immiscible liquids depends upon the relative amounts of two components liquids. The liquid that is in excess forms the dispersion medium.

Thus, the two types of emulsions can be interconverted into each other by changing the concentration of one of the liquids.

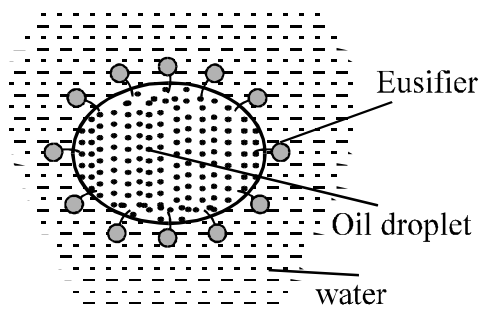
4.9.1 Preparation of emulsion (Emulsification) : Emulsification is the process which involves the preparation of emulsion. Generally, an emulsion is prepared by subjecting a mixture of the immiscible liquid to a distinct layers upon standing. The oil globules rise to form an upper layer while aqueous medium forms lower layers. To prevent the separation of layers and to get the stable emulsion, a small quantity of the third substance is added. This substance which stabilizes the emulsion is called emulsifier or emulsifying agent. The commonly used emulsifying agents are soaps, detergents and lyophilic colloids. Casein, a lyophilic colloid present in milk, acts as an emulsifier as it forms a protective layer around fat molecules dispersed in water. Hence milk is a fairly stable emulsion.

Function of emulsifier : The main function of emulsifier or emulsifying agents is to lower the interfacial tension between oil and water and thus helps the intermixing of two liquids. For example, a molecule of a soap or detergent (emulsifier) gets concentrated at the interface between oil and water. The polar end of the emulsifier is in water and non-polar end is in oil as shown in figure.

In a soap, RCOONa , R is the non-polar end, whereas $\text{COO}^- \text{Na}^+$ is the polar end.

4.9.2 Properties of emulsion :

- (i) The size of particles of the dispersed phase of an emulsion is usually larger than in sols.
- (ii) Like colloidal particles, emulsions exhibit properties such as Tyndall effect, Brownian movement (provided the particles are not too large), electrophoresis, coagulation, etc.



Demulsification : The process which involves the breaking of an emulsion into two separate liquid layers is called demulsification. The following methods may be used to bring demulsification:

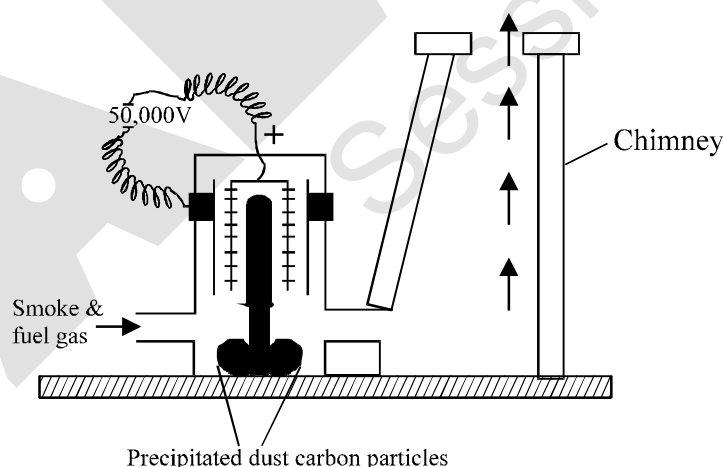
- (1) **Chemical Methods :** An emulsion may be demulsified by adding a chemical substance whose action on the dispersed phase and the dispersion medium is opposite to that of the original emulsifying agent used to produce the stable emulsion.
- (2) **Centrifugation :** Cream is separated from milk by the centrifugal method.
- (3) **Cooling :** Fat can be removed from milk by keeping it in a refrigerator for a few hours.

4.9.3 Oil in water type emulsion (O/W) Use of emulsion :

- (1) Many pharmaceutical preparations-medicines, ointments, creams and various lotions are emulsions. It is believed that medicines are more effective and easily assimilated by the body tissues when they are in colloidal form i.e., emulsion.
- (2) All paints are emulsions.
- (3) The digestion of fat in the intestines is helped by emulsification. A little of the fat forms a medium soap (emulsifier) with the alkaline solution of the intestine and this soap emulsifies the rest of the fats, thus making it easier for the digestive enzymes to do their metabolic functions.
- (4) Soaps and detergents remove dust and dirt from the dirty piece of cloth by making an oil in water emulsion.
- (5) Milk is an emulsion of liquid fats in water.
- (6) In the process of metallurgy, one of the important steps is the concentration of ore which is usually done by froth floatation process in which an oil is added to the finely-divided ore taken in water. The particles of ore go on the surface due to formation of foams while the other impurities are left at the bottom of the vessel.
- (7) The emulsion of asphalt in water is used in road making and building.

4.10 USES OF COLLOIDS :

- (1) **Medicines** : The medicines containing gold, silver or calcium etc. in colloidal form are more effective and easily assimilated by the human systems.
- (2) **Dyes** : In dyeing, mordants colloidal substances are used in textile dyeing industry to fasten dyes.
- (3) **Rubber industry** : Latex is a colloidal solution of negatively charged particles. The article to be rubber plated is made the anode. Under the influence of electric field the rubber particles get deposited on the anode and the article gets rubber plated.
- (4) **Smoke screens** : Smoke screens which consist of titanium dioxide dispersed in air are used in warfare for the purpose of concealment and camouflage.
- (5) **Formation of delta** : The river water carries with it charged clay particles and many other substances in the form of colloidal solution. When the sea water comes in contact with these particles, the colloidal particles in river water are coagulated by the electrolytes present in sea water to form deltas.
- (6) **Purification of water** : The turbidity in water is due to the presence of negatively charged clay particles. The addition of potash alum, i.e., Al^{3+} ions neutralizes the negative charge on the colloidal particles and thus causes their coagulation. The coagulated matter settles down and thus becomes clear.
- (7) **Artificial rain** : Artificial rain can be caused by throwing electrified sand on clouds which are colloidal solutions or charged particles of water in air.
- (8) **Smoke precipitation** : Smoke coming out of the chimney in industrial area is a nuisance and health hazard. It is a colloidal particles are charged particles and thus they are removed from fuel gases by electrical precipitation (Cottrell Precipitator).

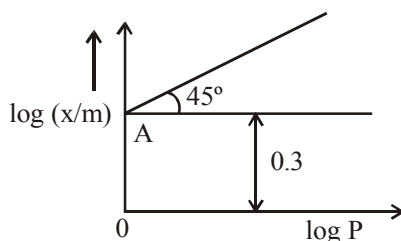


In cottrell precipitator, the smoke is made to pass through chambers fitted with highly electrically charged plates which precipitate the carbon and dust particles leaving in the gases to escape through chimney (figure).

- (9) **Sewage disposal :** Sewage water consists of particles of dirt, rubbish, mud, etc., which are of colloidal dimensions and carry an electric charge and thus do not settle down easily. These particles can thus be removed by cataphoresis. A system of two tanks fitted with metallic electrodes is used for this purpose. When electric field is created, then the dust particles are coagulated on the oppositely charged electrodes. The deposit may be utilized as a manure.
- (10) **Cleansing action of soap and detergent :** Soap solution may be used to wash off the dirt sticking to the fabric, in the presence
- If forms a colloidal solution in water forms (miscelles), removes dirt by simple adsorption of oily substance and thus washes away.
 - It decreases the interfacial tension between water and grease, and it causes the emulsification of grease in water. By mechanical action such as rubbing, the dirt particles are also detached along with the oily material.
- (11) **In Photography :** Various colloidal systems are used in photographic process. In the preparation of photographic plates, the silver bromide in gelatin is coated on thin glass plates. In developing and fixation, various colloidal substances are used. In different kinds of colour printing, gelatin and other colloidal mixtures are used.
- (12) **Blue colour of the sky :** Colloidal particles scatter only blue light and the rest of it is absorbed. In sky there are a number of dust and water particles. They scatter blue light and, therefore, sky looks bluish. If there were no scattering, the sky would have appeared totally dark.

EXERCISE # S-1

1. Graph between $\log \left(\frac{x}{m} \right)$ vs $\log P$ is provided for adsorption of NH_3 gas on metal surface. Calculate weight of NH_3 gas adsorbed by 50 gm of metal surface at 2 atm pressure



2. When a graph is plotted between $\log x/m$ and $\log P$ (where P is in atm), it is straight line with an angle of 45° and intercept 0.3010 on y-axis, what will be the amount of gas adsorbed per gm of adsorbent at pressure 0.3 atm ($\log 2 = 0.3010$)
3. In vessel O_2 gas undergoes physisorption on adsorbent surface causing its partial pressure to decrease from 6 atm to ' P ' atm in a 8.21 L vessel at 500K. If total number of sites is 2×10^{23} on adsorbent surface and every site accommodates effectively three O_2 molecules find $P(\text{atm})$.
[$N_A = 6 \times 10^{23}$]
4. 1.25 mg of Gum Arabica (Gold Number = 0.25) is added to 50 ml of standard gold sol. What maximum volume (in ml) of 10% NaCl solution can be added to this gold sol without causing coagulation.
5. Gold number of haemoglobin is 0.3. Calculate weight (in mg) of haemoglobin for 100 ml of gold sol so that gold sol is not coagulated by 10 ml of 10% NaCl solution.
6. A soap ($\text{C}_{17}\text{H}_{35}\text{COONa}$) solution becomes a colloidal sol at a concentration of 1.2×10^{-3} M. On the average, 2.4×10^{13} colloidal particles are present in 1 mm^3 . What is the average number of stearate ions in one colloidal particle (micelle) : **Take : Avogadro's number = 6×10^{23}**
7. How many of the following phenomenon are observed due to process of coagulation.
- (A) Delta formation at the meeting point of river and sea.
 - (B) Blue colour of sky.
 - (C) Precipitation of coal ash from smoke in chimeney's.
 - (D) destruction of lyophobic solution.
 - (E) Brownian motion.
 - (F) Cleaning action of detergents.
 - (G) Artificial rain.
 - (H) Use of alums in cleaning water.

8. Calculate the number of sols which are negatively charged.

$\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ sol

Sb_2S_3 sol

Ag sol

Cu sol

Blood

basic dye

sol of clay

sol of starch

sol of sodium stearate

sol of sodium laurylsulphate

sol of charcoal.

9. In order to cause coagulation of 200 ml of gold sol, 585 ml of 1% w/w NaCl solution having density 1.2 gm/ml was required. What will be the coagulation value of NaCl?

[express answer in milli-moles /litre]

10. Molecular formula of starch can be represented as $(\text{C}_6\text{H}_{10}\text{O}_5)_n$. If the gold number of one such starch molecule is 6.48 and 0.01 millimoles of the above starch are required to be added to 10ml of gold sol to prevent coagulation by 1 ml of 10% sodium chloride solution, then calculate the value of 'n'.

EXERCISE # S-2

- In a vessel O_2 gas molecules were adsorbed on solid surface causing its partial pressure to decrease from 1 atm to 0.5 atm.
Given : $N_A = 6 \times 10^{23}$
 Volume of vessel = 2.24 L
 Temperature = 273 K
 Total area of solid surface = 10^2 cm^2
 Number of active sites per unit area = 10^{24} m^{-2}
 Find number of O_2 molecules adsorbed per active site.
- A 0.03 M of an acid solution in benzene is dropped on water surface, the benzene evaporates and the acid forms a monomolecular film of solid type. What volume (in ml) of the above solution would be required to cover a 54000 cm^2 surface area of water with monomolecular layer of acid. Area covered by single acid molecule is 0.3 nm^2 ($N_A = 6 \times 10^{23}$)
- The diameter of a colloidal particle is $5000 \text{ \AA}$. If the density of substance formed, dispersed phase, is 4 gm/cc , find the value of surface area per unit mass of colloidal particle in (m^2/gm).
- A container contains 1 litre, 2M solution of benzene in ether. A piece of 3 kg charcoal is dipped in the solution. Molecules of benzene get adsorbed on the surface of charcoal and form monolayer. The molarity of resulting solution decreases to 1M. If surface area available for adsorption on charcoal $12\sqrt{3} \text{ cm}^2/\text{gm}$. Then find distance (in $\text{\AA}$) between two adjacent carbon atoms in a benzene molecule.
 (Assume: Shape of benzene molecule perfectly hexagonal.)
 [Use: $N_A = 6 \times 10^{23}$]
- 112 cm^3 hydrogen gas is adsorbed uniformly at the surface of 5 gm Palladium at 273°C and 2 atm. If the effective surface area of each hydrogen molecule is 0.4 nm^2 , then the specific surface area of Palladium is :
- The desorption of gas molecules from the adsorbent surface obeys Arrhenius equation. The average time upto which a N_2 molecule may remain adsorbed at Pt-surface at 400K is
 [Given: Pre-exponential factor, $A = 1.25 \times 10^8 \text{ s}^{-1}$;
 Activation energy of desorption = 16 Kcal , $e^{20} = 5 \times 10^8$]
- A colloid prepared by the addition of KI to AgNO_3 solution is purified using dialysis. Find the minimum mass in grams of an electrolyte 'AB' (GMM = 60) required to completely coagulate 1 l of the aforementioned colloid

Given :

Active ion causing flocculation	Flocculation value(mmol/l)
A^+	50
B^-	100



8. At 70K, the adsorption of N_2 gas at iron surface obeys Freundlich adsorption isotherm. The experimental data collected is

P(bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

Where $\frac{x}{m}$ is the mass (in gm) of N_2 gas adsorbed per gm of iron at P bar pressure. The moles of N_2

gas adsorbed per gm of iron at 36 bar and 70 K, is

9. 1.9×10^{-4} gm of the metal having density 19 gm/ml is dispersed in one litre of water to give a sol having spherical metal particles of radius 10 nm. The approximate number of metal sol particles per cm^3 of the sol is given by :
10. If unit mass of a solid, taken as cube of volume 8 cm^3 , is powdered into identical 10^{12} cubes, then the specific surface area of the solid increased by :

EXERCISE # O-I

Select the correct alternative. (Only one is correct)

1. Which gas will be adsorbed on a solid to greater extent.
(A) A gas having non polar molecule
(B) A gas having highest critical temperature (T_c)
(C) A gas having lowest critical temperature.
(D) A gas having highest critical pressure.
2. The heat of physisorption lie in the range of
(A) $1 - 10 \text{ kJ mol}^{-1}$
(B) $20 \text{ to } 40 \text{ kJ mol}^{-1}$
(C) $40 \text{ to } 200 \text{ kJ mol}^{-1}$
(D) $200 \text{ to } 400 \text{ kJ mol}^{-1}$
3. Adsorption is multilayer in case of
(A) physical adsorption
(B) chemisorption
(C) in both
(D) none of the these
4. Reversible adsorption is
(A) chemical adsorption
(B) physical adsorption
(C) both
(D) none
5. An emulsion is a colloidal system of
(A) two solids
(B) two liquids
(C) one gas and one solid
(D) one gas and one liquid
6. The nature of bonding forces in chemisorption
(A) purely physical such as Van Der Waal's forces
(B) purely chemical
(C) both chemical and physical simultaneously.
(D) none of these
7. The Tyndall effect associated with colloidal particles is due to
(A) presence of electrical charges
(B) scattering of light
(C) absorption of light
(D) reflection of light
8. Which one of the following is not applicable to chemisorption?
(A) Its heat of adsorption is high
(B) It takes place at high temperature
(C) It is reversible
(D) It forms mono-molecular layers
9. In the colloidal state the particle size ranges
(A) below 1 nm
(B) between 1 nm to 1000 nm
(C) more than 1000 nm
(D) none of the above
10. Colloids can be purified by
(A) condensation
(B) peptization
(C) coagulation
(D) dialysis
11. Milk is an example of
(A) emulsion
(B) suspension
(C) foam
(D) sol.

12. Colloidal particles in a sol. can be coagulated by
(A) heating (B) adding an electrolyte
(C) adding oppositely charged sol (D) any of the above methods
13. Emulsifier is an agent which
(A) accelerates the dispersion (B) homogenizes an emulsion
(C) stabilizes an emulsion (D) aids the flocculation of an emulsion
14. Fog is a colloidal system of
(A) gas in liquid (B) liquid in gas (C) gas in gas (D) gas in solid
15. Given below are a few electrolytes, indicate which one among them will bring about the coagulation of a gold sol. quickest and in the least of molar concentration?
(A) NaCl (B) MgSO_4 (C) $\text{Al}_2(\text{SO}_4)_3$ (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$
16. When a lyophobic colloidal solution is observed, we can see
(A) light scattered by colloidal particle
(B) size of the colloidal particle
(C) shape of the colloidal particle
(D) relative size of the colloidal particle
17. The electrical charge on a colloidal particle is indicated by
(A) Brownian movement (B) electrophoresis
(C) ultra microscope (D) molecular sieves
18. The minimum concentration of an electrolyte required to cause coagulation of a sol is called
(A) flocculation value (B) gold number (C) protective value (D) none of these
19. Smoke precipitator works on the principle of
(A) distribution law (B) neutralization of charge on colloids
(C) Le-Chatelier's principle (D) addition of electrolytes
20. Which one of following statements is not correct in respect of lyophilic sols?
(A) There is a considerable interaction between the dispersed phase and dispersion medium
(B) These are quite stable and are not easily coagulated
(C) They need stabilizing agent
(D) The particle are hydrated
21. As_2S_3 sol is
(A) positive colloid (B) negative colloid (C) neutral colloid (D) none of the above
22. At the critical micelle concentration (CMC) the surfactant molecules
(A) decompose (B) dissociate
(C) associate (D) become completely soluble
23. Small liquid droplets dispersed in another liquid is called
(A) suspension (B) emulsion (C) gel (D) true solution
24. The process which is catalysed by one of the product is called
(A) acid-base catalysis (B) autocatalysis
(C) negative catalysis (D) homogeneous catalysis

25. Tyndal effect would be observed in a
(A) solution (B) solvent (C) precipitate (D) colloidal sol.
26. A liquid is found to scatter a beam of light but leaves no residue when passed through the filter paper. The liquid can be described as
(A) a suspension (B) oil (C) a colloidal sol. (D) a true solution
27. The ability of an ion to bring about coagulation of a given colloid depends upon
(A) its charge (B) the sign of the charge alone
(C) the magnitude of the charge (D) both magnitude and sign of charge
28. An arsenious sulphide sol. carries a negative charge. The maximum precipitating power of this sol. is possessed by
(A) K_2SO_4 (B) $CaCl_2$ (C) Na_3PO_4 (D) $AlCl_3$
29. Which of the following is an example of associated colloid?
(A) Protein + water (B) Soap + water (C) Rubber + benzene (D) $As_2O_3 + Fe(OH)_3$
30. Although nitrogen does not adsorb on surface at room temperature, it adsorbs on surface at 83K. Which one of the following statements is correct -
(A) At 83K, there is formation of monomolecular layer
(B) At 83K, there is formation of multimolecular layer
(C) At 83K, nitrogen molecules are held by chemical bonds
(D) At 83K, nitrogen is adsorbed as atoms.
31. Gold number of a lyophilic sol is such a property that :
(A) The larger its value, the greater is the peptizing power
(B) The lower its value, the greater is the peptizing power
(C) The lower its value, the greater is the protecting power
(D) The larger its value, the greater is the protecting power
32. Which of the following statements is incorrect regarding physisorptions ?
(A) Under high pressure it results into multi molecular layer on adsorbent surface
(B) Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is low and positive
(C) It occurs because of Van der Waal's forces
(D) More easily liquefiable gases are adsorbed readily
33. The volume of a colloidal particle V_c , volume of a solute particle in a true solution V_t , the volume of suspension particle is V_s can be arranged
(A) $V_c = V_t = V_s$ (B) $V_s < V_c < V_t$
(C) $V_s > V_c > V_t$ (D) $V_c > V_s > V_t$
34. Which of the following is not the property of physisorption -
(A) Highly specific in nature (B) Reversible
(C) Multilayer (D) Exothermic
35. Arrange the following electrolytes in the increasing order of coagulation power for the coagulation of As_2S_3 sol -
(I) Na_3PO_4 (II) $MgCl_2$ (III) $AlCl_3$
(A) $I > II > III$ (B) $I < II < III$ (C) $I = III < II$ (D) $III < I < II$

36. The gas, which is most readily adsorbed on the surface of activated charcoal is -
(A) N_2 (B) H_2 (C) O_2 (D) SO_2
37. The migration of colloidal particles under the influence of an electrical field is known as
(A) electro osmosis (B) electrophoresis
(C) electrodialysis (D) None
38. Which reaction show the use of heterogenous catalyst
(A) $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$
(B) $2SO_2(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$
(C) $CH_3COOCH_3(l) + H_2O(l) \xrightarrow{HCl(l)} CH_3COOH(aq.) + CH_3OH(aq.)$
(D) All of the above
39. Promoters and Poison are the substance use in chemical reaction which
(A) Enhance and decrease the activity of catalyst respectively
(B) Decrease and enhance the activity of catalyst respectively
(C) Do not have any effect on catalyst
(D) Can be used in place of catalyst whenever required
40. Peptization process may be defined as
(A) Formation of precipitate by adding two ionic solution
(B) Conversion of colloidal sol into precipitate
(C) Conversion of precipitate into colloidal sol
(D) Enrichment precipitate from solution

EXERCISE # O-II

(More than one may be correct)

1. Which of the following is/are correct statements
(A) Hardy Schulz rule is related to coagulation
(B) Brownian movement and Tyndall effect are shown by colloids
(C) When liquid is dispersed in liquid, it is called gel.
(D) Gold number is a measure of protective power of lyophilic colloid.
2. Which statements is/are correct?
(A) Physical adsorption is multilayer non-directional and non specific
(B) Chemical adsorption is generally monolayer and specific in nature
(C) Physical adsorption is due to free valence of atoms
(D) Chemical adsorption is stronger than physical adsorption
3. Which statement/s is/are correct
(A) A solution is prepared by addition of excess of AgNO_3 solution in KI solution. The charge likely to develop on colloidal particle is positive.
(B) The effects of pressure on physical adsorption is high if temperature is low.
(C) Gold number is the index for extent of gold plating done.
(D) None
4. Colloidal solution can be purified by
(A) Dialysis (B) Electrodialysis (C) Electrophoresis (D) Ultrafiltration
5. Which of the following is not lyophilic
(A) Gelatin sol (B) Silver sol (C) Sulphur sol (D) As_2S_3 sol
6. Colloidal Gold can be prepared by
(A) Bredig's are method (B) Reduction of AuCl_3
(C) Hydrolysis (D) Peptization
7. On adding AgNO_3 solution into KI solution, colloidal sol can be obtained from.
(A) 100 mL of 0.1 M AgNO_3 + 100 of 0.1 M KI
(B) 100 mL of 0.1 M AgNO_3 + 100 of 0.2 M KI
(C) 100 mL of 0.2 M AgNO_3 + 100 of 0.1 M KI
(D) 100 mL of 0.15 M AgNO_3 + 100 of 0.25 M KI

Question No. 8 to 10 (3 questions)

Whenever a mixture of gases is allowed to come in contact with a particular adsorbent under the same conditions, the more strong adsorbate is adsorbed to greater extent irrespective of its amount present, e.g. H_2O is adsorbed to more extent on silica gel than N_2 and O_2 . This shows that some adsorbates are preferentially adsorbed. It is also observed that preferentially adsorbable adsorbate can displace a weakly adsorbed substance from the surface of an adsorbent.

8. Which of the following gases is adsorbed to maximum extent:
(A) He (B) Ne (C) Ar (D) Xe

9. Which of the gas can displace remaining all the gases
(A) O_2 (B) N_2 (C) CO (D) H_2
10. When temperature is increased
(A) extent of adsorption increases
(B) extent of adsorption decreases
(C) no effect on adsorption
(D) extent of adsorption first decreases, then increases

Question No. 11 to 12 (2 questions)

In macromolecular type of colloids, the dispersed particles are themselves large molecules (usually polymers). Since these molecules have dimensions comparable to those of colloidal particles, their dispersions are called macromolecular colloids. Most lyophilic sols belong to this category. There are certain colloids which behave as normal strong electrolytes at low concentrations, but exhibit colloidal properties at higher concentrations due to the formation of aggregated particles. These are known as micelles or associated colloids. Surface active agents like soaps and synthetic detergents belong to this class.

CMC increases with the total surfactant concentration. At concentration higher than CMC, they form extended parallel sheets known as **lamellar micelles** which resemble biological membranes. With two molecules thick, the individual molecule is perpendicular to the sheets such that hydrophilic groups are on the outside in aqueous solution and on the inside is a non-polar medium.

In concentrated solutions, micelles take the form of long cylinders packed in hexagonal arrays and are called lyotropic mesomorphs.

In an aqueous solution (polar medium), the polar group points towards the periphery and the hydrophobic hydrocarbon chains point towards the centre forming the core of the micelle.

-They are capable of forming ions

-Molecules of soaps and detergents consist of lyophilic as well as lyophilic parts which associate together to form micelles.

-Micelles may contain as many as 100 molecules or more.

11. Select incorrect statement(s):
(A) Surface active agent like soaps and synthetic detergents are micelles
(B) Soaps are emulsifying agents
(C) $C_{17}H_{35}$ (hydrocarbon part) and $-COO^-$ (carboxylate) part of stearate ion ($C_{17}H_{35}COO^-$) both are hydrophobic
(D) All are incorrect statements
12. Cleansing action of soap occurs because:
(A) oil and grease can be absorbed into the hydrophobic centres of soap micelles and washed away
(B) oil and grease can be absorbed into hydrophilic centres of soap micelles and washed away
(C) oil and grease can be absorbed into both hydrophilic and hydrophobic centres but not washed away
(D) cleansing action is not related to micelles

Question No.13 to 15 (3 questions)

The protective power of the lyophilic colloids is expressed in terms of gold number a term introduced by Zsigmondy. Gold number is the number of milligram of the protective colloid which prevent the coagulation of 10 ml of red gold sol. when 1 ml of a 10 percent solution of sodium chloride is added to it. Thus, smaller the gold number of lyophilic colloid, the greater is the protective power.

13. On addition of one mL solution of 10% NaCl to 10 mL of red gold sol in presence of 0.025 g of starch, the coagulation is just prevented. The gold number of starch is
(A) 0.025 (B) 0.25 (C) 2.5 (D) 25
14. Which of the following statement(s) is/are correct
(A) Higher the gold number, more protective power of colloid
(B) Lower the gold number, more the protective power
(C) Higher the coagulation value, more the coagulation power
(D) Lower the coagulation value, higher the coagulation power
15. Gold number gives an indication of
(A) protective nature of colloids
(B) purity of gold in suspension
(C) the charge on a colloidal solution of gold
(D) g-mole of gold per litre

Question No. 16 & 19 (4 questions)

These questions consist of two statements each, printed as assertion and reason, while answering these questions you are required to choose any one of the following responses.

- (A) If both assertion and reason are true and the reason is a correct explanation of assertion.
(B) If both assertion and reason are true but reason is not a correct explanation of assertion.
(C) If assertion is true but the reason is false.
(D) If assertion is false but the reason is true.

16. **Assertion :** When AgNO_3 is treated with excess of potassium iodide, colloidal particles gets attracted towards anode.
Reason : Precipitate adsorb common ions (excess) and thus become charged.
17. **Assertion :** For adsorption ΔG , ΔH , ΔS all have -ve values
Reason : Adsorption is a spontaneous exothermic process in which randomness decreases due to force of attraction between adsorbent and adsorbate.
18. **Assertion :** A gas with higher critical temperature gets adsorbed to more extent than a gas with lower critical temperature.
Reason : The easily liquifiable gases get adsorbed to less extent.
19. **Assertion :** Micelles are formed by surfactant molecules above the critical micellar concentration (CMC).
Reason : The conductivity of a solution having surfactant molecules decreases sharply at the CMC.

LIST TYPE :

20.

List-I

- (P) Inversion of can sugar
(Q) Conversion of starch into maltose
(R) Conversion of glucose into ethyle
(S) Conversion of milk into curd

List-II

- (1) Diastase
(2) Lactor bacilli enzyme
(3) Invertase
(4) Zymase

Code:

	P	Q	R	S
(A)	3	4	1	2
(B)	3	4	2	1
(C)	3	1	4	2
(D)	2	4	1	2

EXERCISE # J-MAIN

- Which of the following statements is incorrect regarding physisorptions ? [AIEEE-2009]
 - Under high pressure it results into multi molecular layer on adsorbent surface
 - Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is low and positive
 - It occurs because of Van der Waal's forces
 - More easily liquefiable gases are adsorbed readily
- According to Freundlich adsorption isotherm, which of the following is correct ? [AIEEE-2012]
 - $\frac{x}{m} \propto p^0$
 - $\frac{x}{m} \propto p^1$
 - $\frac{x}{m} \propto p^{1/n}$
 - All the above are correct for different ranges of pressure
- The coagulating power of electrolytes having ions Na^+ , Al^{3+} and Ba^{2+} for arsenic sulphide sol increases in the order :- [J-Main 2013]

(1) $\text{Al}^{3+} < \text{Ba}^{2+} < \text{Na}^+$	(2) $\text{Na}^+ < \text{Ba}^{2+} < \text{Al}^{3+}$
(3) $\text{Ba}^{2+} < \text{Na}^+ < \text{Al}^{3+}$	(4) $\text{Al}^{3+} < \text{Na}^+ < \text{Ba}^{2+}$
- A particular adsorption process has the following characteristics: (i) It arises due to van der Waals forces and (ii) it is reversible. Identify the correct statement that describes the above adsorption process: [J-Main 2015]
 - Enthalpy of adsorption is greater than 100 kJ mol^{-1}
 - Energy of activation is low.
 - Adsorption is monolayer
 - Adsorption increases with increase in temperature.
- For a linear plot of $\log(x/m)$ versus $\log p$ in a Freundlich adsorption isotherm, which of the following statements is correct ? (k and n are constants) [J-Main 2016]
 - $\log (1/n)$ appears as the intercept
 - Both k and $1/n$ appear in the slope term
 - $1/n$ appears as the intercept
 - Only $1/n$ appears as the slope
- The Tyndall effect is observed only when following conditions are satisfied [J- Main (offline)2017]
 - The diameter of the dispersed particles is much smaller than the wavelength of the light used.
 - The diameter of the dispersed particle is not much smaller than the wavelength of the light used.
 - The refractive indices of the dispersed phase and dispersion medium are almost similar in magnitude.
 - The refractive indices of the dispersed phase and dispersion medium differ greatly in magnitude.

(1) (a) and (d) (2) (b) and (d) (3) (a) and (c) (4) (b) and (c)

13. Haemoglobin and gold sol are examples of : [Jee-main(online)-2019(Jan.)]
 (1) negatively charged sols
 (2) positively charged sols
 (3) negatively and positively charged sols, respectively
 (4) positively and negatively charged sols, respectively
14. Among the colloids cheese (C), milk (M) and smoke (S), the correct combination of the dispersed phase and dispersion medium, respectively is : - [Jee-main(online)-2019(Jan.)]
 (1) C : solid in liquid; M : solid in liquid ; S : solid in gas
 (2) C : solid in liquid; M : liquid in liquid ; S : gas in solid
 (3) C : liquid in solid; M : liquid in solid ; S : solid in gas
 (4) C : liquid in solid; M : liquid in liquid ; S : solid in gas
15. Peptization is a : [Jee-main(online)-2019(April)]
 (1) process of converting a colloidal solution into precipitate
 (2) process of converting precipitate into colloidal solution
 (3) process of converting soluble particles to form colloidal solution
 (4) process of bringing colloidal molecule into solution
16. Among the following, the INCORRECT statement about colloids is : [Jee-main(online)-2019(April)]
 (1) They can scatter light
 (2) They are larger than small molecules and have high molar mass
 (3) The range of diameters of colloidal particles is between 1 and 1000 nm
 (4) The osmotic pressure of a colloidal solution is of higher order than the true solution at the same concentration
17. A gas undergoes physical adsorption on a surface and follows the given Freundlich adsorption isotherm equation

$$\frac{x}{m} = kp^{0.5}$$

 Adsorption of the gas increases with : [Jee-main(online)-2019(April)]
 (1) Decrease in p and decrease in T
 (2) Increase in p and increase in T
 (3) Increase in p and decrease in T
 (4) Decrease in p and increase in T
18. The aerosol is a kind of colloid in which : [Jee-main(online)-2019(April)]
 (1) gas is dispersed in solid
 (2) solid is dispersed in gas
 (3) liquid is dispersed in water
 (4) gas is dispersed in liquid
19. 10 mL of 1mM surfactant solution forms a monolayer covering 0.24 cm² on a polar substrate. If the polar head is approximated as cube, what is its edge length? [Jee-main(online)-2019(April)]
 (1) 2.0 pm (2) 2.0 nm (3) 1.0 pm (4) 0.1 nm

20. 0.27 g of a long chain fatty acid was dissolved in 100 cm³ of hexane. 10 mL of this solution was added dropwise to the surface of water in a round watch glass. Hexane evaporates and a monolayer is formed. The distance from edge to centre of the watch glass is 10 cm. What is the height of the monolayer?

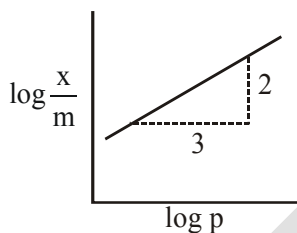
[Density of fatty acid = 0.9 g cm⁻³, $\pi = 3$]

[Jee-main(online)-2019(April)]

- (1) 10⁻⁸ m (2) 10⁻⁶ m (3) 10⁻⁴ m (4) 10⁻² m

21. Adsorption of a gas follows Freundlich adsorption isotherm x is the mass of the gas adsorbed on mass m of the adsorbent. The plot of $\log \frac{x}{m}$ versus $\log p$ is shown in the given graph. $\frac{x}{m}$ is proportional to :

[Jee-main(online)-2019(April)]



- (1) $p^{3/2}$ (2) p^3 (3) $p^{2/3}$ (4) p^2

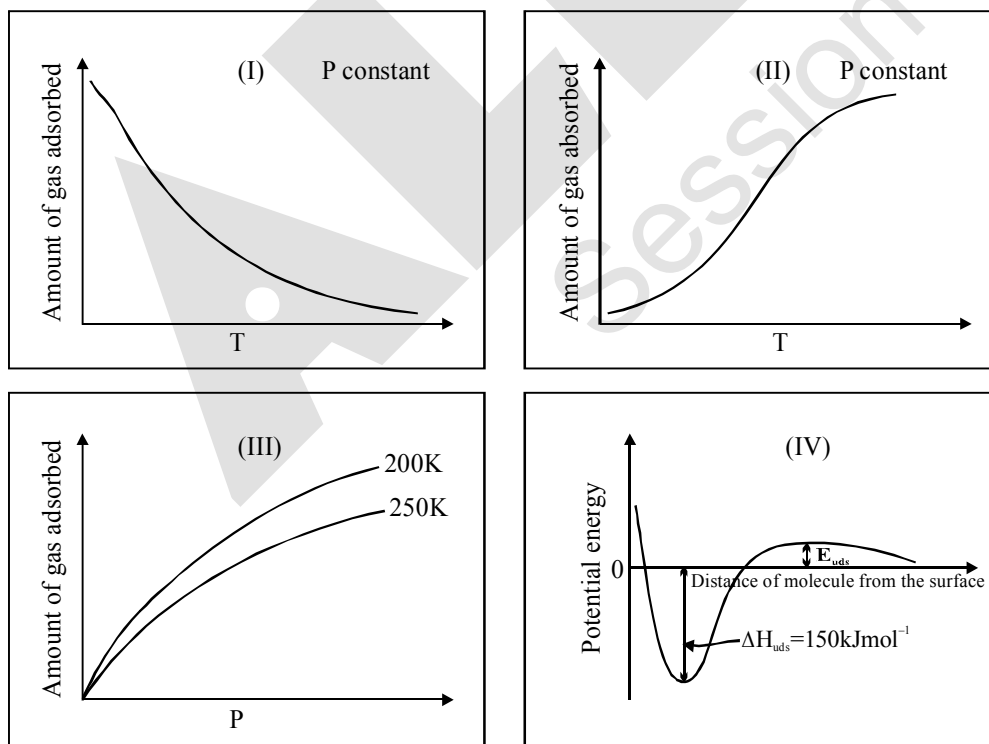
22. The correct option among the following is :

[Jee-main(online)-2019(April)]

- (1) Colloidal particles in lyophobic sols can be precipitated by electrophoresis.
- (2) Brownian motion in colloidal solution is faster the viscosity of the solution is very high.
- (3) Colloidal medicines are more effective because they have small surface area.
- (4) Addition of alum to water makes it unfit for drinking.

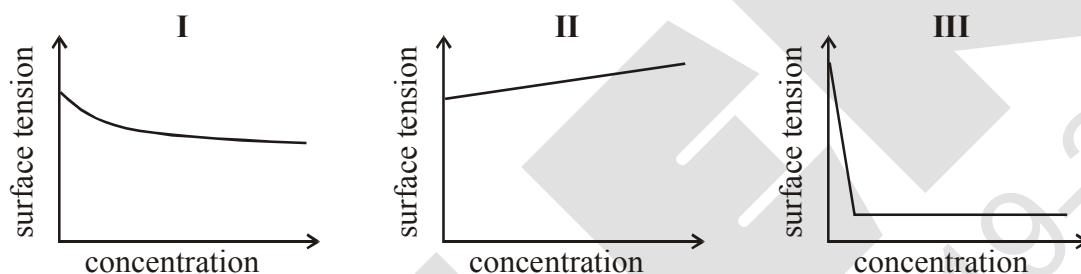
EXERCISE # J-ADVANCED

- Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient conditions is [JEE 2008]
 (A) $\text{CH}_3(\text{CH}_2)_{15}\text{N}^+(\text{CH}_3)_3\text{Br}^-$ (B) $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^-\text{Na}^+$
 (C) $\text{CH}_3(\text{CH}_2)_6\text{COO}^-\text{Na}^+$ (D) $\text{CH}_3(\text{CH}_2)_{11}\text{N}^+(\text{CH}_3)_3\text{Br}^-$
- Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}_2(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulation agent for Sb_2S_3 sol is [JEE 2009]
 (A) Na_2SO_4 (B) CaCl_2 (C) $\text{Al}_2(\text{SO}_4)_3$ (D) NH_4Cl
- The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are) - [JEE 2011]
 (A) Adsorption is always exothermic
 (B) Physisorption may transform into chemisorption at high temperature
 (C) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
 (D) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation
- Choose the correct reason(s) for the stability of the **lyophobic** colloidal particle. [JEE 2012]
 (A) Preferential adsorption of ions on their surface from the solution
 (B) Preferential adsorption of solvent on their surface from the solution
 (C) Attraction between different particles having opposite charges on their surface
 (D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles
- The given graphs / data **I, II, III and IV** represent general trends observed for different physisorption and chemisorption processes under mild conditions of temperature and pressure. Which of the following choice(s) about **I, II, III and IV** is (are) correct ? [JEE 2012]



- (A) **I** is physisorption and **II** is chemisorption (B) **I** is physisorption and **III** is chemisorption
 (C) **IV** is chemisorption and **II** is chemisorption (D) **IV** is chemisorption and **III** is chemisorption

6. Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at 25°C . For this process, the correct statement is - [J. Adv. 2013]
- (A) The adsorption requires activation at 25°C
- (B) The adsorption is accompanied by a decrease in enthalpy
- (C) The adsorption increases with increase of temperature
- (D) The adsorption is irreversible
7. The qualitative sketches I, II and III given below show the variation of surface tension with molar concentration of three different aqueous solutions of KCl, CH_3OH and $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^- \text{Na}^+$ at room temperature. The correct assignment of the sketches is - [J. Adv. 2016]

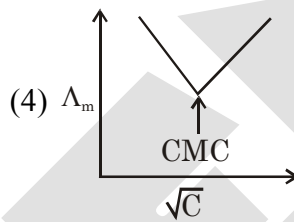
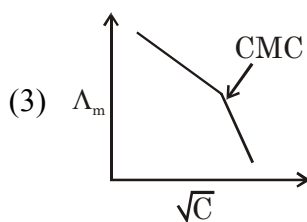
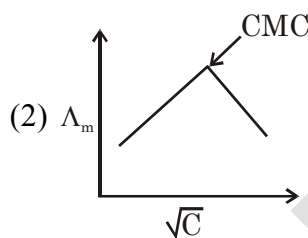
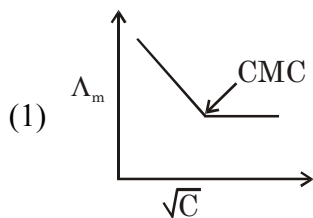


- (A) I : KCl II : CH_3OH III : $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^- \text{Na}^+$
- (B) I : $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^- \text{Na}^+$ II : CH_3OH III : KCl
- (C) I : KCl II : $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^- \text{Na}^+$ III : CH_3OH
- (D) I : CH_3OH II : KCl III : $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^- \text{Na}^+$
8. The correct statement(s) about surface properties is (are) [JEE 2017]
- (A) Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium
- (B) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system.
- (C) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution.
- (D) The critical temperatures of ethane and nitrogen are 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature.

9. Molar conductivity (Λ_m) of aqueous solution of sodium stearate, which behaves as a strong electrolyte, is recorded at varying concentration(c) of sodium stearate. Which one of the following plots provides the correct representation of micelle formation in the solution ?

(Critical micelle concentration (CMC) is marked with an arrow in the figures.)

[JEE 2019]



ANSWER-KEY**EXERCISE # S-I**

- | | | |
|--------------|--------------|--------------|
| 1. Ans.(200) | 2. Ans.(0.6) | 3. Ans (1) |
| 4. Ans.(5) | 5. Ans.(3) | 6. Ans.(30) |
| 7. Ans.(5) | 8. Ans.(9) | 9. Ans.(600) |
| 10. Ans.(4) | | |

EXERCISE # S-II

- | | | |
|-----------------------|--|-------------------------|
| 1. Ans.(3) | 2. Ans. (1) | 3. Ans. (3) |
| 4. Ans.(2) | 5. Ans. $2.4 \times 10^6 \text{ cm}^2/\text{gm}$ | |
| 6. Ans. 4 sec | 7. Ans. (6) | 8. Ans. $\frac{3}{140}$ |
| 9. 2.35×10^9 | 10. Ans. 10^4 times | |

EXERCISE # O-I

- | | | |
|--------------|-------------|-------------|
| 1. Ans. (B) | 2. Ans. (B) | 3. Ans. (A) |
| 4. Ans.(B) | 5. Ans.(B) | 6. Ans.(B) |
| 7. Ans.(B) | 8. Ans.(C) | 9. Ans.(B) |
| 10. Ans.(D) | 11. Ans.(A) | 12. Ans.(D) |
| 13. Ans.(C) | 14. Ans.(B) | 15. Ans.(C) |
| 16. Ans.(A) | 17. Ans.(B) | 18. Ans.(A) |
| 19. Ans.(B) | 20. Ans.(C) | 21. Ans.(B) |
| 22. Ans.(C) | 23. Ans.(B) | 24. Ans.(B) |
| 25. Ans. (D) | 26. Ans.(C) | 27. Ans.(D) |
| 28. Ans.(D) | 29. Ans.(B) | 30. Ans.(B) |
| 31. Ans.(C) | 32. Ans.(B) | 33. Ans.(C) |
| 34. Ans.(A) | 35. Ans.(B) | 36. Ans.(D) |
| 37. Ans.(B) | 38. Ans.(B) | 39. Ans.(A) |
| 40. Ans.(C) | | |

EXERCISE # O-II

- | | | |
|----------------|----------------|---------------|
| 1. Ans.(A,B,D) | 2. Ans.(A,B,D) | 3. Ans.(A, B) |
| 4. Ans.(A,B,D) | 5. Ans.(B,C,D) | 6. Ans.(A, B) |
| 7. Ans.(B,C,D) | 8. Ans.(D) | 9. Ans.(C) |
| 10. Ans.(B) | 11. Ans.(A, C) | 12. Ans.(A) |
| 13. Ans.(D) | 14. Ans.(B, D) | 15. Ans.(A) |
| 16. Ans.(A) | 17. Ans.(A) | 18. Ans.(C) |
| 19. Ans.(B) | 20. Ans.(C) | |

EXERCISE # J-MAIN

- | | | |
|-------------|-------------|--------------|
| 1. Ans.(2) | 2. Ans. (4) | 3. Ans. (2) |
| 4. Ans.(2) | 5. Ans. (4) | 6. Ans. (2) |
| 7. Ans. (2) | 8. Ans. (4) | 9. Ans. (1) |
| 10. Ans.(1) | 11. Ans.(2) | 12. Ans.(4) |
| 13. Ans.(4) | 14. Ans.(4) | 15. Ans.(2) |
| 16. Ans.(4) | 17. Ans.(3) | 18. Ans. (2) |
| 19. Ans.(1) | 20. Ans.(2) | 21. Ans.(3) |
| 22. Ans.(1) | | |

EXERCISE # J-ADVANCED

- | | | |
|--------------|--------------|------------------|
| 1. Ans.(A) | 2. Ans.(C) | 3. Ans.(A, B, D) |
| 4. Ans.(A,D) | 5. Ans.(A,C) | 6. Ans.(B) |
| 7. Ans.(D) | 8. Ans.(B,D) | 9. Ans.(3) |

RADIOACTIVITY

1. NUCLEAR CHEMISTRY

Nuclear chemistry deals with the phenomenon related with nucleus, like Radioactivity, Nuclear fission and fusion reactions, etc.

2. NUCLEAR STABILITY

The number of discovered elements till date is 118. All have some isotopes. The total number of isotopes is about 2000. Among these isotopes, the number of non-radioactive isotopes is even less than 300. Most of the isotopic forms have unstable nucleus due to very strong protonic repulsion. These unstable nuclei undergo spontaneous disintegration, causing radioactivity.

The stability of few nuclei against very strong protonic repulsion may be explained by the following theories :

2.1 NUCLEAR FORCE THEORY :

Nuclear force is an imaginary force which holds the nucleons together.

- The exact nature of force is not known because it does not obey inverse square law.
- It is a very short range force & it drops to 0 at 10^{-14}m . The nuclear force becomes repulsive at very small distance ($8 \times 10^{-16}\text{m}$).
- Nuclear force acts equally between all the nucleons. $p-p = n-n = n-p$
- In the stable nucleus, the nuclear force is stronger than protonic repulsion.
- On increasing atomic number, protonic repulsion increases. As the nuclear force can't hold the particles, nucleus becomes unstable.

2.2 YUKAWA'S MESON THEORY :

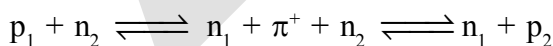
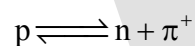
Yukawa suggested that inside the nucleus, interconversion between proton & neutron occur with the help of mesons by which nucleons hold each other.

p-meson or pions $\pi^0 \quad \pi^+ \quad \pi^-$

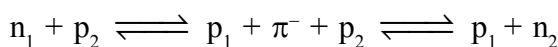
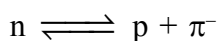
μ -meson or muons $\mu^0 \quad \mu^+ \quad \mu^-$

mass of meson = $(200 \text{ to } 300) \times m_e$

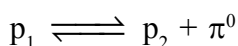
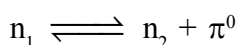
(i) Role of positive meson :



(ii) Role of negative meson :



(iii) Role of neutral meson :



2.3 MASS DEFECT AND BINDING ENERGY :

For all the isotopes (radioactive/non-radioactive), the theoretical mass (sum of masses of protons, neutrons and electrons) is greater than its isotopic (actual) mass. This difference in mass is called **mass defect** (Δm). For example,

For ${}^4_2\text{He}$, isotopic mass = 4.0026 u

Now, Mass due to $e^- = 2 \times 0.00054 = 0.00108\text{u}$

Mass due to p = $2 \times 1.00727 = 2.01454\text{u}$

Mass due to n = $2 \times 1.00867 = 2.01734\text{u}$

Total	4.03296u
-------	----------

The decrease in mass is due to its conversion into energy at the time of atom formation, by Einstein equation :

$$E = \Delta m \cdot c^2$$

As this energy is released in binding the particles together, it is called

binding energy.

If $\Delta m = 1 \text{ a.m.u.}$,

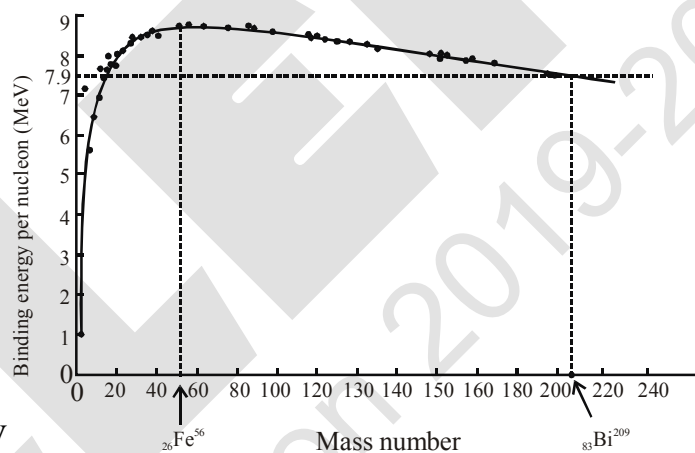
then B.E., $E \simeq 1.5 \times 10^{-10} \text{ J} \simeq 931.5 \text{ MeV}$

* Binding energy per nucleon gives a quantitative measure

of nuclear stability. $\text{B.E. per nucleon} = \frac{\text{B.E.}}{\text{No. of nucleons}}$

Greater the value of B.E. per nucleon, more is the stability of nucleus.

- (i) Nuclear binding energy is maximum for mass number 50 – 60.
- (ii) Fe, Co, Ni have very high binding energy per nucleon.
- (iii) A very heavy nucleus, say $A = 240$, has lower binding energy per nucleon compared to that of a nucleus with $A = 120$. Thus if a nucleus $A = 240$ breaks into two $A = 120$ nuclei, energy would be released in the process. This implies nucleons get more tightly bound. It has very important applications for energy production through fission.
- (iv) Consider two very light nuclei ($A \leq 10$) joining to form a heavier nucleus. The binding energy per nucleon of the heavier nuclei is more than the binding energy per nucleon of the lighter nuclei, again energy would be released in such a process of fusion.
- (v) All the nuclei having BE/nucleon greater than 7.9 MeV are stable. The value 7.9 is not applicable for lighter nuclei because they may be stable even at lower value.



Ex.1. Calculate BE/nucleon for ${}_{17}\text{Cl}^{35}$

$$m_e = 0.00054 \text{ u}, m_p = 1.00727 \text{ u}, m_n = 1.00867 \text{ u}$$

Sol. $\Delta m = (17 \times 0.00054 + 17 \times 1.00727 + 18 \times 1.00867) - 35 = 0.28883 \text{ u}$

$$\text{B.E.} = 0.28883 \times 931.5 = 269.045 \text{ MeV}$$

$$\therefore \text{B.E. nucleon} = \frac{269.045}{35} = 7.687 \text{ MeV}$$

Ex.2. Calculate BE for an additional neutron in ${}_3\text{Li}^6$ nucleus. Nuclear mass of $\text{Li}^6 = 6.0025$, $\text{Li}^7 = 7.0081$, $m_n = 1.0087 \text{ u}$

Sol. ${}_3\text{Li}^6 + {}_0\text{H}^1 \rightarrow {}_3\text{Li}^7$

$$\Delta m = (6.0025 + 1.0087) - 7.0081 = 0.0031 \text{ u}$$

$$\therefore \text{Energy released} = 0.0031 \times 931.5 = 2.888 \text{ MeV}$$

Ex.3. A nucleus A^{240} (BE/nucleon = 7.3 MeV) dissociate into two identical nuclei B^{120} , (BE/nucleon = 8.2 MeV). Calculate the amount of energy absorbed or released.

Sol. Energy released = $2 \times 120 \times 8.2 - 240 \times 7.3 = 216 \text{ MeV}$

2.4 PACKING FRACTION :

'Aston' expressed relation between isotopic mass & mass number in terms of packing fraction.

◆ Packing Fraction

$$= \frac{\text{Atomic mass} - \text{Mass number}}{\text{Mass number}} \times 10^4$$

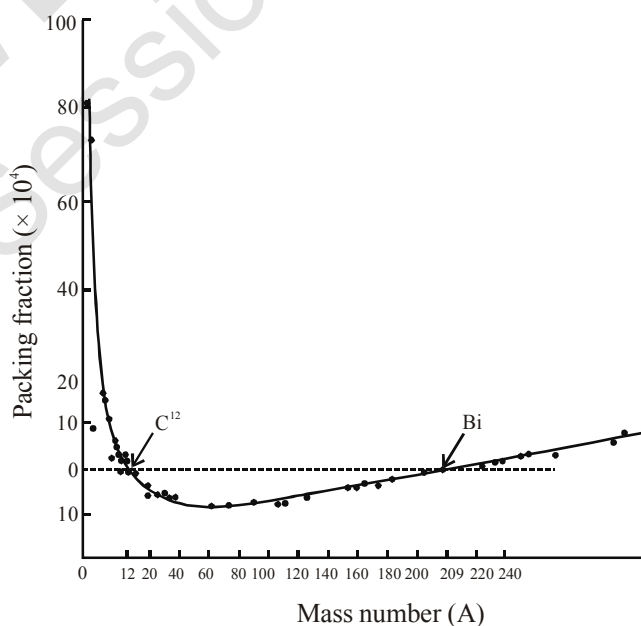
* packing fraction of C-12 is exactly zero.

* packing fraction may be positive or negative

* negative packing fraction implies that nuclei is stable

* positive packing fraction implies that nuclei is unstable

* Some lighter nuclei have positive packing fraction although nucleus is stable



2.5. NEUTRON / PROTON RATIO AND STABILITY BELT :



- ◆ All the stable nuclei lie in a zone or belt of stability, as shown in the figure.
- ◆ None of the stable nucleus have $\frac{n}{p}$ ratio less than 1 (except ${}^1_1\text{H}^1$) and greater than 1.52 ($\frac{n}{p}$ ratio of heaviest stable nucleus ${}^{209}_{83}\text{Bi}^{209}$).
- ◆ In lighter nuclei ($Z < 20$), $\frac{n}{p}$ ratio of stable nuclei is very close to 1.

2.6 EVEN ODD THEORY :

The number of stable nuclides is maximum when both protons and neutrons are even numbers.

p	n	No. of stable nucleus	Examples
even	even	~ 165	${}^4_2\text{He}$, ${}^{12}_6\text{C}$, ${}^{16}_8\text{O}$, etc
even	odd	~ 55	${}^9_4\text{Be}$, ${}^{13}_6\text{C}$, ${}^{17}_8\text{O}$, etc
odd	even	~ 50	${}^7_3\text{Li}$, ${}^{11}_5\text{B}$, ${}^{19}_9\text{F}$, etc
odd	odd	5	${}^2_1\text{H}$, ${}^6_3\text{Li}$, ${}^{10}_5\text{B}$, ${}^{14}_7\text{N}$, ${}^{180}_{73}\text{T}$

- ◆ Nearly 60% of stable nuclei have even number of protons and neutrons.

2.7 MAGIC NUMBERS :

Nuclei with 2, 8, 20, 50, 82 or 126 protons or neutrons are exceptionally stable and have a larger number of stable isotopes than neighboring nuclei in the periodic table. These numbers are called magic numbers. They are supposed to represent completely filled nuclear shells or energy levels.

e.g. ${}_{50}\text{Sn}$ having 10 stable isotopes while ${}_{51}\text{Sb}$ has only two stable isotopes.

- ◆ Nuclei with magic number of protons as well as neutrons have notably high stabilities. [eg. ${}^4_2\text{He}$, ${}^{16}_8\text{O}$, ${}^{40}_{20}\text{Ca}$ and ${}^{208}_{82}\text{Pb}$].

3. RADIO ACTIVITY

The spontaneous emission of particles, electromagnetic radiation or both by unstable nuclei, is known as radioactivity. Only unstable nuclei exhibit this property.

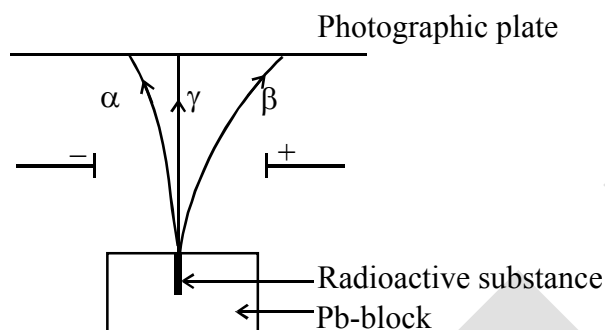
This phenomenon (radioactivity) was discovered by Henry Becquerel. On working with uranium salt, he observed that some radiations having the following properties come out from these salts:

- (i) They blacken the photographic plates.
- (ii) They cause fluorescence on ZnS screen.
- (iii) They ionise the gas through which they pass.

He gave the name **becquerel rays** to these radiations.

Later on, **Madam Curie** discovered two elements Radium (Ra) and Polonium (Po). She found that these elements as well as their salts emit similar radiations. She generalised the name of Becquerel rays as **radioactive rays** and the phenomenon, as **radioactivity**.

4. NATURE OF RADIOACTIVE RAYS



4.1 PROPERTIES OF α , β -PARTICLES AND γ -RAYS :

	Properties	Alpha	Beta	Gamma
1.	Nature	Fast moving He nuclei	Fast moving electrons	Electromagnetic radiation of very high energy.
2.	Representation	${}_2\text{He}^4$ or α	${}_{-1}\text{e}^0$ or ${}_{-1}\beta^0$ or β^-	γ or ${}_0^0\gamma$
3.	Charge	2 unit (+ve)	1 unit (-ve)	no charge
4.	Velocity	10-20% of speed of light	33% to 90% of speed of light	Same as light waves
5.	Relative penetrating power	1 (0.01 mm of Al foil)	100 (0.1 cm Al foil)	10000 (8 cm lead or 25 cm steel)
6.	Travel distance in air (Range)	2 – 4 cm	200 – 300 cm	–
7.	Kinetic energy	high	low	–
8.	Effect on ZnS plate	Luminosity	Little effect	–
9.	Mass g/particle	6.64×10^{-27} kg	9.109×10^{-31} kg	–
10.	Relative ionising power	10000	100	1

4.2. α -DECAY :

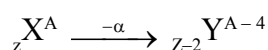
If any α -emitter is taken in a closed vessel, after sufficient time, presence of He gas is observed in the vessel. It is experimental fact considering α -particle as He^{2+} ion or He nucleus.

It is assumed that inside the nuclei, some p & n exists as a α -particle due to the following facts :

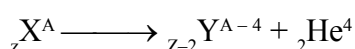
- Initially all the lighter nuclides ($Z \leq 19$) were transmuted by α -particles except ${}^4_2\text{He}$, ${}^8_4\text{Be}$, ${}^{12}_6\text{C}$. These nuclides were supposed to have 1, 2 and 3 α -particles respectively.
- Among the lighter nucleides, BE/nucleon for He is exceptionally high ($\approx 7\text{MeV}$). Hence the combination of 2p & 2n as α -particles is considered stable combination.
- Among p, n, deuteron (d) & α -decay, Q value is positive only for α -decay (energy is released). Hence among these decay, α -decay is only spontaneous decay.

(i) Effects of α -decay :

- The atomic number decreases by two units and mass number by 4 units per decay.



- n/p ratio increases on α -decay.
- Isodiaphers are formed. Isodiaphers are the atoms of different elements having the same number of excess neutron over proton. [Same $[N - P]$ difference.]

(ii) Q-value for α -emission :

$$\Delta m = (\text{nucleidic mass } {}_Z\text{X}^A) - (\text{nucleidic mass } {}_{Z-2}\text{Y}^{A-4} + {}^4_2\text{He})$$

$$= \text{Atomic mass of } {}_Z\text{X}^A - \text{At mass of } ({}_{Z-2}\text{Y}^{A-4} + {}^4_2\text{He})$$

$$\text{Now, } Q = \Delta mc^2 = \Delta m \times 931.5 \text{ MeV.}$$

(iii) KE of emitted α -particle :

$$Q \text{ value} = (\text{KE})_{\alpha} + (\text{KE})_{\text{Y}}$$

$$\& \quad (mV)_{\alpha} = (mV)_{\text{Y}}$$

$$(\text{KE})_{\alpha} = \frac{m_{\text{Y}}}{m_{\text{Y}} + m_{\alpha}} \times Q = \frac{A-4}{A} Q$$

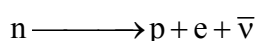


All the α -particle emitted out from a particular radio-nuclide must have same KE, if the contribution of γ -photon is neglected.

4.3 β -DECAY :

On β -decay, the atomic species get changed and hence β decay is always due to some nuclear change.

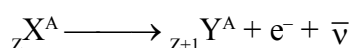
- A neutron breaks into proton, electron & anti-neutrino. As e and anti-neutrino can never exist in the nucleus, they come out.



(ii) Effect of β -decay :

- The atomic number increases by one unit but the mass number remains unchanged.
- n/p ratio decreases.
- Isobars are formed. (Atom of different elements having same mass number)
- Isotopes are formed by the sequential decay of 1α & 2β particles. (Atoms of same element having different mass numbers.)
- Normally a radio nuclide either undergoes α or β decay. There are very few radio nucleides which undergo simultaneous α & β decay.

(iii) Q-value



$$\Delta m = \text{nucleidic mass } [{}_Z^AX^A - ({}_{Z+1}Y^A + m_e + m_{\bar{\nu}})]$$

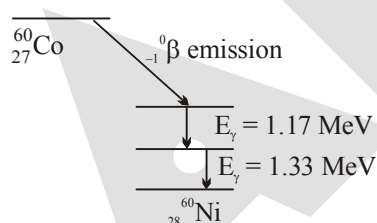
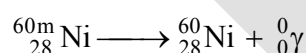
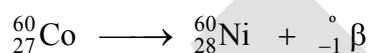
$$= \text{At mass } ({}_Z^AX^A - {}_{Z+1}Y^A)$$

$$Q = \Delta mc^2$$

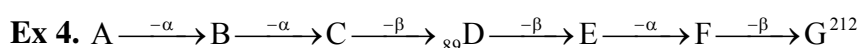
- (iv) **K.E. of β -particle :** The Q value will be shared between emitted β -particles & antineutrino. Hence KE of β -particles may have any value from 0 to Q value. The β -particle emitted out from a particular radionuclide may have different KE.

4.4. γ -DECAY :

This is a secondary phenomenon. When an α or β decay takes place, the daughter nucleus generally formed is in excited state & comes to ground state by a single or successive transitions by emitting electromagnetic radiations i.e. γ rays.



- The atomic number & mass number remain unchanged.
- Nuclear isomers are formed. They have same nuclear composition but different energy.



- (i) Identify isotopic pairs

Sol. (B, E) ; (C, F) ; (D, G)

- (ii) What is the atomic & mass number of A

Sol. 92, 224

Ex 5. Which of the following may be the disintegration product of ${}_{92}\text{U}^{238}$

- I. (a) ${}_{90}\text{Th}^{234}$ (b) ${}_{88}\text{Ra}^{232}$ (c) ${}_{85}\text{Ra}^{235}$ (d) ${}_{87}\text{Fr}^{227}$
 (a)
 II. (a) ${}_{88}\text{Ra}^{230}$ (b) ${}_{89}\text{Ac}^{230}$ (c) ${}_{87}\text{Fr}^{230}$ (d) ${}_{87}\text{Fr}^{226}$
 (e) ${}_{86}\text{Rn}^{226}$

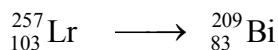
Sol. (a, b, d, e)

Ex 6. Russian and American scientist have artificially prepared elements with atomic number greater than

100. The number of alpha and beta particles produced when one atom of ${}_{103}^{257}\text{Lr}$ decay after being produced by artificial means to its stable product is -

- (A) 4 α and 5 β (B) 4 α and 12 β (C) 12 α and 4 β (D) 12 α and 5 β

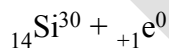
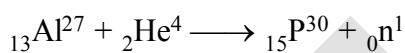
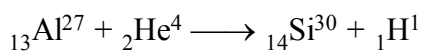
Sol. 6. Ans. (C)



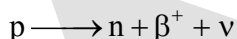
4.5 POSITRON DECAY :

Given by Irene Curie and F. Juliot.

When Mg^{24} , Al^{27} or B^{10} were bombarded by α -particle, emission of p, n & β^+ (positron) occur. On stopping the bombardment, p & n emission stop but β^+ emission continues obeying 1st order kinetics. The isotope responsible for such β^+ emission is called artificial **radio isotope** and the phenomenon as artificial radioactivity.



In positron decay, a proton (inside the nucleus) breaks into neutron, positron and neutrino.

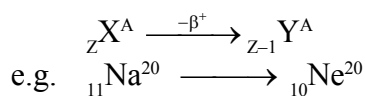


As β^+ and ν can never exist in the nucleus, they come out, resulting positron decay.

Note: A free neutron may break into proton (mass decreases) but a free proton can never break into neutron (mass increases). Inside the nucleus, both conversions are possible because of involvement of nucleus.

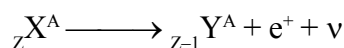
(i) **Effect of β^+ -decay :**

- (a) The atomic number decreases by one unit but the mass number remain unchanged.



- (b) Isobars are formed.
 (c) n/p ratio increase.

(ii) **Q-value :**

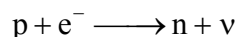


$$\begin{aligned} \Delta m &= (\text{nucleidic mass of } {}_Z X^A) - (\text{nucleidic mass of } {}_{Z-1} Y^A + \text{mass of } \beta^+ + \text{mass of } \nu) \\ &= \text{At mass } ({}_Z X^A - {}_{Z-1} Y^A) - 2m_e \end{aligned}$$

(iii) **K.E. of β^+ :** The KE of emitted e^+ may have any value from 0 to Q-value.

4.6. K-CAPTURE OR ELECTRON CAPTURE :

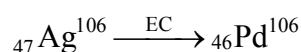
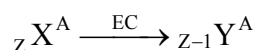
In electron capture, nucleus captures an electron of K-shell. Proton converts into neutron inside the nucleus.



As neutrino can not exist in the nucleus, it comes out.

(i) **Effects of electron capture :**

(a) Atomic number decreases by one unit but mass number remains unchanged.

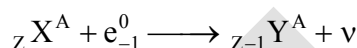


(b) $\frac{n}{p}$ ratio increases

(c) Isobar is formed.

(d) As one electron from K-shell is captured by nucleus, electron from higher shell de-excite to K-shell, emitting X-rays.

(ii) **Q-value :**



$$\Delta m = (\text{nuclear mass of } {}_Z X^A + \text{mass of electron}) - (\text{nuclear mass of } {}_{Z-1} Y^A + \text{mass of neutrino})$$

$$= \text{Atomic mass of } {}_Z X^A - \text{Atomic mass of } {}_{Z-1} Y^A$$

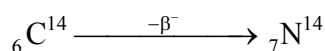
$$\text{and Q-value} = \Delta m \cdot C^2$$

(iii) **K.E. of emitted neutrino** = Q-value

5. PREDICTION OF KIND OF RADIOACTIVE DECAY:

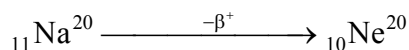
(i) $\frac{n}{p}$ ratio increases in α -decay, β^+ -decay and electron capture but $\frac{n}{p}$ ratio decreases in β^- -decay.

If the $\frac{n}{p}$ ratio of any radioisotope is greater than that of non-radioactive (stable) isotope of the same element, the possible mode of decay is β^- -decay. For example, ${}_6 C^{12}$ is non-radioactive and ${}_6 C^{14}$ is β^- emitter.



$$\frac{n}{p} = \frac{8}{6} > \frac{6}{6} \text{ of } {}_6 C^{12} \quad \frac{n}{p} = \frac{7}{7}$$

If the $\frac{n}{p}$ ratio is smaller, the possible mode of decay may be α -decay or β^+ -decay or electron capture. For example, ${}_{11}\text{Na}^{23}$ is non-radioactive and ${}_{11}\text{Na}^{20}$ is β^+ -emitter.



$$\frac{n}{p} = \frac{9}{11} < \frac{12}{11} \text{ of } {}_{11}\text{Na}^{23} \quad \frac{n}{p} = \frac{10}{10}$$

- (ii) Normally, α -decay occurs only in heavier nucleus
- (iii) If (atomic mass of ${}_Z\text{X}^A$ - Atomic mass of ${}_{Z-1}\text{Y}^A$) is less than mass of two electrons, β^+ -decay is not possible (Q-value will become negative)

Ex.7. Predict the possible mode of radioactive decay by (i) ${}_1\text{H}^3$ (ii) ${}_8\text{O}^{18}$ (iii) ${}_2\text{He}^3$ (iv) ${}_2\text{He}^6$

Sol. (i) $\left(\frac{n}{p}\right) \text{ of } {}_1\text{H}^3 = \frac{2}{1} > \left(\frac{n}{p}\right) \text{ of } {}_1\text{H}^1 \text{ or } {}_1\text{H}^2$

hence, ${}_1\text{H}^3$ is β^- -emitter.

(ii) $\left(\frac{n}{p}\right) \text{ of } {}_8\text{O}^{18} = \frac{10}{8} > \left(\frac{n}{p}\right) \text{ of } {}_8\text{O}^{16}$

hence, ${}_8\text{O}^{18}$ is β^- -emitter

(iii) $\left(\frac{n}{p}\right) \text{ of } {}_2\text{He}^3 < \left(\frac{n}{p}\right) \text{ of } {}_2\text{He}^4$

hence, ${}_2\text{He}^3$ may be β^+ -emitter

(iv) $\left(\frac{n}{p}\right) \text{ of } {}_2\text{He}^6 > \left(\frac{n}{p}\right) \text{ of } {}_2\text{He}^4$

hence, ${}_2\text{He}^6$ is β^- -emitter

Ex.8. The isotope ${}_{71}\text{Lu}^{173}$ is neutron deficient for stability. The isotopic masses of ${}_{71}\text{Lu}^{173}$ and stable isotope ${}_{70}\text{Yb}^{173}$ are 172.9390 and 172.9383 amu, respectively. Predict whether ${}_{71}\text{Lu}^{173}$ decays by β^+ -emission or electron capture or both.

Sol. $\Delta m = 172.9390 - 172.9383 = 0.0007 \text{ amu} < 2 \times \text{mass of electron}$

Hence, ${}_{71}\text{Lu}^{173}$ can not undergo β^+ -decay. The decay mode must be electron capture.

6. RADIOACTIVE DISINTEGRATION SERIES

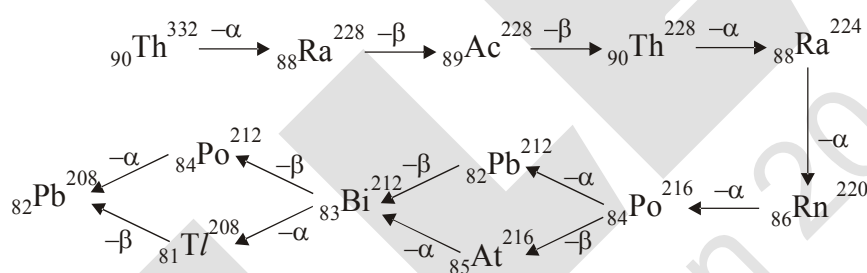
Many radioactive nucleus ($Z > 82$) are obtained as a member of decay series.

The series of nuclei is known as radioactive disintegration series.

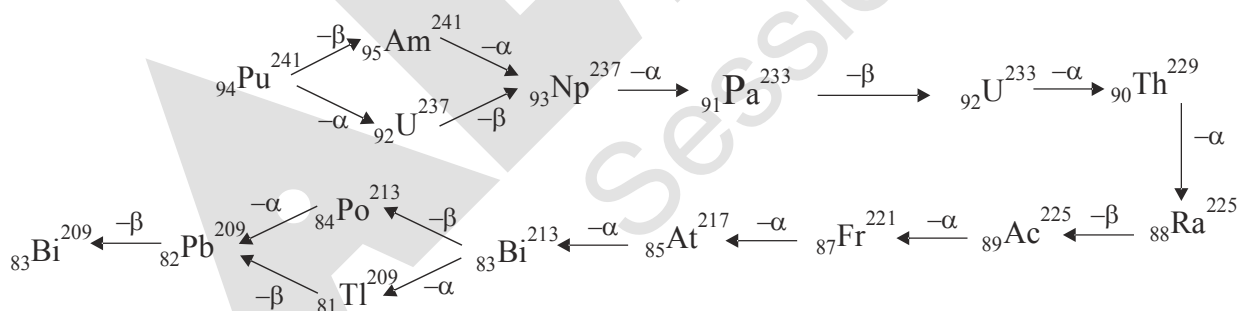
Series	Parent Nucleus	Last Nucleus	No. of α -particles	No. of β -particles
4n or Thorium series	${}_{90}\text{Th}^{232}$	${}_{82}\text{Pb}^{208}$	6	4
(4n+1) or Neptunium series	${}_{94}\text{Pu}^{241}$	${}_{83}\text{Bi}^{209}$	8	5
4n+2 or Uranium series	${}_{92}\text{U}^{238}$	${}_{82}\text{Pb}^{206}$	8	6
4n + 3 or Actinium series	${}_{92}\text{U}^{235}$	${}_{82}\text{Pb}^{207}$	7	4

4n, 4n + 2 and 4n + 3 series are natural while (4n + 1) is artificial.

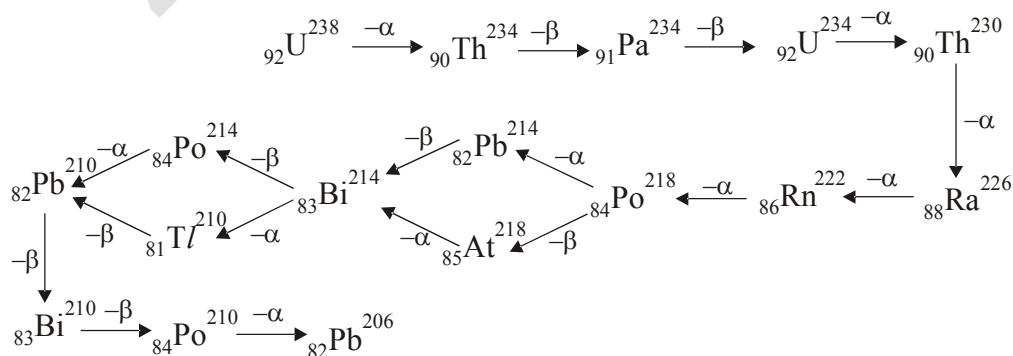
(A) The thorium series :



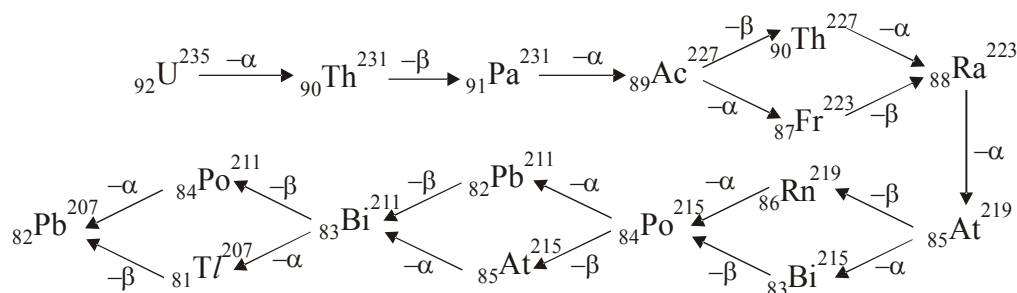
(B) The neptunium series :



(C) The uranium series :



(D) The actinium series :



7. RATE LAW

• Rutherford and Soddy's law :

At an instant, rate of decay or disintegration of active nuclei is directly proportional to the number of active nuclei at that instant.

$-\frac{dN}{dt}$ = rate of decay or activity (A) at time t and N = active nuclei at time t

$$-\frac{dN}{dt} \propto N \quad \text{or} \quad -\frac{dN}{dt} = \lambda N \quad \dots(i)$$

Here λ is the decay constant or disintegration constant, which is characteristic of that radioisotope, independent from any chemical or physical condition.

Integral rate law : $N = N_0 e^{-\lambda t}$

7.1 UNITS OF RATE OF DECAY OR ACTIVITY :

1 Becquerel (1 Bq) = 1 dps (SI unit)

1 Curie (Ci) = 3.70×10^{10} dps

1 Rutherford (1 Rd) = 10^6 dps

• **Specific activity :** Activity of 1 gm sample of radioactive substance. Its unit is (dps per gram)

Specific activity of radium (226) is 1 Ci/gm.

• **Geiger – Muller counter** is used for detecting and counting the α and β -particles.7.2 HALF LIFE ($T_{1/2}$) :

It is the time in which half of the active nuclei decay

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

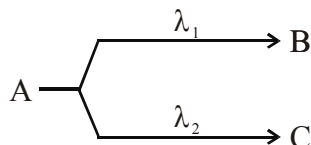
After 'n' half lives, the activity as well as the number of active nuclei reduced to $\frac{1}{2^n}$ times of initial value.

7.3 MEAN OR AVERAGE LIFE (T_{av}) :

It is the average age of all active nuclei i.e.

$$T_{av} = \frac{\text{sum of times of existence of all nuclei in a sample}}{\text{initial number of active nuclei in that sample}} = \frac{1}{\lambda}$$

7.4 PARALLEL DECAY:



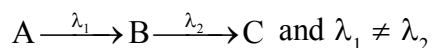
(a) The overall decay constant of A, $\lambda = \lambda_1 + \lambda_2$

(b) $\frac{N_B}{N_C} = \frac{\lambda_1}{\lambda_2}$

(c) $\lambda_1 = \text{fractional yield of B} \times \lambda$

(d) $\lambda_2 = \text{fractional yield of C} \times \lambda$

7.5 SEQUENTIAL DECAY :



(a) At any time, t:

$$N_A = N_A^0 \cdot e^{-\lambda_1 t}$$

$$N_B = \frac{\lambda_1 \cdot N_A^0}{\lambda_2 - \lambda_1} \cdot (e^{-\lambda_1 t} - e^{-\lambda_2 t})$$

$$N_C = N_A^0 - (N_A + N_B)$$

(b) Time for maximum nuclei of B,

$$T_{\max} = \frac{1}{\lambda_2 - \lambda_1} \cdot \ln \frac{\lambda_2}{\lambda_1}$$

(c) Maximum nuclei of B,

$$N_{B\max} = N_A^0 \cdot \left(\frac{\lambda_2}{\lambda_1} \right)^{\frac{-\lambda_2}{\lambda_2 - \lambda_1}}$$

(d) For steady state of B (radioactive equilibrium $\lambda_1 \cdot N_A = \lambda_2 \cdot N_B$)

Ex.9 A free neutron is unstable against β -decay with a half life of about 600 seconds.–

(i) Write the expression of this decay process.

(ii) If there are 600 free neutrons initially, calculate the time by which 450 of them have decayed. Also determine the initial decay rate of the sample.

Sol. (i) $n \rightarrow p + e^- + \bar{\nu}$

(ii) The number of undecayed neutron would be 150 by using $N = N_0 e^{-\lambda t}$

$$150 = 600e^{-\lambda t} \Rightarrow t = 2T_{1/2} = 1200 \text{ sec}$$

$$\text{Decay rate (initial)} R = \lambda N_0 = \frac{0.693}{600} \times 600 = 0.693 \text{ dps}$$

Ex.10. Obtain the amount of polonium necessary to provide a radioactivity source of 5.0 mili curie strength. The half life of polonium is 138 days.

Given : 1 curie = 3.7×10^{10} disintegration/sec., Avogadro number = 6.02×10^{26} per k-mole

Sol. Given : $r = 5 \times 10^{-3} \times 3.7 \times 10^{10}$ disint./sec. & $t_{1/2} = 138 \times 24 \times 3600$ sec and $r = \lambda N$

$$= \frac{0.693}{t_{1/2}} N$$

$$\Rightarrow N = \frac{138 \times 24 \times 3600 \times 5 \times 3.7 \times 10^7}{0.693} = 3.18 \times 10^{15} \text{ atoms}$$

$$\therefore \text{Amount of } {}_{84}\text{Po}^{210} \text{ in grams required} = \frac{210 \times 3.18 \times 10^{15}}{6.02 \times 10^{23}} = 1.11 \times 10^{-6}$$

Ex.11. Calculate the radioactive disintegration constant if 3.7×10^{10} alpha particles are emitted by 1 gram of radium per second. Avogadro's number is 6.03×10^{23} and the mass number of radium is 226.

Sol. Activity = $N\lambda = \left(\frac{N_A}{M_w} \times m \right) \lambda$

$$\Rightarrow 3.7 \times 10^{10} = \left(\frac{6.03 \times 10^{23}}{226} \times 1 \right) \lambda$$

$$\lambda = \frac{3.7 \times 10^{10} \times 226}{6.03 \times 10^{23}} = 1.38 \times 10^{-11} \text{ per second}$$

Ex.12. The half lives of X and Y are 3 minutes and 27 minutes respectively. At some instant activity of both are same, then the ratio of active nuclei of X and Y at that instant is ?

Sol. $A_1 = \lambda_1 N_1$ and $A_2 = \lambda_2 N_2$

$$A_1 = A_2 \Rightarrow \frac{0.693}{T_1} N_1 = \frac{0.693}{T_2} N_2$$

$$\Rightarrow \frac{N_1}{T_1} = \frac{N_2}{T_2} \Rightarrow \frac{N_1}{N_2} = \frac{3}{27} = \frac{1}{9}$$

$$\Rightarrow N_1 : N_2 = 1 : 9$$

Ex.13. Decay constant of two radioactive samples is λ and 3λ respectively. At $t = 0$, they have equal number of active nuclei. Calculate when will be the ratio of active nuclei becomes $e : 1$.

Sol. Number of active nuclei of two radioactive sample is

$$N_1 = N_{01}e^{-\lambda t} \text{ and } N_2 = N_{02}e^{-3\lambda t}$$

$$\therefore \frac{N_1}{N_2} = \frac{e}{1} = \frac{N_{01}e^{-\lambda t}}{N_{02}e^{-3\lambda t}} = e^{2\lambda t} \quad [\because N_{01} = N_{02}]$$

$$\therefore 1 = 2\lambda t \Rightarrow t = \frac{1}{2\lambda}$$

Ex.14. The fraction of a radioactive sample which remains active after time t is $\frac{9}{16}$. What fraction remains active after $\frac{t}{2}$ time ?

Sol. Active fraction $= \frac{N}{N_0} = e^{-\lambda t}$ At time t , $\frac{9}{16} = e^{-\lambda t}$

$$\text{At time } t/2 \text{ active fraction} = x = e^{-\lambda t/2} = \left(e^{-\lambda t}\right)^{\frac{1}{2}}$$

$$\text{So } x = \left(\frac{9}{16}\right)^{\frac{1}{2}} = \frac{3}{4}$$

Ex.15. Two radioactive nuclides A and B have half life 150 min and 15 mins respectively. A fresh sample contains nuclides of B to be 32 times that of A. How much time should elapse so that number of nuclides of A becomes twice that of B ?

Sol. (100 sec)

N_0 = initial nuclides of A

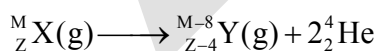
$$\text{After } t \text{ mins, } N_0 e^{-\left(\frac{\ln 2}{150}\right)t} = 2 \left[32 N_0 e^{-\left(\frac{\ln 2}{15}\right)t} \right] \Rightarrow t = 100 \text{ sec.}$$

Ex.16. ${}^M_Z X(g) \longrightarrow {}^{M-8}_{Z-4} Y(g) + \alpha - \text{particles}$

The radioactive disintegration follows first order kinetics starting with one mole of X in a 12-litre closed flask at 27°C . Find the total pressure (in atm) after two half lives.

$$[R = 0.08 \text{ L atm mol}^{-1} \text{ K}^{-1}]$$

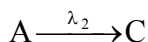
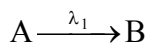
Ans. 5



$t = 0$	1 mol	0	0
$t = 2t_{1/2}$	0.25 mol	0.75 mol	1.5 mol
total gaseous moles = 0.25 + 0.75 + 1.5			
= 2.5 mol			

$$p = \frac{nRT}{V} = \frac{2.5 \times 0.08 \times 300}{12} = 5 \text{ atm}$$

Ex.17. In the parallel radioactive decay,



the time when number of radioactive nuclei of A, B & C becomes equal is

[Given $\lambda_1 = \ln 3 \text{ hr}^{-1}$, $\lambda_2 = \ln 3 \text{ hr}^{-1}$]

(1) 0.5 min

(2) 30 min

(3) 60 min

(4) 90 min

Ex.17. Ans.(2)

Sol. $C_A = C_{A_0} - C_B - C_C$

$$x = C_{A_0} - x - x$$

$$\text{or } 3x = C_{A_0} \Rightarrow x = \frac{C_{A_0}}{3}$$

$$\therefore C_A = C_{A_0} e^{-(\lambda_1 + \lambda_2)t}$$

$$\text{or } \frac{C_{A_0}}{3} = C_{A_0} e^{-(\lambda_1 + \lambda_2)t}$$

$$\text{or } 3 = e^{(\lambda_1 + \lambda_2)t}$$

$$\text{or } t = \frac{\ln 3}{2 \ln 3} = \frac{1}{2} \text{ hr} = 30 \text{ min}$$

8. APPLICATION OF RADIOACTIVITY AND RADIO ISOTOPES

8.1 IN MEDICINE :

- | | | |
|----------------------------------|---|--|
| (i) Testing of blood circulation | — | Cr^{57} |
| (ii) Brain tumour detecting | — | Hg^{203} |
| (iii) Thyroid testing (cancer) | — | I^{131} |
| (iv) Cancer cure | — | Co^{60} |
| (v) Blood cancer cure | — | $\text{Au}^{189}, \text{Na}^{24}, \text{P}^{32}$ |

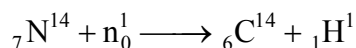
8.2 IN AGRICULTURE :

- | | | |
|--|---|------------------|
| (i) For protecting potato from earthworm | — | Co^{60} |
| (ii) Fertilizers | — | P^{32} |

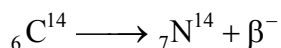
8.3 IN ARCHAEOLOGY :

8.3.1 Carbon dating (age of fossils) :

Radiocarbon (${}_6\text{C}^{14}$) dating of historical wooden derived objects is based on the knowledge that the cosmic ray intensity (responsible for C^{14} production) has been practically constant for thousands of years. C^{14} is formed in the upper atmosphere by the action of cosmic radiation of N^{14} .



The C^{14} so produced is eventually converted into CO_2 , which in turn is incorporated into plants and trees by the process of photosynthesis and then finds way into animals, which eat plants. Because of the natural plant-animal cycle, an equilibrium is set up and all living matter contains the same proportion of C^{14} as it occurs in the atmosphere. Once the plant or animal dies, the uptake of the CO_2 by it ceases and the level of C^{14} in the dead begins to fall due to β^- decay.



A comparison of the β^- -activity of the dead matter with that of the carbon still in circulation enables measurement of the period of isolation of the material from the living cycle (age of fossil).

Ex.18. The beta activity of 1 g of carbon made from green wood is 15.3 counts per minute. If the activity of 1 g of carbon derived from the wood of an Egyptian mummy case is 9.4 counts per minute under the same conditions, how old is the wood of the mummy case? $t_{1/2}$ of $C^{14} = 5770$ yrs.

Sol. $t = \frac{t_{1/2}}{\log 2} \cdot \log \frac{r_0}{r} = \frac{5770}{\log 2} \cdot \log \frac{15.3}{9.4} \approx 3920$ yrs.

8.3.2 Rock dating (Age of minerals, rocks, earth, etc) :

The age of the rocks and minerals may be determined by analysing the sample for a radionuclide (say U^{238}) and its decay product (Pb^{206}). Assuming that no decay product was present initially and all the intermediate products (if any) have achieved steady state, we can determine the age.

	A radio nuclide	$\longrightarrow$	B decay product
t = 0	a mole		0
t = present	(a-x) mole		x mole
$\therefore$	age of rock, $t = \frac{1}{\lambda}$,		$\ln \frac{a}{a-x}$

In the determination of age of earth, moon, etc, it is assumed that the rock was present from the time of evolution of earth, moon, etc.

Ex.19. A rock was found to contain U^{238} and Pb^{206} in 1.19: 1.03 mass ratio. What is the age of rock? $t_{1/2}$ of $U^{238} = 4.5 \times 10^9$ yrs.

Sol.

	U^{238}	$\longrightarrow$	Pb^{206}
t = 0	a mole		0
t = t	(a-x) mole		x mole
	$= \frac{1.19}{238}$		$= \frac{1.03}{206}$

$\therefore x = 5 \times 10^{-3}$ and $a = 10 \times 10^{-3}$

Now, $t = \frac{t_{1/2}}{\log 2} \cdot \log \frac{a}{a-x} = 4.5 \times 10^9$ yrs

Ex.20. To determine age of a stone it was analysed and 10^{20} nuclei of A & 7×10^{20} nuclei of B were obtained. If nuclei B are assumed to be obtained only due to radioactive decay of A and no B was present initially. Find age of stone in days. [Half life of A = 1000 days]

Sol. $A \longrightarrow B$

$$a - x = 10^{20} \quad x = 7 \times 10^{20}$$

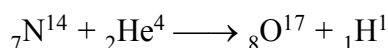
$$a = 10^{20} + 7 \times 10^{20} = 8 \times 10^{20}$$

$$\frac{a}{a-x} = \frac{8}{1}$$

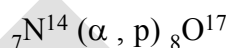
$$t = 3 \times t_{1/2} = 3 \times 1000 = 3000 \text{ days}$$

9. NUCLEAR REACTIONS

The reactions in which nuclei of atoms interact with other nuclei or elementary particles such as α -particle, proton, neutron, deuteron, etc, resulting in the formation of new nuclei with or without liberation of one or more elementary particles, are called nuclear reactions. The particles causing nuclear reactions are also called projectiles. In all the nuclear reactions, the sum of number of protons and neutrons and the total charge are conserved. Nuclear reactions may be expressed as similar as chemical reactions, like



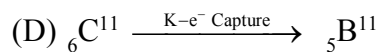
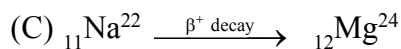
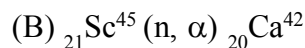
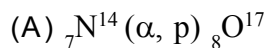
Here, the nucleus of nitrogen atom is converted in to the nucleus of oxygen atom by α -particle and proton is also produced as a by-product. These reactions may be expressed by short hand notation, in which the projectile and the liberating particle are expressed by their symbols, in a small bracket in between the parent and the product nucleus. For example, the above reaction may also be expressed as :



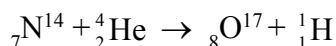
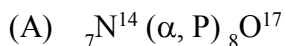
9.1 SOME DIFFERENCES BETWEEN NUCLEAR AND CHEMICAL REACTIONS :

No.	Chemical reaction	Nuclear reaction
1.	No new element is formed	New element is formed
2.	Valence electrons of atoms participates	Only the nucleus of atoms participates
3.	Balanced by the conservation of atoms	Balanced by the conservation of nuclear charge and mass number (total number of neutrons and protons)
4.	Mass conservation is obeyed	Disobey mass conservation
5.	May be exothermic or endothermic, liberating or absorbing relatively small amount of energy	May be exothermic or endothermic, liberating or absorbing relatively very high amount of energy
6.	May be reversible	Irreversible
7.	May obey kinetics of any order	Obeys only first order kinetics
8.	Rate depends on external factors like temperature and the catalytic conditions	Rate is independent of any external condition

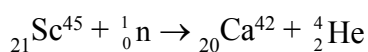
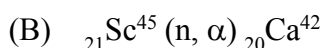
Ex.21. Select the incorrect nuclear reaction -



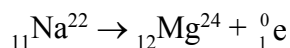
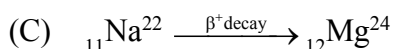
Ans (B, C)



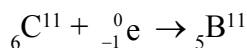
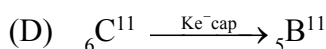
(A) is correct



(B) is incorrect



(C) is incorrect

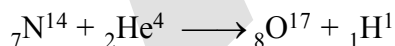


(D) is correct

hence Ans is (B) & (C)

9.2 ARTIFICIAL TRANSMUTATION :

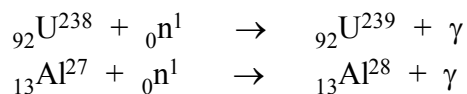
It is the method of conversion of atom of one element in to the atom of other element with the help of some particles like alpha particle, proton, deuteron, neutron, etc (called projectiles). The first such transmutation was performed by Rutherford. When N^{14} atoms were bombarded by very fast moving α -particles, the nitrogen atom has changed in to oxygen atom and proton is produced simultaneously



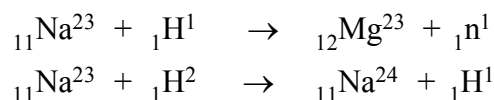
Later on, Rutherford and Chadwick showed that most of the nuclei may be transmuted by the suitable projectile. After the discovery of cyclotron, a particle accelerating machine, such transmutations became easier.

9.3 TYPES OF NUCLEAR REACTIONS :

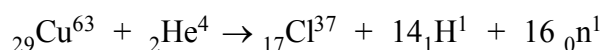
(A) Projectile Capture Reactions :



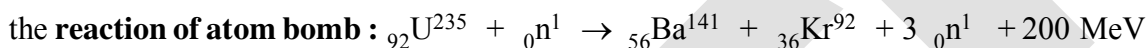
(B) Particle - particle reactions :



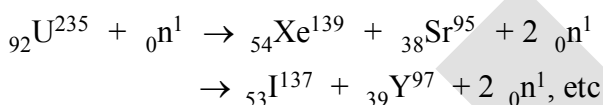
(C) Spallation reactions : High speed projectiles with 400MeV bombarded on high nucleus giving smaller nucleus.



(D) Fission reactions : It is the nuclear reaction in which a heavy nucleus is broken down by a slow or thermal neutron (energy about 0.04 eV) into two relatively smaller nuclei with the emission of two or more neutrons and large amount of energy. For example,



It is also found that the products of nuclear fission reactions are not unique. Some more products are formed. The most probable mass numbers of the two nuclides formed are around 95 and 140 and an average of 2.5 neutrons is emitted out per fission.



The destructive action of atom bomb is due to the following reasons:

- (i) As some neutrons are produced in each fission, they may collide efficiently with the other U^{235} nuclei to produce more neutrons and thus the reaction occurs in chain like fashion. It results the emission of a large amount of energy in very small time.
- (ii) Each product of fission is radioactive and hence increases the intensity of radiation in that region, causing the problems due to radiations.

(E) Fusion reactions: It is the nuclear reaction in which two or more light nuclei fused together to form heavier nuclei, with the evolution of tremendous amount of energy. In such reactions, relatively more stable nucleus having higher binding energy per nucleon is formed. Such reaction is difficult to occur because when the nuclei of different atoms come closer, they repel each other strongly. This is why, very high temperature of the order 10^6K is needed for the occurrence of such reactions. However, the overall reaction is highly exothermic due to large mass defect. Some examples of nuclear fusion reactions are :

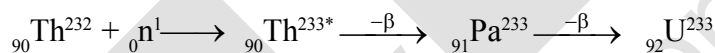
10. NUCLEAR REACTOR

A nuclear reactor is the furnace, place where nuclear fission reaction is performed to get energy. The essentials of a nuclear reactor are:



(A) **Fuel** : Nuclear fuels are of two types:

- (i) **Fissile materials** : These are the nuclides which directly results chain reaction on bombardment with slow neutrons. Such nuclides are U^{235} , Pu^{239} , U^{233} , etc.
- (ii) **Fertile material** : These are the nuclides which are non-fissile, but they may be converted in to a fissile material by the action of neutrons. Such nuclides are U^{238} and Th^{232} .



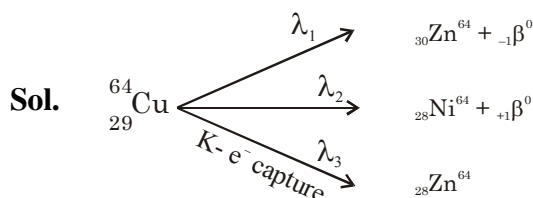
Such conversions are performed in a special type of nuclear reactor called **Breeder Reactor**.

- (B) **Moderator**: It is used to slow down the fast neutrons without absorbing them. Example: water, graphite, helium, D_2O etc.
- (C) **Control rods**: These are the rods of material which can absorb neutrons and hence control the fission reaction. Example: Cadmium, boron, etc.
- (D) **Coolant** : These are the material which transforms the energy produced in the fission reaction in to heat energy. Example: Liquid alloy of sodium and potassium, heavy water, polyphenyls, etc

PREVIOUS MISCELLANEOUS QUESTIONS

Q.1 ^{64}Cu (half-life = 12.8 hr) decays by β^- emission (38%), β^+ emission (19%) and electron capture (43%). Write the decay product and calculate partial half-lives for each of the decay processes. [JEE 2002]

Ans. $^{64}_{30}\text{Zn}$, $^{64}_{28}\text{Ni}$, $(t_{1/2})_1 = 33.68$ hr, $(t_{1/2})_2 = 67.36$ hr, $(t_{1/2})_3 = 29.76$ hr



$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} \times 100 = 38 \quad \text{.....(i)}$$

$$\frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} \times 100 = 18 \quad \text{.....(ii)}$$

$$\frac{\ln 2}{\lambda_1 + \lambda_2 + \lambda_3} = 12.8 \quad \text{.....(iii)}$$

On solving equations :

$$\lambda_1 = 0.38 \quad (t_{1/2})_I = 33.68 \text{ hr} ; (t_{1/2})_{II} = 67.36 ; (t_{1/2})_{III} = 29.7 \text{ hr}$$

Q.2 A radioactive sample emits n β -particles in 2 sec. In next 2 sec it emits $0.75n$ β -particles, what is the mean life of the sample? [JEE 2003]

Ans. $1.75n = N_0(1 - e^{-4\lambda})$, 6.95 sec, $\frac{2}{\ln\left(\frac{4}{3}\right)}$

Sol. $n = N_0[1 - e^{-2\lambda}] \quad \text{.....(i)}$

$1.75n = N_0[1 - e^{-4\lambda}] \quad \text{.....(ii)}$

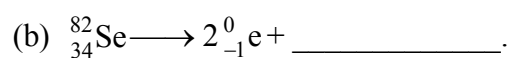
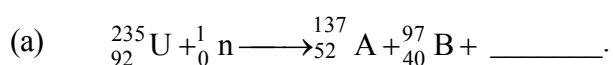
$$\Rightarrow 1.75 = 1 + e^{-2\lambda}$$

$$\frac{3}{4} = e^{-2\lambda}$$

$$\lambda = \frac{1}{2} \ln\left(\frac{4}{3}\right)$$

Q.3 Fill in the blanks

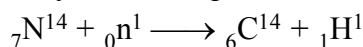
[JEE 2005]



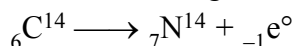
Ans. (a) $2 {}^1_0\text{n}$, (b) $^{82}_{36}\text{Kr}$

Question No. 4 to 6 (3 questions)

Carbon 14 is used to determine the age of organic material. The procedure is based on the formation of ^{14}C by neutron capture in the upper atmosphere.



^{14}C is absorbed by living organisms during photosynthesis. The ^{14}C content is constant in living organism once the plant or animal dies, the uptake of carbon dioxide by it ceases and the level of ^{14}C in the dead being falls due to the decay which C^{14} undergoes.



The half life period of ^{14}C is 5770 years. The decay constant (λ) can be calculated by using the

following formula $\lambda = \frac{0.693}{t_{1/2}}$

The comparison of the β^- activity of the dead matter with that of carbon still in circulation enables measurement of the period of the isolation of the material from the living cycle. The method however, ceases to be accurate over periods longer than 30,000 years. The proportion of ^{14}C to ^{12}C in living matter is 1 : 10^{12}

[JEE 2006]

Q.4 Which of the following option is correct ?

- (A) In living organisms, circulation of ^{14}C from atmosphere is high so the carbon content is constant in organism
- (B) Carbon dating can be used to find out the age of earth crust and rocks
- (C) Radioactive absorption due to cosmic radiation is equal to the rate of radioactive decay, hence the carbon content remains constant in living organism
- (D) Carbon dating cannot be used to determine concentration of ^{14}C in dead beings.

Ans. (C)

Q.5 What should be the age of fossil for meaningful determination of its age?

- (A) 6 years
- (B) 6000 years
- (C) 60000 years
- (D) it can be used to calculate any age

Ans. (B)

Q.6 A nuclear explosion has taken place leading to increase in concentration of C^{14} in nearby areas. C^{14} concentration is C_1 in nearby areas and C_2 in areas far away. If the age of the fossil is determined to be t_1 and t_2 at the places respectively, then

(A) The age of the fossil will increase at the place where explosion has taken place and

$$t_1 - t_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

(B) The age of the fossil will decrease at the place where explosion has taken place and

$$t_1 - t_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

(C) The age of fossil will be determined to be the same

$$(D) \frac{t_1}{t_2} = \frac{C_1}{C_2}$$

Ans. (A)

Sol. Age in nearby areas :

$$\lambda t_1 = \ln \frac{C_1}{C_t} \dots\dots(i)$$

age in for away area

$$\lambda t_2 = \ln \frac{C_2}{C_t} \dots\dots(ii)$$

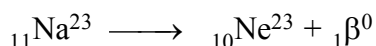
as $C_1 > C_2$ so age of fossil increase is nearby areas by value.

$$(t_1 - t_2) = \frac{1}{\lambda} \ln \left(\frac{C_1}{C_2} \right)$$

Q.7 A positron is emitted from ${}_{11}^{23}\text{Na}$. The ratio of the atomic mass and atomic number of the resulting nuclide is **[JEE 2006]**

- (A) 22/10 (B) 22/11 (C) 23/10 (D) 23/12

Ans. (C)



$$\frac{n}{p} = \frac{23}{10}$$

Q.8 **STATEMENT-1 :** The plot of atomic number (y-axis) versus number of neutrons (x-axis) for stable nuclei shows a curvature towards x-axis from the line of 45° slope as the atomic number is increased.

and

STATEMENT-2 : Proton-proton electrostatic repulsions begin to overcome attractive forces involving protons and neutrons in heavier nuclides. **[JEE 2008]**

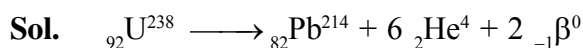
- (A) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1
 (B) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
 (C) STATEMENT-1 is True, STATEMENT-2 is False
 (D) STATEMENT-1 is False, STATEMENT-2 is True

Ans. (A)

Q.9 The total number of α and β particles emitted in the nuclear reaction ${}_{92}^{238}\text{U} \rightarrow {}_{82}^{214}\text{Pb}$ is.

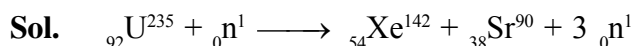
[JEE 2009]

Ans. (8)



Q.10 The number of neutrons emitted when ${}_{92}^{235}\text{U}$ undergoes controlled nuclear fission to ${}_{54}^{142}\text{Xe}$ and ${}_{38}^{90}\text{Sr}$ is - **[JEE 2010]**

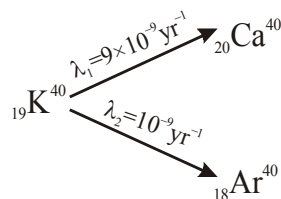
Ans. (4)



EXERCISE (S-1)

- Q.1 Of the three isobars $^{114}_{48}\text{Cd}$, $^{114}_{49}\text{In}$ and $^{114}_{50}\text{Sn}$, which is likely to be radioactive? Explain your choice.
- Q.2 Classify each of the following nuclides as "beta ($^0_{-1}\beta$) emitter", or "positron ($^0_{+1}\beta$) emitter":
 $^{49}_{20}\text{Ca}$, $^{195}_{80}\text{Hg}$, ^8_5B , $^{150}_{67}\text{Ho}$, $^{30}_{13}\text{Al}$, $^{94}_{36}\text{Kr}$. Note: $^{84}_{36}\text{Kr}$, $^{200}_{80}\text{Hg}$ and $^{165}_{67}\text{Ho}$ are stable
- Q.3 (a) Calculate number of α and β -particles emitted when $^{238}_{92}\text{U}$ changes into radioactive $^{206}_{82}\text{Pb}$.
 (b) $^{235}_{92}\text{U}$ disintegrates and emits 4β - and 7α -particles to form a stable element. Find the atomic number and mass number of the stable product. Also identify the element.
- Q.4 A radioactive isotope disintegrates and emits 6β and 7α particles to form a stable product. Find the difference between atomic number of parent radioactive isotope and the stable product.
- Q.5 Find the α -activity of $1\text{ gm }^{200}\text{Ra}$. ($\lambda = 4.8 \times 10^{-7}\text{min}^{-1}$) in dpm. ($N_A = 6 \times 10^{23}$)
- Q.6 The half life of the nuclide Rn^{220} is 54.5 sec. Find mass (in kg) of radon is equivalent to 1 millicurie.
- Q.7 At any given time a piece of radioactive material ($t_{1/2} = 30$ days) contains 10^{12} atoms. Calculate the activity of the sample in dps.
- Q.8 The activity of the radioactive sample drops to $1/64$ of its original value in 2 hr. Find the decay constant (λ).
- Q.9 A radioactive substance decays 20% in 10 min. If at start there are 5×10^{20} atoms present, find time after which the number of atoms will be reduced to 10^{18} . ($\log 2 = 0.3$)
- Q.10 The half life period of $^{125}_{53}\text{I}$ is 60 days. What % of radioactivity would be present after 240 days.
- Q.11 The activity of a radioactive sample decreases to $1/3$ of the original activity (A_0) in a period of 9 years. After 9 years more, its activity is A_0/x . Find the value of x .
- Q.12 5 moles of a radio-active isotope $^A_Z\text{X}(\text{s})$ were taken in a 24.6 litre vessel at 600K and the following decay started : $^A_Z\text{X}(\text{s}) \longrightarrow ^{A-8}_{Z-4}\text{Y}(\text{s})$
- If pressure developed after 16 hours was 15 atm, calculate half life of sample 'X' in hours.
 Use : $R = 0.082\text{ L-atm/mol-K}$.
- Q.13 Calculate the age of a vegetarian beverage whose tritium content is only 15% of the level in living plants. Given $t_{1/2}$ for ^3_1H = 12.3 years. ($\log 2 = 0.3$, $\log 3 = 0.48$)
- Q.14 The volume of the blood in a patient is estimated by recording the activity of Na^{24} administered into the patient's blood. Find the volume (in ℓ) of the blood if the activity after 25 hrs is $8\text{ dpm m}\ell^{-1}$. Given the initial activity of Na^{24} sample is $2 \times 10^3\text{ Bq}$ when administered & the $t_{1/2}$ for $\text{Na}^{24} = 15$ hrs. (use $32^{1/3} \approx 3$)
- Q.15 A sample of U^{238} (half life = 4.5×10^9 yr) ore is found to contain 23.8 g of U^{238} and 20.6 g of Pb^{206} . Calculate the age of the ore.
- Q.16 An isotopes of Potassium $^{40}_{19}\text{K}$ has a half life of 1.4×10^9 year and decays to Argon $^{40}_{18}\text{Ar}$ which is stable.
- (i) Write down the nuclear reaction representing this decay.
 (ii) A sample of rock taken from the moon contains both potassium and argon in the ratio $1/3$. Find age of rock.

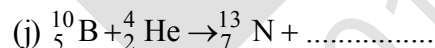
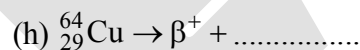
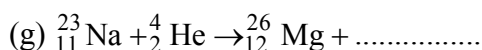
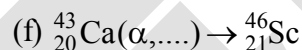
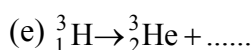
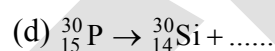
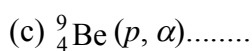
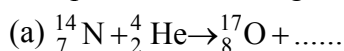
Q.17 K^{40} decays into Ca^{40} & Ar^{40} simultaneously as follows



Ratio of atoms of 'Ar' to 'K' in a rock sample is 3 : 10. Calculate age of rock sample assuming that source of 'Ar' is radioactive decay of ' K^{40} ' only and no 'Ar' was present at the time of formation of rock.

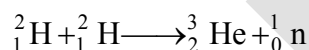
If your answer is $(\ln x) \times 10^8$ years, then the value of 'x' is.

Q.18 Complete the following nuclear equations:



Q.19 Consider the following reaction; ${}^2_1\text{H}_1 + {}^2_1\text{H}_1 \rightarrow {}^4_2\text{He} + Q$: Mass of the deuterium atom = 2.0141 u ; Mass of the helium atom = 4.0024 u. Find approximate value of 'Q' (in MeV)

Q.20 Calculate the energy released in MeV in the following nuclear reaction:



Assume that the masses of ${}^2_1\text{H}$, ${}^3_2\text{He}$ and neutron (n) respectively are 2.020, 3.016 and 1.008 in amu.

EXERCISE (S-2)

- Q.1 In the nuclear reaction $X^{200} \longrightarrow A^{110} + B^{90}$. If the binding energy per nucleon for X, A and B is 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, Find the energy released.
- Q.2 The binding energies per nucleon for ${}_1\text{H}^2$ and ${}_2\text{He}^4$ are 1.1 MeV and 7.0 MeV respectively. Find the energy released when two deuterons fuse to form a helium nucleus.
- Q.3 When ${}_c^a\text{X}$ changes to ${}_d^b\text{Y}$. Find number of α and β particle.
- Q.4 At a given instant there are 25 % undecayed radioactive nuclei in a sample. After 10 sec, the number of undecayed nuclei remains 12.5 % . Calculate :

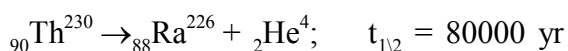
(i) mean – life of the nuclei and

(ii) The time in which the number of undecayed nuclear will further reduce to 6.25 % of the reduced number.

- Q.5 One of the hazards of nuclear explosion is the generation of Sr^{90} and its subsequent incorporation in bones. This nuclide has a half life of 40 year. Suppose one microgram was absorbed by a new-born child, how much Sr^{90} will remain in his bones after 20 years?
- Q.6 During the decay of 928 gm ${}_{90}\text{Th}^{232}$ to ${}_{82}\text{Pb}^{208}$, total 9.3×10^{24} β -particle are emitted upto 't' time. Then calculate t/T.

$$[\lambda \text{ of "Th"} = \frac{\ln 2}{T} ; N_A = 6 \times 10^{23}].$$

- Q.7 The β count given by carbon obtained from an ancient fossil was 12 counts per second, while the β -count given by carbon sample from freshly cut tree was found to be 15 count per second. What is the age of fossil in years. Given : ($t_{1/2}$ of $\text{C}^{14} = 5770$ yrs, $\ln 5 = 1.6$, $\ln 2 = 0.7$)
- Q.8 The nuclidic ratio ${}_1^3\text{H}$ to ${}_1^1\text{H}$ in a sample of water is 8.0×10^{-18} : 1 Tritium undergoes decay with a half-life period of 12 years. How many tritium atoms would 10.0 g of such a sample contain 36 years after the original sample is collected.
- Q.9 Ac^{227} has a half life of 22 year w.r.t radioactive decay. The decay follows two parallel paths, one leading the Th^{227} and the other leading to Fr^{223} . the percentage yields of these two daughters nucleides are 2% and 98% respectively. What is the rate constant in yr^{-1} , for each of the separate paths?
- Q.10. Consider the following process of decay,



After the above process has occurred for a long time, a state is reached where for every two thorium atoms formed from ${}_{92}\text{U}^{234}$, one decomposes to form ${}_{88}\text{Ra}^{226}$ and for every two ${}_{88}\text{Ra}^{226}$ formed, one decomposes. Calculate the ratio of number of atoms of ${}_{90}\text{Th}^{230}$ to ${}_{88}\text{Ra}^{226}$ at this state.

EXERCISE (O-1)

- Q.1 ${}^{14}_6\text{C}$ decays by emission of
 (A) β^- (B) β^+ (C) n (D) α
- Q.2 When ${}^{30}_{15}\text{P}$ emits a positron, the daughter nuclide formed is
 (A) ${}^{29}_{15}\text{P}$ (B) ${}^{30}_{16}\text{Si}$ (C) ${}^{30}_{14}\text{Si}$ (D) ${}^{30}_{16}\text{P}$
- Q.3 ${}^{27}_{13}\text{Al}$ is a stable isotope. ${}^{29}_{13}\text{Al}$ is expected to disintegrated by
 (A) α emission (B) ${}^0_{-1}\beta$ emission (C) Positron emission (D) Proton emission
- Q.4 Loss of a β^- particle is equivalent to
 (A) Increase of one proton only
 (B) Decrease of one neutron only
 (C) Increase of one proton and decrease of one neutron
 (D) None of these.
- Q.5 Which of the following nuclear reactions will generate an isotope?
 (A) neutron emission (B) positron emission
 (C) α -emission (D) β -emission
- Q.6 The S^{35} is neutron-rich, therefore, it is likely to undergo radioactive decay by
 (A) electron capture (B) beta emission (C) positron emission (D) alpha emission
- Q.7 The number of α and β -particles emitted, when the following nuclear transformation takes place are _____ and _____ respectively.

$${}^{238}_{92}\text{X} \longrightarrow {}^{206}_{82}\text{Y}$$

 (A) 6, 2 (B) 5, 6 (C) 8, 4 (D) 8, 6
- Q.8 ${}^{88}_{35}\text{X}$ an unstable isotope, decays in two successive steps to produce stable isotope ${}^{84}_{32}\text{Z}$ as

$${}^{88}_{35}\text{X} \xrightarrow{\text{I}} \text{Y} \xrightarrow{\text{II}} {}^{84}_{32}\text{Z}$$

 The correct statement is (possible emission are α , β , positron, neutron, and K-capture)
 (A) I may involve a β -emission. (B) II may involve a neutron emission
 (C) Y and Z may be isodiaphers (D) X and Z may be isodiaphers
- Q.9 Which of the following can not be natural decay product (only α , β & γ decay) of ${}^{232}_{90}\text{Th}$
 (A) ${}^{228}_{89}\text{Ac}$ (B) ${}^{220}_{86}\text{Rn}$ (C) ${}^{226}_{88}\text{Ra}$ (D) ${}^{216}_{84}\text{Po}$
- Q.10 The half-life of a radioactive isotope is three hours. If the initial mass of the isotope were 256 g, the mass of it remaining undecayed after 18 hours would be
 (A) 16.0 g (B) 4.0 g (C) 8.0 g (D) 12.0 g
- Q.11 The half-life of a radioisotope is four hours. If the initial rate of the isotope was 200 dpm, the rate after 24 hours is
 (A) 6.25 dpm (B) 2.084 dpm (C) 3.125 dpm (D) 4.167 dpm

- Q.12 The half life of Tc^{99} is 6.0 hr. The delivery of a sample of Tc^{99} from the reactor to the nuclear medicine lab of a certain hospital takes 3.0 hr. What is the minimum amount of Tc^{99} that must be shipped in order for the lab to receive 10.0 mg? ($\sqrt{2} = 1.41$)
 (A) 20.0 mg (B) 15.0 mg (C) 14.1 mg (D) 12.5 mg
- Q.13 Two radioactive nuclides A and B have half lives of 50 min and 10 min respectively. A fresh sample contains the nuclides of B to be eight time that of A. How much time should elapse so that the number of nuclides of A becomes double of B
 (A) 30 min. (B) 40 min. (C) 50 min. (D) 100 min.
- Q.14 A radioactive sample had an initial activity of 56 dpm (disintegration per min). After 69.3 min it was found to have an activity of 28 dpm. Find the number of atoms in a sample having an activity of 10 dpm.
 (A) 693 (B) 1000 (C) 100 (D) 10,000
- Q.15. The radioactivity of a sample is R_1 at a time T_1 and R_2 at a time T_2 . If the half life of the specimen is T , the number of atoms that have disintegrated in the time $(T_2 - T_1)$ is equal to
 (A) $(R_1 T_1 - R_2 T_2)$ (B) $(R_1 - R_2)$ (C) $(R_1 - R_2) / T$ (D) $(R_1 - R_2) T / 0.693$
- Q.16 The analysis of a mineral of uranium reveals that ratio of mole of ^{206}Pb and ^{238}U in sample is 0.2. If effective decay constant of process $^{238}\text{U} \longrightarrow ^{206}\text{Pb}$ is λ then age of rock is
 (A) $\frac{1}{\lambda} \ln \left(\frac{5}{4} \right)$ (B) $\frac{1}{\lambda} \ln \left(\frac{5}{1} \right)$ (C) $\frac{1}{\lambda} \ln \left(\frac{4}{1} \right)$ (D) $\frac{1}{\lambda} \ln \left(\frac{6}{5} \right)$
- Q.17 Wooden article and freshly cut tree show activity of 7.6 and 15.2 $\text{min}^{-1} \text{gm}^{-1}$ of carbon ($t_{1/2} = 5760$ years) respectively. The age of article in years, is
 (A) 5760 (B) $5760 \times \left(\frac{15.2}{7.6} \right)$ (C) $5760 \times \left(\frac{7.6}{15.2} \right)$ (D) $5760 \times (15.2 - 7.6)$
- Q.18 Symbol is needed to complete the nuclear equation $^{63}_{29}\text{Cu}(p, \dots) ^{62}_{29}\text{Cu}$
 (A) $^1_1\text{H}^2$ (B) $^1_0\text{n}^1$ (C) $^4_2\text{He}^4$ (D) $^1_{-1}\text{n}^0$
- Q.19 Consider the following nuclear reactions:
 $^{238}_{92}\text{M} \rightarrow ^X_Y\text{N} + 2\ ^4_2\text{He};$
 $^X_Y\text{N} \rightarrow ^A_B\text{L} + 2\beta^+$
 The number of neutrons in the element L is
 (A) 142 (B) 144 (C) 140 (D) 146
- Q.20 The number of neutrons accompanying the formation of $^{139}_{54}\text{X}$ and $^{94}_{38}\text{Sr}$ from the absorption of slow neutron by $^{235}_{92}\text{U}$ followed by nuclear fission is
 (A) 0 (B) 2 (C) 1 (D) 3

EXERCISE (O-2)

- Q.1 Helium nuclei combines to form an oxygen nucleus. The energy released per nucleon of oxygen nucleus is if $m_0 = 15.834$ amu and $m_{\text{He}} = 4.0026$ amu
 (A) 10.27 MeV (B) 0 MeV (C) 5.24 MeV (D) 164.3 MeV
- Q.2 A radioactive element gets spilled over the floor of a room. Its half-life period is 30 days. If the initial activity is ten times the permissible value, after how many minimum days will it be safe to enter the room?
 (A) 1000 days (B) 300 days (C) 10 days (D) 100 days
- Q.3 The radioactive sources A and B of half lives of t hr and $2t$ hr respectively, initially contain the same number of radioactive atoms. At the end of t hours, their rates of disintegration are in the ratio :
 (A) $2\sqrt{2} : 1$ (B) $1 : 8$ (C) $\sqrt{2} : 1$ (D) $1 : \sqrt{2}$
- Q.4 The average (mean) life at a radio nuclide which decays by parallel path is
 $A \xrightarrow{\lambda_1} B; \quad \lambda_1 = 1.8 \times 10^{-2} \text{ sec}^{-1}$
 $A \xrightarrow{\lambda_2} C; \quad \lambda_2 = 2 \times 10^{-3} \text{ sec}^{-1}$
 (A) 52.63 sec (B) 500 sec (C) 50 sec (D) None
- Q.5 A sample of $^{14}\text{CO}_2$ was to be mixed with ordinary CO_2 for a biological tracer experiment. In order that 10 cm^3 of diluted gas at STP should have 10^4 dis/min, what activity (in μCi) of radioactive carbon is needed to prepare 60 L of diluted gas at STP. [$1 \text{ Ci} = 3.7 \times 10^{10} \text{ dps}$]
 (A) 270 μCi (B) 27 μCi (C) 2.7 μCi (D) 2700 μCi

Multiple correct :

- Q.6 Select **correct** statement(s):
 (A) The emission of gamma radiation involves transition between energy levels within the nucleus.
 (B) ^4_2He is formed due to emission of beta particle from tritium ^3_1H .
 (C) When positron ($^0_{+1}\text{e}$) is emitted, $\frac{n}{p}$ ratio increases.
 (D) Decay constant of radioactive substance is independent of temperature.
- Q.7 Select the correct statements
 (A) A radioactive element decays by emitting one α and two β -particles. The daughter element formed is an isotope of the parent element.
 (B) The daughter product formed by the emission of α -particle has mass number less by 4 units than the parent nuclide.
 (C) $^{27}_{13}\text{Al}$ is a stable isotope hence $^{29}_{13}\text{Al}$ is expected to disintegrate by β -emission.
 (D) Emission of a β -particle by a radioactive nuclide results in decrease in n/p ratio.
- Q.8 Select the correct statements
 (A) The decay constant of the end product of a radioactive series is zero
 (B) Positron has same mass as that of an electron.
 (C) $^{14}_9\text{N}$ and $^{16}_8\text{O}$ are isotones.
 (D) The S.I. unit of activity is Curie (Ci).

Q.9 Select the correct statements

- (A) Half-life period of a radioactive substance can be changed by using some suitable catalyst.
- (B) The nuclides with same difference of number of neutrons and number of protons are called isodiaphers
- (C) Half life for certain radioactive element is 15 min. Four nuclei of that element are observed at a certain instant of time. After fifteen minutes, it can be definitely said that two nuclei will be left undecayed.
- (D) 5α and $4\beta^-$ are emitted during the radioactive decay chain starting from ${}^{226}_{88}\text{Ra}$ and ending at ${}^{206}_{82}\text{Pb}$

Q.10 Select the correct statement(s) -

- (A) ${}_{15}\text{P}^{29}$ may emit positron to increase n / p ratio
- (B) During β^- emission, neutron changes into proton in nucleus
- (C) Energy liberated during nuclear fission or fusion is mainly due to mass defect
- (D) Binding energy per nucleon increases continuously with mass number

Assertion & Reason

Q.11. **Statement-1** : ${}^{238}\text{UF}_6$ and ${}^{238}\text{U}$ both have same specific activity.

Statement-2 : ${}^{238}\text{U}$ has same half life whether in free state or bonded state.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Q.12. **Statement-1** : An element may belong to more than one disintegration series.

Statement-2 : Mass number of an element decides the disintegration series to which it belongs.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Q.13 **STATEMENT-1** : The plot of atomic number (y-axis) versus number of neutrons (x-axis) for stable nuclei shows a curvature towards x-axis from the line of 45° slope as the atomic number is increased.

and

STATEMENT-2 : Proton-proton electrostatic repulsions begin to overcome attractive forces involving protons and neutrons in heavier nuclides.

[JEE 2008]

- (A) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1
- (B) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
- (C) STATEMENT-1 is True, STATEMENT-2 is False
- (D) STATEMENT-1 is False, STATEMENT-2 is True

☐ **COMPREHENSION :**

Paragraph for Q.14 to 15

Mass defect in the nuclear reactions may be expressed in terms of the atomic masses of the parent and daughter nuclides in place of their nuclear masses.

- Q.14 The mass defect of nuclear reaction: ${}_4\text{Be}^{10} \rightarrow {}_5\text{B}^{10} + e^-$ is
 (A) $\Delta m = \text{At. mass of } {}_4\text{Be}^{10} - \text{At. mass of } {}_5\text{B}^{10}$
 (B) $\Delta m = \text{At. mass of } {}_4\text{Be}^{10} - \text{At. mass of } {}_5\text{B}^{10} - \text{mass of one electron}$
 (C) $\Delta m = \text{At. mass of } {}_4\text{Be}^{10} - \text{At. mass of } {}_5\text{B}^{10} + \text{mass of one electron}$
 (D) $\Delta m = \text{At. mass of } {}_4\text{Be}^{10} - \text{At. mass of } {}_5\text{B}^{10} - \text{mass of two electrons}$

- Q.15 The mass defect of the nuclear reaction: ${}_5\text{B}^8 \rightarrow {}_4\text{Be}^8 + e^+$ is
 (A) $\Delta m = \text{At. mass of } {}_5\text{B}^8 - \text{At. mass of } {}_4\text{Be}^8$
 (B) $\Delta m = \text{At. mass of } {}_5\text{B}^8 - \text{At. mass of } {}_4\text{Be}^8 - \text{mass of one electron}$
 (C) $\Delta m = \text{At. mass of } {}_5\text{B}^8 - \text{At. mass of } {}_4\text{Be}^8 + \text{mass of one electron}$
 (D) $\Delta m = \text{At. mass of } {}_5\text{B}^8 - \text{At. mass of } {}_4\text{Be}^8 - \text{mass of two electrons}$

- Q.16 **Match the Column-I with Column-II**

Column-I

- (A) ${}_1^2\text{D} + {}_1^3\text{T} \rightarrow {}_2^4\text{He} + {}_0^1\text{n} + \text{energy}$
 (B) ${}_4^9\text{Be} + {}_2^4\text{He} \rightarrow {}_6^{12}\text{C} + {}_0^1\text{n}$
 (C) ${}_{12}^{24}\text{Mg} + {}_2^4\text{He} \rightarrow {}_{14}^{27}\text{Si} + {}_0^1\text{n}$
 (D) ${}_0^1\text{n} \longrightarrow {}_1^1\text{H} + {}_{-1}^0\text{e}$

Column-II

- (P) β -emission
 (Q) Artificial transmutation
 (R) Discovery of neutrons
 (S) Hydrogen bomb

- Q.17

Column-I

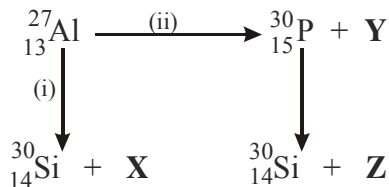
- (A) $\text{n} \longrightarrow \text{p} + \dots\dots\dots$
 (B) $\text{p} \longrightarrow \text{n} + \dots\dots\dots$
 (C) X-ray emission
 (D) $4[{}_1^1\text{H}] \longrightarrow \dots\dots + 2\beta^+ + \text{energy}$

Column-II

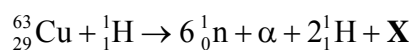
- (P) Positron emission
 (Q) β -emission
 (R) K-electron capture
 (S) α -emission

EXERCISE : J-ADVANCED

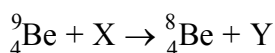
- Q.1 Bombardment of aluminium by α -particle leads to its artificial disintegration in two ways, (i) and (ii) as shown. Products **X**, **Y** and **Z** respectively are : [JEE 2011]



- (A) proton, neutron, positron
(B) neutron, positron, proton
(C) proton, positron, neutron
(D) positron, proton, neutron
- Q.2 The periodic table consists of 18 groups. An isotope of copper, on bombardment with protons, undergoes a nuclear reaction yielding element **X** as shown below. To which group, element **X** belongs in the periodic table ? [JEE 2012]



- Q.3 In the nuclear transmutation [JEE 2013]



(X, Y) is(are)

- (A) (γ , n) (B) (p, D) (C) (n, D) (D) (γ , D)
- Q.4 A closed vessel with rigid walls contains 1 mol of ${}^{238}_{92}\text{U}$ and 1 mol of air at 298 K. Considering complete decay of ${}^{238}_{92}\text{U}$ to ${}^{206}_{82}\text{Pb}$, the ratio of the final pressure to the initial pressure of the system at 298 K is - [JEE 2015]

- Q.5 A plot of the number of neutrons (N) against the number of protons (P) of stable nuclei exhibits upwards deviation from linearity for atomic number, $Z > 20$. For an unstable nucleus having N/P ratio less than 1, the possible mode(s) of decay is(are) - [JEE 2016]

- (A) β^- decay (β emission) (B) orbital or K-electron capture
(C) Neutron emission (D) β^+ decay (positron emission)

ANSWER KEY**EXERCISE (S-1)**

- Q.1 Ans. ${}_{49}^{114}\text{In}$, odd number of nucleons
- Q.2 Ans. beta emitter : ${}^{49}\text{Ca}$, ${}^{30}\text{Al}$, ${}^{94}\text{Kr}$, positron emitter : ${}^{195}\text{Hg}$, ${}^8\text{B}$, ${}^{150}\text{Ho}$
- Q.3 Ans. (a) No. of α -particles = 8, No. of β -particles = 6 ; (b) ${}_{82}^{207}\text{Pb}$
- Q.4 Ans. (8) Q.5 Ans. 1.44×10^{15} dpm.
- Q.6 Ans. 1.06×10^{-15} Q.7 Ans. 2.67×10^5 dps
- Q.8 Ans. $\lambda = 2.079 \text{ hr}^{-1}$ Q.9 Ans. 4.5 hr
- Q.10 Ans. 6.25 % Q.11 Ans. 9
- Q.12 Ans. (8) Q.13 Ans. 33.62 years
- Q.14 Ans.(5) Q.15 Ans. 4.5×10^9 year
- Q.16 Ans. (i) ${}_{19}^{40}\text{K} \longrightarrow {}_{18}^{40}\text{Ar} + {}_{+1}^0\beta + \nu$ (ii) 2.8×10^9 years
- Q.17 Ans (4)
- Q.18 (a) ${}_1^1\text{H}$, (b) ${}_0^1\text{n}$, (c) ${}_3^6\text{Li}$, (d) ${}_{+1}^0\text{e}$, (e) ${}_{-1}^0\text{e}$, (f) p (proton)
 (g) ${}_1^1\text{H}^1$ (h) ${}_{28}^{64}\text{Ni}$ (i) ${}_{-1}^0\text{e}$ (j) ${}_0^1\text{n}^1$
- Q.19 Ans. 24
- Q.20 Ans. $\Delta E = 14.904 \text{ MeV}$

EXERCISE (S-2)

- Q.1 Ans. 160 MeV.
- Q.2 Ans. 23.6 MeV
- Q.3 Ans. $\alpha = \frac{a-b}{4}$; $\beta = d + \frac{(a-b)}{2} - c$
- Q.4 Ans. (i) $t_{\text{means}} = 14.43 \text{ s}$ (ii) 40 seconds
- Q.5 Ans. $7.07 \times 10^{-7} \text{ gm}$
- Q.6 Ans. (5)
- Q.7 Ans. 1648.6
- Q.8 Ans. 6.67×10^5
- Q.9 Ans. ($6.30 \times 10^{-4} \text{ yr}^{-1}$, $3.087 \times 10^{-2} \text{ yr}^{-1}$)
- Q.10. Ans. 100

EXERCISE (O-1)

- | | | | |
|--------------|--------------|---------------|--------------|
| Q.1 Ans.(A) | Q.2 Ans.(C) | Q.3 Ans.(B) | Q.4 Ans.(C) |
| Q.5 Ans.(A) | Q.6 Ans.(B) | Q.7 Ans.(D) | Q.8. Ans.(C) |
| Q.9 Ans.(C) | Q.10 Ans.(B) | Q.11 Ans.(C) | Q.12 Ans.(C) |
| Q.13 Ans.(C) | Q.14 Ans.(B) | Q.15. Ans.(D) | Q.16 Ans.(D) |
| Q.17 Ans.(A) | Q.18 Ans.(A) | Q.19 Ans.(B) | Q.20 Ans.(D) |

EXERCISE (O-2)

- | | | | | | | | |
|------|---|-------|-------------|-------|---------------|-------|-----------|
| Q.1 | Ans.(A) | Q.2 | Ans.(D) | Q.3 | Ans.(C) | Q.4 | Ans.(C) |
| Q.5 | Ans.(B) | Q.6 | Ans.(A,C,D) | Q.7 | Ans.(A,B,C,D) | Q.8 | Ans.(A,B) |
| Q.9 | Ans.(B,D) | Q.10. | Ans (A,B,C) | Q.11. | Ans.(D) | Q.12. | Ans.(A) |
| Q.13 | Ans.(A) | Q.14 | Ans.(A) | Q.15 | Ans.(D) | | |
| Q.16 | Ans. (A) - S ; (B) - Q, R ; (C) - Q ; (D) - P | | | | | | |
| Q.17 | Ans. (A) - Q ; (B) - P ; (C) - R ; (D) - S | | | | | | |

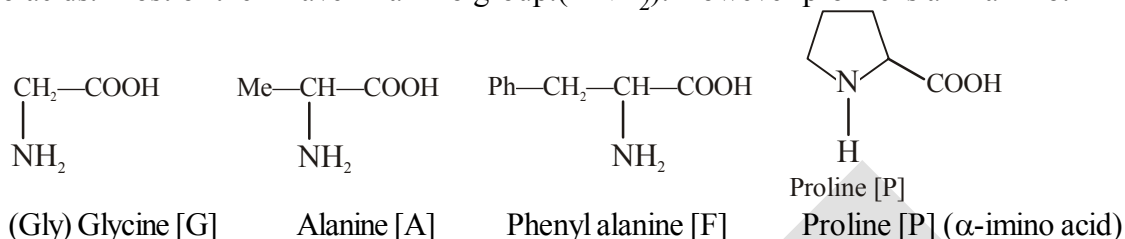
EXERCISE (J-ADVANCED)

- | | | | |
|-----|------------|-----|----------|
| Q.1 | Ans. (A) | Q.2 | Ans. (8) |
| Q.3 | Ans. (A,B) | Q.4 | Ans. (9) |
| Q.5 | Ans.(B, D) | | |

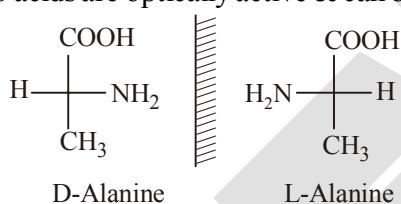
AMINO ACIDS AND PROTEINS

Bifunctional compounds having an acidic carboxylic group & a basic amino group are known as amino acid.

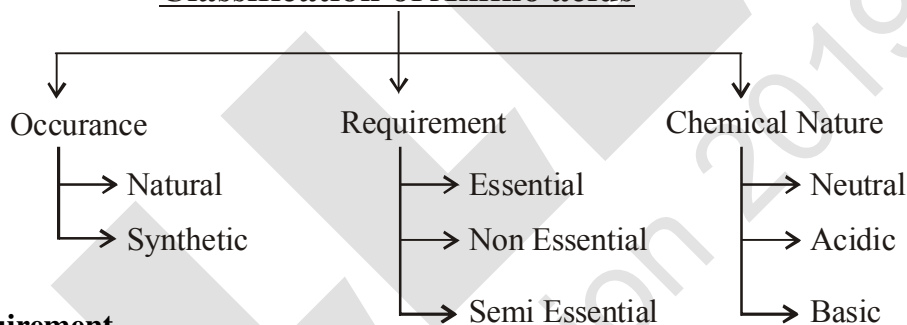
There are 20 amino acids commonly found in proteins and are standard amino acids. All are α amino acids. Most of them have 1° amino group ($-\text{NH}_2$). However proline is a 2° amino.



All amino acids are chiral molecules with at least one chiral carbon (except glycine, $\text{H}_3\text{N}^+\text{CH}_2\text{COO}^-$). Except Glycine all other amino acids are optically active & can be assigned D & L configuration.



Classification of Amino acids



Based on requirement.

- Essential amino acids can not be synthesized in human body so dietary intake is required. For any human being 1 gm a day is required.
- Semi essential amino acids can be synthesized in human body but dietary intake is required during growing stages (when more of cell division is required).
For example : Early childhood, pregnancy and lactating mother.
- Non essential amino acid - Body can synthesize them.

Based on chemical nature

Neutral - Amino acid having equal number of NH_2 and COOH .

Neutral amino acids are further classified as polar and nonpolar depending on whether their side chains have polar substituents (for example, asparagine with an NH_2CO group) or are completely hydrocarbon in nature (for example alanine, valine etc.).

Acidic - Amino acid having more COOH than NH_2 group.

For example : Aspartic acid and glutamic acids, each with a second CO_2H in their side chain are acidic amino acids.

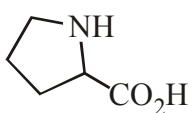
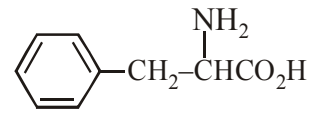
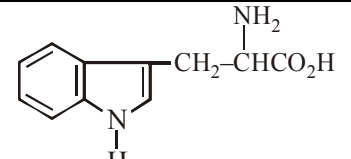
Basic - Amino acid having more NH_2 than COOH group.

For example : Lysine, arginine and histidine)

Proteins: The name protein is taken from the Greek word "proteios", which means "first". Of all chemical compounds, proteins must almost certainly be ranked first, for they are the substance of life.

Proteins make up a large part of the animal body, they hold it together and they run it. They are found in all living cells.

Chemically, proteins are high polymers. They are polyamides and the monomers from which they are derived are the α - amino carboxylic acids. A single protein molecule contains hundreds or even thousands of amino acid units. These units can be of twenty-odd different kinds. The number of different combinations, i.e., the number of different protein molecules that are possible, is almost infinite.

<i>A -1. Neutral amino acids (with nonpolar side chains)</i>			
NAME	ABBREVIATIONS	STRUCTURAL FORMULAE	ISOELECTRIC POINT[pI]
@Glycine	Gly(G)	$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}_2\text{C} \\ \\ \text{C}=\text{O} \\ \\ \text{OH} \end{array}$	6.0
Alanine	Ala(A)	$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}_3\text{C}-\text{CH} \\ \\ \text{C}=\text{O} \\ \\ \text{OH} \end{array}$	6.0
Valine*	Val(V)	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{C}-\text{NH}_2 \\ \quad \quad \\ \text{CH}_3 \quad \quad \text{C}=\text{O} \\ \quad \quad \quad \\ \quad \quad \quad \text{OH} \end{array}$	6.0
Leucine*	Leu(L)	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{CH}_2-\text{C}-\text{NH}_2 \\ \quad \quad \quad \\ \text{CH}_3 \quad \quad \quad \text{C}=\text{O} \\ \quad \quad \quad \quad \\ \quad \quad \quad \quad \text{OH} \end{array}$	6.0
Isoleucine*	Ile(I)	$\begin{array}{c} \text{CH}_3 \quad \text{NH}_2 \\ \quad \quad \\ \text{CH}_3\text{CH}_2\text{CH}-\text{CHCO}_2\text{H} \end{array}$	6.0
Methionine*	Met(M)	$\begin{array}{c} \text{NH}_2 \\ \\ \text{CH}_3\text{SCH}_2\text{CH}-\text{CHCO}_2\text{H} \end{array}$	5.7
@@Proline	Pro(P)		6.3
Phenylalanine*	Phe(F)		5.5
Tryptophan*	Trp(W)		5.9

A -2. Neutral amino acids (with polar, but nonionized side chains)

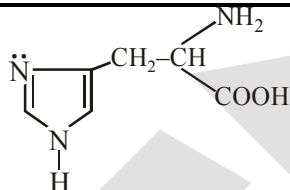
NAME	ABBREVIATIONS	STRUCTURAL FORMULAE	ISOELECTRIC POINT [pI]
Asparagine	Asn(N)	$\text{H}_2\text{N}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\underset{\text{C}=\text{O}}{\overset{\text{NH}_2}{\text{CH}}}-\text{OH}$	5.4
Glutamine	Gln(Q)	$\text{H}_2\text{N}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_2-\underset{\text{C}=\text{O}}{\overset{\text{NH}_2}{\text{CH}}}-\text{OH}$	5.7
Serine	Ser(S)	$\text{HO}-\text{CH}_2-\underset{\text{C}=\text{O}}{\overset{\text{NH}_2}{\text{CH}}}-\text{OH}$	5.7

A -3. Neutral amino acids (with polar, but nonionized side chains)

NAME	ABBREVIATIONS	STRUCTURAL FORMULAE	ISOELECTRIC POINT [pI]
Threonine*	Thr	$\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\underset{\text{NH}_2}{\text{CH}}-\text{CO}_2\text{H}$	5.6
Tyrosine	Tyr(Y)	$\text{HO}-\text{C}_6\text{H}_4-\underset{\text{NH}_2}{\text{CH}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$	5.7
Cysteine	Cys	$\text{HSCH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{CO}_2\text{H}$	5.1
± Cystine	Cys-Cys	$\text{HOOC}-\underset{\text{NH}_2}{\text{CH}}-\text{CH}_2-\text{S}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$	

B - Acidic amino acids (side chain with carboxylic acid group)

NAME	ABBREVIATIONS	STRUCTURAL FORMULAE	ISOELECTRIC POINT [pI]
Aspartic acid	Asp(D)	$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$	2.8
Glutamic Acid	Glu(E)	$\text{O}=\overset{\text{OH}}{\text{C}}-\text{CH}_2-\text{CH}_2-\underset{\text{C}=\text{O}}{\overset{\text{NH}_2}{\text{CH}}}-\text{OH}$	3.2

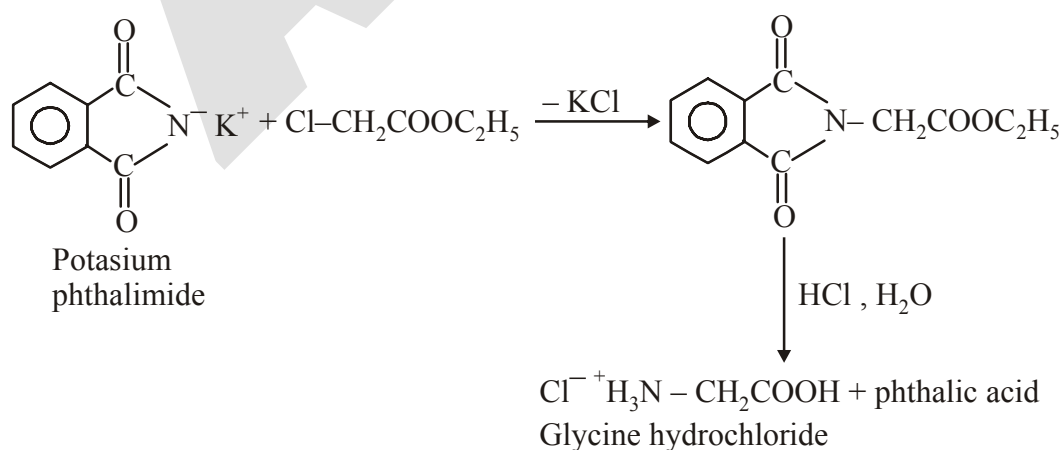
<i>C - Basic amino acids (side chain with nitrogenous basic group)</i>			
NAME	ABBREVIATIONS	STRUCTURAL FORMULAE	ISOELECTRIC POINT[pI]
Lystine*	Lys(K)	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH} \begin{array}{l} \nearrow \text{NH}_2 \\ \searrow \text{C}=\text{O} \\ \quad \text{OH} \end{array}$	9.7
Arginine*	Arg(R)	$\text{H}_2\text{N}-\overset{\text{NH}}{\underset{\text{ }}{\text{C}}}-\text{NH}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH} \begin{array}{l} \nearrow \text{NH}_2 \\ \searrow \text{C}=\text{O} \\ \quad \text{OH} \end{array}$	10.8
Histidine*	His(H)		7.6

Note:

- * Amino acids with an asterisk are essential amino acids.
- † At pH = 7, Asp and Glu have a net negative charge and exist as anions. At pH = 7, Lys and Arg have a net positive charge and exist as cations. Rest of the amino acids at this pH exist in the neutral form.
- ‡ Structurally, in cystine, the two cysteine molecules are joined through sulfur (disulfide linkage).
- @@ Proline is an α -imino acid, all amino acids are primary amines except proline and 4-hydroxyproline, which are 2° amines.
- @ Except Glycine all other amino acids are optically active.

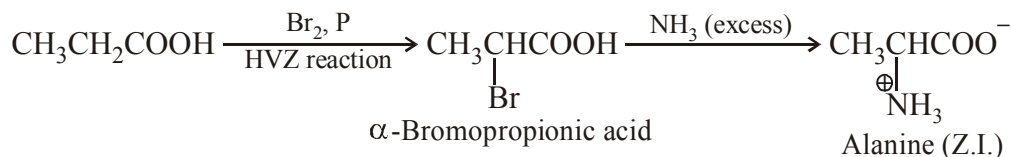
Preparation of amino acids**(a) Gabriel Phthalimide synthesis**

Good yields of amino acids are generally obtained by the Gabriel phthalimide synthesis ; In this method α - halo esters are used instead of α - halo acids .

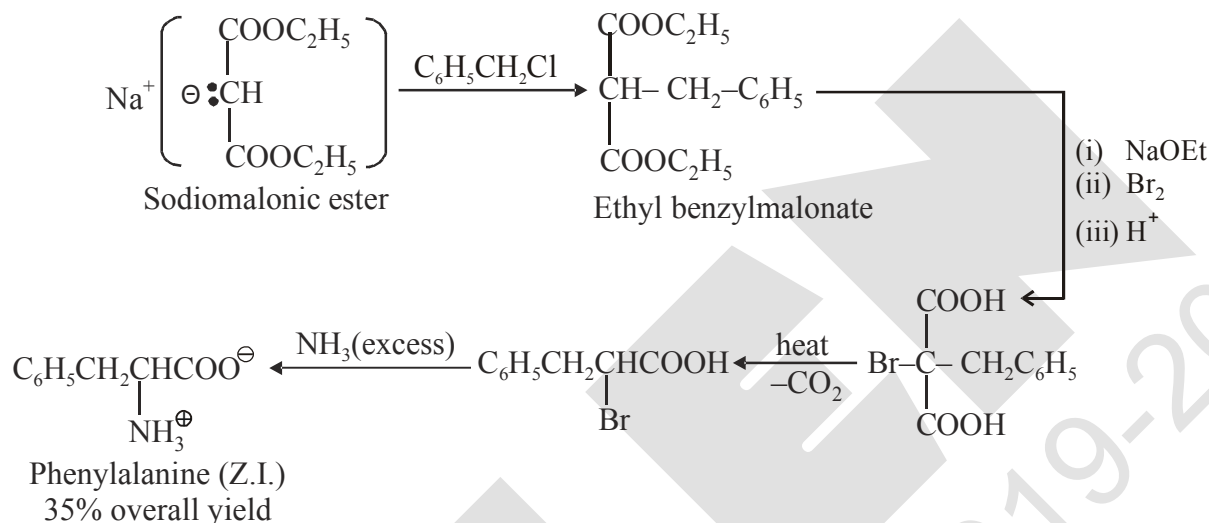


(b) **Amination of α - Halo acids**

Sometimes an α - chloro or α - bromo acid is subjected to direct ammonolysis with excess of concentrated ammonia.

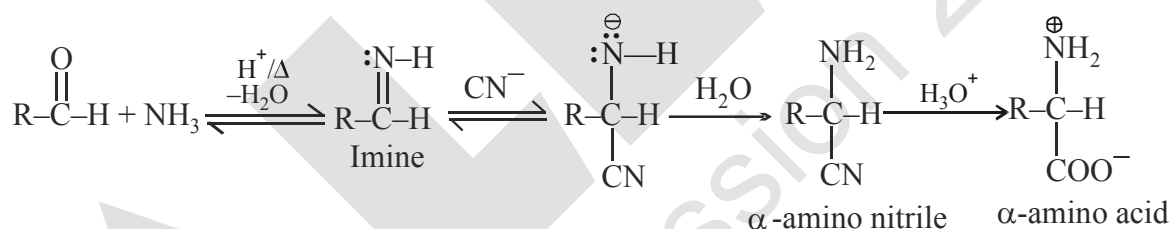


(c) **From diethyl malonate**

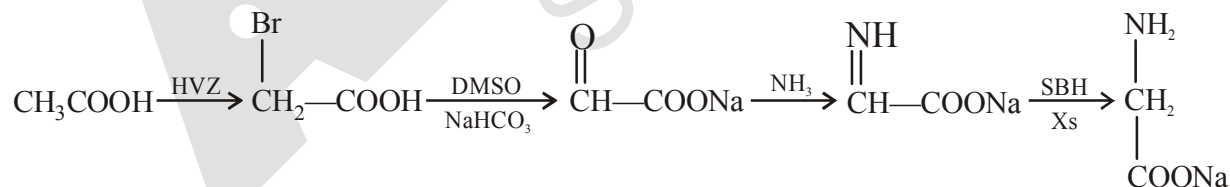


(d) **Strecker's synthesis**

Strecker's synthesis is also used for preparing α - amino acids



(e) **Using KOOP synthesis**



Properties of Amino acids

Although the amino acids are commonly shown as containing an amino group and a carboxyl group, $\text{H}_2\text{NCH(R)COOH}$, certain properties, (both physical and chemical) are not consistent with this structure

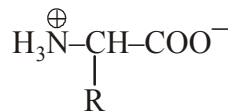
I. Physical properties

In contrast to amines and carboxylic acids, the amino acids are nonvolatile crystalline solids, which melt with decomposition at fairly high temperatures. They are insoluble in non-polar solvents like petroleum ether, benzene or ether and are appreciably soluble in water. Their aqueous solutions behave like solutions of substances of high dipole moment due to existence.

Amino acids as dipolar ions as zwitter ion

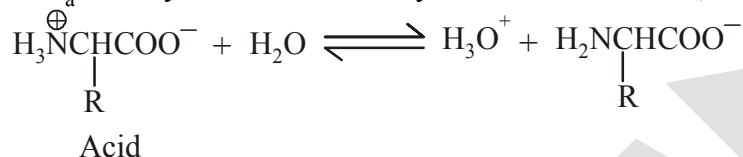
Acidity and basicity constant are ridiculously low for $-\text{COOH}$ and $-\text{NH}_2$ groups. Glycine, for example, has $K_a = 1.6 \times 10^{-10}$ and $K_b = 2.5 \times 10^{-12}$, whereas most carboxylic acids have K_a values of about 10^{-5} and most aliphatic amines have K_b values of about 10^{-4} .

All these properties are quite consistent with a dipolar ion structure for the amino acids (I)



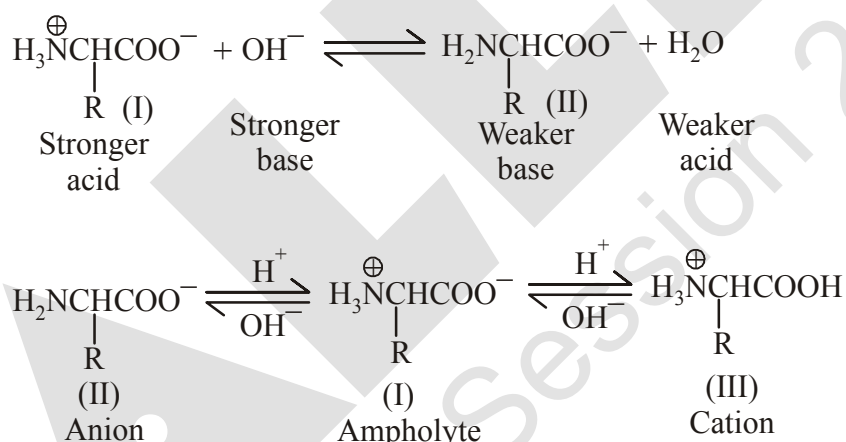
Amino acids : dipolar ions (Zwitter ion)

Physical properties - melting point, solubility, high dipole moment - are just what would be expected of such a salt. The acid-base properties also become understandable when it is realized that the measured K_a actually refers to the acidity of an ammonium ion, RNH_3^+ ,



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{H}_2\text{NCHRCOO}^-]}{[\text{H}_3\text{NCHRCOO}^+]}$$

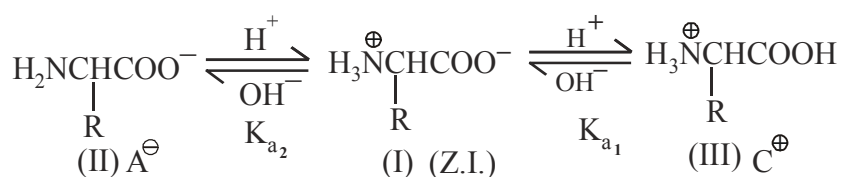
When the solution of an amino acid is made alkaline, the dipolar ion (I) is converted into the anion (II). The stronger base, hydroxide ion, removes a proton from the ammonium ion and displaces the weaker base, the amine.



Wherever feasible, we can speed up a desired reaction by adjusting the acidity or basicity of the solution in such a way as to increase the concentration of the reactive species.

Isoelectric point of amino acids

What happens when a solution of an amino acid is placed in an electric field depends upon the of the solution.



In quite alkaline solution, anions (II) exceed cations (III), and there is a net migration of amino acid toward the anode. In quite acidic solution, cations (III) are in excess, and there is a net migration of amino acid toward the cathode. If (II) and (III) are exactly balanced, there is no net migration; under such conditions any one molecule exists as a positive ion and as a negative ion for exactly the same amount of time, and any small movement in the direction of one electrode is subsequently cancelled by an equal movement back towards the other electrode. **The hydrogen ion concentration of the solution in which a particular amino acid does not migrate under the influence of an electric field is called the isoelectric point (pI) of that amino acid. The isoelectric point (pI) is the pH at which the amino acid exists only as a dipolar ion with net charge zero.**

For glycine, for example, the isoelectric point is at pH 6.1.

An amino acid usually shows its lower solubility in a solution at the isoelectric point, since here there is the highest concentration of the dipolar ion. As the solution is made more alkaline or more acidic, the concentration of one of the more soluble ions, (II) or (III) increases.

$$K_{a1} = \frac{[Z.I.] [H^+]}{[C^+]} \quad K_{a2} = \frac{[A^-] [H^+]}{[DI]} \quad \text{at pI } [A^\ominus] = [C^\oplus]$$

$$\frac{[Z.I.] [H^+]}{K_{a1}} = \frac{K_{a2} [DI]}{[H^+]} \quad [H^+]^2 = K_{a1} \& K_{a2}$$

$$\text{on taking antilog pI} = \frac{pK_{a1} + pK_{a2}}{2}$$

An amino acid having $-\text{COOH}$ group more than NH_2 group or such amino acid have pI less than 7.

An amino acid having more $-\text{NH}_2$ more than COOH group such amino acid have pI more than 7.

Q. Write the structure of alanine at pH 2.5, 10.5 and 6.

Electrophoresis

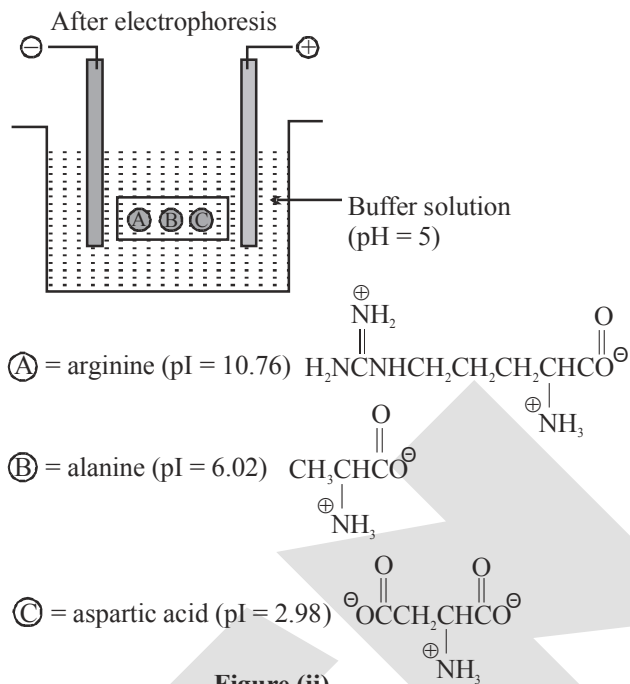
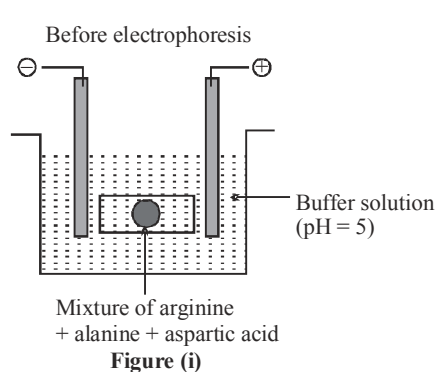
The movement of charged molecules (like amino acid) under the influence of an electric field is called electrophoresis. Electrophoresis separates amino acids on the basis of their pI values.

Amino acid is positively charged (moves towards cathode) if pH of the solution $<$ pI

Amino acid is negatively charged (moves towards anode) if pH of the solution $>$ pI

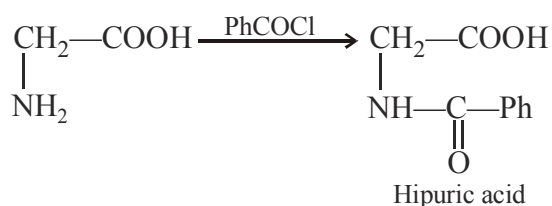
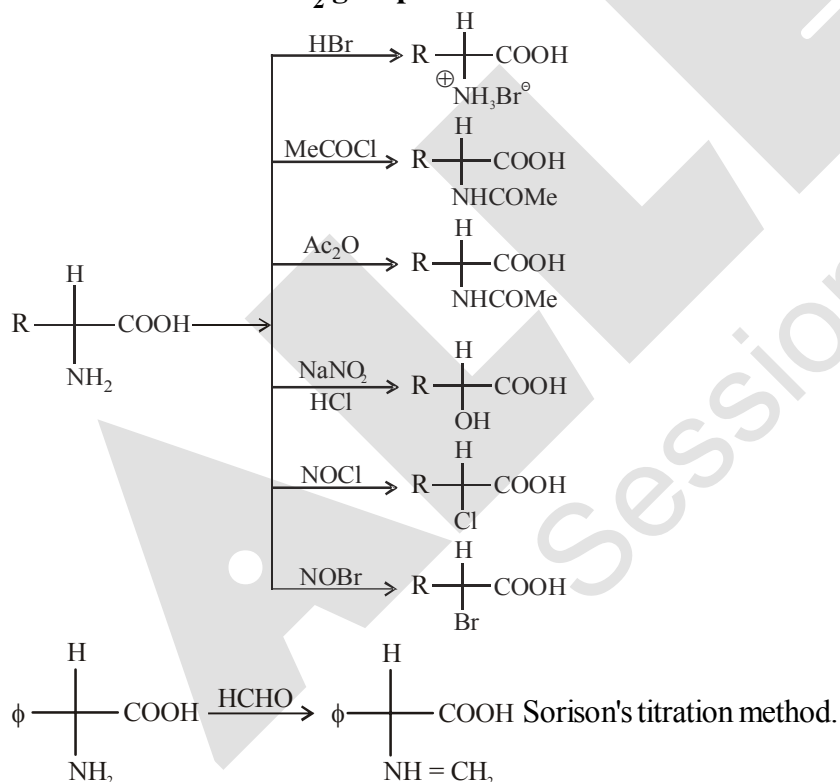
Q. How will you separate a ternary mixture of arginine, alanine & aspartic acid?

Ans. A few drops of a solution of an amino acid mixture are applied to the middle of a piece of filter paper. When the paper is placed in a buffer solution (pH = 5) between the two electrodes and an electric field is applied then arginine & alanine with $pI > pH$ move towards the cathode and aspartic acid with $pI < pH$ moves towards the anode. Out of arginine & alanine, alanine will move slowly towards the cathode due to lesser positive charge.



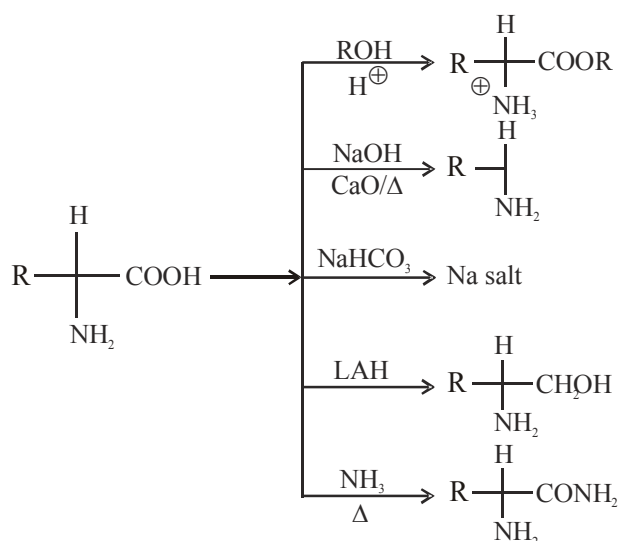
General reactions of amino acids

(1) Reactions due to $-\text{NH}_2$ group



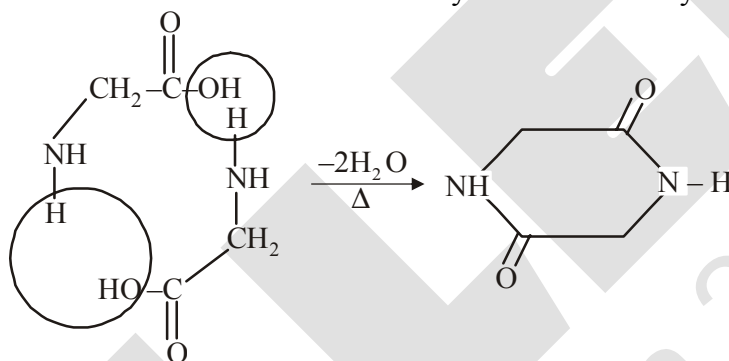
Reactions is used to block $-\text{NH}_2$ group during volumetric analysis in.

(2) Reactions due to – COOH group.

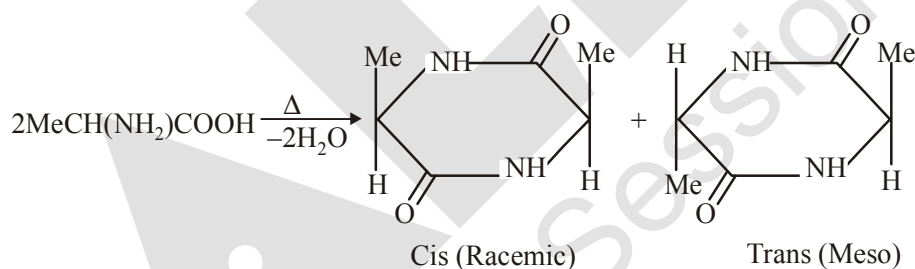


(3) **Heating Effect**

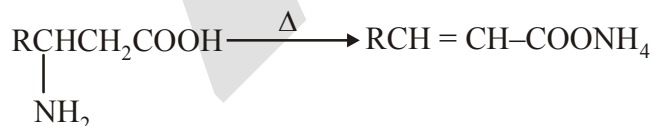
(i) Heating of amino acids leads to intermolecular dehydration to form cyclic diamides.



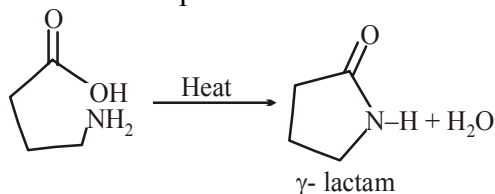
(ii) When alanine is heated, then two diastereomers are obtained. One of them (trans) is not resolvable.

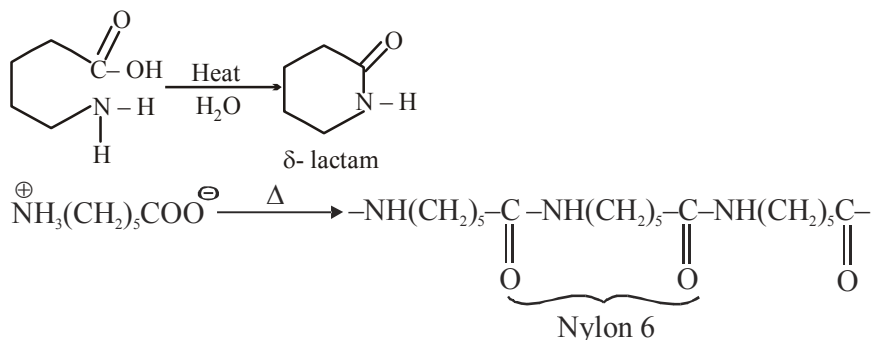


(iii) When β - amino acids are heated, α , β - unsaturated salt are formed.



(iv) γ, δ, ϵ - amino acids when heated alone gives γ, δ - lactam and polymer respectively. The reason for the formation of polymer is that when ϵ - amino cyclises intramolecularly, it leads to large angle strain within the compound



(4) **Peptide**

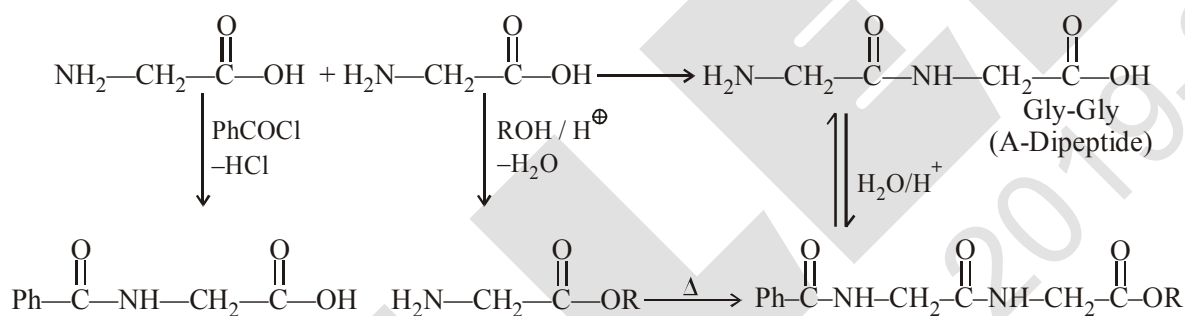
A peptide is a kind of amide formed by intermolecular reaction of the amino group of one amino acid and the carboxyl group of a second amino acid. Dipeptides are made from two amino acids, tripeptides from three amino acids, etc, which may be the same or different. If there are three to ten amino acid residues, the peptide is also called an oligopeptide.

If they give 3 to 10 amino acid they are oligopeptide

If they give 11 to 100 amino acid they are Polypeptide

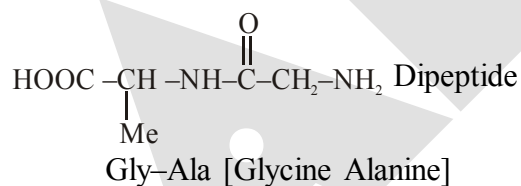
For more than 100 it is Macropeptide

- Peptides can be prepared by blocking technique

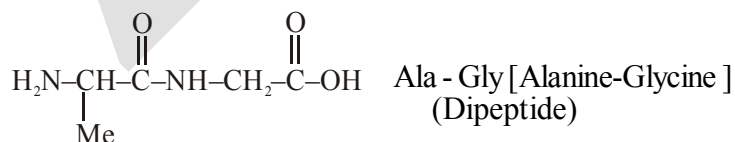


Firstly, the amino and carboxyl groups that are not to be linked in peptide bonds must be blocked to make unreactive.

Abbreviated name of amino acid with free NH_2 is written first.



By convention, the amino acid with the free amino group (N-terminal) is written at the left end and the one with the unreacted carboxyl group (C-terminal) at the right end.



- When different amino acids are involved in peptide formation.

Then total number of polypeptide possible = X^n

[X = type of amino acid interacting,

n = number of amino acid molecule are interacting.]

- Q. Glycine can form how many Dipeptide ? [Ans. One]
 Q. Glycine can form how many Tripeptide ? [Ans. One]
 Q. Glycine and Ala can form how many Dipeptide ? [Ans. Four]
 Q. Gly, Ala, and Phenyl Ala can form how many Dipeptide ? [Ans. Nine]
 Q. Gly, Ala, can form how many Tripeptide ? [Ans. Eight]

A polypeptide with more than hundred amino acid residues, having molecular mass higher than 10,000 is called a protein. However, the distinction between a polypeptide and a protein is not very sharp. Polypeptides with fewer amino acids are likely to be called proteins they ordinarily have a well defined conformation of a protein such as insulin which contains 51 amino acids.

Proteins can be classified into two types on the basis of their molecular shape.

(a) **Fibrous proteins**

When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed. Such proteins are generally insoluble in water. Some common examples are keratin (present in hair, wool, silk) and myosin (present in muscles), etc.

(b) **Globular proteins**

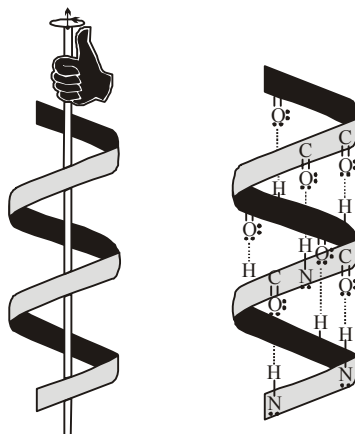
This structure results when the chains of polypeptides coil around to give a spherical shape. These are usually soluble in water. Insulin and albumins are the common examples of globular proteins.

Structure of Proteins

Structure and shape of proteins can be studied at four different levels, i.e., primary, secondary, tertiary and quaternary, each level being more complex than the previous one.

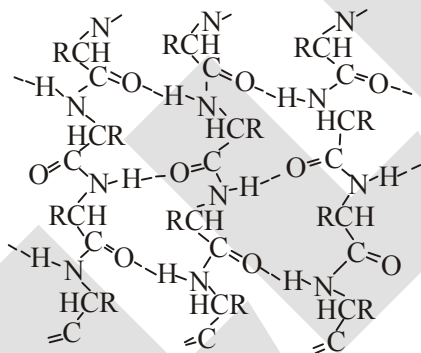
- (i) **Primary structure of proteins :** Proteins may have one or more polypeptide chains. Each polypeptide in a protein has amino acids linked with each other in a specific sequence and it is this sequence of amino acids that is said to be the primary structure of the protein. Any change in this primary structure i.e., the sequence of amino acids creates a different protein.
- (ii) **Secondary structure of proteins :** The secondary structure of protein refers to the shape in which a long polypeptide chain can exist. They are found to exist in two different types of structures viz. α -helix and β -pleated sheet structure. These structures arise due to the regular folding of the backbone

of the polypeptide chain due to hydrogen bonding between $\text{—}\overset{\text{O}}{\parallel}{\text{C}}\text{—}$ and —NH— groups of the peptide bond.

 α -Helix structure of proteins

α -Helix is one of the most common ways in which a polypeptide chain forms all possible hydrogen bonds by twisting into a right handed screw (helix) with the —NH group of each amino acid residue hydrogen bonded to the $\text{C}=\text{O}$ of an adjacent turn of the helix as shown in figure.

In β -structure all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together, by intermolecular hydrogen bonds. The structure resembles the pleated folds of drapery and therefore is known as β -pleated sheet.

 β -Pleated sheet structure of proteins

- (i) Ionic bonding : between COO^- and NH_3^+ at different sites.
- (ii) H-bonding : mainly between side-chain NH_2 and COOH , also involving OH's (Of serine, for example) and the N—H of tryptophan.
- (iii) Weakly hydrophobic Van der Waal's attractive forces engendered by side-chain R groups and
- (iv) Disulfide cross linkages between loops of the polypeptide chain.

The same kind of attractive and repulsive forces responsible for the tertiary structure operate to hold together and stabilize the subunits of the quaternary structure.

- (iii) **Tertiary structure of proteins :** The tertiary structure of proteins represents overall folding of the polypeptide chains i.e., further folding of the secondary structure. It gives rise to two major molecular shapes viz. fibrous and globular. The main forces which stabilise the 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages, van der Waals and electrostatic forces of attraction.

- (iv) **Quaternary structure of proteins :** Some of the proteins are composed of two or more polypeptide chains referred to as sub-units. The spatial arrangement of these subunits with respect to each other is known as quaternary structure. Example : Haemoglobin, Chlorophyll.

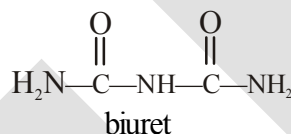
According to their biological action, they are classified as enzymes, hormones, antibodies, etc.

Protein found in living system with definite configuration and biological activity is termed as native protein. If a native protein is subjected to physical or chemical treatment, which may disrupt its higher structures (conformations) without affecting its primary structure, the protein is said to be denatured. During denaturation, the protein molecule uncoils from an ordered and specific conformation into a more random conformation leading to precipitation. Thus denaturation leads to increase in entropy and loss of biological activity of the protein. The denaturation may be reversible or irreversible. Thus, the coagulation of egg white on boiling of egg protein is an example of irreversible protein denaturation. However, in certain cases it is found that if the disruptive agent is removed the protein recovers its original physical and chemical properties and biological activity the reverse of denaturation is known as renaturation.

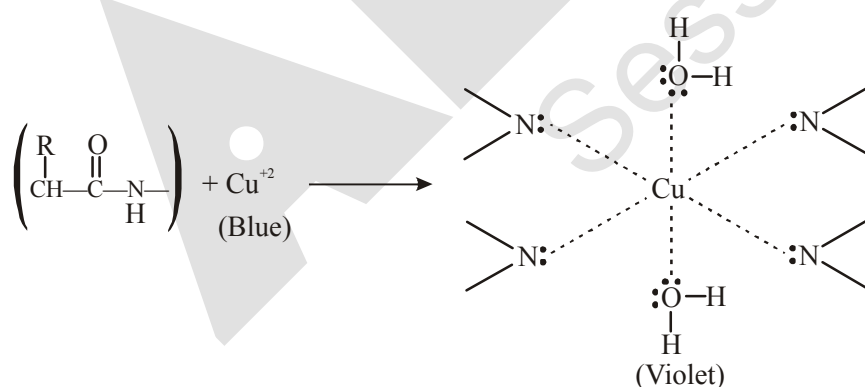
Tests of Proteins:

Biuret test : Addition of a very dilute solution of CuSO_4 to an alkaline solution of a protein is done. A positive test is indicated by the formation of a pink violet to purple violet color.

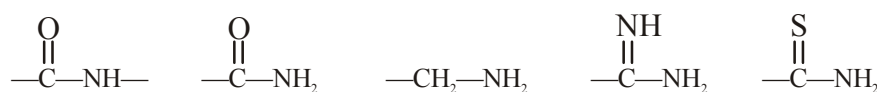
The name of test is derived from a specific compound, biuret, which gives a positive test with this reagent



When a protein reacts with copper (II) sulfate (blue), the positive test is the formation of a violet colored complex.

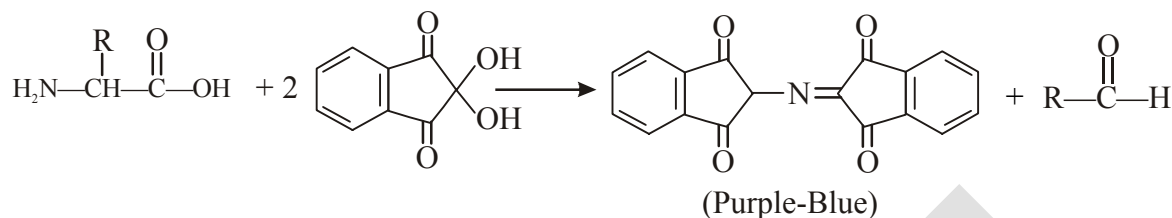


The biuret test works for any compound containing two or more of the following groups.



Ninhydrin Test : The ninhydrin test is a test for amino acids and proteins with a free $-\text{NH}_2$ group.

Amino acids are detected by ninhydrin test. All amino acids give violet - coloured product with ninhydrin (triketo hydroindene hydrate) except proline and 4 - hydroxy proline, which gives yellow colour with it. When such an -NH_2 group reacts with ninhydrin, a purple-blue complex is formed.



The same violet coloured dye forms from all α - AA's with 1° amino groups because only their nitrogen is incorporated into it. The 2° amines proline and 4 - hydroxyproline give different adducts that absorb light at a different and thus have a different yellow colour.

CARBOHYDRATES

Carbohydrates (hydrates of carbon) are naturally occurring compounds having general formula $\text{C}_x(\text{H}_2\text{O})_y$, which are constantly produced in nature & participate in many important bio-chemical reactions.

Ex. - Glucose	$\text{C}_6\text{H}_{12}\text{O}_6$	$\text{C}_6(\text{H}_2\text{O})_6$
Fructose	$\text{C}_6\text{H}_{12}\text{O}_6$	$\text{C}_6(\text{H}_2\text{O})_6$
Cellulose and Starch	$(\text{C}_6\text{H}_{10}\text{O}_5)_n$	$(\text{C}_6(\text{H}_2\text{O})_5)_n$

- Sucrose (Cane sugar) - $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, and
- Maltose (Malt Sugar) $\text{C}_{12}(\text{H}_2\text{O})_{11}$

But some compounds which have formula according to $\text{C}_x(\text{H}_2\text{O})_y$ are not known as carbohydrate

Ex.	CH_2O	Formaldehyde
	$\text{C}_2(\text{H}_2\text{O})_2$	Acetic acid
	$\text{C}_3(\text{H}_2\text{O})_3$	Lactic acid

There are many compounds, which show chemical behaviour of carbohydrate but do not confirm the general formula $\text{C}_x(\text{H}_2\text{O})_y$ such as - $\text{C}_5\text{H}_{10}\text{O}_4$ (2-deoxyribose), $\text{C}_6\text{H}_{12}\text{O}_5$ (Rahmnose)
 $\text{C}_7\text{H}_{14}\text{O}_6$ (Rahmnohexose)

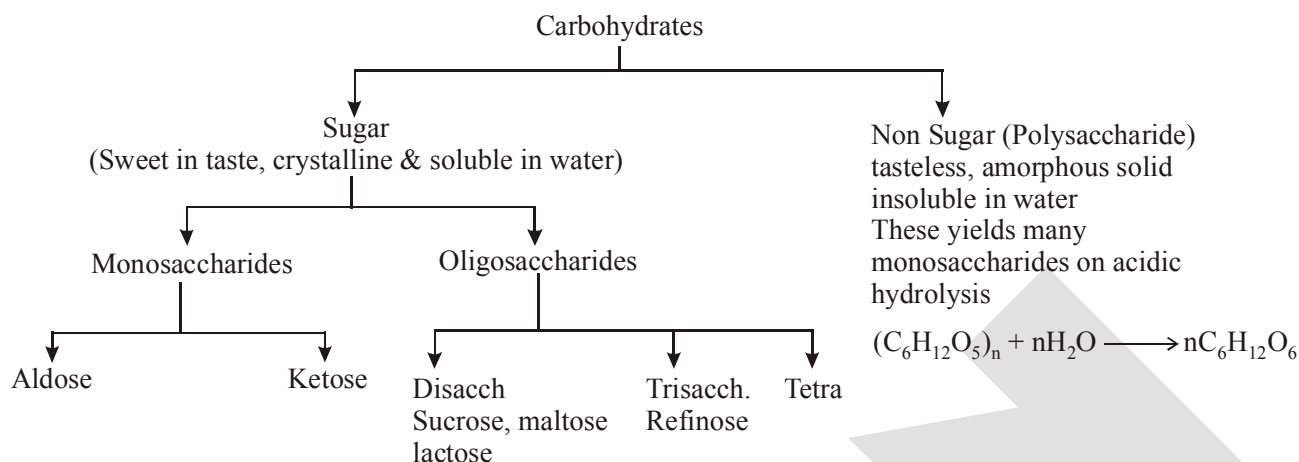
Modern Concept : Carbohydrates are polyhydroxy aldehyde or ketone

or

Substances which yield these (polyhydroxy aldehyde or ketone) on hydrolysis

- Carbohydrates $\xrightarrow{\text{H}_2\text{O}/\text{H}^+}$ Polyhydroxy aldehyde or ketone
- Carbohydrates are also known as Saccharides.
- In plants carbohydrates are synthesised by photosynthesis

Classification of carbohydrates :



Monosaccharides : (simple sugars)

These are the sugars which cannot be hydrolysed into smaller molecules.

General formula is $C_nH_{2n}O_n$

Ex. - Glucose, Fructose, Ribose, Deoxyribose

- If aldehyde group is present in monosaccharide, then it is known as aldose.
- If ketone group is present in monosaccharide, then it is known as ketose.

Oligosaccharides :-

These are the sugars which yield 2–10 monosaccharide units on hydrolysis. such as.

These are of following types

(a) *Disaccharides* :- Gives two monosaccharide unit on hydrolysis (may or may not be same).

Ex. - Sucrose, Maltose, Lactose

(b) *Trisaccharides* :- Gives Three monosaccharide unit on hydrolysis.

Polysaccharides :- These are the non sugars which yield a large no of monosaccharide units on hydrolysis

General formula - $(C_6H_{10}O_5)_n$. Ex.- Starch, Cellulose, Glycogen

Note :- A group of polysaccharides which are not so widely used in nature is pentosans $(C_5H_8O_4)_n$

Monosaccharides, General formula $C_x(H_2O)_y$ $x = 3 - 8$. Nomenclature of monosaccharides are given according to the no. of carbons present in them.

On the basis of C-atom monosaccharides can be farther classified as, Trioses, Tetras, Pentoses, Hexoses.

Table

		Aldoses	Ketoses
3C	Triose or Triose	Aldotriose	Ketotriose
4C	Tetrose	Aldotetrose	Ketotetrose
5C	Pentose	Aldopentose	Ketopentose
5C	Including $-\text{CHO}$	Aldopentose (Ribose)	$ \begin{array}{c} \text{H}-\text{C}=\text{O} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} $
5C	Including $-\text{C}(=\text{O})-$	Ketopentose	$ \begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{C}=\text{O} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} \text{ (Ributose)} $
6C	Hexose	Aldohexose	Ketohexose
6C	Including $-\text{CHO}$	Aldohexose (Glucose)	$ \begin{array}{c} \text{CHO} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{HO}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} $ D-glucose $ \begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{C}=\text{O} \\ \\ \text{HO}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} $ D-fructose

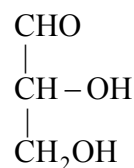
Stereochemistry of carbohydrates :

D & L-Sugars : The series of aldoses or ketoses in which the configuration of the penultimate C-atom (C-next to $\text{CH}_2\text{-OH}$ group) is described as D-sugars if -OH is towards RHS & L-sugars if it is towards LHS.

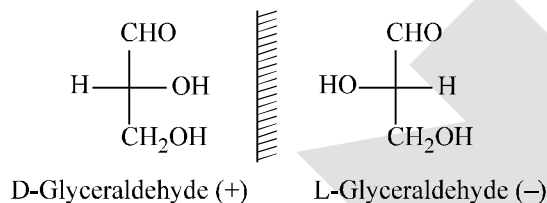
Smallest carbohydrate

* Aldotriose

* Glyceraldehyde



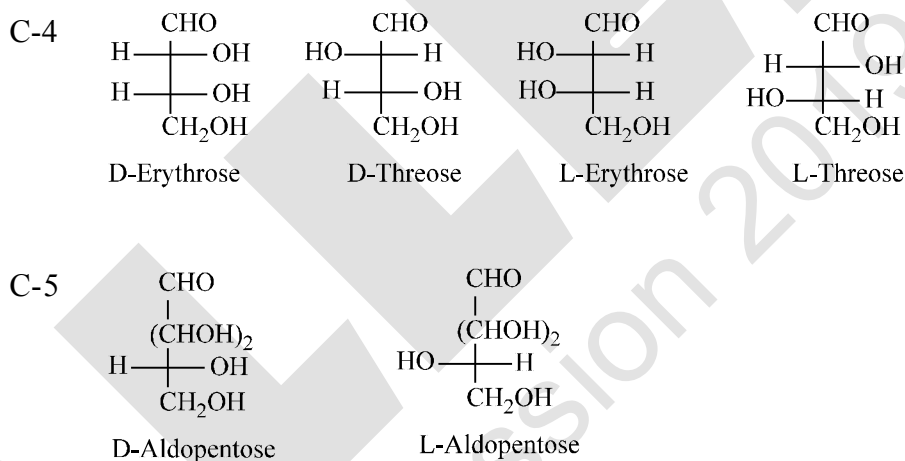
Fischer projection



Classification of Aldotetros :

(i) Erythrose

(ii) Threose



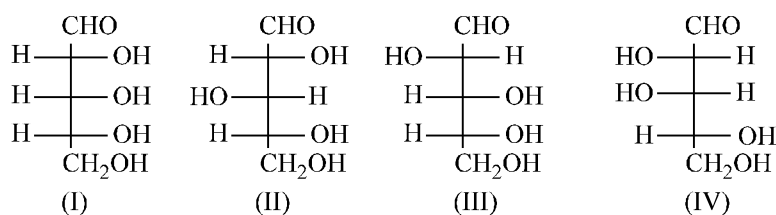
No. of $\text{C}^* = 3$ (in Aldopentose)

No. of optical isomers $2^3 = 8$

No. of D Sugars 4

No. of L Sugars 4

D-Aldopentose :



All Isomeric D-sugars are diastereomers.

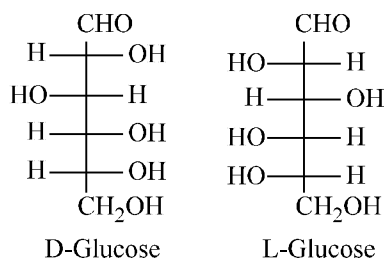
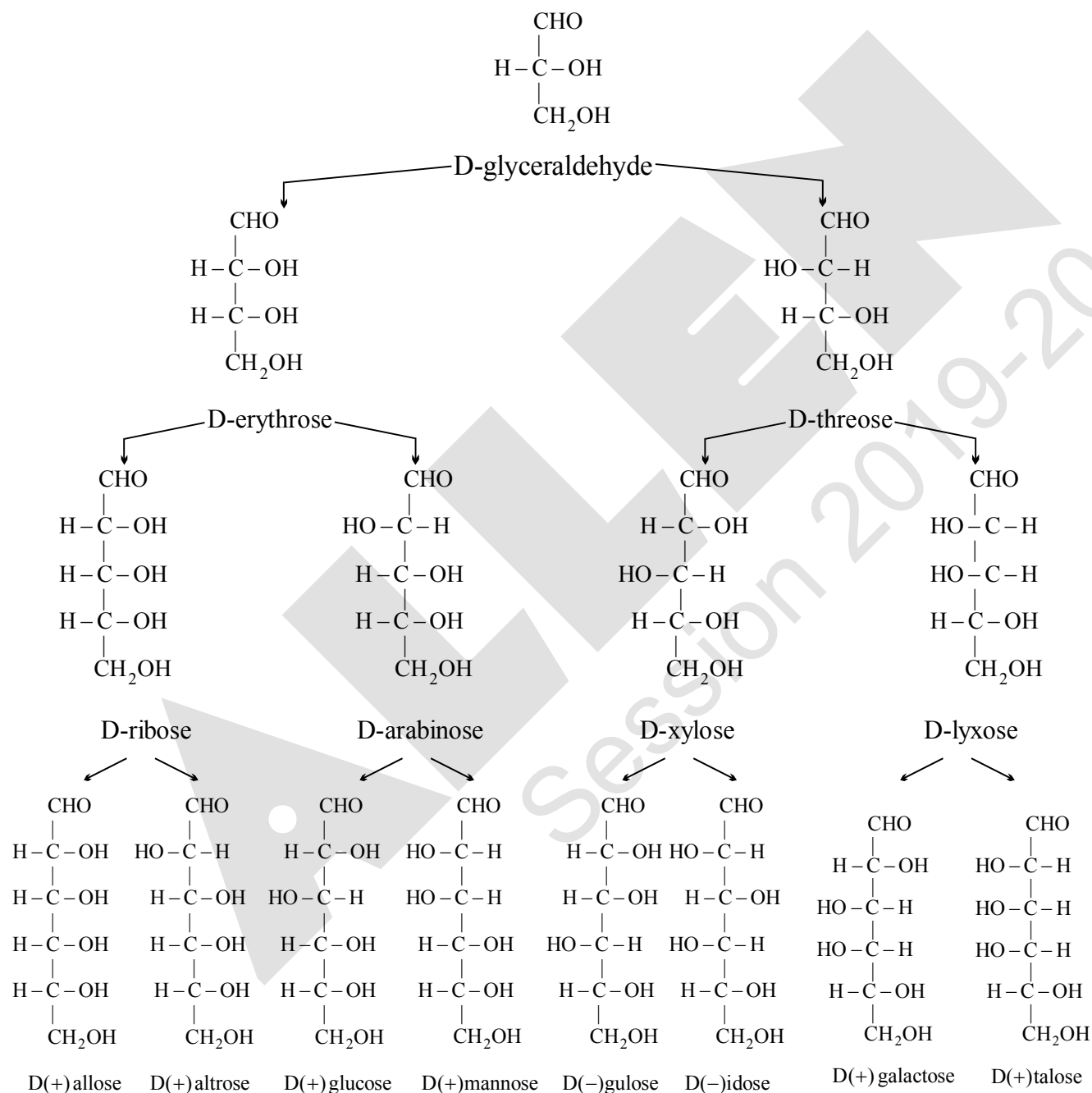
Aldohexose :

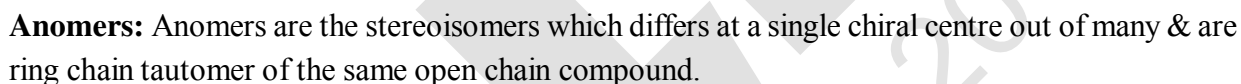
No. of C* = 4

No. of stereoisomers = $2^4 = 16$

No. of D-sugars = 8

No. of L-sugars = 8

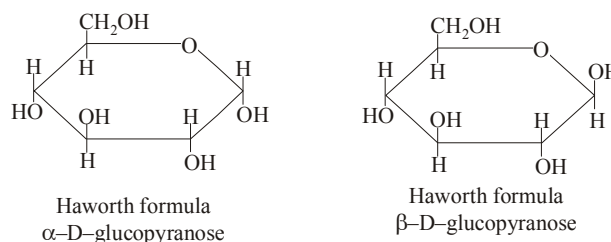
**The D-family aldoses**



The two sugars that differs in configuration only on the carbon that was the carbonyl carbon in the open chain form is called as anomers α glucose and β glucose are known as anomers their equilibrium mixture contains 36 % α -D-glucose , 63.8 % β -D-glucose and 0.2 % open chain form.

C₁ Carbon is known as anomeric carbon.

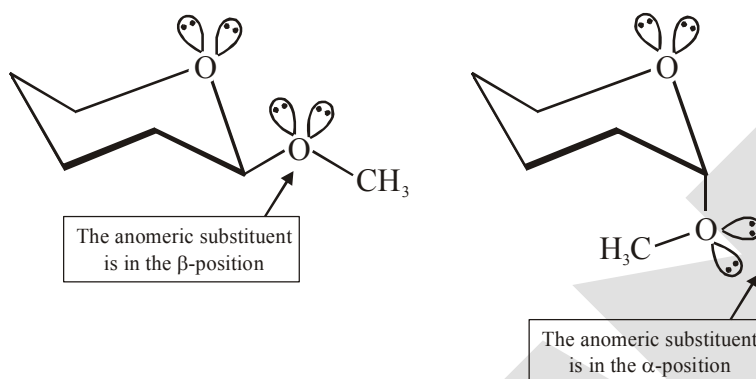
Haworth suggested to write α glucose and β glucose in pyran structure.



Anomers are epimers but epimers may not be anomers.

Anomeric effect :

β -D-glucose is more stable than α -D glucose because there is more room for a substituent in the equatorial position. However when glucose reacts with an alcohol to form a glucoside, the major product is the α -glucoside. The preference for the axial position by certain substituents bonded to the anomeric carbon is called anomeric effect.



What is responsible for the anomeric effect ? One clue is that all the substituents that prefer the axial position have lone pair electrons on the atom bonded to the ring. The lone pair electrons of the anomeric substituent have repulsive interaction with the lone pair electron of the ring oxygen if the anomeric substituent is the β -position, but not if it is the α -position.

Apparently attractive interaction of the hydrogen of the anomeric OH group of D-glucose with the lone pair electron of the ring oxygen decreases the importance of the anomeric effect making β -D glucose more stable than α -D glucose. However, when the hydrogen is replaced by an alkyl group, the anomeric effect decreases the stability of the β -position so, α -glycosides are more stable than β -glycosides.

Cyclic structures of monosaccharides

Many five membered and six membered monosaccharides occur in cyclic form. Cyclic structures of monosaccharides are established by many experiments. The cyclic structure is due to intramolecular hemiacetal formation between aldo / keto group and OH of any one carbon. The ring formed are generally six membered (pyranose) or five membered (furanose). Each cyclization results in creation of a new asymmetric centre apart from the existing ones. The isomers resulting from cyclizations are called anomers. example, when D-glucose (open structure) cyclise, it gives α -D-glucose and β -D-glucose.

Haworth projection :-

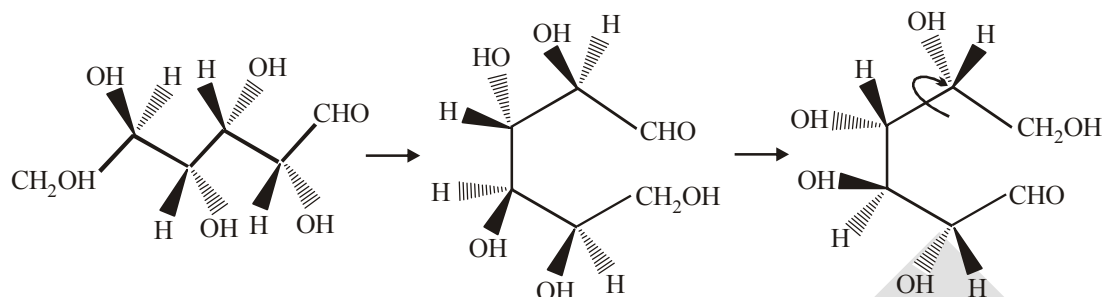
Many of monosaccharides form cyclic structures. The actual structure is almost planar and be represented by Haworth projection, which is a way of depicting three - dimensional cyclic structure.

Rule -1 :- In a Haworth projection draw a Fischer projection in which ring oxygen is in a down position.

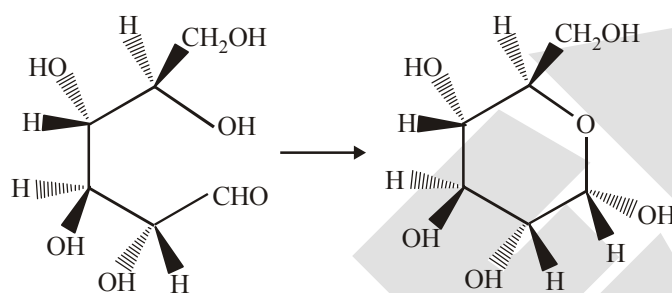
Rule -2 :- Imagine that carbon chain of Fischer projection is folded around a barrel or drum, which provides a ring lies in a plane $\perp$ to the page.

Rule -3 :- Now plane of ring is turned 90° so that anomeric carbon is on the right and the ring oxygen is in the rear. Obtained projection is a Haworth projection.

Example : (D-glucose)

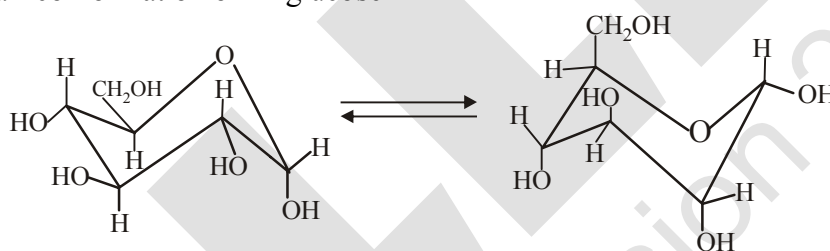


Projection :

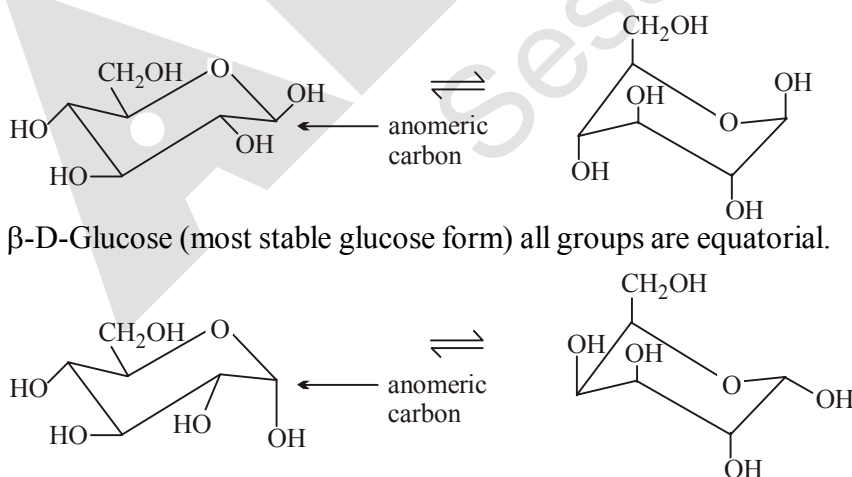


Hawarth projection

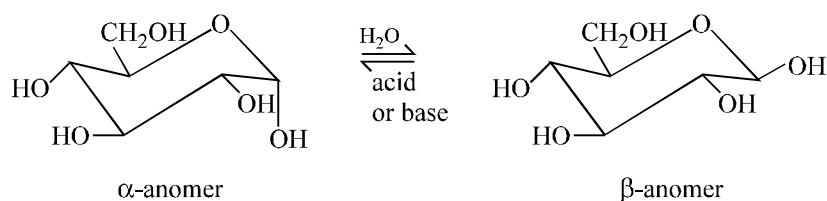
$\Rightarrow$ Chair conformation of D-glucose



Chair forms of (conformation) α and β D-Glucose :



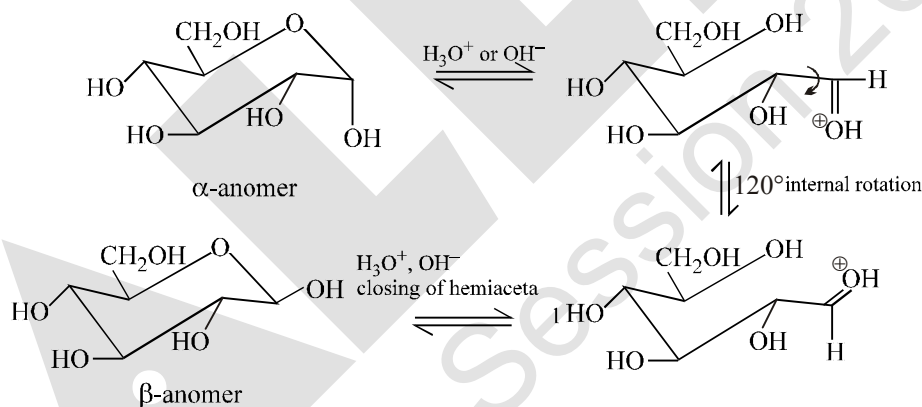
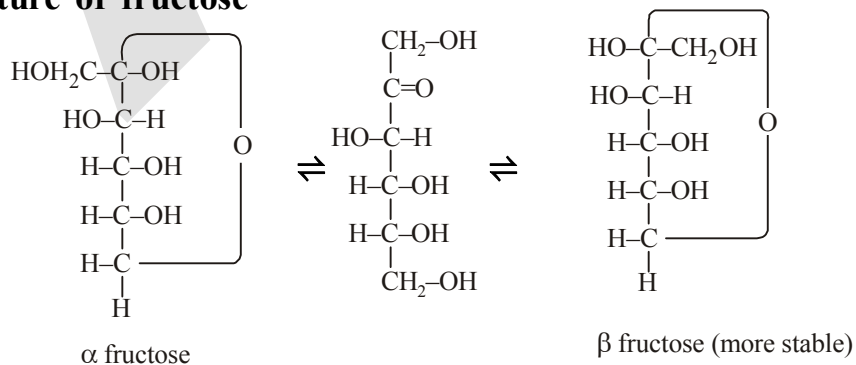
α -D-Glucose -OH group at anomeric carbon is axial.

MutarotationSpecific rotation of α glucose $+112^\circ$ Specific rotation of β glucose $+19^\circ$ Equilibrium mixture $[\alpha]_D = 52.5 \text{ degree mL g}^{-1} \text{ dm}^{-1}$ Fresh α -glucose $\longrightarrow 52.5 \longleftarrow$ Fresh β -glucose 112° 36 % α glucose 19° 63.8 % β glucose

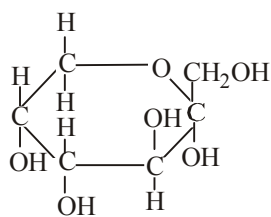
When pure α -D glucose is dissolved in water its specific rotation is found to be $+112^\circ$ with time, however the specific rotation of the solution decreases ultimately reaches stable value of $+52.5^\circ$. When β -D-glucose is dissolved in water, it has a specific rotation of 19° . The specific rotation of this solution increases with time also to $+52.5^\circ$.

This change of optical rotation with time is called mutarotation. It is caused by the conversion of α and β glucopyranose anomers into an equilibrium mixture of both. Mutarotation is catalyzed by both acid and base, but also occurs even in pure water. Mutarotation is characteristic of the cyclic hemiacetal form of glucose.

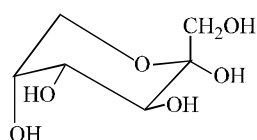
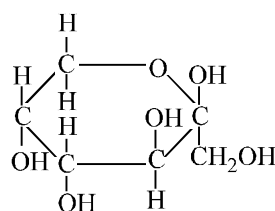
Mutarotation occurs first by opening of the pyranose ring to the free aldehyde form.

**Structure of fructose**

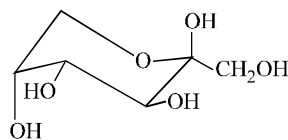
Ring structure of fructose C, Pyranose structures 6 membered ring, $\text{C}_2 - \text{C}_6$ linkage



Chair conformations

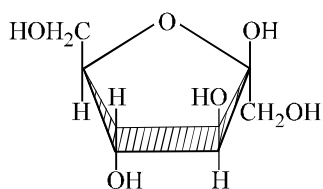


α -D-fructopyranose

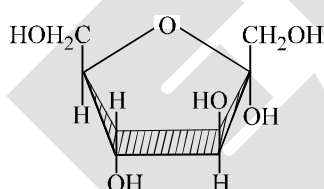


β -D-fructopyranose

Furanose structure (5 membered ring)

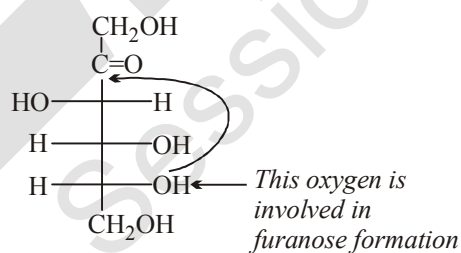
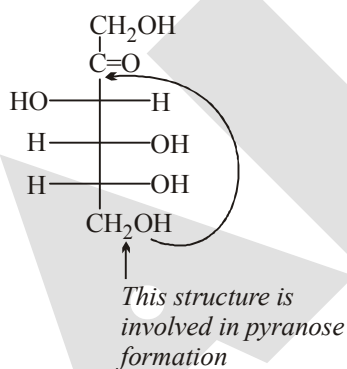


β -D-fructofuranose

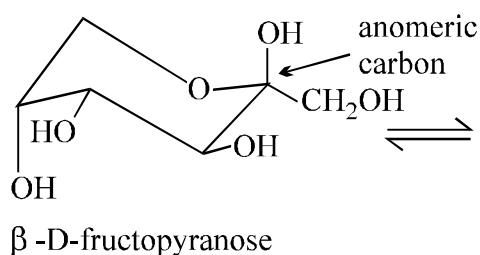


α -D-fructofuranose

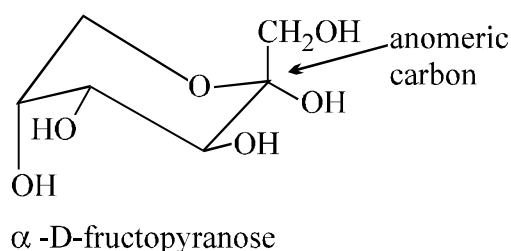
Mutarotation: Fructose undergo complex mutarotation. The structure of the cyclic hemiacetal form of d-fructose can be derived from its carbonyl (Ketone) form using the methods described as follows.



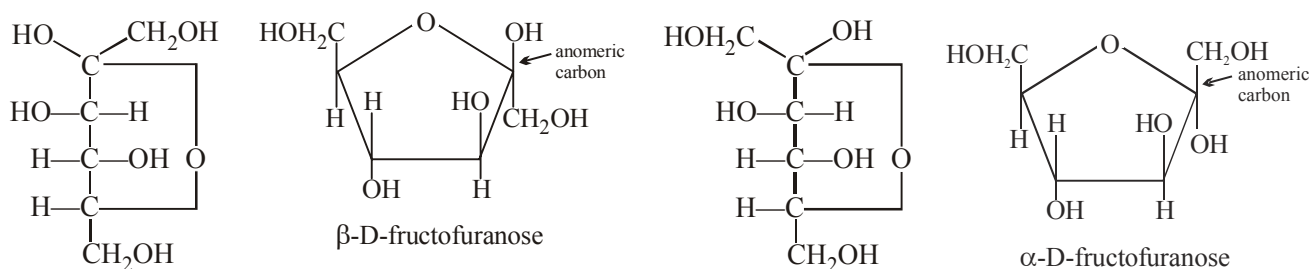
It happens that the crystalline form of D-fructose is β -D-Fructopyranose. When crystals of this form are dissolved in water, it equilibrates to both pyranose & furanose forms.



β -D-fructopyranose



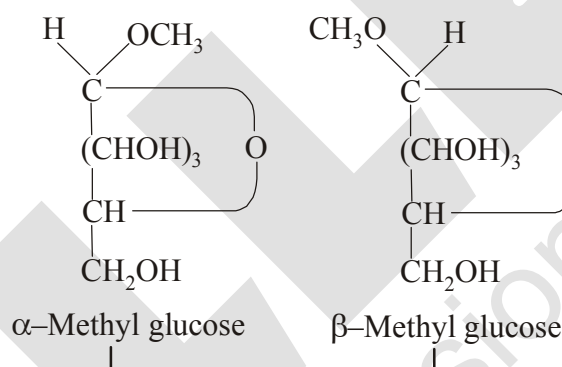
α -D-fructopyranose



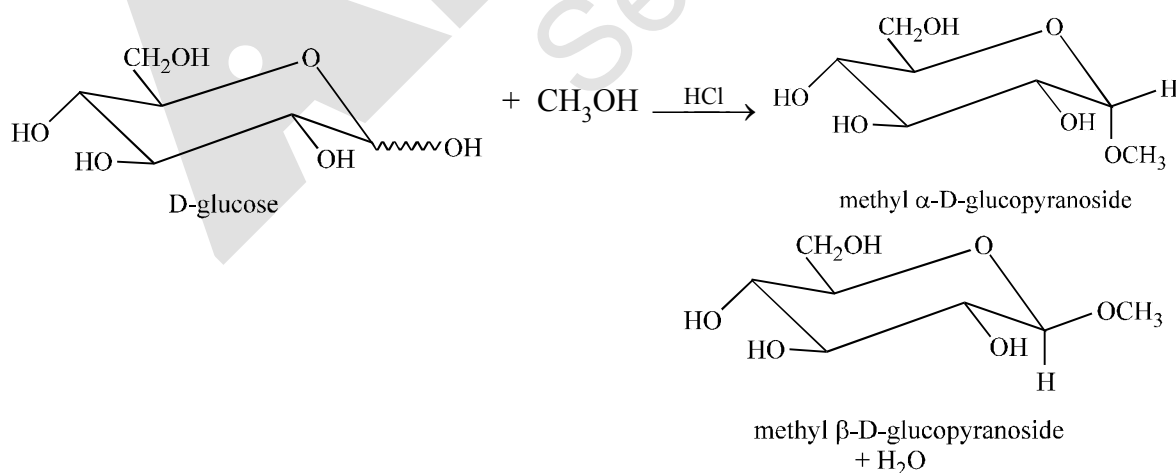
- * All monosaccharides are reducing sugars and they show mutarotation.
- * Starch, cellulose are Polymers of Glucose
- * Lactose and sucrose are disaccharides
- * Sucrose is a non reducing sugar, gives negative test for Benedict and tollen's reagent, they do not form osazone and do not show mutarotation.
- * Acetals of carbohydrates are called as GLYCOSIDE

Formation of Glycosides

Glucose reacts with methyl alcohol in presence of dry HCl to form α and β -methyl glycoside of glucose. The reaction takes place only on OH of hemi-acetylic carbon. Other hydroxyl groups are unreactive.



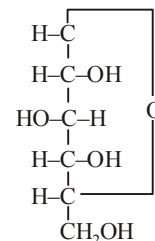
To methylate all the OH groups, methylating agent used is dimethyl sulphate.



Such compounds are called glucoside (cyclic acetals). They are special type of acetals in which one of the oxygen of the acetal linkage is the ring oxygen of the pyranose or furanose.

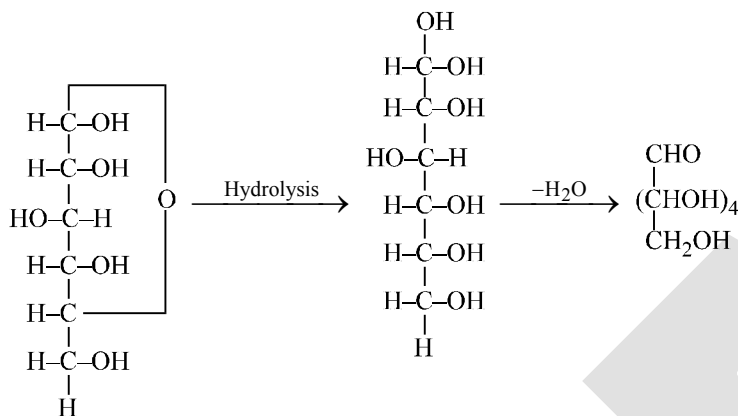
Ring structure of glucose :

- (i) Glucose does not give pink colour with schief reagent.
- (ii) Does not form adduct with NaHSO_3 , NH_3
- (iii) Glucose exist in two isomeric form
- (iv) It show mutarotation

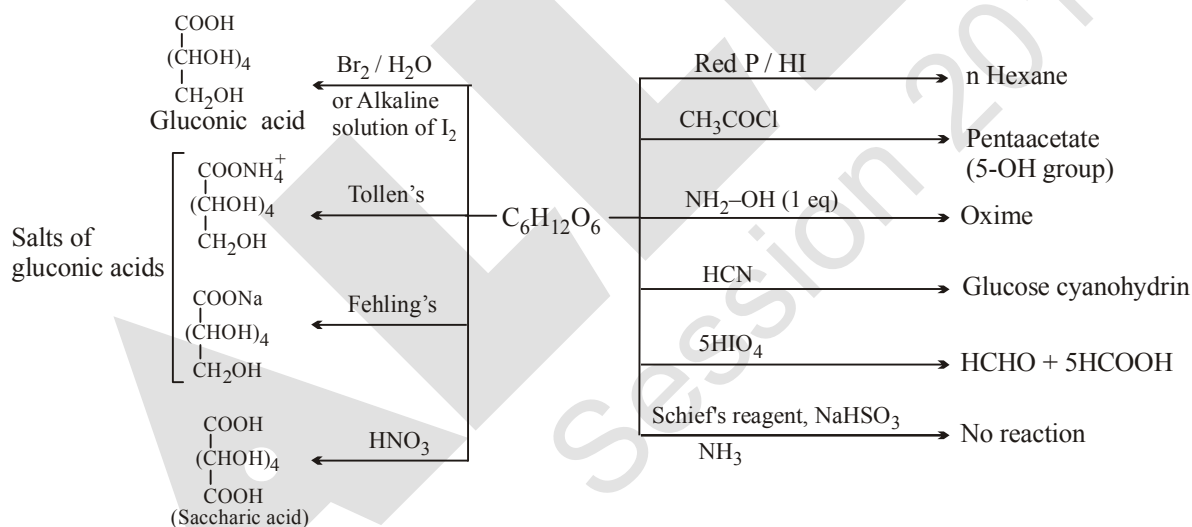


Since there is no free aldehyde group, so it does not react with weak reagent

(NH_3 , NaHSO_3) but strong reagent (HCN , NH_2OH , $\text{C}_6\text{H}_5\text{NH}-\text{NH}_2$) break up ring

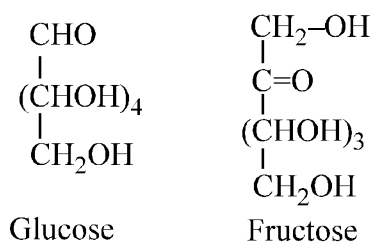


REACTIONS OF GLUCOSE

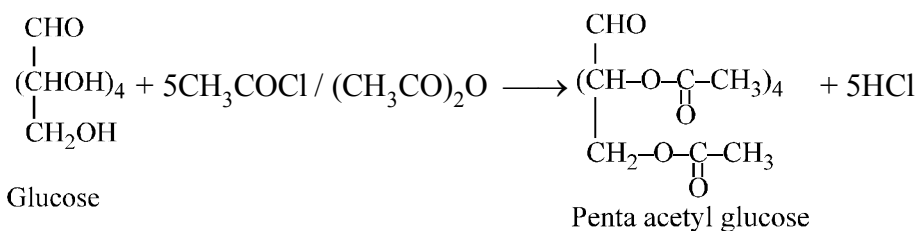


These reactions indicate that glucose has 6-C straight chain with one $-\text{CHO}$ group & 5-OH group.

General reactions of monosaccharies

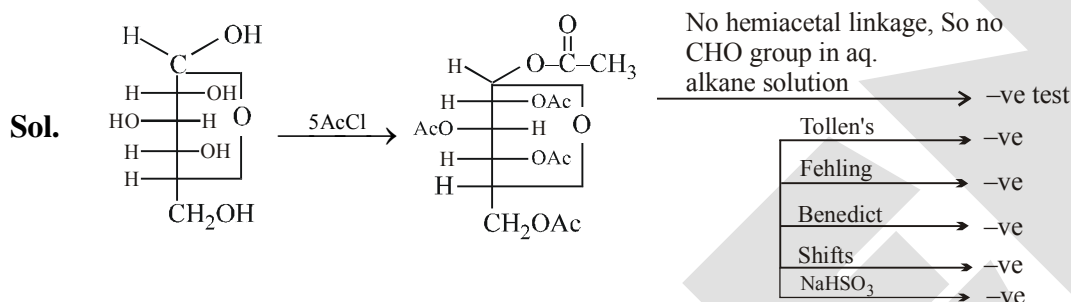


1. Acetylation :



This reaction suggests presence of 5(-OH) group.

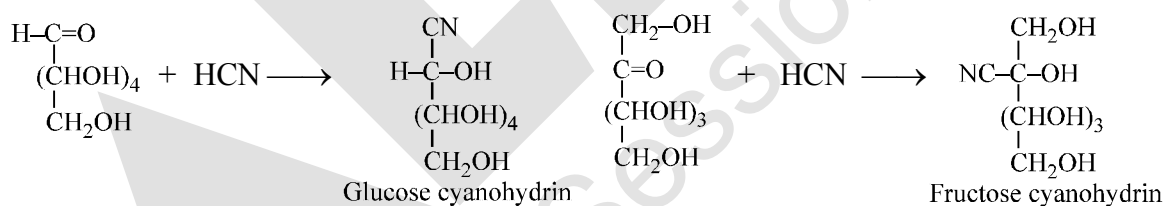
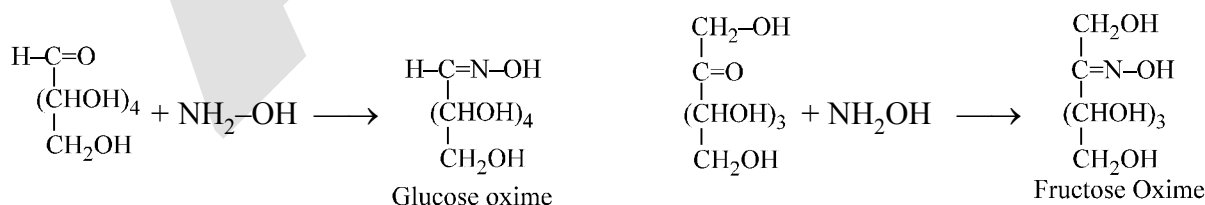
Q. The penta acetate of glucose give -ve test with Tollen's reagent & Fehling solution, explain?



2. Red by HI / Red P :

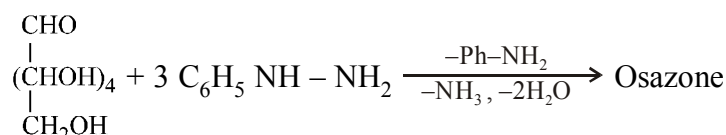


3. Reaction with HCN:

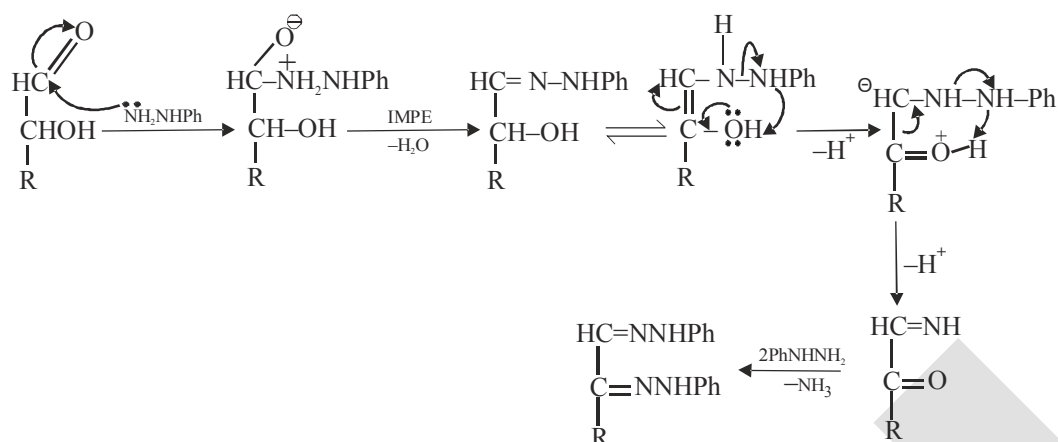
4. Reaction with NH₂-OH (hydroxyl amine):

5. Reaction with phenyl hydrazine: Both glucose and fructose give "osazone".

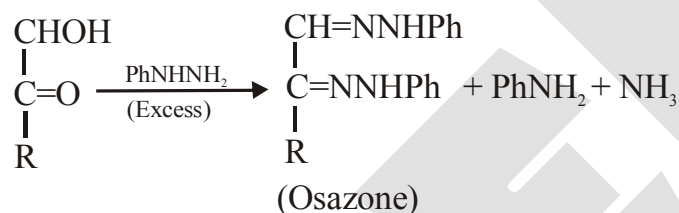
Reaction with glucose :



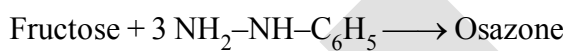
Mechanism :



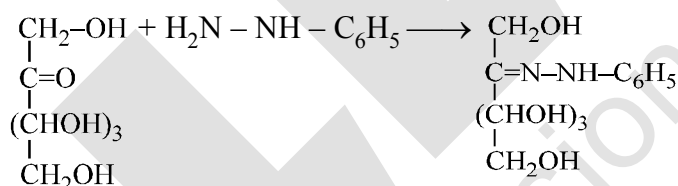
Propose Mechanism for :



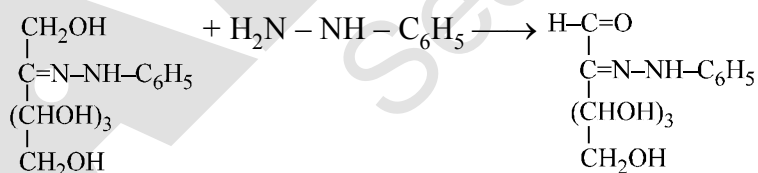
Reaction with fructose :



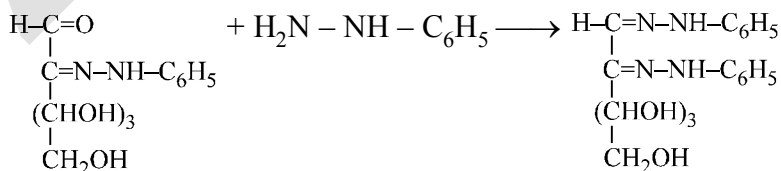
Step (i)



Step (ii)



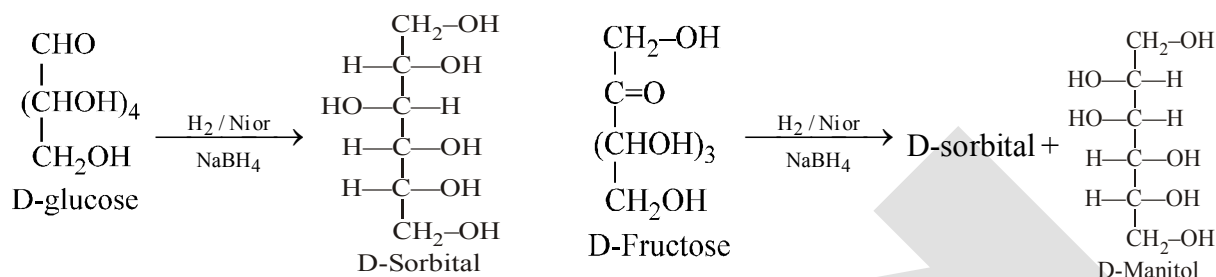
Step (iii)



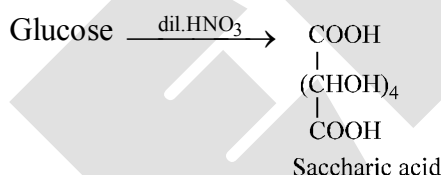
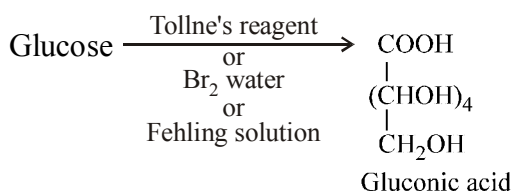
Glucose , Mannose & Fructose give same osazone because structure of last four carbons in these carbohydrates is same.

Only C-1 and C-2 in glucose and fructose are involved in osazone formation addition reaction do not run through out the chain. The failure to undergo further reaction has been explained by stabilization of the osazone by chelation.

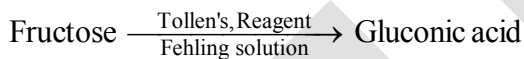
6. Catalytic reduction:



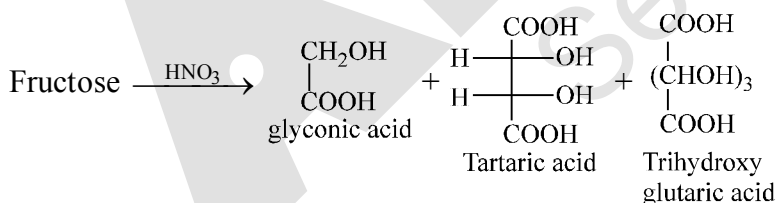
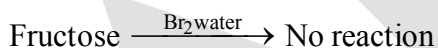
7. Oxidation:



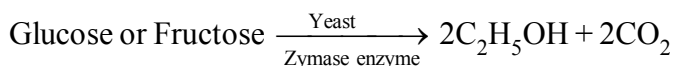
Oxidation of fructose :



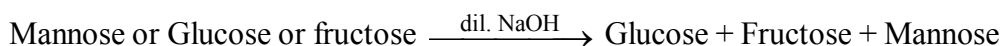
Fructose also reduces tollen's & fehling reagent because in basic medium fructose isomerises to glucose.



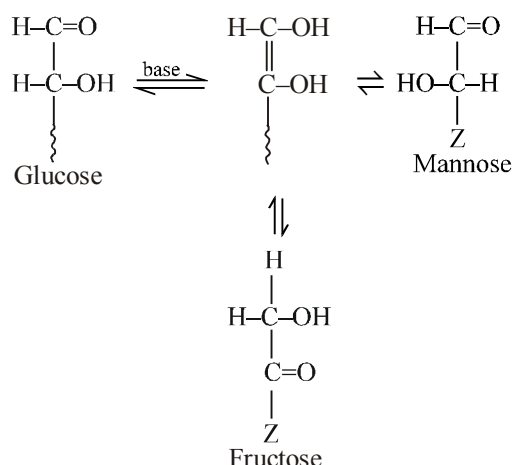
8. Reaction with enzyme:



9. Reaction with dil NaOH / Ca(OH)₂

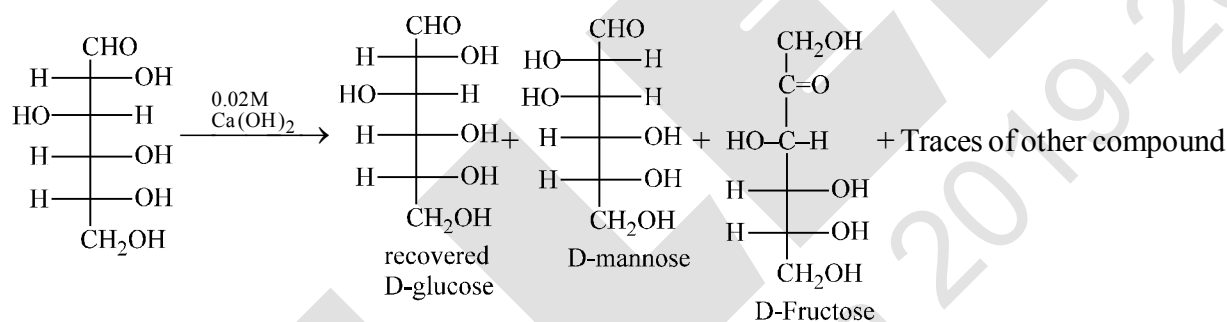


Mechanism :



Base-catalyzed isomerisation of aldoses and Ketoses:

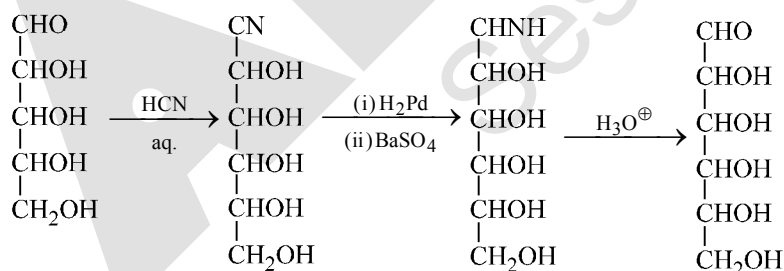
Although glucose in solution exists mostly in its cyclic hemiacetal forms it is also in equilibrium with a small amount of its acyclic aldehyde form.



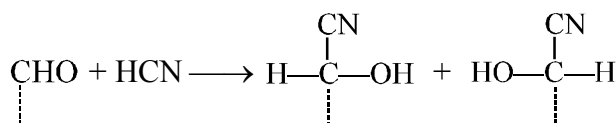
Method of ascending the sugar series:

An aldose may be converted into its next higher aldose eg. an aldopentose into an aldohexose.

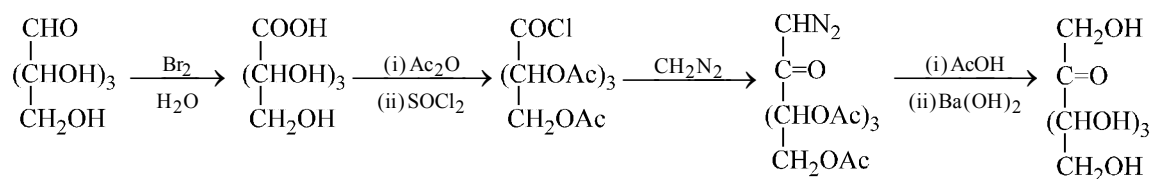
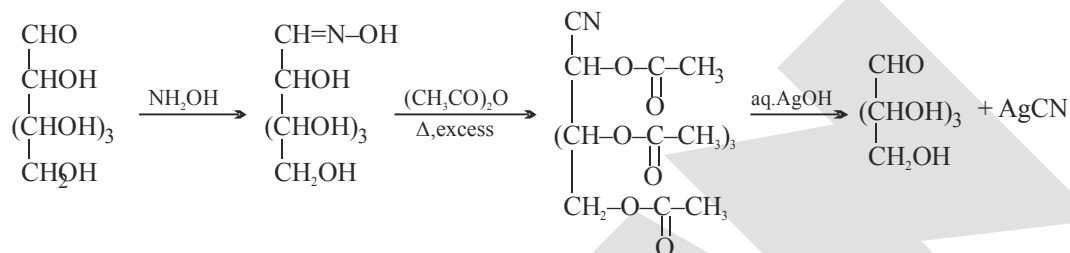
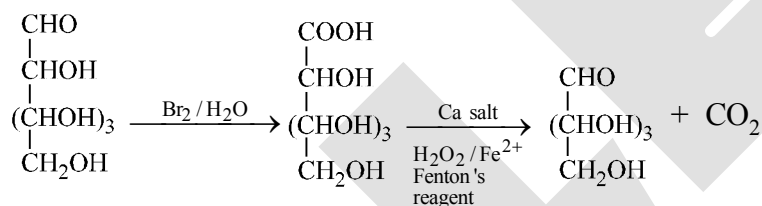
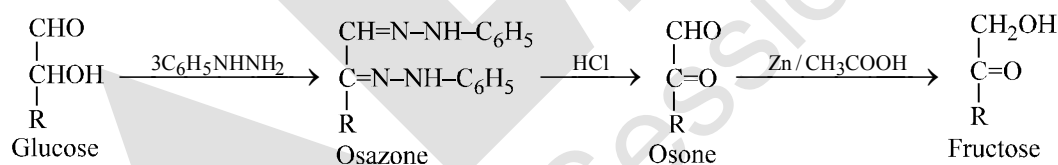
By Kiliani Fischer upgradation:



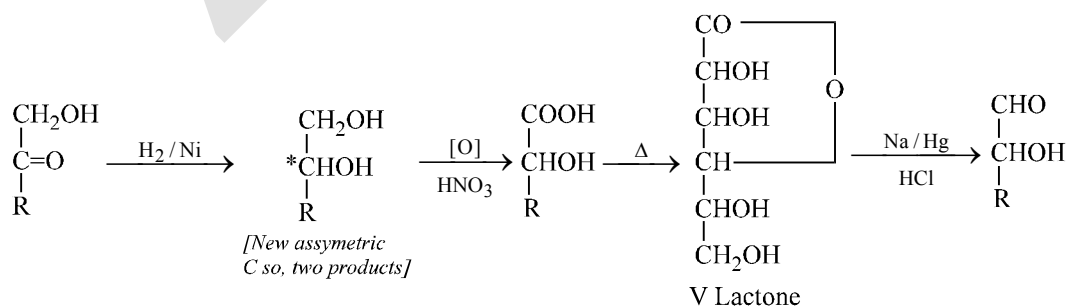
Theoretically two lactones are possible, since two cyanohydrin may be formed when hydrogen cyanide adds on to the aldopentose (a new asymmetrical carbon is produced)



Wolf from reaction : Wolf from have stepped up an aldose to a ketose with one more carbon atom by a modified Arndt-Eistere reaction.

**Method of descending the sugar series :****Wohl's method:****Ruff's method:****Conversion of an aldose into a ketose :**

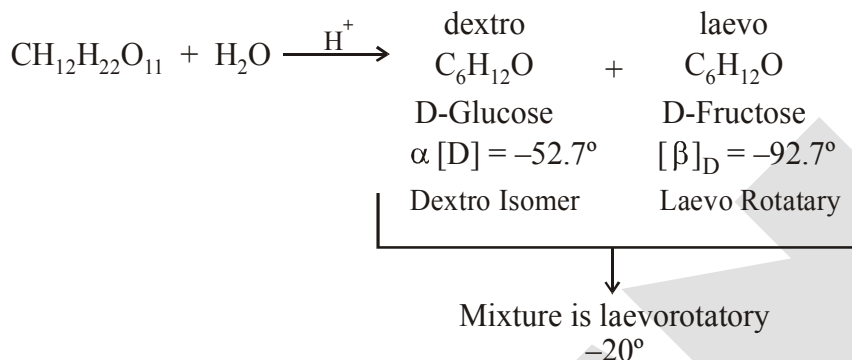
An aldehyde group is reduced more readily than a ketonic group.

Conversion of a Ketose into an aldose :

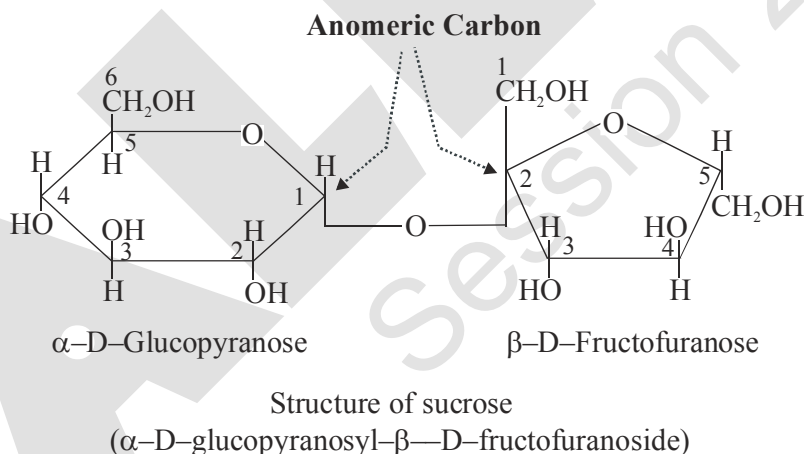
SOME IMPORTANT CARBOHYDRATES

- Sucrose** ($C_{12}H_{22}O_{11}$): → A dimer of α -D-Glucose & β -D-Fructose. It is white, crystalline & sweet substance soluble in water obtained from the sugar cane. When heated above its melting point. It forms a brown substance known as caramel.

It's aqueous solution is dextrorotatory $[\alpha]_D = +66.5^\circ$



- Thus hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to leavo (−) & such a change is known as inversion of sugar and the mixture obtained on acidic hydrolysis is known as *invert sugar*.
- The inversion of cane-sugar may also be done by the enzyme invertase which is found in yeast.
- Sucrose is non-reducing sugar because it has stable acetal linkage & in aq. solution it does not give free carbonyl group and so it does not reduces Tollen's & Fehling's solution.
- This indicates that neither the aldehyde group of glucose nor the ketonic group of fructose is free in sucrose.



Do not show mutarotation .

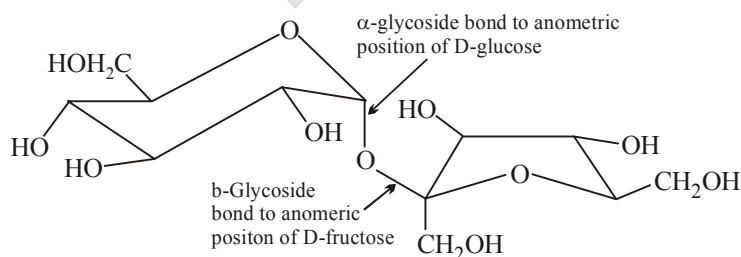


Fig. Showing Sucrose in other structure

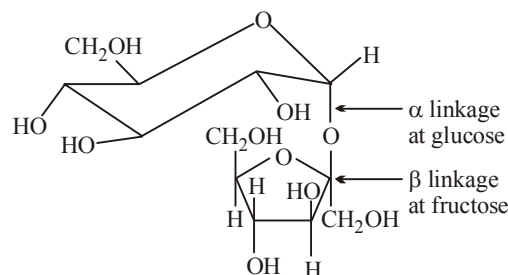
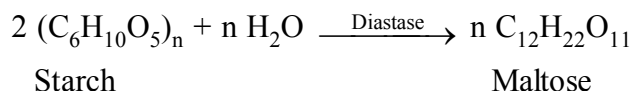


Fig. Showing Sucrose in other structure

2. Maltose :

- It is dimer of α -D-Glucose

It is obtained by partial hydrolysis of starch by the enzyme diastase present in malt i.e., sprouted barely seeds.

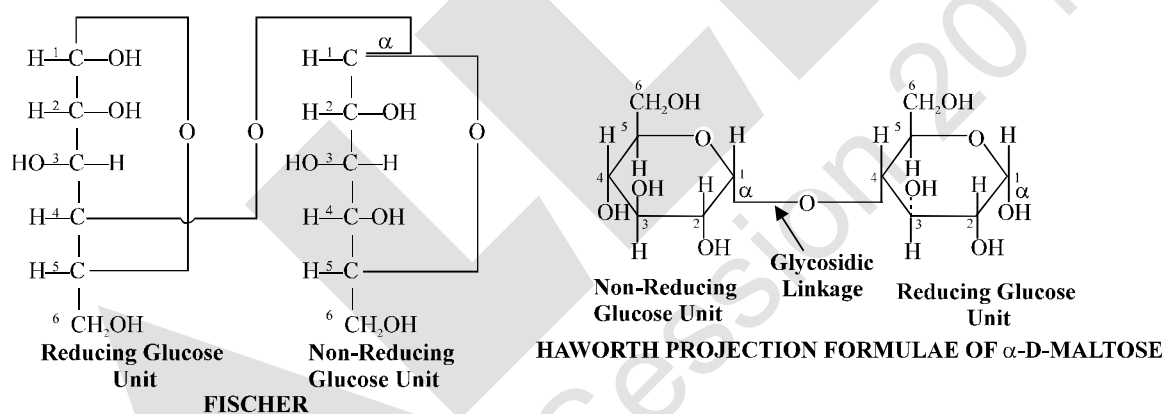


- Hydrolysis of one mole of maltose yields two moles of D-glucose.
- Maltose is a reducing sugar since it forms an osazone, undergoes mutarotation and also reduces Tollen's reagents and Fehling's solutions, Methylation studies have revealed that

(i) Both glucose units are present in the pyranose form.

(ii) C_1 of one glucose unit is linked to C_4 of the other

Further since maltose is hydrolysed by the enzyme maltase which specifically hydrolyses α -glycosidic linkage, therefore, the non-reducing glucose unit in maltose must be present in the α -form. In other words, $\text{C}_1 - \alpha$ of non-reducing glucose unit is attached to C_4 of the reducing glucose unit as shown in the figure on next page.

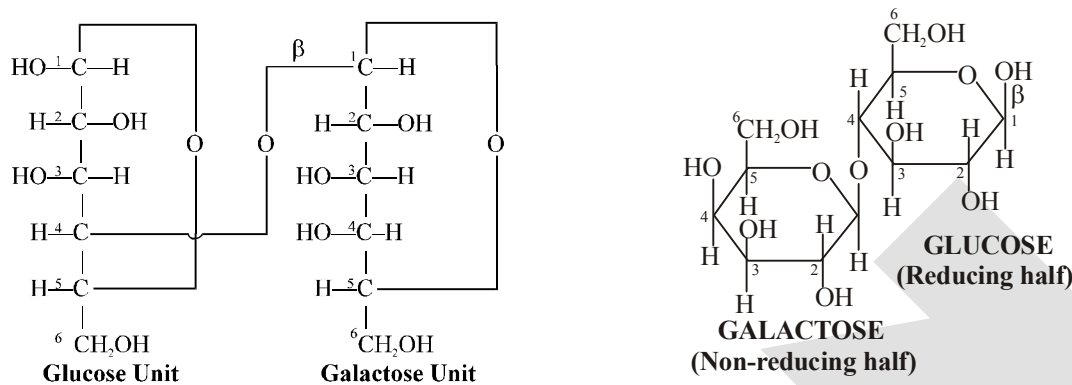
3. Lactose (Milk sugar) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$

Lactose occurs in milk and that is why it is called milk sugar.

Lactose on hydrolysis with dilute acid or by enzyme lactase, yields an equimolar mixture of D-glucose and D-galactose. It is a reducing sugar it forms an osazone, undergoes mutarotation and also reduces Tollen's or Fehling's solution. Methylation studies have revealed that

- both glucose and galactose are present in the pyranose form.
- glucose is the reducing half while β -galactose is the non-reducing half.
- C_1 of galactose unit is connected to C_4 of glucose unit.

Further since emulsin, as, enzyme which specifically hydrolyses β -glycosidic linkages also hydrolyses lactose, therefore, galactose must be present in the β -form. In other words, in lactose, $C_1 - \beta$ of galactose is attached to C_4 of glucose as shown in figure.

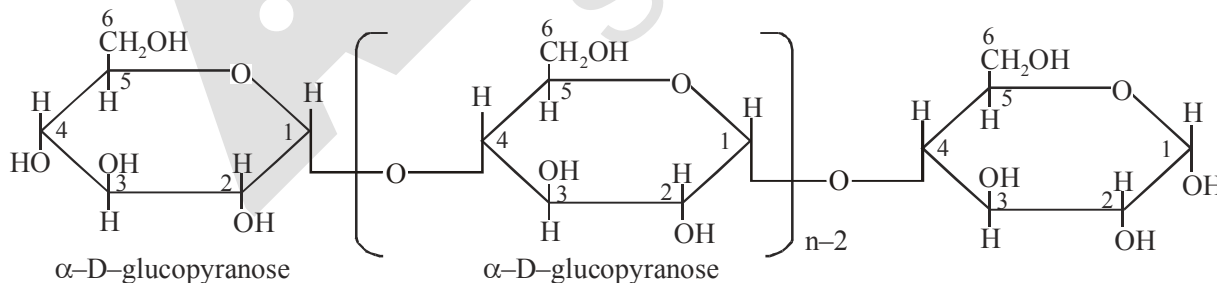


FISCHER

4. Starch Amylum, $(C_6H_{10}O_5)_n$

Occurrence : The value of n (100 – 3000) generally however it may varies from source to source. It is the chief food reserve material or storage polysaccharide of plants and is found mainly in seeds, roots tubers wheat, maize, rice, potatoes, barley, bananas and sorghum are the main sources of starch. Starch occurs in the form of granules, which vary in shape and size depending upon their plant source.

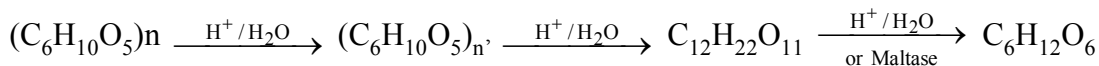
Occurs in all green plants. Starch consists of two fractions, one being known as (amylose), which gives blue colour with iodine. This blue colour is believed to be due to the formation of an inclusion complex. An aqueous solution of α -amylose slowly forms a precipitate, since α -amylose has a strong tendency to 'revert' to the insoluble state in solution. Amylopectin is insoluble in water and is stable towards both hydrolysis to maltose by the enzyme diastase and to D(+)-glucose by dilute acids (amylopectin gives about 50 percent of maltose).



Structure of Starch (α -D-glucoamylose)

α -amylose consists of an unbranched chain, with a molecular weight varying between 10,000($n \approx 60$) and 10,00,000($n \approx 6,000$), The value of n depends on the source and treatment of α -amylose.

Properties : (i) Starch is a white amorphous powder sparingly soluble in water. Its aqueous solution gives a blue colour with iodine solution due to the formation of an inclusion complex. The blue pearls on cooling. (ii) On hydrolysis with dilute mineral acids or enzymes, starch breaks down first to smaller molecules ($n > n'$), then to maltose and finally to D-glucose.



Starch

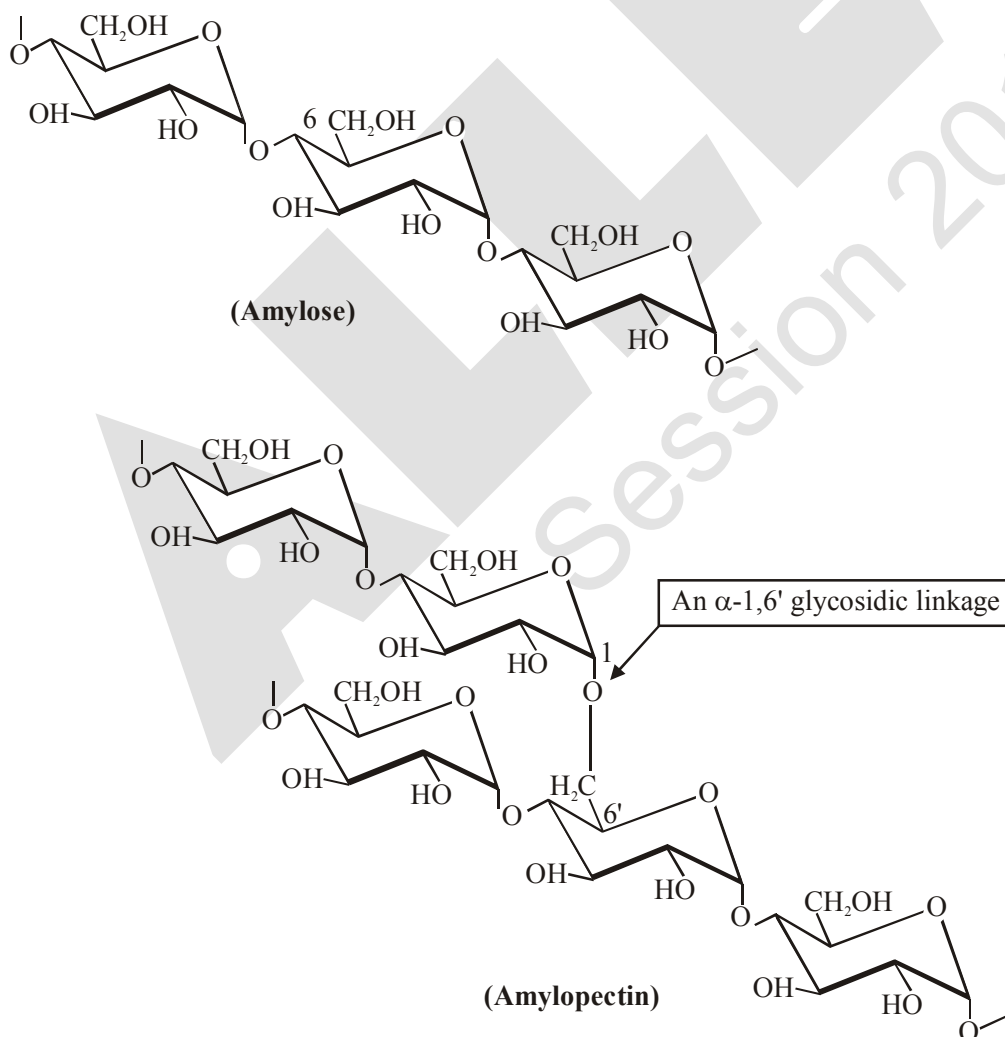
Maltose

D-Glucose

(iii) Starch is a non-reducing saccharide. It neither reduces Tollen's reagent or Fehling's solution nor forms an osazone. This suggests that all hemiacetal OH groups of glucose units at C_1 are not free but are involved in glycosidic linkages.

Composition : Starch is not a single compound but is a mixture of two components—a water soluble component called amylose (10-20%) and a water insoluble component called amylopectin (80-90%). Both amylose and amylopectin are polymers of α -D-glucose.

Structure of amylose : Amylose is water soluble and gives blue colour with iodine solution. It may have 100-3000 glucose units, i.e., its molecular mass can vary from 10,000 to 500,000. It is a linear polymer of α -D-glucose in which C_1 of one glucose unit is attached to C_4 of the other through α -glycosidic linkage as shown in figure.



Pectins

Pectins are found in plant and fruit juices. Their characteristic property is the ability of their solutions to gelate, i.e. form jellies. They have a high molecular weight and are polygalacturonic acid (linked 1,4) with the carboxyl groups partially esterified with methanol.

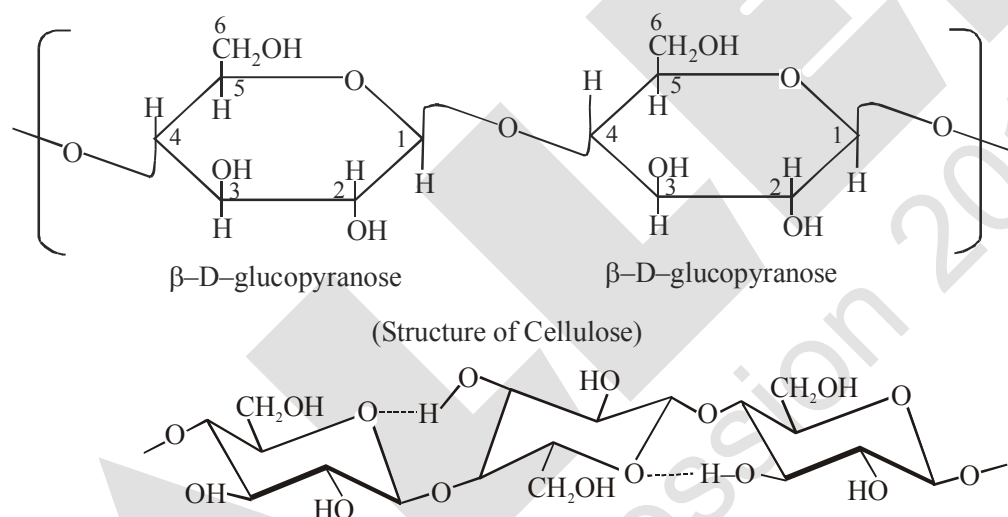
Glycogen ($C_6H_{10}O_5$)_n :

Glycogen is found in nearly all animals cells, occurring mainly in liver. It is the reserve carbohydrate of animals and so is often known as 'animal starch'. It has also been isolated from plant sources.

Glycogen is a white powder, soluble in water, the solution giving a purplish-red colour with iodine. On hydrolysis with dilute acid, glycogen gives D(+)-glucose. The molecular weight of glycogen has been given as 10,00,000 to 50,00,000 and glycogen contains highly branched chains. Glycogen has a structure similar to amylopectin, except that it has more cross-linking.

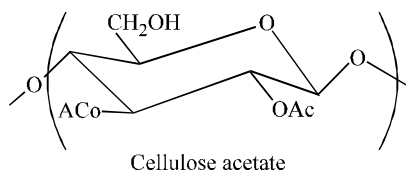
5. Cellulose:

Cellulose is colourless, solid which is insoluble in water & organic solvents. But it is soluble in ammonical cupric hydroxide (Schweizer's reagent) or in conc. HCl cellulose is a regular polymer of d-glucopyranose residues connected by β -1,4 glycosidic linkages. It is straight chain polymer.



Some points about cellulose :

1. General empirical formula ($C_6H_{10}O_5$)
2. $\text{Cellulose} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} 96\% \text{ of crystalline D-glucose}$
3. No. of monomer units in cellulose are 1000 – 1500 in one molecule.
4. Cellulose doesn't show mutarotation (like starch)
5. It is non reducing sugar because there is no hemiacetal linkage.
6. Acetylation, nitration & methylation of cellulose give trisubstituted cellulose which suggest that only three – OH groups are free.

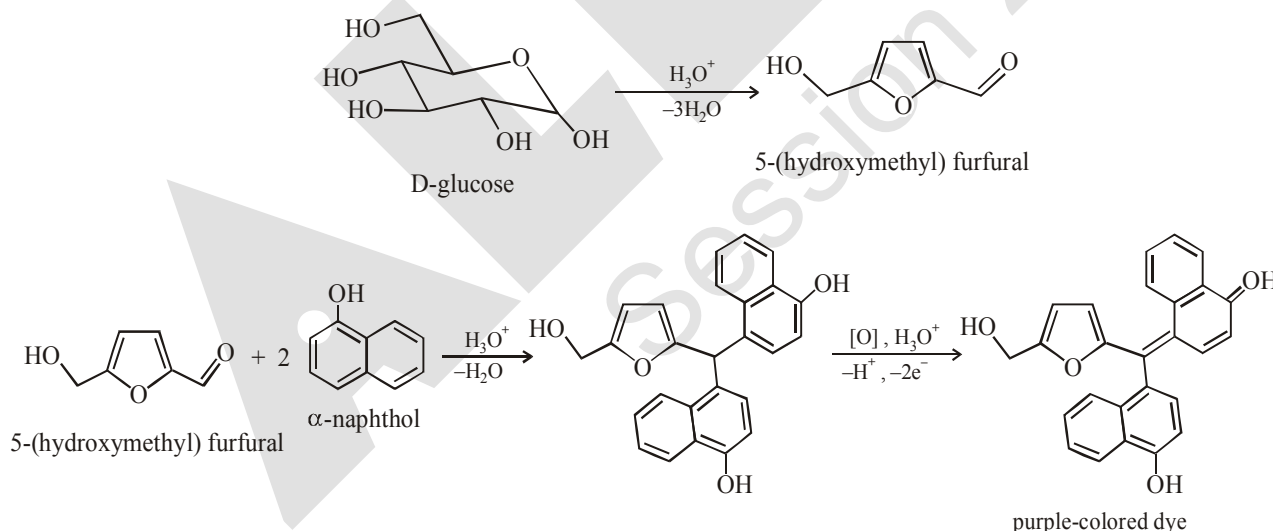


Tests for carbohydrates:

- (i) When heated in a dry test tube, it melts, turns brown and finally black, giving a characteristic smell of burning sugar.
- (ii) When warmed with a little concentrated H_2SO_4 , it leaves a charred residue of carbon.
- (iii) **Molisch's Test** (named after Austrian botanist Hans Molisch) is a sensitive chemical test for the presence of carbohydrates, based on the dehydration of the carbohydrate by sulfuric acid to produce an aldehyde, which condenses with two molecules of phenol (usually α -naphthol, though other phenols (e.g. resorcinol, thymol) also give colored products) resulting in a red- or purple-colored compound.

The test solution is combined with a small amount of Molisch's reagent (α -naphthol dissolved in ethanol) in a test tube. After mixing, a small amount of concentrated sulfuric acid is slowly added down the sides of the sloping test-tube, without mixing, to form a bottom layer. A positive reaction is indicated by appearance of a purple ring at the interface between the acid and test layers.

All carbohydrates — monosaccharides, disaccharides, and polysaccharides — should give a positive reaction, and nucleic acids and glycoproteins also give a positive reaction, as all these compounds are eventually hydrolyzed to monosaccharides by strong mineral acids. Pentoses are then dehydrated to furfural, while hexoses are dehydrated to 5-hydroxymethylfurfural. Either of these aldehydes, if present, will condense with two molecules of naphthol to form a purple-colored product, as illustrated below by the example of glucose:

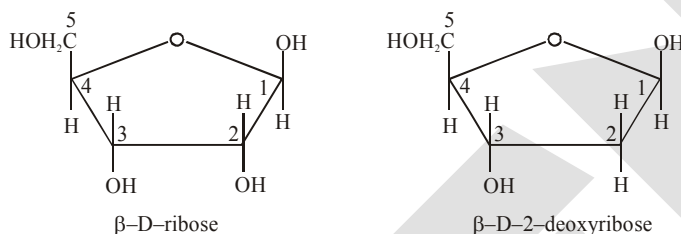


NUCLEIC ACIDS

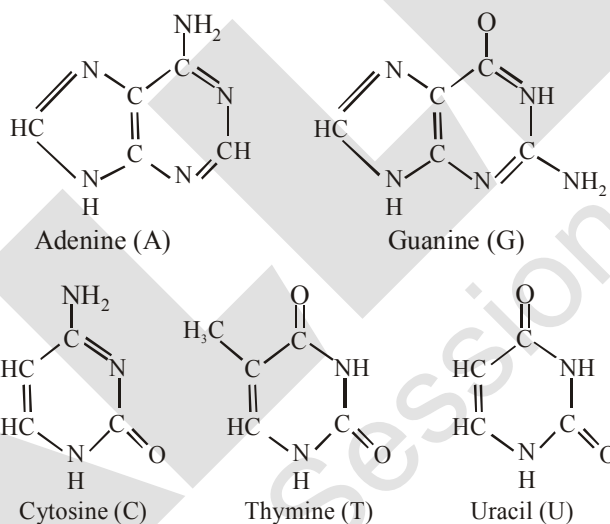
The particles in nucleus of the cell, responsible for heredity, are called chromosomes which are made up of proteins and another type of biomolecules called **nucleic acids**. These are mainly of two types, the **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**. Since nucleic acids are long chain polymers of **nucleotides**, so they are also called polynucleotides.

Chemical Composition of Nucleic Acids

Complete hydrolysis of DNA (or RNA) yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compounds (called bases). In DNA molecules, the sugar moiety is β -D-2-deoxyribose whereas in RNA molecule, it is β -D-ribose.

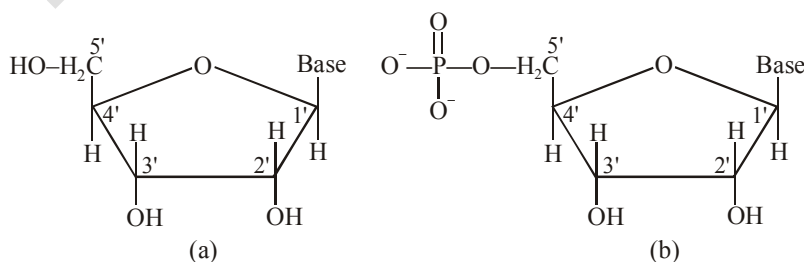


DNA contains four bases viz. adenine (A), guanine (G), cytosine (C) and thymine (T). RNA also contains four bases, the first three bases are same as in DNA but the fourth one is uracil (U).



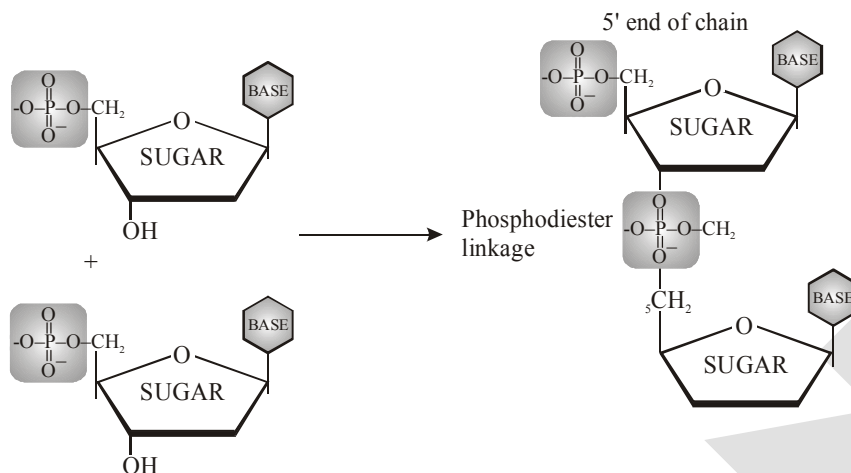
Structure of Nucleic Acids

A unit formed by the attachment of a base to 1' position of sugar is known as **nucleoside**. In nucleosides, the sugar carbons are numbered as 1', 2', 3', etc. in order to distinguish these from the bases. When nucleoside is linked to phosphoric acid at 5'-position of sugar moiety, we get a nucleotide



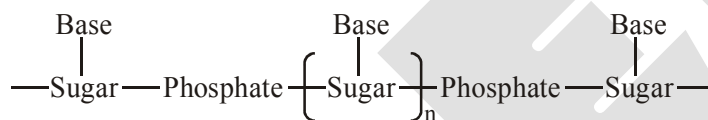
Structure of (a) a nucleoside and (b) a nucleotide

Nucleotides are joined together by phosphodiester linkage between 5' and 3' carbon atoms of the pentose sugar. The formation of a typical dinucleotide is shown in Fig.



Formation of dinucleotide

A simplified version of nucleic acid chain is as shown below.

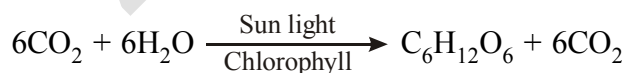


Information regarding the sequence of nucleotides in the chain of a nucleic acid is called its primary structure. Nucleic acids have a secondary structure also. James Watson and Francis Crick gave a double strand helix structure for DNA (Fig. 14.7). Two nucleic acid chains are wound about each other and held together by hydrogen bonds between pairs of bases. The two strands are complementary to each other because the hydrogen bonds are formed between specific pairs of bases. Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine.

In secondary structure of RNA, helices are present which are only single stranded. Sometimes they fold back on themselves to form a double helix structure. RNA molecules are of three types and they perform different functions. They are named as **messenger RNA (m-RNA)**, **ribosomal RNA (r-RNA)** and **transfer RNA (t-RNA)**.

Fig. : Double strand helix structure for DNA

Photosynthesis



* 3 ATP are required to fix 1 mole of CO_2 . Therefore 18 ATP are required to fix 6 mole of CO_2 .

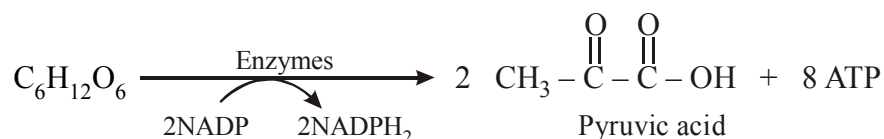
Cellular Respiration



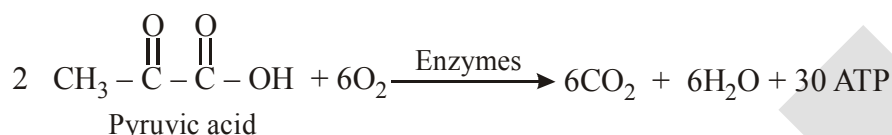
* 38 ATP are generated by complete oxidation of 1 mole of glucose. But net gain of ATP are 36, because 2 ATP are consumed when pyruvic acid enters mitochondria.

* Complete oxidation takes place in two steps

Step - 1 : Glycolysis or EMP or HMP (Completes in cytoplasm)



Step - 2 : Kreb cycle (Completes in mitochondria)

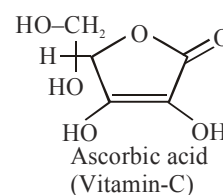


VITAMINES

Table : Some important Vitamins, their Sources and their Deficiency Diseases

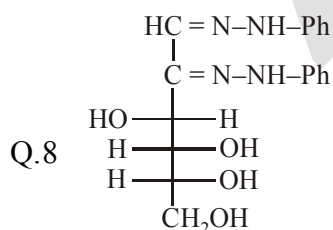
Sl. No.	Name of Vitamins	Sources	Deficiency diseases
1	Vitamin A	Fish liver oil, carrots, butter and milk	Xerophthamla (hardening of cornea of eye) Night blindness
2	Vitamin B ₁ (Thiamine)	Yeast, milk, green vegetables and cereals	Beri beri (loss of appetite, retarded growth)
3	Vitamin B ₃ (Riboflavin)	Milk, eggwhite, liver, kidney	Cheilosis (fissuring at corners of mouth and lips), digestive disorders and burning sensation of the skin.
4	Vitamin B ₃ (Pyridoxine)	Yeast, milk, egg yolk, cereals and grams	Convulsions
5	Vitamin B ₁₂	Meat, fish, egg and curd	Pernicious anaemia (RBC deficient in haemoglobin)
6	Vitamin B ₅ (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
7	Vitamin D	Exposure to sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteomalacia (soft bones and joint pain in adults)
8	Vitamin E	Vegatable oils like wheat germ oil, sunflower oil, etc.	Increased fragility of RBCs and muscular weakness
9	Vitamin K	Green leafy vegetables	Increased blood clotting time

NOTE : Vitamin-D , E , K , A are fat soluble vitamins.
Vitamin-B , C are water soluble vitamins.



EXERCISE # I

- Q.1 Iso-electric point of alanine is (pH = 6). At which pH, maximum concentration of zwitter ion of alanine will be present ?
 (A) pH > 6 (B) pH < 6 (C) pH = 6 (D) pH = 7
- Q.2 At iso-electric point :
 (A) Concentration of cation is equal to concentration of anion
 (B) Net charge is zero
 (C) Maximum conc. of di-pole ion (Zwitter ion) will be present
 (D) All of the above
- Q.3 Which of following amino acid has lowest iso-electric point ?
 (A) Glycine (B) Alanine (C) Aspartic acid (D) Lysine
- Q.4 $\text{H}-\text{C}\equiv\text{C}-\text{H} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} (\text{A}) \xrightarrow[(2) \text{H}_3\text{O}^+]{(1) \text{NH}_3+\text{HCN}} (\text{B})$; Product (B) of given reaction is :
 (A) Glycine (B) Alanine (C) Valine (D) Leucine
- Q.5 Which amino acid does not contain chiral centre ?
 (A) Valine (B) Leucine (C) Glycine (D) Iso-leucine
- Q.6 Which of the following is Sanger reagent ?
 (A) 2,4-Di-nitro fluorobenzene (B) Phenyl isocyanate
 (C) 2,4-Di-nitro chlorobenzene (D) 1,2,4-Di-nitro-iodobenzene
- Q.7 A D-carbohydrate is :
 (A) Always dextrorotatory
 (B) Always laevorotatory
 (C) Always the mirror of the corresponding L-carbohydrate
 (D) None of these



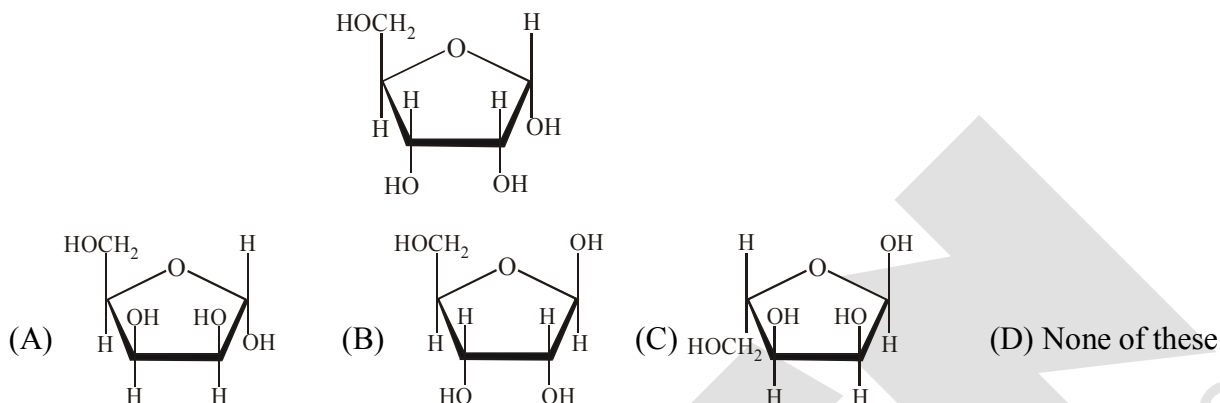
The given osazone can be obtained by :

- (A) D-glucose (B) D-mannose (C) D-Idose (D) Both (A) & (B)

Q.9 Which of the following pair gives same phenyl osazone ?

- (A) D-Glucose and D-Allose (B) D-Glucose and D-Altrose
(C) D-Glucose and D-Mannose (D) D-Glucose and D-Talose

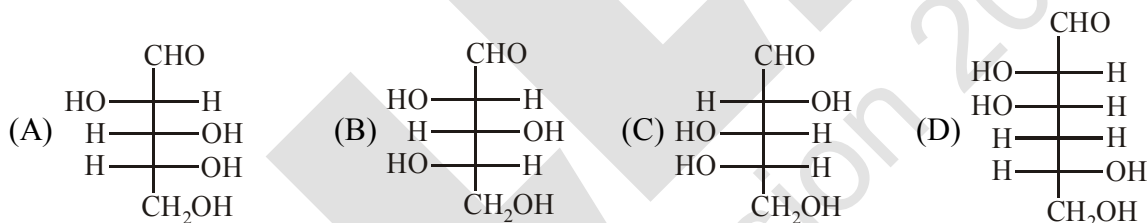
Q.10 Which of the following represents the anomer of the compound shown ?



Q.11 For the complex conversion of D-glucose into the corresponding osazone, the minimum number of equivalents of phenyl hydrazine required is :

- (A) Two (B) Three (C) Four (D) Five

Q.12 Which of the following structure is L-arabinose ?

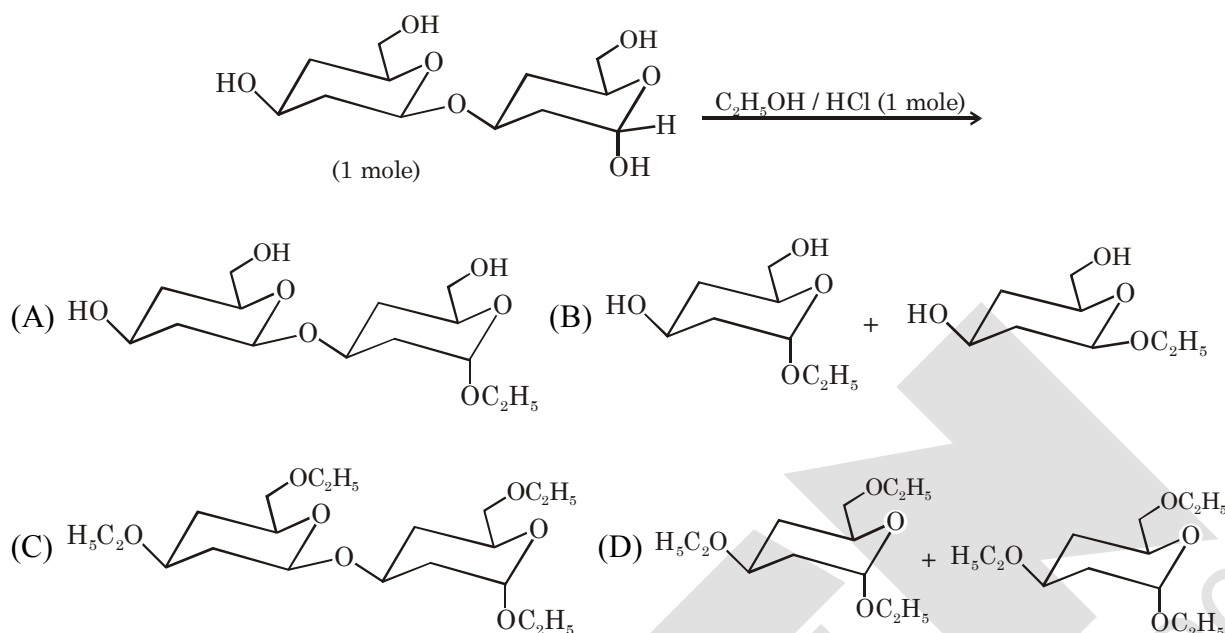


Q.13 Which one of the statements concerning the equilibrium shown is true ?

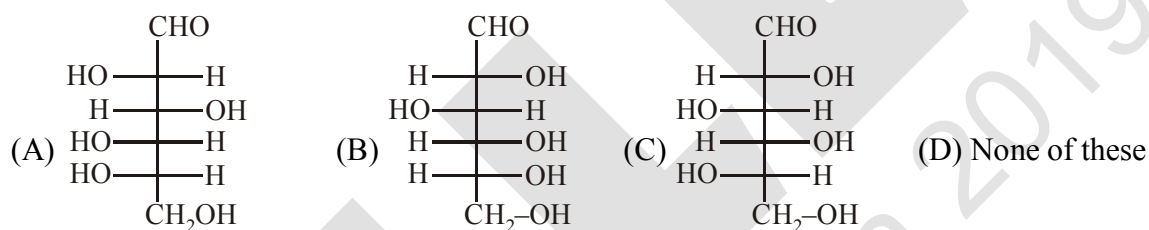


- (A) The two structures are enantiomers of each other. They have equal but opposite optical rotations and racemize slowly at room temperature
(B) The two structures are enantiomers of each other. They racemize too rapidly at room temperature for their optical rotations to be measured
(C) The two structures are diastereomers of each other. Their interconversion is called mutarotation
(D) The two structures are diastereomers of each other. Their interconversion does not require breaking and making bonds, only a change in conformation

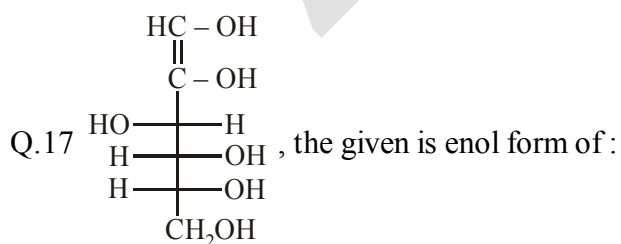
Q.14 Major product of following reaction is :



Q.15 What is the structure of L-glucose ?



Q.16 What is the structure of L-glyceraldehyde ?



- (A) D-glucose (B) D-mannose (C) D-fructose (D) All of these

Q.18 D-glucose & D-fructose can be differentiated by :

- (A) Fehling solution (B) Tollen's reagent (C) Benedict test (D) $\text{Br}_2 / \text{H}_2\text{O}$

Q.19 D-Glucose exist in x different forms. The value of x (stereoisomer) is :

- (A) 2 (B) 3 (C) 4 (D) 5

Q.20 $\text{D-Mannose} \xrightleftharpoons{\text{HO}^-} \text{D-Glucose} \xrightleftharpoons{\text{HO}^-} \text{(A)}$;

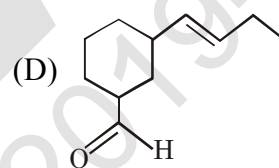
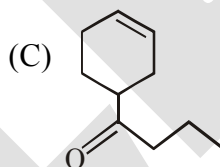
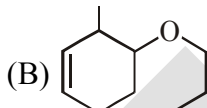
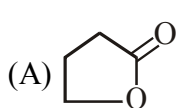
Product (A) of above reaction is :

- (A) D-glucose (B) D-fructose (C) D-talose (D) D-idose

Q.21 Which of the following statement is not correct for maltose.

- (A) It is a disaccharide (B) It undergoes mutarotation
(C) It is a reducing sugar (D) It does not have hemiacetal group.

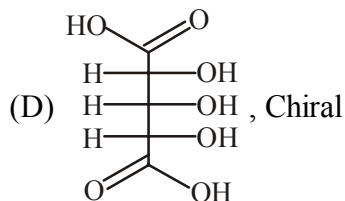
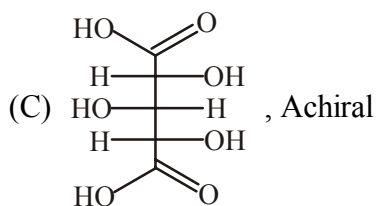
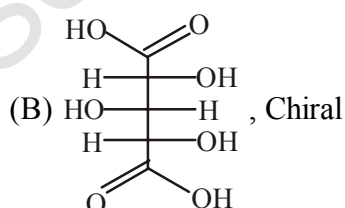
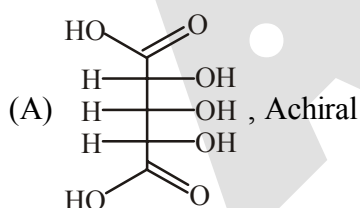
Q.22 A compound which does not react with Brady's reagent but decolourise $\text{Br}_2 / \text{H}_2\text{O}$ solution is :



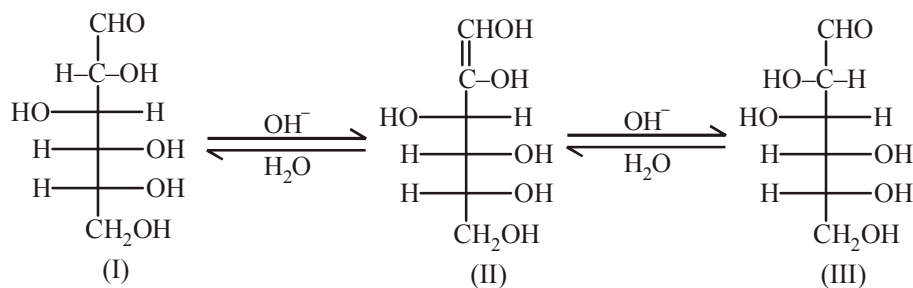
Q.23 Maltose is made up of to two sugars—

- (A) Glucose (B) Fructose
(C) Glucose & galactose (D) Glucose & fructose

Q.24 D-Ribose when treated with dilute HNO_3 forms.



Q.25 Consider the given process



and identify the incorrect statement.

- (A) Configuration at C-2 is lost on enolisation
 (B) I and III are epimers
 (C) Proton transfer from water to C-1 converts ene diol to an aldose.
 (D) D-glucose can isomerise to D-fructose through enol intermediate.

Q.26 When methyl D-glucopyranoside is treated with HIO_4 its number of mole consumed per mole of the sugare is -

- (A) 2 (B) 3 (C) 4 (D) 5

Q.27 α - amino acid when heated with NaOH/CaO forms -

- (A) α, β - unsaturated acid (B) α, β - unsaturated amine
 (C) Carboxylic acid (D) Amine

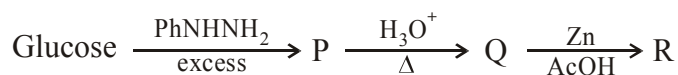
Q.28 The configuration of the C-2 epimer of D-glucose is-

- (A) 2R, 3S, 4R, 5S (B) 2S, 3S, 4R, 5R
 (C) 2S, 3R, 4S, 5R (D) 2R, 3S, 4R, 5R

Q.29 Mutarotation involve-

- (A) Racemisation (B) Diastereomerisation
 (C) Optical resolution (D) Conformational inversion

Q.30 Consider the reaction sequence -



The product R is -

- (A) Arbinose (B) Sorbitol (C) Fructose (D) Mannose

Q.31 The pH of the solution containing following zwitter ion species is $\text{NH}_3^+ - \text{C}(\text{COO}^-)(\text{H})(\text{R})$

- (A) 4 (B) 5 (C) 7 (D) 9

Q.32 Peptide linkage is -

- (A) $\text{O}=\text{C}-\text{O}-$ (B) $\text{O}=\text{C}-\text{NH}_2$ (C) $\text{O}=\text{C}-\text{NH}-$ (D) $\text{O}=\text{C}-\text{NH}-\text{NH}_2$

Q.33 Same osazone derivative is obtained in case of D-glucose, D-Mannose and D-Fructose due to

- (A) The same configuration at C-5
(B) The same constitution.
(C) The same constitution at C-1 and C-2
(D) The same constitution and acid configuration at C-3, C-4, C-5 and C-6 but different constitution and configuration at C-1 and C-2 which becomes identical by osazone formation.

Q.34 $\text{D}(-)\text{-Erythrose} \xrightarrow{\text{NaBH}_4} (\text{P})$
 $\text{D}(-)\text{-Threose} \xrightarrow{\text{NaBH}_4} (\text{R})$

Which of the following statement is correct about P and R ?

- (A) Both are optically active (B) Both are optically inactive
(C) P is optically inactive and R is optically active (D) Neither P nor R has asymmetric carbon.

Q.35 The monomer of nucleic acids are held together by

- (A) Phosphodiester linkage (B) Amide linkage
(C) Glycosidic linkage (D) Ester linkage

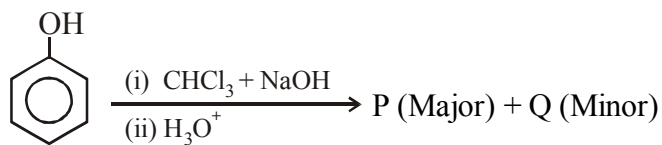
Q.36 Select the incorrect statement about Nylon 2-nylon-6.

- (A) It is a copolymer
(B) It is biodegradable
(C) It is an alternating polyamide
(D) It is made up of $\text{CH}_3-\text{CH}(\text{NH}_2)-\text{COOH}$ and $\text{H}_2\text{N}(\text{CH}_2)_5\text{COOH}$

Q.37 The monomer that can undergo radical, cationic and anionic polymerisation with equal ease-

- (A) $\text{Me}-\text{C}(\text{Me})=\text{CH}_2$ (B) $\text{Ph}-\text{CH}=\text{CH}_2$ (C) $\text{CH}_2=\text{CH}_2$ (D) $\text{CH}_2=\text{CH}-\text{CN}$

Q.38 Consider the reaction-



Mixture of A and B can be best separated by -

- (A) Steam distillation (B) Vacuum distillation
(C) Fractional distillation (D) Crystallisation

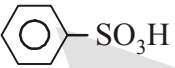
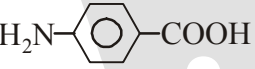
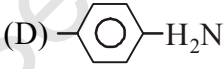
Q.39 Which of the following statements are incorrect :

- (A) Copolymer of 1,3-butadiene & acrylonitrile is Buna-S
(B) HDPE is obtained by Ziegler-Natta polymerisation
(C) Polymer dacron can be polyester fabric
(D) Phenol & formaldehyde resin is called novolac

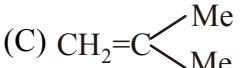
Q.40 A segment (X) of cellulose obtained on partial hydrolysis has molecular mass 1476 gm. On complete acidic hydrolysis, mass of the product obtained is 1620 gm. Find out the number of glycosidic linkage(s) present in segment (X) :

- (A) 8 (B) 6 (C) 4 (D) 10

Q.41 When solidum extract is treated with FeCl_3 solution a blood red coloured is obtained due to the presence of -

- (A)  (B) 
(C)  (D) 

Q.42 The monomer that undergo radical polymerisation most easily is

- (A) $\text{CH}_2=\text{CH}_2$ (B) $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$ (C)  (D) $\text{CH}_3-\text{CH}=\text{CH}_2$

EXERCISE # II

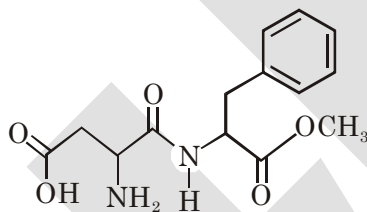
Q.1 Carbohydrates may be :

- (A) Sugars (B) Starch
(C) Polyhydroxy aldehyde/ ketones (D) Compounds that can be hydrolysed to sugar

Q.2 Select the correct statement :

- (A) Glycosides do not undergo mutarotation
(B) All OH groups of a cyclic monosaccharides are converted to ethers by treatment with base and an alkyl halide
(C) α -D glucose reacts with Ag_2O and excess CH_3I to form tetramethyl ether
(D) D-glucose upon treatment with warm HNO_3 forms D-glucaric acid

Q.3 "Aspartame" is roughly 100 times as sweet as cane sugar. On complete hydrolysis of aspartame, products obtained is/are :



- (A) $\text{PhCH}_2\text{-CH(NH}_2\text{)-CO}_2\text{H}$ (B) $\text{H}_2\text{N-CH(CH}_2\text{CO}_2\text{H)-CO}_2\text{H}$ (C) CH_3OH (D) $\text{CH}_3\text{-CH(NH}_2\text{)-CO}_2\text{H}$

Q.4 Starch molecules are polymer with repeating glucose units. Select the correct statement(s).

- (A) Glucose units are joined through α -glycosidic linkage
(B) The branches of amylopectin are linked to the chain with α -1,6'-glycosidic linkages
(C) The linear linkages of amylopectin are formed by α -1,6'-glycosidic bond
(D) Amylose has an unbranched skeleton of glucose molecules with α -1,4'-glycosidic linkages

Q.5 Select the correct option.

- (A) Isoelectric point is the pH at which an amino acid exists primarily in its neutral form.
(B) Isoelectric point is the average of pK_a values of α -COOH amino α - NH_3^+ groups [valid only for neutral amino acid]
(C) Glycine is characterised by two pK_a values
(D) For neutral amino acid the concentration of zwitter ion is maximum at its isoelectric point

Q.6 Amino acids are synthesised from

- (A) α -Halo acids by reaction with NH_3
- (B) Aldehydes by reaction with NH_3 and cyanide ion followed by hydrolysis
- (C) Pyruvic acid is treated with NH_3 followed by addition of $\text{H}_2(\text{Ni})$
- (D) Alcohols by reaction with NH_3 and CN^- ion followed by hydrolysis.

Q.7 Which of the following carbohydrates develops blue colour on treatment with iodine solution ?

- (A) Glucose
- (B) Amylose
- (C) Starch
- (D) Fructose

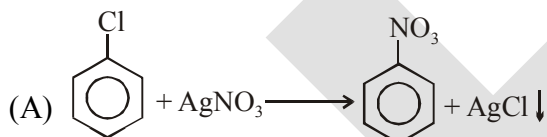
Q.8 Select the **correct** statement

- (A) High density polythene is a linear polymer.
- (B) Low density polythene is a branched chain polymer.
- (C) Chain growth polymers are also known as addition polymer.
- (D) Step growth polymer is also known as condensation polymer.

Q.9 Select the correct statement.

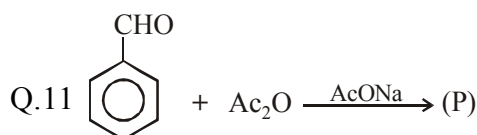
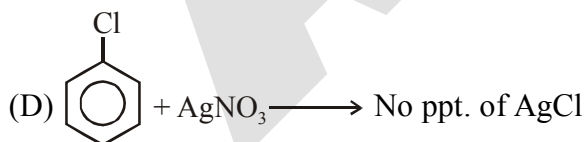
- (A) Elastomers have the weakest intermolecular forces
- (B) Buna-N is example of synthetic copolymer
- (C) Some fibres have crystalline nature
- (D) Thermoplastic polymers have stronger intermolecular forces than fibres

Q.10 Which of the following options are correct :



(B) Na extract of sample containing chlorobenzene $\xrightarrow{+\text{AgNO}_3}$ white ppt. of AgCl

(C) Na extract of sample containing chlorobenzene $\xrightarrow{+\text{AgNO}_3}$ No ppt. of AgCl



Before isolating (P) unreacted Ph-CHO is removed first. Select the correct statement.

- (A) P is cinnamaldehyde
- (B) P is crotonic acid
- (C) Removal is done by simple distillation
- (D) P is cinnamic acid.

Q.12 If on a strand of DNA the base sequence is ATTGACGCAT then the sequence transcription on RNA would be -

- (A) UAACUGCGUA (B) AUUCUGCGUA
(C) UAACTGCGUA (D) TAACTGCGTA

Q.13 Select the correct statement among following

- (A) Number of chiral atom in α -D-glucose is less than D-glucose
(B) D-glucose and D-fructose give same product with HIO_4
(C) D-glucose and D-fructose give same product with H_2NOH
(D) D-glucose and D-fructose form same product with $\text{H}_2\text{N}-\text{NH}-\text{Ph}$

Q.14 The N-base which differentiate DNA with RNA is :

- (A) Cytosine (B) Uracil (C) Adenine (D) Guanine

Q.15 Correct statement

- (A) Nylon-66 is example of addition polymer
(B) Alanin having pH 9 at isoelectric point
(C) Fructose when reacts with fehling solution reddish brown ppt. formed
(D) All of these

Q.16 Match the column :

Column I

- (A) Sucrose
(B) Maltose
(C) Lactose
(D) Cellulose

Column II

- (P) Two acetals
(Q) No hemiacetal
(R) β -1,4'-glycosidic bond
(S) One of the hydrolysis product is glucose

Q.17 Match the column

Column I

(Component of mixture)

- (A) Crystalline Na_2CO_3 +
Sodium citrate + CuSO_4 (aq. sol.)
(B) CuSO_4 + Rochelle Salt + NaOH (Aq. sol.)
(C) 10% α - naphthol in alcohol
(D) HgCl + KI + KOH (aq. sol.)

Column II

(Reagent)

- (P) Fehling solution
(Q) Nessler's Reagent
(R) Benedict's solution
(S) Molisch's Reagent

EXERCISE # III

Assertion Reason

Q.1 **Statement 1** : Furanose ring, like pyranose rings are not planar.

Statement 2 : The most stable conformation of furanose is envelope form.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Q.2 **Statement 1** : Bromine water changes glucose to gluconic acid.

Statement 2 : Bromine water acts as oxidising agent.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

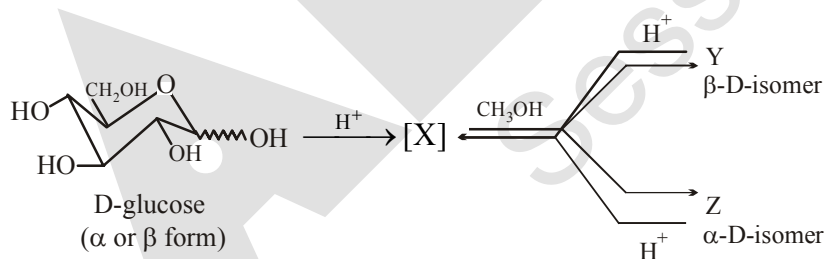
Q.3 **Statement 1** : All monosaccharide ketoses are reducing sugars.

Statement 2 : Monosaccharide ketose give positive Tollen's and Fehling's test.

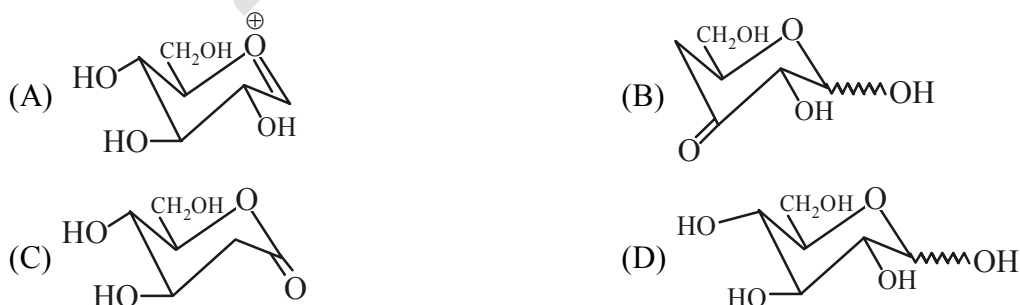
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Comprehension (Q.4 to Q.6)

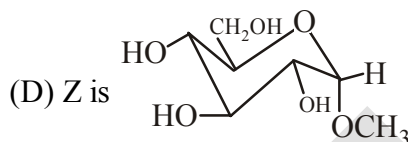
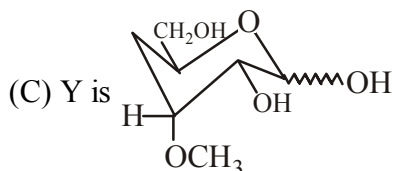
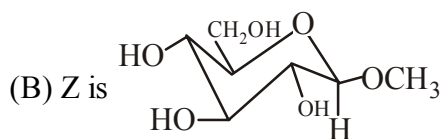
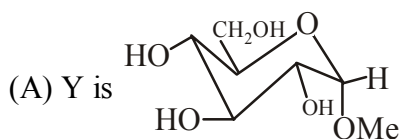
Consider the following reversible process for a reaction of D-glucose.



Q.4 The structure of intermediate [X] is



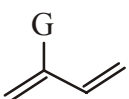
Q.5 Select the correct option.

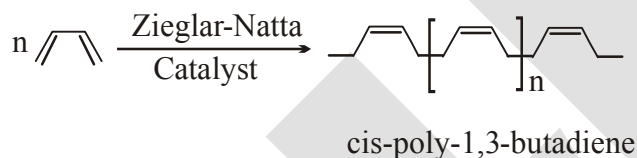


Q.6 Select the correct statement.

- (A) Y is more stable than Z due to H-bond (B) Y is less stable than Z due to anomeric effect.
(C) Y is more stable than Z due to anomeric effect (D) Y is less stable than Z due to H-bond.

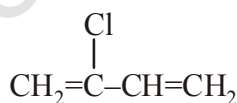
Comprehension (Q.7 to Q.9)

The monomer  (G = Me or Cl) when treated with Ziegler - Natta catalyst undergo polymerisation in the manner given below -



Q.7 The Ziegler-Natta catalyst is

- (A) TiCl_4 (B) R_3Al (C) $\text{R}_3\text{Al} / \text{TiCl}_4$ (D) $\text{R}_3\text{B} / \text{TiCl}_4$

Q.8 The polymer obtained when monomeric unit used is 

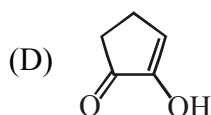
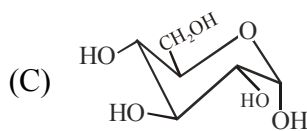
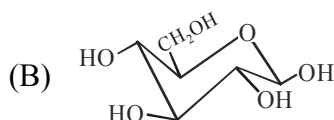
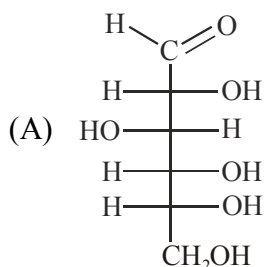
- (A) Neoprene (B) Stilbene (C) Styrene (D) Chloropirrin

Q.9 Which of the following statement is not true considering the process given above.

- (A) The general class of polymer formed is known as homopolymer
(B) The polymer obtained is stereoregular
(C) Buna-N can be prepared using above process
(D) Synthetic rubber can be formed by above process using 1,3- butadiene.

Match the Column

Q.10 Match the compounds given below existing in equilibrium mixture with their percentage given in the right hand side.

Column I**Column II**

(P) 36 %

(Q) 63.8 %

(R) 99.9 %

(S) 0.2 %

Q.11 Match the column :

Column I

(Carbohydrate)

(A) Starch

(B) Sucrose

(C) Lactose

(D) Maltose

Column II

(Properties)

(P) Mutarotation

(Q) Non reducing sugar

(R) β -glycosidic bond

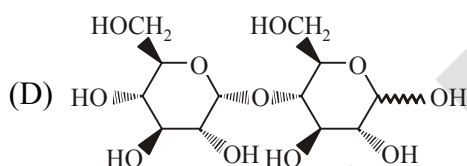
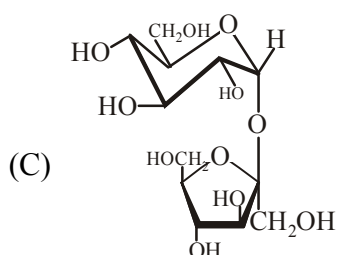
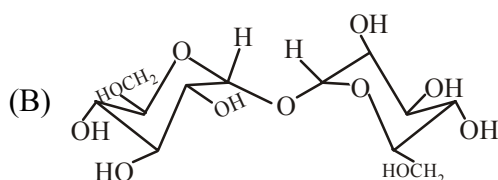
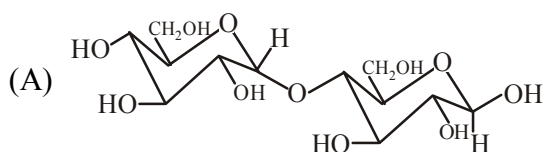
(S) α -glycosidic bond

(T) Reducing sugar

(U) Hemiacetal

Q.12 Match the column :

Column I



Column II

(P) α -glycoside bonds

(Q) Reducing sugar

(R) Forms enediol intermediate

(S) β -glycoside bond

Q.13 Match the column

Column I

(Functional group)

(A) Aldehydic

(B) Phenolic

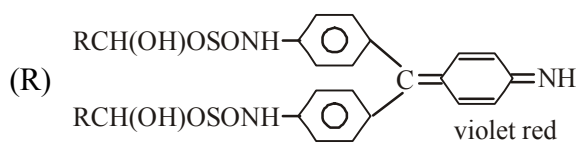
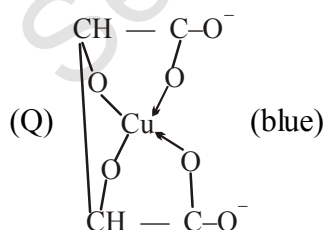
(C) Alcohol

(D) α -D- Glucopyranose

Column II

(Test used or complex formed during confirmatory test)

(P) $[(C_6H_5O)_6Fe]^{-3}$ (violet)



(S) $(ROH)_2Ce(NO_3)_4$ (Red)

(T) Molisch's Test

Q.14 Match the column :

Column I

- (A) Addition polymer
(B) Condensation polymer
(C) Homopolymer
(D) Copolymers

Column II

- (P) Buna-S
(Q) Buna-N
(R) Polythene
(S) Nylon 6,6

Q.15 Match the column

Presence of the element in**Na - extract**

- (A) N
(B) S
(C) N and S together

Complex formed in one of the test

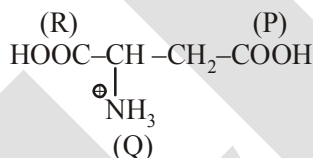
- (P) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$
(Q) $\text{Fe}(\text{CNS})_3$
(R) PbS
(S) $\text{Fe}_4[\text{Fe}(\text{CN})_6]$

Colour of the complex

- (W) Prussian Blue
(X) Black
(Y) Violet
(Z) Blood Red

Subjective Type :

Q.16 The pKa values for the three acidic group P,Q,R are 4.3, 9.7 and 2.2 respectively



Calculate the isoelectric point of the amino acid ?

Q.17 How will you separate?

- | | |
|---|--|
| (a) Ethane & Ethyne | (b) 1-Butyne & 2-Butyne |
| (c) Phenol & Propanol | (d) 2-Propanol & Propanone |
| (e) CH_3COOH & HCOOCH_3 | (f) PhOH & PhCOOH |
| (g) C_2H_4 & C_2H_2 | (h) EtNH_2 & Me_2NH |
| (i) EtOH & $\text{Et}-\text{O}-\text{Et}$ | |

Q.18 How will you differentiate?

- | | |
|--------------------------------------|---|
| (a) Propane & Propene | (b) 1,1-Dichloroethane & 1,2-Dichloroethane |
| (c) Chloroethane & Chloroethene | (d) Pure & Oxidized CHCl_3 |
| (e) p-chlorotoluene & benzylchloride | (f) n-propylchloride & isopropylchloride |
| (g) Methanol & ethanol | (h) Isobutanol & tert-butanol |

(i) 2-Pentanol & 3-pentanol

(k) Ethanol & Propanol

(m) HCHO & PhCHO

(o) HCOOH & CH₃COOH

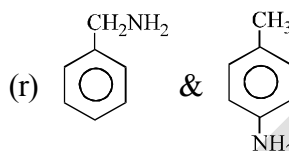
(q) MeNH₂ & Me₂NH

(j) O-cresol & benzylalcohol

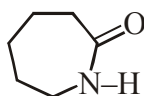
(l) Propanone & Ethanol

(n) Glucose & Fructose

(p) HCOOH & HCHO



Q.19 (a) What is the structure of nylon-6, made by alkaline polymerisation of caprolactam.



(b) Suggest mechanism for the process. Is polymerisation of the chain reaction or step reaction type.

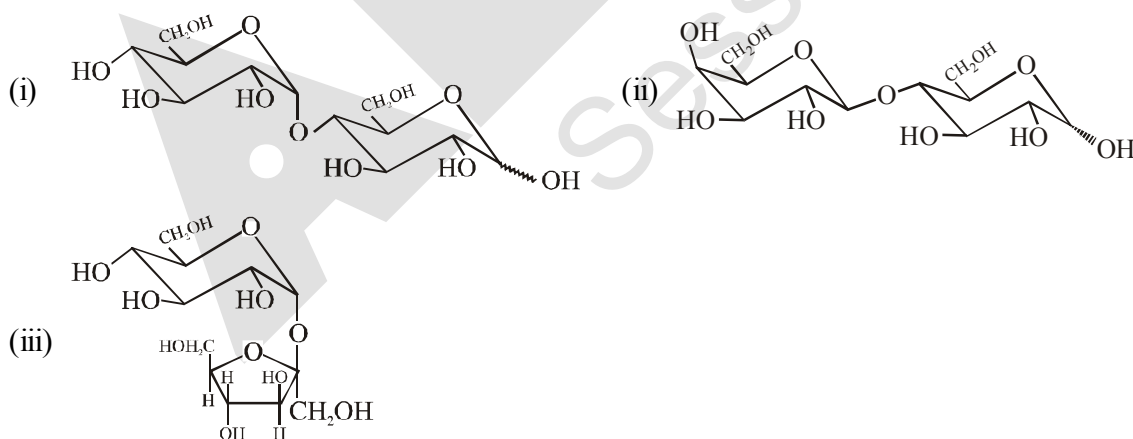
Q.20 Compound (A) C₅H₁₀O₅ give a tetra-acetate with Ac₂O & oxidation of A with Br₂/H₂O gives an acid C₅H₁₀O₆. Reduction of A with HI & Red P. Give 2-methyl butane. What is structure of 'A'?

Q.21 (i) Sulphanilic acid although has acidic as well as basic group. It is soluble in alkali but insoluble in mineral acid. Explain.

(ii) Explain why sulphonic acid is not soluble in organic solvents.

Q.22 Account for the fact that 2-amino ethanoic acid (glucine) exist as a dipolar ion as does p-aminobenzene sulphonic acid but p-amino benzoic acid does not.

Q.23 For given compound



(a) Which glycosidic linkage is present

(b) Name of the compound

(c) Products obtained on hydrolysis

(d) Is it reducing sugar

(e) Number of moles of HIO₄ consumed

(f) Is mutarotation possible

(g) Number of moles of CH₃COCl consumed per mole of compound.

EXERCISE # IV (J-MAIN)

1. Which of the following pairs give positive Tollen's test? [J-Main-2004]

(1) Glucose, sucrose	(2) Glucose, fructose
(3) Hexanal, Acetophenone	(4) Fructose, sucrose

2. Two forms of D – glucopyranose, are called. [J-Main-2005]

(1) Enantiomers	(2) Anomers	(3) Epimers	(4) Diastereomers
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3. When benzene sulfonic acid and p-nitrophenol are treated with NaHCO_3 , the gases released respectively are [J-Main-2006]

(1) SO_2 , NO_2	(2) SO_2 , NO	(3) SO_2 , CO_2	(4) CO_2 , CO_2
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4. **Statement-1** : Glucose gives a reddish-brown precipitate with Fehling's solution.
because [J-Main-2007]

Statement-2 : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.

 - (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 - (3) Statement-1 is True, Statement-2 is False.
 - (4) Statement-1 is False, Statement-2 is True.

5. Which one of the following bases is **not present** in DNA ? [J-Main-2014]

(1) Cytosine	(2) Thymine	(3) Quinoline	(4) Adenine
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6. Which one is classified as a condensation polymer? [J-Main- 2014]

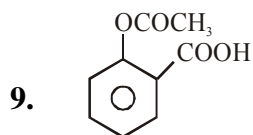
(1) Teflon	(2) Acrylonitrile	(3) Dacron	(4) Neoprene
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7. Which of the vitamins given below is water soluble? [J-Main- 2015]

(1) Vitamin E	(2) Vitamin K	(3) Vitamin C	(4) Vitamin D
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8. Which of the following compounds is not an antacid? [J-Main- 2015]

(1) Phenelzine	(2) Ranitidine
(3) Aluminium hydroxide	(4) Cimetidine



is used as :

[J-Main-2015]

- (1) Insecticide (2) Antacid (3) Antihistamine (4) Analgesic

10. Thiol group is present in :

[J-Main-2016]

- (1) Methionine (2) Cytosine (3) Cystine (4) Cysteine

11. The distillation technique most suited for separating glycerol from spent-lye in the soap industry is

[J-Main-2016]

- (1) Distillation under reduced pressure (2) Simple distillation
(3) Fractional distillation (4) Steam distillation

12. Which of the following statements about low density polythene is FALSE ?

[J-Main-2016]

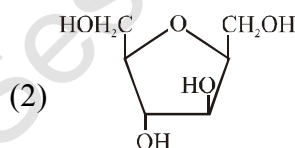
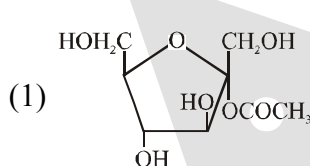
- (1) It is used in the manufacture of buckets, dust-bins etc
(2) Its synthesis requires high pressure
(3) It is a poor conductor of electricity
(4) Its synthesis requires dioxygen or a peroxide initiator as a catalyst.

13. Which of the following is an anionic detergent ?

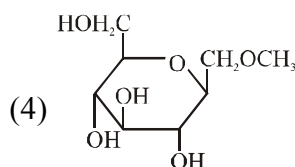
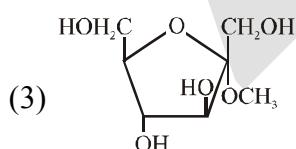
[J-Main-2016]

- (1) Glyceryl oleate (2) Sodium stearate
(3) Sodium lauryl sulphate (4) Cetyltrimethyl ammonium bromide

14. Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution



[J-Main-2017]



15. The formation of which of the following polymers involves hydrolysis reaction ? [J-Main-2017]

- (1) Nylon 6 (2) Bakelite (3) Nylon 6, 6 (4) Terylene

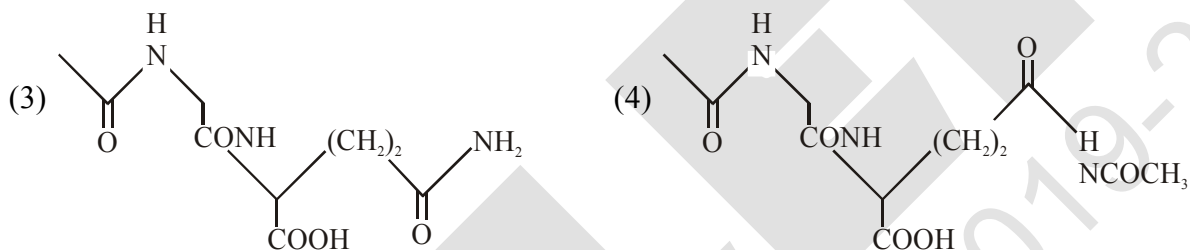
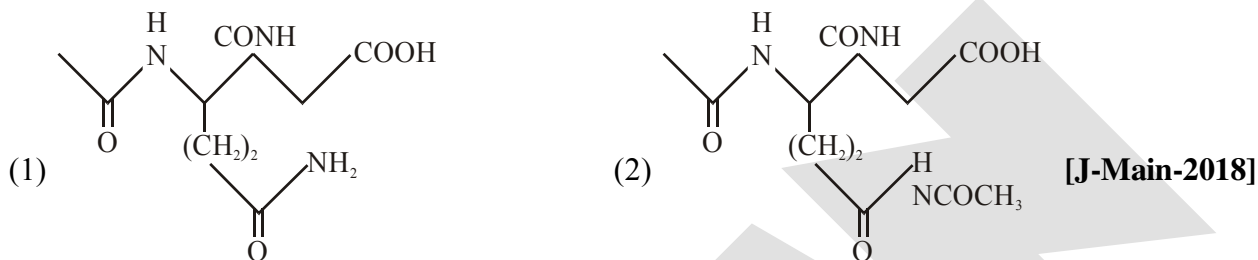
16. Glucose on prolonged heating with HI gives : [J-Main-2018]

- (1) 1-Hexene (2) Hexanoic acid (3) 6-iodohexanal (4) n-Hexane

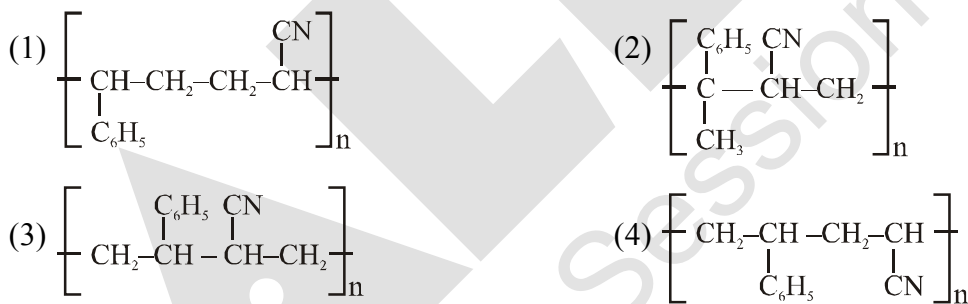
17. Which of the following statement is not true :- [J-Main-2018]

- (1) Nylon 6 is an example of step-growth polymerisation
 (2) Chain growth polymerisation involves homopolymerisation only
 (3) Step growth polymerisation requires a bifunctional monomer
 (4) Chain growth polymerisation includes both homopolymerisation and copolymerisation

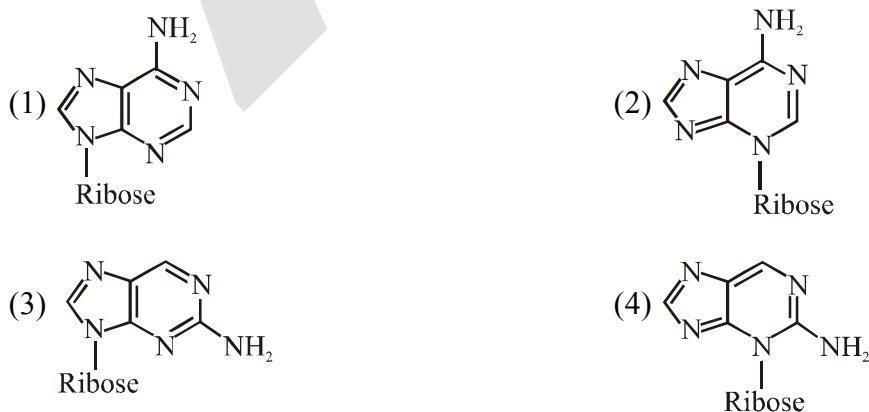
18. The dipeptide, Gln-Gly, on treatment with CH_3COCl followed by aqueous work up gives :-



19. The copolymer formed by addition polymerization of styrene and acrylonitrile in the presence of peroxide is : [J-Main-2018]



20. Which of the following is the correct structure of Adenosine ? [J-Main-2018]



21. Among the following, the incorrect statement is:-

[J-Main-2018]

- (1) Cellulose and amylose has 1, 4-glycosidic linkage.
- (2) Lactose contains β -D-galactose and β -D-glucose.
- (3) Maltose and lactose has 1, 4-glycosidic linkage.
- (4) Sucrose and amylose has 1, 2-glycosidic linkage.

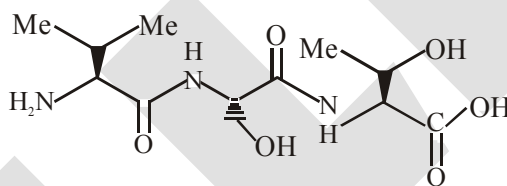
22. The correct match between items of List-I and List-II is :-

[J-Main-2018]

	List-I		List-II
(A)	Phenelzine	(P)	Pyrimidine
(B)	Chloroxylenol	(Q)	Furan
(C)	Uracil	(R)	Hydrazine
(D)	Ranitidine	(S)	Phenol

- (1) (A)-(S), (B)-(R), (C)-(P), (D)-(Q)
- (2) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)
- (3) (A)-(S), (B)-(R), (C)-(Q), (D)-(P)
- (4) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)

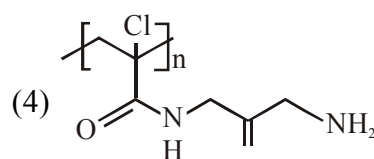
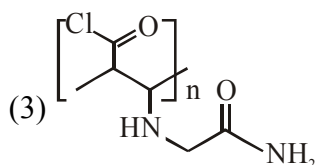
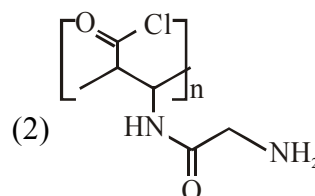
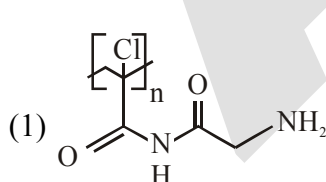
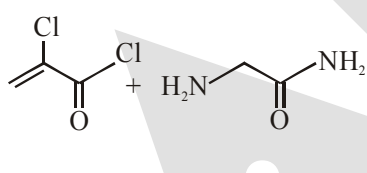
23. The correct sequence of amino acids present in the tripeptide given below is : [J-Main-(Jan)-2019]



- (1) Leu - Ser - Thr
- (2) Thr - Ser - Leu
- (3) Thr - Ser - Val
- (4) Val - Ser - Thr

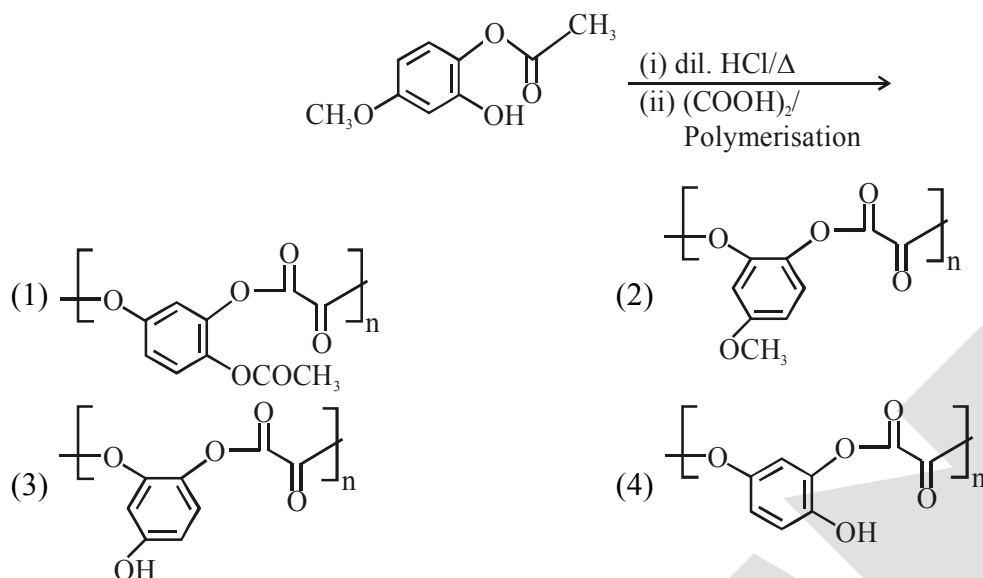
24. Major product of the following reaction is :

[J-Main-(Jan)-2019]



25. The major product of the following reaction is:

[J-Main-(Jan)-2019]



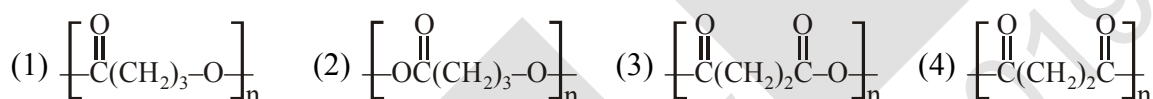
26. Which of the following tests cannot be used for identifying amino acids ?

[J-Main-(Jan)-2019]

- (1) Biuret test (2) Xanthoproteic test (3) Barfoed test (4) Ninhydrin test

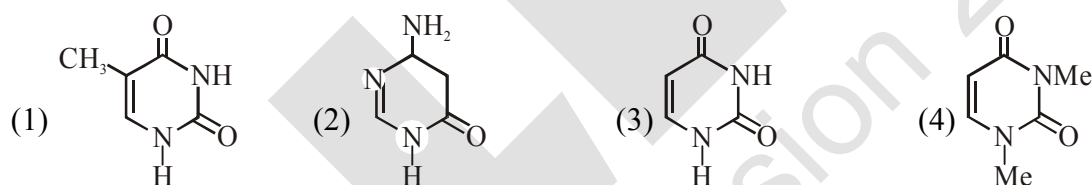
27. The homopolymer formed from 4-hydroxy-butanoic acid is :-

[J-Main-(Jan)-2019]



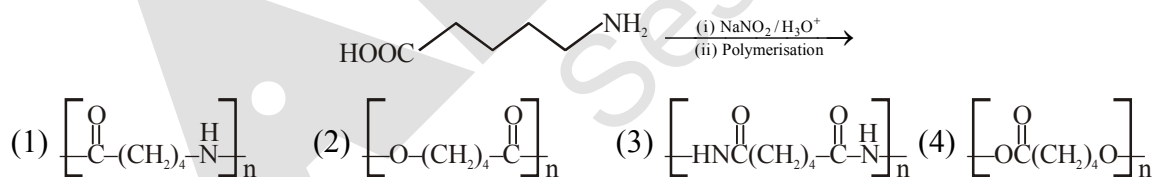
28. Among the following compound which one is found in RNA ?

[J-Main-(Jan)-2019]



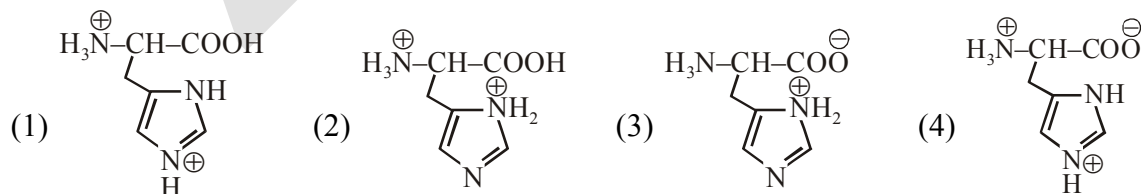
29. The polymer obtained from the following reactions is :

[J-Main-(Jan)-2019]



30. The correct structure of histidine in a strongly acidic solution (pH=2) is

[J-Main-(Jan)-2019]



31. The two monomers for the synthesis of Nylon 6, 6 are :

[J-Main-(Jan)-2019]

- (1) $\text{HOOC}(\text{CH}_2)_6\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (2) $\text{HOOC}(\text{CH}_2)_4\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_4\text{NH}_2$
 (3) $\text{HOOC}(\text{CH}_2)_6\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_4\text{NH}_2$ (4) $\text{HOOC}(\text{CH}_2)_4\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$

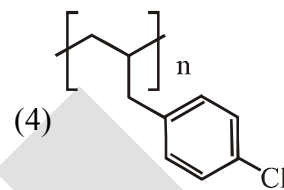
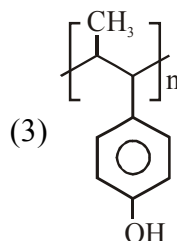
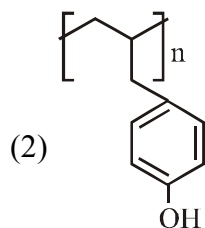
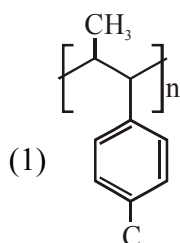
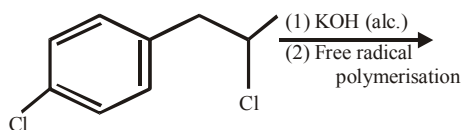
32. Maltose on treatment with dilute HCl gives : [J-Main-(April)-2019]
 (1) D-Galactose (2) D-Glucose
 (3) D-Glucose and D-Fructose (4) D-Fructose
33. Fructose and glucose can be distinguished by : [J-Main-(April)-2019]
 (1) Fehling's test (2) Barfoed's test
 (3) Benedict's test (4) Seliwanoff's test
34. The peptide that gives positive ceric ammonium nitrate and carbylamine tests is : [J-Main-(April)-2019]
 (1) Lys-Asp (2) Ser-Lys (3) Gln-Asp (4) Asp-Gln
35. Which of the following statements is not true about sucrose? [J-Main-(April)-2019]
 (1) On hydrolysis, it produces glucose and fructose
 (2) The glycosidic linkage is present between C₁ of α-glucose and C₁ of β-fructose
 (3) It is also named as invert sugar
 (4) It is a non reducing sugar
36. Number of stereo centers present in linear and cyclic structures of glucose are respectively : [J-Main-(April)-2019]
 (1) 4 & 5 (2) 5 & 5
 (3) 4 & 4 (4) 5 & 4
37. Amylopectin is composed of : [J-Main-(April)-2019]
 (1) α-D-glucose, C₁-C₄ and C₁-C₆ linkages
 (2) α-D-glucose, C₁-C₄ and C₂-C₆ linkages
 (3) β-D-glucose, C₁-C₄ and C₂-C₆ linkages
 (4) β-D-Glucose, C₁-C₄ and C₁-C₆ linkages
38. Which of the following statements is not true about RNA ? [J-Main-(April)-2019]
 (1) It has always double stranded α-helix structure
 (2) It usually does not replicate
 (3) It is present in the nucleus of the cell
 (4) It controls the synthesis of protein
39. Which of the following compounds is a constituent of the polymer [J-Main-(April)-2019]

$$\left[\text{HN} - \overset{\text{O}}{\parallel} \text{C} - \text{NH} - \text{CH}_2 \right]_n$$

 (1) Formaldehyde (2) Ammonia (3) Methylamine (4) N-Methyl urea

40. The major product of the following reaction is :

[J-Main-(April)-2019]



41. Which of the following is a condensation polymer ?

[J-Main-(April)-2019]

(1) Buna - S

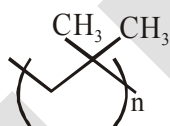
(2) Nylon 6, 6

(3) Teflon

(4) Neoprene

42. The correct name of the following polymer is:

[J-Main-(April)-2019]



(1) Polyisoprene

Polytert-butylene

(3) Polyisobutane

(4) Polyisobutylene

43. Which of the following is a thermosetting polymer?

[J-Main-(April)-2019]

(1) Buna-N

(2) PVC

(3) Bakelite

(4) Nylon 6

EXERCISE # V (A) (J-ADVANCE) (OBJECTIVE)

Q.1 **Statement-1** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

because

[JEE 2007]

Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

Q.2 Match the chemical substances in Column I with type of polymers / type of bonds in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

[JEE 2007]

Column I

- (A) Cellulose
- (B) Nylon-6, 6
- (C) Protein
- (D) Sucrose

Column II

- (P) Natural polymer
- (Q) Synthetic polymer
- (R) amide linkage
- (S) Glycoside linkage

Q.3 Match the compounds/ion in column I with their properties/ reaction in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2007]

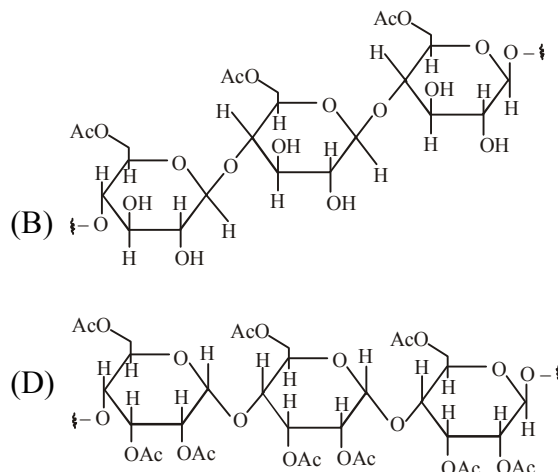
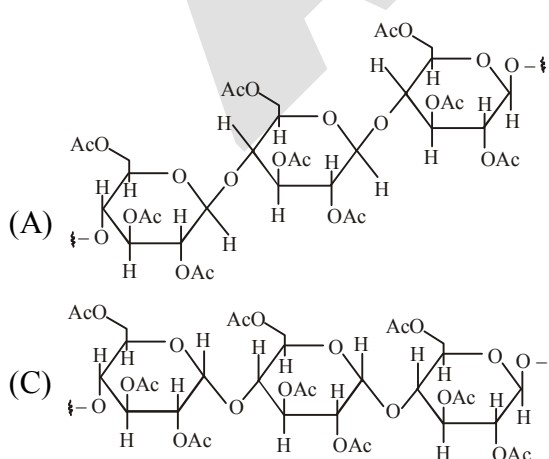
Column I

- (A) C_6H_5CHO
- (B) $CH_3C \equiv CH$
- (C) CN^-
- (D) I^-

Column II

- (P) gives precipitate with 2,4-dinitrophenylhydrazine
- (Q) gives precipitate with $AgNO_3$
- (R) is a nucleophile
- (S) is involved in cyanohydrin formation

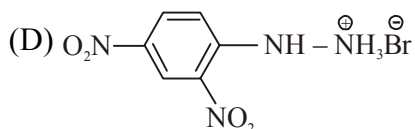
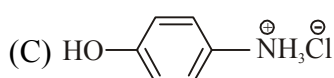
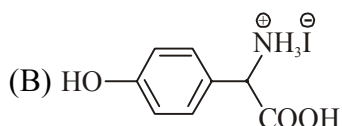
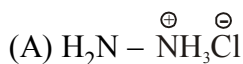
Q.4 Cellulose upon acetylation with excess acetic anhydride / H_2SO_4 (catalytic) gives cellulose triacetate whose structure is [JEE 2008]



Q.5 Match the compounds in **Column I** with their characteristic test(s)/reaction(s) given in **Column II**.
Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

[JEE 2008]

Column-I



Column II

(P) Sodium fusion extract of the compound gives Prussian blue colour with FeSO_4

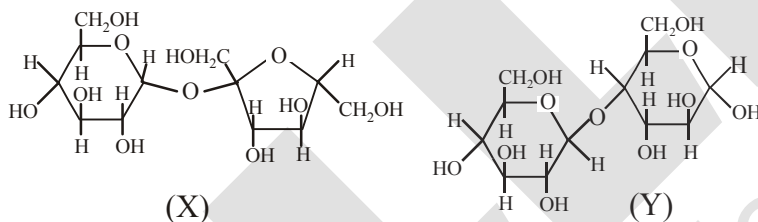
(Q) Gives positive FeCl_3 test

(R) Gives white precipitate with AgNO_3

(S) Reacts with aldehydes to form the corresponding Hydrazone derivative

Q.6 The correct statement(s) about the following sugars X and Y is(are)

[JEE 2009]



- (A) X is a reducing sugar and Y is a non-reducing sugar
(B) X is a non-reducing sugar and Y is a reducing sugar
(C) The glucosidic linkages in X and Y are α and β , respectively.
(D) The glucosidic linkages in X and Y are β and α , respectively

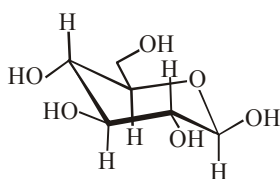
Q.7 Among cellulose, poly(vinyl chloride), nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is

[JEE 2009]

- (A) Nylon (B) Poly (vinyl chloride)
(C) Cellulose (D) Natural Rubber

Q.8 The following carbohydrate is

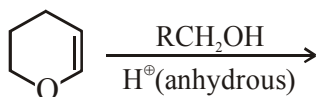
[JEE 2011]



- (A) a ketohexose (B) an aldohexose (C) an α -furanose (D) an α -pyranose

Q.9 The major product of the following reaction is

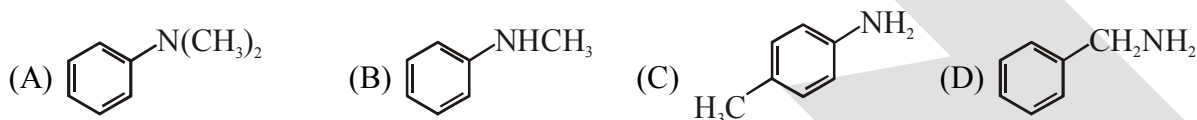
[JEE 2011]



- (A) a hemiacetal (B) an acetal (C) an ether (D) an ester

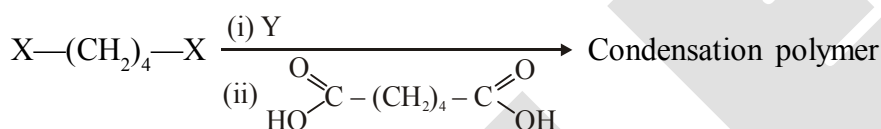
Q.10 Amongst the compounds given, the one that would form a brilliant coloured dye on treatment with NaNO_2 in dil. HCl followed by addition to an alkaline solution of β -naphthol is -

[JEE 2011]



Q.11 The correct functional group X and the reagent/reaction conditions Y in the following scheme are

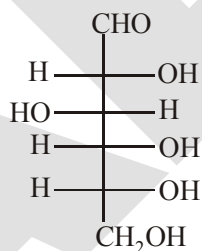
[JEE 2011]



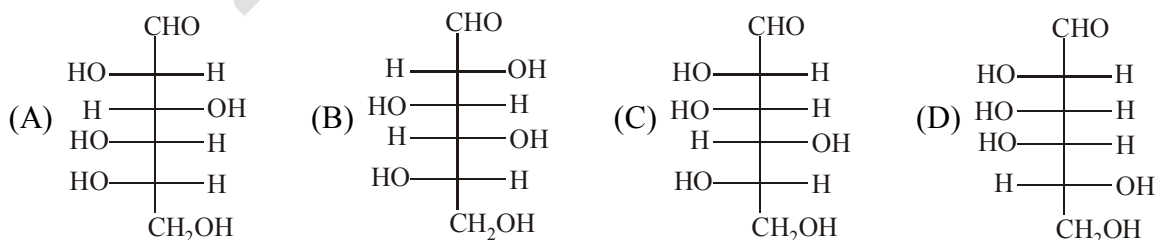
- (A) $\text{X} = \text{COOCH}_3$, $\text{Y} = \text{H}_2 / \text{Ni} / \text{heat}$ (B) $\text{X} = \text{CONH}_2$, $\text{Y} = \text{H}_2 / \text{Ni} / \text{heat}$
 (C) $\text{X} = \text{CONH}_2$, $\text{Y} = \text{Br}_2 / \text{NaOH}$ (D) $\text{X} = \text{CN}$, $\text{Y} = \text{H}_2 / \text{Ni} / \text{heat}$

Q.12 The structure of D-(+)-glucose is

[IIT 2011]



The structure of L(-)-glucose is



Q.13 For 'invert sugar', the correct statement(s) is (are) :

[JEE 2016]

(Given : specific rotations of (+)-sucrose, (+)-maltose, L-(−)-glucose and L-(+)-fructose in aqueous solution are $+66^\circ$, $+140^\circ$, -52° and $+92^\circ$, respectively)

- (A) 'invert sugar' is prepared by acid catalyzed hydrolysis of maltose
- (B) 'invert sugar' is an equimolar mixture of D-(+) glucose and D-(−)-fructose
- (C) specific rotation of 'invert sugar' is -20°
- (D) on reaction with Br_2 water, 'invert sugar' forms saccharic acid as one of the products

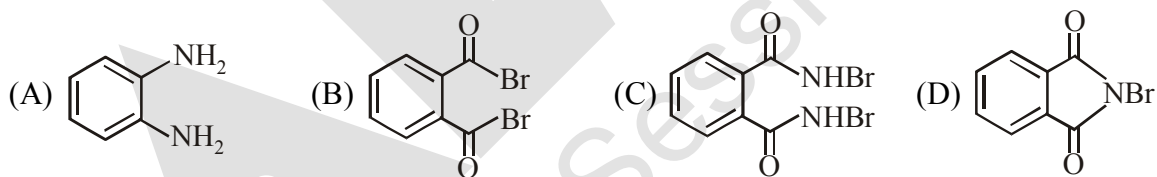
Comprehension : Q.No. 14 to 15

Treatment of compound **O** with $\text{KMnO}_4 / \text{H}^+$ gave **P**, which on heating with ammonia gave **Q**. The compound **Q** on treatment with $\text{Br}_2 / \text{NaOH}$ produced **R**. On strong heating, **Q** gave **S**, which on further treatment with ethyl 2-bromopropionate in the presence of KOH following by acidification, gave a compound **T**.



Q.14 The compound **R** is :

[JEE 2016]



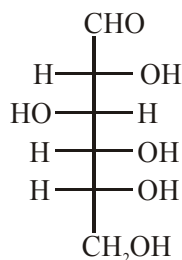
Q.15 The compound **T** is :

[JEE 2016]

- (A) Glycine
- (B) Alanine
- (C) Valine
- (D) Serine

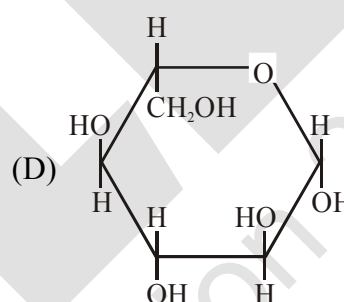
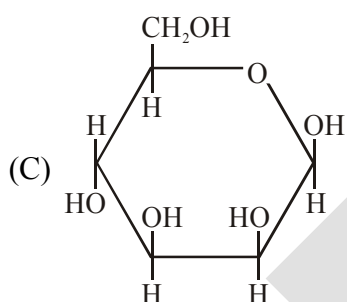
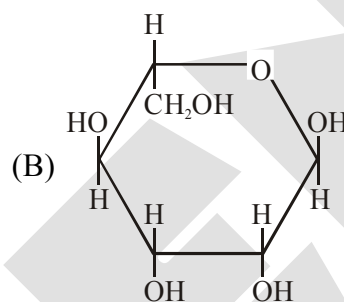
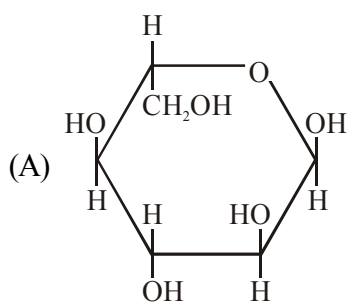
16. The Fischer presentation of D-glucose is given below.

[JEE 2018]



D-glucose

The correct structure(s) of β -L-glucopyranose is (are) :-



17. Which of the following statement(s) is(are) true ?

[JEE 2019]

- (A) Oxidation of glucose with bromine water gives glutamic acid
- (B) The two six-membered cyclic hemiacetal forms of D-(+)-glucose are called anomers
- (C) Hydrolysis of sucrose gives dextrorotatory glucose and laevorotatory fructose
- (D) Monosaccharides **cannot** be hydrolysed to give polyhydroxy aldehydes and ketones

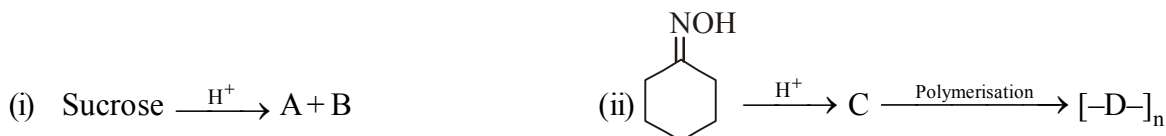
18. Choose the correct option(s) from the following

[JEE 2019]

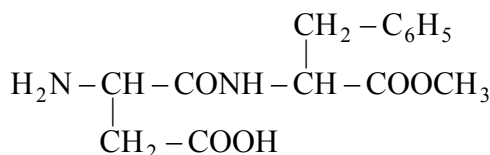
- (A) Natural rubber is polyisoprene containing *trans* alkene units
- (B) Nylon-6 has amide linkages
- (C) Cellulose has only α -D-glucose units that are joined by glycosidic linkages
- (D) Teflon prepared by heating tetrafluoroethene in presence of a persulphate catalyst at high pressure

EXERCISE # V (B) (J-ADVANCE) (SUBJECTIVE)

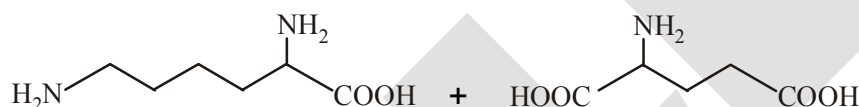
Q.1 Give the structures of the products in each of the following reactions. [JEE 2000]



Q.2 Aspartame, an artificial sweetener, is a peptide and has the following structure: [JEE 2001]

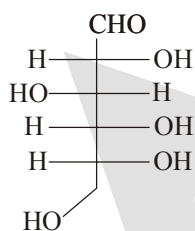


- Identify the four functional groups
 - Write the zwitterionic structure
 - Write the structures of the amino acids obtained from the hydrolysis of aspartame
 - Which of the two amino acids is more hydrophobic?
- Q.3 Following two amino acids lysine and glutamine form dipeptide linkage. What are two possible dipeptides? [JEE 2003]



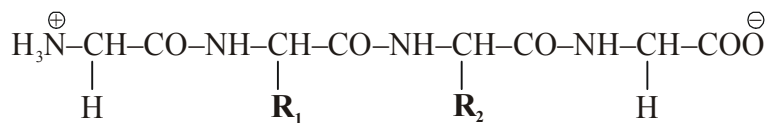
Q.4 The structure of D-Glucose is as follows- [JEE 2004]

- Draw the structure of L -Glucose
- Give the reaction of L - Glucose with Tollens reagent.



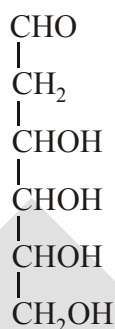
- Q.5 Monomer A of a polymer on ozonolysis yields two moles of HCHO and one mole of CH_3COCHO . [JEE 2005]
- Deduce the structure of A.
 - Write the structure of “all cis” – form of polymer of compound A.
- Q.6 A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is [JEE 2011]

- Q.7 The substitutes R_1 and R_2 for nine peptides are listed in the table given below. How many of these peptides are positively charged at pH = 7.0 ? [JEE 2012]

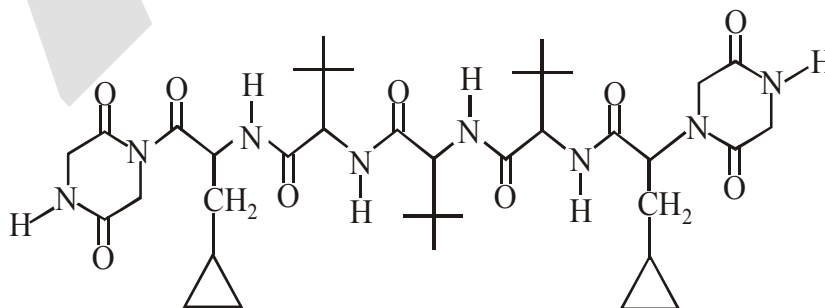


Peptide	R_1	R_2
I	H	H
II	H	CH_3
III	CH_2COOH	H
IV	CH_2CONH_2	$(\text{CH}_2)_4\text{NH}_2$
V	CH_2CONH_2	CH_2CONH_2
VI	$(\text{CH}_2)_4\text{NH}_2$	$(\text{CH}_2)_4\text{NH}_2$
VII	CH_2COOH	CH_2CONH_2
VIII CH_2OH	$(\text{CH}_2)_4\text{NH}_2$	
IX	$(\text{CH}_2)_4\text{NH}_2$	CH_3

- Q.8 When the following aldohexose exists in its d-configuration, the total number of stereoisomers in its pyranose form is - [JEE 2012]



- Q.9. A tetrapeptide has $-\text{COOH}$ group on alanine. This produces glycine (Gly), valine (Val), phenyl alanine (Phe) and alanine (Ala), on complete hydrolysis. For this tetrapeptide, the number of possible sequences (Primary structures) with $-\text{NH}_2$ group attached to a chiral center is [JEE 2013]
- Q.10. The total number of lone-pairs of electrons in melamine is : [JEE 2013]
- Q.11 The total number of distinct naturally occurring amino acids obtained by complete acidic hydrolysis of the peptide shown below is : [JEE 2014]



ANSWER-KEY**EXERCISE # I**

Q.1 Ans. (C)	Q.2 Ans. (D)	Q.3 Ans. (C)	Q.4 Ans. (B)	Q.5 Ans. (C)
Q.6 Ans. (A)	Q.7 Ans. (C)	Q.8 Ans. (D)	Q.9 Ans. (C)	Q.10 Ans. (B)
Q.11 Ans. (B)	Q.12 Ans. (C)	Q.13 Ans. (C)	Q.14 Ans. (A)	Q.15 Ans. (A)
Q.16 Ans. (D)	Q.17 Ans. (D)	Q.18 Ans. (D)	Q.19 Ans. (B)	Q.20 Ans. (B)
Q.21 Ans. (D)	Q.22 Ans. (B)	Q.23 Ans. (A)	Q.24 Ans. (A)	Q.25 Ans. (C)
Q.26 Ans. (A)	Q.27 Ans. (D)	Q.28 Ans. (B)	Q.29 Ans. (B)	Q.30 Ans. (C)
Q.31 Ans. (C)	Q.32 Ans. (C)	Q.33 Ans. (D)	Q.34 Ans. (C)	Q.35 Ans. (A)
Q.36 Ans. (D)	Q.37 Ans. (B)	Q.38 Ans. (C)	Q.39 Ans. (A)	Q.40 Ans. (A)
Q.41 Ans. (B)	Q.42 Ans. (B)			

EXERCISE # II

Q.1 Ans. (A,B,C,D)	Q.2 Ans. (A,B,D)	Q.3 Ans. (A,B,C)	Q.4 Ans. (A,B,D)	Q.5 Ans. (A,B,D)
Q.6 Ans. (A,B,C)	Q.7 Ans. (B,C)	Q.8 Ans. (A,B,C,D)	Q.9 Ans. (A,B,C)	Q.10 Ans. (B,D)
Q.11 Ans. (D)	Q.12 Ans. (A)	Q.13 Ans. (D)	Q.14 Ans. (B)	Q.15 Ans. (C)
Q.16 Ans. (A) → P, Q, S ; (B) → S ; (C) → R, S ; (D) → Q, R, S				
Q.17 Ans. (A) → R ; (B) → P ; (C) → S ; (D) → Q				

EXERCISE # III**Assertion Reason**

Q.1 Ans. (A)	Q.2 Ans. (A)	Q.3 Ans. (A)	Q.4 Ans. (A)	Q.5 Ans. (D)
Q.6 Ans. (B)	Q.7 Ans. (C)	Q.8 Ans. (A)	Q.9 Ans. (C)	

Match the Column

Q.10 Ans. (A) → S ; (B) → Q ; (C) → P ; (D) → R				
Q.11 Ans. (A) → Q, S ; (B) → Q, R, S ; (C) → P, R, T, U (D) → P, S, T, U				
Q.12 Ans. (A) → P, Q, R ; (B) → P ; (C) → P, S ; (D) → P, Q, R				
Q.13 Ans. (A) → Q, R (B) → P, (C) → S ; (D) → T				
Q.14 Ans. (A) → P, Q, R ; (B) → S ; (C) → R ; (D) → P, Q, S				
Q.15 Ans. (A) → S-W (B) → P-Y, R-X (C) → Q-Z				

Subjective Type :

Q.16 Ans. (3.25)

EXERCISE # IV (J-MAIN)

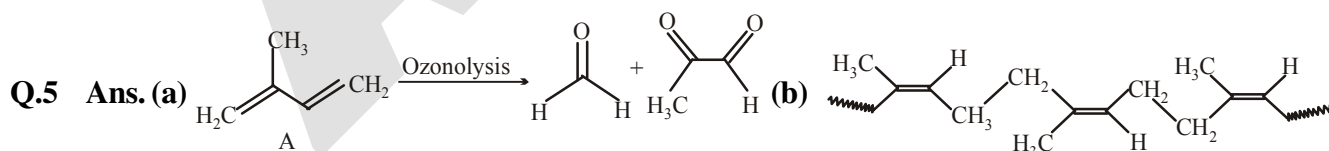
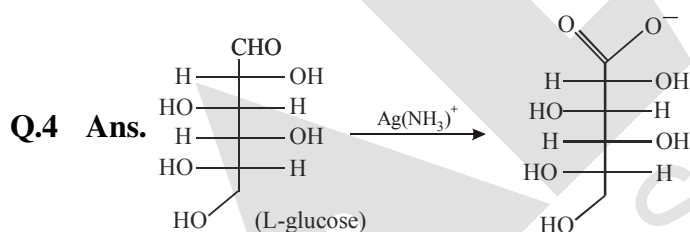
Q.1 Ans. (2)	Q.2 Ans. (2)	Q.3 Ans. (4)	Q.4 Ans.(3)	Q.5 Ans.(3)
Q.6 Ans. (3)	Q.7 Ans. (3)	Q.8 Ans. (1)	Q.9 Ans.(4)	Q.10 Ans.(4)
Q.11 Ans. (1)	Q.12 Ans. (1)	Q.13 Ans. (3)	Q.14 Ans.(1)	Q.15 Ans.(1)
Q.16. Ans. (4)	Q.17. Ans. (2)	Q.18. Ans. (1)	Q.19. Ans. (4)	Q.20. Ans.(1)
Q.21. Ans. (4)	Q.22. Ans. (2)	Q.23. Ans. (4)	Q. 24. Ans. (4)	Q.25. Ans.(2)
Q.26. Ans.(3)	Q.27. Ans. (1)	Q.28. Ans. (3)	Q. 29. Ans. (2)	Q. 30. Ans.(1)
Q.31. Ans.(4)	Q. 32. Ans.(2)	Q. 33. Ans.(4)	Q. 34. Ans. (2)	Q. 35. Ans.(2)
Q.36. Ans. (1)	Q. 37. Ans.(1)	Q. 38. Ans. (1)	Q. 39. Ans. (1)	Q. 40. Ans.(1)
Q.41. Ans. (2)	Q. 42. Ans. (4)	Q. 43. Ans. (3)		

EXERCISE # V (A) (J-ADVANCE) (OBJECTIVE)

Q.1 Asn. (D)	Q.2 Ans. (A) → P, S ; (B) → Q, R ; (C) → P, R ; (D) → S	
Q.3 Ans. (A) → P, S ; (B) → Q ; (C) → Q, R, S ; (D) → Q, R		Q.4 Ans. (A)
Q.5 Ans. (A) → R, S ; (B) → P, Q ; (C) → P, Q, R ; (D) → P, S		Q.6 Ans. (B,C)
Q.7 Ans. (D)	Q.8 Ans. (B)	Q.9 Ans.(B)
	Q.10 Ans. (C)	Q.11 Ans. (A,B,C,D)
Q.12 Ans. (A)	Q.13 Ans. (B,C)	Q.14 Ans. (A)
	Q.15 Ans. (B)	Q.16 Ans. (D)
Q.17. Ans. (B,C,D)	Q. 18 Ans. (B,D)	

EXERCISE # V (B) (J-ADVANCE) (SUBJECTIVE)

Q.2 Ans. (i) Amine, carboxylic acid, Amide, Ester



Q.6 Ans. (6)	Q.7 Ans. (4)	Q.8 Ans. (8)	Q.9. Asn. (4)	Q.10. Ans. (6)
Q.11 Ans. (1)				

POLYMERS AND PRACTICAL ORGANIC CHEMISTRY NOTES

- The term polymer is used to describe a very large molecule that is made up of many small repeating molecular units. These small molecular units from which the polymer is formed are called monomers.
- The chemical reaction that joins the monomers together is called polymerisation.
- Starting from n molecules of a compound M , linking in a linear manner will form polymer $x-M-(M)_{n-2}-M-y$. The nature of linkages at the terminal units i.e. $M-x$ and $M-y$ depends upon the mode of reaction used in making the polymers.

Homopolymers and Copolymers

Polymers which are formed by only one type of monomer are called homopolymers. Some examples of homopolymers and their monomers are given below :

Homopolymer	Monomer
Starch	Glucose
Cellulose	Glucose
Glycogen	Glucose
Dextrin	Glucose
Inulin	Fructose
Polyethylene	Ethylene
Polyvinyl chloride	Vinyl chloride
Teflon	Tetrafluoro ethylene
Nylon-6	Caprolactam
Polystyrene	Styrene
Orlon (Acrilan)	Acrylonitrile
Plexiglas (Lucite)	Methyl methacrylate
Polyvinyl acetate	Vinyl acetate

Polymers, which are formed by more than one type of monomers are known as copolymers. Some examples are given below in the table :

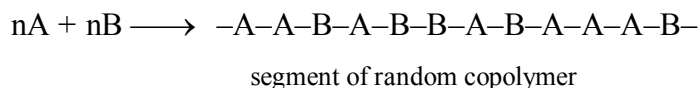
Copolymer	Monomers
Saran	Vinyl chloride and vinylidene chloride
SAN	Styrene and acrylonitrile
ABS	Acrylonitrile, butadiene and styrene
Butyl rubber	Isobutylene and Isoprene
Buna-S, SBR	Styrene and Butadiene
Buna-N, NBR	Acrylonitrile and Butadiene
Nylon-66	Hexamethylenediamine and Adipic acid
Terylene	Terephthalic acid and ethylene glycol

Types of copolymers

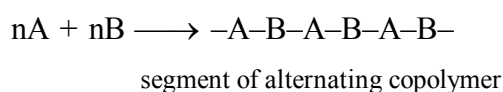
Depending upon the distribution of monomer units, the following types of copolymers are possible.

(1) Random Copolymer

If the monomer units have random distribution throughout the chain, it is called random copolymer. For example, if the monomer A and monomer B undergo copolymerisation then the structure of the random copolymer is

**(2) Alternating Copolymer**

If the two monomer units present alternatively throughout the polymer chain, it is said to be alternating copolymer. For example,

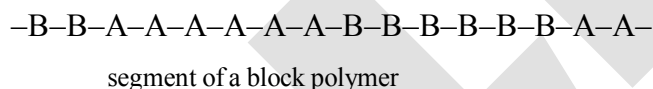


The exact distribution depends upon the proportion of the two reactant monomers and their relative reactivities. In practice neither perfectly random nor perfectly alternating copolymers are usually formed. However, most copolymers tend more towards alternating type but have many random imperfections.

(3) Block copolymer

Polymers in which different blocks of identical monomer units alternate with each other are called block copolymers.

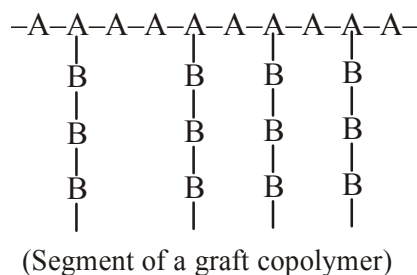
For example,



Block copolymer can be prepared by initiating the radical polymerisation of one monomer to grow homopolymer chains, followed by addition of an excess of the second monomer.

(4) Graft copolymer

Polymers in which homopolymer branches of one monomer unit are grafted onto a homopolymer chain of another monomer unit are called graft co-polymers. For example:



Graft copolymers are prepared by γ -irradiation of a homopolymer chain in the presence of a second monomer. the high energy radiation knock out H-atoms from the homopolymer chain at random points thus generating radical sites that can initiate polymerisation of the second monomer.

CLASSIFICATION OF POLYMERS

Polymers are classified in following ways :

(I) CLASSIFICATION BASED UPON SOURCE**(1) Natural polymers**

Polymers which are obtained from animals and plants are known as natural polymers. Examples of natural polymers are given below.

Natural polymer

1. Polysaccharide
2. Proteins
3. Nucleic acid
4. Silk
5. Natural Rubber (cis polyisoprene)
6. Gutta purcha (trans polyisoprene)

Monomers

- Monosaccharide
 α -L-Amino acids
Nucleotide
Amino acids
Isoprene (2-Methyl-1,3-butadiene)
Isoprene

Natural polymers which take part in metabolic processes are known as biopolymers. Examples are polysaccharides, proteins, RNA and DNA.

(2) Semisynthetic polymers

Polymers which are prepared from natural polymers are known as semisynthetic polymers. Most of the semisynthetic polymers are prepared from cellulose. Examples are: cellulose acetate, cellulose nitrate, cellulose xanthate and Rayon.

(3) Synthetic polymers

Man-made polymers, i.e. polymers prepared in laboratory are known as synthetic polymers. Example are : PVC, polyethylene, polystyrene, nylon-6, nylon-66, nylon-610, terylene, synthetic rubbers etc.

(II) CLASSIFICATION BASED UPON SHAPE**(1) Linear polymers**

Polymer whose structure is linear is known as linear polymer. The various linear polymeric chains are stacked over one another to give a well packed structure.



The chains are highly ordered with respect to one another. The structure is close packed in nature, due to which they have high densities, high melting point and high tensile (pulling) strength. Linear polymers can be converted into fibres.

Note :

- (i) All fibers are linear polymers. Examples are cellulose, silk, nylon, terylene etc.
- (ii) Linear polymers may be condensation as well as addition polymers. Examples are cellulose, polypeptide, nucleic acid, nylon, terylene etc.

(2) Branched chain polymers

Branched chain polymers are those in which the monomeric units constitute a branched chain. Due to the presence of branches, these polymers do not pack well. As a result branched chain polymers have lower melting points, low densities and tensile strength as compared to linear polymers. Branched chain polymers may be formed due to addition as well as condensation polymerisation. Examples are amylopectin, glycogen, low density polyethylene.

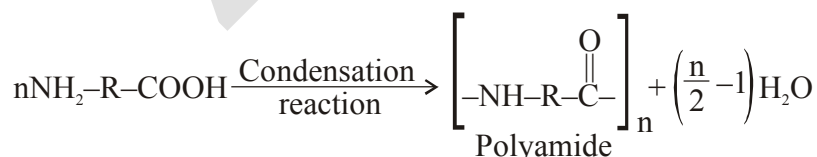
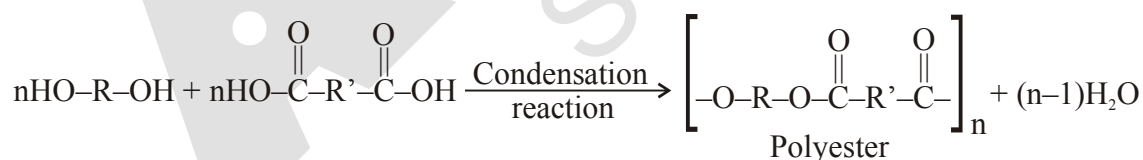
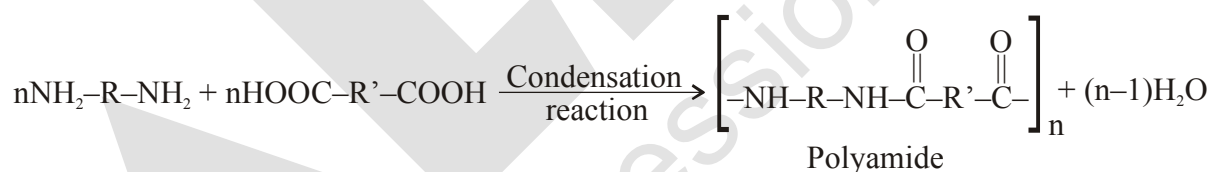
**(3) Cross-linked or Three Dimensional network polymers**

In these polymers the initially formed linear polymeric chains are joined together to form a three dimensional network structure. These polymers are hard, rigid and brittle. Cross-linked polymers are always condensation polymers. Resins are cross linked polymers, Urea-formaldehyde resin, phenol-formaldehyde resin.

**CLASSIFICATION BASED UPON SYNTHESIS****(1) Condensation polymerisation**

They are formed due to condensation reactions. Condensation polymerisation is also known as step growth polymerisation. For condensation polymerisation, monomers should have at least two functional groups. Both functional groups may be same or different. Monomers having only two functional group always give linear polymer.

For example,

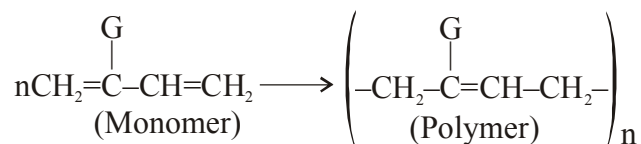


Condensation polymers do not contain all atoms initially present in the monomers. Some atoms are lost in the form of small molecules. Monomer having three functional groups always gives cross-linked polymer.

Examples are : Urea-formaldehyde resin, phenol-formaldehyde resin.

(3) Addition polymerisation

Polymers which are formed by addition reaction are known as addition polymers. If monomer is ethylene or its derivative, then addition polymer is either linear polymer or branch-chain polymer. Examples are : polystyrene, polytetrafluoroethylene, polyacrylonitrile etc. If monomer is 1, 3-butadiene or 2-substituted-1, 3-butadiene $\left(\begin{array}{c} \text{CH}_2 = \text{C} - \text{CH} = \text{CH}_2 \\ | \\ \text{G} \end{array} \right)$, then polymer is always branched chain polymer.



(Monomer)

- | | |
|---|----------------------------------|
| (i) $G = H$: 1,3 - Butadiene | (i) Polybutadiene |
| (ii) $G = CH_3$; 2 - Methyl-1,3-butadiene or isoprene | (ii) Polyisoprene |
| (iii) $G = Cl$; 2 -Chloro-1,3-butadiene or chloroprene | (iii) Polychloroprene (Neoprene) |

(Polymer)

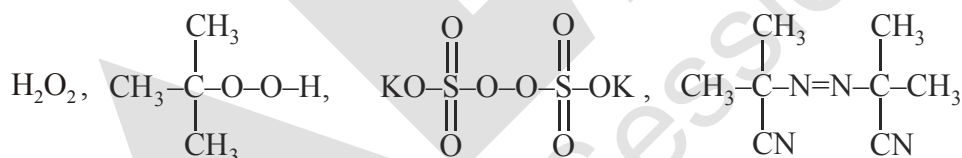
- (i) Polybutadiene
- (ii) Polyisoprene
- (iii) Polychloroprene (Neoprene)

Addition polymers retain all the atoms of the monomer units in the polymer. Addition polymerisation takes place in three steps. Initiation, chain propagation and chain termination. Addition polymers are called as chain growth polymers.

Types of Addition Polymerisation

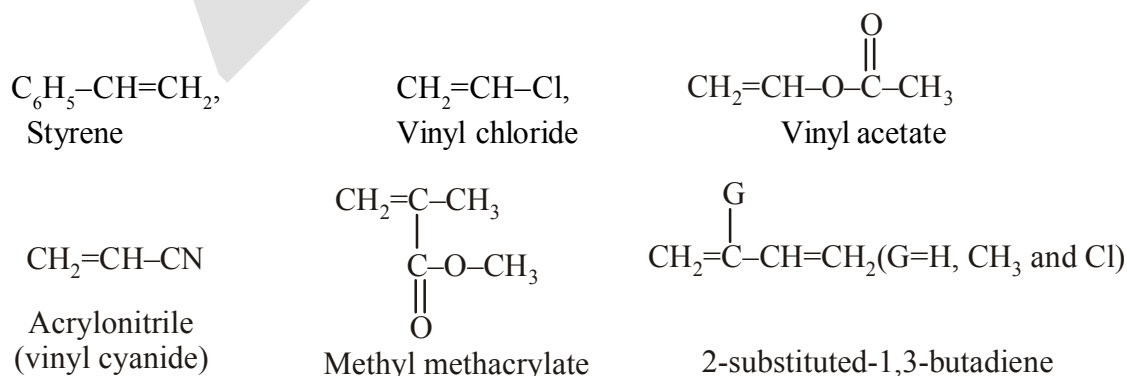
(A) Radical Polymerisation :

Radical polymerisation takes place in the presence of radical initiators. The radical initiator may be any of the following :



Reaction intermediate of radical polymerization is a free radical. Radical polymerization has more chance for those monomers whose free radicals are more stable.

Examples are :

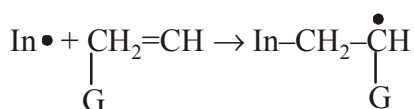
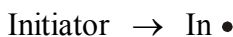


Radical polymer has linear as well as branched chain structure.

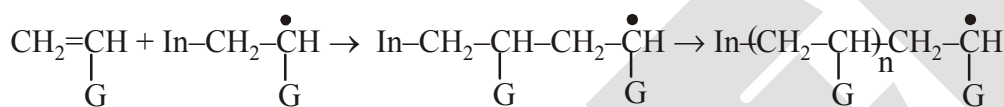
Most of the commercial addition polymers are vinyl polymers obtained from alkenes and their derivatives $\text{CH}_2=\underset{\text{G}}{\text{CH}}$ [G is H, C_6H_5 , R etc.]. This type of polymerisation is performed by heating the

monomer with only a very small amount of the initiator or by exposing the monomer to light. the general mode of radical polymerisation of vinyl monomers is depicted below :

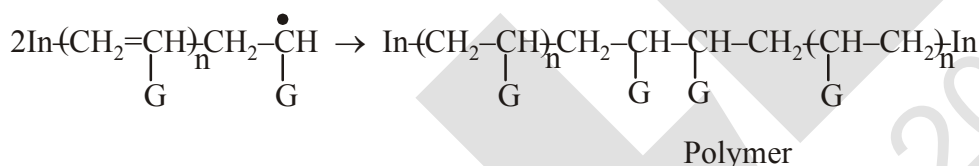
Chain initiation step :



Chain propagating step :

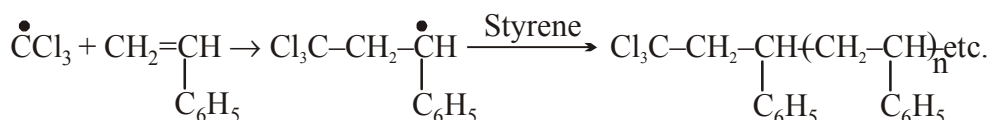
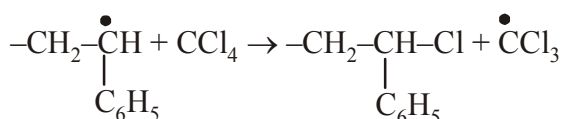


Chain terminating step :



In vinylic polymerisation, various other reaction of free radicals with some other compounds present may compete with the parent addition chain reactions. One such reaction takes place with molecules that can react with the growing chain to interrupt the further growth. This leads to the lowering of the average molecular mass of the polymer. Such reagents are called as chain transfer agents and include CCl_4 , CBr_4 etc.

For example, in the presence of CCl_4 , styrene polymerises to form polystyrene of a lower average molecular mass which also contains some chlorine what happens here is that growing polystyrene radical which normally would add on a monomer reacts with the chain transfer agent to end the original chain and produces a new radical. The latter initiates a new polymerisation chain and thereby forms a new polymer as depicted below.

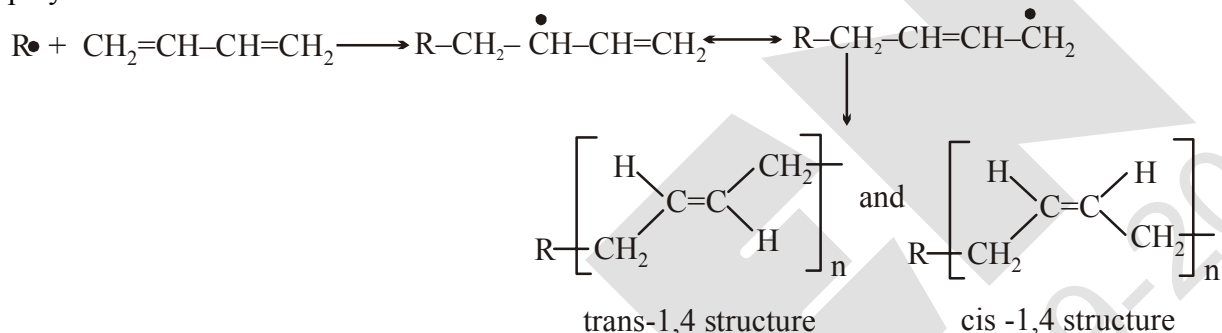


If the chain transfer agent a radical, which is highly unreactive, the reaction chain gets terminated such a compound thus inhibits or arrests polymerisation. Many amines, phenols, quinones etc. act as inhibitors. So, even traces of certain impurities, which can act as chain transfer agent or an inhibitor can interfere with the original polymerisation chain reaction. Hence, the monomers could be free from such inhibitors.

In case the alkene is a (diene), the following kinds of polymerisation is possible :

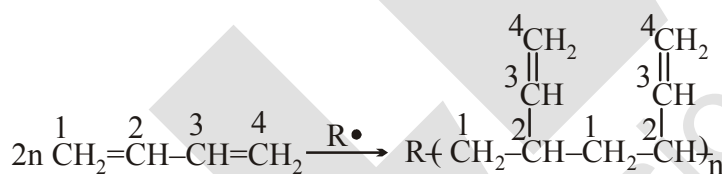
(1) **1,4 - polymerisation.**

When the polymerisation takes place at C_1 and C_4 of butadiene, an unbranched polymer is formed. This product is different from that formed from an alkene having a double bond, which at each of its carbons is substituted by different groups and hence can exist either as trans-polybutadiene or cis-polybutadiene or a mixture as shown below.



(2) **1,2-Polymerisation**

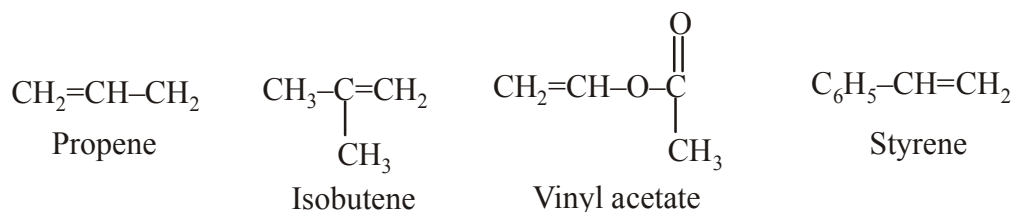
Alternatively, 1,3-butadiene can undergo polymerisation at C_1 and C_2 to yield the polymeric product, polyvinyl polythene.



The double bonds in these initial polymers can be linked by further treatment with chemicals to modify the properties of the polymers. These reactions form the basis of the formation of rubber.

(B) **Cationic Polymerisation :**

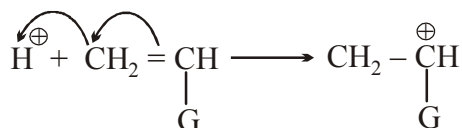
Polymerisation which is initiated by an electrophile is known as cationic polymerisation. Reaction intermediate of cationic polymerisation is a carbocation. Carbocations can undergo rearrangement leading to the formation of a more stable carbocation. The electrophile commonly used for initiation is $BF_3 \cdot OEt_2$. Monomers that are best able to undergo polymerisation by a cationic mechanism are those with electron - donating substituents that can stabilise the carbocation. Some examples are:



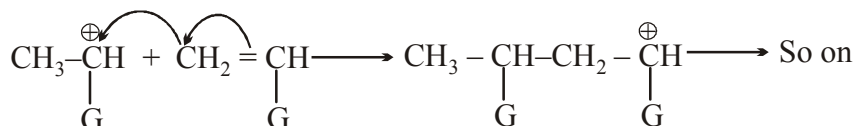
It is terminated by a base.

Thus, when the initiator is cationic in nature, it would generate a cationic intermediate on addition to the double bond for propagating the addition chain process and is termed as cationic addition polymerisation. The process is initiated by an acid. The stages of polymerisation are depicted below.

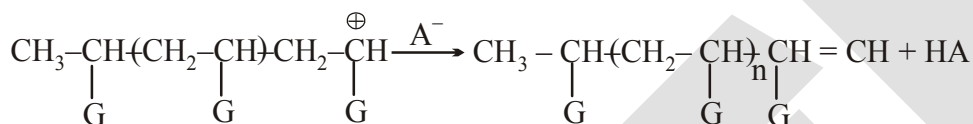
Chain initiation step :



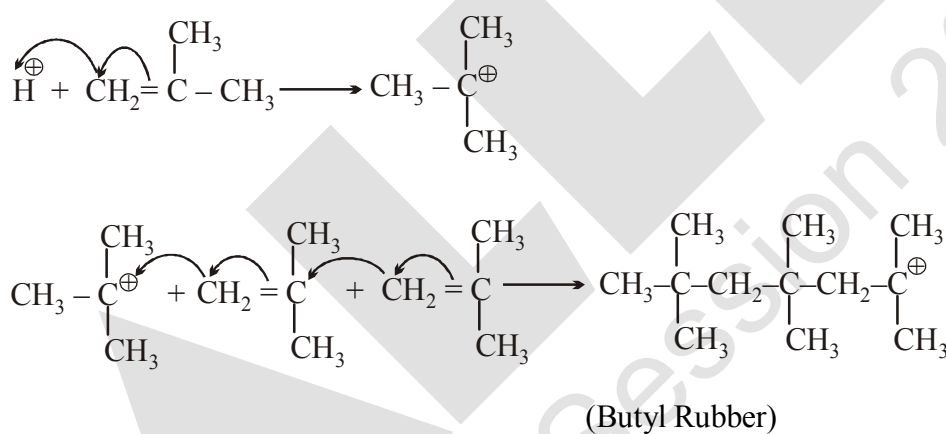
Chain propagating step :



Chain terminating step :



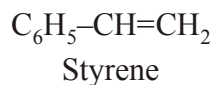
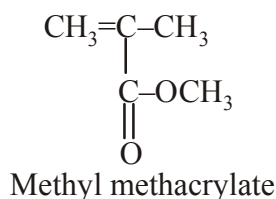
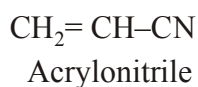
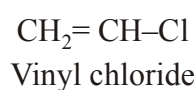
Cationic polymerisation is facilitated in monomers containing electron - releasing groups. Thus, isobutylene undergoes cationic polymerisation easily as it has two electron releasing $-\text{CH}_3$ groups that will stabilize the intermediate carbocation.



(C) Anionic Polymerisation :

Anionic polymerisation takes place in the presence of base or nucleophile, which is initiator in this polymerization. Reaction intermediate in propagation steps are carboanion. The suitable initiator can be NaNH_2 or RLi . Those monomers undergo anionic polymerisation reaction whose anion is stable.

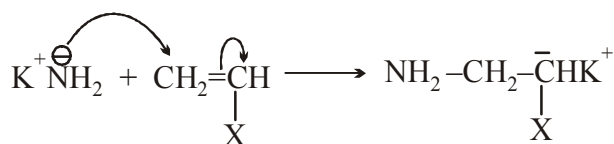
Example of monomers are :



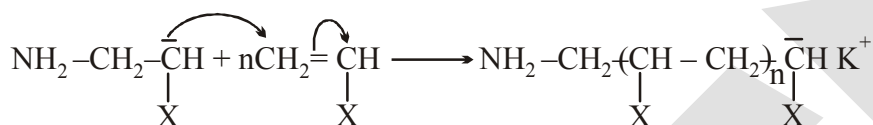
Anionic polymerisation always gives linear polymer. Anionic polymerisation terminated by an acid.

The formation of polystyrene from styrene in the presence of potassium amide is an important example of this category of polymerisation. The mode of anionic polymerisation is depicted below:

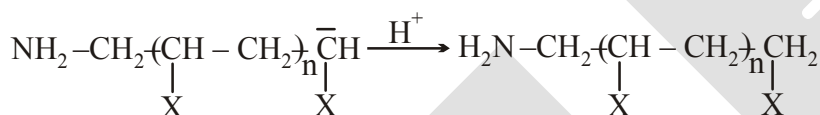
Chain initiation step :



Chain propagating step :



Chain terminating step :



(D) Ziegler- Natta polymerisation :

Addition polymerisation which takes place in the presence of Ziegler- Natta catalyst $[(\text{C}_2\text{H}_5)_3\text{Al}$ and $\text{TiCl}_4]$ is known as Ziegler- Natta polymerisation or coordination polymerisation. Ziegler- Natta polymerisation always gives linear, stereo-regular polymers. Ziegler- Natta catalyst revolutionised the field of polymer chemistry because they allow the synthesis of stronger and stiffer polymers (due to linear geometry) that have greater resistance to cracking and heat.

High density polyethylene is prepared using a Ziegler- Natta catalyst.

CLASSIFICATION BASED ON INTERMOLECULAR FORCES

(SECONDARY FORCES)

Intermolecular forces present between polymeric chains are (a) Van der waals forces (b) Hydrogen bonds and (c) Dipole - dipole attractions. Mechanical properties such as tensile strength, elasticity, toughness etc. depend upon the secondary forces present between the polymeric chains. Magnitude of secondary forces depends upon the size of the molecule and the number of functional groups along the polymeric chains. Magnitude of secondary forces is directly proportional to the length of the polymeric chain. On the basis of magnitude of secondary forces, polymers can be divided into the following categories.

(1) Elastomes

An elastomer is a plastic that stretches and then reverts back to its original shape. It is randomly oriented amorphous polymer. It must have some cross-links so that the chains do not slip over one another. Very weak Vander Waal forces are present in between polymeric chains.

When elastomers are stretched, the random chains stretch out, but there are insufficient Vander Waal forces to maintain them in that configuration and position. When the stretching force is removed, they go back to their random shape. Elastomers have the ability to stretch out over ten times their normal length. Important examples are vulcanized rubbers.

Note : Addition polymers obtained from butadiene and its derivatives are elastomers.

(2) Fibres

Fibres are linear polymers in which the individual chains of a polymer are held together by hydrogen bonds or dipole-dipole attraction. In the fibres, the polymeric chains are highly ordered with respect to one another. Due to strong intermolecular forces of attraction and highly ordered geometry, fibres have high tensile strength and least elasticity. They have crystalline character and have high melting points and low solubility. Examples are cellulose, nylon, terylene, wool, silk etc.

Note :

- (i) Condensation polymers formed from bifunctional monomers are fibres in character.
- (ii) Addition polymers of alkene derivatives having strong -I group are fibres in character.

(3) Thermoplastic Polymers

Thermoplastic polymers are polymers that have both ordered crystalline regions (the regions of the polymer in which the chains are highly ordered with respect to one another) and amorphous, non-crystalline regions (the regions of the polymer in which the chains are randomly oriented). The intermolecular forces of attraction are in between elastomers and fibres. There are no cross links between the polymeric chains. Thermoplastic polymers are hard at room temperature, but when they are heated, the individual chains can slip past one another and the polymer becomes soft and viscous. This soft and viscous material becomes rigid on cooling. The process of heating softening and cooling can be repeated as many times as desired without any change in chemical composition and mechanical properties of the plastic. As a result, these plastics can be moulded into toys, buckets, telephone and television cases. Some common examples are : polyethene, polypropylene, polystyrene, polyvinyl chloride, teflon etc.

Note : Addition polymers obtained from ethylene and ethylene derivatives are thermoplastic polymers.

(4) Thermosetting Polymers

Polymers which become hard on heating are called thermosetting polymers. Thermosetting polymers can be heated only once when they permanently set into a solid, which cannot be remelted by heating. Thermosetting polymers are cross-linked polymers. Greater the degree of cross-linking that exists, the more rigid is the polymer. Cross-linking reduces the mobility of the polymer chains, causing them to be relatively brittle materials. The hardening on heating is due to the extensive cross-linking between different polymer chains to give a three dimensional network solid. Examples are : phenol formaldehyde resin, urea-formaldehyde resin, melamine-formaldehyde resin.

DIFFERENCE BETWEEN THERMOPLASTIC AND THERMOSETTING POLYMERS

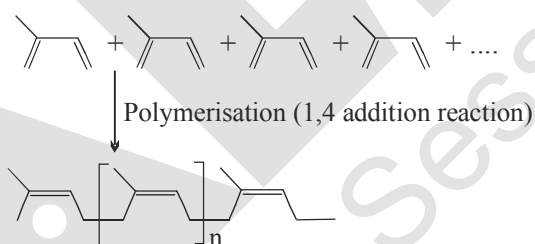
S.No.	Thermoplastic polymers	Thermosetting polymers
1.	Soften and melt on heating and become hard on cooling i.e. process is reversible	Become hard on heating and process is irreversible.
2.	Can be moulded and remoulded and reshaped.	They can be moulded once and cannot be remoulded or reshaped.
3.	They are addition polymers	They are condensation polymers.
4.	Structure is generally linear	Structure is cross - linked.

RUBBER

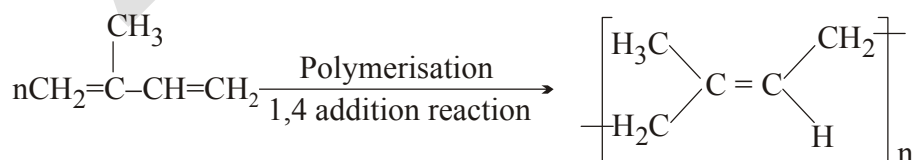
1. Natural Rubber

Natural rubber is obtained from nearly five hundred different plants but the main source is a *braziliensis* tree. It is obtained in the form of milky sap known as latex. This latex is coagulated with acetic acid and formic acid. The coagulated mass is then squeezed.

The raw natural rubber is a soft gummy and sticky mass. It is insoluble in water, dilute acids and alkalis but soluble in non-polar solvents. It has low elasticity and low tensile strength. Natural rubber is a polymer of 2-methyl-1,3-butadiene (isoprene). On average, a molecule of rubber contains 5000 isoprene units held together by head to tail. All the double bonds in rubber are *cis*, hence natural rubber is *cis*-polyisoprene.

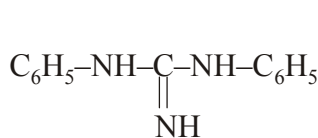


Gutta - percha is a naturally occurring isomer of rubber in which all the double bonds are *trans*. Thus, gutta-percha is *trans*-polyisoprene.

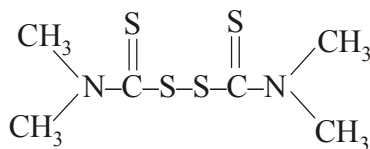


It is harder and more brittle than rubber. It is the filling material that dentists use in root canal treatment.

In order to give strength and elasticity to natural rubber, it is vulcanized. Heating of rubber with sulphur or sulphur containing compound at 150°C for few hours is known as vulcanisation. The essential feature of the vulcanisation is the formation of cross-linking between the polymeric chains. This cross-linking gives mechanical strength to the rubber. Vulcanisation process can be enhanced in the presence of certain organic compounds known as accelerators. The common accelerators are:



Diphenyl guanidine



Tetramethylthioureadisulph

In addition, fillers such as carbon black and zinc oxide are usually added to the crude rubber before vulcanisation in order to improve its wearing characteristics.

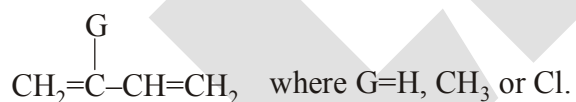
Natural rubber is used for making shoes, water - proof coats and golf balls. Vulcanised rubber is used for manufacture of rubber bands, gloves tubing and car tyres.

SYNTHETIC RUBBER OR POLYMERISATION OF DIENES

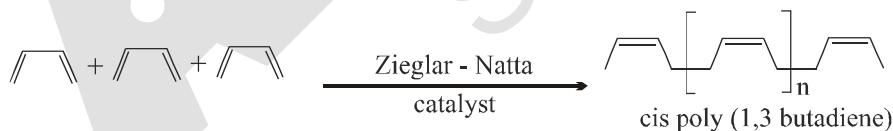
Polymers of 1,3 - butadienes are called synthetic rubbers because they have some of the properties of natural rubbers including the fact that they are water proof and elastic. Synthetic rubbers have some improved properties. They are more flexible, tougher and more durable than natural rubber.

1. Homopolymers

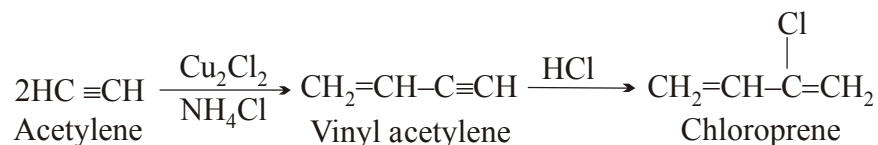
Monomer of this class is 2 - substituted - 1,3- butadienes.



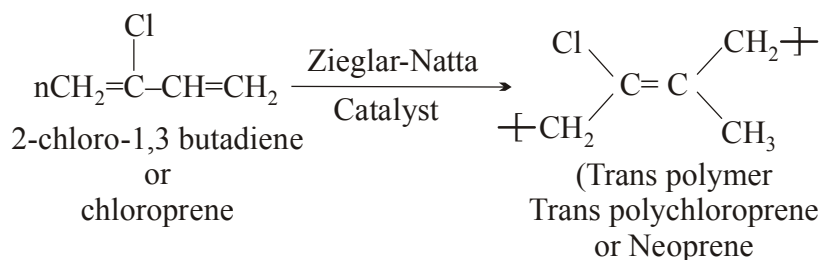
Polymerisation is always carried out in the presence of Ziegler-Natta catalyst which gives stereo regular polymers.



Neoprene was the first synthetic rubber manufactured on large scale. It is also called dieprene. Its monomer, chloroprene(2-chlorobutadiene) is prepared from acetylene.



Chloroprene undergoes free radical polymerisation to form neoprene (polychloroprene).



Many of the properties of neoprene are similar to natural rubber but neoprene is more resistant to action of oils, gasoline and other hydrocarbons. It is also resistant to sunlight, oxygen, ozone and heat. It is non - inflammable. It is used for making automobile and refrigerator parts, hoses for petrol and oil containers, insulation of electric wires and conveyor belts.

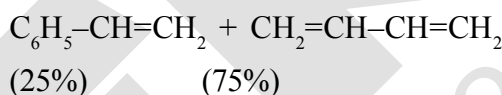
2. Copolymers

The following synthetic rubbers are example of copolymers.

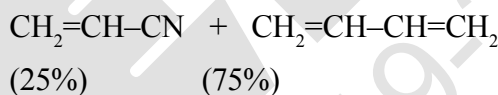
Synthetic rubber

Monomers

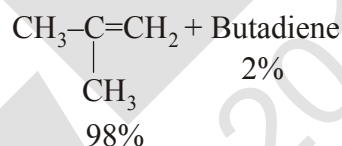
1. Buna-S, SBR (Styrene-butadiene rubber)



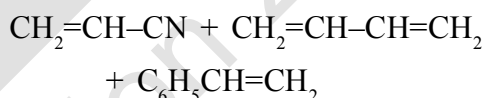
2. Buna-N, NBR (Nitrile-butadiene rubber)



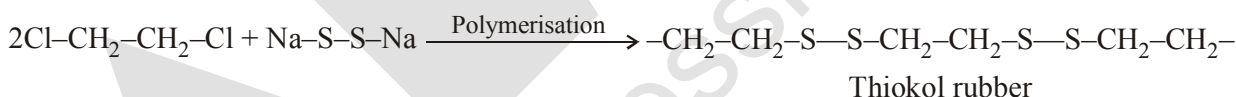
3. Butyl rubber



4. ABS; Acrylonitrile, Butadiene, Styrene

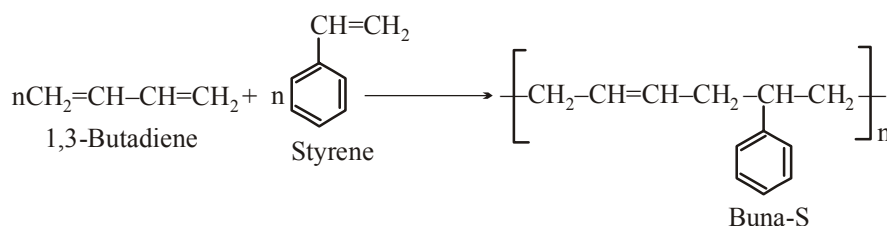


(a) **Thiokol** : Thiokol is made by polymerising ethylene chloride and sodium polysulphide.



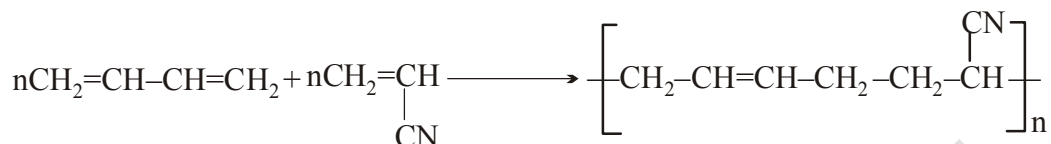
The repeating unit is $-\text{CH}_2-\text{S}-\text{S}-\text{CH}_2-$. Thiokol is chemically resistant polymer. It is used in the manufacture of hoses and tank linings, engine gaskets and rocket fuel when mixed with oxidising agents.

(b) **Buna -S (SBR : Styrene-butadiene rubber)** : Buna-S rubber is a copolymer of three moles of butadiene and one mole of styrene. In Buna-S, 'Bu' stands for butadiene, 'na' for symbol of sodium (Na) which is a polymerizing agent and 'S' stands for styrene. It is an elastomer (General purpose styrene Rubber or GRS).



Buna-S is generally compounded with carbon black and vulcanised with sulphur. It is extremely resistant to wear and tear and therefore used in the manufacture of tyres and other mechanical rubber goods. It is obtained as a result of free radical copolymerisation of its monomers.

- (c) **Buna-N** : It is obtained by copolymerisation of butadiene and acrylonitrile (General Purpose Rubber acrylonitrile or GRA).



It is very rigid and is very resistant to action of petrol, lubricating oil and many organic solvents. It is mainly used for making fuel tanks.

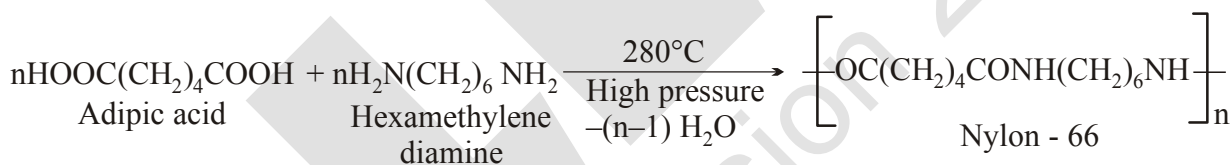
- (d) **Cold Rubber** : Cold rubber is obtained by polymerisation of butadiene and styrene at -18° to 5°C temperature in the presence of redox system. Cold rubber has a greater tensile strength and greater resistance to abrasion than SBR.

NYLON

Nylon is used as a general name for all synthetic fibres forming polyamides, i.e., having a protein like structure. A number is usually suffixed with the name 'nylon' which refers to the number of carbon atoms present in the monomers.

- (1) **NYLON - 66** (Nylon six, six)

It is obtained by the condensation polymerisation of hexamethylenediamine (a diamine with six carbon atoms) and adipic acid (a dibasic acid having 6 carbon atoms).



- (2) **NYLON-610** (Nylon six, ten)

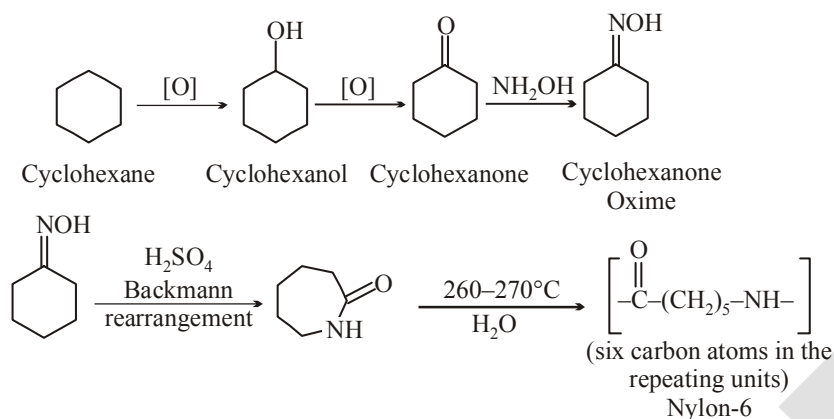
It is obtained by condensation polymerisation of hexamethylenediamine (six carbon atoms) and sebacic acid (a dibasic acid with 10 carbon atoms.)

Nylon fibres are stronger than natural fibres and so are used in making cords and ropes. The fibres are elastic, light, very strong and flexible. They have drip dry property and retain creases. It is inert towards chemicals and biological agents. It can be blended with wool. Nylon fibres are used in making garments, carpets, fabrics, tyre cords, ropes, etc.

- (3) **NYLON-6 (Perlon L)**

A polyamide closely related to nylon is known as perlon L (Germany) or Nylon - 6 (USA). It is prepared by prolonged heating of caprolactam at $260-270^\circ\text{C}$. It is formed by self condensation of a large number of molecules of amino caproic acid. Since, caprolactam is more easily available, it is used for polymerization, which is carried out in the presence of H_2O that first hydrolyses the lactam to amino acid. Subsequently, the amino acid can react with the lactam and the process goes on and on to form the polyamide polymer.

Carpolactam is obtained by Backmann rearrangement of cyclohexanone oxime.



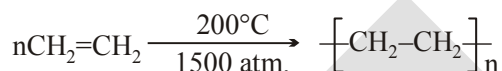
(4) NYON-2-NYLON-6

It is in alternating polyamide copolymer of glycine and amino caproic acid and is biodegradable.

POLYETHYLENE

Polyethylene is of two types :

- (a) **Low Density Poly Ethylene (LDPE)** : It is manufactured by heating ethylene at $200^\circ C$ under a pressure of 1500 atmospheres and in the presence of traces of oxygen. This polymerisation is a free radical polymerisation.



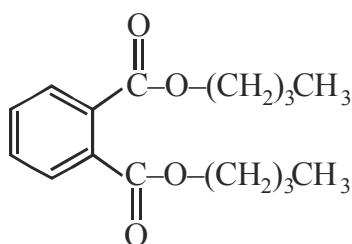
The polyethylene produced has a molecular mass of about 20,000 and has a branched structure. Due to this, polyethylene has a low density (0.92) and low melting point ($110^\circ C$). That is why polyethylene prepared by free radical polymerisation is called low density polyethylene. It is a transparent polymer of moderate tensile strength and high toughness. It is widely used as a packing material and as insulation for electrical wires and cables.

- (b) **High Density Poly Ethylene (HDPE)** : It is prepared by the use of Ziegler - Natta catalyst at $160^\circ C$ under pressure of 6 to 7 atmospheres.

The polymer is linear chain, hence it has high density (0.97) and has high melting point ($130^\circ C$). That is why it is called high density polyethylene. It is a translucent polymer. It has greater toughness, hardness and tensile strength than low density polyethylene. It is used in the manufacture of containers (buckets, tubes), house wares, bottles and toys.

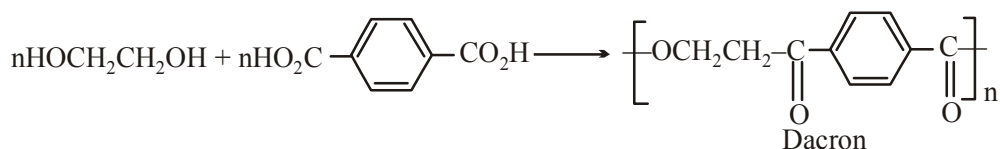
PLASTICISER

A plasticiser is an organic compound that dissolves in the polymer and allows the polymer chains to slide past one another. This makes polymer more flexible. Dibutylphthalate is a commonly used plasticiser.



POLYESTERS

Dacron is a common polyester, prepared using ethylene glycol and terephthalic acid. The reaction is carried out at 140° to 180° C in the presence of zinc acetate and Sb₂O₃ as catalyst.



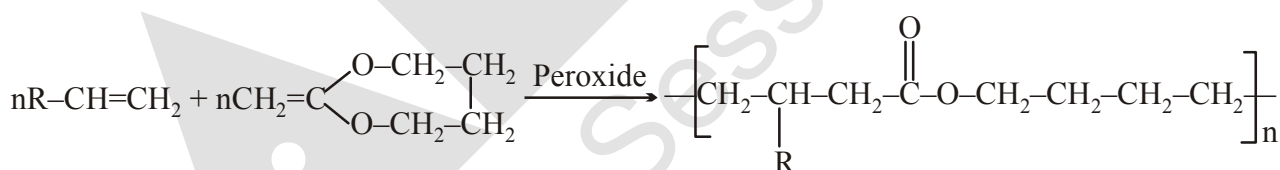
The terylene fibre (Dacron) is crease resistant and has low moisture absorption. It has high tensile strength. It is mainly used in making wash and wear garments, in blending with wood to provide better crease and wrinkle resistance.

BIODEGRADABLE POLYMERS

By far the largest use of synthetic polymers is as plastic. A major portion of it is used as throwaway containers and packing materials. Since plastics do not disintegrate by themselves, they are not biodegradable over a period of time. Non - biodegradability is due the carbon-carbon bonds of addition polymers which are inert to enzyme catalysed reaction. These polymers create pollution problem.

Biodegradable polymers are the polymers that can be broken into small segments by enzyme catalysed reactions using enzymes produced by microorganisms. In biodegradable polymers, bonds that can be broken by the enzymes are inserted into the polymers. Therefore, when they are buried as waste, enzymes present in the ground can degrade the polymer.

One method involves inserting hydrolysable ester group into the polymer. For example, when acetal (I) is added during the polymerization of alkene, ester group is inserted into the polymeric chains.

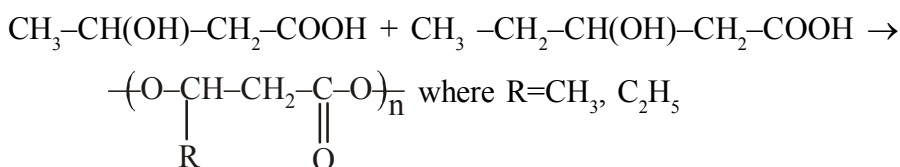


Ester linkage attacked by enzyme

Aliphatic polyesters are important class of biodegradable polymers. some examples are described below:

(1) Poly - Hydroxybutyrate-CO-β-Hydroxyvalerate (PHBV)

It is a copolymer of 3 - hydroxybutanoic acid and 3 hydroxypentanoic acid, in which the monomer units are connected by ester linkages.



The properties of PHBV vary according to the ratio of both the acids. 3-Hydroxybutanoic acid provides stiffness and 3-hydroxypentanoic acid imparts flexibility to the co-polymer. It is used in specialty packaging, orthopaedic devices and even in controlled drug release. When a drug is put in a capsule of PHBV, It is released only after the polymer is degraded. PHBV also undergoes bacterial degradation in the environment.

(2) POLY (GLYCOLIC ACID) AND POLY (LACTIC ACID)

They constitute commercially successful biodegradable polymers such as sutures. Dextron was the first bioadsorbable suture made for biodegradable polyesters for post - operative stitches.

MOLECULAR MASS OF POLYMER

Normally, a polymer contains chains of varying lengths and therefore, its molecular mass is always expressed as an average. In contrast, natural polymers such as protein contains chain of identical length and hence, have definite molecular mass.

The molecular mass of a polymer is expressed as

- (a) Number average molecular mass ($\overline{M}_n$)

$$\overline{M}_n = \frac{\sum N_i M_i}{\sum N_i}$$

where N_i is the number of molecules of molecular mass M_i

- (b) Weight average molecular mass ($\overline{M}_w$)

$$\overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i}$$

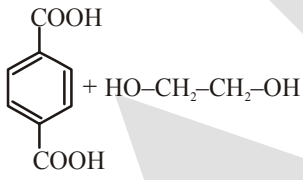
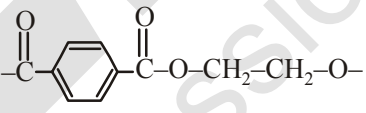
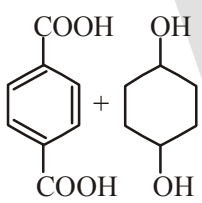
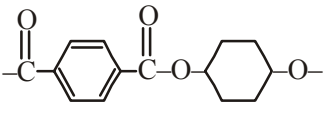
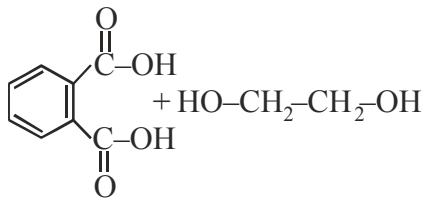
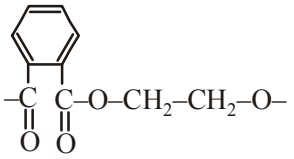
where N_i is the number of molecules of molecular mass M_i . Methods such as light scattering and ultracentrifuge depend on the mass of the individual molecules and yield weight average molecular masses. $\overline{M}_n$ is determined by employing methods which depend upon the number of molecules present in the polymer sample viz. colligative properties like osmotic pressure.

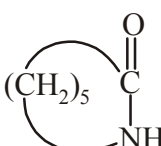
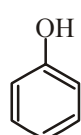
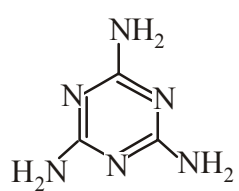
The ratio of the weight and number average molecular masses ($\overline{M}_w / \overline{M}_n$) is called Poly Dispersity Index (PDI). Some natural polymers, which are generally monodispersed, the PDI is unity (i.e. $\overline{M}_w = \overline{M}_n$).

In synthetic polymers, which are always polydispersed, $PDI > 1$ because $\overline{M}_w$ is always higher than $\overline{M}_n$.

COMMON POLYMERS

Monomer	Repeating unit	Polymer
1. $\text{CH}_2=\text{CH}_2$ Ethylene	$-\text{CH}_2-\text{CH}_2-$	Polyethylene
2. $\text{CH}_3-\text{CH}=\text{CH}_2$ Propene	$-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-$	Polypropene
3. $\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$ Styrene	$-\text{CH}_2-\underset{\text{C}_6\text{H}_5}{\text{CH}}-$	Polystyrene
4. $\text{CF}_2=\text{CF}_2$ Tetrafluoroethylene	$-\text{CF}_2-\text{CF}_2-$	Polytetrafluoroethylene (PTFE), Teflon
5. $\text{CH}_2=\text{CH}-\text{Cl}$ Vinyl chloride	$-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-$	Polyvinyl Chloride (PVC)
6. $\text{CH}_2=\text{CH}-\text{CN}$ Vinyl cyanide or Acrylonitrile	$-\text{CH}_2-\underset{\text{CN}}{\text{CH}}-$	Polyvinyl cyanide, polyacrylonitrile, Orlon.
7. $\begin{array}{c} \text{H}_3\text{C} \quad \text{O} \\ \quad \\ \text{CH}_2=\text{C}-\text{C}-\text{O}-\text{CH}_3 \end{array}$ Methyl methacrylate	$-\text{CH}_2-\underset{\text{CH}_3}{\overset{\text{COOCH}_3}{\text{C}}}-$	Polymethyl methacrylate, Plexiglas, Lucite
8. $\begin{array}{c} \text{O} \\ \\ \text{CH}_2=\text{CH}-\text{O}-\text{C}-\text{CH}_3 \end{array}$ Vinyl acetate	$-\text{CH}_2-\underset{\text{OCOCH}_3}{\text{CH}}-$	Polyvinyl Acetate
9. $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ 1,3-butadiene	$-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2-$	Polybutadiene, Buna rubber
10. $\text{CH}_2=\text{CH}-\text{Cl}$ (vinyl chloride) + $\text{CH}_2=\text{CCl}_2$ (Vinylidene chloride)	$-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-\text{CCl}_2-\text{CH}_2-$	Saran

11.	$\begin{array}{c} \text{C}_6\text{H}_5-\text{CH}=\text{CH}_2 \text{ (Styrene)} \\ + \\ \text{CH}_2=\text{CH}-\text{CN} \text{ (acrylonitrile)} \end{array}$	$\begin{array}{c} \text{C}_6\text{H}_5 \quad \text{CN} \\ \quad \\ -\text{CH}_2-\text{CH}-\text{CH}-\text{CH}_2- \end{array}$	SAN
12.	$\begin{array}{c} \text{CH}_2=\text{CH}-\text{CN} + \text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2 \\ + \\ \text{C}_6\text{H}_5-\text{CH}=\text{CH}_2 \end{array}$	—	ABS
13.	$\begin{array}{c} \text{CH}_2=\text{C}-\text{CH}_2 + \text{CH}_2=\text{C}-\text{CH}=\text{CH}_2 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$	—	Butyl rubber
14.	$\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2 + \text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$	—	Buna -S, SBR
15.	$\text{CH}_2=\text{CH}-\text{CN} + \text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$	—	Buna-N, NBR
16.	$\begin{array}{c} \text{Cl} \\ \\ \text{CH}_2=\text{C}-\text{CH}=\text{CH}_2 \\ \text{Cloroprene} \end{array}$	$\begin{array}{c} \text{Cl} \\ \\ -\text{CH}_2-\text{C}=\text{CH}-\text{CH}_2- \end{array}$	Neoprene
17.			Poly(ethylene terephthalate, Terylene, Dacron or Mylar
18.			Kodel Polyester
19.			Polyethylene phthalate alkyd resin (Glyptal)

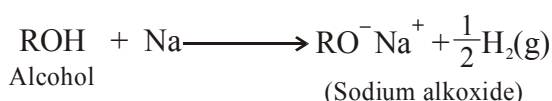
20.  Caprolactam	$\text{—NH(CH}_2)_5\text{—C(=O)—}$	Nylon-6
21. $\text{NH}_2(\text{CH}_2)_6\text{—NH}_2$ Hexamethylenediamine + $\text{HO—C(=O)—(CH}_2)_4\text{—C(=O)—OH}$ Adipic Acid	$\text{—NH—(CH}_2)_6\text{—NH—C(=O)—(CH}_2)_4\text{—C(=O)—}$	Nylon - 66
22.  + HCHO	—	Bakelite or resol
23. $\text{NH}_2\text{—C(=O)—NH}_2$ + HCHO	—	Urea- formaldehyde resin
24.  + HCHO Melamine	—	Malamine formaldehyde resin

CHEMICAL TESTS FOR FUNCTIONAL GROUPS

- Elemental analysis, physical, spectral and solubility test give a reasonable idea regarding the identity of unknown compound.
- Despite the tremendous importance and ease of spectral analysis, chemical tests are indispensable to complete characterization.

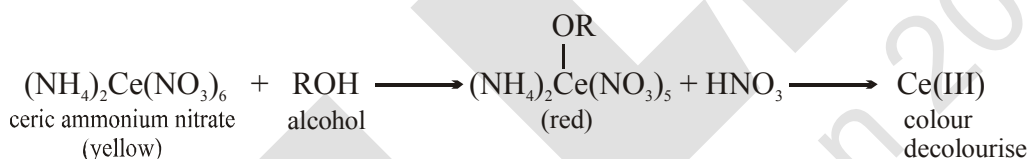
1. Test For Alcohols :

(i) Sodium Metal



- When Na-metal reacted with alcohol $\text{H}_2(\text{g})$ evolve
- It occurs in 1° , 2° & 3°
- The rate is highly variable and depends upon the alcohol structure.
- Other Functional groups that evolve $\text{H}_2(\uparrow)$
 R_2NH , RSH , $\text{RC}\equiv\text{C-H}$, RCO_2H etc.

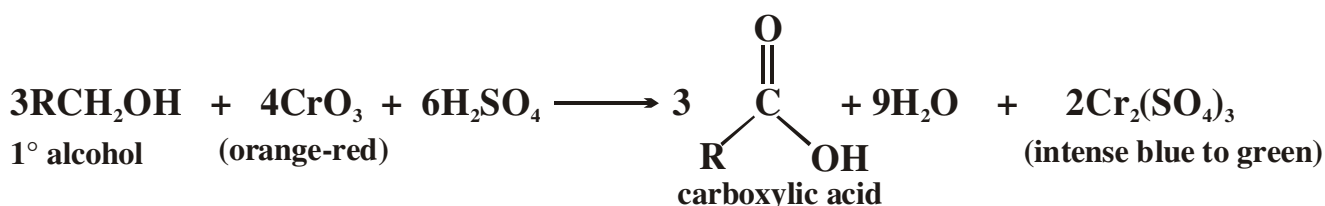
(ii) Ceric Ammonium Nitrate Oxidation (CAN Test)

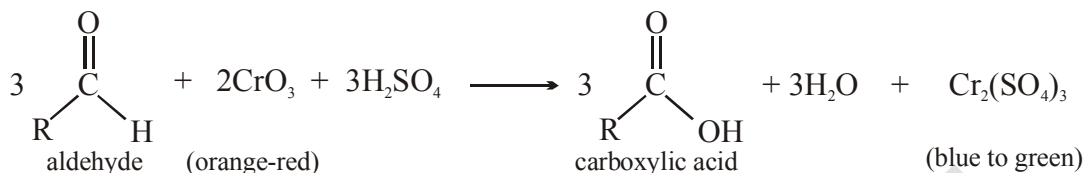
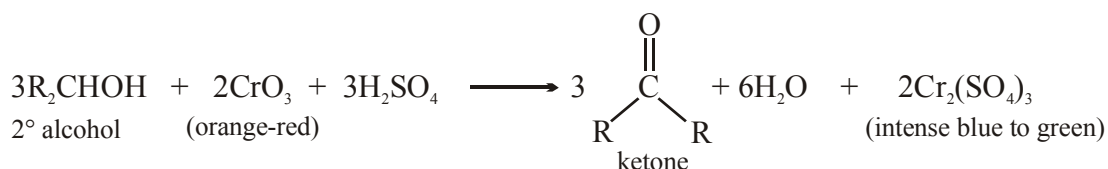


- Positive Test:** Color changes from yellow to red first then to colorless solution. (1 min. to 12 hrs)
- A positive test includes successively the formation, and then the disappearance of the red color.
- Very good test for 1° , 2° alcohol, but slow for 3° alcohol
- Note:** Phenols gives brown or black colour.

(iii) Jones Oxidation: Chromic anhydride or Chromium Trioxide (CrO_3)

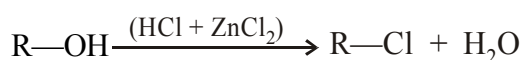
- A positive test for 1° , and 2° alcohols consists in the changes from an orange-red color (Cr^{6+}) to opaque suspension with green to blue color (Cr^{3+}) in 2 sec.
- 3° alcohols give no visible reaction within 2 sec. remaining orange
- Note:** Aldehydes give positive result.





(iv) **Lucas Test**

- This test distinguishes 1°, 2° & 3° alcohol from each other
- It uses, anhydrous ZnCl_2 in conc. HCl (lucas reagent)



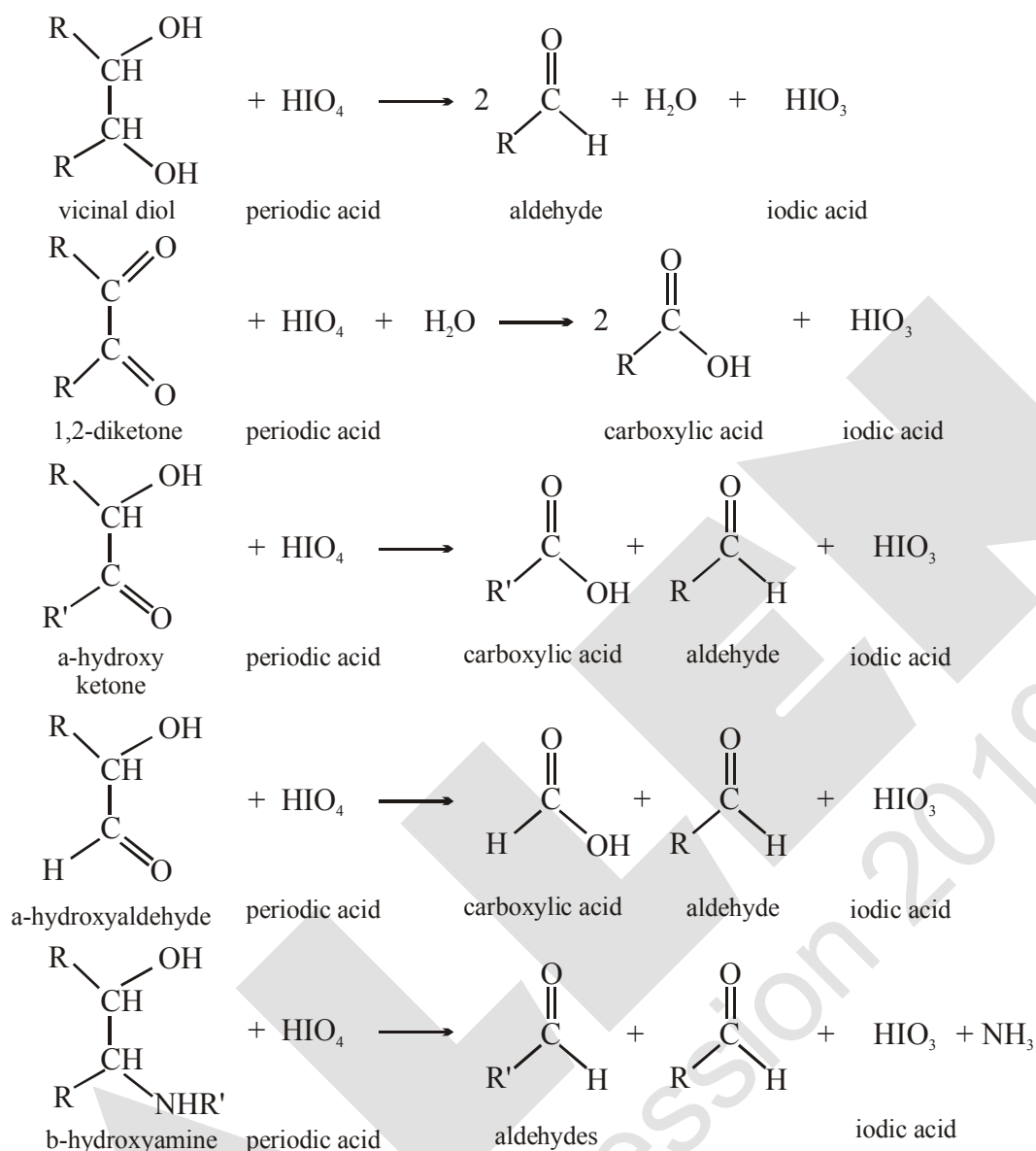
- 3° alcohols give white turbidity solution immediately
- 2° alcohols give white turbidity solution within 5 to 10 minutes
- 1° alcohols does not give white turbidity at room temperature

Note : Benzyl alcohol also react immediately

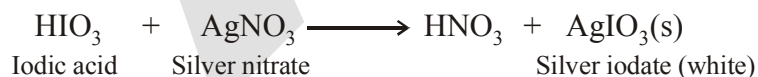
(v) **Victor-Mayer Test**

- This test also distinguishes 1°, 2° & 3° alcohol

1° - Alcohol	2° - Alcohol	3° - Alcohol
$\begin{array}{c} \text{R}-\text{CH}_2-\text{OH} \\ \downarrow \text{Red P} + \text{I}_2 \\ \text{R}-\text{CH}_2-\text{I} \\ \downarrow \text{NaNO}_2 \\ \text{R}-\text{CH}_2-\text{N} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O} \end{array} \\ \downarrow \text{HNO}_2 \\ \text{R}-\text{C} \begin{array}{l} \nearrow \text{N}-\text{OH} \\ \searrow \text{O} \end{array} \\ \text{(Nitrolic Acid)} \\ \downarrow \text{NaOH} \\ \text{R}-\text{C} \begin{array}{l} \parallel \\ \text{N}-\text{O}^\ominus \text{Na}^\oplus \end{array} \\ \text{(Red colour)} \end{array}$	$\begin{array}{c} \text{R} \\ \\ \text{R}-\text{CH}-\text{OH} \\ \downarrow \text{Red P} + \text{I}_2 \\ \text{R} \\ \\ \text{R}-\text{CH}-\text{I} \\ \downarrow \text{NaNO}_2 \\ \text{R} \\ \\ \text{R}-\text{CH}-\text{NO}_2 \\ \downarrow \text{HNO}_2 \\ \text{R} \\ \\ \text{R}-\text{C} \begin{array}{l} \nearrow \text{N}=\text{O} \\ \searrow \text{O} \end{array} \\ \text{(Pseudo nitro)} \\ \downarrow \text{NaOH} \\ \text{(Blue colour)} \end{array}$	$\begin{array}{c} \text{R} \\ \\ \text{R}-\text{C}-\text{OH} \\ \downarrow \text{Red P} + \text{I}_2 \\ \text{R} \\ \\ \text{R}-\text{C}-\text{I} \\ \downarrow \text{NaNO}_2 \\ \text{R} \\ \\ \text{R}-\text{C}-\text{NO}_2 \\ \downarrow \text{HNO}_2 \\ \text{No Reaction} \end{array}$

(vi) Periodic Acid (HIO_4) Test for detection of Vicinal Diols and Related Compounds

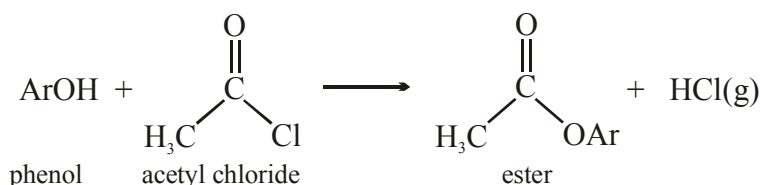
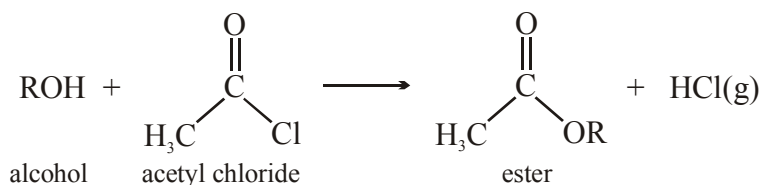
- The iodic acid is detected with 5% AgNO_3 solution -an immediate precipitation of silver iodate occurs.



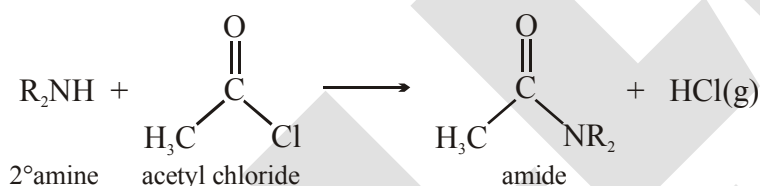
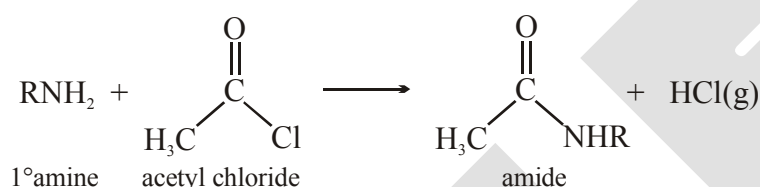
- Olefins, 2° alcohols, 1,3-glycols, ketones and aldehydes are not affected by HIO_4 under the test condition.

(vii) Acetyl Chloride

- Positive Test: Evolution of HCl gas and formation of ester as a top layer

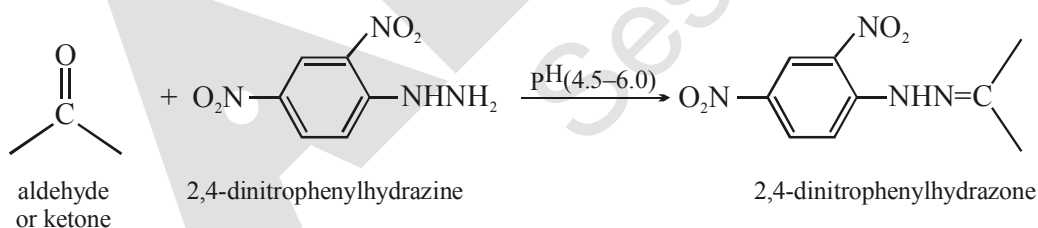


- 3° alcohols form primarily alkyl chloride due to the reaction of the liberated HCl on another molecule of the alcohol.
- Other functional groups that give positive test: 1° and 2° amines



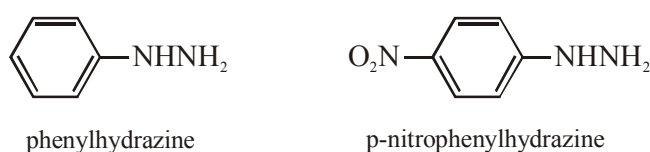
2. Classification Tests For Aldehydes and Ketone

(i) 2,4-Dinitrophenyl Hydrazine



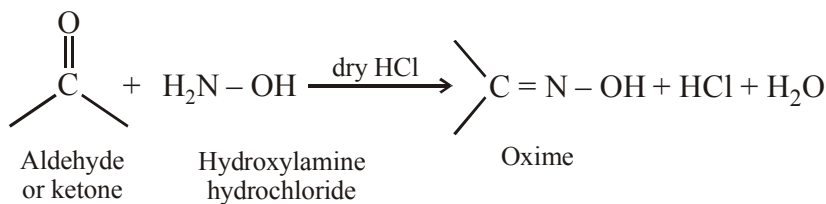
- Positive Test:** formation of yellow, orange or red ppt.
- The precipitate may be oily at first and become crystalline on standing.

(ii) Phenyl hydrazine and p-Nitrophenyl hydrazine.



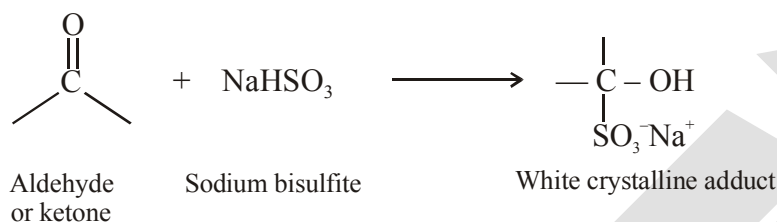
Positive Test: formation of yellow ppt.

(iii) Hydroxylamine Hydrochloride



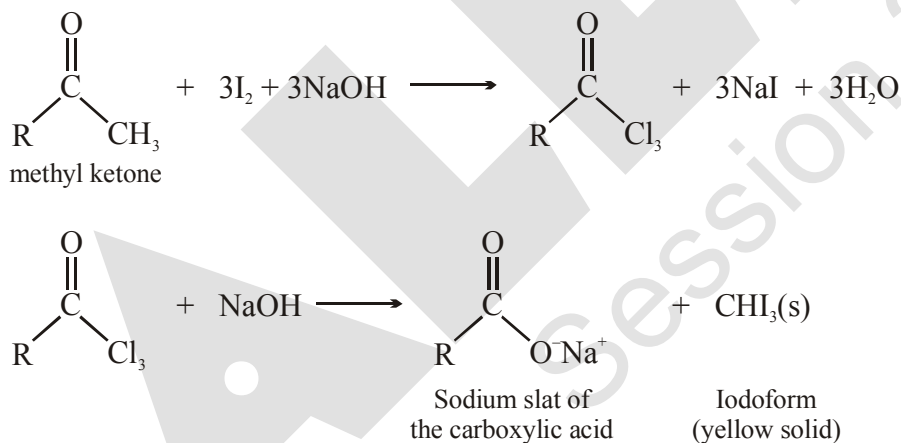
The liberation of hydrogen chloride can be detected by the change in colour from orange to red of a pH indicator.

(iv) Sodium Bisulfite (NaHSO₃)

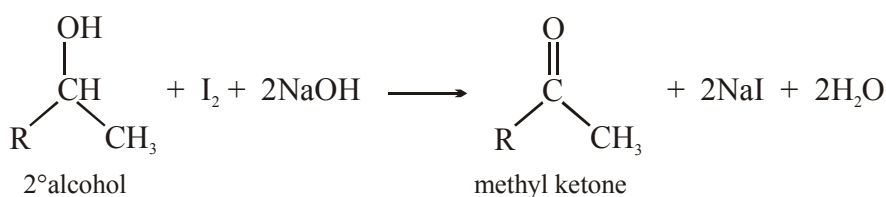


- **Positive Test:** By aldehydes and methyl ketone
- Only some cyclic ketones give positive results (ppt)
- This reaction is greatly inhibited by the steric constraints about the carbonyl group.

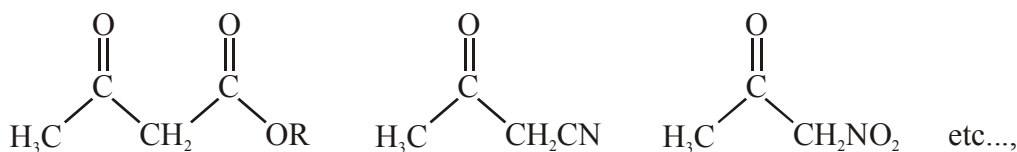
(v) Iodoform Test (For Methyl Ketones)



- **Positive Test:** Yellow ppt for methyl ketones
- **Disadvantages:** Some compounds that can be easily oxidized to methyl ketones give also positive results
- The principal types of compounds that give a positive test:

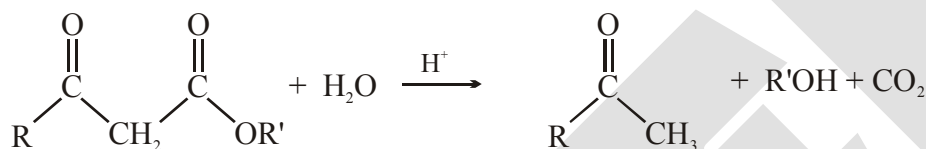
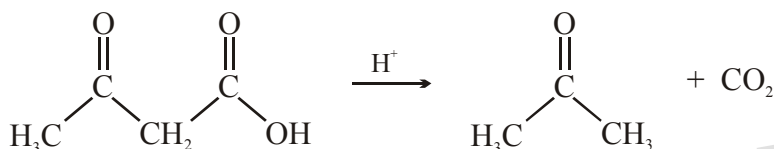


- The test is negative for the following type:



In such compounds the reagent removes the acetyl group and converts it to acetic acid, which resists iodination.

Acetoacetic acid is unstable, acidic aqueous solutions decompose to give CO_2 and acetone.



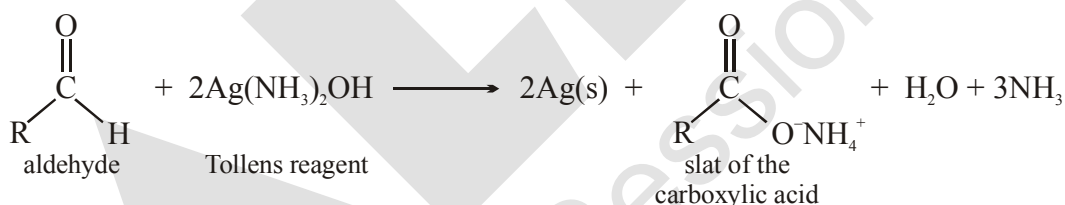
give positive iodoform test

3. Tests that give positive results with aldehydes and negative results with Ketones

- (i) **Jones Oxidation: Chromic anhydride or Chromium Trioxide (CrO_3)**

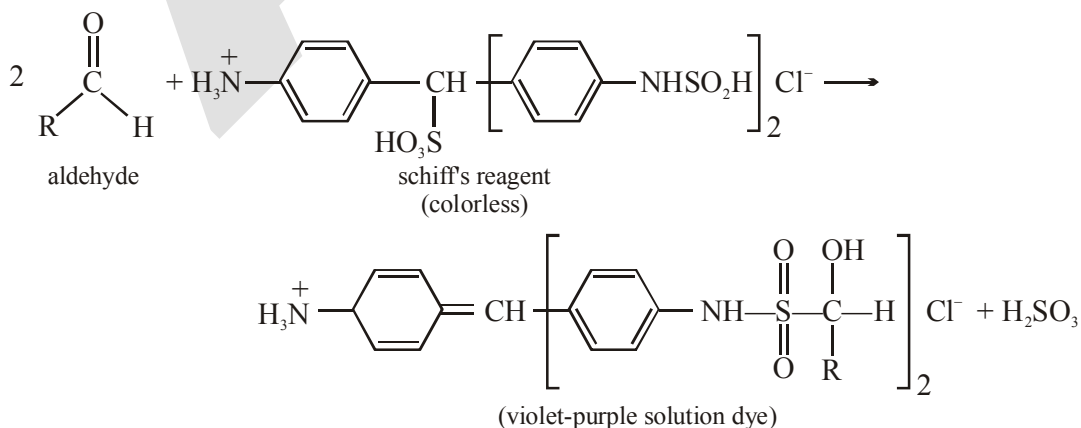
(Refer Jones Oxidation for alcohols at Page No. 22)

- (ii) **Tollens Reagents $\text{Ag}(\text{NH}_3)_2\text{OH}$**



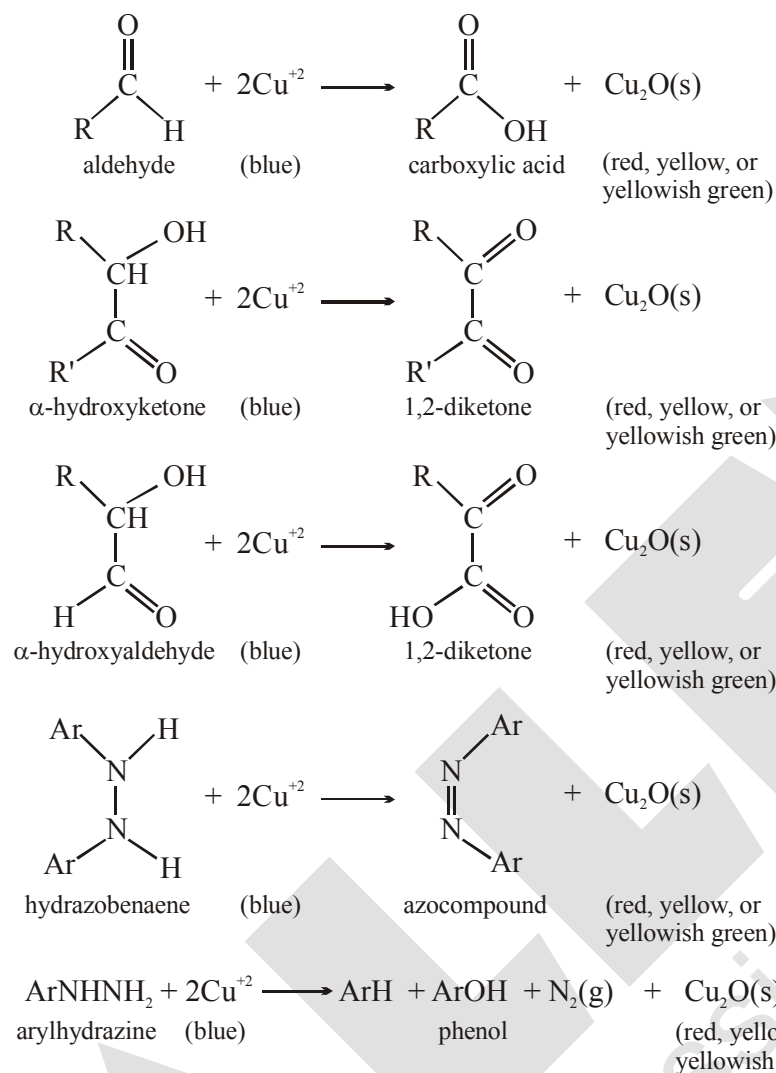
- Positive Test:** Formation of silver mirror (Ag) or colloidal (granular) gray or black Ag precipitate.

- (iii) **Schiff's Reagent**

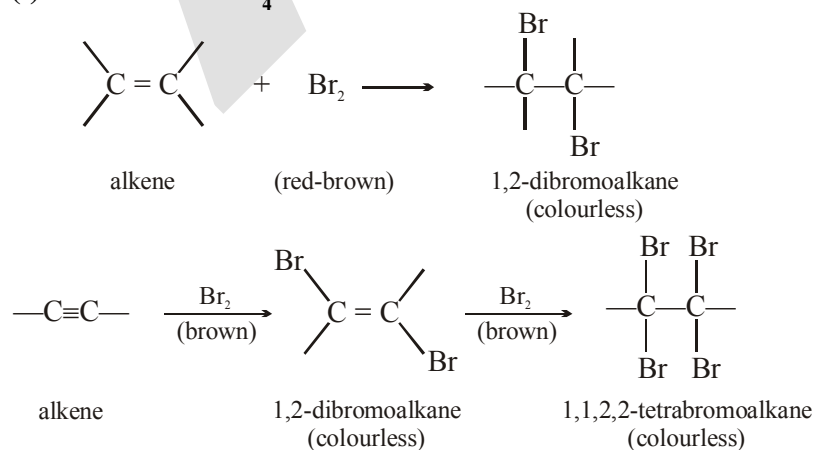


(iv) **Bendicts solution and Fehling's Solution**

- **Positive Test:** yellow or yellowish green ppt.
- All aldehydes give positive result except aromatic aldehydes (negative)

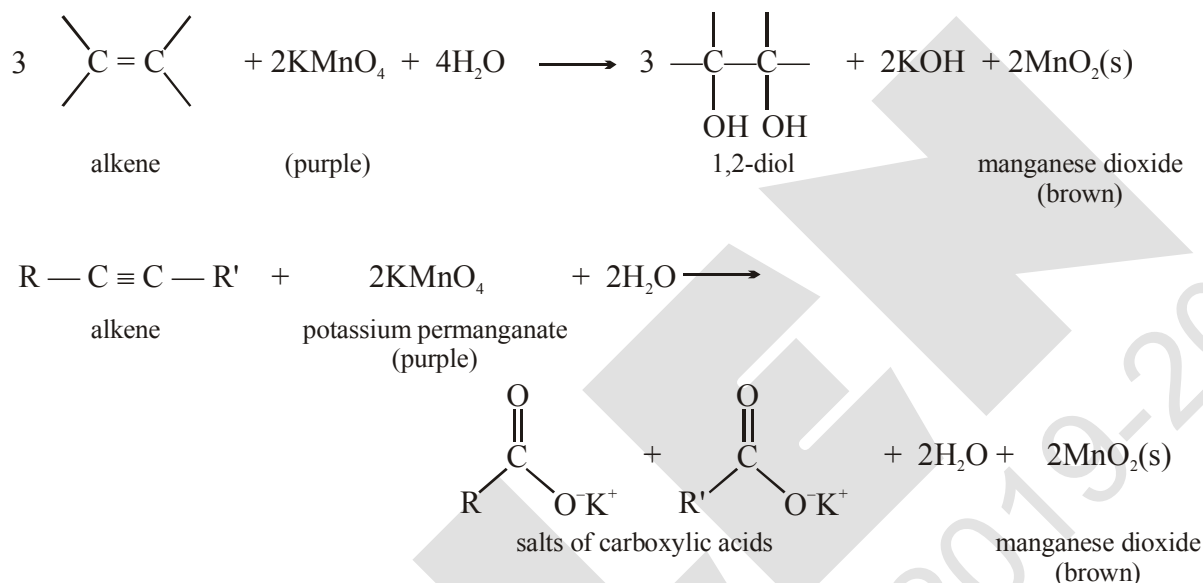
**4. Classification Tests for Unsaturation "alkenes & alkynes"**

- Alkanes are not usually characterized chemically because they are quite inert to most classification reactions

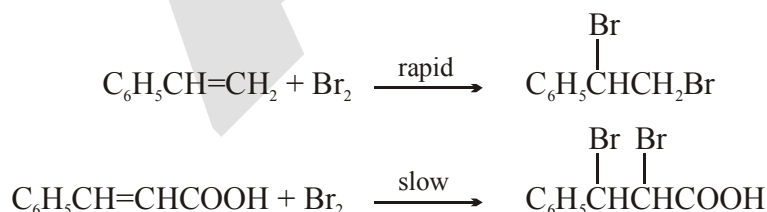
(i) Bromine in CCl₄

- **Positive Test:** Bromine color discharged without evolution of gas (HBr)
- Alkenes & alkynes give positive results
- If HBr is evolved, it indicates phenols, enols & enolizable compounds, -
- Bromine color is discharged by amines to produce a salt, which could be mistaken for addition.

(ii) **Baeyer Test (KMnO₄ aqueous)**



- **Positive Test:** Purple color discharges, and brown color ppt (MnO₂) appears
- **Note :** Aldehydes and alcohols also give positive result
- Since the ionic characters of the Br₂ and KMnO₄ reactions are very different, there is some complementary character between the two tests.
- For example, some alkenes bearing electron-withdrawing groups undergo rapid reaction with KMnO₄ but often slow or negligible reaction with Br₂.

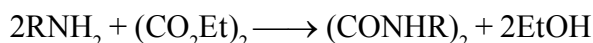


- A few tetrasubstituted olefins such as C₆H₅CBr=CBrc₆H₅ and (C₆H₅)₂C=C(C₆H₅)₂ fail to give positive tests with Br₂ in CCl₄ or KMnO₄ solution.

6. Tests for Amines and Amine Salts

(i) Diethyl oxalate test :

- This method helps in the separation of primary, secondary and tertiary amines.
- Primary amines on reaction with diethyl oxalate gives solid oxamide, secondary amine gives liquid oxamic ester and tertiary amine remains unreacted in gaseous form.

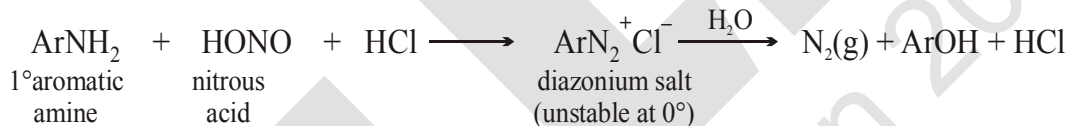
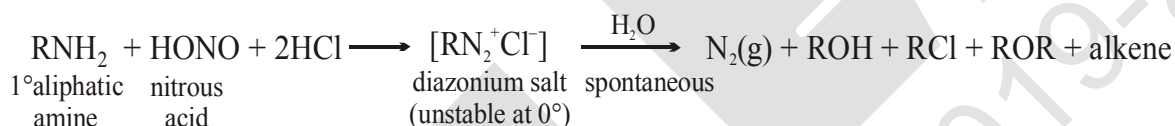


- The gaseous tertiary amine is separated out.
- The solid oxamide is filtered out of the liquid oxamic ester.
- These products are added to the alkaline solution of NaOH.

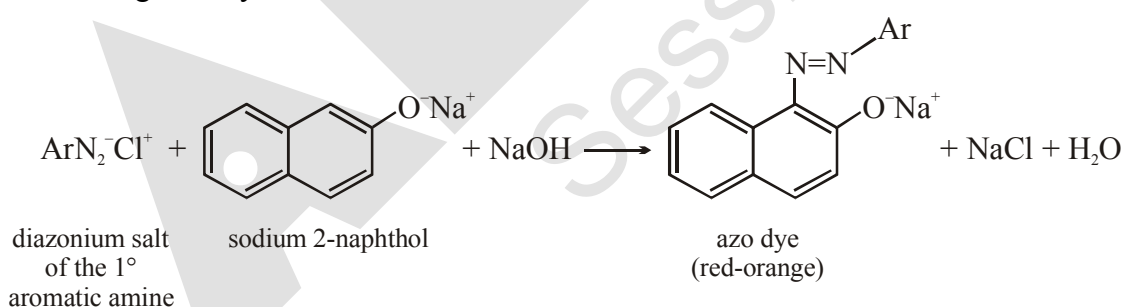
In this way, Primary, secondary and tertiary amines are separated by Hoffmann's Method

(ii) Nitrous Acid

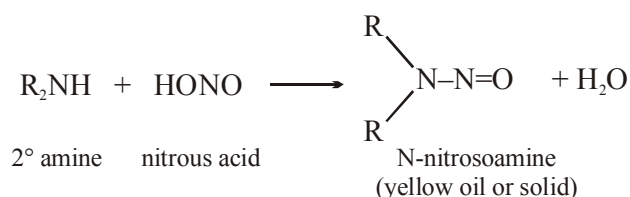
- Reaction of amines with nitrous acid (HONO) classifies the amine not only as 1°, 2°, or 3°, but also as aliphatic or aromatic.



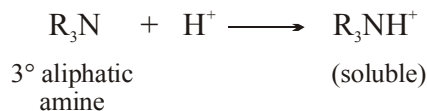
- The diazonium salt of the primary aromatic amine reacts with sodium 2-naphthol to produce a red-orange azo dye.



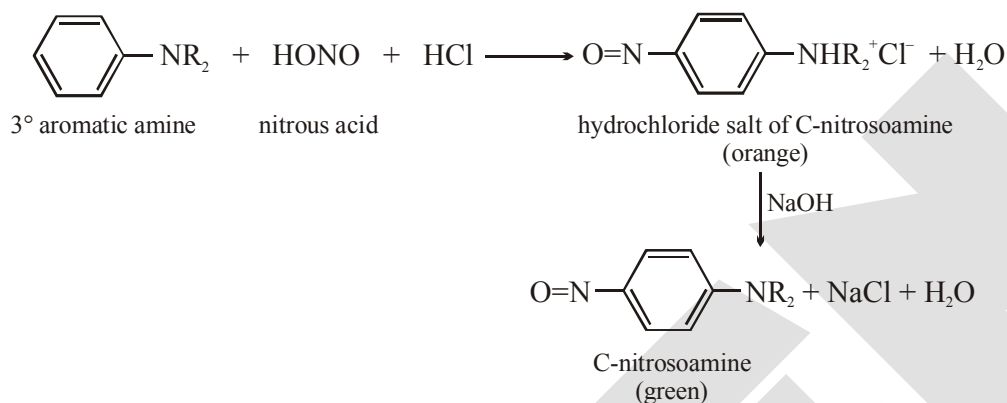
- 2° amines undergo a reaction with nitrous acid to form N-nitrosoamine, which are usually yellow solids.



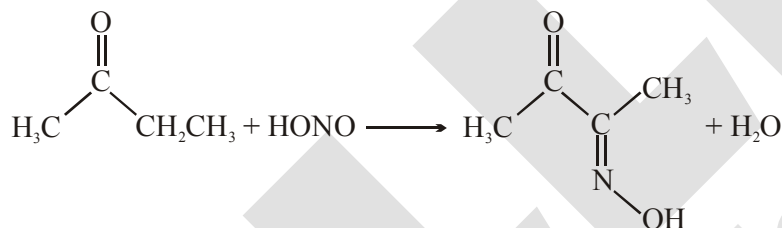
- 3° aliphatic amines do not react with nitrous acid, but they form a soluble salt. The reaction mixture gives an immediate positive test on the starch-iodide paper for nitrous acid.



- 3° aromatic amines react with nitrous acid to form the orange-colored hydrochloride salt of the C-nitrosoamine. Treating the solution with base liberates the blue or green C-nitrosoamine.

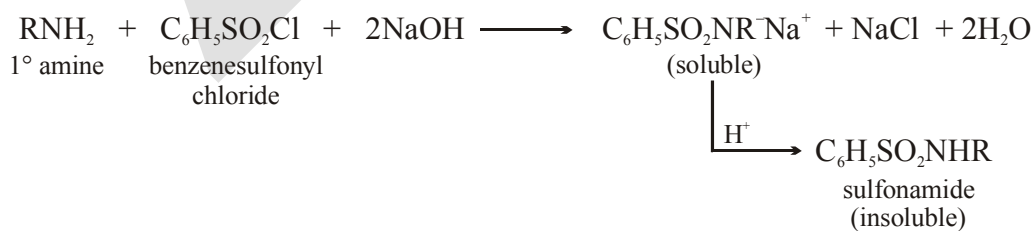


- Although nitrous acid is useful for characterizing amines, other functional groups also react.
- A methylene group adjacent to a keto group is converted to an oximino group and alkyl mercaptans yield red S-alkyl thionitrites.

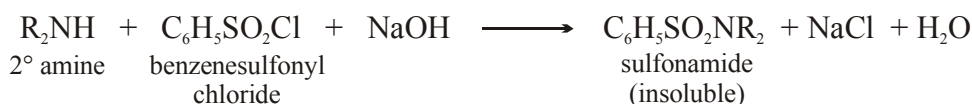


(iii) Hinsberg Test

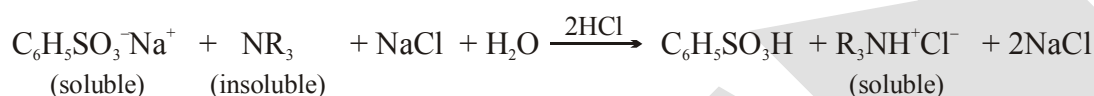
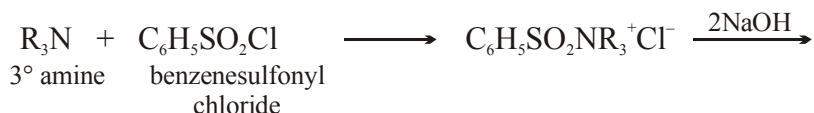
- Based on the reaction of the amine with benzenesulfonyl chloride (Hinsberg reagent), it can be used to separate 1°, 2°, and 3° amines.
- 1° amines: give solution that produce ppt after addition of HCl



2° amines: give ppt - acidification of the solution does not dissolve the sulfonamide (i.e., the PPT is not soluble in NaOH or HCl)



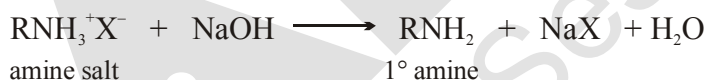
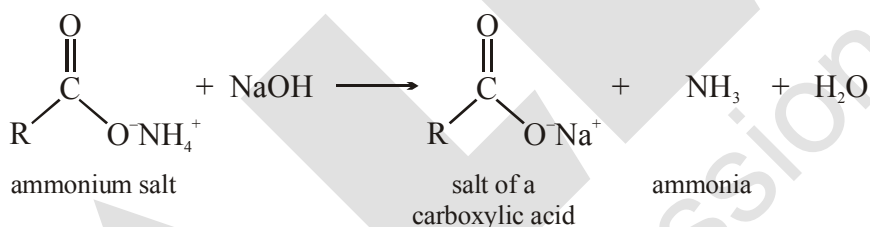
- 3° amines: undergo reaction with benzenesulfonyl chloride to produce quaternary ammonium sulfonate salts, which yield sodium sulfonate and insoluble 3° amines in basic solution.
- Acidification of the reaction mixture results in the formation of sulfonic acids and soluble amine salts



- The results of the Hinsberg test must not be used alone in classifying amines. The solubility of the original compounds must also be considered.
- If the original compound is amphoteric, which means that it is soluble in both acids and alkalies, the Hinsberg method fails to distinguish among the types of amines.

(iv) Sodium Hydroxide Treatment of Ammonium Salt and Amine Salts

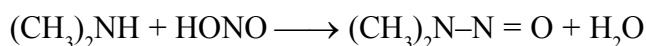
- Amine salts can be detected by treating the salt with NaOH to liberate ammonia or amine.



- Moistened pink litmus paper placed in the vapor above the solution will turn blue if ammonia or a volatile amine is present.

(v) Libermann's nitroso test :

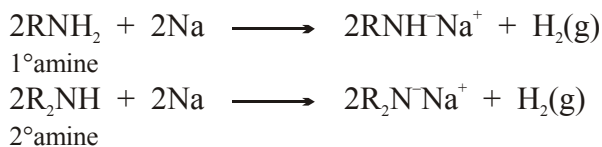
It is used as a test for secondary amines. Secondary amines (aliphatic as well as aromatic) reacts with nitrous acid to form N-nitrosoamines.



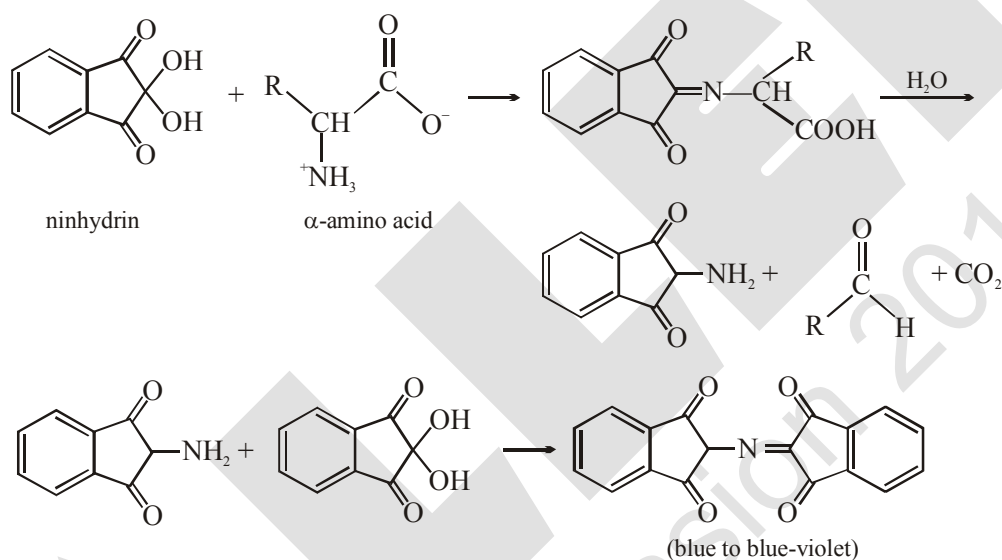
- Nitrosoamines are water soluble yellow oils and when warmed with phenol and few drops of conc. H_2SO_4 produce a green colour solution which turns blue on adding alkali. This reaction is called Libermann's nitroso reaction.
- Tertiary amine do not react with nitrous acid.

Sodium Metal :

- The active hydrogen on 1° and 2° amines undergo reaction with sodium to form salt and liberate hydrogen gas

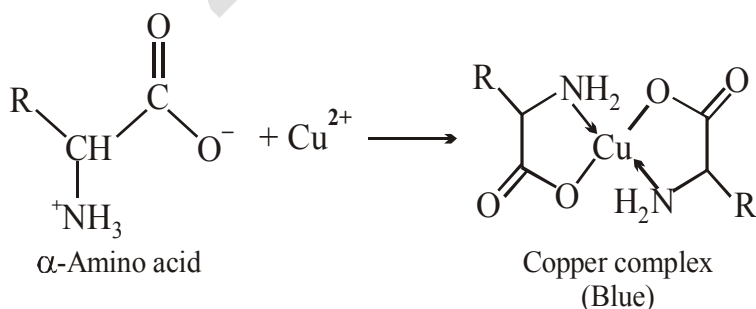
**7. Tests for Amino Acids****(i) Ninhydrin Test :**

- Amino acids and α -amino acids react with ninhydrin to give a positive test which is blue or blue-violet color.



- Ammonium salts ($\text{NH}_4^+ \text{Cl}^-$) give a positive test.
- Some amines, such as aniline, yield orange to red colors, which is considered a negative test.
- Proline, hydroxyproline, and 2-, 3-, and 4-aminobenzoic acids fail to give

Blue colour but produce a yellow color instead.

(ii) Copper Complex Formation

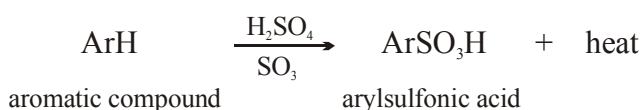
- Reaction with copper(II) sulfate solution yields a moderate-to deep-blue liquid or a dark-blue solid.

8. Tests for Aromatics

- If the molecule already contains reactive chemical substituents (acids, amines, ethers, carbonyl compounds, etc.,) use the corresponding classification test for that particular group.
- New substituents can be added onto the aromatic ring or existing substituents can be modified, such that the new

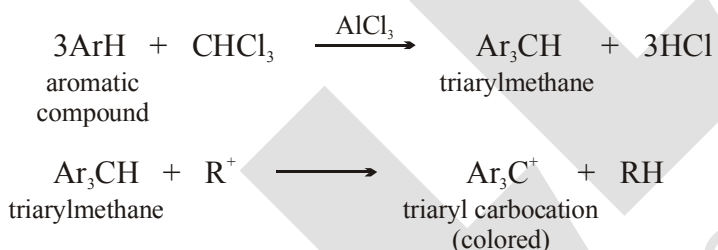
Compound May be More Readily Characterized.

(i) Fuming Sulfuric Acid



- This test is good for aromatics with no other functional groups.
- Positive Test: the aromatic compound dissolves completely in H_2SO_4 with the evolution of heat (Fuming)

(ii) Chloroform and Aluminum Chloride



- Aromatics give colored solution or powder.
- Positive Test: (Orange, red, blue, purple, green)
- Non aromatics give yellow color (Negative result)

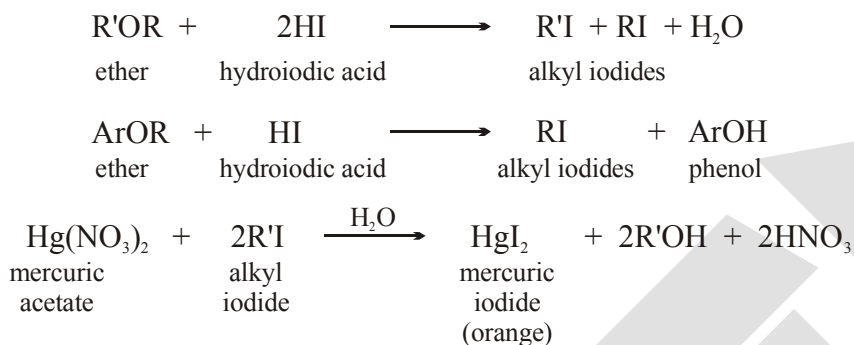
(iii) Azoxybenzene and Aluminum Chloride

- The color of the solution or precipitate is dependent on the functional groups present on the aryl group.
- Aromatic hydrocarbons and their halogen derivatives produce a deep-orange to dark-red color in solution or give a precipitate.
- Fused aromatic ring (such as naphthalene, anthracene, and phenanthrene) produce brown color.
- Aliphatic hydrocarbons give no color or, at most, a pale yellow.

9. Tests for Ethers

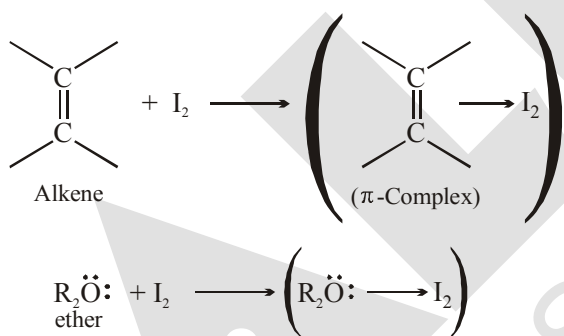
- Ethers are only a little more polar and slightly more reactive than either saturated hydrocarbons or alkyl halides
- Ethers form extremely explosive peroxides upon standing, specially when exposed to air and/or light. Liquid ethers that shows solid precipitates should not be handled.

(i) Hydroiodic Acid (Zeisel's, Alkoxy method)



- Positive Test: Orange or Orange-red color indicating.
- Note: Ethyl and methyl esters give also positive result.

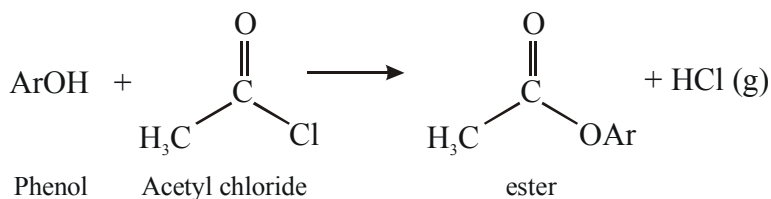
(ii) Iodine Test for Ethers and Unsaturated Hydrocarbons



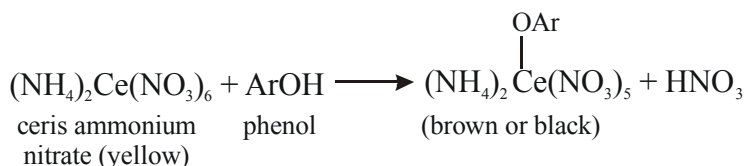
- Positive test: the color of the solution changes from purple to tan.
- Aromatic hydrocarbons, saturated hydrocarbons, fluorinated hydrocarbons and chlorinated hydrocarbons do not react.
- Unsaturated hydrocarbons produce a light-tan solid, while retaining the purple color of the iodine solution.

10. Tests for Phenols

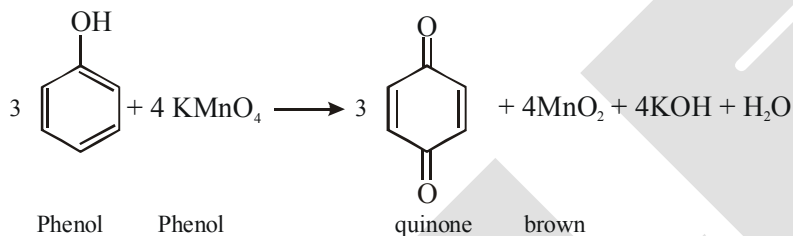
- As with alcohols, the acidic hydrogen in phenol can be detected with sodium (hydrogen gas is evolved) or with acetyl chloride (an ester layer is formed).



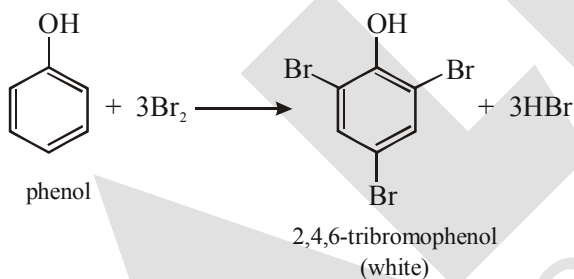
- Phenols undergo reaction with yellow ceric ammonium nitrate to produce brown or black products.



- Phenols reduce potassium permanganate solution and undergo oxidation to quinones.
- The manganese is reduced from +7 which gives a purple solution to +4 which is brown.

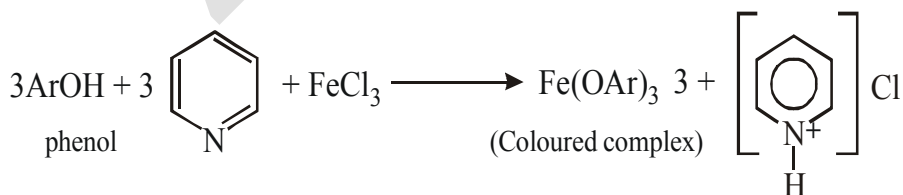


(i) **Bromine water**



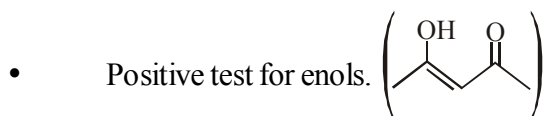
- Positive Test: decolorization of bromine.
- This is good for water soluble phenols

(ii) **Ferric Chloride - Pyridene Reagent**



- Positive Test: Production of blue, violet, purple, green, or red-brown colors

- Good for all types of Ar-OH.

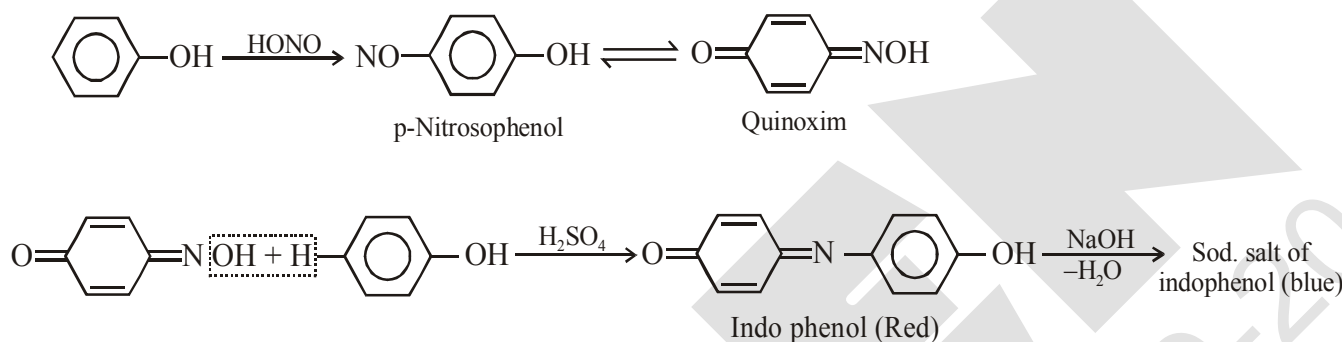


- Carboxylic acid with FeCl_3 gives red colour only when saturating with NH_3 .

(iii) **Libermann's nitroso test :**

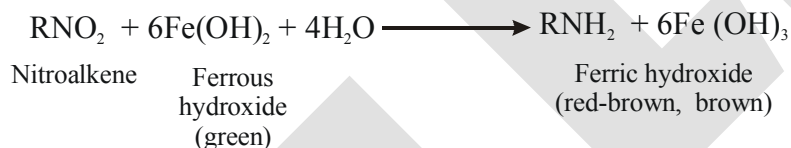
While phenol is reacted with NaNO_2 and concentrated H_2SO_4 , it provides a deep green colour which changes to red on dilution with water. while generated alkaline along with NaOH , blue colour is restored.

This reaction is termed as Liebermann's nitroso reaction and is employed as a test of phenol.



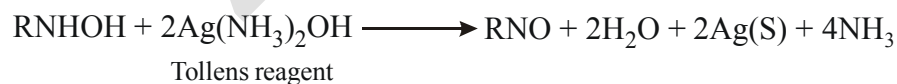
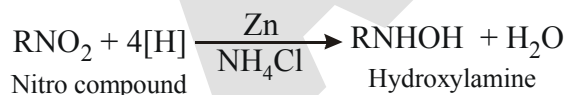
11. Test For Nitro Compounds

(i) Ferrous Hydroxide Reduction



- Positive Test: indicated by the change in color from green to red-brown or brown due to the oxidation of iron from +2 to +3.
- A negative test is indicated by a greenish precipitate.
- Note: Nitroso compounds, quinones, hydroxylamines alkyl nitrates give also positive results

(ii) Zinc and Ammonium Chloride Reduction

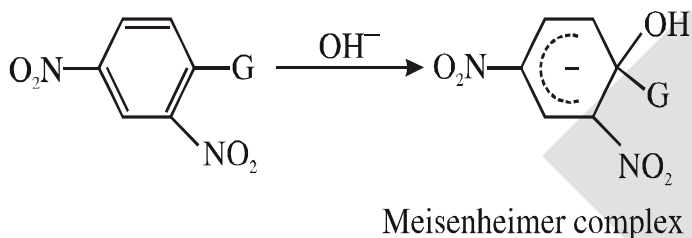


- Test the solution with Tollens Reagent
- **Positive Test:** formation of metallic silver
- Only 3° aliphatic nitro compounds and aromatic nitro compounds are reduced by $\text{Zn/NH}_4\text{Cl}$ to the hydroxylamine.

- The hydroxylamine is then detected by the formation of metallic silver in the Tollens test (or formation of a black or grey precipitate). This is known as Mulliken–baker test..

(iii) Treatment of Aromatic Compounds with Sodium Hydroxide

- The number of nitro groups on an aromatic ring can be determined by the reaction with NaOH
- In the reaction with NaOH:
- Mononitro aromatic compounds yield no color change or a very light yellow
- Dinitro aromatic compounds produce a bluish-purple color
- Trinitro aromatic compounds give a red color
- The color of the solution is due to a Meisenheimer complex



**CARBONYL COMPOUND,
ALDOL & SIMILAR NAME REACTIONS**

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Session 2019-20

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CARBONYL COMPOUND

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CARBONYL COMPOUND

EXERCISE # I

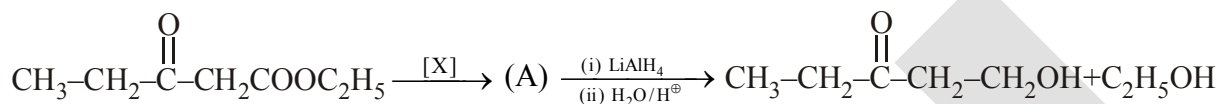
Q.1 Arrange these compounds in decreasing order of reactivity for the nucleophilic attack :

- (I) Acid chloride (II) Aldehyde (III) Ketone (IV) Ester

Select the correct answer from the codes given below :

- (A) I > II > III > IV (B) IV > III > II > I (C) III > II > I > IV (D) I > IV > II > III

Q.2 In the given reaction



[X] will be :

- (A) HCHO (B) $\begin{array}{c} \text{CH}_2\text{---OH} \\ | \\ \text{CH}_2\text{---OH} \end{array} + \text{H}^+$ (C) $\begin{array}{c} \text{CH}_2\text{---OH} \\ | \\ \text{CH}_2\text{---OH} \end{array} + \overset{\ominus}{\text{O}}\text{H}^-$ (D) HCN

Q.3 In the given reaction : $\text{C}_6\text{H}_5\text{---}\overset{\text{O}}{\parallel}\text{C---H} \xrightarrow{\text{NH}_2\text{OH/H}^+} [\text{X}]$

[X] will be:

- (A) Only syn oxime (B) Only anti oxime
(C) Mixture of syn and anti oxime (D) Secondary amide

Q.4 Acetophenone can be obtained by the distillation of :

- (A) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ (B) $(\text{CH}_3\text{COO})_2\text{Ca}$
(C) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ and $(\text{CH}_3\text{COO})_2\text{Ca}$ (D) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ and $(\text{HCOO})_2\text{Ca}$

Q.5 Gem dihalide on hydrolysis gives :

- (A) Vic diol (B) Gem diol (C) Carbonyl compound (D) Carboxylic acid

Q.6 Acetal or ketal is:

- (A) Vic dialkoxy compound (B) α, ω -dialkoxy compound
(C) α -alkoxy alcohol (D) Gem dialkoxy compound

Q.7 Cross cannizzaro reaction is example of :

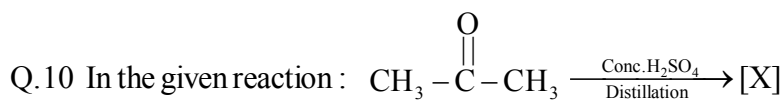
- (A) Redox reaction (B) Disproportionation (C) Both (A) and (B) (D) Only oxidation

Q.8 Acetaldehyde can be converted into $\text{HOCH}_2\text{---}\overset{\text{CH}_2\text{OH}}{\underset{\text{CH}_2\text{OH}}{\text{C}}}\text{---CH}_2\text{OH}$ by which reagent ?

- (A) KOH (B) KOH followed by LAH
(C) excess of HCHO and KOH (D) KCN followed by SBH

Q.9 Which one of the combinations will give propanaldehyde on dry distillation ?

- (A) $(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$ and $(\text{HCOO})_2\text{Ca}$ (B) $(\text{CH}_3\text{COO})_2\text{Ca}$ and $(\text{CH}_3\text{CH}_2\text{---COO})_2\text{Ca}$
(C) $(\text{CH}_3\text{---CH}_2\text{---COO})_2\text{Ca}$ and $(\text{HCOO})_2\text{Ca}$ (D) $(\text{CH}_3\text{COO})_2\text{Ca}$ and $(\text{CH}_3\text{COO})_2\text{Ca}$



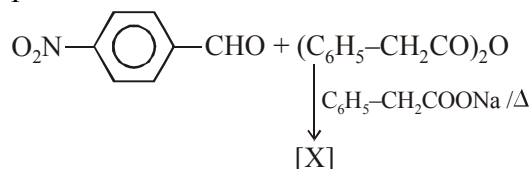
[X] will be :

- (A) Methyl oxide (B) Phorone
(C) 1, 3, 5-Trimethylbenzene (D) 2-Butyne

Q.11 Grignard reagents can not give carbonyl compounds with :

- (A) CO_2 (B) RCOCl (C) RCN (D) RCOOR

Q.12 The product of the reaction :



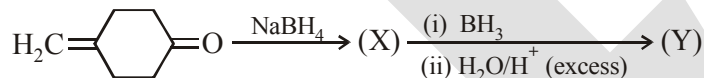
will be :

- (A) $\text{C}_6\text{H}_5-\text{CH}=\text{CH}-\text{COOH}$ (B) $\text{NO}_2-\text{C}_6\text{H}_4-\text{CH}=\text{CH}-\text{COOH}$
(C) $\text{C}_6\text{H}_5-\text{CH}=\underset{\text{CH}_3}{\text{C}}-\text{COOH}$ (D) $\text{NO}_2-\text{C}_6\text{H}_4-\text{CH}=\underset{\text{C}_6\text{H}_5}{\text{C}}-\text{COOH}$

Q.13 Cyanohydrin of which compound on hydrolysis will give lactic acid ?

- (A) $\text{C}_6\text{H}_5\text{CHO}$ (B) HCHO (C) CH_3CHO (D) $\text{CH}_3-\text{CH}_2-\text{CHO}$

Q.14 In the given reaction :

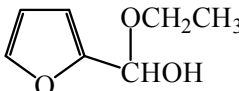
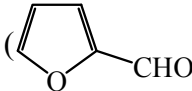


(X) and (Y) are :

- (A) $\text{CH}_2=\text{C}_6\text{H}_{10}-\text{OH}$ and $\text{HOCH}_2-\text{C}_6\text{H}_{10}=\text{O}$
(B) $\text{CH}_3-\text{C}_6\text{H}_{10}=\text{O}$ and $\text{HOCH}_2-\text{C}_6\text{H}_{10}=\text{O}$
(C) $\text{CH}_2=\text{C}_6\text{H}_{10}-\text{OH}$ and $\text{CH}_3-\text{C}_6\text{H}_{10}-\text{OH}$
(D) $\text{CH}_2-\text{C}_6\text{H}_{10}-\text{OH}$ and $\text{CH}_3-\text{C}_6\text{H}_{10}-\text{OH}$

Q.15 Acetaldehyde cannot give :

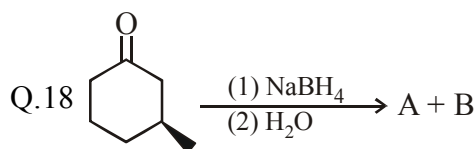
- (A) Iodoform test (B) Lucas test (C) Benedict test (D) Tollens test

Q.16 Compound  formed by the reaction of furfural () with ethanol is :

- (A) an aldol (B) an acetal (C) a ketal (D) a hemiacetal

Q.17 A compound with molecular formula $C_8H_{18}O_4$ does not give litmus test and does not give colour with 2,4-DNP. It reacts with excess $MeCOCl$ to give a compound whose vapour density is 152. Compound A contains how many hydroxy groups ?

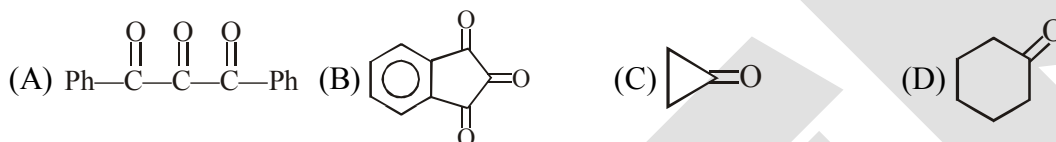
- (A) 1 (B) 2 (C) 3 (D) 4



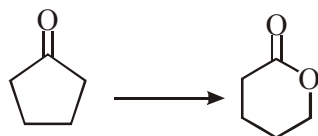
Identify relationship between A & B products ?

- (A) Diastereoisomers (B) Enantiomers (C) Positional isomer (D) Identical

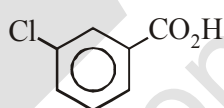
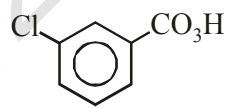
Q.19 Which of the following does not form a stable hydrate by the addition of H_2O ?



Q.20 The conversion

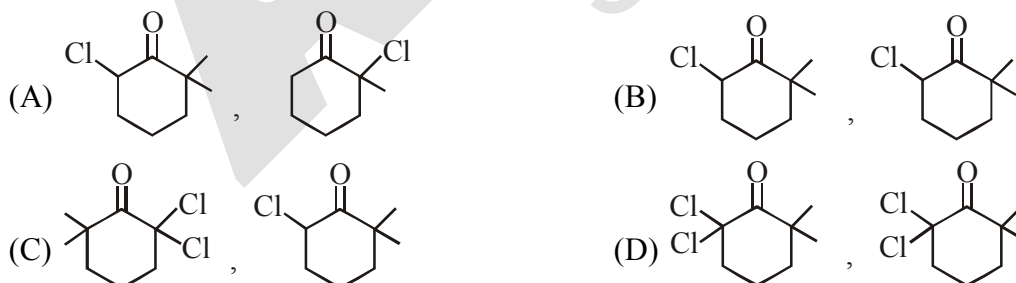


can be effected by using the reagent

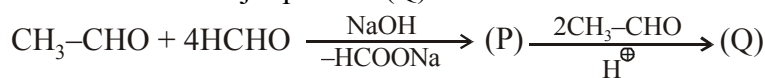
- (A) Tollen's reagent (B) O_3 (C)  (D) 



Organic product P & Q are respectively -

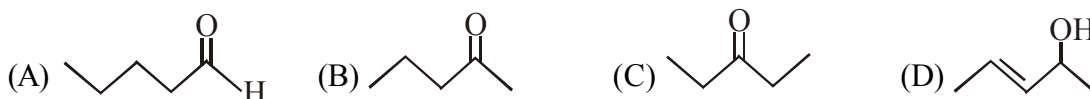


Q.22 Total number of stereoisomers of major product (Q) are :

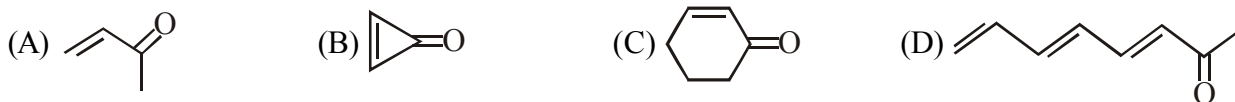


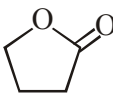
- (A) 0 (B) 4 (C) 8 (D) 2

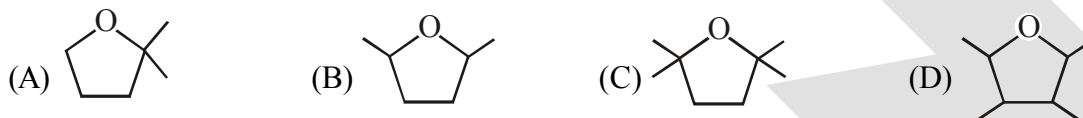
Q.23 An organic compound (A), $C_5H_{10}O$, reacts with hydrazine to form a hydrazone derivative (B). The hydrazone (B) on being heated with KOH at about $180^\circ C$, gives n-pentane. The compound (A) does not respond positively to Tollen's reagent and to the iodoform test. The compound (A) is



Q.24 The compound having the highest dipole moment is :



Q.25  $\xrightarrow[(2) H_2O]{(1) \text{ Excess MeMgCl}}$ (A) $\xrightarrow[H_2SO_4]{\text{conc.}}$ 'B' Identify 'B' product ?



Question No. 26 to 28 (3 questions)

An alkene (A) $C_{16}H_{16}$ on ozonolysis gives only product (B) C_8H_8O . (B) also can be obtained by hydrolysis of the product obtained by reaction between cyano benzene and CH_3MgBr . (A) can show geometrical isomerism and it can decolourise Br_2 water. (B) on treatment with SeO_2 produces (C)

Q.26 Which is not correct about (A) ?

- (A) A is optically inactive
(B) On catalytic hydrogenation 'trans' form of A produces racemic mixture
(C) A can be prepared by Wittig reaction on acetophenone with $Ph_3P=C(CH_3)Ph$.
(D) On treatment with per acid followed by hydrolysis 'trans' form of A produces racemic mixture

Q.27 Which is not correct about B ?

- (A) It gives iodoform test
(B) On treatment with $LiAlH_4$, H_2O it produces a compound which also responds to iodoform test.
(C) It gives Tollen's test
(D) On treatment with NH_2NH_2 followed by alc. KOH at high temperature, it produces ethyl benzene

Q.28 Which is not correct about C?

- (A) On treatment with $NaBH_4$ it will produce a diol.
(B) On treatment with OH^- (conc.) followed by acidification racemic mixture of a carboxylic acid is obtained
(C) It gives Tollen's test
(D) It can take part in aldol condensation

(Question No. 29 & 30)

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

- (A) If both (A) and (R) are true and (R) is the correct explanation of (A)
(B) If both (A) and (R) are true but (R) is not correct explanation of (A)
(C) If (A) is true but (R) is false
(D) If (A) is false and (R) is true

Q.29 **Assertion** : Benzaldehyde with HCN gives two isomeric compounds

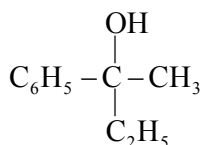
Reason : Both nitrile and isonitrile compounds are possible when HCN reacts with carbonyl group.

Q.30 **Assertion** : $Cl_3C-C(=O)H \xrightarrow{NaOH} Cl_3C-CH_2OH + Cl_3C-COONa$

Reason : There are no $\alpha-H$ in this compound, so it can't give aldol.

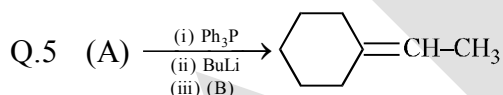
EXERCISE # II

- Q.1 Two isomeric ketones, 3-pentanone and 2-pentanone can be distinguished by :
 (A) $I_2 / NaOH$ (B) $NaSO_3H$ (C) $NaCN / HCl$ (D) 2,4-DNP
- Q.2 An optically inactive alcohol (A) $C_6H_{12}O$ is oxidized by MnO_2 to produce optically inactive carbonyl compound while reduction of (A) by H_2/Ni produces optically active compound. Possible structure(s) of alcohol is/are
 (A) Hex-2-ene-1-ol (B) Hex-3-ene-2-ol
 (C) 2-Methyl pent-2-ene-1-ol (D) 3-Methyl pent-2-ene-1-ol
- Q.3 Consider the structure of given alcohol:



This alcohol can be prepared from:

- (A) $C_6H_5 - \overset{O}{\parallel} C - CH_3$ and C_2H_5MgBr (B) $CH_3 - CH_2 - \overset{O}{\parallel} C - CH_3$ and C_6H_5MgBr
 (C) $C_6H_5 - \overset{O}{\parallel} C - C_2H_5$ and CH_3MgBr (D) $C_6H_5 - \overset{O}{\parallel} C - Cl$ and C_2H_5MgCl
- Q.4 Which of the following compounds will not give aldol condensation :
 (A) Acetaldehyde (B) Formaldehyde (C) Pivaldehyde (D) Crotonaldehyde

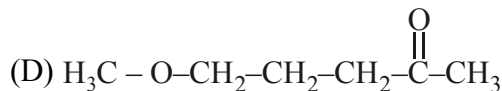
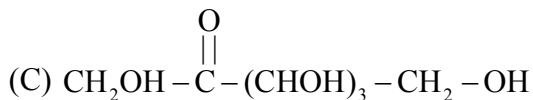
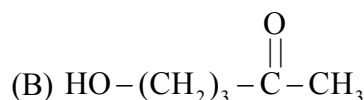
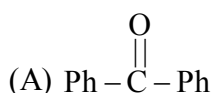


In above reaction (A) and (B) will respectively be

- (A) & CH_3CHO (B) CH_3CH_2Cl &
 (C) & $HCHO$ (D) $Cl - CH_2 - CH_2 - CH_2 - CH_2 - CH_2 - Cl$ & CH_3CHO

- Q.6 Stability of hydrates of carbonyl compounds depends on:
 (A) Steric hindrance (B) Presence of $-I$ group on gemdiol carbon
 (C) Intramolecular hydrogen bonding (D) angle strain in carbonyl compound
- Q.7 Which of the following can be used for protection of carbonyl group
 (A) $CH_2OH - CH_2OH / H^+$ (B) $CH_2OH - CH_2 - CH_2OH / H^+$
 (C) $HS - (CH_2)_3 - SH$ (D) $CH_2OH - CH_2 - CHO$

Q.8 Which of the following(s) will form stable hemiketal :



Q.9 Mixture of $\text{Ph}-\text{CHO}$ & HCHO is treated with NaOH then Cannizzaro reaction involves:

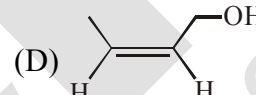
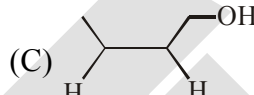
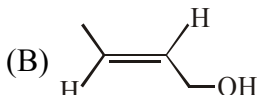
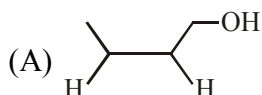
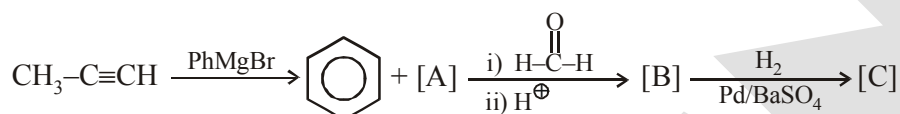
(A) Oxidation of HCHO

(B) Reduction of HCHO

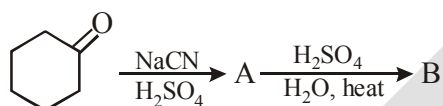
(C) Oxidation of $\text{Ph}-\text{CHO}$

(D) Reduction of $\text{Ph}-\text{CHO}$

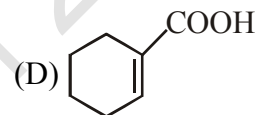
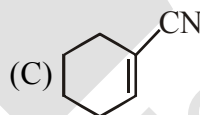
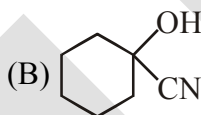
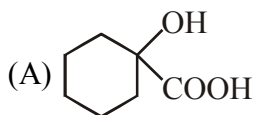
Q.10 Final product in the given reaction sequence is :



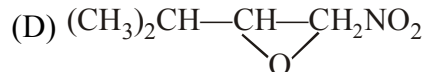
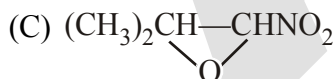
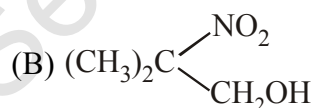
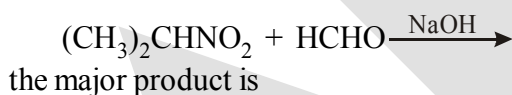
Q.11 Consider the following sequence of reactions.



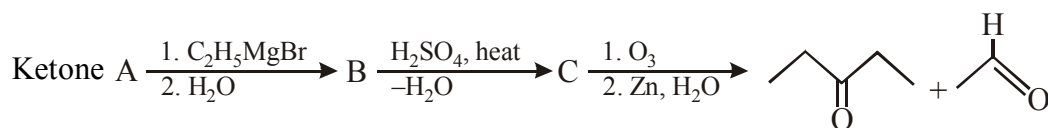
The major product (B) is :



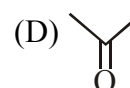
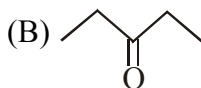
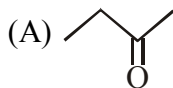
Q.12 In the reaction



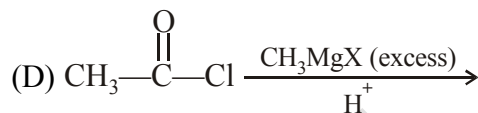
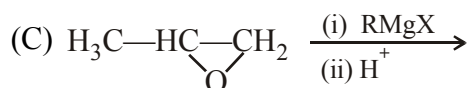
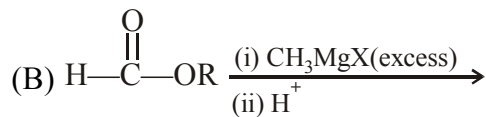
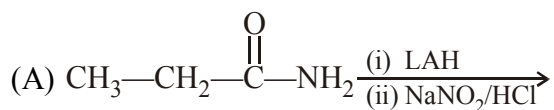
Q.13 Consider the following sequence of reactions.



The ketone (A) is :

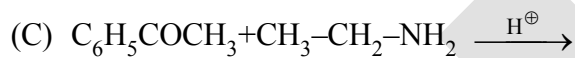
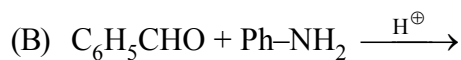
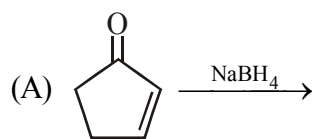


Q.14 Which of the following reactions will give(s) 2° alcohol as a major product :



Q.15 Match list-I with list-II :

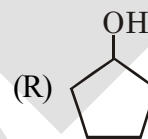
List - I



List - II

(P) Acetal

(Q) Schiff's base

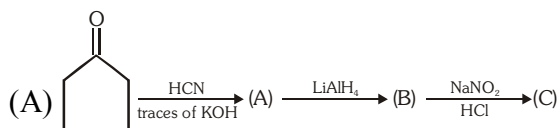


(S) Imine

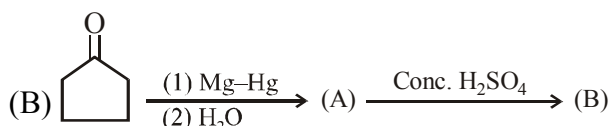
EXERCISE # III

Q.1 Column - I

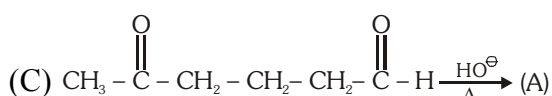
Column - II



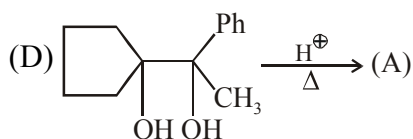
(P) Formation of six member ring takes place



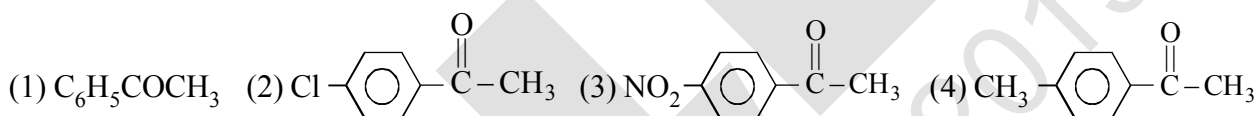
(Q) Final product is Ketone



(R) Final product formed will give positive Iodoform test

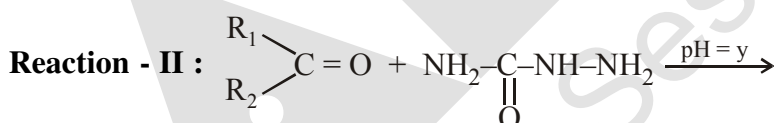
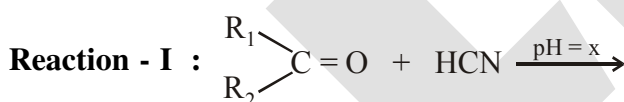


(S) Final product formed will react with 2,4-DNP. (2,4-Di-nitrophenyl hydrazine)

Q.2 Arrange the following compounds in decreasing order of K_{eq} for hydrate formation.

Paragraph for Q. 03 to 04

Two reactions which are example of nucleophilic attack are given as below .



Q.3 Value of x is :

(A) $x \leq 4.5$ (B) $x = 6$ (C) $x > 7$

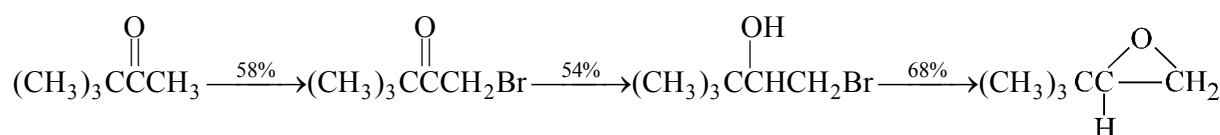
(D) Can't decide

Q.4 Value of y is :

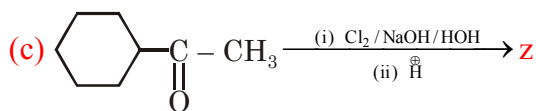
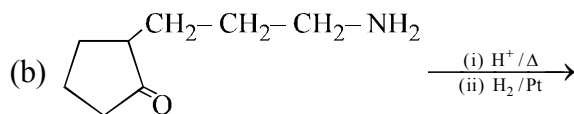
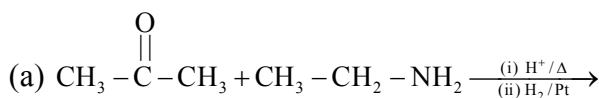
(A) $x = 4.5$ (B) $x = 1.5$ (C) $x = 7$ (D) $x = 9$

Q.5 Some Grignard reagents react with ethyl orthoformate, followed by acidic hydrolysis, to give aldehydes. Propose mechanisms for the two steps in this synthesis.

Q.6 A synthesis that begins with 3,3-dimethyl-2-butanone gives the epoxide shown. Suggest reagents appropriate for each step in the synthesis.

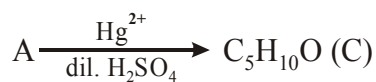


Q.7 Predict the organic products :

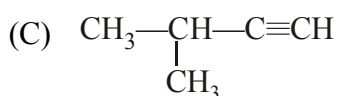
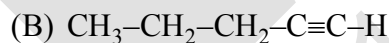
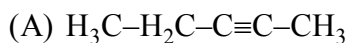


Paragraph for Q.No.8 to 9

A (Hydrocarbon) (C $\Rightarrow$ 88.24%) [Molecular weight of A = 68]



Q.8 'A' can be :



Q.9 Correct statement regarding C is.

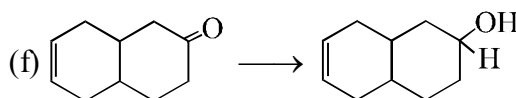
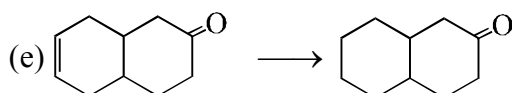
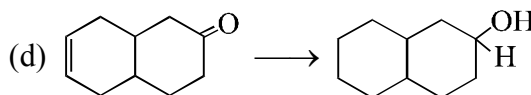
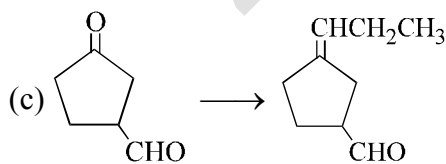
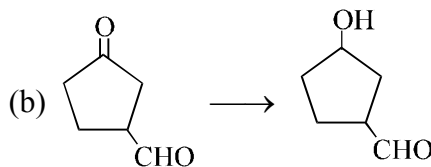
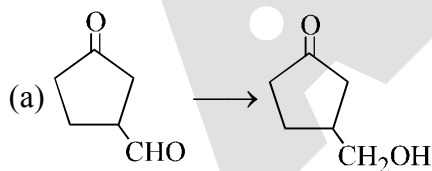
(A) C reacts with fehling solution to give red ppt.

(B) C gives +ve iodoform test

(C) C give -ve 2, 4, D.N.P test

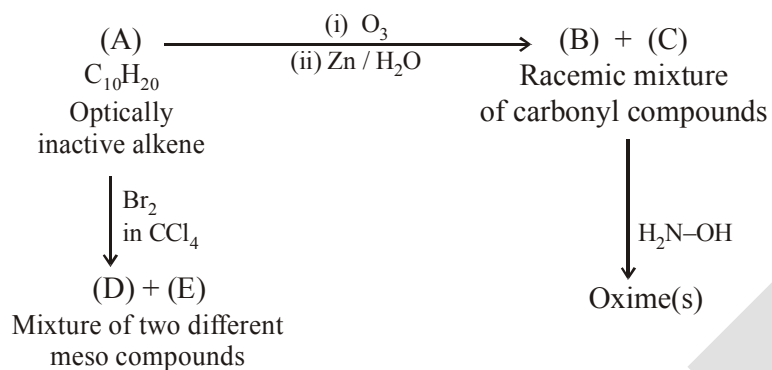
(D) C is aldehyde

Q.10 Show how you would accomplish the following syntheses efficiently and in good yield. You may use any necessary reagents.

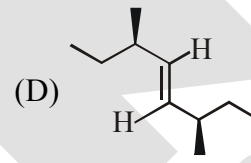
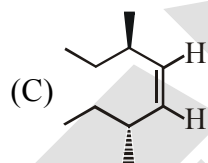
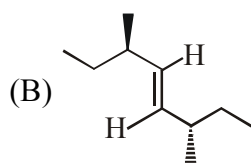
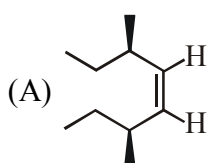


Paragraph for Q.No. 11 to 12

In given reaction sequence



11. Alkene A is :



12. How many total oxime(s) is/are obtained

(A) 1

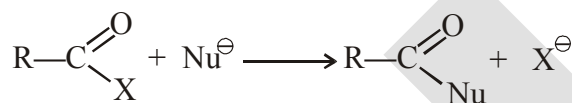
(B) 2

(C) 3

(D) 4

EXERCISE # IV (MAINS)

- Q.1 When $\text{CH}_2 = \text{CH}-\text{COOH}$ is reduced with LiAlH_4 , the compound obtained will be -
 (1) $\text{CH}_3-\text{CH}_2-\text{CH}_2\text{OH}$ (2) $\text{CH}_3-\text{CH}_2-\text{CHO}$ [AIEEE-2003]
 (3) $\text{CH}_3-\text{CH}_2-\text{COOH}$ (4) $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$
- Q.2 Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid ? [AIEEE-2004]
 (1) Phenol (2) Benzaldehyde (3) Butanal (4) Benzoic acid
- Q.3 Which one of the following is reduced with Zn-Hg/HCl to give the corresponding hydrocarbon [AIEEE-2004]
 (1) Butan-2-one (2) Acetic acid (3) Acetamide (4) Ethyl acetate
- Q.4 On mixing ethyl acetate with aqueous sodium chloride, the composition of the resultant solution is
 (1) $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaCl}$ (2) $\text{CH}_3\text{COONa} + \text{C}_2\text{H}_5\text{OH}$ [AIEEE-2004]
 (3) $\text{CH}_3\text{COCl} + \text{C}_2\text{H}_5\text{OH} + \text{NaOH}$ (4) $\text{CH}_3\text{Cl} + \text{C}_2\text{H}_5\text{COONa}$
- Q.5 The best reagent to convert pent-3-en-2-ol into pent-3-en-2-one is - [AIEEE-2005]
 (1) Acidic dichromate (2) Acidic permanganate
 (3) Pyridinium chloro-chromate (4) Chromic anhydride in glacial acetic acid
- Q.6 Rate of the reaction- [AIEEE-2005]



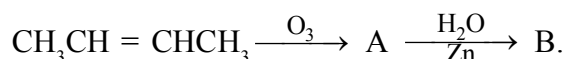
is fastest when X is -

- (1) NH_2 (2) Cl (3) OCOR (4) OC_2H_5
- Q.7 Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is-
 (1) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$ [AIEEE-2006]
 (3) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_2\text{OH}$ (4) PhCHOHCH_3

- Q.8 In the following sequence of reactions



- (1) Butanal (2) n-Butyl alcohol
 (3) n-Propyl alcohol (4) Propanal
- Q.9 In the following sequence of reactions, the alkene affords the compound 'B' :- [AIEEE-2008]



The compound B is

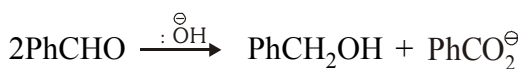
- (1) $\text{CH}_3\text{CH}_2\text{CHO}$ (2) CH_3COCH_3 (3) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (4) CH_3CHO
- Q.10 Bakelite is obtained from phenol by reacting with [AIEEE-2008]
 (1) $(\text{CH}_2\text{OH})_2$ (2) CH_3CHO (3) CH_3COCH_3 (4) HCHO

Q.11 Which of the following on heating with aqueous KOH, produces acetaldehyde ? [AIEEE-2009]

- (1) $\text{CH}_2\text{ClCH}_2\text{Cl}$ (2) CH_3CHCl_2 (3) CH_3COCl (4) $\text{CH}_3\text{CH}_2\text{Cl}$

Q.12 In Cannizzaro reaction given below :-

[AIEEE-2009]



the slowest step is :-

- (1) The abstraction of proton from the carboxylic group
 (2) The deprotonation of PhCH_2OH
 (3) The attack of $:\ddot{\text{O}}\text{H}^-$ at the carboxyl group
 (4) The transfer of hydride to the carbonyl group

Q.13 One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is :-

[AIEEE-2010]

- (1) Ethene (2) Propene (3) 1-Butene (4) 2-Butene

Q.14 Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of :-

[AIEEE-2011]

- (1) An isopropyl group (2) An acetylenic triple bond
 (3) Two ethylenic double bonds (4) A vinyl group

Q.15 Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify 'A' from the following compounds :-

[AIEEE-2011]

- (1) 2-Methyl - 1- pentene (2) 1-Pentene
 (3) 2-Pentene (4) 2-Methyl-2-pentene

Q.16 Trichloroacetaldehyde was subjected to assumed Cannizzaro's reaction by using NaOH. The mixture of the products contains sodium trichloroacetate and another compound. The other compound is :-

[AIEEE-2011]

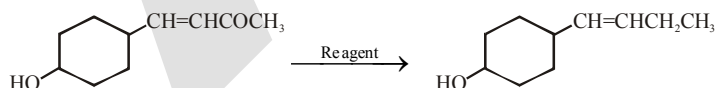
- (1) 2,2,2-Trichloropropanol (2) Chloroform
 (3) 2,2,2-Trichloroethanol (4) Trichloromethanol

Q.17 Silver Mirror test is given by which one of the following compounds?

[AIEEE-2011]

- (1) Formaldehyde (2) Benzophenone (3) Acetaldehyde (4) Acetone

Q.18 In the given transformation, which of the following is the most appropriate reagent ? [AIEEE-2012]



- (1) NaBH_4 (2) $\text{NH}_2\text{NH}_2, \text{OH}^-$ (3) $\text{Zn} - \text{Hg} / \text{HCl}$ (4) $\text{Na}, \text{Liq. NH}_3$

Q.19 Iodoform can be prepared from all except :-

[AIEEE-2012]

- (1) Isobutyl alcohol (2) Ethyl methyl ketone
 (3) Isopropyl alcohol (4) 3-Methyl-2-butanone

Q.20 A compound with molecular mass 180 is acylated with CH_3COCl to get a compound with molecular mass 390. The number of amino groups present per molecule of the former compound is :-

- (1) 2 (2) 5 (3) 4 (4) 6 [JEE(Main)-2013]

Q.21 The major organic compound formed by the reaction of 1, 1, 1-trichloroethane with silver powder is :-

[JEE(Main)-2014]

- (1) 2-Butyne (2) 2-Butene (3) Acetylene (4) Ethene

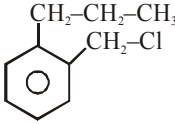
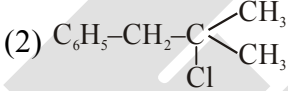
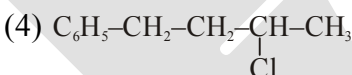
Q.22 The most suitable reagent for the conversion of $R-CH_2-OH \rightarrow R-CHO$ is :-

[JEE(Main)-2014]

- (1) CrO_3 (2) PCC (Pyridinium chlorochromate)
(3) $KMnO_4$ (4) $K_2Cr_2O_7$

Q.23 A compound A with molecular formula $C_{10}H_{13}Cl$ gives a white precipitate on adding silver nitrate solution. A on reacting with alcoholic KOH gives compound B as the main product. B on ozonolysis gives C and D. C gives Cannizzaro reaction but not aldol condensation. D gives aldol condensation but not Cannizzaro reaction. A is :

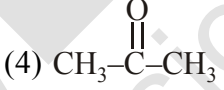
[JEE(Main)-2015]

- (1)  (2) 
(3) $C_6H_5-CH_2-CH_2-CH_2-CH_2-Cl$ (4) 

Q.24 In the reaction sequence

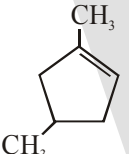
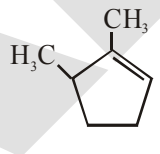
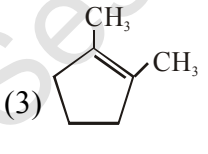
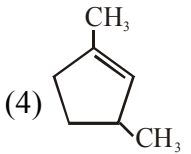
[JEE(Main)-2015]

$2CH_3CHO \xrightarrow{OH^-} A \xrightarrow{\Delta} B$; the product B is:-

- (1) $CH_3-CH=CH-CHO$ (2) $CH_3-CH_2-CH_2-CH_3$
(3) $CH_3-CH_2-CH_2-CH_2-OH$ (4) 

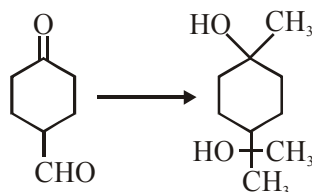
Q.25 Which compound would give 5-keto-2-methyl hexanal upon ozonolysis?

[JEE(Main) 2015]

- (1)  (2)  (3)  (4) 

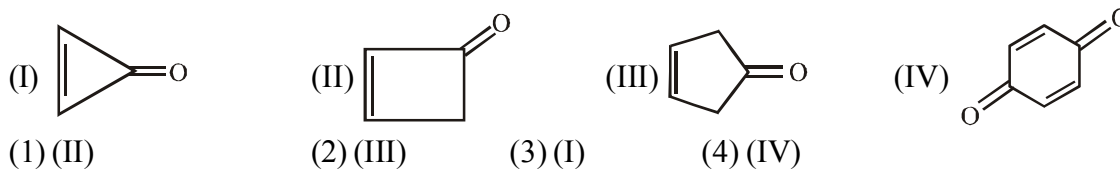
Q.26 The correct sequence of reagents for the following conversion will be :-

[JEE(Main)-2017]



- (1) $[Ag(NH_3)_2]^+ OH^-$, H^+/CH_3OH , CH_3MgBr
(2) CH_3MgBr , H^+/CH_3OH , $[Ag(NH_3)_2]^+ OH^-$
(3) CH_3MgBr , $[Ag(NH_3)_2]^+ OH^-$, H^+/CH_3OH
(4) $[Ag(NH_3)_2]^+ OH^-$, CH_3MgBr , H^+/CH_3OH

Q.27 Which of the following compounds will show highest dipole moment ? [JEE(Main-on-line)-2017]



Q.28 Glucose on prolonged heating with HI gives :

[JEE(Main)-2018]

- (1) 1-Hexene (2) Hexanoic acid (3) 6-iodohexanal (4) n-Hexane

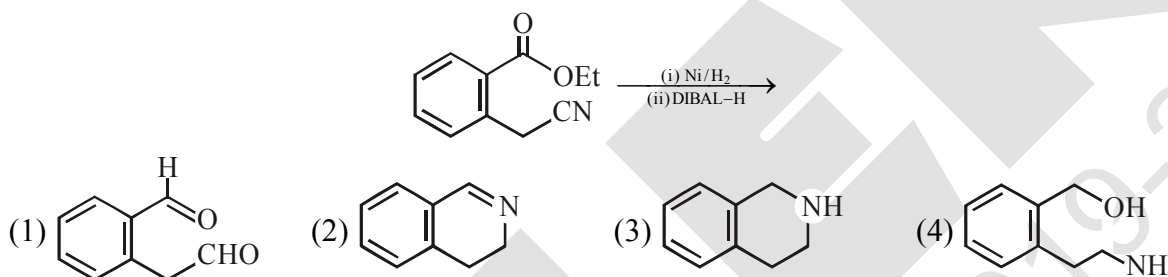
Q.29 Which of the following compounds will most readily be dehydrated to give alkene under acidic condition?

[JEE(Main-on-line)-2018]

- (1) 4-Hydroxypentan-2-one (2) 2-Hydroxycyclopentanone
(3) 3-Hydroxypentan-2-one (4) 1-Pentanol

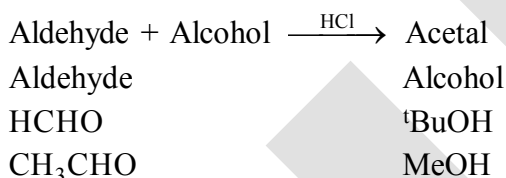
Q.30. The major product of the following reaction is:

[JEE-Main(January)-2019]



Q.31 In the following reaction

[JEE-Main(January)-2019]

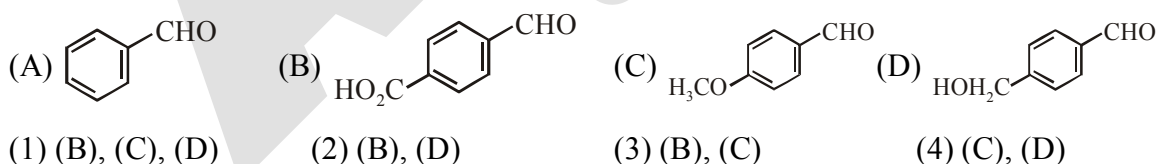


The best combinations is :

- (1) HCHO and MeOH (2) HCHO and ^tBuOH
(3) CH₃CHO and MeOH (4) CH₃CHO and ^tBuOH

Q.32 The aldehydes which will not form Grignard product with one equivalent Grignard reagents are :

[JEE-Main(January)-2019]



EXERCISE-IV # (A) (OBJECTIVE QUESTIONS)

Q.1 Which of the following has the most acidic hydrogen: [IIT 2000]

- (A) 3-hexanone (B) 2,4-hexanedione
(C) 2,5-hexanedione (D) 2,3-hexanedione

Q.2 A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives:

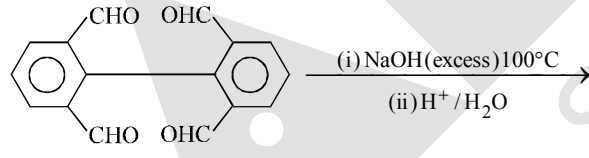
- (A) benzyl alcohol and sodium formate (B) sodium benzoate and methyl alcohol
(C) sodium benzoate and sodium formate (D) benzyl alcohol and methyl alcohol [IIT 2001]

Q.3 1-propanol & 2-propanol can be best distinguished by : [IIT 2001]

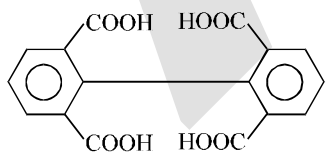
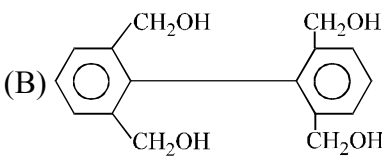
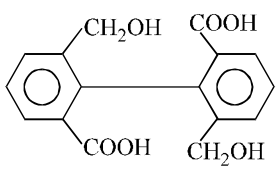
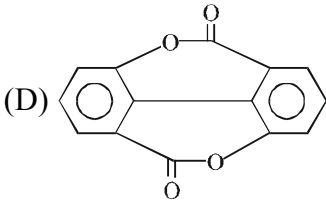
- (A) Oxidation with alkaline KMnO_4 followed by reaction with Fehling solution
(B) Oxidation with acidic dichromate followed by reaction with Fehling solution
(C) Oxidation by heating with copper followed by reaction with Fehling solution
(D) Oxidation with concentrated H_2SO_4 followed by reaction with Fehling solution

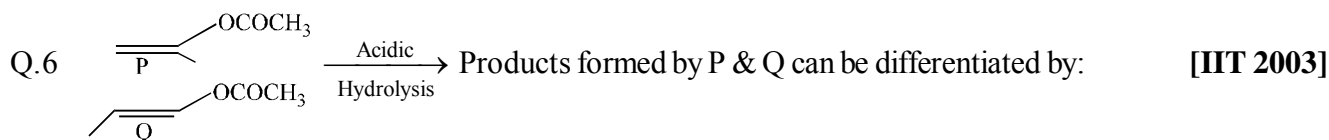
Q.4 Compound A (molecular formula $\text{C}_3\text{H}_8\text{O}$) is treated with acidified potassium dichromate to form a product B (molecular formula $\text{C}_3\text{H}_6\text{O}$). B forms a shining silver mirror on warming with ammonical silver nitrate. B when treated with an aqueous solution of $\text{H}_2\text{NCONHNH}_2$, HCl and sodium acetate gives a product C. Identify the structure of C. [IIT 2002]

- (A) $\text{CH}_3\text{CH}_2\text{CH} = \text{NNHCONH}_2$ (B) $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} = \text{NNHCONH}_2$
(C) $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} = \text{NCONHNH}_2$ (D) $\text{CH}_3\text{CH}_2\text{CH} = \text{NCONHNH}_2$

Q.5  [IIT 2003]

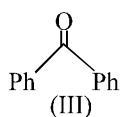
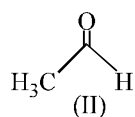
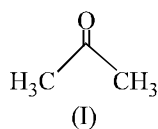
any one of the products formed is :

- (A)  (B) 
(C)  (D) 

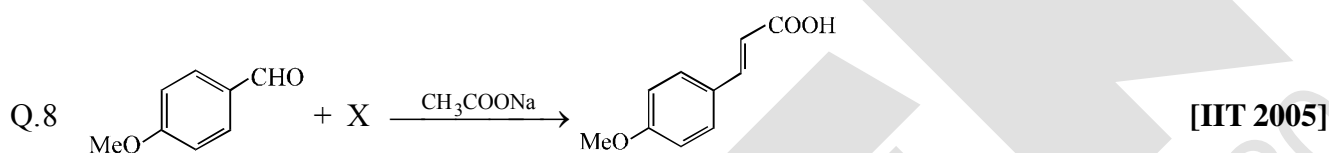


- (A) 2, 4 DNP (B) Lucas reagent (ZnCl_2) conc. HCl
(C) NaHSO_3 (D) Fehlings solution

Q.7 The order of reactivity of phenyl Magnesium Bromide with the following compounds is [IIT 2004]



- (A) $\text{II} > \text{III} > \text{I}$ (B) $\text{I} > \text{III} > \text{II}$ (C) $\text{II} > \text{I} > \text{III}$ (D) All react with the same rate



What is X?

- (A) CH_3COOH (B) BrCH_2COOH (C) $(\text{CH}_3\text{CO})_2\text{O}$ (D) $\text{CHO}-\text{COOH}$

Q.9 The smallest ketone and its next homologue are reacted with NH_2OH to form oxime.

- (A) Two different oximes are formed (B) Three different oximes are formed
(C) Two oximes are optically active (D) All oximes are optically active [IIT 2006]

Q.10 Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound E. Compound E on further treatment with aqueous KOH yields compound F. Compound F is



Q.11 **Statement-1** : Glucose gives a reddish-brown precipitate with Fehling's solution.

because

Statement-2 : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

[IIT-JEE(ADV.)- 2007]

Q.12 Match the compounds/ion in column I with their properties/ reaction in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

[IIT-JEE(ADV.)- 2007]

Column I

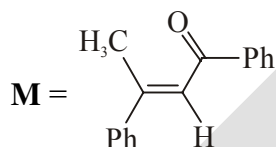
- (A) C_6H_5CHO
(B) $CH_3C \equiv CH$
(C) CN^-
(D) I^-

Column II

- (P) gives precipitate with 2,4-dinitrophenylhydrazine
(Q) gives precipitate with $AgNO_3$
(R) is a nucleophile
(S) is involved in cyanohydrin formation

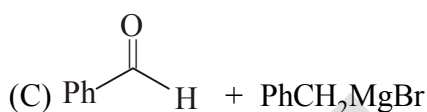
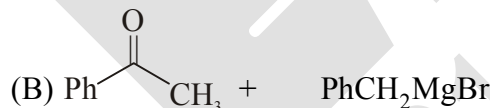
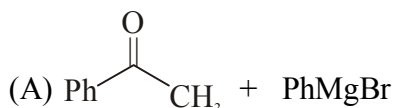
Paragraph for Question No. 13 to 15

A tertiary alcohol **H** upon acid catalysed dehydration gives a product **I**. Ozonolysis of **I** leads to compounds **J** and **K**. Compound **J** upon reaction with KOH gives benzyl alcohol and a compound **L**, whereas **K** on reaction with KOH gives only **M**.



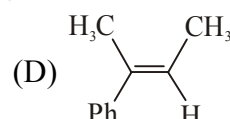
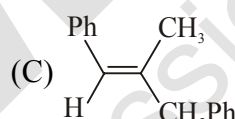
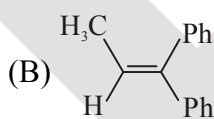
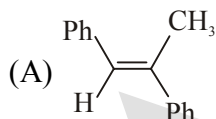
Q.13 Compound **H** is formed by the reaction of

[IIT-JEE(ADV.)- 2008]



Q.14 The structure of compound **I** is

[IIT-JEE(ADV.)- 2008]



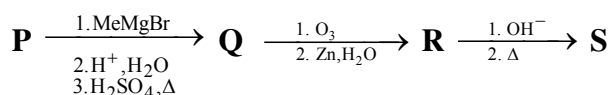
Q.15 The structures of compounds **J**, **K** and **L**, respectively, are

[IIT-JEE(ADV.)- 2008]



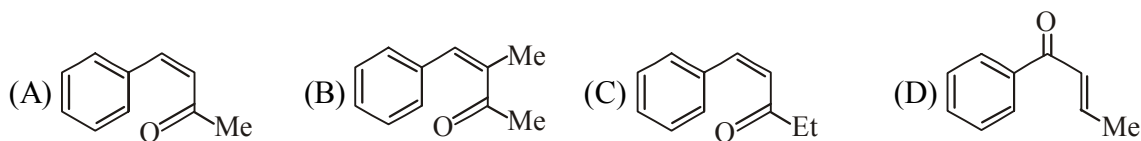
Paragraph for Question Nos. 16 to 38

A carbonyl compound **P**, which gives positive iodoform test, undergoes reaction with $MeMgBr$ followed by dehydration to give an olefin **Q**. Ozonolysis of **Q** leads to a dicarbonyl compound **R**, which undergoes intramolecular aldol reaction to give predominantly **S**.

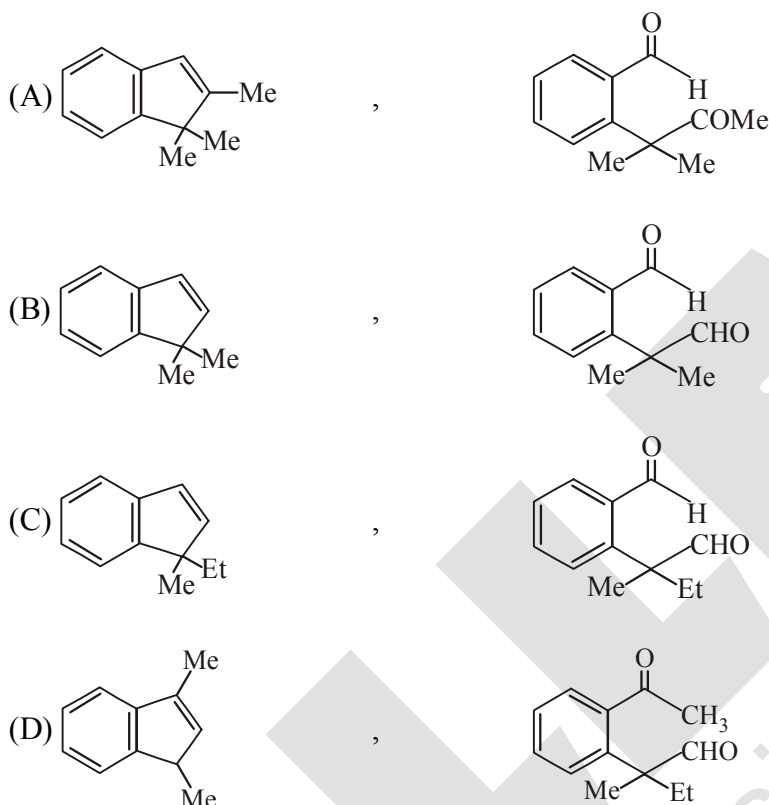


Q.16 The structure of the carbonyl compound **P** is

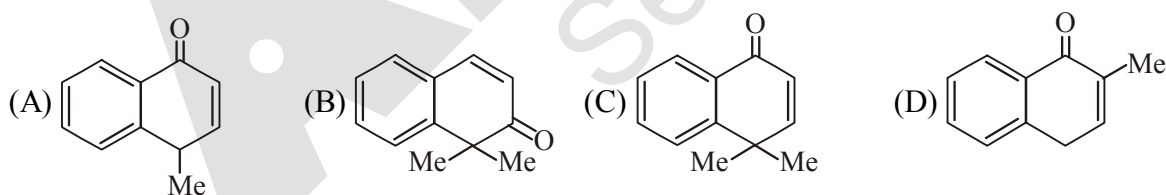
[IIT-JEE(ADV.)- 2009]

Q.17 The structure of the products **Q** and **R**, respectively, are

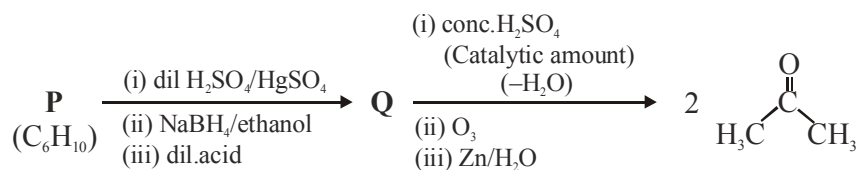
[IIT-JEE(ADV.)- 2009]

Q.18 The structure of the product **S** is

[IIT-JEE(ADV.)- 2009]

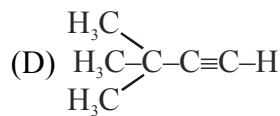
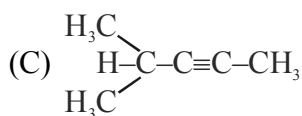
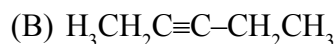
**Paragraph for Questions Nos. 19 to 20**

An acyclic hydrocarbon **P**, having molecular formula C_6H_{10} , gave acetone as the only organic product through the following sequence of reactions, in the which **Q** is an intermediate organic compound.



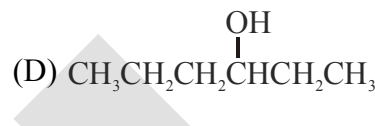
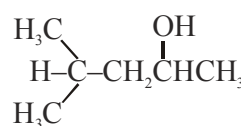
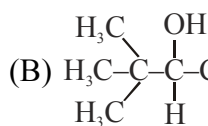
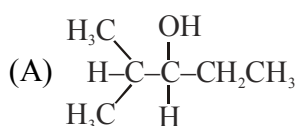
Q.19 The structure of compound P is -

[IIT-JEE(ADV.)- 2011]

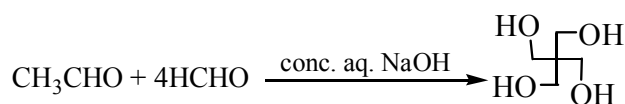


Q.20 The structure of the compound Q is -

[IIT-JEE(ADV.)- 2011]



Q.21 The number of aldol reaction(s) that occurs in the given transformation is [IIT-JEE(ADV.)- 2012]



(A) 1

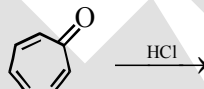
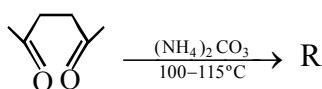
(B) 2

(C) 3

(D) 4

Q.22 Among P, Q, R and S, the aromatic compound(s) is / are :

[IIT-JEE(ADV.)- 2013]



(A) P

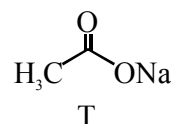
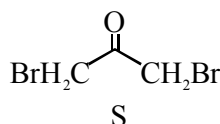
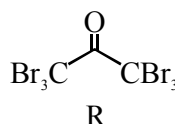
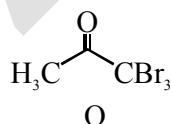
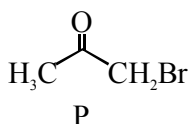
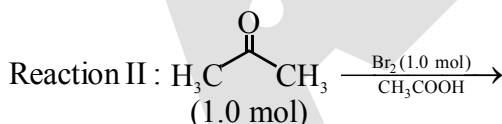
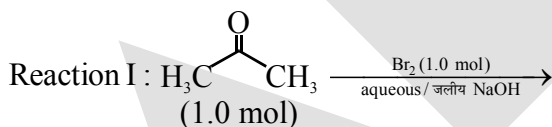
(B) Q

(C) R

(D) S

Q.23 After completion of the reactions (I and II), the organic compound(s) in the reaction mixtures is(are)

[IIT-JEE(ADV.)- 2013]



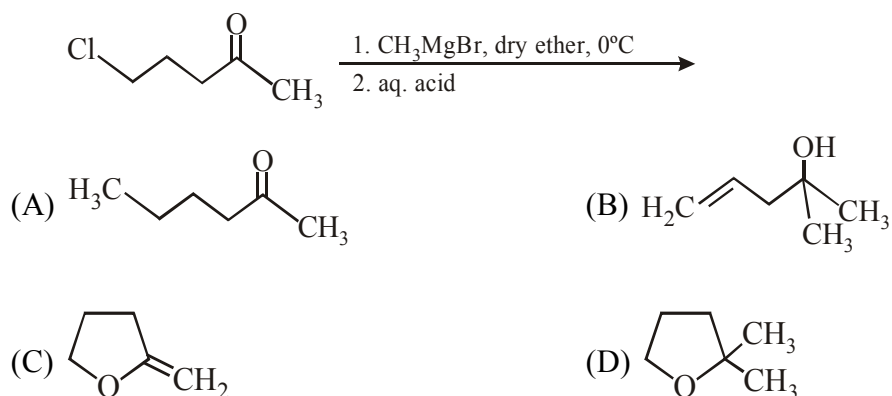
(A) Reaction I : P and Reaction II : P

(B) Reaction I : U, acetone and Reaction II : Q acetone

(C) Reaction I : T, U, acetone and Reaction II : P

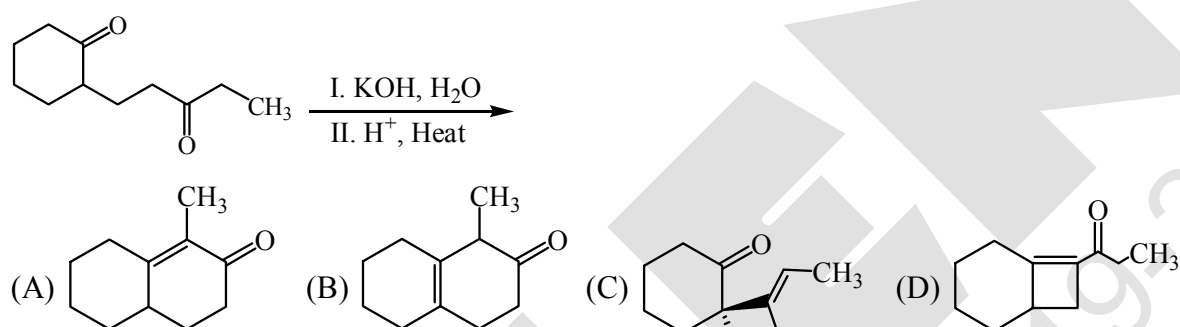
(D) Reaction I : R, acetone and Reaction II : S acetone

Q.24 The major product in the following reaction is [IIT-JEE(ADV.)- 2014]



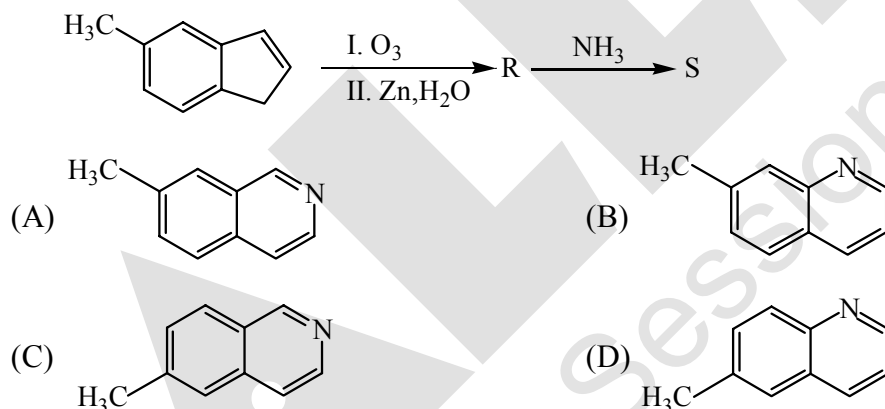
Q.25 The major product of the following reaction is -

[IIT-JEE(ADV.)- 2015]



Q.26 In the following reactions, the product S is -

[IIT-JEE(ADV.)- 2015]



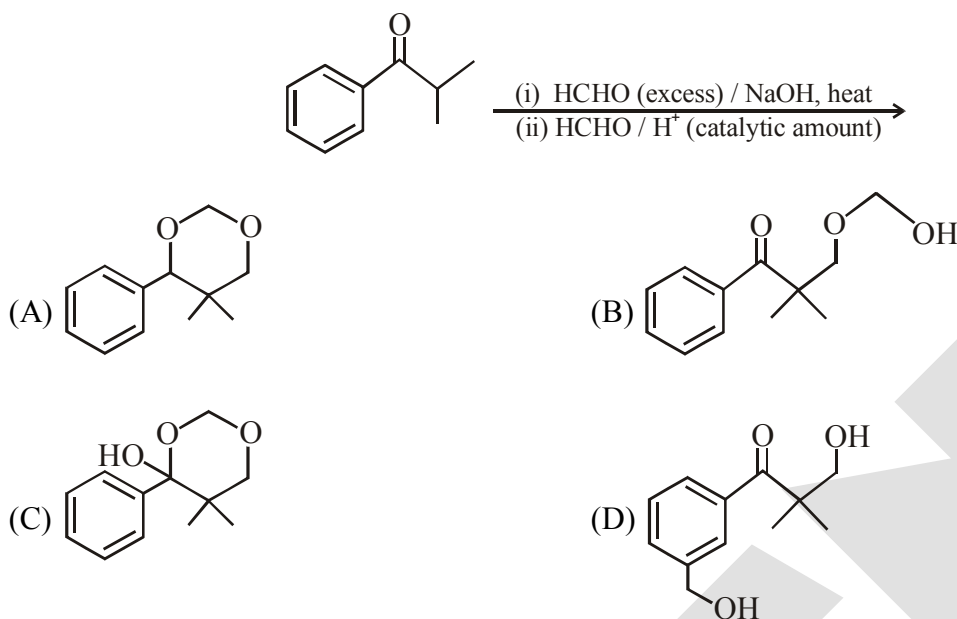
Q.27 Positive Tollen's test is observed for :

[IIT-JEE(ADV.)- 2016]



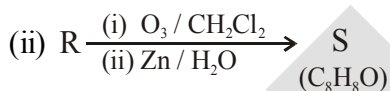
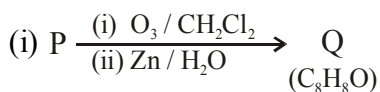
Q.28 The major product of the following reaction sequence is :

[IITJEE(ADV.)-2016]

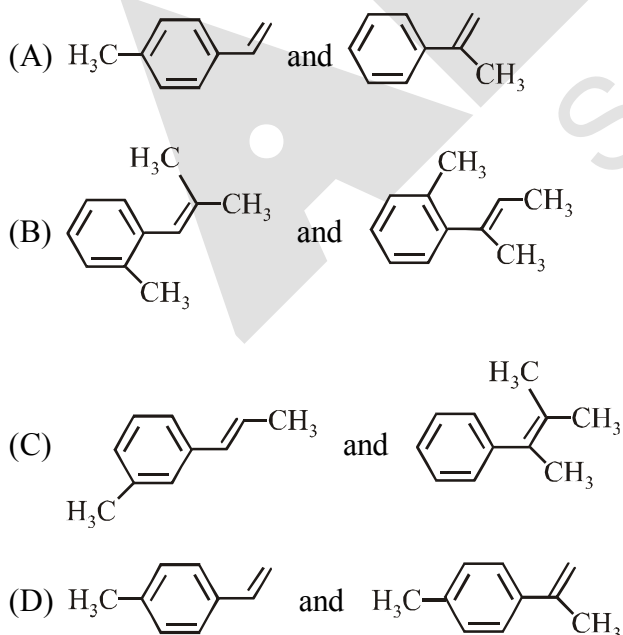


Q.29 Compound **P** and **R** upon ozonolysis produce **Q** and **S**, respectively. The molecular formula of **Q** and **S** is C_8H_8O . **Q** undergoes Cannizzaro reaction but not haloform reaction, whereas **S** undergoes haloform reaction but not Cannizzaro reaction :

[IIT-JEE(ADV.)- 2017]

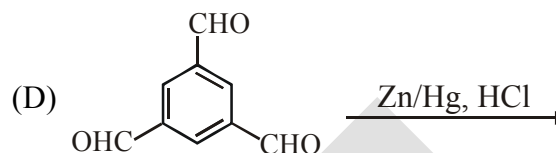
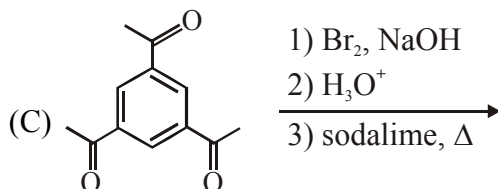


The option(s) with suitable combination of **P** and **R**, respectively, is(are)



30. The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is (are)

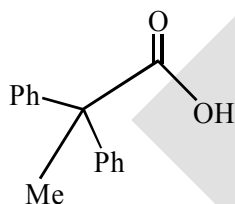
[IIT-JEE(ADV.)- 2018]



31. The desired product X can be prepared by reacting the major product of the reactions in LIST-I with one or more appropriate reagents in LIST-II.

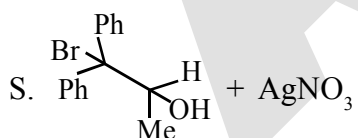
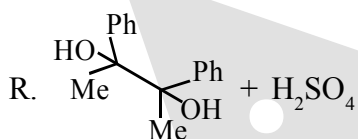
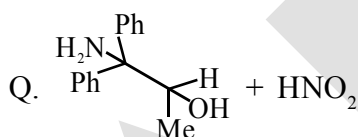
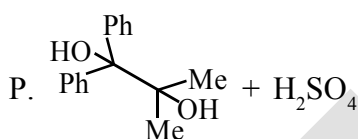
[IIT-JEE(Adv.)-2018]

(given, order of migratory aptitude: aryl > alkyl > hydrogen)



X

LIST-I



LIST-II

1. I_2, NaOH

2. $[\text{Ag}(\text{NH}_3)_2]\text{OH}$

3. Fehling solution

4. HCHO, NaOH

5. NaOBr

The correct option is

(A) $\text{P} \rightarrow 1; \text{Q} \rightarrow 2,3; \text{R} \rightarrow 1,4; \text{S} \rightarrow 2,4$

(B) $\text{P} \rightarrow 1,5; \text{Q} \rightarrow 3,4; \text{R} \rightarrow 4,5; \text{S} \rightarrow 3$

(C) $\text{P} \rightarrow 1,5; \text{Q} \rightarrow 3,4; \text{R} \rightarrow 5; \text{S} \rightarrow 2,4$

(D) $\text{P} \rightarrow 1,5; \text{Q} \rightarrow 2,3; \text{R} \rightarrow 1,5; \text{S} \rightarrow 2,3$

ANSWER-KEY

EXERCISE # I

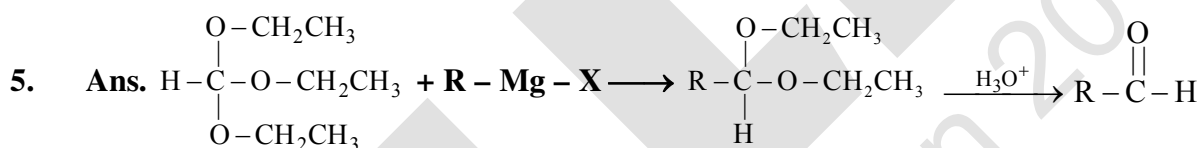
1	Ans. (A)	2	Ans. (B)	3	Ans. (C)	4	Ans. (C)
5	Ans. (C)	6	Ans. (D)	7	Ans. (A)	8	Ans. (C)
9	Ans. (C)	10	Ans. (C)	11	Ans. (A)	12	Ans. (D)
13	Ans. (C)	14	Ans. (C)	15	Ans. (B)	16	Ans. (D)
17	Ans. (C)	18	Ans. (A)	19	Ans. (D)	20	Ans. (D)
21	Ans. (C)	22	Ans. (D)	23	Ans. (C)	24	Ans. (B)
25	Ans. (A)	26	Ans. (D)	27	Ans. (C)	28	Ans. (D)
29	Ans. (C)	30	Ans. (D)				

EXERCISE # II

1	Ans. (A,B)	2	Ans. (C,D)	3	Ans. (A,B,C)	4	Ans. (B,C)
5	Ans. (A,B)	6	Ans. (A,B,C,D)	7	Ans. (A,B,C)	8	Ans. (B,C)
9	Ans. (A,D)	10	Ans. (D)	11	Ans. (D)	12	Ans. (B)
13	Ans. (B)	14	Ans. (A,B,C)	15	Ans. (A)→R ; (B)→Q,S ; (C)→S ; (D)→P		

EXERCISE # III

- Ans. (A) P,Q,S ; (B) P,Q,S ; (C) P,Q,S ; (D) P,Q,S
- Ans. $3 > 2 > 1 > 4$
- Ans. (C)
- Ans. (A)

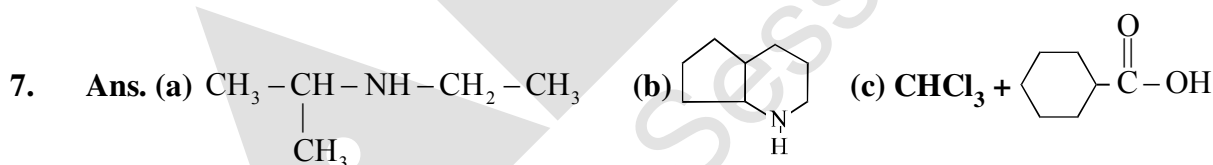


Ethyl orthoformate

Acetal

aldehyde

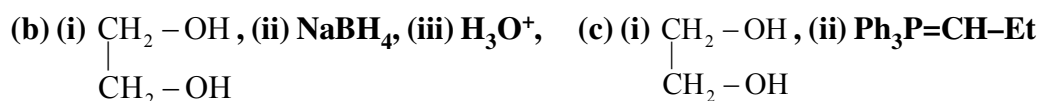
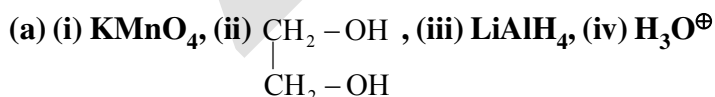
- Ans. H^+/Br_2 ; H_2/Ni ; NaOH



- Ans. (B,C)

- Ans. (B)

- Ans.

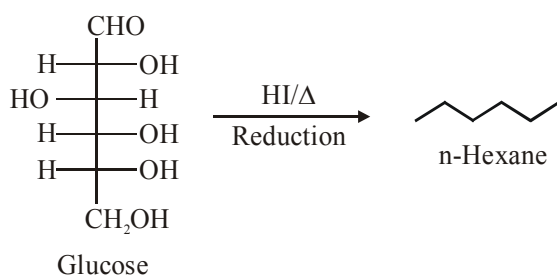


- Ans. (B)
- Ans. (D)

EXERCISE # IV (MAINS)

- | | | | |
|-----------------|--------------|--------------|--------------|
| 1. Ans. (4) | 2. Ans. (2) | 3. Ans. (1) | 4. Ans. (2) |
| 5. Ans. (4) | 6. Ans. (2) | 7. Ans. (4) | 8. Ans. (3) |
| 9. Ans. (4) | 10. Ans. (4) | 11. Ans. (2) | 12. Ans. (4) |
| 13. Ans. (4) | 14. Ans. (4) | 15. Ans. (4) | 16. Ans. (3) |
| 17. Ans. (1, 3) | 18. Ans. (2) | 19. Ans. (1) | 20. Ans. (2) |
| 21. Ans. (1) | 22. Ans. (2) | 23. Ans. (2) | 24. Ans. (1) |
| 25. Ans. (4) | 26. Ans. (1) | 27. Ans. (3) | |
| 28. Ans. (4) | | | |

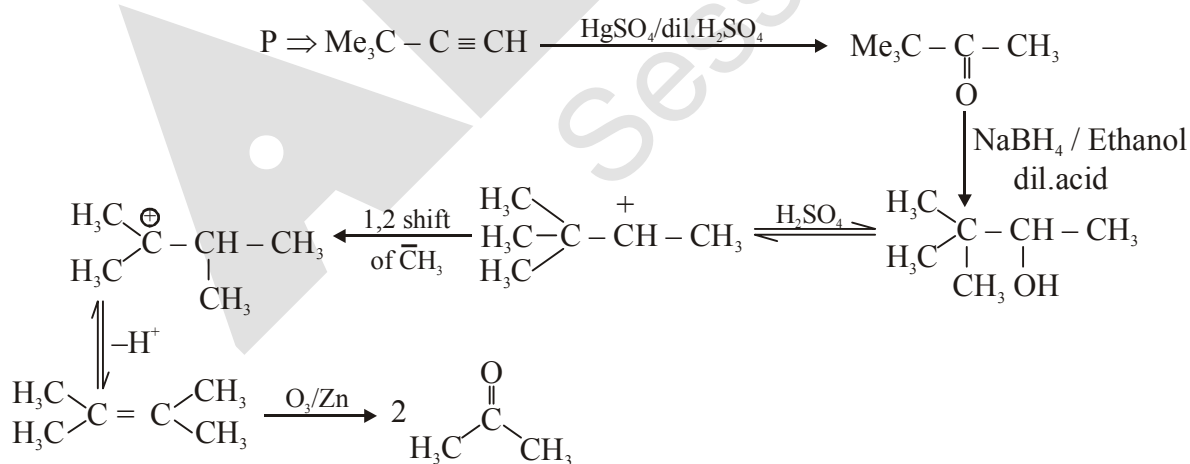
Sol.



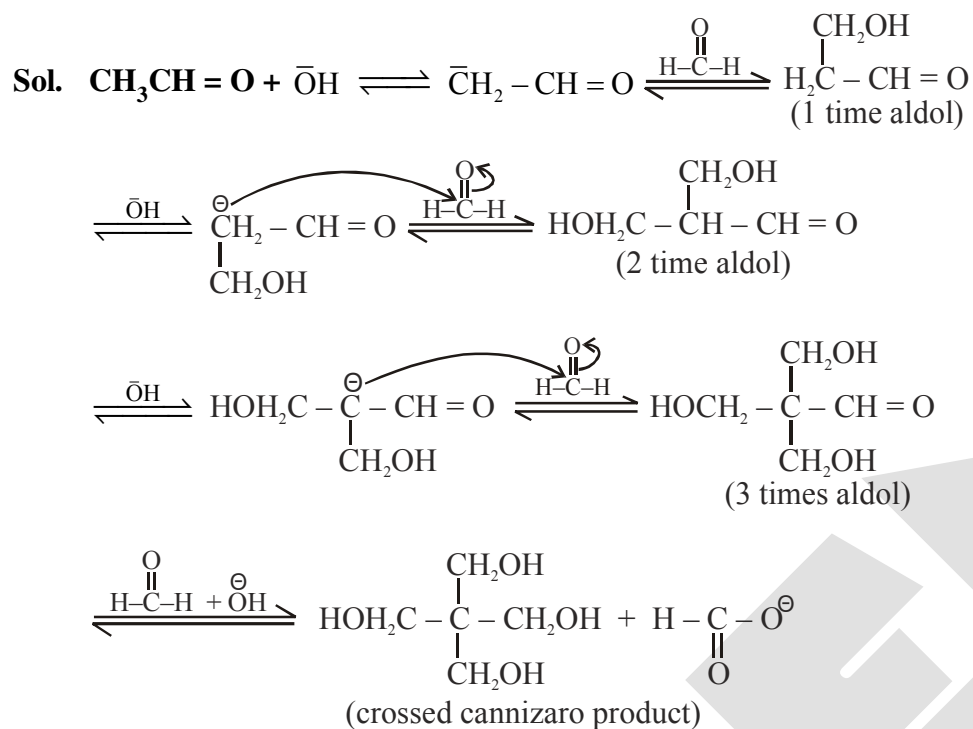
- | | | | |
|--------------|--------------|--------------|--------------|
| 29. Ans. (1) | 30. Ans. (2) | 31. Ans. (1) | 32. Ans. (2) |
|--------------|--------------|--------------|--------------|

EXERCISE-IV # (A) (OBJECTIVE QUESTIONS)

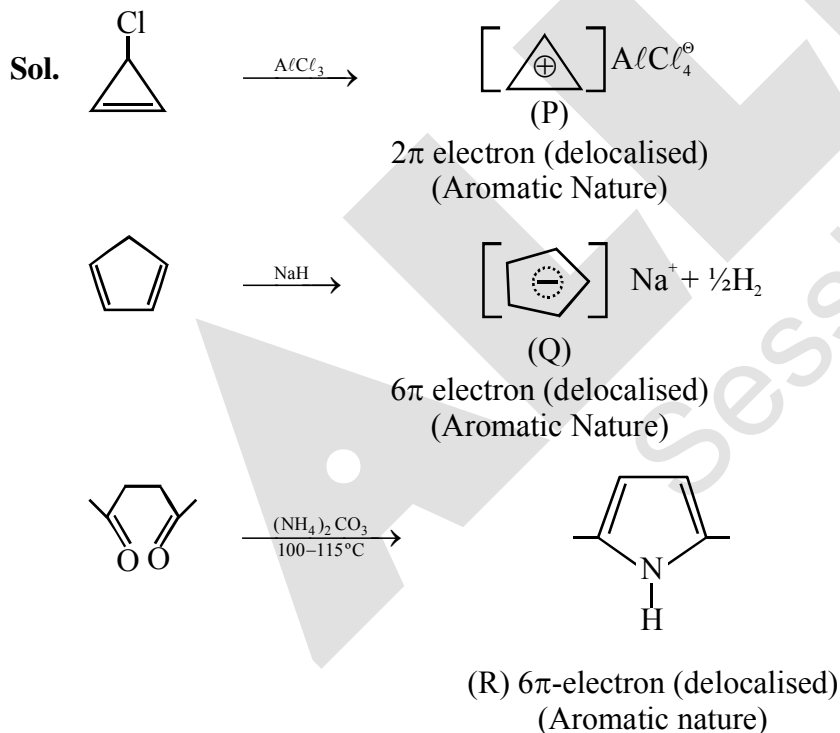
- | | | | |
|---|--------------|--------------|--------------|
| 1. Ans. (B) | 2. Ans. (A) | 3. Ans. (C) | 4. Ans. (A) |
| 5. Ans. (C) | 6. Ans. (D) | 7. Ans. (C) | 8. Ans. (C) |
| 9. Ans. (B) | 10. Ans. (A) | 11. Ans. (C) | |
| 12. Ans. (A) P,S; (B) Q; (C) Q,R,S; (D) Q,R | 13. Ans. (B) | 14. Ans. (A) | |
| 15. Ans. (D) | 16. Ans. (B) | 17. Ans. (A) | 18. Ans. (B) |
| 19. Ans. (D) | | | |
| 20. Ans. (B) | | | |



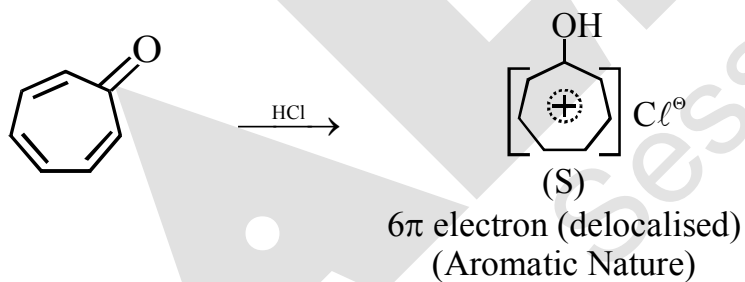
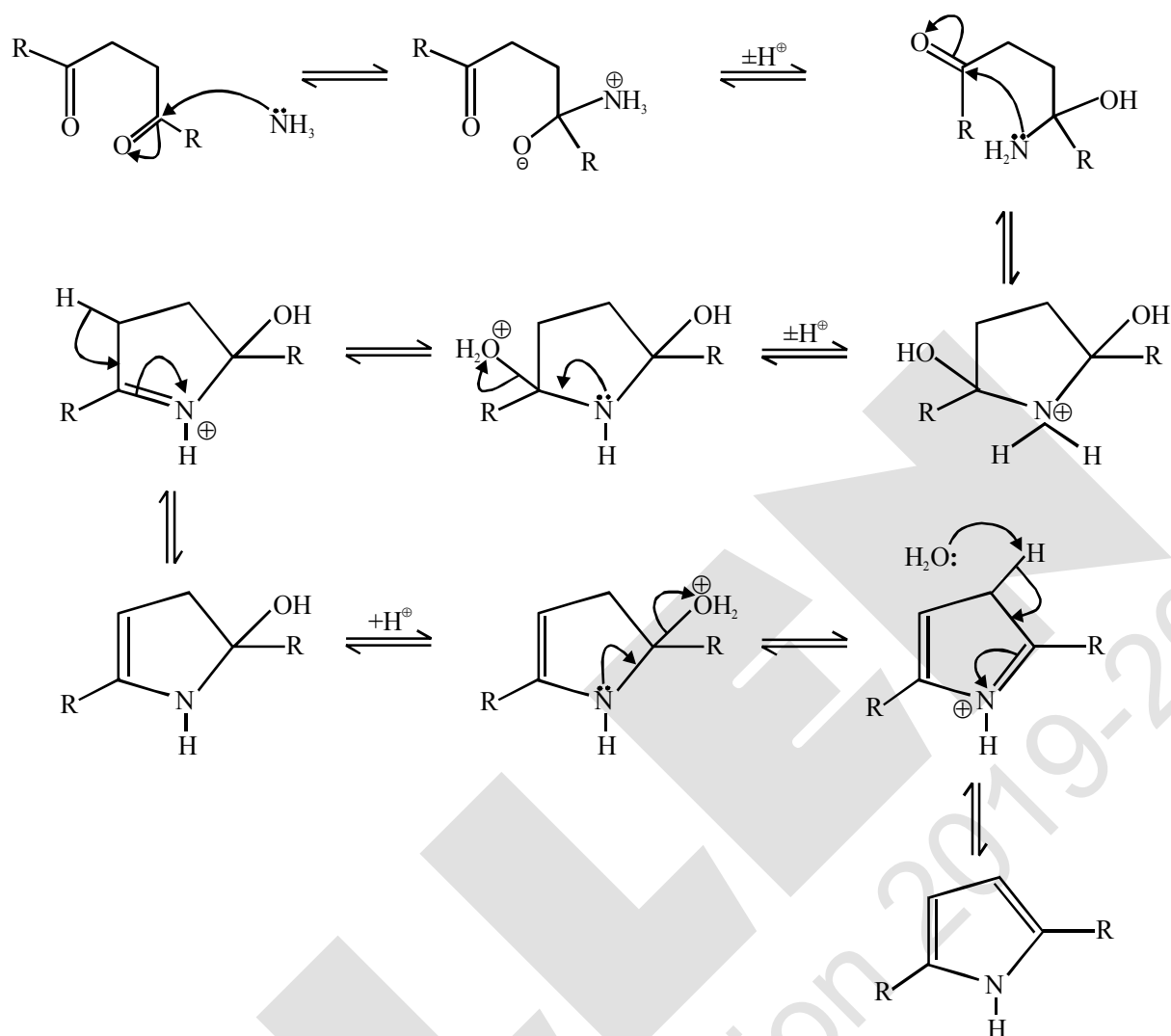
21. Ans. (C)



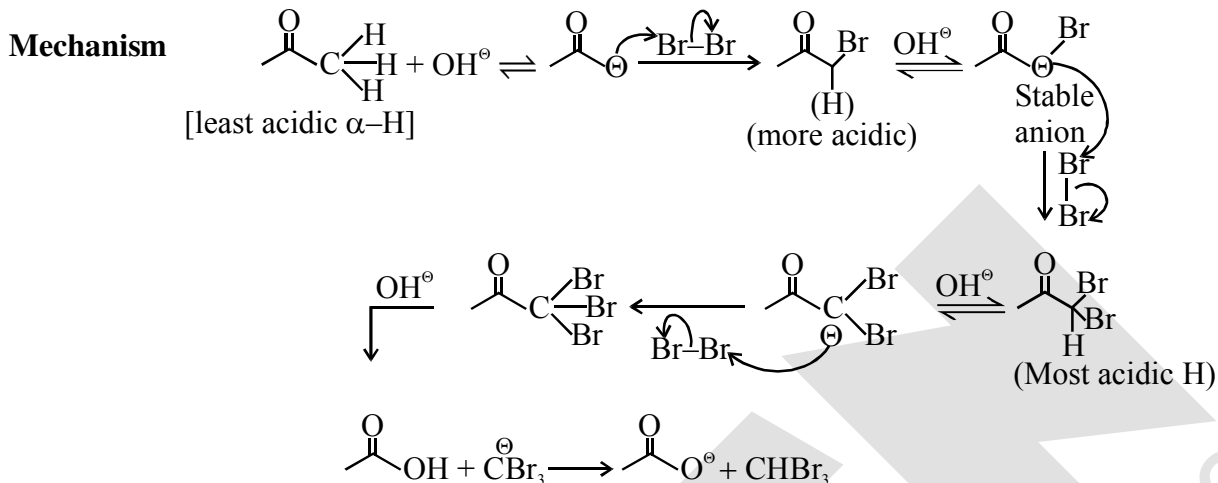
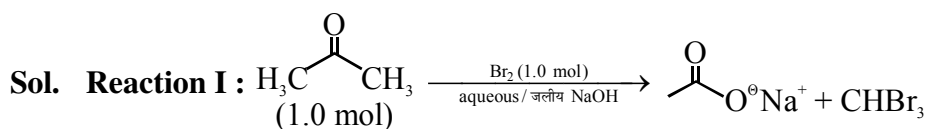
22. Ans. (A,B,C,D)



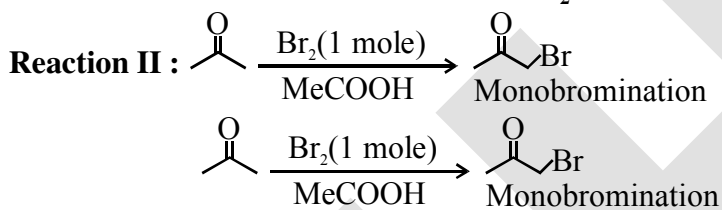
Mechanism :



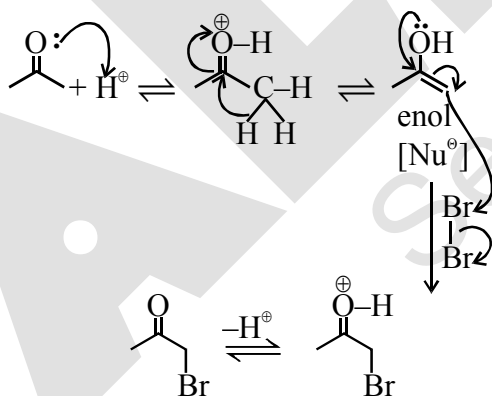
23. Ans. (C)



In basic medium halogenation does not stop with replacement of just one hydrogen and poly halogenation takes place because α -haloketones are more reactive towards base and haloform reaction takes place. In above reaction Br_2 is limiting agents.

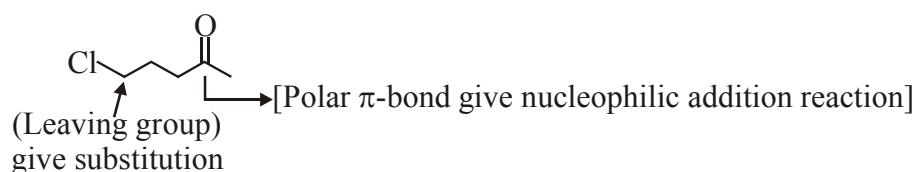


Mechanism :

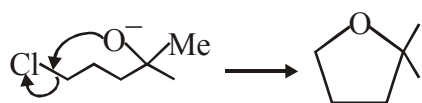


Further bromination is less favourable because of less amount of Br_2

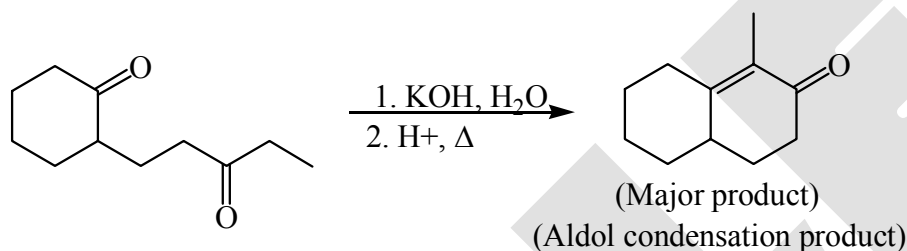
24. Ans. (D)

(i) Grignard prefer to give nucleophilic addition on polar π -bond and form anion intermediate.

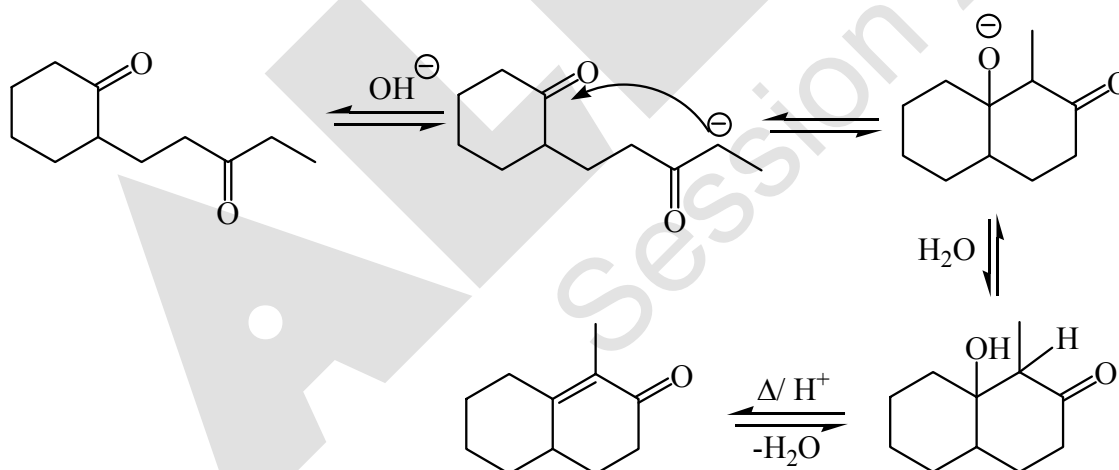
(ii) In next step anion give intramolecular nucleophilic substitution reaction & form 5 membered ring.



25. Ans. (A)

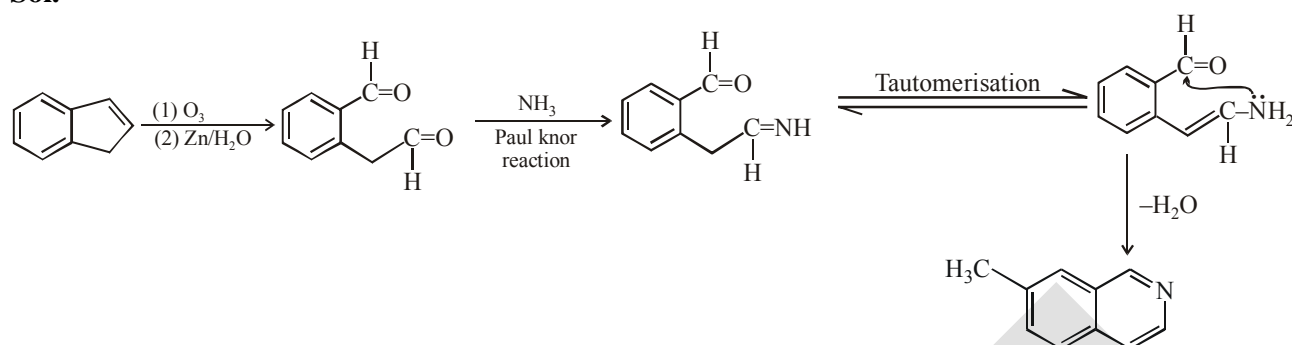


Mechanism :



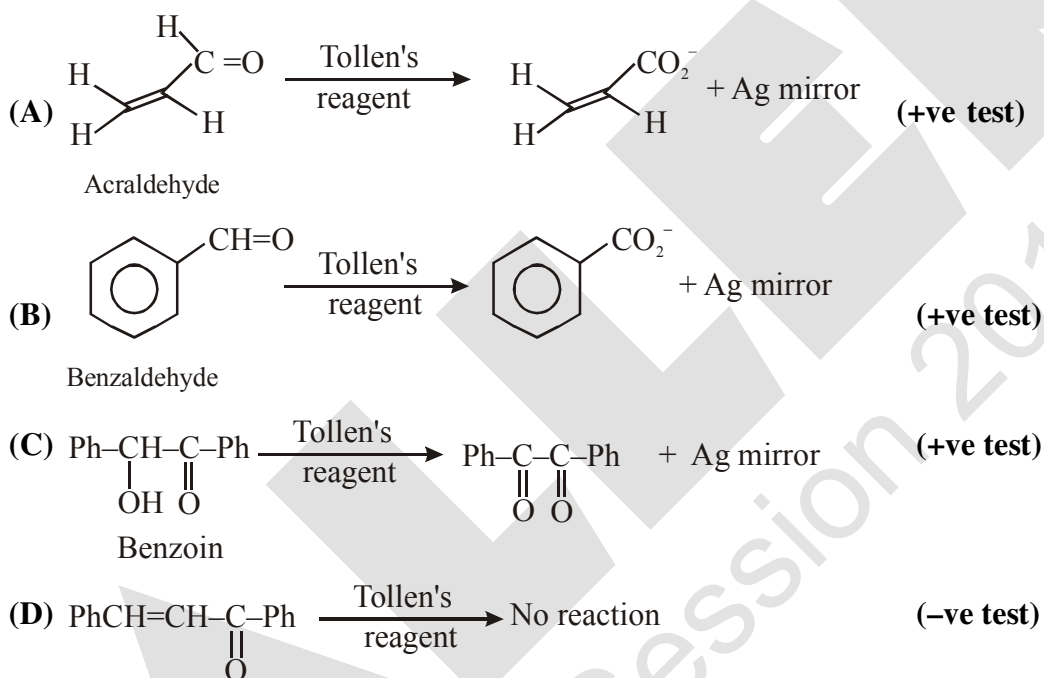
26. Ans.(A)

Sol.



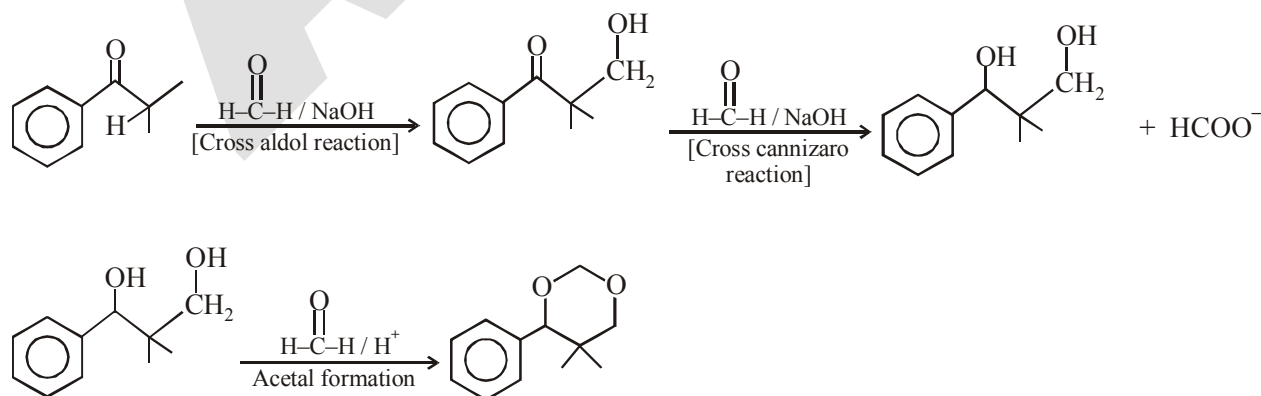
27. Ans. (A,B,C)

Sol. Tollens's test is given by compounds having aldehyde group. Also α -hydroxy carbonyl gives positive tollens's test.

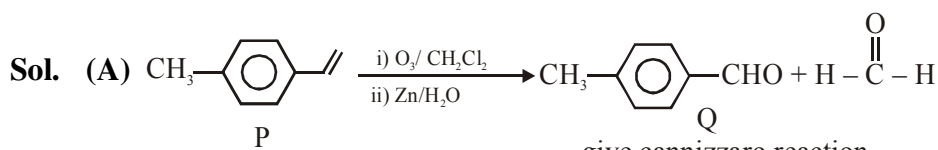


28. Ans. (A)

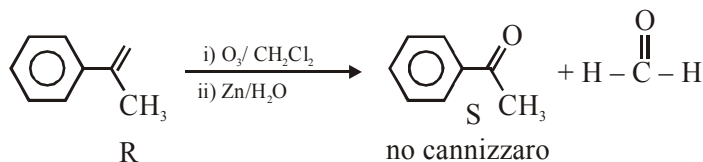
Sol.



29. Ans. (A,C)

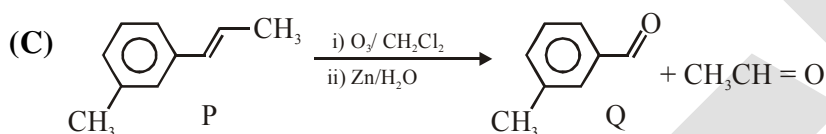


give cannizzaro reaction
but no haloform

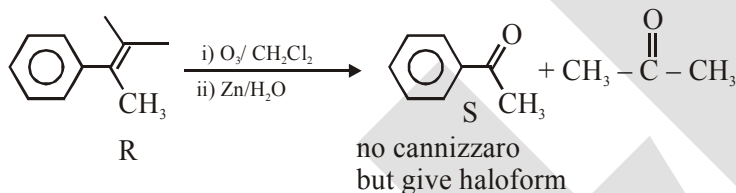


no cannizzaro
but give haloform

(B) Product of ozonolysis of R is having 9 carbon.



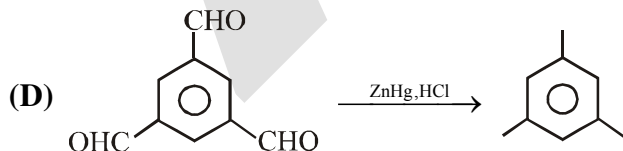
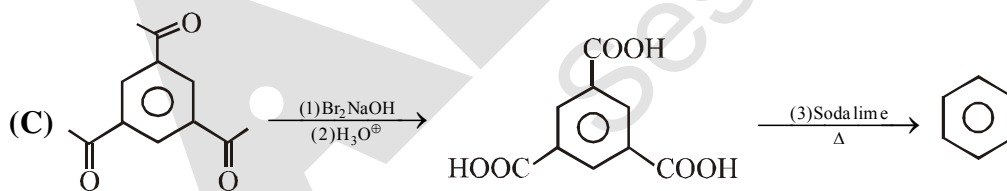
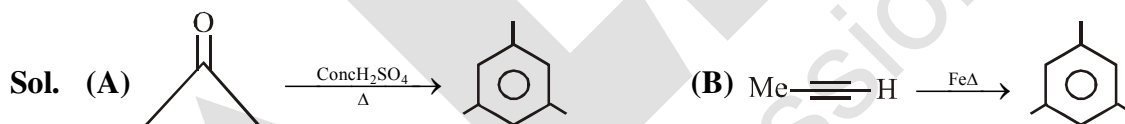
give cannizzaro reaction
but no haloform



no cannizzaro
but give haloform

(D) Product of ozonolysis of R is having 9 carbon.

30. Ans. (A,B,D)

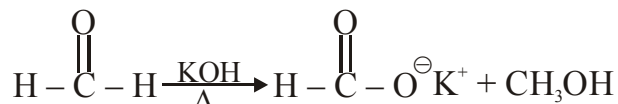


31. Ans. (D)

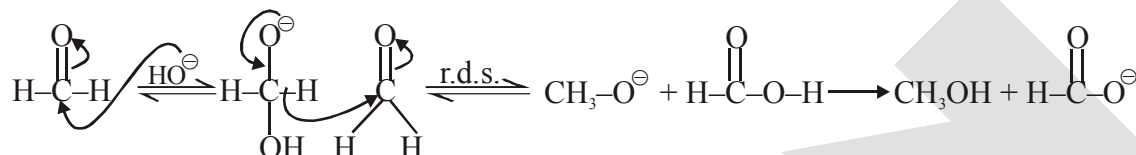
ALDOL & SIMILAR NAME REACTIONS

(1) CANNIZARO REACTION

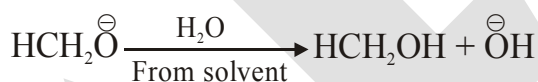
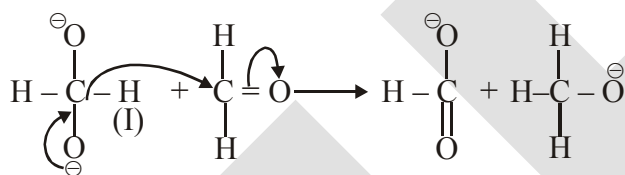
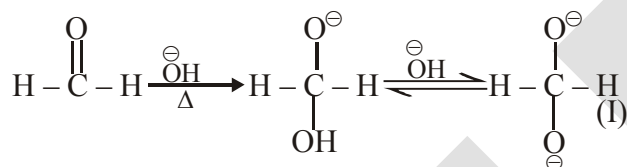
This reaction is given by aldehyde having no α -hydrogen in the presence of conc. NaOH/Δ or KOH/Δ .



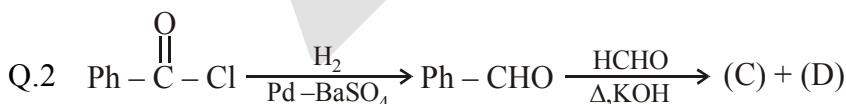
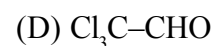
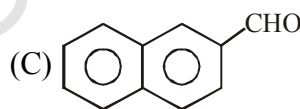
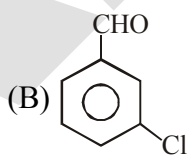
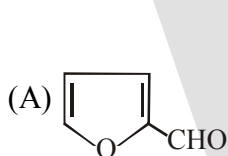
Mechanism :



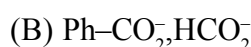
In the presence of a very strong concentration of alkali, aldehyde first forms a doubly charged anion (I) from which a hydride anion is transferred to the second molecule of the aldehyde to form acid and an alkoxide ion. Subsequently, the alkoxide ion acquires a proton from the solvent.

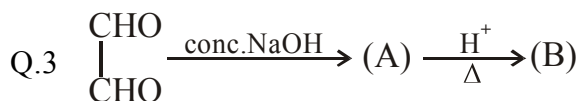


Q.1 Which of following will not undergo Cannizzaro reaction

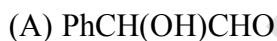
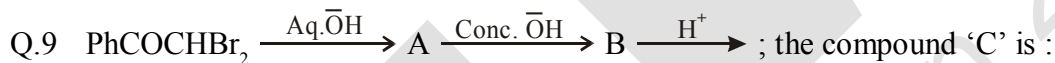
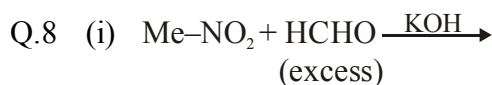
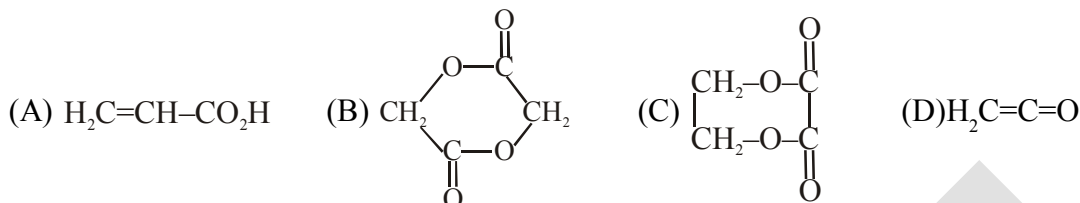


Product (C) & (D) are :



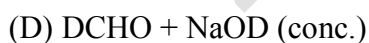
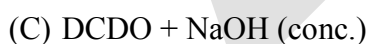


Product (B) is :

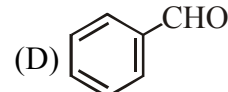
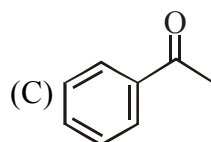
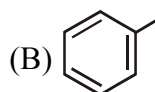
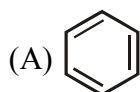
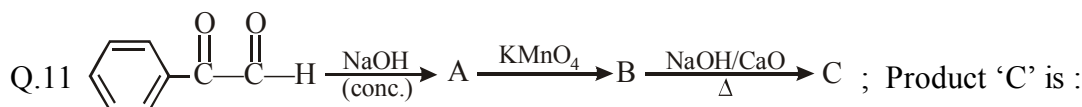
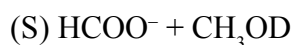
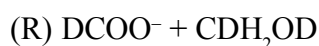
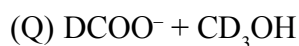
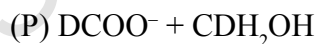


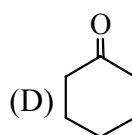
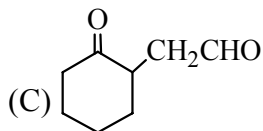
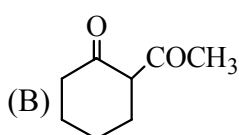
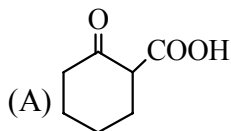
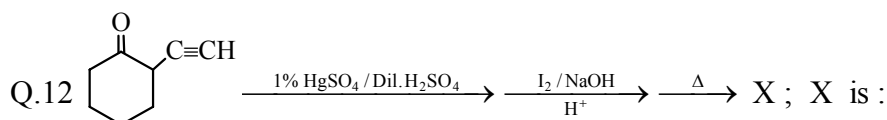
Q.10 Match the column :

Column - I

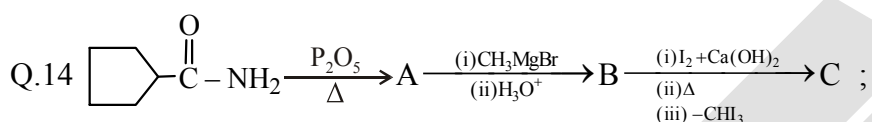
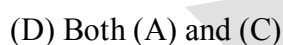
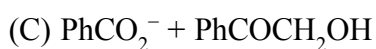
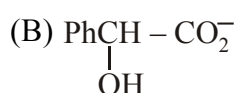


Column - II

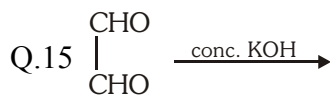
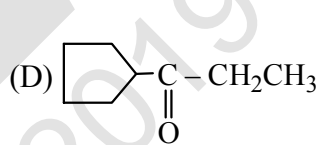
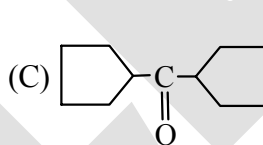
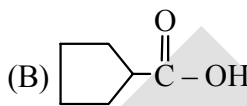
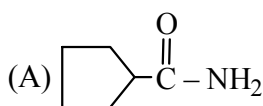




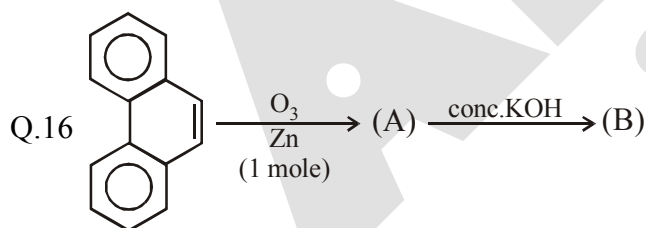
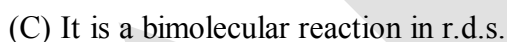
Q.13 The cannizzaro reaction of Ph COCHO forms the product(s)



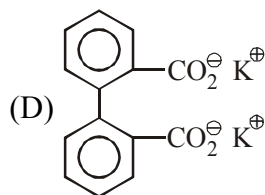
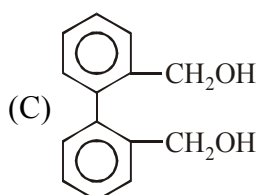
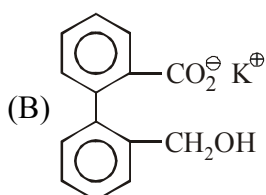
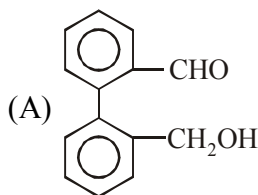
Product is:



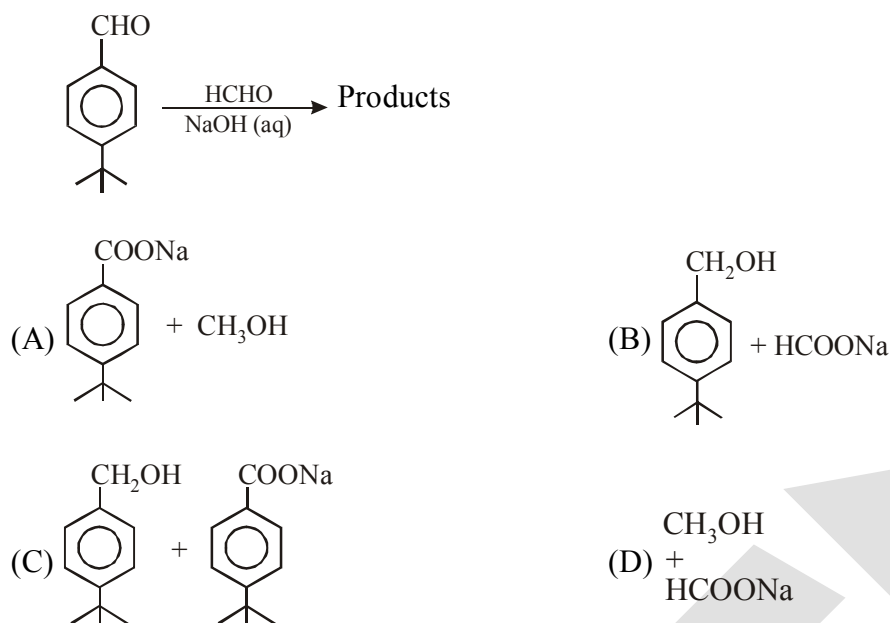
True about this reaction is / are



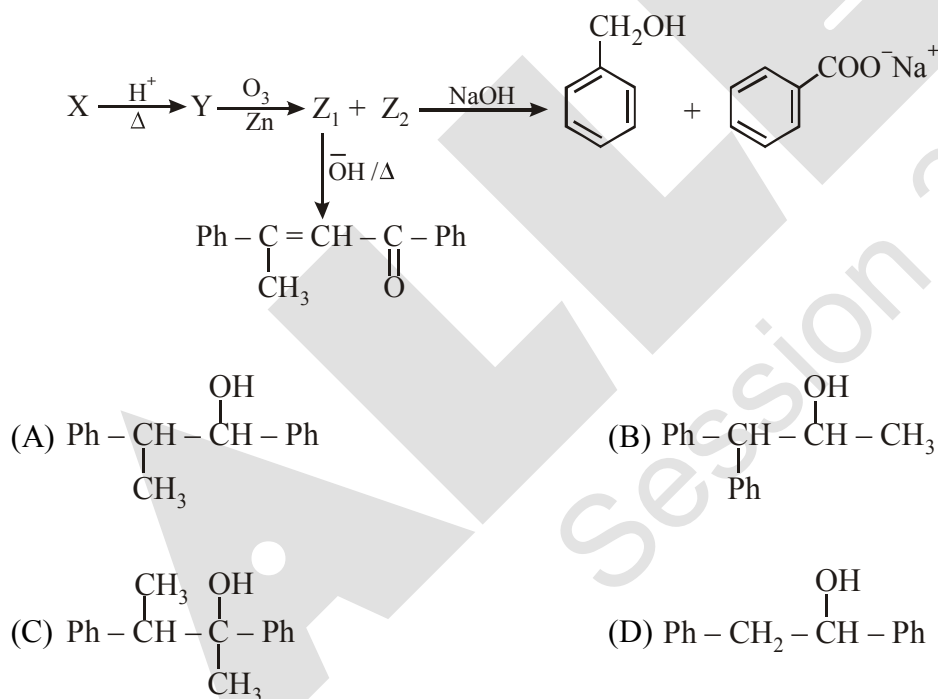
End product (B) of above reaction is :



Q.17 The major product pair of the following reaction will be :

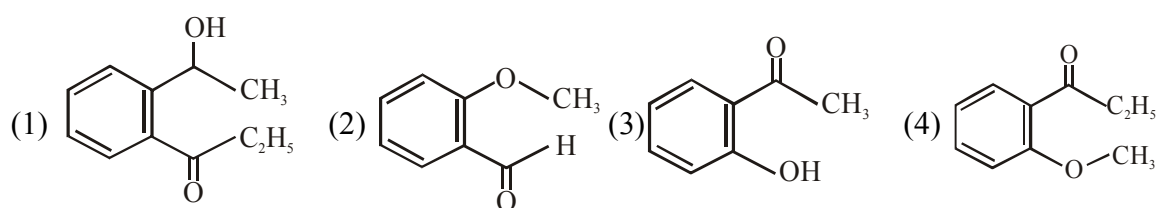


Q.18 Reactant 'X' will be :



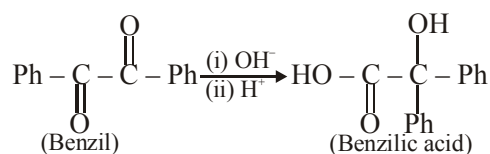
Q.19 An organic compound neither reacts with neutral ferric chloride solution nor with Fehling solution, It however, reacts with Grignard reagent and gives positive iodoform test. The compound is -

[JEE-MAIN (APRIL)-2019]

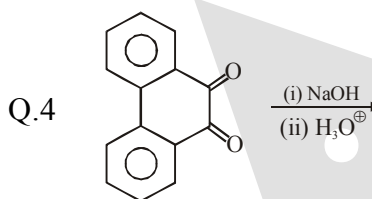
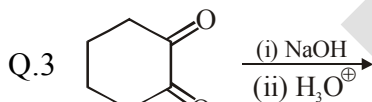
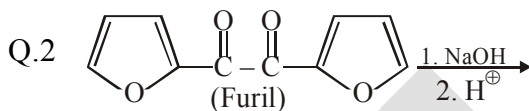
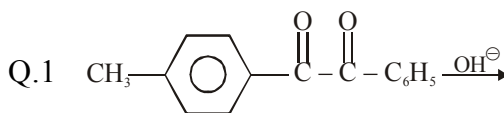
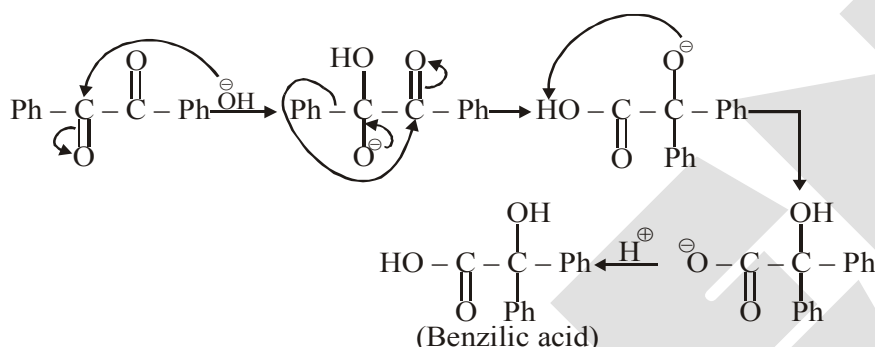


(2) BENZIL-BENZILIC REARRANGEMENT OR BENZILIC ACID REARRANGEMENT

The base catalysed reaction of 1,2-diketones to a salt of α -hydroxy carboxylic acid is known as Benzilic acid rearrangement, this reaction is mainly applicable when aryl group is present on both carbonyl carbons.

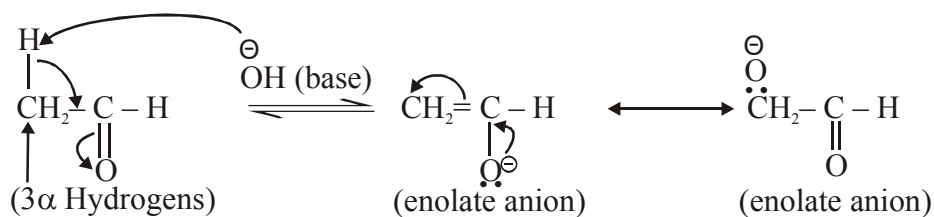


Mechanism :



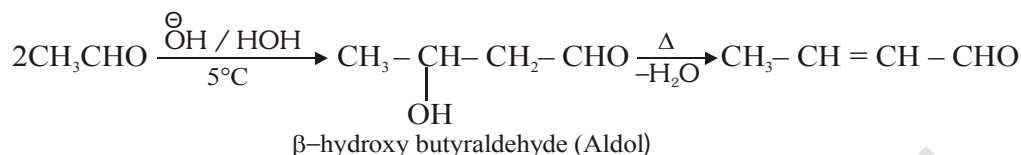
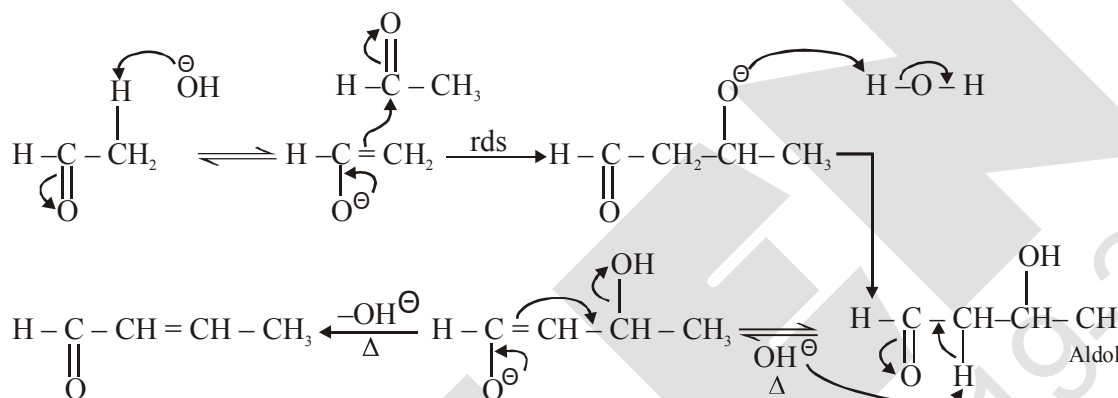
(3) ALDOL CONDENSATION

The α -hydrogen of carbonyl compounds are acidic due to the fact that the anion (enolate ion) is stabilized by resonance.



Base catalysed Aldol

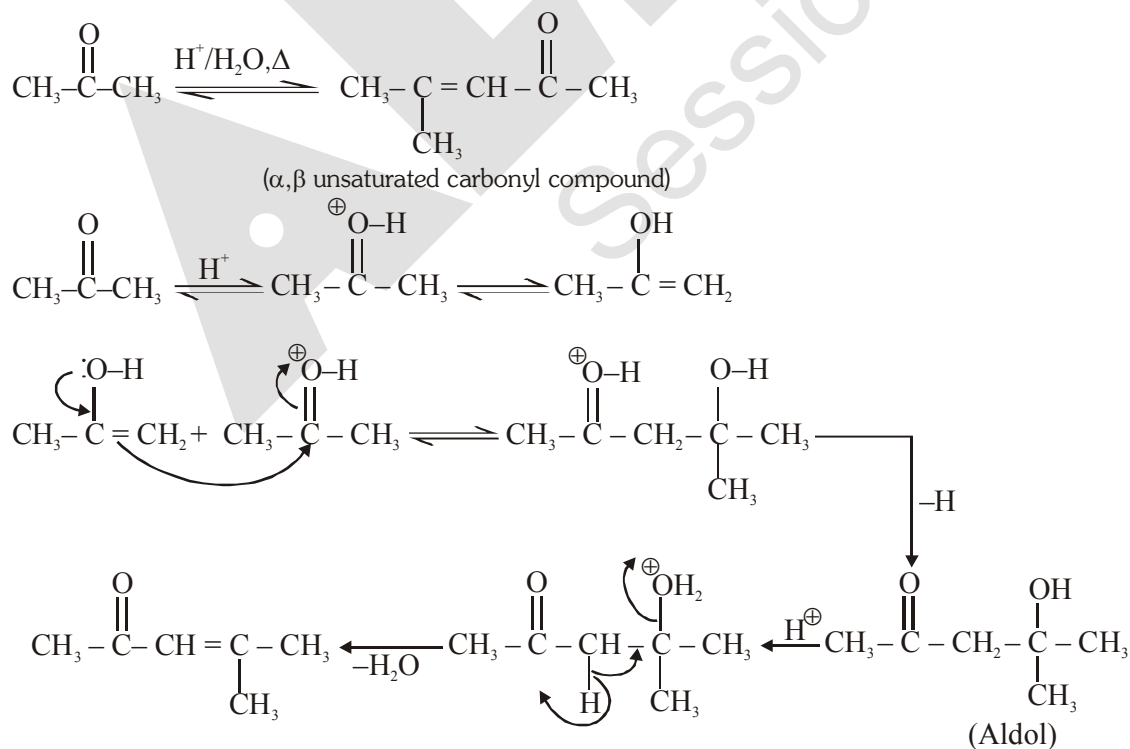
In aqueous base, two acetaldehyde molecules react to form β -hydroxy aldehyde called aldol. The reaction is called Aldol condensation. The enolate ion is the intermediate in the aldol condensation of aldehyde and ketone. Acetaldehyde for instance, forms a dimeric product aldol in presence of a dilute base ($\approx 10\%$ NaOH)

**Mechanism :**

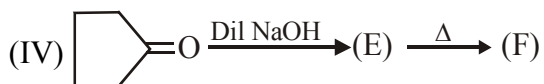
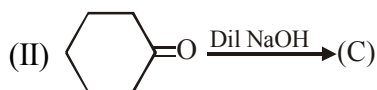
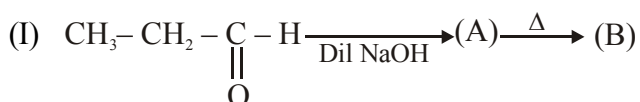
Aldols are stable and may be isolated. They, however can be dehydrated easily by heating the basic reaction mixture or by a separate acid catalyzed reaction. Thus if the above reaction is heated the product is dehydrated to 2-butenal (crotonaldehyde).

Acid catalysed Aldol

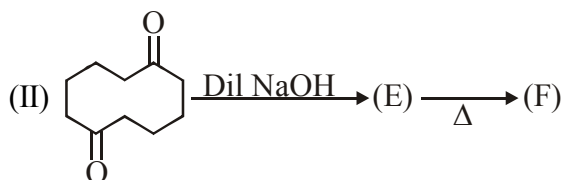
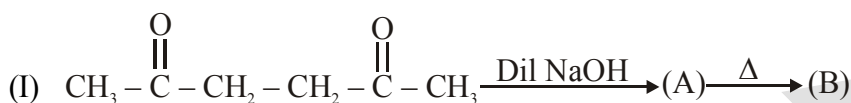
In acid catalysed aldol condensation enol form of carbonyl is the nucleophile in place of enolate.

Mechanism :

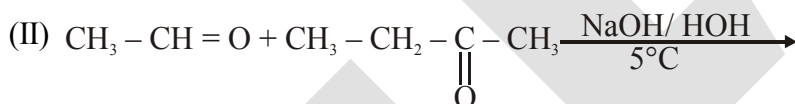
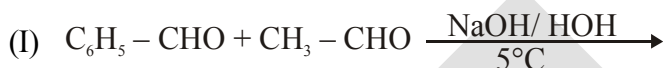
Q.1 Write the product and mechanism for given reactions.



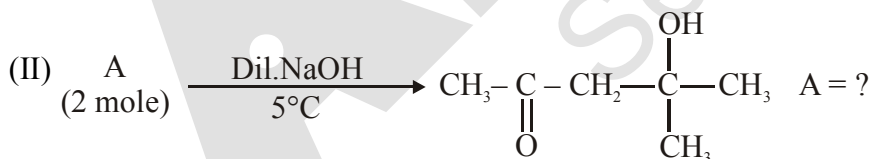
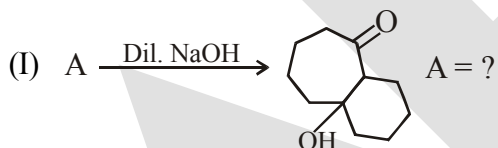
Q.2 Identify the intramolecular aldol product ?



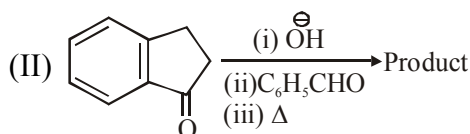
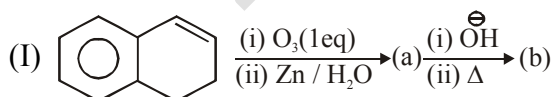
Q.3 Find out the total number of possible aldol products (including and excluding stereo products).



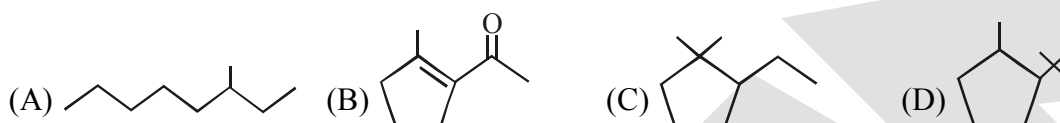
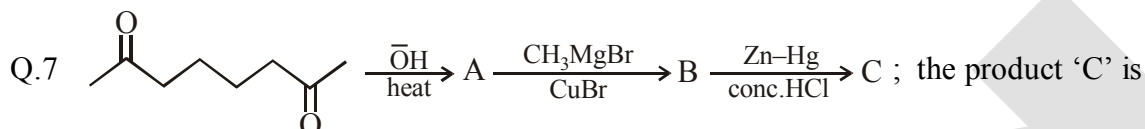
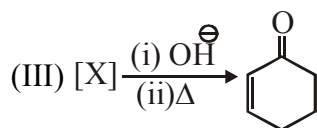
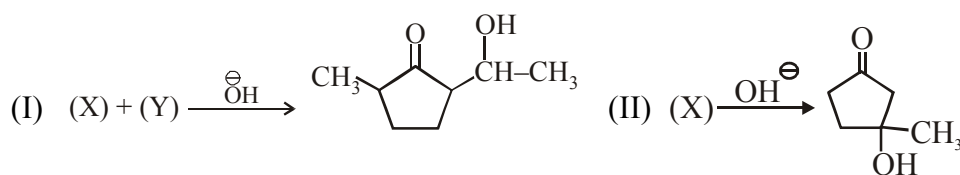
Q.4 Identify the structure of substrate ?



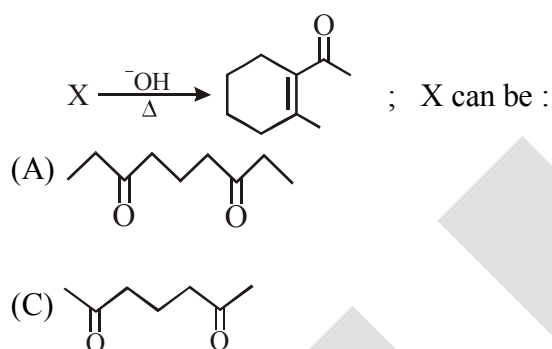
Q.5 Complete reaction sequence :



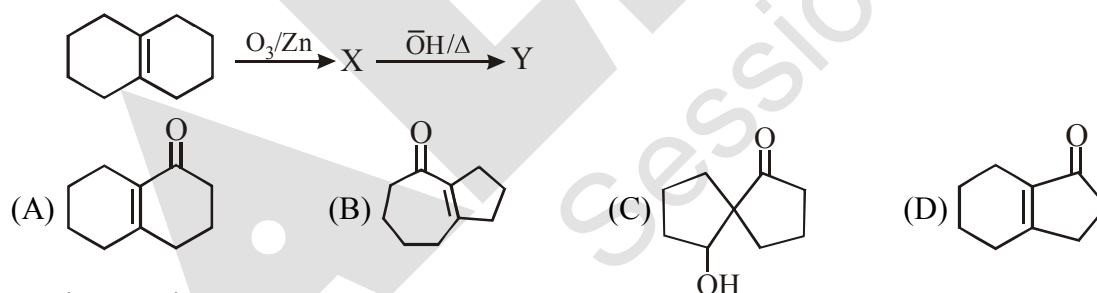
Q.6 Complete the following reactions :



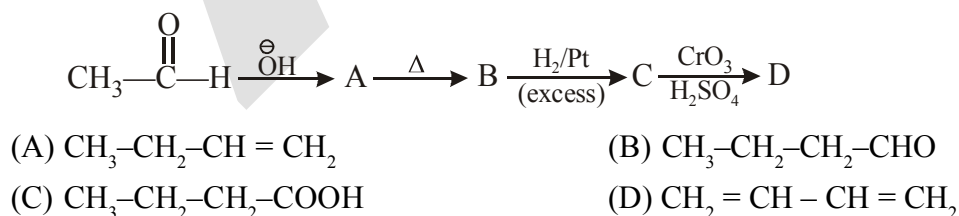
Q.8 Consider following intramolecular aldol condensation reaction :



Q.9 Product 'Y' formed in the given reaction is :



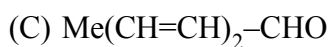
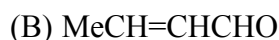
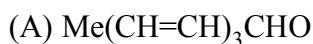
Q.10 Product 'D' is :



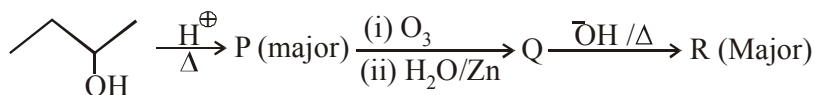
Q.11 $\text{CH}_3\text{CH}=\text{CHCHO} \xrightarrow[\text{aldol}]{\text{OH}^\ominus} \xrightarrow{\Delta} \text{A}$, A is :



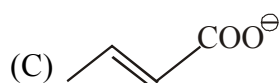
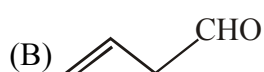
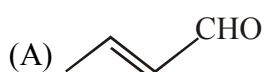
Q.12 $\text{MeCHO} \xrightarrow{\text{NaOH}, \Delta} \text{A} \xrightarrow{\text{NaOH}, \Delta} \text{B}$; B is :



Q.13 For the given reaction :

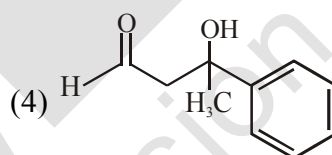
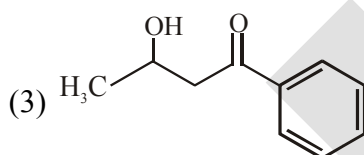
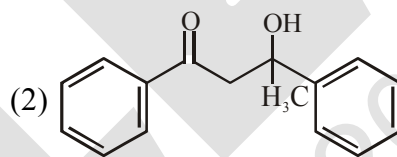
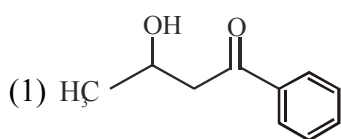
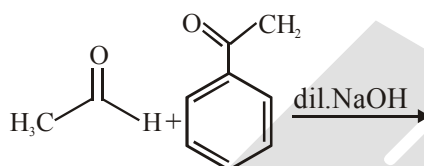


The (R) is :



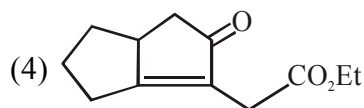
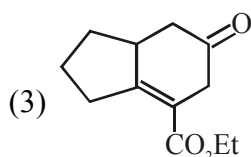
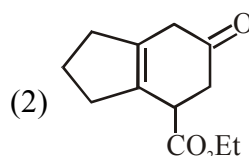
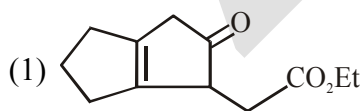
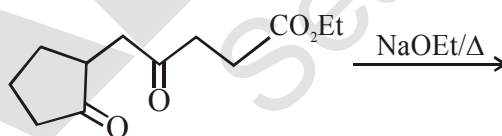
Q.14 The major product formed in the following reaction is:

[JEE-MAIN (JANUARY)-2019]



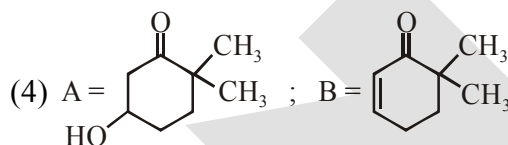
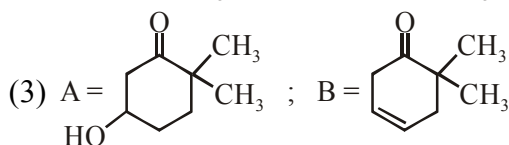
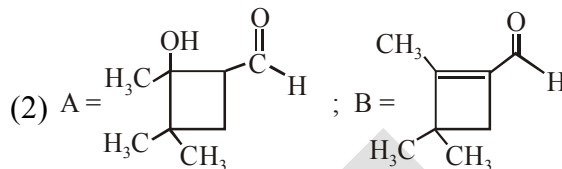
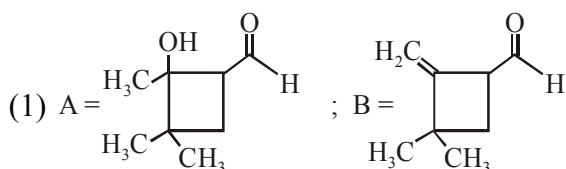
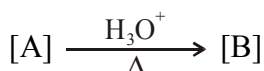
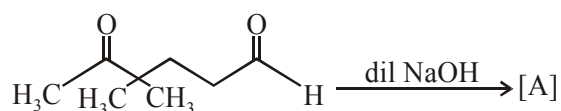
Q.15 The major product obtained in the following reaction is :

[JEE-MAIN (JANUARY)-2019]



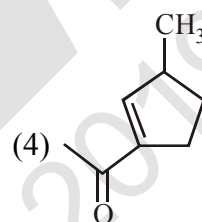
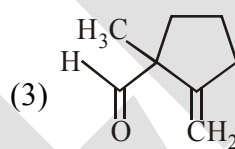
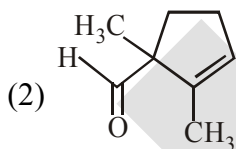
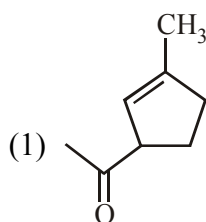
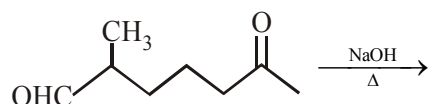
Q.16 In the following reactions, products A and B are :

[JEE-MAIN (JANUARY)-2019]



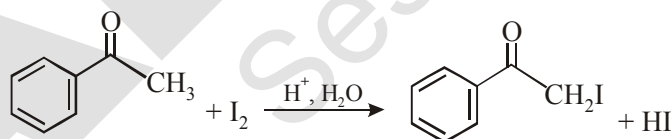
Q.17 The major product obtained in the following reaction is

[JEE-MAIN (APRIL)-2019]

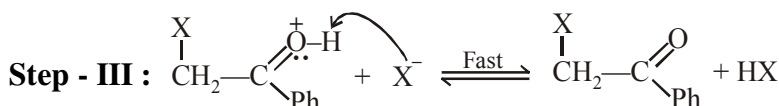
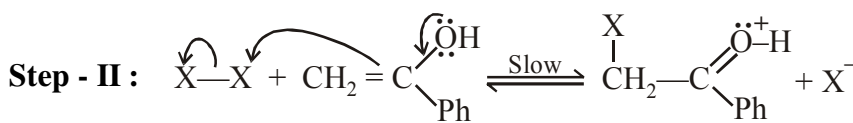
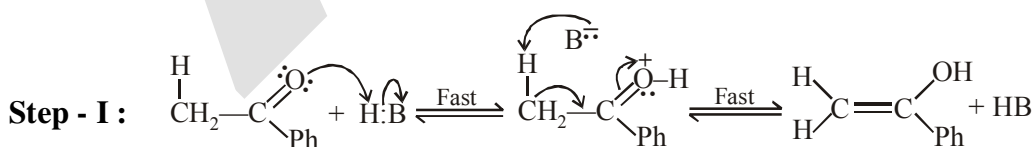


(4) HALOGENATION OF THE α -CARBON OF ALDEHYDES AND KETONES

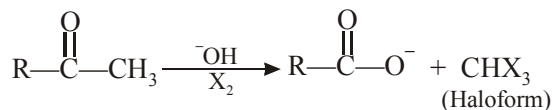
1st Acid-Catalyzed halogenation : - When Br_2 , Cl_2 and I_2 is added to an acidic solution of an aldehyde or a ketone, a halogen replaces one of the α -hydrogens of the carbonyl compound. Halogenation takes place through the slow formation of an enol followed by rapid reaction of the enol with the halogen.



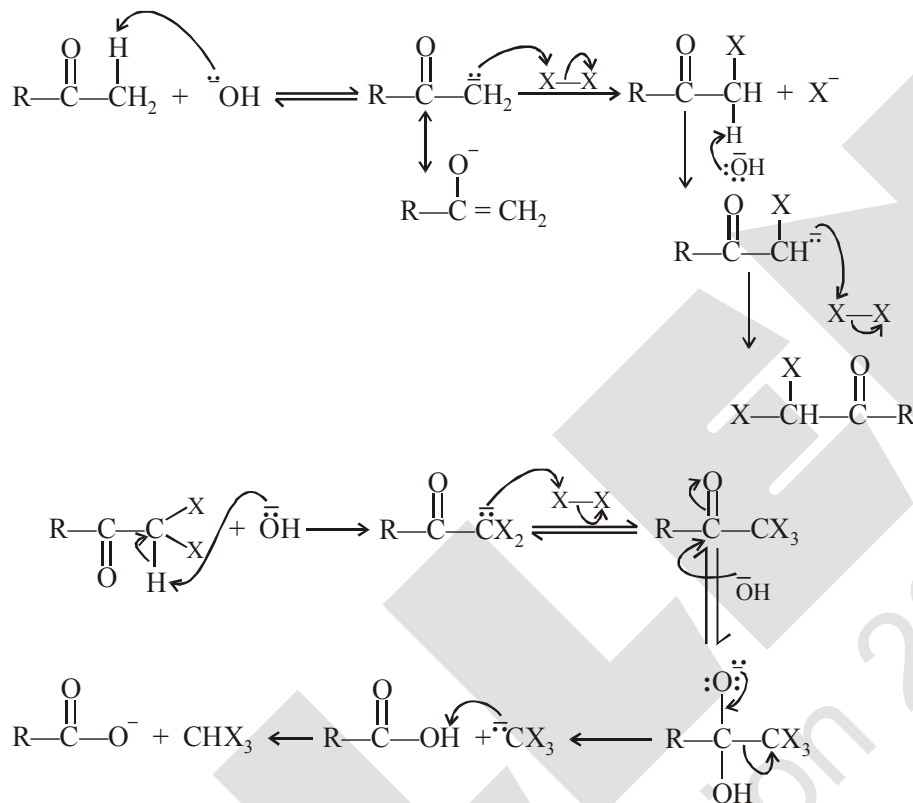
Mechanism :



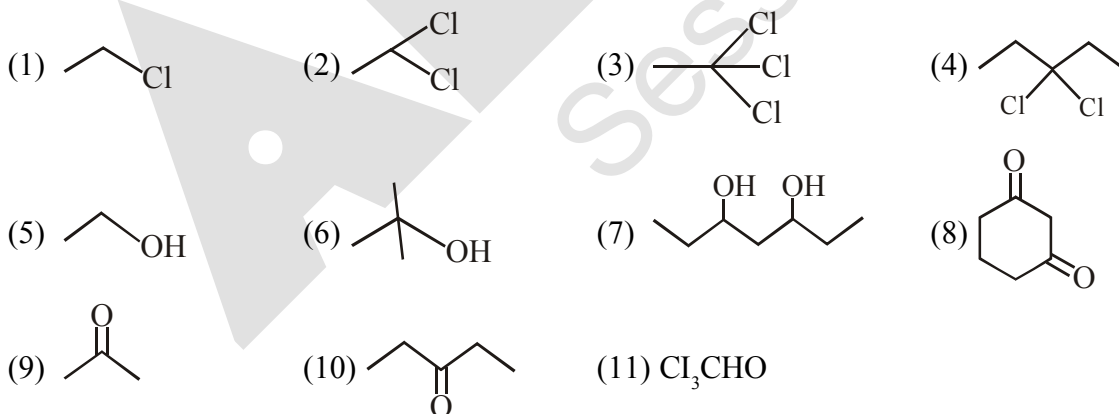
IInd Base-Catalyzed halogenation : - In the presence of excess base and excess halogen, a methyl ketone is converted first into a trihalo-substituted ketone and then into a carboxylic acid.



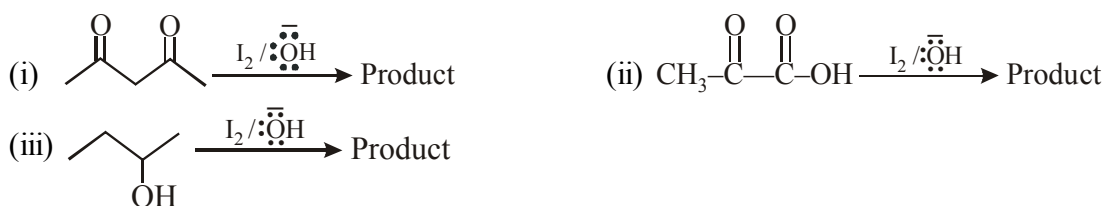
Mechanism :



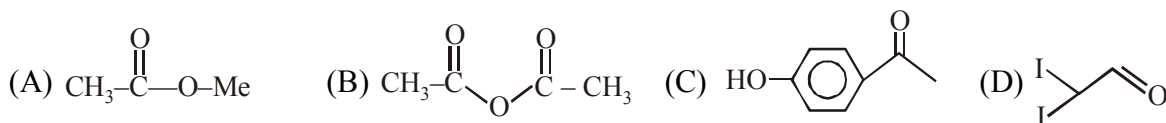
Q.1 Identify the compounds which can show iodoform test and complete the reaction ?



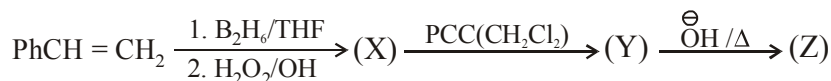
Q.2 Complete the following given reaction :



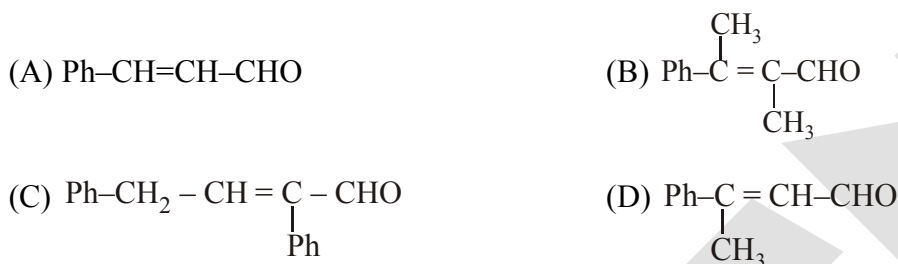
Q.3 Select the compound which does not show haloform reaction is/are :



Q.4 For the reaction



(Z) is :



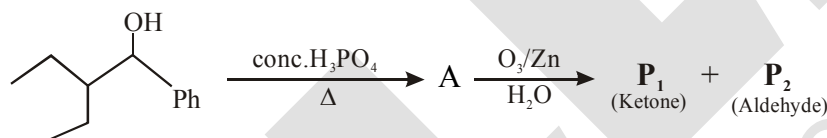
Q.5 Two isomeric ketones, 3-pentanone and 2-pentanone can be distinguished by :



Q.6 Which of the reagent is used to convert 2-Butanone into propanoic acid -



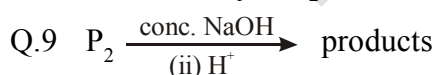
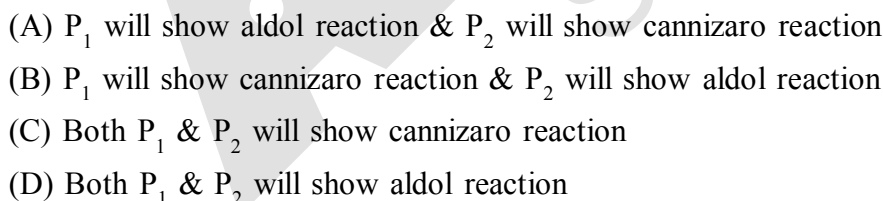
Paragraph for Q.07 to Q.09



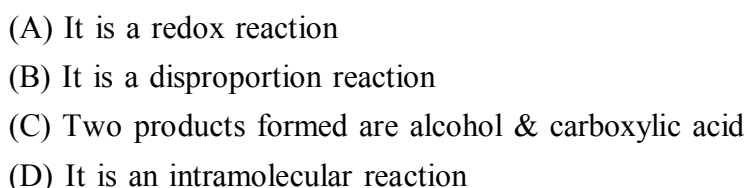
Q.7 Which one of the following reagent is best suitable for distinction between P_1 and P_2 -



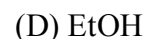
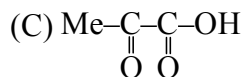
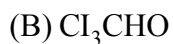
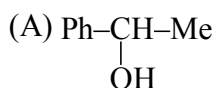
Q.8 Select the correct statement among the following -



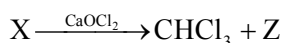
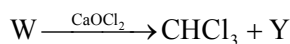
Select incorrect statement for the above reaction -



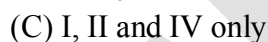
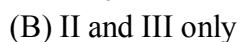
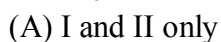
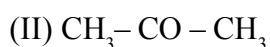
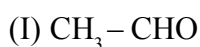
Q.10 Which of the following will give yellow precipitate with NaOH / I₂



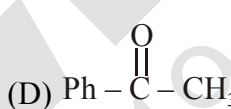
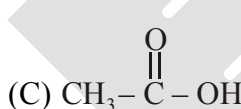
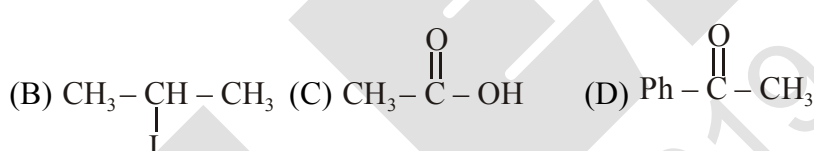
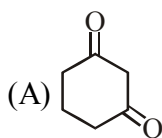
Q.11 Ethylmethanoate $\xrightarrow[\text{(ii)H}_3\text{O}^+]{\text{(i)MeMgBr(2eq)}}$ $\xrightarrow{\text{W + X}}$ organic product



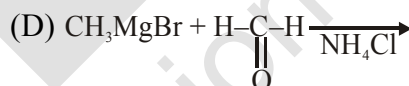
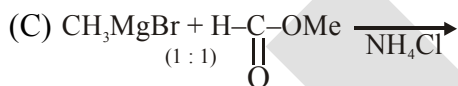
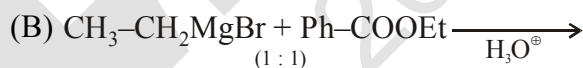
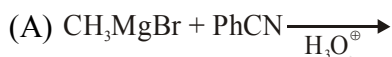
Which of the following organic product can be formed on dry distillation reaction



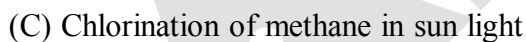
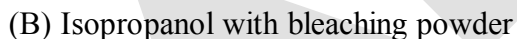
Q.12 Which of the following compounds(s) give a sweet smelling product having anesthetic use in presence of Cl_2 , NaOH, Δ .



Q.13 Reaction in which product can show positive haloform test :-



Q.14 In which reaction haloform is obtained as one product :-



Q.15. Fructose and glucose can be distinguished by :

[JEE-MAIN (APRIL)-2019]

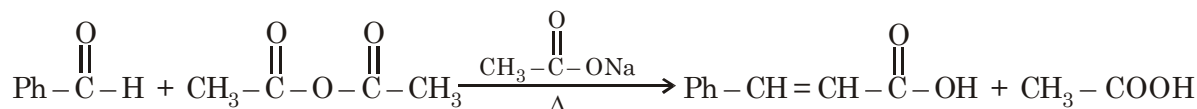


(5) PERKIN CONDENSATION

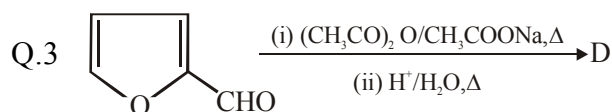
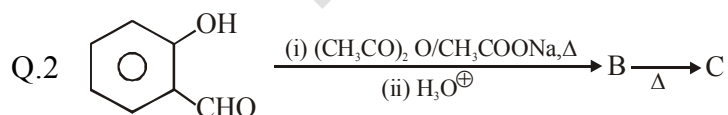
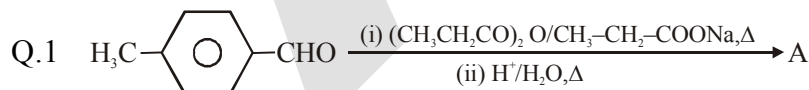
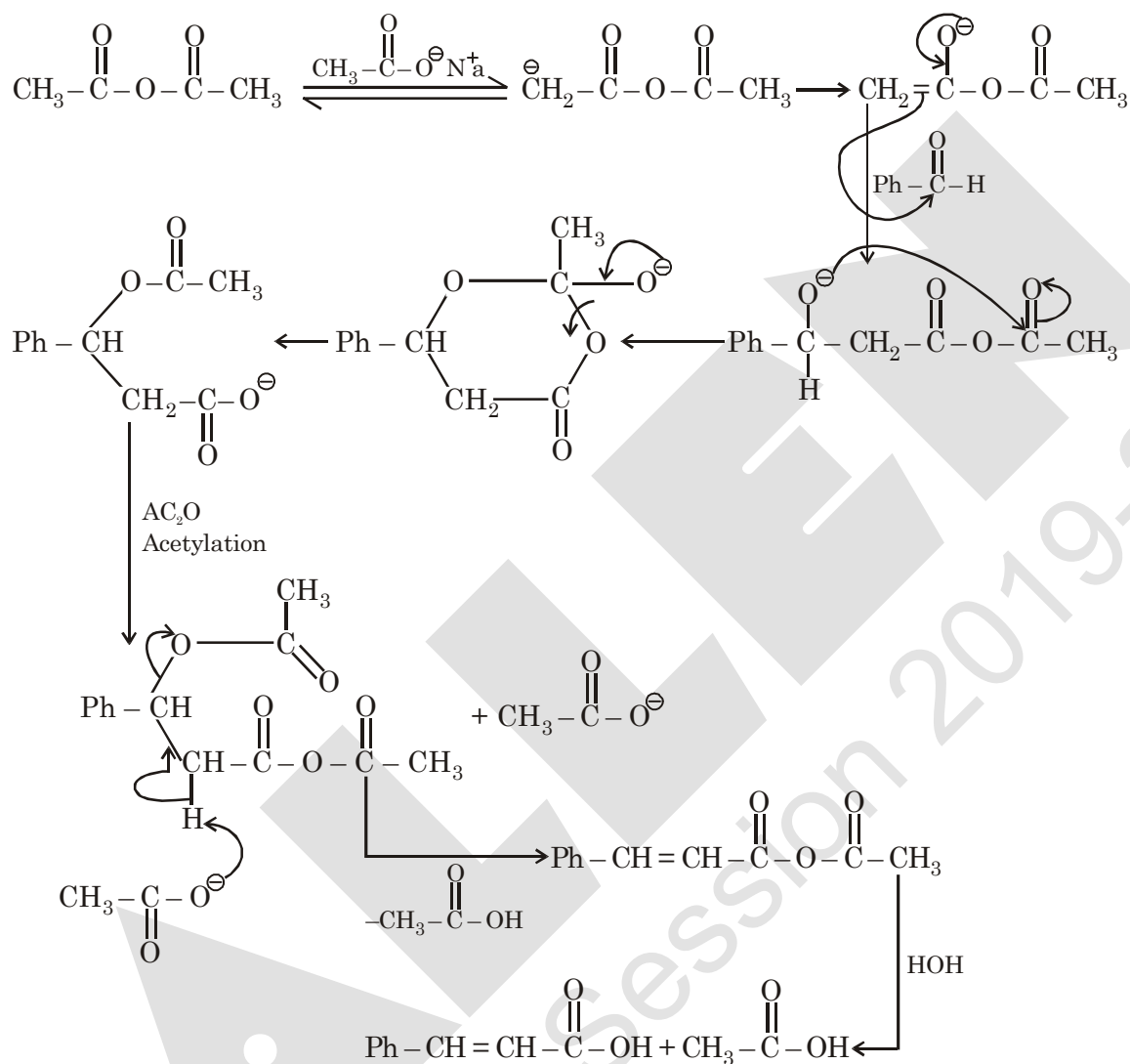
Perkin Reaction :

Condensation reaction between aromatic aldehyde and aliphatic acid anhydride having at least two alpha hydrogen in the presence of a base is known as **Perkin reaction**.

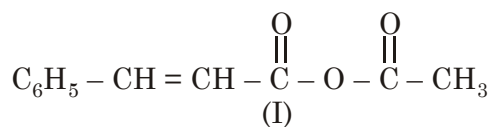
In this reaction, the anion of an acid anhydride adds to an aromatic aldehyde to produce an β aryl α, β , unsaturated acid. In order to prevent side reactions, the base that is used to make the anion of the anhydride is usually the sodium salt of the acid corresponding to the anhydride.



Mechanism :

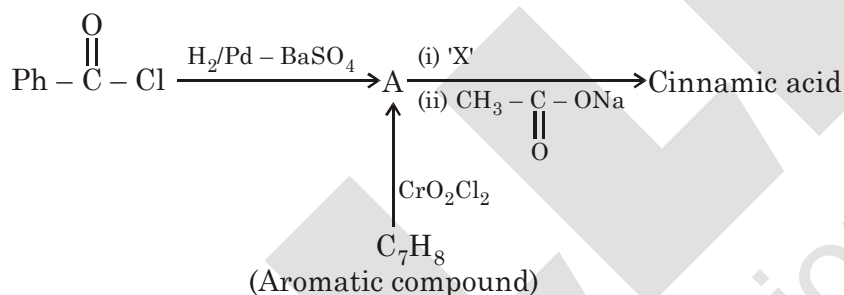


4. In the perkin reaction which one of the following intermediates gives compound (I)



- (A) $\text{C}_6\text{H}_5 - \overset{\text{O}^\ominus}{\text{CH}} - \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{O} - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3$
- (B) $\text{C}_6\text{H}_5 - \text{CH} \begin{cases} \text{O} - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3 \\ \text{CH}_2 - \text{COO}^\ominus \end{cases}$
- (C) $\text{C}_6\text{H}_5 - \text{CH} \begin{cases} \text{O} - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_2^\ominus \\ \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{OH} \end{cases}$
- (D) $\text{C}_6\text{H}_5 - \text{CH} \begin{cases} \text{O} - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3 \\ \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{O} - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3 \end{cases}$

Paragraph for Q.no. 5-6

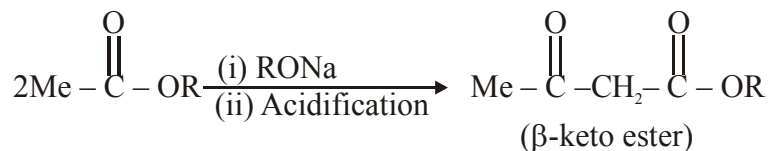


5. Compound 'A' on treatment with conc. NaOH followed by acidification can give :
- (A) $\text{Ph} - \text{CH}_2 - \text{COOH}$ (B) $\text{Ph} - \text{OH}$
- (C) $\text{Ph} - \text{COOH}$ (D) $\text{Ph} - \text{CH}_2 - \text{OH}$
6. In the above reaction the reagent "X" can be :

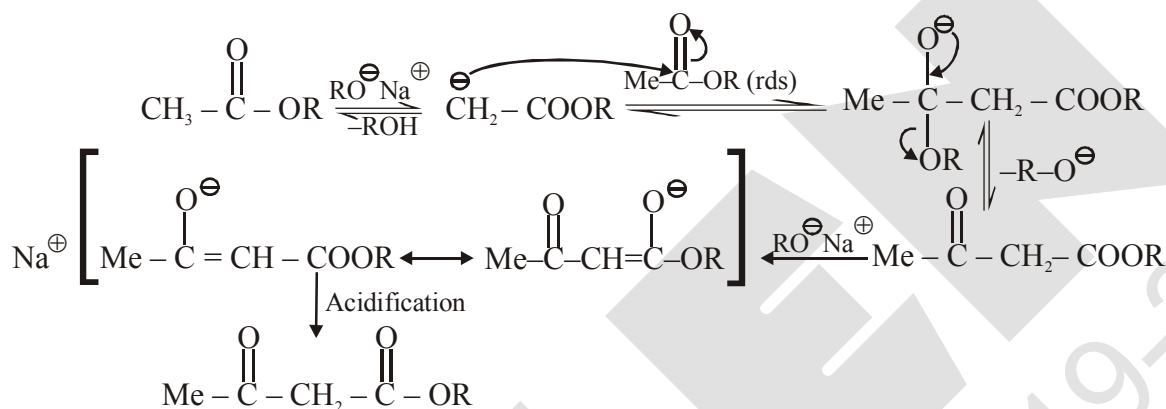
- (A) $\text{CH}_2 - \overset{\text{Br}}{\text{C}} - \overset{\text{O}}{\parallel}{\text{C}} - \text{OEt}$
- (B) $\text{CH}_2 \begin{cases} \overset{\text{O}}{\parallel}{\text{C}} - \text{OEt} \\ \text{C} - \text{OEt} \\ \parallel \\ \text{O} \end{cases}$
- (C) $\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{O} - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3$
- (D) $\text{CH}_3 - \text{CH} - \overset{\text{O}}{\parallel}{\text{C}} - \text{O} - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH} - \text{CH}_3$

(6) CLAISEN CONDENSATION

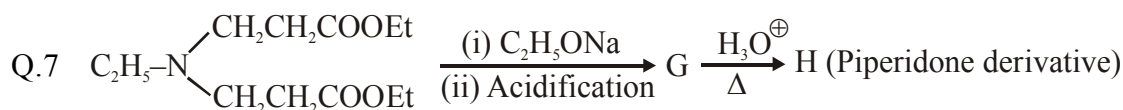
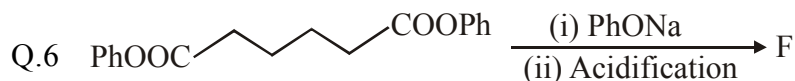
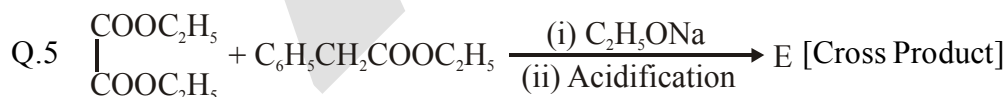
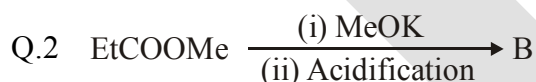
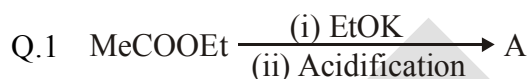
Esters undergo S_NAE Reaction, when attacked by a Nu^- generated by the interaction of a base (usually base related to the Alkoxy anion of ester) with one of the molecule of ester and this Nu^- attacks on another molecule. The reaction over all is considered as condensation of ester known as claisen ester condensation.



Mechanism :



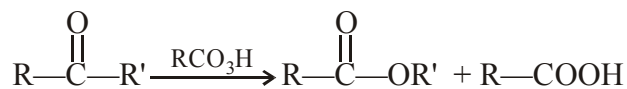
Some times, when two ester group are present within the molecule then the condensation occurs intramolecular then cyclization caused thus is known as Dieckmann cyclization or Dieckmann's condensation.



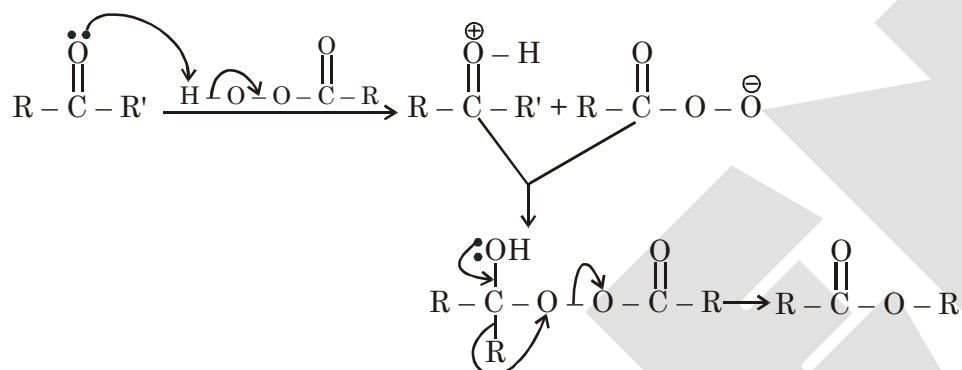
(7) Beayer villiger oxidation

Reaction involve the oxidation of ketone to ester by treatment with per acid such as per acetic acid, MCPBA, per trifluoroacetic acid etc.

- It is a redox reaction
- Cyclic ketones are converted to lactone with ring expansion.

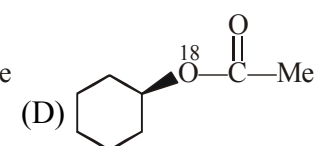
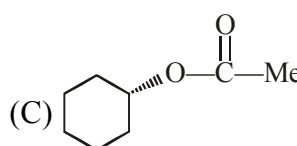
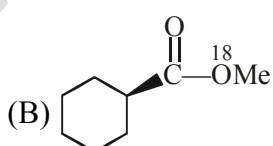
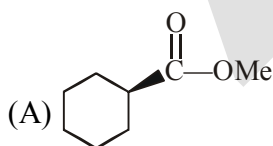
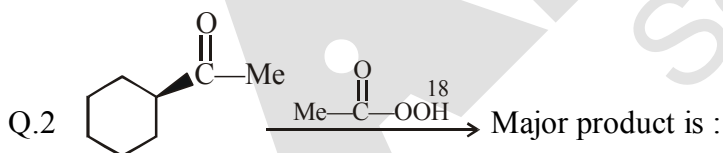
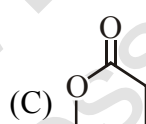
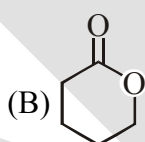
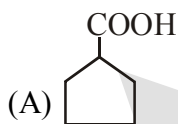
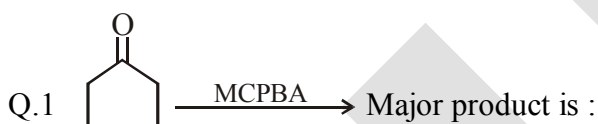
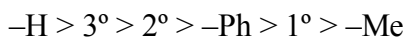


Mechanism



Note : In above reaction for shifting of groups migratory aptitude is applicable

Migratory order for above reaction



ANSWER KEY

(1) CANNIZARO REACTION

Q.1 Ans. (D)

As $\text{Cl}_3\text{C}-\text{CHO}$ given chloroform

Q.2 Ans. (C)

In crum cannizzaro reaction more reactive carbonyl compound is oxidised & less in reduced.

Q.3 Ans. (B)

Intramolecular cannizzaro followed by heating effect

Q.4 Ans. (i) $\text{CH}_3\text{OD} + \text{HCOONa}$ (ii) $\text{DCH}_2\text{OD} + \text{DCOONa}$ Q.5 Ans. (i) $\text{PhCH}_2\text{OD} + \text{PhCOONa}$ (ii) $\text{Ph}-\text{CH}_2\text{OH} + \text{PhCOONa}^{18}$ Q.6 Ans. (i) $\text{Ph}-\text{CH}_2\text{OH} + \text{HCOOK}$ (ii) $\text{Ph}-\underset{\text{OH}}{\text{CH}}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{OK}$ Q.7 Ans. (i) $\text{MeCH}_2-\text{CH}=\underset{\text{Me}}{\text{C}}-\overset{\text{O}}{\text{C}}-\text{H}$ (ii) $\text{Me}_2\text{CH}-\text{CH}_2\text{OH} + \text{Me}_2\text{CHCOOK}$ Q.8 Ans. (i) $\text{HO}-\underset{\text{OH}}{\text{C}}(\text{NO}_2)-\text{CH}_2\text{OH} + \text{CH}_3\text{OH} + \text{HCOOK}$ (ii) $\text{HO}-\underset{\text{OH}}{\text{C}}(\text{OH})-\text{CH}_2\text{OH} + \text{HCOOK}$

Q.9 Ans. (B)

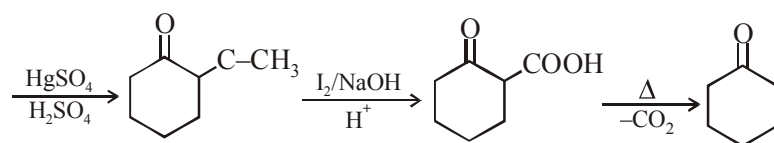
A : $\text{Ph}-\overset{\text{O}}{\text{C}}-\text{CH}=\text{O}$ B : $\text{Ph}-\underset{\text{H}}{\overset{\text{OH}}{\text{C}}}-\text{COO}^\ominus$ C : $\text{Ph}-\underset{\text{OH}}{\text{CH}}-\text{COOH}$

Q.10 Ans. (A)→S ; (B)→P ; (C)→Q ; (D)→R

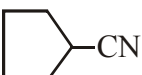
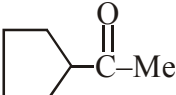
Q.11 Ans. (A)

A : $\text{Ph}-\underset{\text{OH}}{\text{CH}}-\text{COONa}$ B : PhCOOH C : 

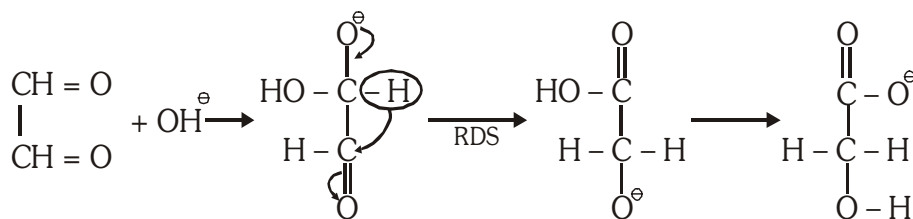
Q.12 Ans. (D)



Q.13 Ans. (B)

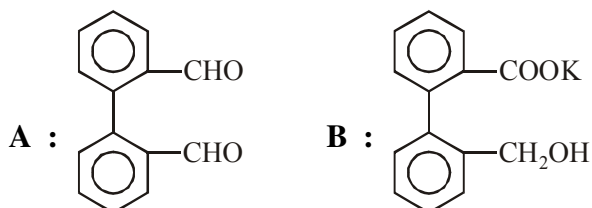
Q.14 Ans. (C) A : B : 

Q.15 Ans. (A)



option A is correct.

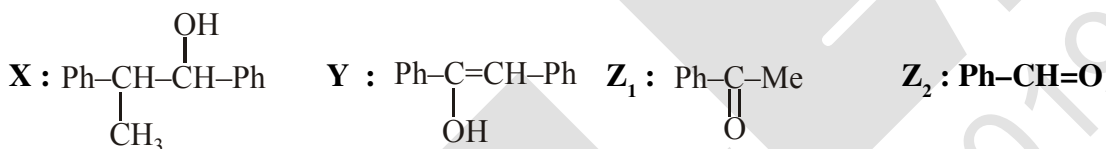
Q.16 Ans. (B)



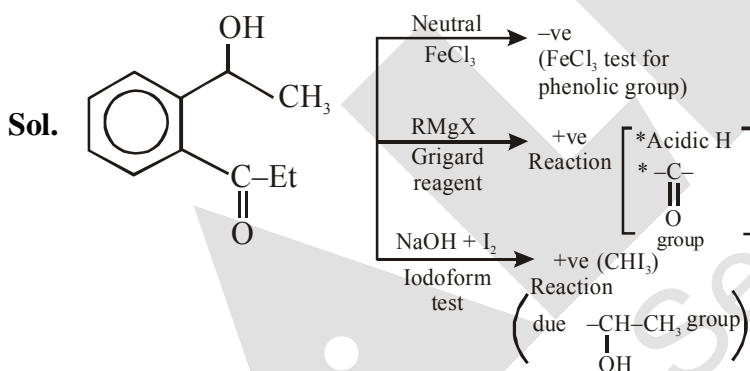
Q.17 Ans. (B)

Cron cannizaro reaction

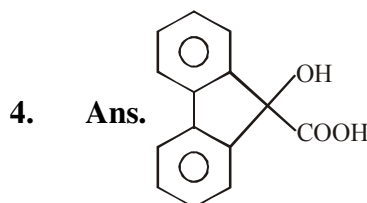
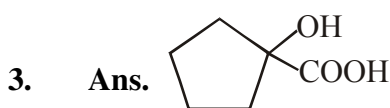
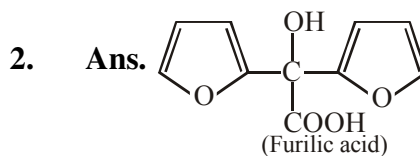
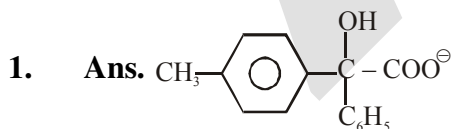
Q.18 Ans. (A)

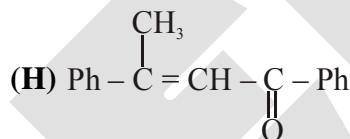
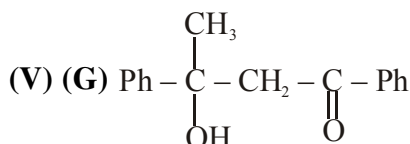
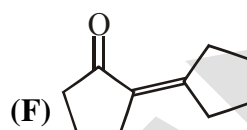
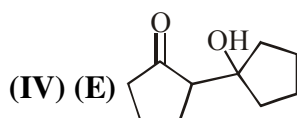
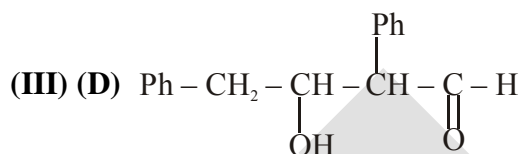
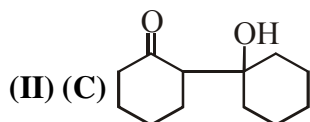
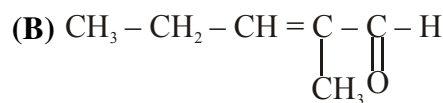
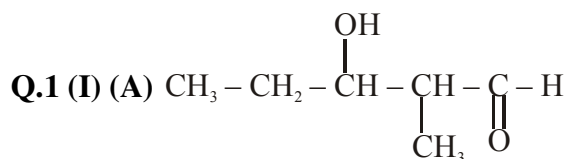
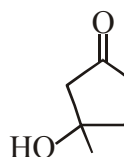
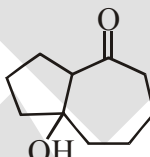
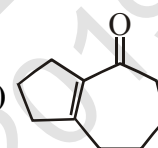
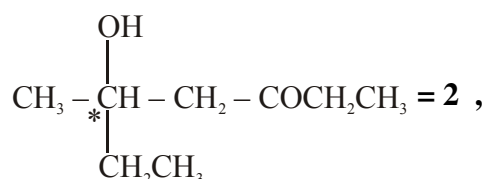
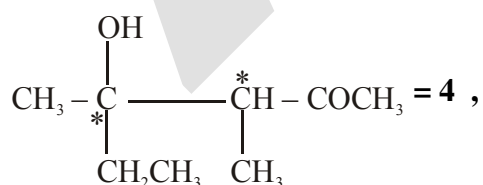
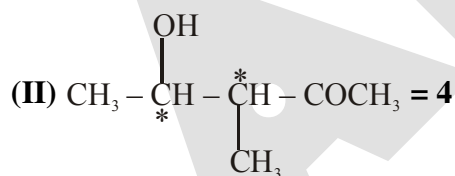
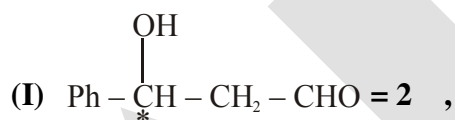
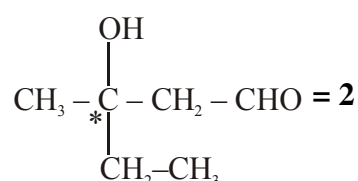
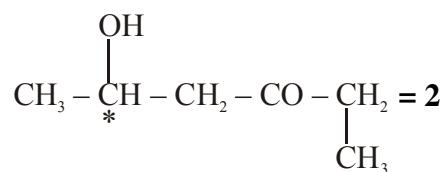
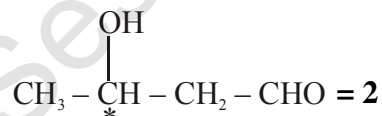
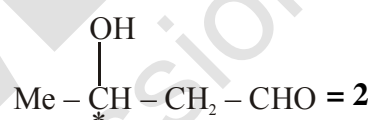


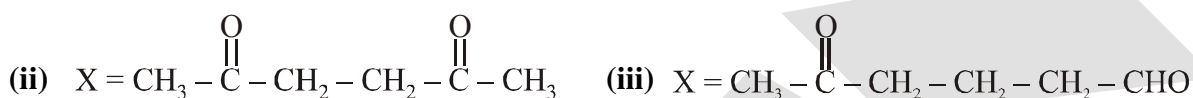
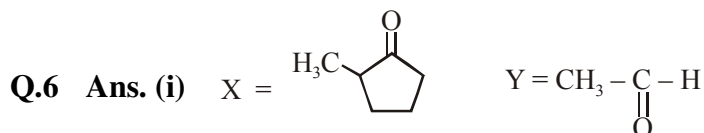
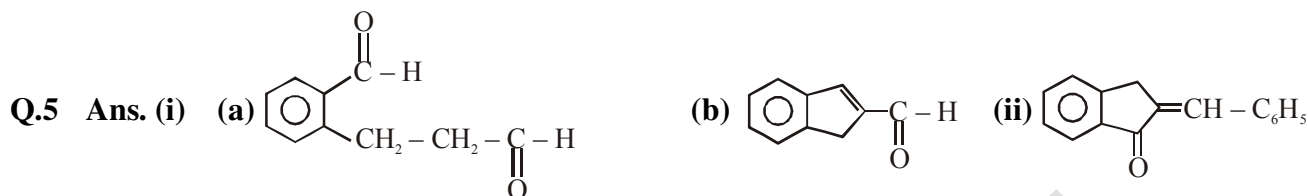
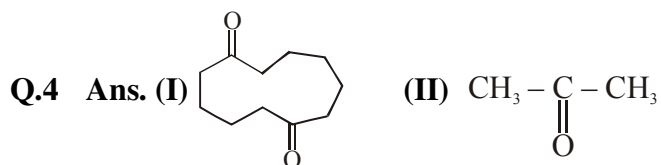
Q.19 Ans. (1)



(2) BENZIL-BENZILIC REARRANGEMENT OR BENZILIC ACID REARRANGEMENT



(3) ALDOL CONDENSATION**Q.1 Ans.****Q.2 Ans. (I) (A)****(B)** **(II) (E)****(F)****Q.3****Ans.****Excluding Stereo****Including Stereo**



Q.7 Ans. (C)

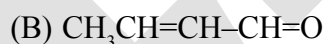
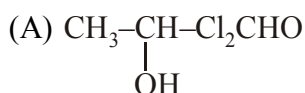
Intramolecular aldol condensation the 1,4-addition and finally Clemmensen reduction.

Q.8 Ans. (D)

Q.9 Ans. (B)

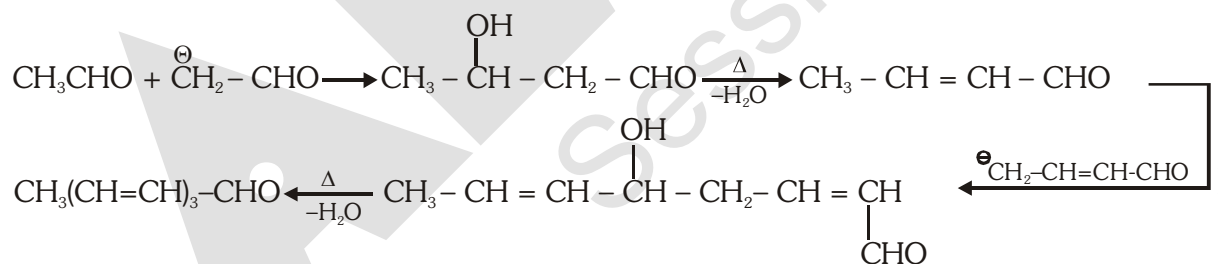
Ozonolysis followed by intramolecular aldol.

Q.10 Ans. (C)

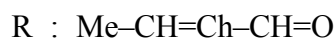
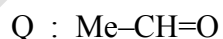


Q.11 Ans. (A)

Q.12 Ans. (A)



Q.13 Ans. (A)



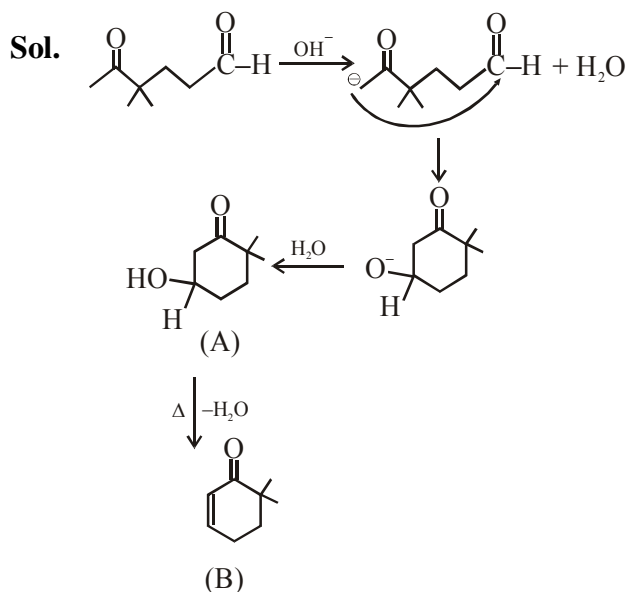
Q.14 Ans. (1)

Sol. Aldehyde reacts at a faster rate than ketone during aldol and sterically less hindered anion will be

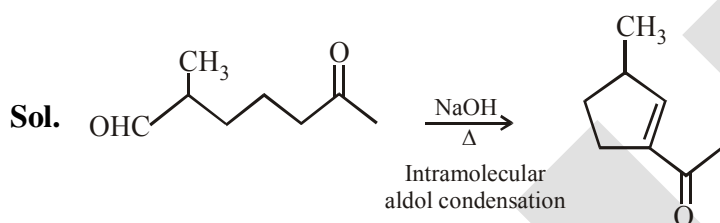
a better nucleophile so self aldol of $\text{CH}_3 - \text{C}(=\text{O}) - \text{H}$ will be the major product.

Q.15 Ans. (4)

Q.16 Ans. (4)



Q.17 Ans. (4)

**(4) HALOGENATION OF THE α -CARBON OF ALDEHYDES AND KETONES**

Q.1 Ans. (1, 2, 5, 7, 8, 9, 11)

Q.2 Ans. (i) $\text{CH}_3\text{-COO}^- + \text{CHI}_3$; (ii) $^-\text{O}-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{O}^- + \text{CHI}_3$; (iii) $\text{CH}_3\text{CH}_2\text{COO}^- + \text{CHI}_3$

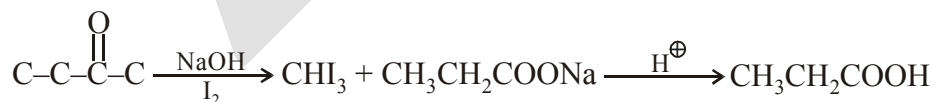
Q.3 Ans. (A,B,C)

Q.4 Ans. (C)

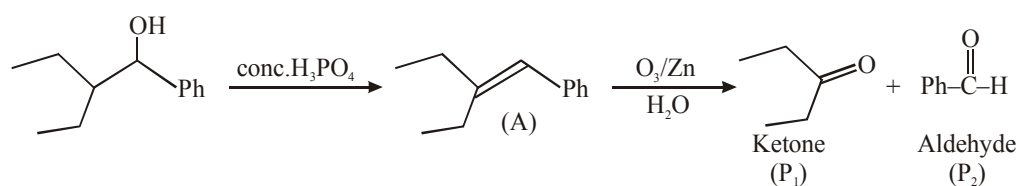


Q.5 Ans. (A)

Q.6 Ans. (A)

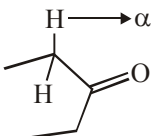


Q.7 Ans. (B)

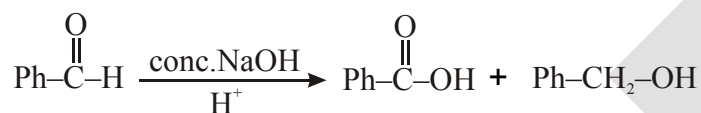


- (A) Both ketone & aldehyde react with braddy's reagent $\text{H}_2\text{N}-\text{NH}-\text{C}_6\text{H}_3(\text{NO}_2)_2$, 2,4 dinitrophenyl hydrazine.
- (B) With NaHSO_3 only aldehyde give crystalline bisulphite adduct whereas hindered ketone is unable react.
- (C) NaHCO_3 will not react any of them.
- (D) Both P_1 & P_2 are unable to form iodoform with NaOH/I_2

Q.8 Ans. (A)

For aldol αH should be present i.e.  P_1 where as P_2 $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ will give cannizaro's as there is no. αH in P_2 .

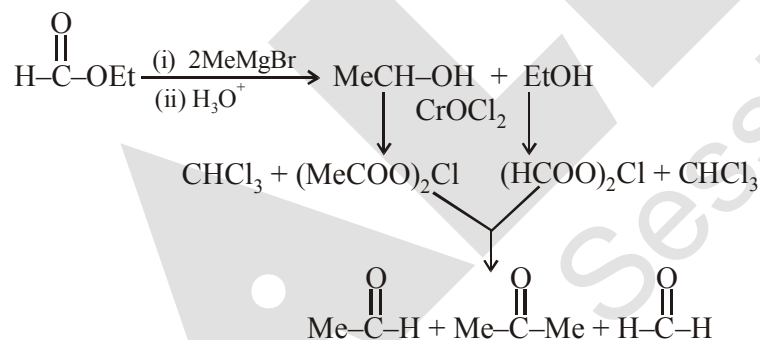
Q.9 Ans. (D)



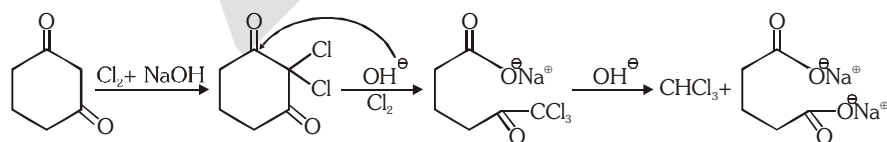
The above reaction is intermolecular redox reaction. In which desprotonation is taking place as a result of which 2 different products are formed from singlet reactant.

Q.10 Ans. (A,B,C,D)

Q.11 Ans. (C)



Q.12 Ans. (A,B,D)

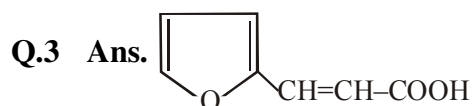
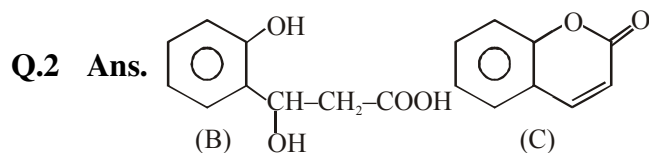
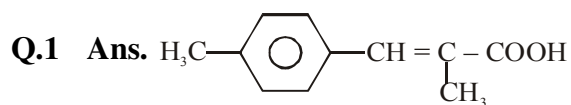


Q.13 Ans. (A,B,C,D)

Q.14 Ans. (A,B,C,D)

Q.15 Ans. (4)

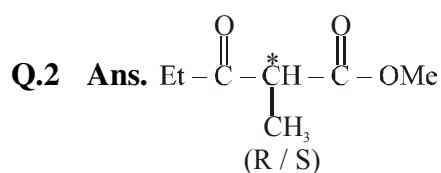
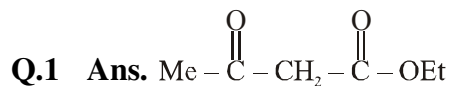
Sol. Seliwanoff's test is used to distinguished aldose and ketose group.

(5) PERKIN CONDENSATION

Q.4 Ans. (A)

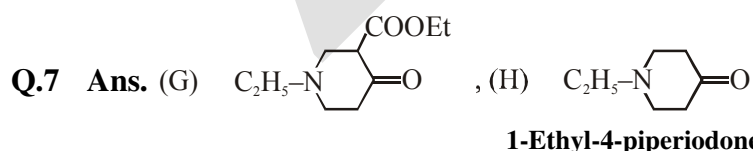
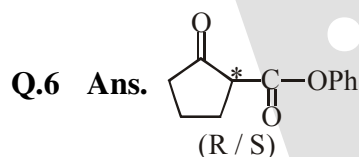
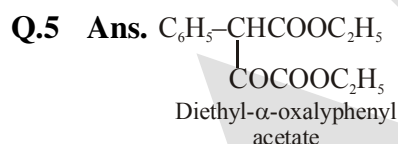
Q.5 Ans. (C, D)

Q.6 Ans. (C)

(6) CLAISEN CONDENSATION

Q.3 Ans. 4 products

Q.4 Ans. $\text{C}_6\text{H}_5\text{COCH}_2\text{COOC}_2\text{H}_5$, $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5$
Ethyl benzoylacetate

**(7) Beayer villiger oxidation**

Q.1 Ans. (B)

Q.2. Ans. (D)

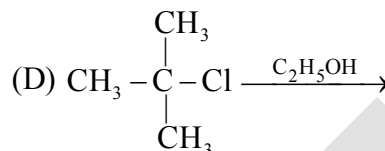
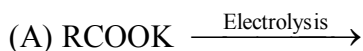
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Session 2019-20

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Session 2019-20

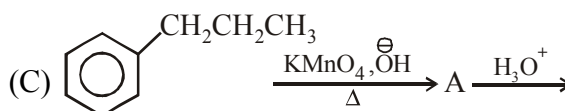
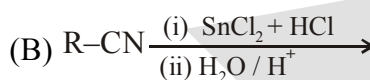
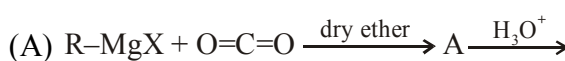
CARBOXYLIC ACIDS AND IT'S DERIVATIVE, ALIPHATIC AMINES

EXERCISE # I

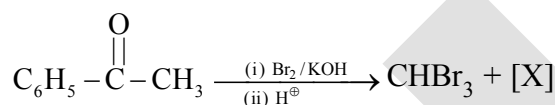
1. In which reaction product is hydrocarbon ?



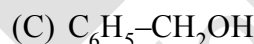
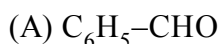
2. Which of the following set of reaction can not prepare carboxylic acid as the final product :



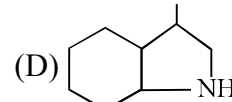
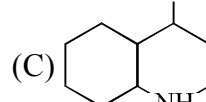
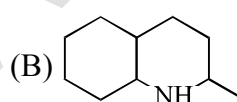
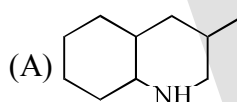
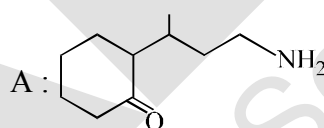
3. In the given reaction,



[X] will be :



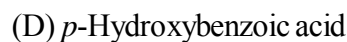
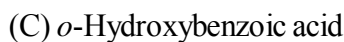
4. Reductive amination of A forms:



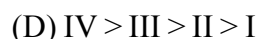
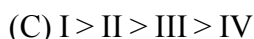
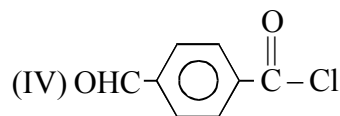
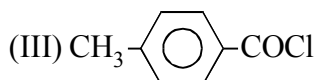
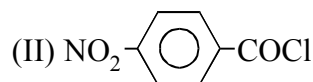
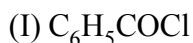
5. In the given reaction :



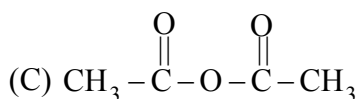
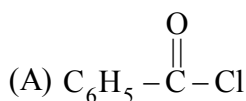
[X] will be :



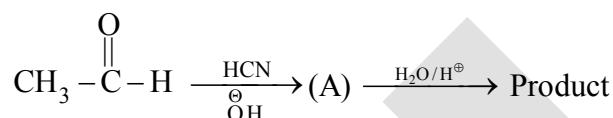
6. Arrange following compounds in decreasing order of reactivity for hydrolysis reaction :



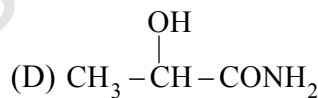
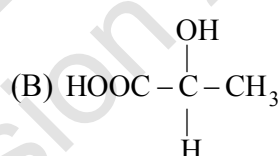
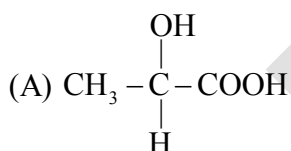
7. Which one of the following compounds gives carboxylic acid with HNO_2 ?



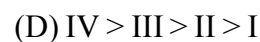
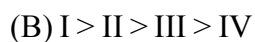
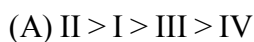
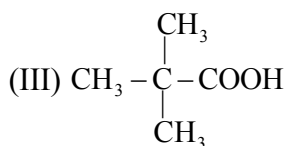
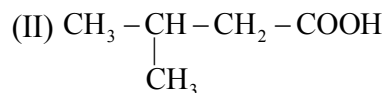
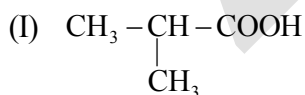
8. In the reaction sequence ,



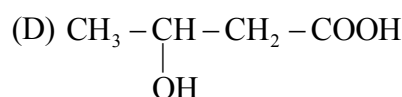
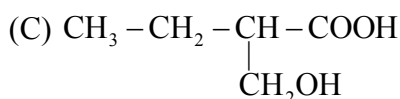
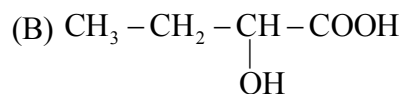
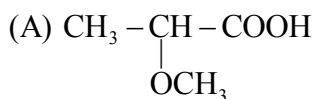
Product will be :



9. Arrange these esters in decreasing order of ease of esterification with $\text{CH}_3\text{OH}/\text{H}^+$:



10. Which optically active compound on reduction with LiAlH_4 will give optically inactive compound?



11. Which will form lactone on treatment with NaOH ?

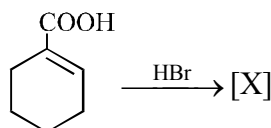
(A) α -Bromo acid

(B) β -Bromo acid

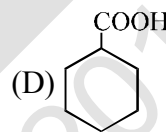
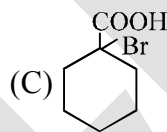
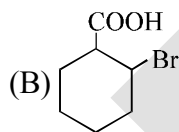
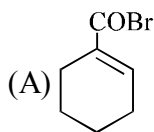
(C) β -Hydroxy acid

(D) δ -Bromo acid

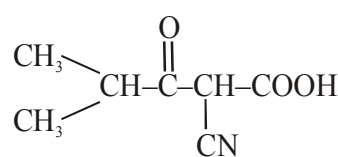
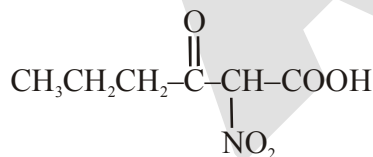
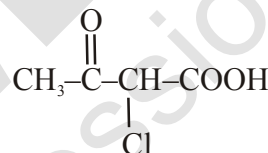
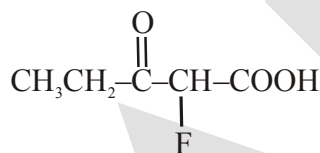
12. In the given reaction:



[X] will be :



13. Correct order of decarboxylation



(A) $a > b > c > d$

(B) $c > d > b > a$

(C) $c > d > a > b$

(D) $d > c > a > b$

14. N-Ethyl phthalimide on hydrolysis gives:

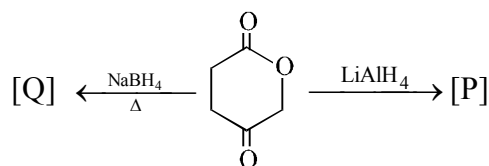
(A) Methyl alcohol

(B) Ethyl amine

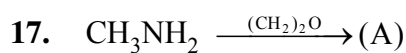
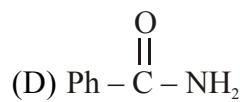
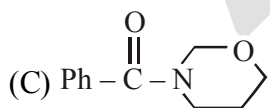
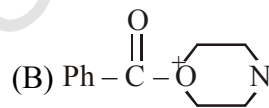
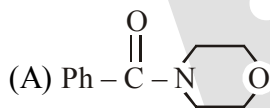
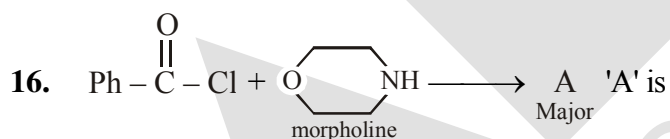
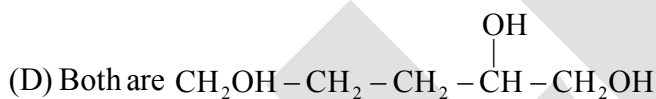
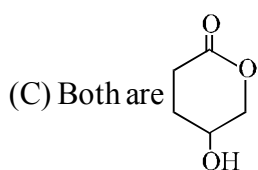
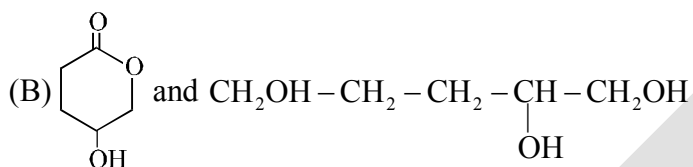
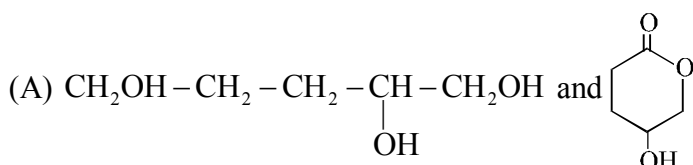
(C) Dimethyl amine

(D) Diethyl amine

15. In the given reaction:



[P] and [Q] respectively be :



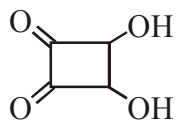
(A) 'A' is more basic than CH_3NH_2

(B) 'A' is less basic than CH_3NH_2

(C) 'A' is Ter-amine

(D) None

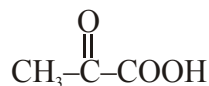
18. Which of the following can release CO_2 with NaHCO_3 .



(i)



(ii)



(iii)

(A) (i), (ii) & (iii)

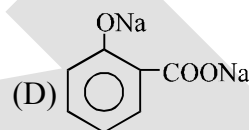
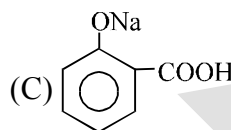
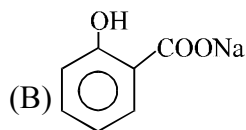
(B) (i) & (ii)

(C) (ii) & (iii)

(D) (i) & (ii)

19. Sodium bicarbonate reacts with salicylic acid to form :

(A) $\text{C}_6\text{H}_5\text{ONa}$



20. Which of the following diazonium salt is relatively stable at $0-5^\circ\text{C}$:

(A) $\text{CH}_3\text{N}_2^+\text{Cl}^-$

(B) $(\text{CH}_3)_2\text{CHN}_2^+\text{Cl}^-$

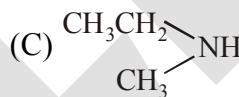
(C) $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$

(D) $(\text{CH}_3)_3\text{CN}_2^+\text{Cl}^-$

21. Which is most volatile ?

(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$

(B) $(\text{CH}_3)_3\text{N}$



(D) CH_3OH

22. $\text{C}_6\text{H}_5\text{CONH}_2 \xrightarrow{\text{Br}_2/\text{OD}^\ominus} \text{P}$, 'P' is :

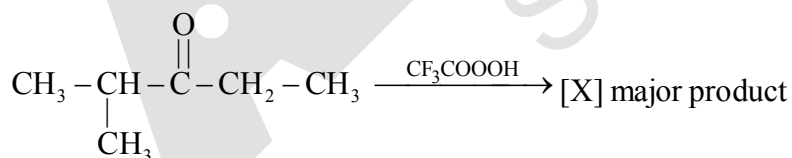
(A) $\text{C}_6\text{H}_5\text{COND}_2$

(B) $\text{C}_6\text{H}_5\text{ND}_2$

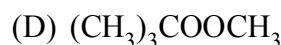
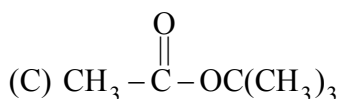
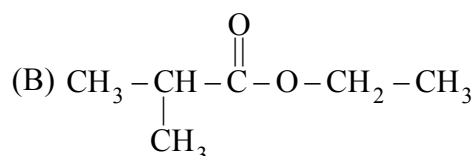
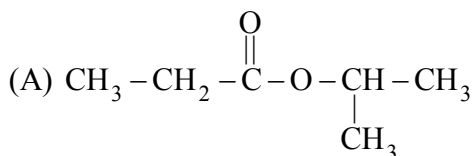
(C) $\text{C}_6\text{H}_5\text{NHD}$

(D) $\text{C}_6\text{H}_5\text{NH}_2$

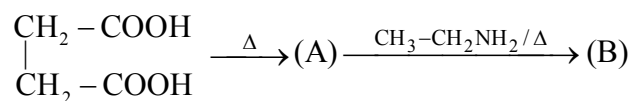
23. In the given reaction:



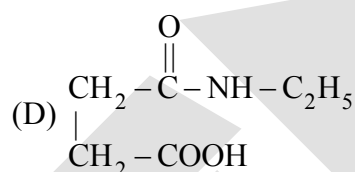
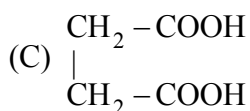
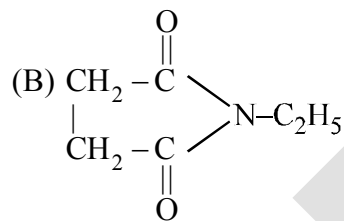
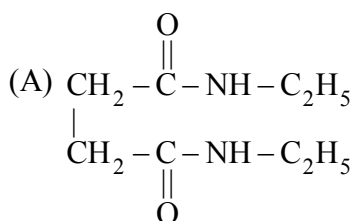
[X] will be:



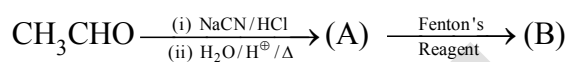
24. In the given reaction sequence:



(B) will be:



25. In the given reaction :



(B) will be :

(A) Acetic acid

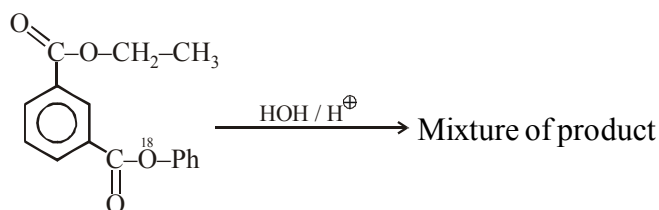
(B) Oxalic acid

(C) Pyruvic acid

(D) Citric acid

EXERCISE # II

1. Which of the following is/are present in mixture of product :



- (A) $\text{CH}_3\text{CH}_2\text{OH}$ (B) (C) $\text{Ph}-^{18}\text{OH}$ (D)

2. Mixture of 1° , 2° and 3° amines can be separated by:

- (A) Hinsberg's method (B) Hofmann's isocyanide test
(C) Fractional distillation (D) NaNO_2/HCl

3. RCOOR' can be prepared by:

- (A) Esterification of RCOOH
(B) Reaction of $\text{CH}_3\text{CH}=\text{CH}_2$ with methanol
(C) Baeyer-Villiger oxidation of RCOR' with peroxy acid
(D) reaction of RCOCl with $\text{R}'\text{OH}$

4. Which of the following amine reacts with Hinsberg reagent to give base soluble product :-

- (A) Neopentyl amine (B) sec propyl amine (C) diethyl amine (D) ethyl methyl amine

5. Which is/are correct reaction(s):

- (A) (B)
(C) (D)

6. Which of the following will form acetyl chloride with PCl_5 ?

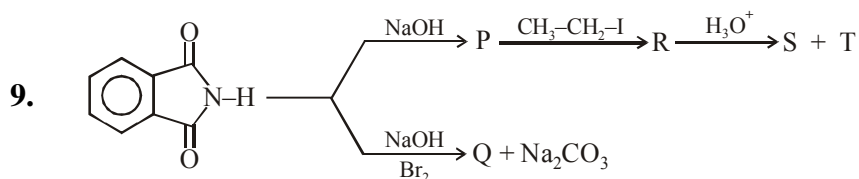
- (A) MeCOOH (B) MeCOOMe (C) MeCOOCOMe (D) MeCONH_2

7. Sodium salt of which compound on electrolysis does not give hydrocarbon:

- (A) $\text{C}_6\text{H}_5\text{COOH}$ (B) HCOOH (C) $\text{Me}_3\text{C-COOH}$ (D) COOH-CH=CH-COOH

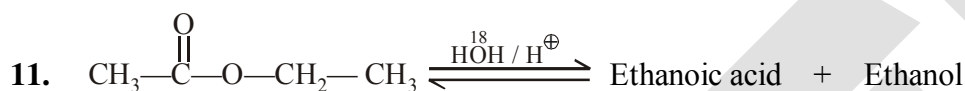
8. Among the following, which statement is not correct ?

- (A) $\text{H}_3\text{C-C(=O)-OH}$ will not respond to haloform test
(B) Schiff's reagent and Schiff's base are different compounds
(C) Fehling's solution is a good reagent to detect aromatic aldehydes
(D) Both aldehyde and ketone can react with 2,4-dinitrophenylhydrazine reagent



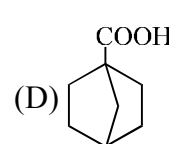
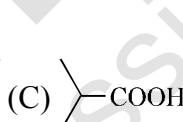
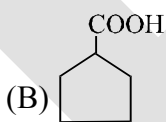
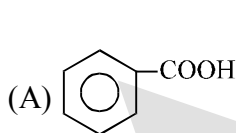
If T can evolve effervescence of CO_2 with a NaHCO_3 , then correct statement(s) is/are :

- (A) S & Q can be distinguished by dye azo test
 (B) T is most acidic among all isomeric benzenoid dicarboxylic acid
 (C) Q & S can be distinguished by mustered oil test
 (D) P, Q & T all are soluble in a NaHCO_3
10. Acetic anhydride and ammonia gives the product:
 (A) CH_3CONH_2 (B) $\text{CH}_3\text{CONHCH}_3$ (C) CH_3CN (D) $\text{CH}_3\text{COONH}_4$



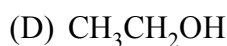
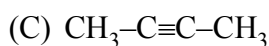
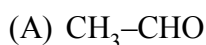
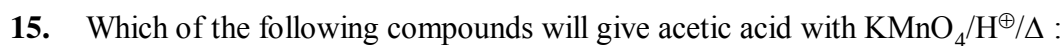
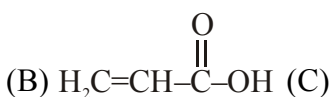
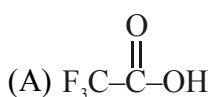
Isotopic oxygen of water will be present with

- (A) Ethanoic acid
 (B) Ethanol
 (C) After some time it will also be present in some molecules of ester
 (D) None of these
12. Which one of the following compounds will give HVZ reaction?

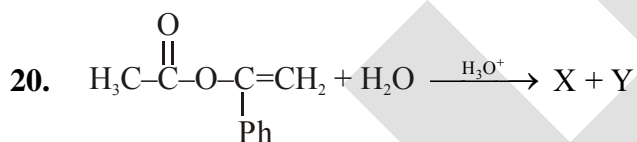
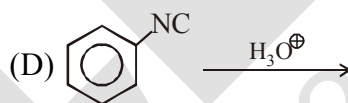
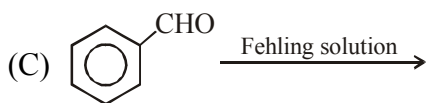
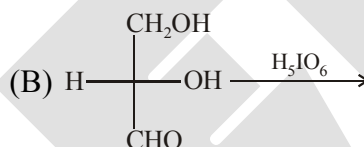
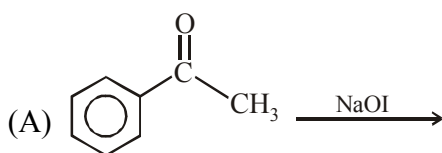


Rate of reaction will be faster if 'R' is

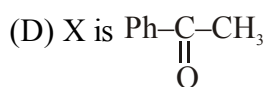
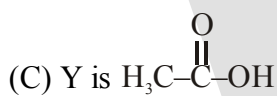
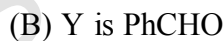
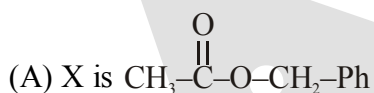
- (A) CH_3- (B) C_2H_5- (C) NO_2- (D) $\text{CN}-$
14. Which of the following carboxylic acids do not undergo decarboxylation simply on heating ?



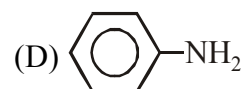
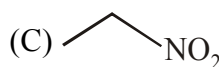
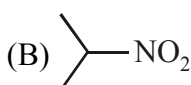
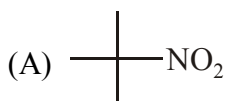
16. Hofmann degradation is given by:
 (A) Succinimide (B) Acid chloride (C) Acid anhydride (D) Acetamide
17. The presence of primary amine can be confirmed by its reaction with :
 (A) HNO_2 (B) $\text{CHCl}_3 + \text{NaOH}$ (C) CS_2 & HgCl_2 (D) H_2SO_4
18. Total number of compounds which are soluble in hot a NaOH are :
 (i) Salicyclic acid (ii) Aspirine (iii) Carbolic acid (iv) Acetic acid
 (v) Succinic anhydride (vi) Cyclohexanone (vii) Benzene sulphonamide (viii) Cyclohexene
 (A) 5 (B) 6 (C) 7 (D) 8
19. Number of oxidation reactions in which organic reactant gets oxidised & one of the major product is carboxylic acid/salt :



X and Y are :



21. Which of the following compound react with HNO_2 :



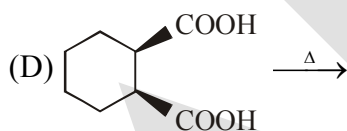
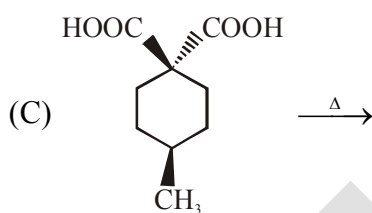
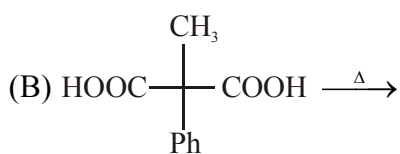
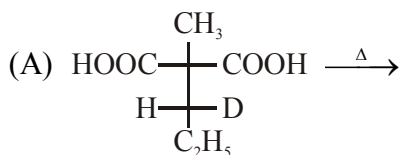
EXERCISE # III

Matching Type Questions

1. Match the following question :

Column - I

(Reaction)



Column-II

(Products)

(P) Diastereomers

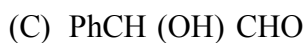
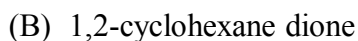
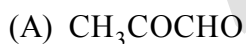
(Q) Racemic mixture

(R) Meso comp.

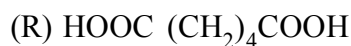
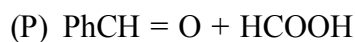
(S) CO₂ gas will evolve

2. Match the following question :

Column I

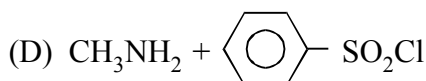
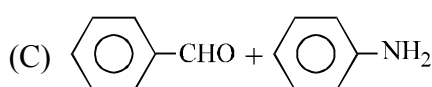
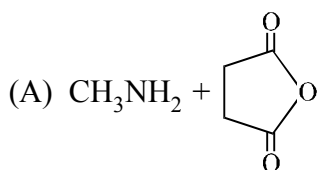
(Organic compounds oxidised by HIO₄)

Column II

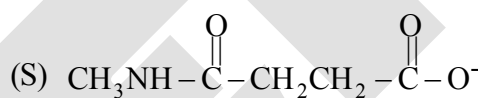
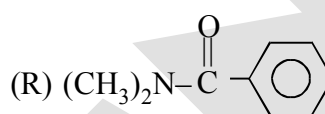
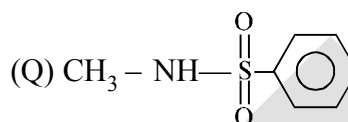
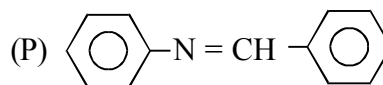
(Products of HIO₄ oxidation)

3. Match the following question :

Column I (Reactions)



Column II (Products)



4. Match the following question :

Column I

(Correct about product)



Column II

(P) Product is yellow oily liquid

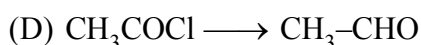
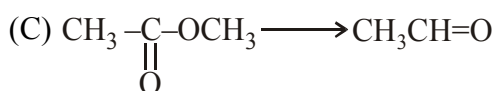
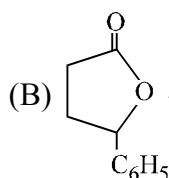
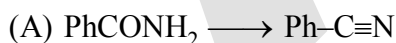
(Q) Gives red colour with CAN

(R) Gives fruity smell with CH_3OH

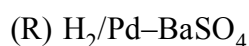
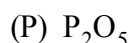
(S) Foul smelling compound is formed.

5. Match the following question :

Column I

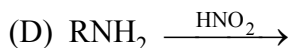
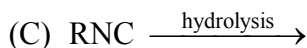
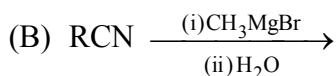
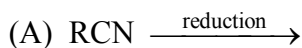


Column II



6. Match the following question :

Column I



Column II

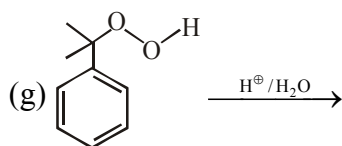
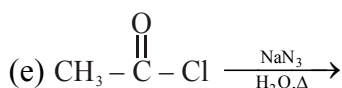
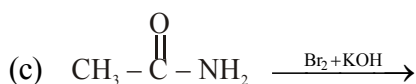
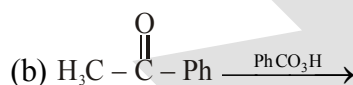
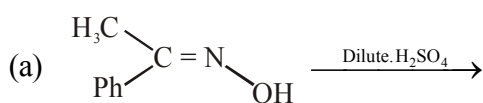
(P) 1° Amine

(Q) Alcohol

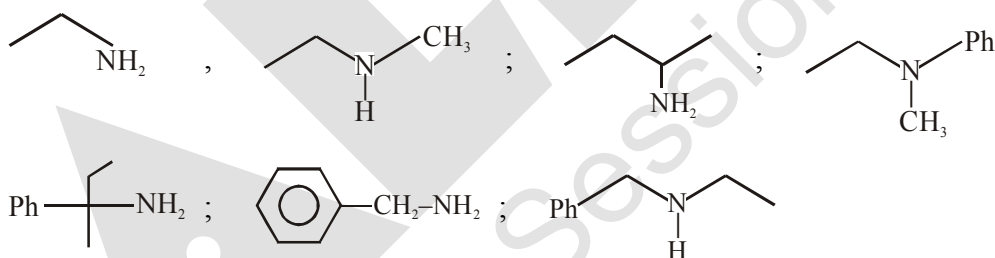
(R) Ketone

(S) Acid

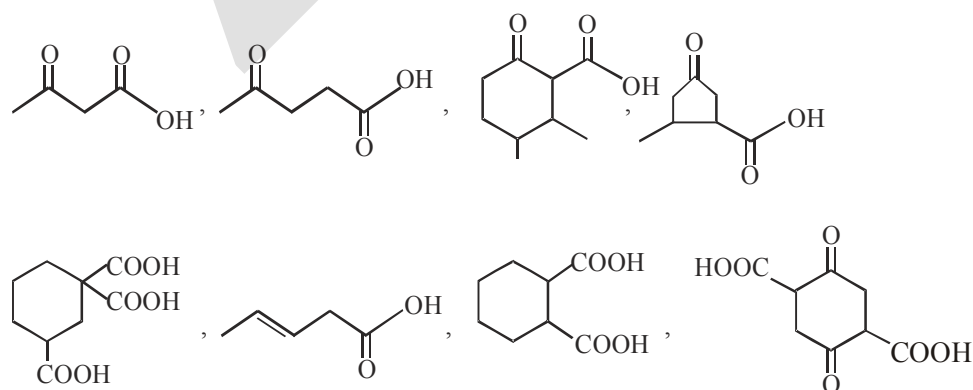
7. Find out number of reactions which involve electron deficient nitrogen [Nitrene character] during reaction mechanism.

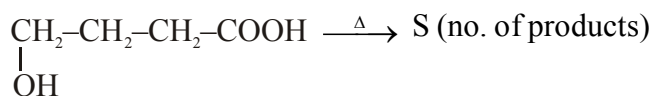
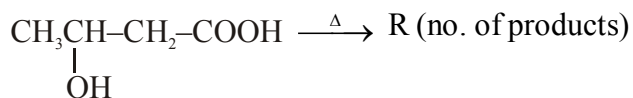
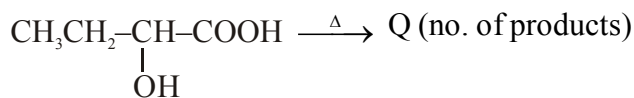
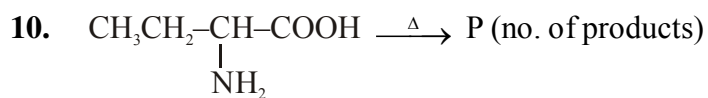


8. Of the following amines how many can be separated by Hoffmann's mustard oil reaction.



9. Examine the structure of following compounds, and find out number of compounds that will undergo decarboxylation in presence of heat.





In all reactions the sum of product is.

How will you bring about the following transformation:

11. Propanoic acid into lactic acid.
12. Ethyl benzene to 2-phenyl propionic acid.
13. Acetamide from acetone.

EXERCISE # IV (A) (JEE-MAIN)

1. Reaction -

[AIEEE-2002]

Primary amine + $\text{CHCl}_3 + \text{KOH} \rightarrow$ product, here product will be -

- (1) Cyanide (2) Isocyanide (3) Amine (4) Alcohol

2. The compound formed in the positive test for nitrogen with the Lassaigne solution of an organic compound is- [AIEEE-2004]

- (1) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (2) $\text{Na}_3[\text{Fe}(\text{CN})_6]$ (3) $\text{Fe}(\text{CN})_3$ (4) $\text{Na}_4[\text{Fe}(\text{CN})_5]\text{NOS}$

3. Which one of the following methods is neither meant for the synthesis nor for separation of amines-

- (1) Hofmann method (2) Hinsberg method [AIEEE-2005]
(3) Curtius reaction (4) Wurtz reaction

4. In the chemical reaction, $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \rightarrow (\text{A}) + (\text{B}) + 3\text{H}_2\text{O}$, the compounds (A) and (B) are respectively - [AIEEE-2007]

- (1) $\text{C}_2\text{H}_5\text{CN}$ and 3KCl (2) $\text{CH}_3\text{CH}_2\text{CONH}_2$ and 3KCl
(3) $\text{C}_2\text{H}_5\text{NC}$ and K_2CO_3 (4) $\text{C}_2\text{H}_5\text{NC}$ and 3KCl

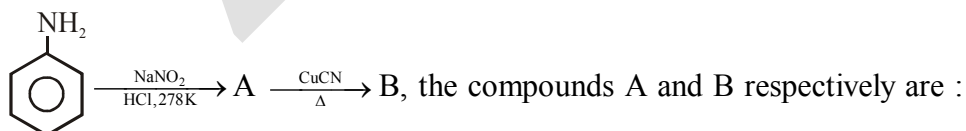
5. In the chemical reactions,

[AIEEE-2010]



- (1) Nitrobenzene and chlorobenzene
(2) Nitrobenzene and fluorobenzene
(3) Phenol and benzene
(4) Benzene diazonium chloride and fluorobenzene

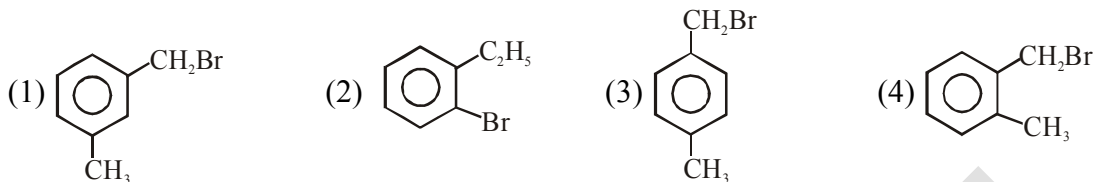
6. In the chemical reactions



[AIEEE-2011]

- (1) Fluorobenzene and phenol (2) Benzene diazonium chloride and benzonitrile
(3) Nitrobenzene and chlorobenzene (4) Phenol and bromobenzene

7. Compound (A), C_8H_9Br , gives a white precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives an acid (B), $C_8H_6O_4$. (B) easily forms anhydride on heating. Identify the compound (A): [AIEEE-2013]



8. An organic compound A upon reacting with NH_3 gives B. On heating, B gives C. C in presence of KOH reacts with Br_2 to give $CH_3CH_2NH_2$. A is :- [AIEEE-2013]

- (1) CH_3COOH (2) $CH_3CH_2CH_2COOH$
(3) $CH_3-CH(CH_3)-COOH$ (4) CH_3CH_2COOH

9. On heating an aliphatic primary amine with chloroform & ethenolic potassium hydroxide the organic compound formed is [AIEEE-2014]

- (1) An alkyl cyanide (2) An alkyl isocyanide
(3) an alkanol (4) an alkanediol

10. In the reaction $CH_3COOH \xrightarrow{LiAlH_4} A \xrightarrow{PCl_5} B \xrightarrow{alc.KOH} C$ 'C' is [AIEEE-2014]
(1) Ethylene (2) Acetyl chloride (3) Acetaldehyde (4) Acetylene.

11. In the presence of a small amount of phosphorous, aliphatic carboxylic acids react with chlorine or bromine to yield a compound in which α - hydrogen has been replaced by halogen. This reaction is known as : [JEE(Main)-2015]

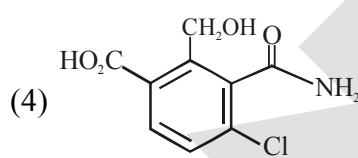
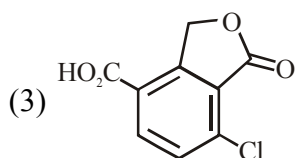
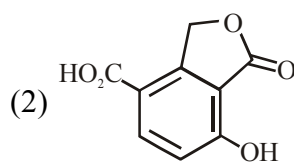
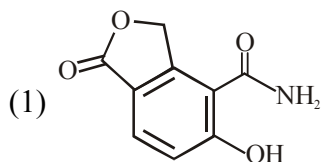
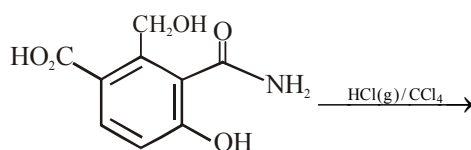
- (1) Etard reaction (2) Hell - Volhard - Zelinsky reaction
(3) Wolff - Kischner reaction (4) Rosenmund reaction

12. In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br_2 used per mole of amine produced are : [JEE(Main)-2016]

- (1) Four moles of NaOH and one mole of Br_2
(2) One mole of NaOH and one mole of Br_2
(3) Four moles of NaOH and two moles of Br_2
(4) Two moles of NaOH and two moles of Br_2

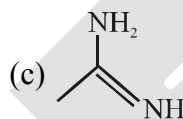
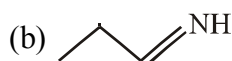
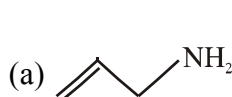
13. The major product expected from the following reaction is :

[JEE(Main On-Line)-2017]



14. The increasing order of basicity of the following compounds is :

[JEE(Main)-2018]



(1) (b) < (a) < (c) < (d)

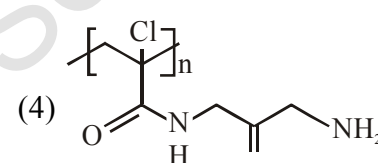
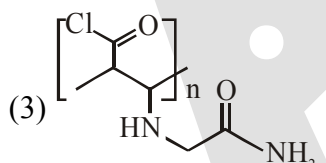
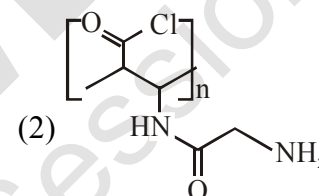
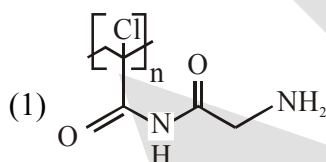
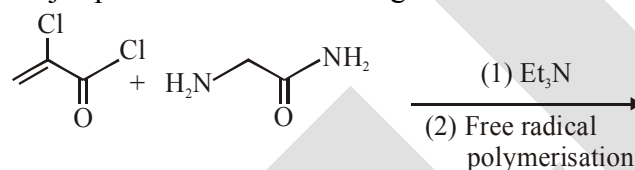
(2) (b) < (a) < (d) < (c)

(3) (d) < (b) < (a) < (c)

(4) (a) < (b) < (c) < (d)

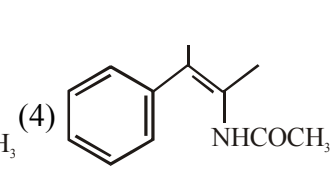
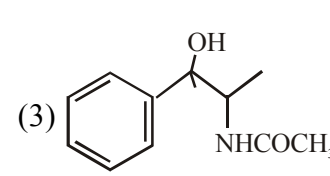
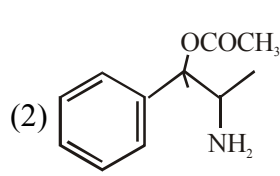
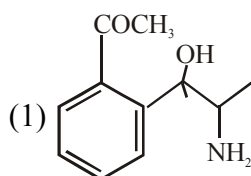
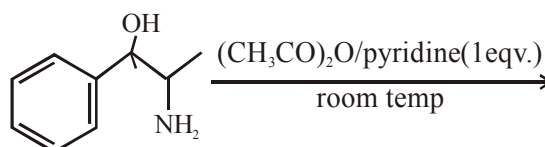
15. Major product of the following reaction is :

[JEE Main (Jan)-2019]



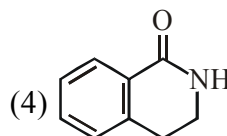
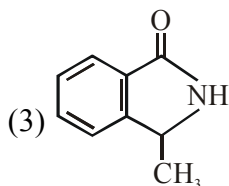
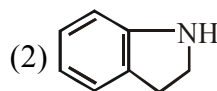
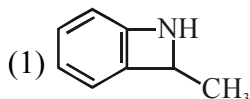
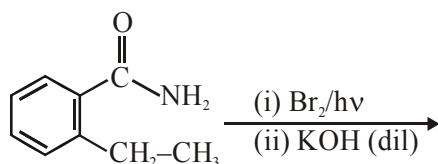
16. The major product obtained in the following reaction is :

[JEE Main (Jan)-2019]



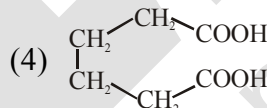
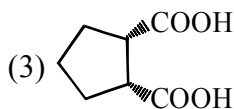
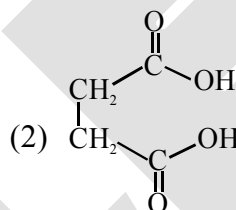
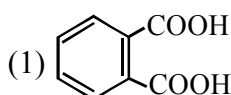
17. The major product of the following reaction is :

[JEE Main (Jan)-2019]

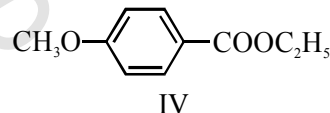
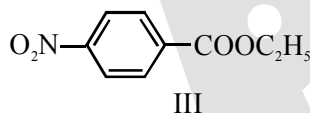
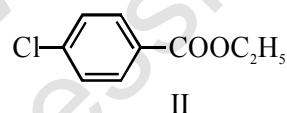
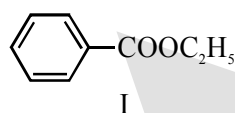


18. Which dicarboxylic acid in presence of a dehydrating agent is least reactive to give an anhydride :

[JEE Main (Jan)-2019]



19. The decreasing order of ease of alkaline hydrolysis for the following esters is :

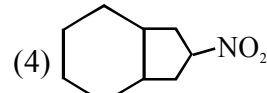
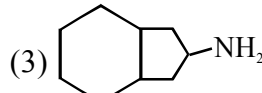
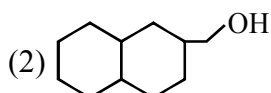
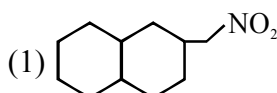
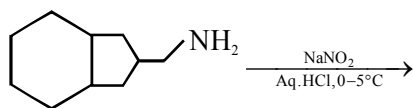


[JEE Main (Jan)-2019]

- (1) IV > II > III > I (2) III > II > I > IV (3) III > II > IV > I (4) II > III > I > IV

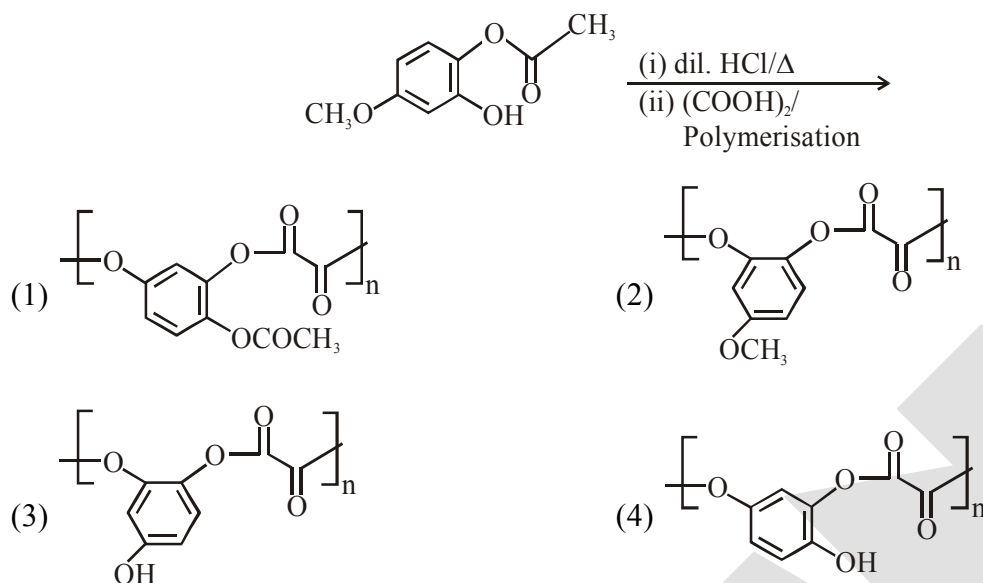
20. The major product formed in the reaction given below will be :

[JEE Main (Jan)-2019]



21. The major product of the following reaction is:

[JEE Main (Jan)-2019]



22. An aromatic compound 'A' having molecular formula $C_7H_6O_2$ on treating with aqueous ammonia and heating forms compound 'B'. The compound 'B' on reaction with molecular bromine and potassium hydroxide provides compound 'C' having molecular formula C_6H_7N . The structure of 'A' is :

[JEE Main (Jan)-2019]



23. A compound 'X' on treatment with $Br_2/NaOH$, provided C_3H_9N , which gives positive carbylamine test. Compound 'X' is :-

[JEE Main (Jan)-2019]

- (1) $CH_3COCH_2NHCH_3$ (2) $CH_3CH_2COCH_2NH_2$
 (3) $CH_3CH_2CH_2CONH_2$ (4) $CH_3CON(CH_3)_2$

24. The increasing order of reactivity of the following compounds towards reaction with alkyl halides directly is :

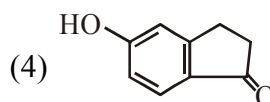
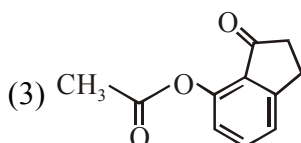
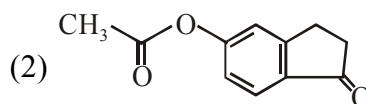
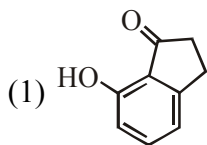
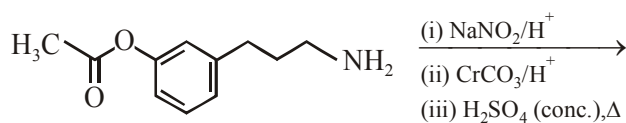
[JEE Main (Jan)-2019]



- (1) (B) < (A) < (D) < (C) (2) (B) < (A) < (C) < (D)
 (3) (A) < (C) < (D) < (B) (4) (A) < (B) < (C) < (D)

25. The major product of the following reaction is:

[JEE Main (Jan)-2019]



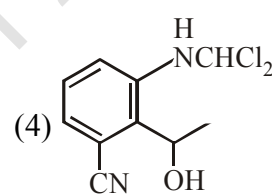
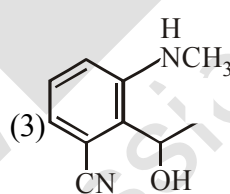
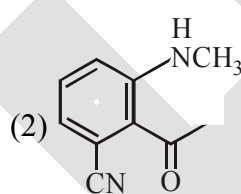
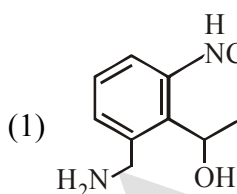
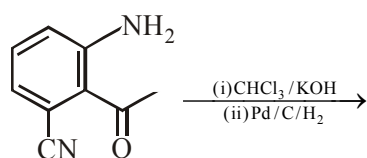
26. Which of the following amines can be prepared by Gabriel phthalimide reaction ?

[JEE Main (Apr)-2019]

- (1) Neo-pentylamine (2) n-butylamine (3) triethylamine (4) t-butylamine

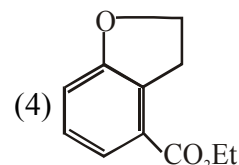
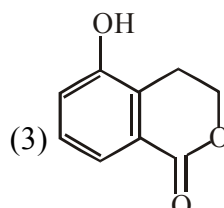
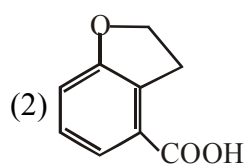
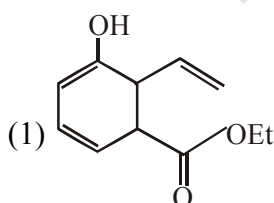
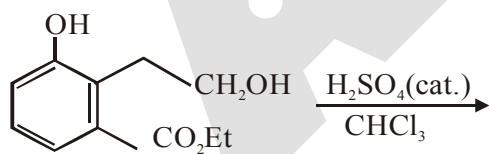
27. The major product obtained in the following reaction is :

[JEE Main (Apr)-2019]



28. The major product of the following reaction is:

[JEE Main (Apr)-2019]



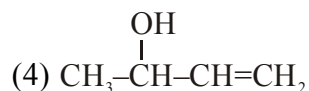
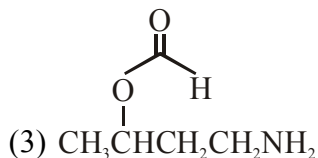
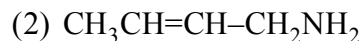
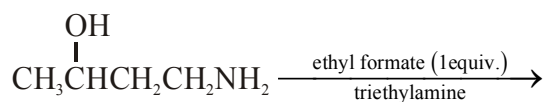
29. Hinsberg's reagent is :

[JEE Main (Apr)-2019]

- (1) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ (2) $\text{C}_6\text{H}_5\text{COCl}$ (3) SOCl_2 (4) $(\text{COCl})_2$

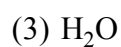
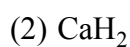
30. The major product of the following reaction is :

[JEE Main (Apr)-2019]



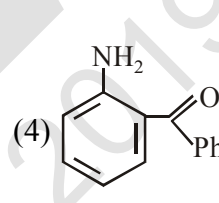
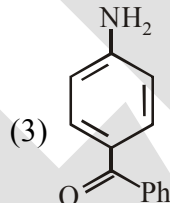
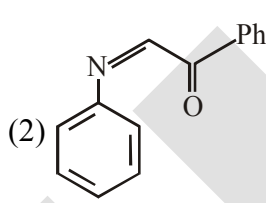
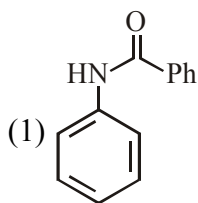
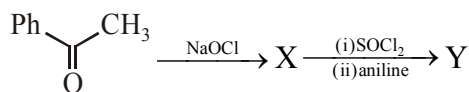
31. Ethylamine ($\text{C}_2\text{H}_5\text{NH}_2$) can be obtained from N-ethylphthalimide on treatment with :

[JEE Main (Apr)-2019]



32. The major product 'Y' in the following reaction is:-

[JEE Main (Apr)-2019]



EXERCISE # IV (B) (JEE ADVANCED)

- Which of the following carboxylic acids undergo decarboxylation easily: [IIT 1995]

(A) $\text{C}_6\text{H}_5\text{CO}-\text{CH}_2\text{COOH}$	(B) $\text{C}_6\text{H}_5\text{COCOOH}$
(C) $\text{C}_6\text{H}_5\underset{\text{OH}}{\text{CH}_2}-\text{COOH}$	(D) $\text{C}_6\text{H}_5\underset{\text{NH}_2}{\text{CH}_2}-\text{COOH}$
- The molecular weight of benzoic acid in benzene as determined by depression in freezing point method corresponds to : [IIT 1996]

(A) Ionization of benzoic acid

(B) Dimerisation of benzoic acid

(C) Trimerisation of benzoic acid

(D) Solvation of benzoic acid
- When propionic acid is treated with aqueous NaHCO_3 , CO_2 is liberated. The 'C' of CO_2 comes from. [IIT 1999]

(A) Methyl group	(B) Carboxylic acid group
(C) methylene group	(D) bicarbonate
- Benzoyl chloride is prepared from benzoic acid by: [IIT 2000]

(A) $\text{Cl}_2, h\nu$	(B) SO_2Cl_2	(C) SOCl_2	(D) $\text{Cl}_2, \text{H}_2\text{O}$
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- Which of the following acids has the smallest dissociation constant? [IIT 2002]

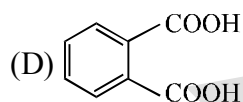
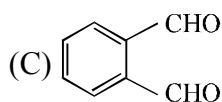
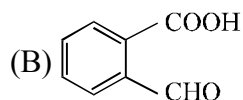
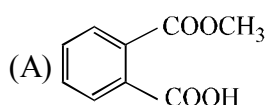
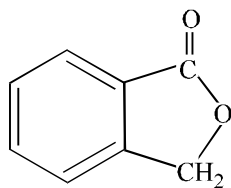
(A) $\text{CH}_3\text{CHF}\text{COOH}$	(B) $\text{FCH}_2\text{CH}_2\text{COOH}$
(C) $\text{BrCH}_2\text{CH}_2\text{COOH}$	(D) $\text{CH}_3\text{CHBr}\text{COOH}$
- When benzamide is treated with POCl_3 , the product is: [IIT 2004]

(A) Benzonitrile	(B) Aniline	(C) Chlorobenzene	(D) Benzylamine
------------------	-------------	-------------------	-----------------
- $$\text{MeO}-\text{C}_6\text{H}_4-\text{CHO} + (\text{X}) \xrightarrow[\text{H}_3\text{O}^+]{\text{CH}_3\text{COONa}} \text{MeO}-\text{C}_6\text{H}_4-\text{CH}=\text{CH}-\text{COCH}_3$$

The compound (X) is [IIT 2005]

(A) $(\text{CH}_3\text{CO})_2\text{O}$	(B) $\text{BrCH}_2-\text{COOH}$
(C) CH_3COOH	(D) $\text{CHO}-\text{COOH}$

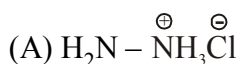
8. Which of the following reactants on reaction with conc. NaOH followed by acidification gives the following lactone as the only product? [IIT 2006]



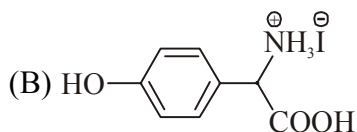
9. Match the compounds in **Column I** with their characteristic test(s)/reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. **Column-I**

Column-II

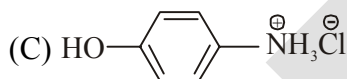
[IIT 2008]



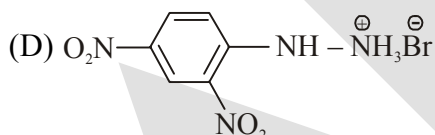
(P) Sodium fusion extract of the compound gives Prussian blue colour with FeSO_4



(Q) Gives positive FeCl_3 test



(R) Gives white precipitate with AgNO_3



(S) Reacts with aldehydes to form the corresponding hydrazone derivative

10. Match each of the compound in Column I with its characteristic reaction(s) in Column II.

Column-I

Column-II

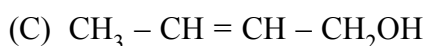
[IIT 2009]



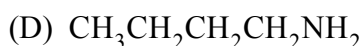
(P) Reduction with $\text{Pd-C} / \text{H}_2$



(Q) Reduction with $\text{SnCl}_2 / \text{HCl}$



(R) Development of foul smell on treatment with chloroform and alcoholic KOH

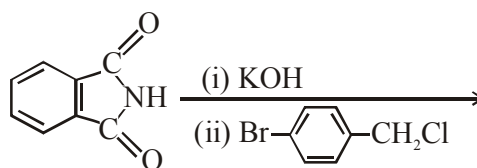


(S) Reduction with diisobutylaluminium hydride (DIBAL-H)

(T) Alkaline hydrolysis

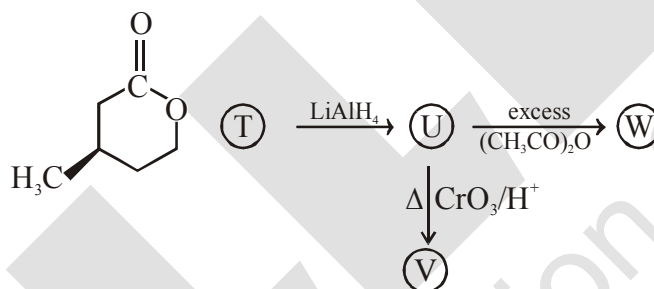
11. The major product of the following reaction is

[IIT 2011]

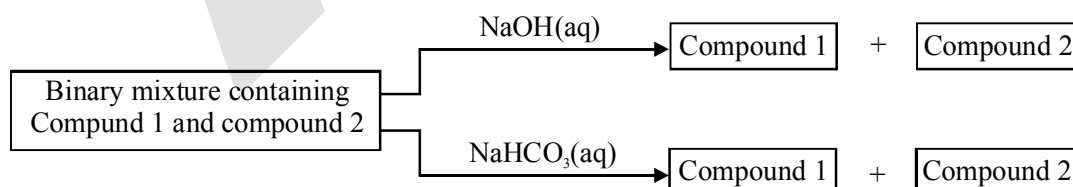


- (A)
- (B)
- (C)
- (D)

12. With reference the scheme given, which of the given statement(s) about T, U, V & W is/are correct [IIT 2012]



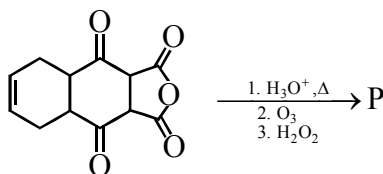
- (A) 'T' is soluble in hot aq NaOH
- (B) 'U' is optically active
- (C) mol formula of W is $C_{10}H_{18}O_4$
- (D) V gives effervescence with aq $NaHCO_3$
13. Identify the binary mixtures (s) that can be separated into the individual compounds, by differential extraction, as shown in the given scheme - [IIT 2012]



- (A) C_6H_5OH and C_6H_5COOH
- (B) C_6H_5COOH and $C_6H_5CH_2OH$
- (C) $C_6H_5CH_2OH$ and C_6H_5OH
- (D) $C_6H_5CH_2OH$ and $C_6H_5CH_2COOH$

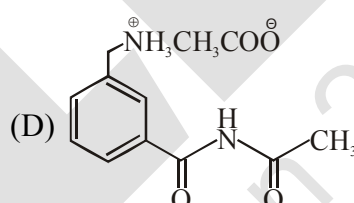
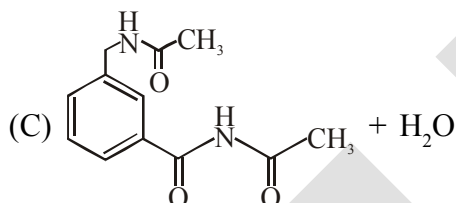
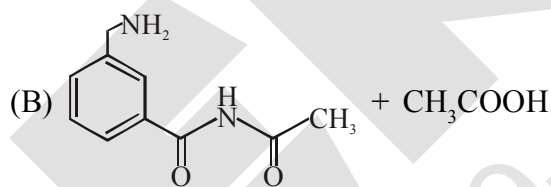
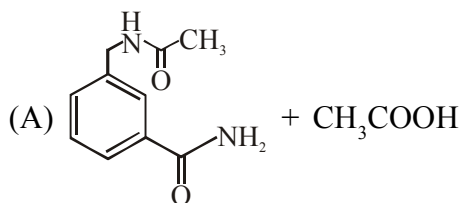
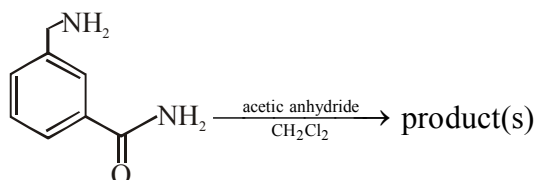
14. The total number of carboxylic acid groups in the product P is

[IIT 2013]



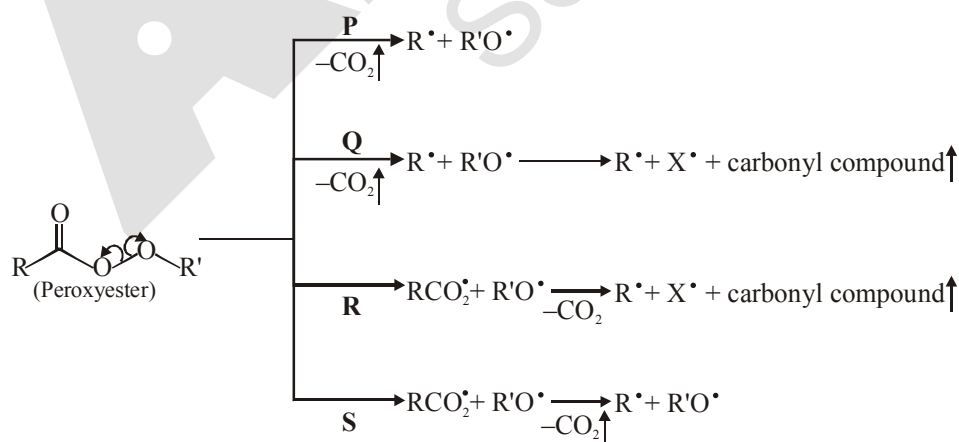
15. In the reaction shown below, the major product(s) formed is / are :

[IIT 2014]



16. Different possible **thermal** decomposition pathways for peroxyesters are shown below. Match each pathway from List-I with an appropriate structure from List-II and select the correct answer using the code given below the lists.

[IIT 2014]



List-I

(P) Pathway P

(Q) Pathway Q

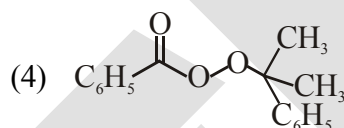
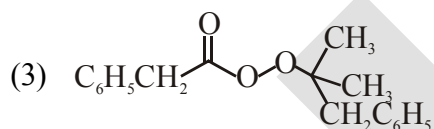
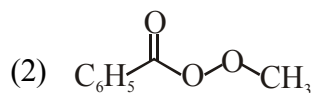
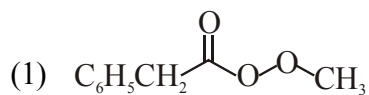
(R) Pathway R

(S) Pathway S

Code :

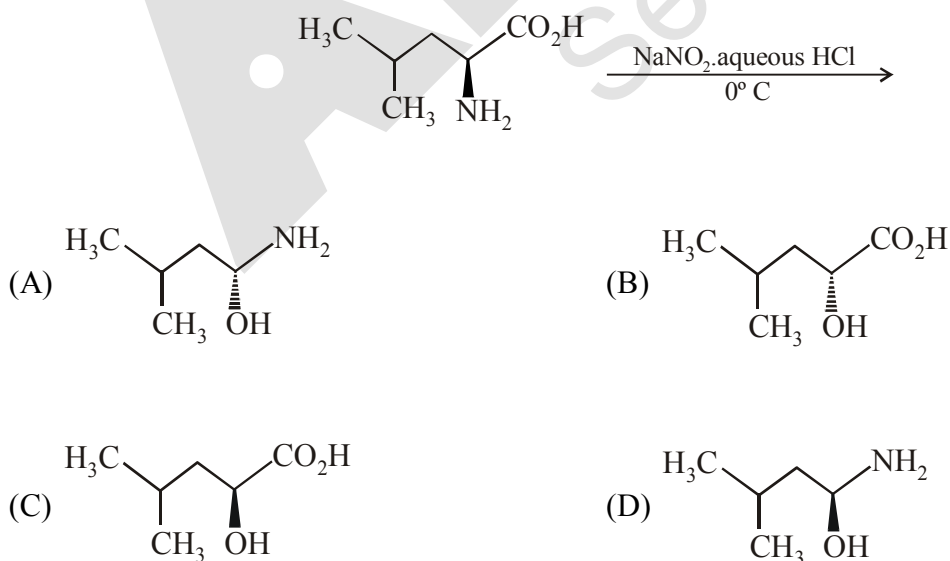
P	Q	R	S
(A) 1	3	4	2
(B) 2	4	3	1
(C) 4	1	2	3
(D) 3	2	1	4

List-II



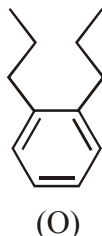
17. The major product of the reaction is :

[IIT 2015]

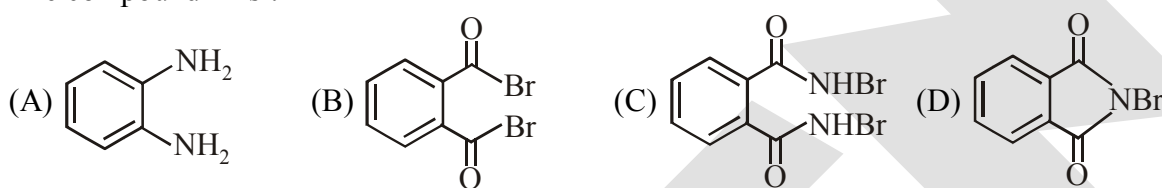


PARAGRAPH FOR NO. 18 & 19

Treatment of compound **O** with KMnO_4/H^+ gave **P**, which on heating with ammonia gave **Q**. The compound **Q** on treatment with Br_2/NaOH produced **R**. On strong heating, **Q** gave **S**, which on further treatment with ethyl 2-bromopropionate in the presence of KOH followed by acidification, gave a compound **T**. [IIT-JEE-2016]



18. The compound **R** is :

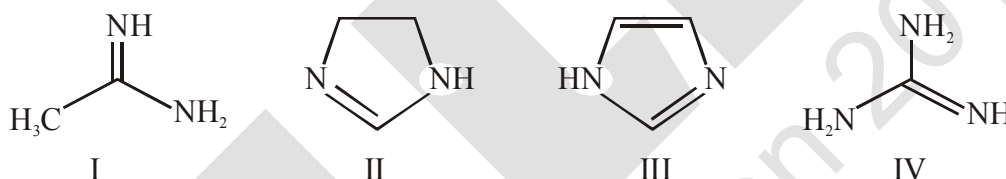


19. The compound **T** is :

- (A) Glycine (B) Alanine (C) Valine (D) Serine

20. The order of basicity among the following compounds is

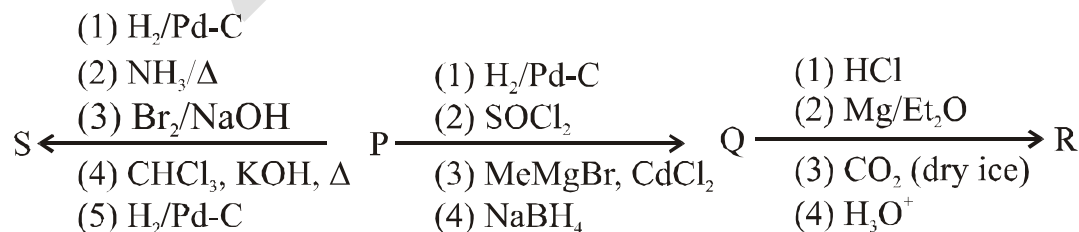
[IIT-JEE(Adv.)-2017]



- (A) II > I > IV > III (B) IV > II > III > I
(C) I > IV > III > II (D) IV > I > II > III

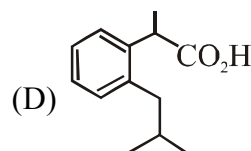
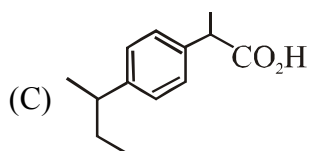
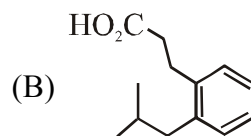
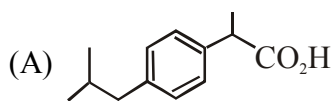
PARAGRAPH FOR NO. 21 & 22

An organic acid **P** ($\text{C}_{11}\text{H}_{12}\text{O}_2$) can easily be oxidized to a dibasic acid which reacts with ethyleneglycol to produce a polymer dacron. Upon ozonolysis, **P** gives an aliphatic ketone as one of the products. **P** undergoes the following reaction sequences to furnish **R** via **Q**. The compound **P** also undergoes another set of reactions to produce **S**. [IIT-JEE(Adv.)-2018]

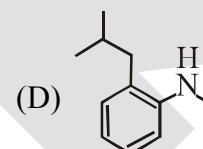
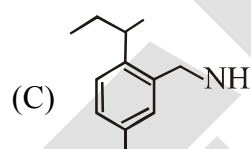
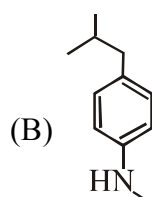
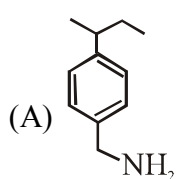


(There are two questions based on PARAGRAPH "A", the question given below is one of them)

21. The compound **R** is



22. The compound **S** is



ANSWER-KEY

EXERCISE # I

- | | | | |
|--------------|--------------|--------------|--------------|
| 1. Ans. (A) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (C) |
| 5. Ans. (C) | 6. Ans. (A) | 7. Ans. (B) | 8. Ans. (C) |
| 9. Ans. (A) | 10. Ans. (C) | 11. Ans. (D) | 12. Ans. (B) |
| 13. Ans. (B) | 14. Ans. (B) | 15. Ans. (A) | 16. Ans. (A) |
| 17. Ans. (B) | 18. Ans. (C) | 19. Ans. (B) | 20. Ans. (C) |
| 21. Ans. (B) | 22. Ans. (B) | 23. Ans. (A) | 24. Ans. (B) |
| 25. Ans. (C) | | | |

EXERCISE # II

- | | | | |
|------------------|------------------|--------------------|----------------|
| 1. Ans. (A,C,D) | 2. Ans. (A,C) | 3. Ans. (A,C,D) | 4. Ans. (A,B) |
| 5. Ans. (A,C,D) | 6. Ans. (A,B,C) | 7. Ans. (B,C) | 8. Ans. (C) |
| 9. Ans. (A,B,D) | 10. Ans. (A) | 11. Ans. (A,C) | 12. Ans. (B,C) |
| 13. Ans. (B) | 14. Ans. (A,B,C) | 15. Ans. (A,B,C,D) | 16. Ans. (A,D) |
| 17. Ans. (A,B,C) | 18. Ans. (B) | 19. Ans. (A,B,D) | 20. Ans. (C,D) |
| 21. Ans. (B,C,D) | | | |

EXERCISE # III

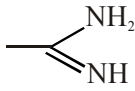
- | | |
|--|--|
| 1. Ans. (A)→P, S ; (B)→Q, S ; (C)→P, S ; (D)→R | |
| 2. Ans. (A)→S ; (B)→R ; (C)→P ; (D)→Q | 3. Ans. (A)→S ; (B)→R ; (C)→P ; (D)→Q |
| 4. Ans. (A)→R ; (B)→S ; (C)→Q ; (D)→P | 5. Ans. (A)→P ; (B)→Q ; (C)→S ; (D)→R, S |
| 6. Ans. (A)→P ; (B)→R ; (C)→P, S ; (D)→Q | |
| 7. Ans. (4) | 8. Ans. (4) |
| 9. Ans. (5) | 10. Ans. (9) |

EXERCISE # IV (A) (JEE-MAIN)

- | | | | |
|--------------|--------------|--------------|--------------|
| 1. Ans. (2) | 2. Ans. (1) | 3. Ans. (4) | 4. Ans. (4) |
| 5. Ans. (4) | 6. Ans. (2) | 7. Ans. (4) | 8. Ans. (4) |
| 9. Ans. (2) | 10. Ans. (1) | 11. Ans. (2) | 12. Ans. (1) |
| 13. Ans. (1) | | | |
| 14. Ans. (2) | | | |

Sol. Order of base nature depends on electron donation tendency.

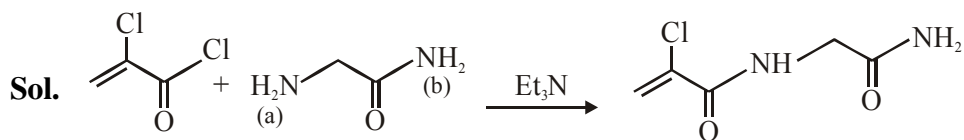
In compound  nitrogen is sp^2 hybridized so least basic among all given compound.

compound  is very strong nitrogenous organic base as lone pair of one nitrogen delocalize in resonance and make another nitrogen negatively charged and conjugate acid have two equivalent resonating structure.

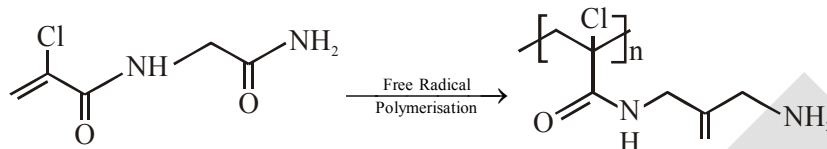
Thus it is most basic in given compounds.

 NHCH_3 (secondary amine) more basic than  NH_2 (primary amine)

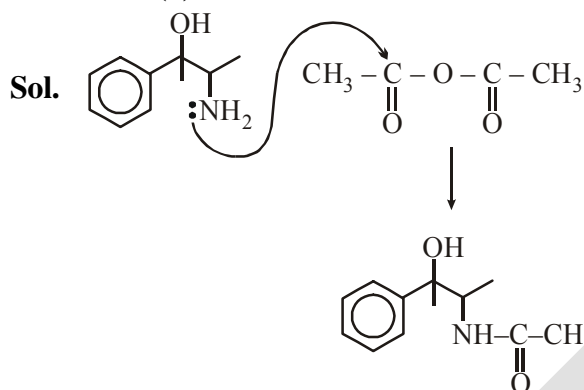
15. Ans. (4)



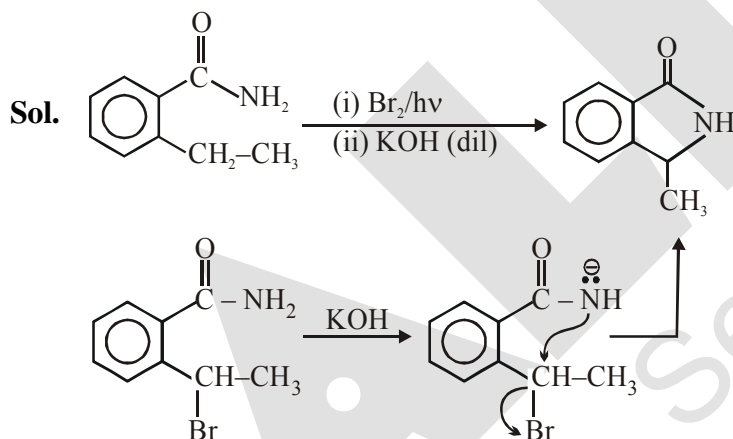
NH₂(a) will act as nucleophile as (b) is having delocalised lonepair.



16. Ans. (3)



17. Ans. (3)



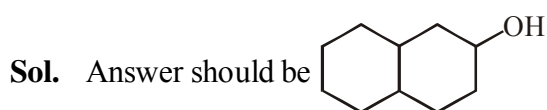
18. Ans. (4)

Sol. Adipic acid $\text{CO}_2\text{H}-(\text{CH}_2)_4-\text{CO}_2\text{H} \xrightarrow[\text{agent}]{\text{dehydrating}}$ 7 membered cyclic anhydride (Very unstable)

19. Ans. (2)

Sol. More is the electrophilic character of carbonyl group of ester faster is the alkaline hydrolysis.

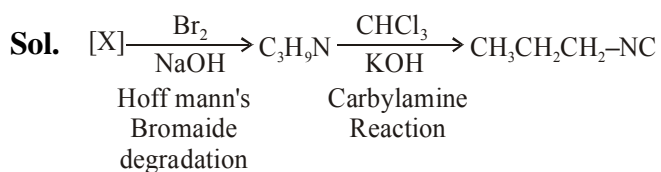
20. Ans. (Bonus)



21. Ans. (2)

22. Ans. (3)

23. Ans. (3)

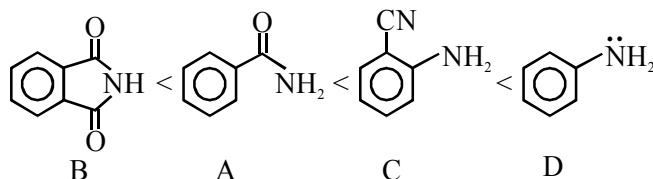


Thus [X] must be an alkyl halide with one carbon more than is amine.

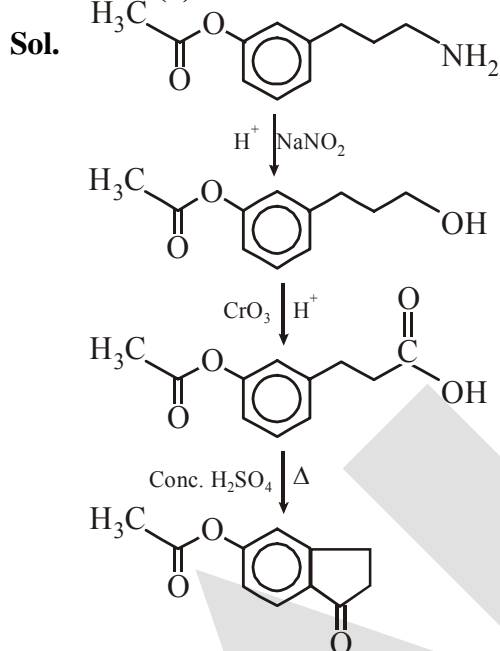
Thus [X] is $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONH}_2$

24. Ans. (2)

Sol. Nucleophilicity order

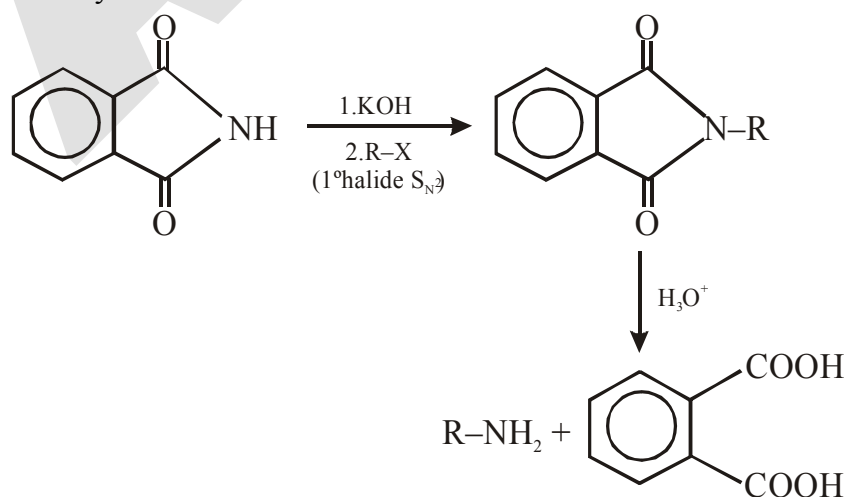


25. Ans. (4)

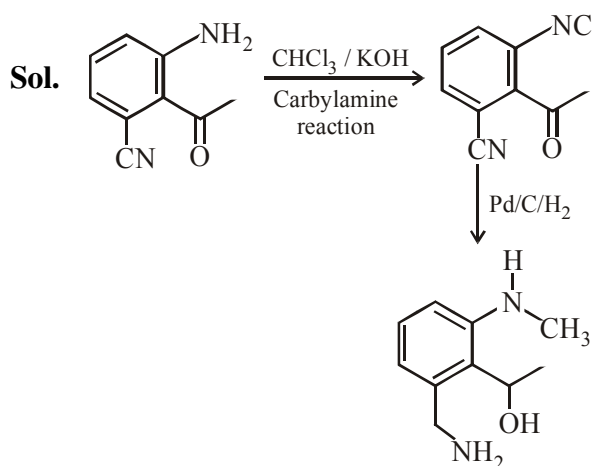


26. Ans. (2)

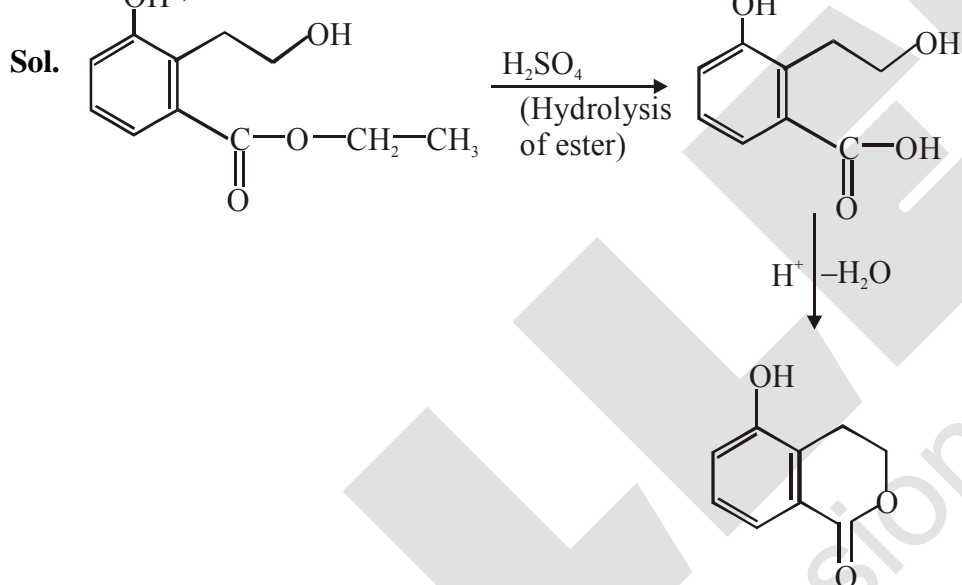
Sol. Gabriel phthalimide synthesis :



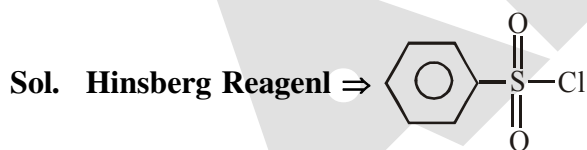
27. Ans. (1)



28. Ans. (3)

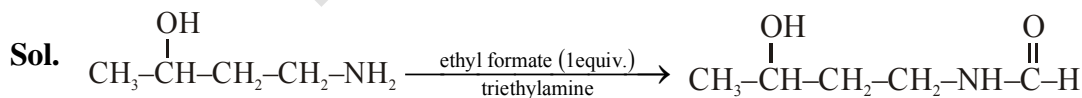


29. Ans. (1)



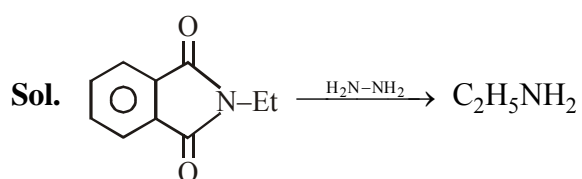
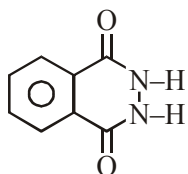
[Benzene Sulphonyl chloride]

30. Ans. (1)

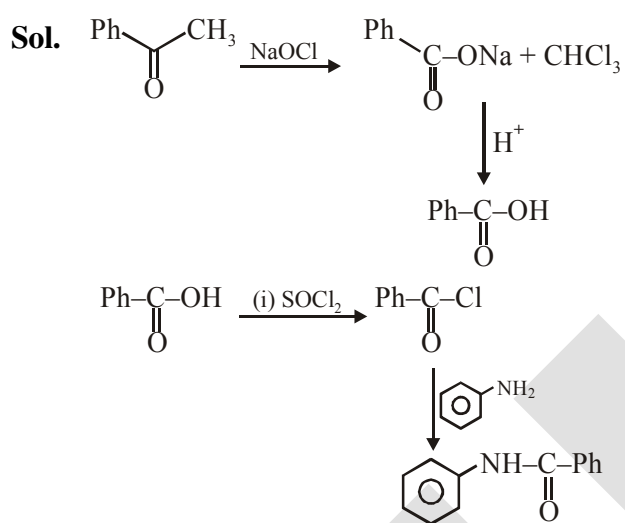


as NH_2 is a better nucleophile than OH .

31. Ans. (4)

reagent is $\text{NH}_2\text{-NH}_2$ byproduct will be

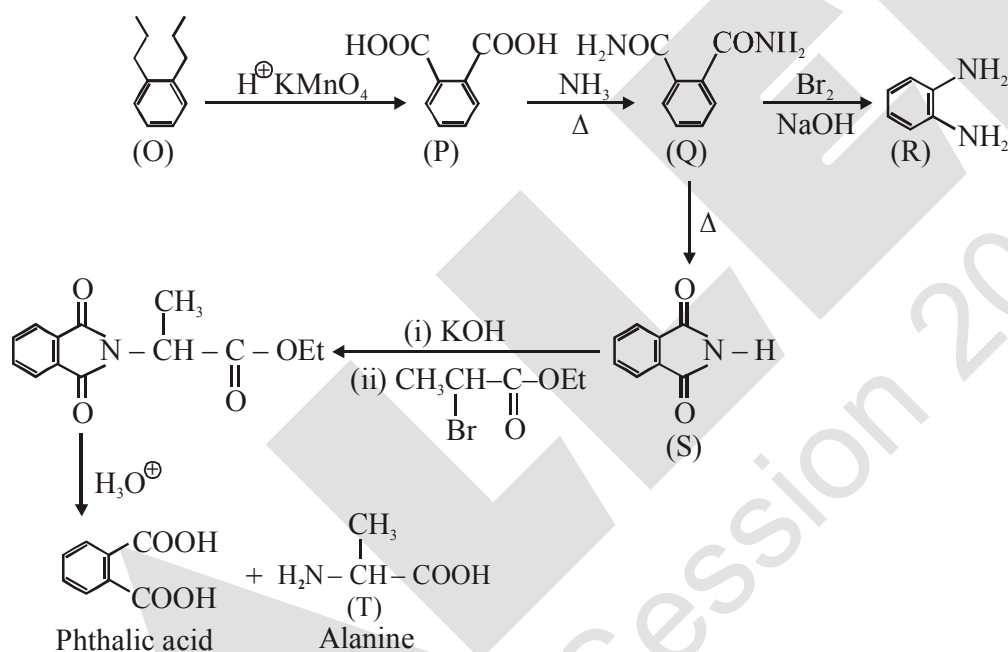
32. Ans. (1)



EXERCISE # IV (B) (JEE ADVANCED)

1. Ans. (A) 2. Ans. (B) 3. Ans. (D)
4. Ans. (C) 5. Ans. (C) 6. Ans. (A) 7. Ans. (A)
8. Ans. (C) 9. Ans. (A)→R, S ; (B)→P, Q ; (C)→P, Q, R ; (D)→P, S
10. Ans. (A)→P, Q, S, T ; (B)→P, S, T ; (C)→P ; (D)→R 11. Ans. (A)
12. Ans. (A, C, D) 13. Ans. (B, D) 14. Ans. (2) 15. Ans. (A)
16. Ans. (A) 17. Ans. (C)
18. Ans. (A)
19. Ans. (B)

Solution 18 & 19.



Q to R is Hoffmann's bromamide degradation reaction

S to T is Gabriel's phthalimide synthesis

20. Ans. (D) IV > I > II > III

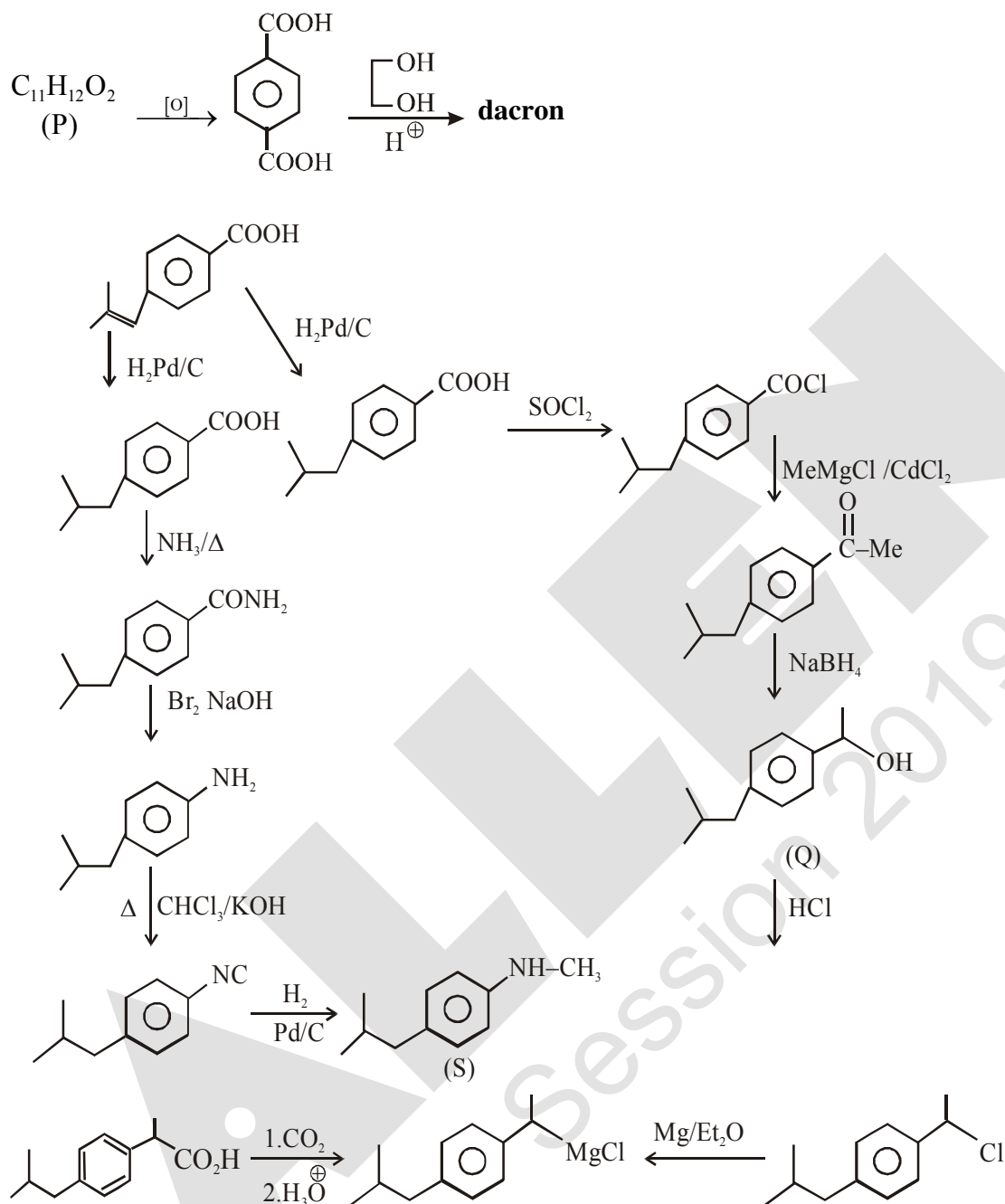
Sol. Basic strength $\propto$ stability of conjugated acid.

$$\propto +M / +H / +I$$

21. Ans. (A)

22. Ans. (B)

Solution 21 & 22.



Important Notes

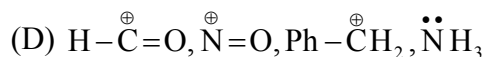
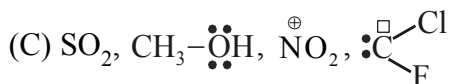
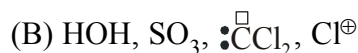
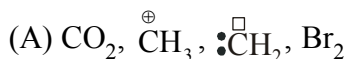
HALOGEN DERIVATIVES

ALLIED
Session 2019-20

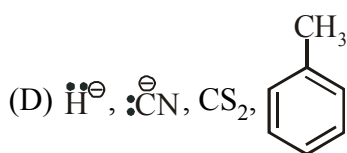
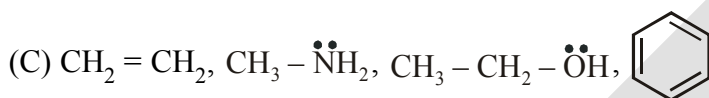
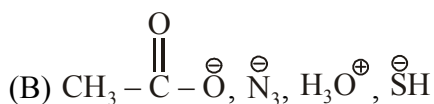
HALOGEN DERIVATIVES

EXERCISE # I (MAINS ORIENTED)

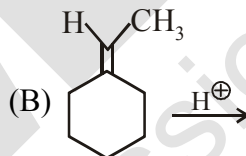
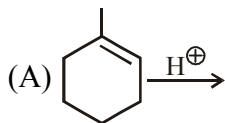
1. Identify set of electrophiles :



2. Identify set of nucleophiles :



3. Which of the following will form 2° carbocation?



4. Incorrect statement about carbocation is :

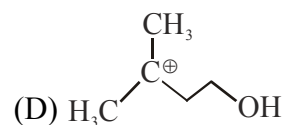
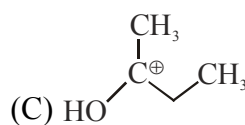
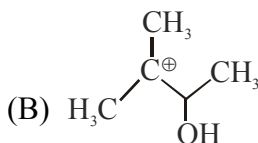
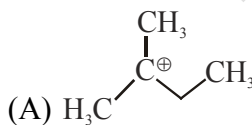
(A) It is lewis acid

(B) It has 6 electrons in valency shell

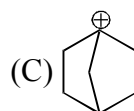
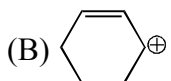
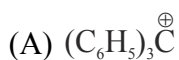
(C) It is electrophile

(D) It is always trigonal planer

5. Which of the following carbocation is most stable ?



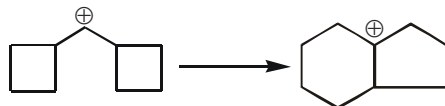
6. Which carbocation is least likely to be formed as an intermediate ?



7. Which one of the following carbocation would you expect to rearrange :

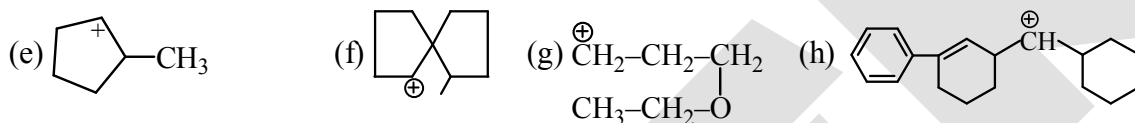


8. How many 1,2-shifts are involved during the course of following reaction :



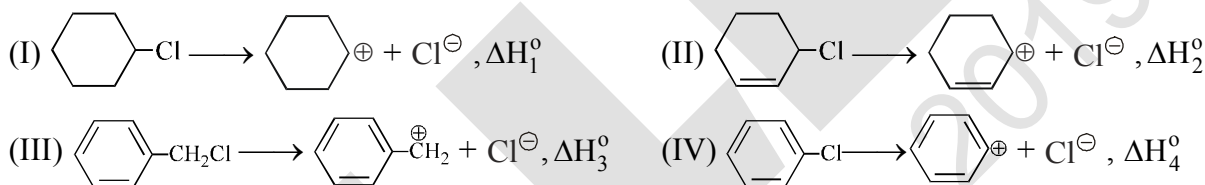
(A) 1 (B) 2 (C) 3 (D) 4

9. How many following carbocation undergo re-arrangement -



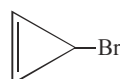
(A) 5 (B) 8 (C) 6 (D) 7

10. For the reactions

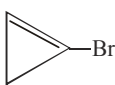


The correct decreasing order of enthalpies of reaction for producing carbocation is :

(A) $\Delta H_1^0 > \Delta H_2^0 > \Delta H_3^0 > \Delta H_4^0$ (B) $\Delta H_4^0 > \Delta H_1^0 > \Delta H_2^0 > \Delta H_3^0$
 (C) $\Delta H_3^0 > \Delta H_2^0 > \Delta H_1^0 > \Delta H_4^0$ (D) $\Delta H_2^0 > \Delta H_1^0 > \Delta H_4^0 > \Delta H_3^0$

11. , which is not the correct statement :

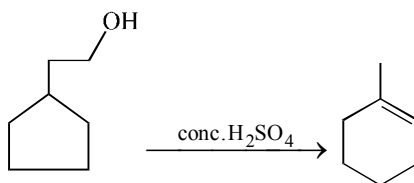
(A) I is more soluble in water than bromocyclopropane
 (B) I gives pale yellow ppt. on addition with aq. AgNO_3
 (C) I is having lower dipole moment than bromocyclopropane

(D) I is more ionic than 

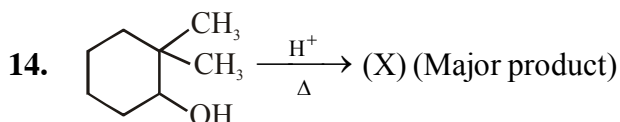
12. A solution of (–) –1-chloro–1-phenylethane in toluene racemises slowly in the presence of a small amount of SbCl_5 , due to the formation of :-

(A) carbanion (B) Carbene (C) carbocation (D) free radical

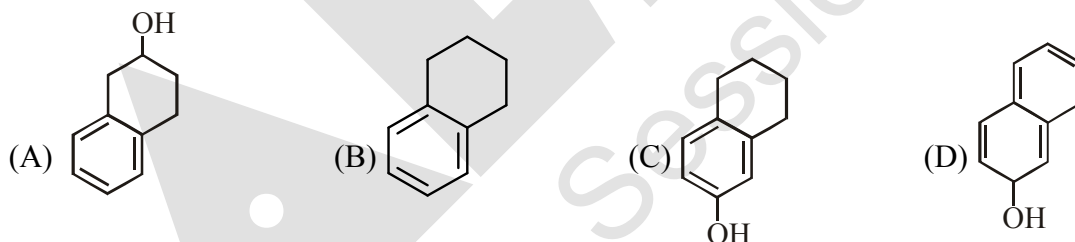
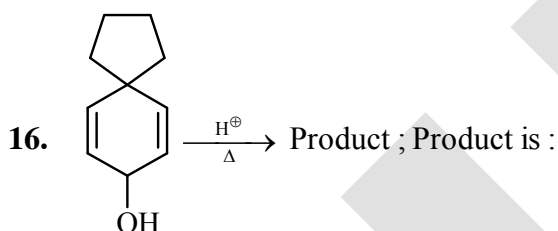
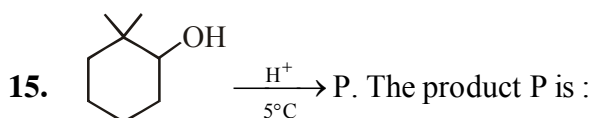
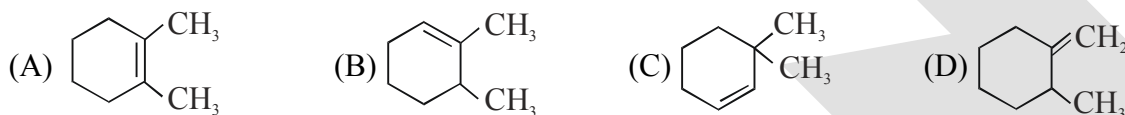
13. How many 1,2-Shifts of carbocation intermediate are involved during the course of following reaction :



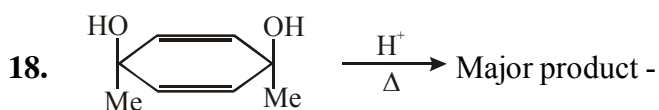
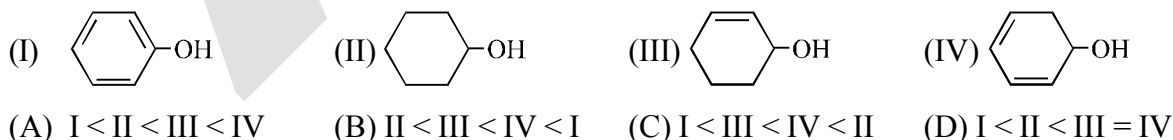
- (A) 1 (B) 2 (C) 3 (D) 4



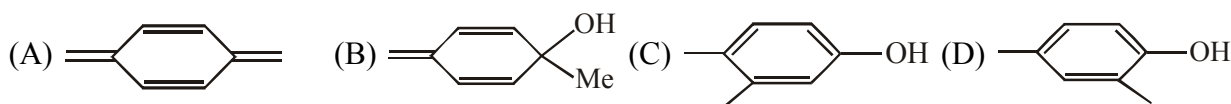
Major product (X) is :



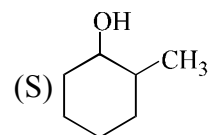
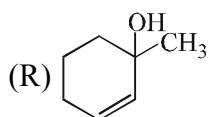
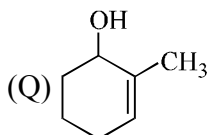
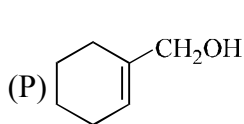
17. Among the given compounds, the correct order of rate of dehydration is :



Major product is :



19. Identify the correct order of rate of dehydration when given compounds are treated with conc. H_2SO_4 :

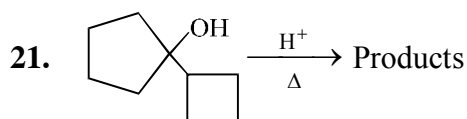
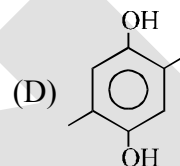
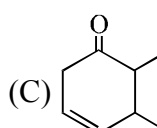
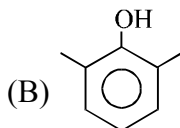
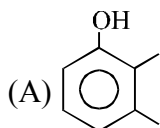
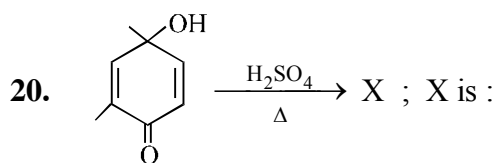


(A) $P > Q > R > S$

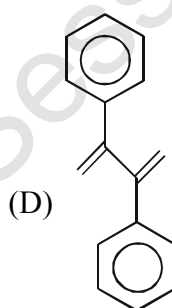
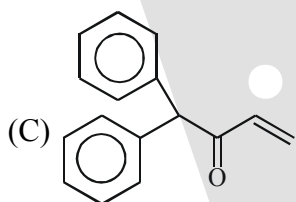
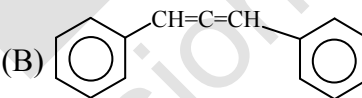
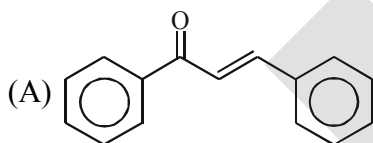
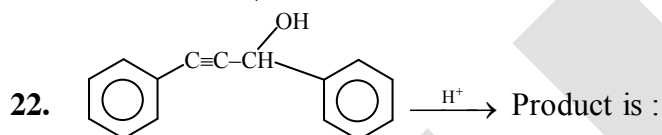
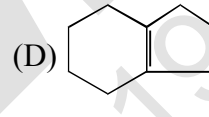
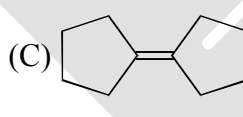
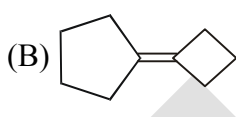
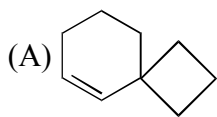
(B) $Q > P > R > S$

(C) $R > Q > P > S$

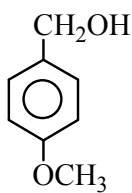
(D) $R > Q > S > P$



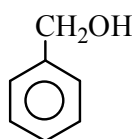
Major products is :



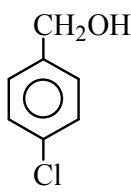
23. What is the decreasing order of rate of reaction with HBr for the following benzyl alcohol and its derivative :



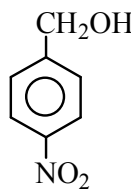
(A)



(B)



(C)



(D)

(A) $A > C > D > B$

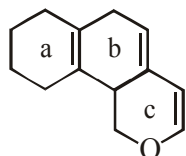
(B) $A > B > D > C$

(C) $D > C > B > A$

(D) $A > B > C > D$

24. Which will dehydrate at fastest rate by H_3PO_4 :

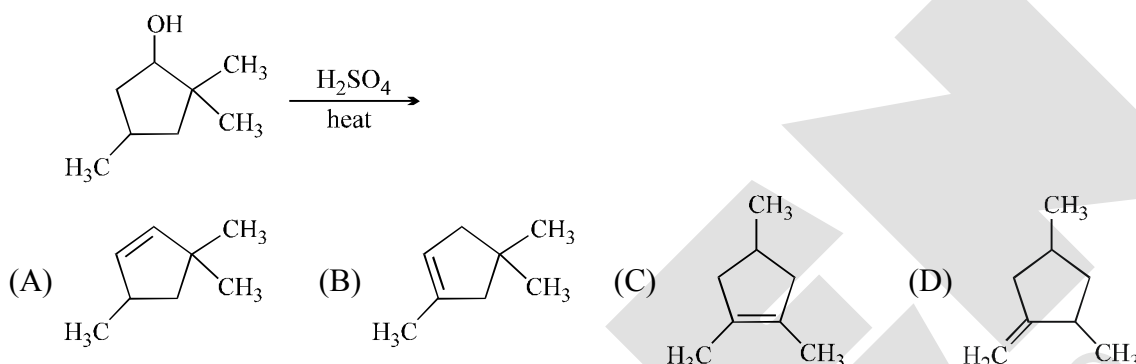
- (A) 2-methyl butan-2-ol (B) 3-methyl butan-2-ol
(C) Butan-1-ol (D) 2-methyl butan-1-ol



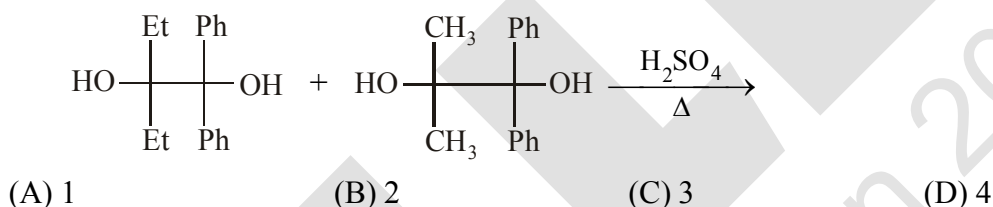
25. The double bond which is most reactive towards attack of electrophile :

- (A) a (B) b (C) c (D) None

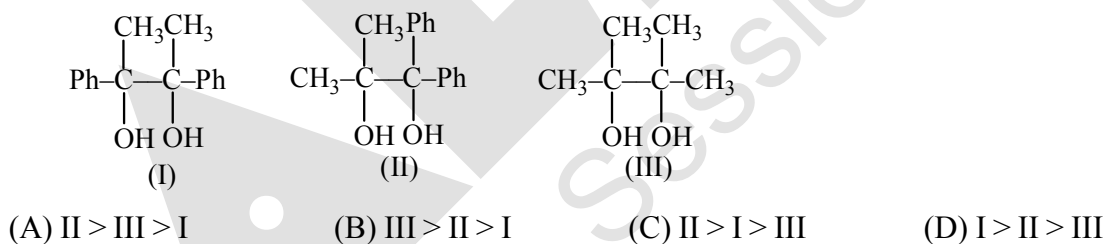
26. The major product formed in the following reaction is :



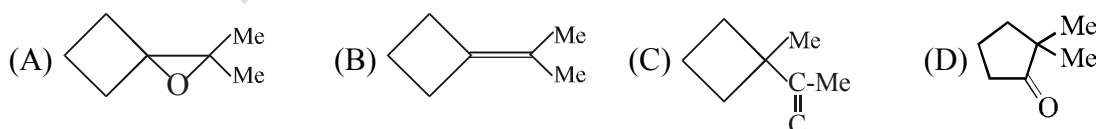
27. How many products are obtained in the given reaction :



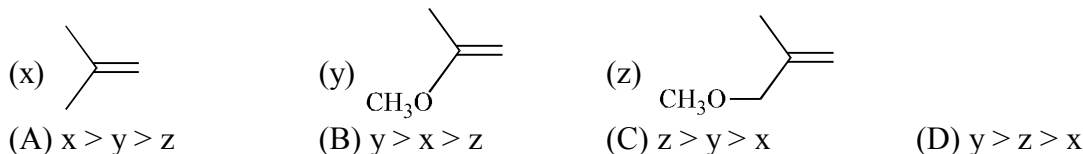
28. Compare rate of reaction towards pinacol pinacolone rearrangement.



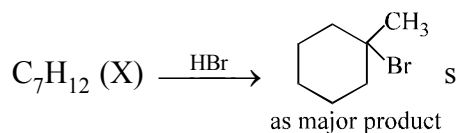
Product A is :



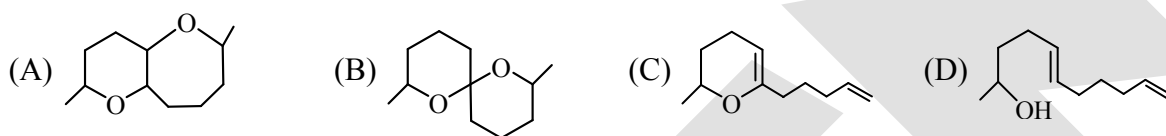
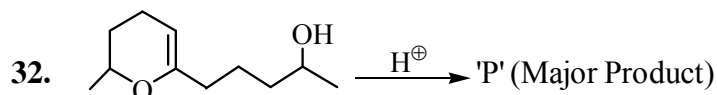
30. What is the order of reactivity with HBr :



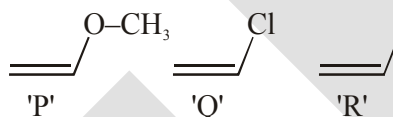
31. In the given reaction



(X) can not be :



33. Arrange the following compounds in decreasing order of electrophilic addition :

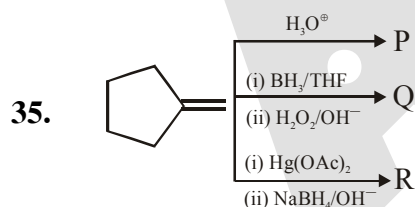


(A) $P > Q > R$ (B) $P > R > Q$ (C) $R > P > Q$ (D) $R > Q > P$



What is stereochemistry of product :

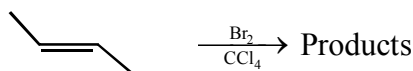
(A) Racemic mixture (B) Optically inactive
(C) Mixture of diastereomers (D) Meso product



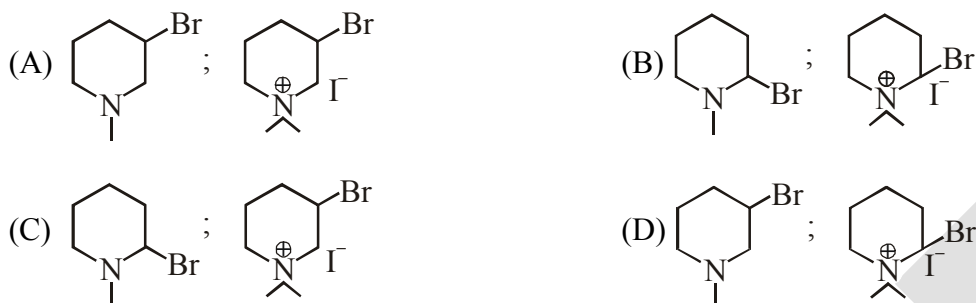
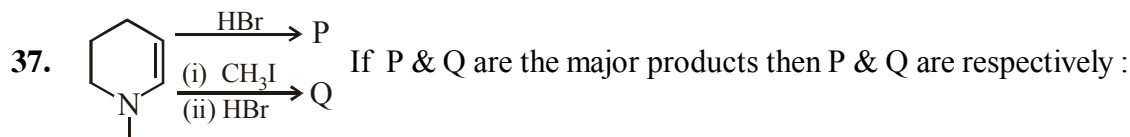
Correct statement regarding products P, Q & R

(A) Product P & R are same (B) Product Q & R are same
(C) P & Q are functional isomers (D) Product P, Q & R all are different

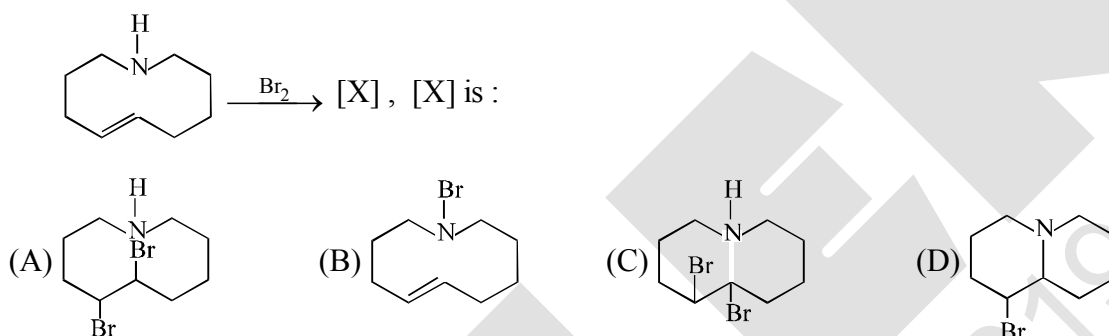
36. Select the incorrect statement about the product mixture in the following reaction :



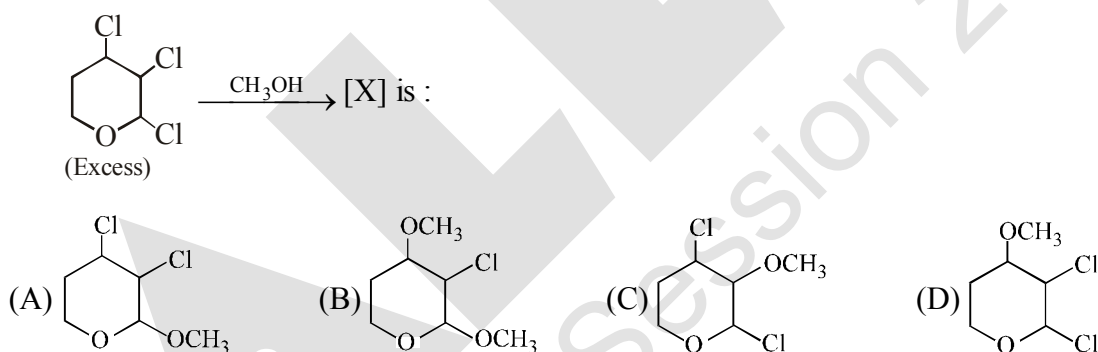
(A) It is optically active (B) It is racemic mixture
(C) It is a resolvable mixture (D) It is a mixture of erythro compounds



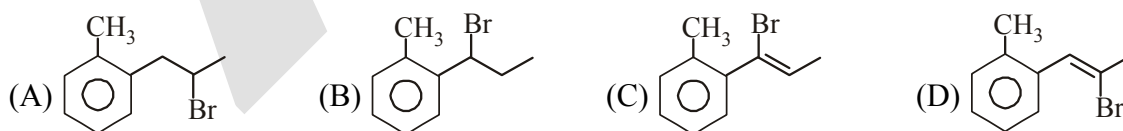
38. In the given reaction :



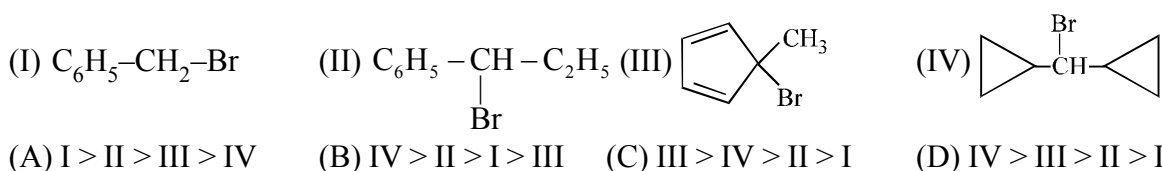
39. In the given reaction:



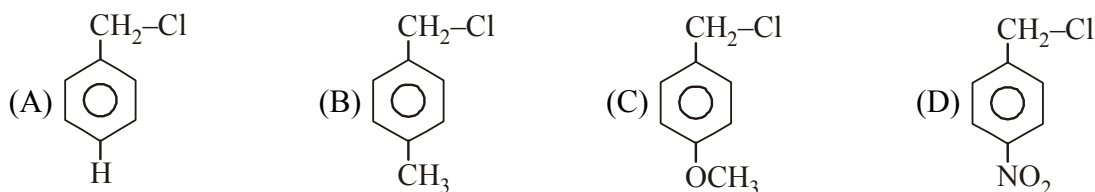
40. Which compound undergoes hydrolysis by the S_N1 mechanism at the fastest rate?



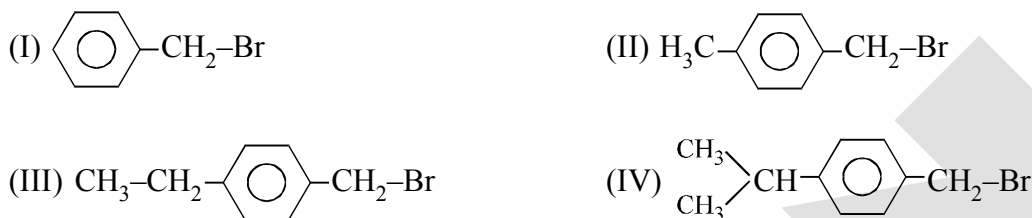
41. Arrange the following compounds in decreasing order of their reactivity for hydrolysis reaction



42. Which of the following is most reactive toward S_N1 reaction.

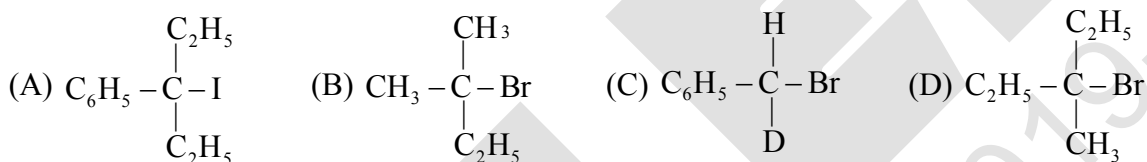


43. Arrange the following compounds in order of decreasing rate of hydrolysis for S_N1 reaction:

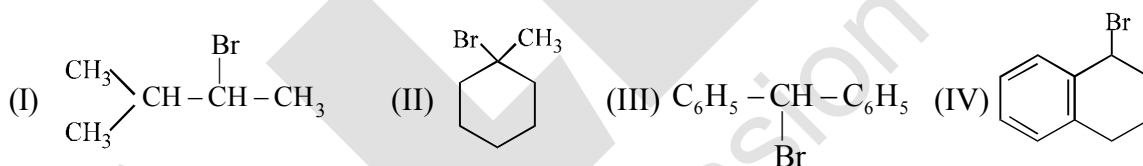


(A) II > III > IV > I (B) IV > III > II > I (C) III > IV > II > I (D) I > II > III > I

44. Which one of the following compounds will give enantiomeric pair on treatment with HOH?



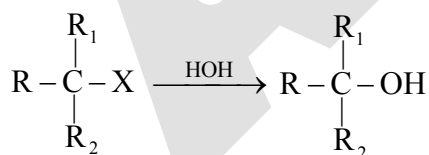
45. Consider the S_N1 solvolysis of the following halides in aqueous formic acid:



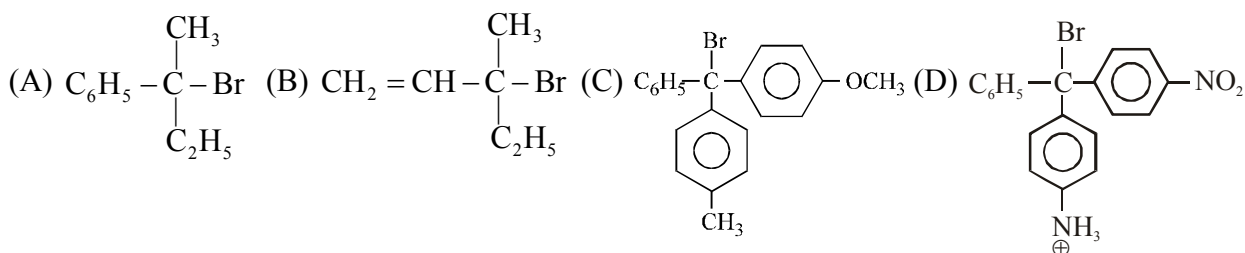
Decide decreasing order of reactivity of above alkyl halide?

(A) III > IV > II > I (B) II > IV > I > III (C) I > II > III > IV (D) III > I > II > IV

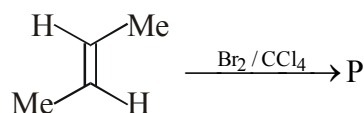
46. For the given reaction



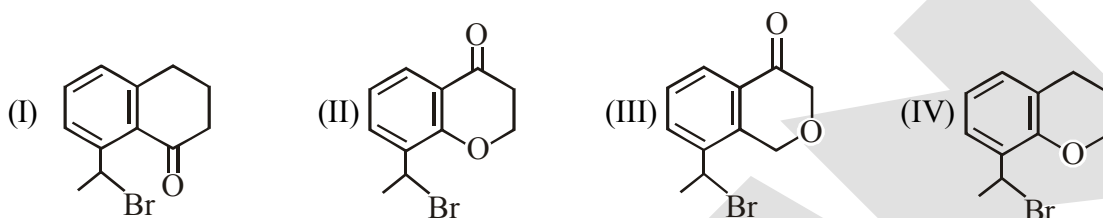
Which substrate will give maximum racemisation?



47. Select incorrect statements about the product (P) of the reaction :

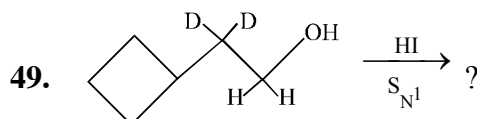


- (A) P is optically inactive due to internal compensation
 (B) P is optically inactive due to the presence of plane of symmetry in the molecule
 (C) The structure of P can have three optical isomers possible.
 (D) P can have four possible optical isomers.
48. Consider the following molecules :



The correct decreasing ease of hydrolysis of alkyl halide is :

- (A) II > III > IV > I (B) II > IV > III > I (C) II > I > III > IV (D) IV > II > III > I

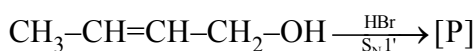


Major product is:



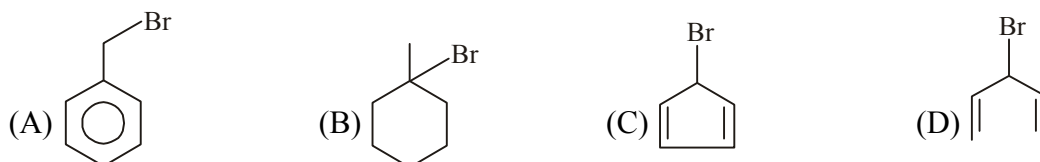
Find out the correct statement about the reaction.

- (A) Among the products 48% S and 48% R configuration containing molecules are present
 (B) Among the products 50% S and 50% R configuration containing molecules are present
 (C) Among the products 48% S and 52% R configuration containing molecules are present
 (D) Among the products 52% S and 48% R configuration containing molecules are present
51. In the given reaction the product [P] can be :



- (A) $\text{CH}_3\text{--CH=CH--CH}_2\text{--Br}$ (B) $\text{CH}_3\text{--}\overset{\text{Br}}{\text{CH}}\text{--CH=CH}_2$
 (C) $\text{CH}_2\text{=CH--CH=CH}_2$ (D) $\text{CH}_3\text{--}\overset{\text{Br}}{\text{CH}}\text{--CH}_2\text{--CH}_2\text{--OH}$

52. Which of the following can not give S_N1 reaction easily?



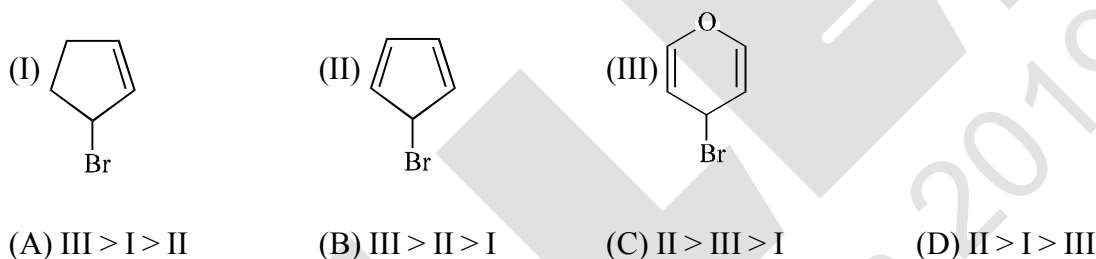
53. Which one of the following compounds will be most reactive for S_N1 reactions ?



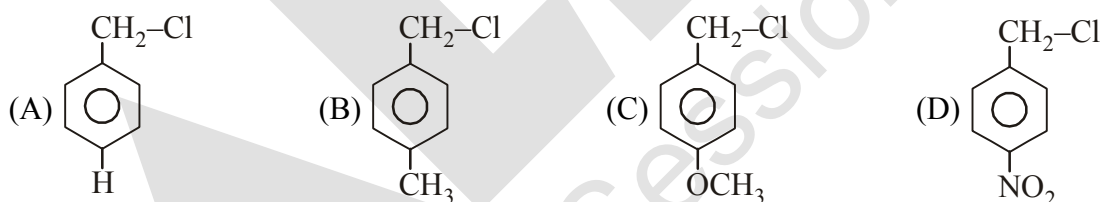
54. Which of the following compounds is most rapidly hydrolysed by S_N1 mechanism ?



55. Among the bromides I–III given below, the order of reactivity in S_N1 reaction is:



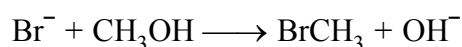
56. Which of the following is most reactive toward S_N2 .



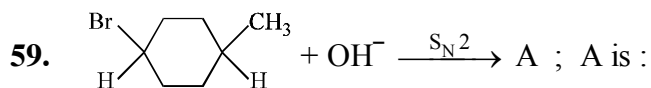
57. For reaction $CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$ the rate of reaction is given by the expression :

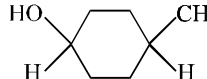
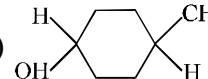


58. Select suitable reason for non-occurrence of the following reaction.



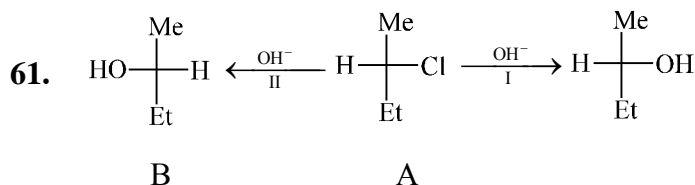
- (A) Attacking nucleophile is stronger one
(B) Leaving group is a stronger base than nucleophile
(C) Alcohols are not good substrate for S_N reaction
(D) Hydroxide ions are weak bases



- (A)  (B)  (C) Both (D) None

60. The reactivity of 2-bromo-2-methylbutane (I), 1-bromopentane (II) and 2-bromopentane (III) towards S_N2 displacement is such that :

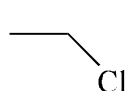
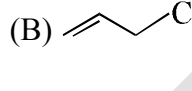
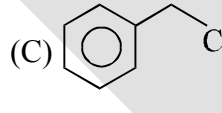
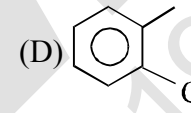
- (A) I > II > III (B) I > III > II (C) II > III > I (D) II > I > III



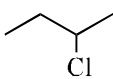
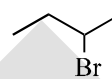
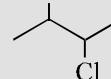
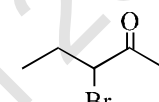
Mechanism involved :

- (A) I can't be S_N1 (B) II can't be S_N2 (C) I S_N1 & II S_N2 (D) I S_N2 & II S_N1

62. In which of the following replacement of Cl⁻ is most difficult ?

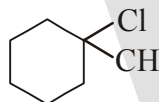
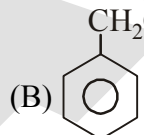
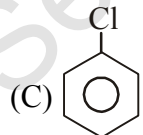
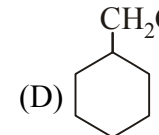
- (A)  (B)  (C)  (D) 

63. Arrange these compounds in order of increasing S_N2 reaction rate :

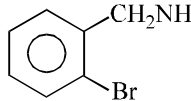
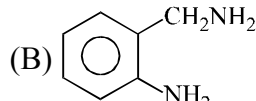
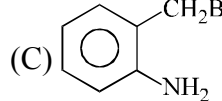
- (I)  (II)  (III)  (IV) 

- (A) III < I < II < IV (B) III < II < I < IV (C) IV < III < I < II (D) III < IV < I < II

64. Which reaction proceeds faster with NaI in DMSO ?

- (A)  (B)  (C)  (D) 

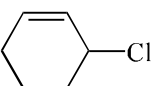
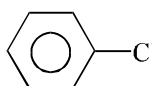
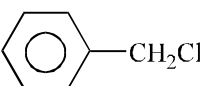


- (A)  (B)  (C)  (D) All of these

66. The given compound CH₃-O-CH₂-Br gives which one of the following reactions:

- (A) Only S_N1 (B) Only S_N2
(C) S_N1 as well as S_N2 (D) E₁ only

67. Which will give white ppt. with AgNO_3 ?

- (A)  (B)  (C)  (D) Both A & C

68. Consider the following groups :

- (I) $-\text{OAc}$ (II) $-\text{OMe}$ (III) $-\text{OSO}_2\text{Me}$ (IV) $-\text{OSO}_2\text{CF}_3$

The order of leaving group nature is:

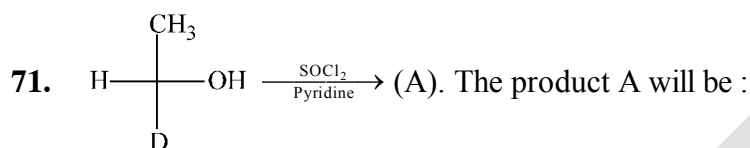
- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{I} > \text{II}$ (C) $\text{III} > \text{II} > \text{I} > \text{IV}$ (D) $\text{II} > \text{III} > \text{IV} > \text{I}$

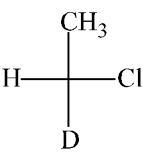
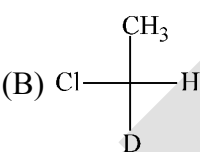
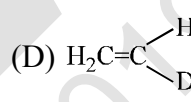
69. When ethyl bromide is treated with moist Ag_2O , the main product is:

- (A) Ethyl ether (B) Ethanol (C) Ethoxy ethane (D) All of these

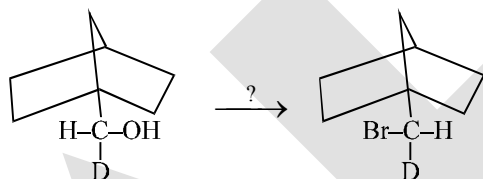
70. When ethyl bromide is treated with dry Ag_2O , the main product is:

- (A) Ethyl ether (B) Ethanol (C) Ethoxy ethane (D) All of these



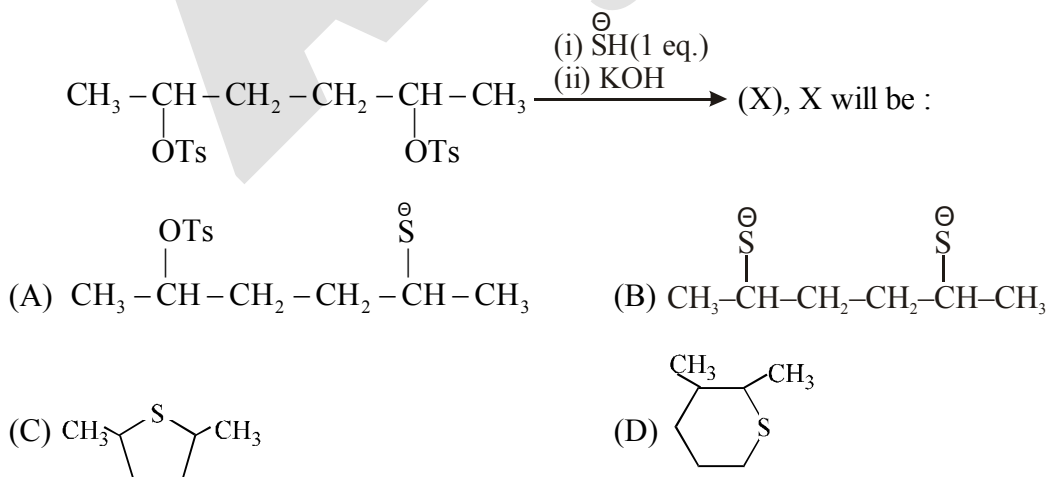
- (A)  (B)  (C) $\text{H}_2\text{C} = \text{CH}_2$ (D) 

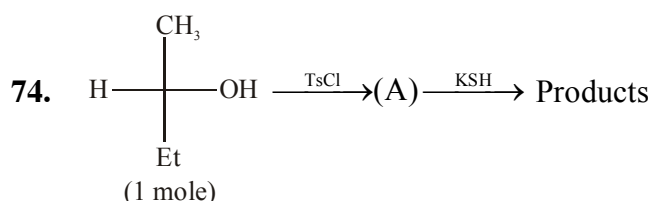
72. Which reaction conditions (reagents) is suitable for the following reaction:



- (A) $\text{Br}_2 / \text{CCl}_4$ (B) SOBr_2 (C) PBr_3 (D) $\text{HBr} / \text{conc H}_2\text{SO}_4$

73. In the given reaction



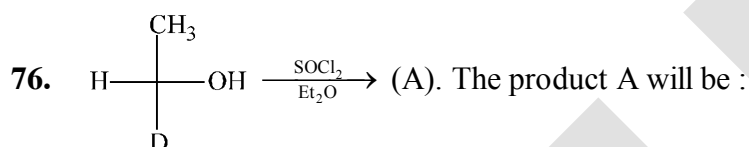


(Assuming all the substrate convert into substitution products containing 0.05 mole of S-configuration) Calculate the percentage of $\text{S}_{\text{N}}2$ mechanism.

- (A) 90% (B) 80% (C) 70% (D) 95%

75. The reaction of SOCl_2 on alkanols to form alkyl chlorides gives good yields because

- (A) Alkyl chlorides are immiscible with SOCl_2
(B) The other products of the reaction are gaseous and escape out
(C) Alcohol and SOCl_2 are soluble in water
(D) The reaction does not occurs via intermediate formation of an alkyl chloro sulphite

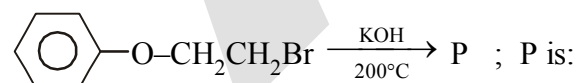


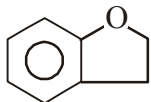
- (A)
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}-\text{C}-\text{Cl} \\ | \\ \text{D} \end{array}$$
 (B)
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{Cl}-\text{C}-\text{H} \\ | \\ \text{D} \end{array}$$
 (C) $\text{H}_2\text{C} = \text{CH}_2$ (D)
$$\begin{array}{c} \text{H} \\ \diagup \\ \text{H}_2\text{C} = \text{C} \\ \diagdown \\ \text{D} \end{array}$$

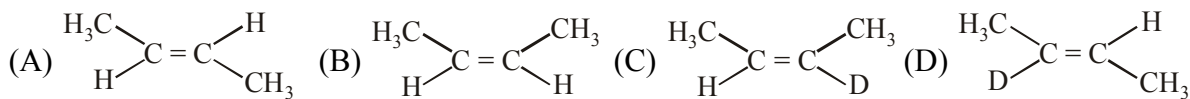
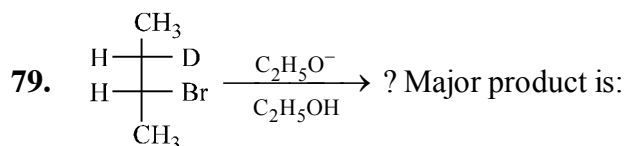
77. In the given pairs, which pair represent correct order of rate dehydrohalogenation reaction.

- (A) $\text{CH}_3(\text{CH}_2)_4\text{Cl} < (\text{CH}_3)_2\text{CHCH}_2\text{Cl}$ (B) $\text{CH}_3\text{CH}=\text{CHCH}_2\text{Cl} < (\text{CH}_3)_2\text{CHCH}_2\text{Cl}$
(C) $(\text{CH}_3)_2\text{CHI} < (\text{CH}_3)_2\text{CHBr}$ (D) $\text{CH}_3 - \text{CH}_2 - \text{Cl} < \text{CD}_3 - \text{CD}_2 - \text{Cl}$

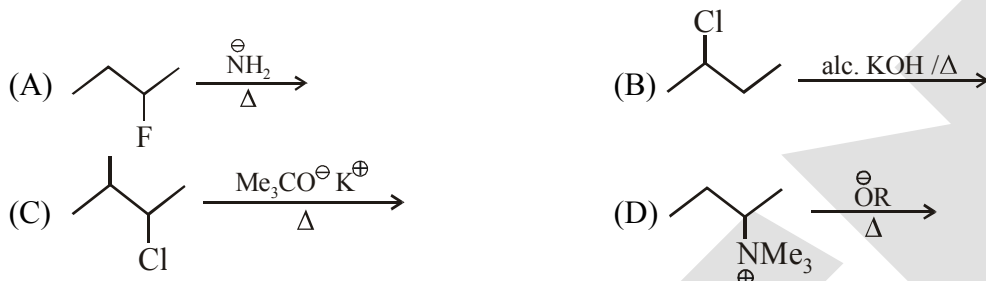
78. The product of the reaction



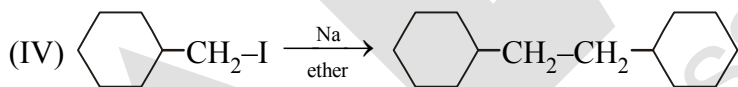
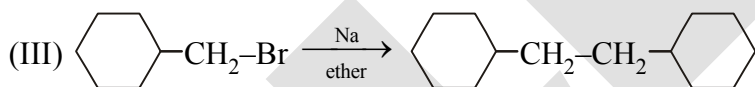
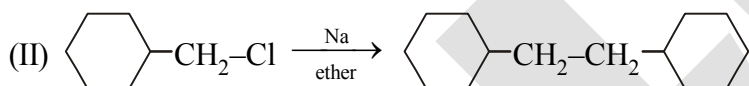
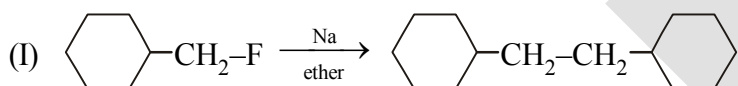
- (A) $\text{C}_6\text{H}_5\text{OH}$ (B) $\text{C}_6\text{H}_5\text{OCH}_2\text{CH}_2\text{OH}$
(C)  (D) $\text{C}_6\text{H}_5\text{OCH}=\text{CH}_2$



80. Which of following reaction(s) produce Saytzeff product as a major product :



81. The correct order of rate of following Wurtz reactions :

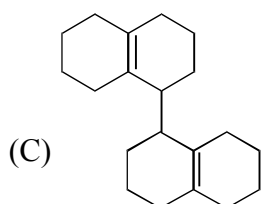
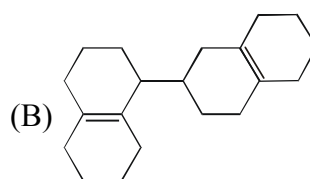
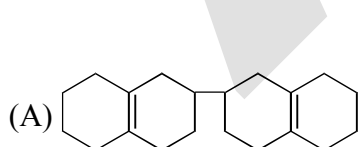
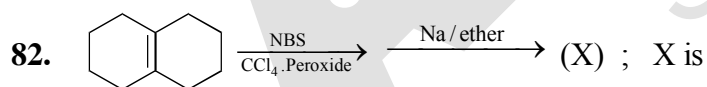


(A) I > II > III > IV

(B) II > I > III > IV

(C) IV > III > II > I

(D) In all rate of Wurtz reaction is same



(D) None of these

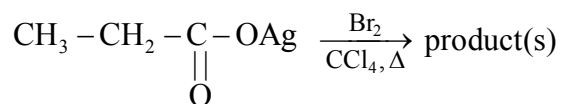
83. Find out the correct order of rate of reaction towards free radical allylic substitution :

- (I) $\text{CH}_3-\text{CH}=\text{CH}_2$ (II) $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}_2$ (III) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}=\text{CH}_2$
 (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$ (C) $\text{III} > \text{II} > \text{I}$ (D) $\text{III} > \text{I} > \text{II}$

84. What will be the major product, when 2-methyl butane undergoes bromination in presence of light ?

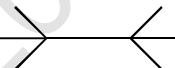
- (A) 1-Bromo-2-methyl butane (B) 2-Bromo-2-methyl butane
 (C) 2-Bromo-3-methyl butane (D) 1-Bromo-3-methyl butane

85. Which can not be the possible product of the given reaction ?



- (A) $\text{CH}_3-\text{CH}_2-\text{Br}$ (B) $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\underset{||}{\text{C}}}-\text{O}-\text{CH}_2-\text{CH}_3$
 (C) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$ (D) $\text{CH}_3-\text{CH}_2-\text{CH}_3$

86. Choose that alkane which cannot give only one monochloro derivative upon reaction with chlorine in sun light :

- (A)  (B)  (C)  (D) 

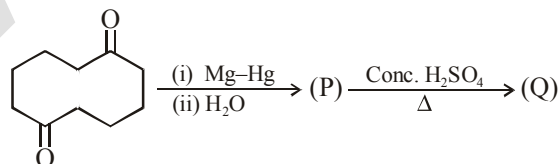
87. 2-chloropentane on halogenation with chlorine gives 2,3, dichloropentane. What will be the structure of free radical species formed in the reaction ?

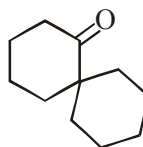
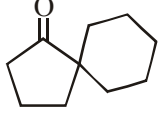
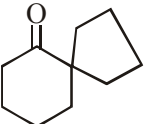
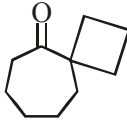
- (A) Tetrahedral (B) Trigonal planar (C) Square planar (D) Pyramidal

88. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be -

- (A) neopentane (B) propane (C) pentane (D) isopentane

89. Major product (Q) of following reaction is :

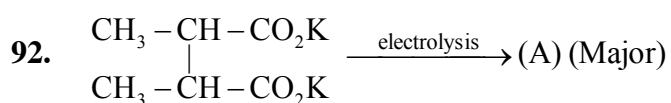
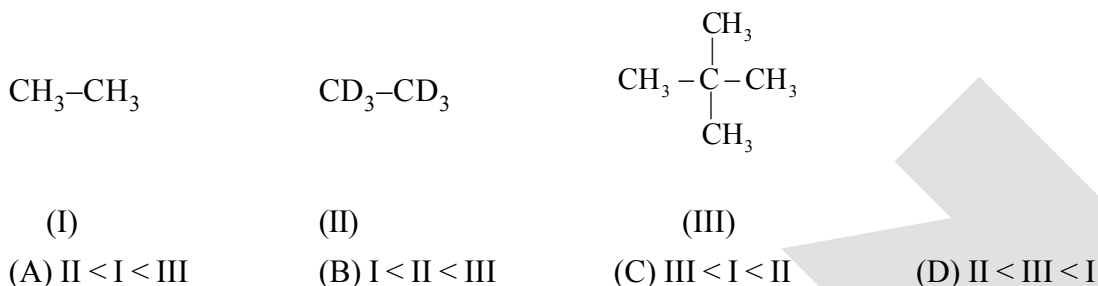


- (A)  (B)  (C)  (D) 

90. 1-Bromo-3-chloro cyclobutane on reaction with 2-equivalent of sodium in ether gives



91. Correct order of rate of photochlorination for following compounds is :



Major product (A) of above reaction :



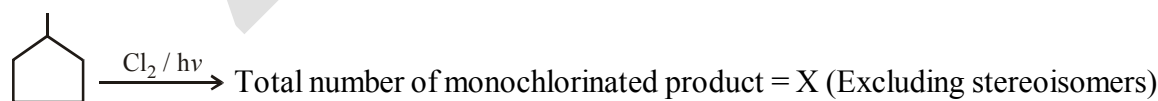
93. During the preparation of ethane by Kolbe's electrolytic method using inert electrode the pH of the electrolyte

- (A) Decreases progressively as the reaction proceeds
 (B) Increases progressively as the reaction proceeds
 (C) Remains constant throughout the reaction
 (D) May decrease if concentration of the electrolytes is not very high

94. When isobutane is chlorinated in the presence of diffused sunlight, then the product formed is :

- (A) Tertiary butyl chloride in major amount
 (B) Isobutyl chloride in major amount
 (C) Both 50 % each
 (D) n-Butyl chloride, isobutyl chloride and sec-butyl chloride are formed

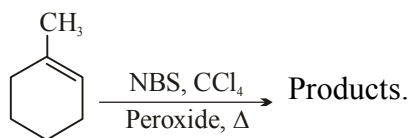
95. Consider the following reactions :



Identify value of X + Y.

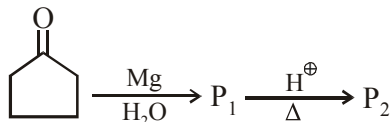
- (A) 8 (B) 9 (C) 11 (D) 10

96. Find out the total no. of products (including stereo) in the given reaction :



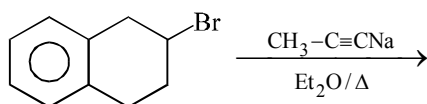
- (A) 8 (B) 9 (C) 10 (D) 11

97. Which of the following is not correct about P_2 :



- (A) It is a spiro compound (B) It is a Ketone
(C) It can show tautomerism (D) Its double bond equivalent is 4
98. On heating glycerol with excess amount of HI, the product formed is—
(A) Allyl iodide (B) Isopropyl iodide
(C) Propylene (D) 1,2,3-tri-iodopropane

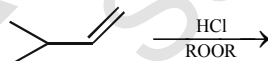
99. In the given reaction:



the products are:

- (A) and (B)
(C) (D)

100. Major product of the reaction -

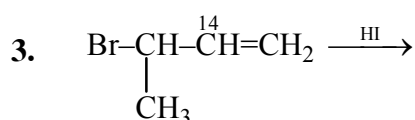
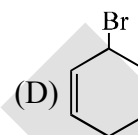
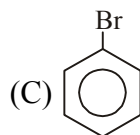
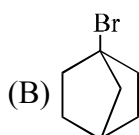
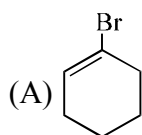


- (A) (B) (C) (D)

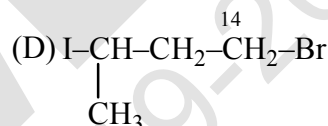
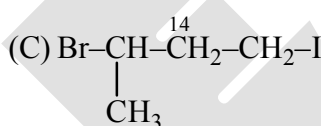
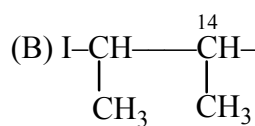
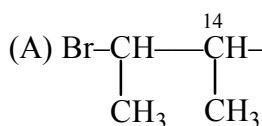
EXERCISE # II (A)

(Choose the correct option. One or more than one are correct)

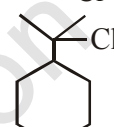
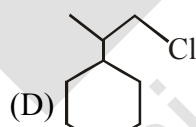
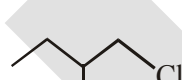
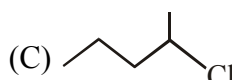
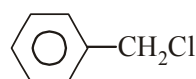
1. Rate of S_N2 depends on :
 (A) Conc of Nucleophile (B) Conc of substrate
 (C) Nature of leaving group (D) Nature of solvent
2. S_N2 reaction will be negligible in



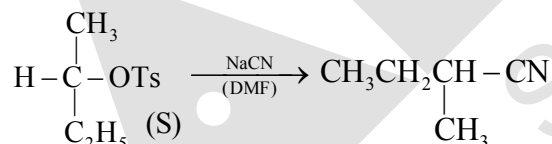
Products which can be obtained during the reaction in good yield :



4. In the given pair in which pair the first compound is more reactive than second towards S_N2 reaction.



5. Consider the given reaction



which of following statements is/are correct for the above reaction.

- (A) Product formation takes place due to the breaking of O-Ts
 (B) The reaction is S_N2
 (C) The reaction is S_N1
 (D) Configuration of product is (R)

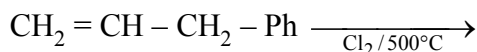
6. Which of the following statements is / are true?
- (A) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--I}$ will react more readily than $(\text{CH}_3)_2\text{CHI}$ for $\text{S}_\text{N}2$ reactions.
- (B) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--Cl}$ will react more readily than $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--Br}$ for $\text{S}_\text{N}2$ reaction.
- (C) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH}_2\text{--Br}$ will react more readily than $(\text{CH}_3)_3\text{C--CH}_2\text{--Br}$ for $\text{S}_\text{N}2$ reactions
- (D) $\text{CH}_3\text{--O--C}_6\text{H}_4\text{--CH}_2\text{Br}$ will react more readily than $\text{NO}_2\text{--C}_6\text{H}_5\text{--CH}_2\text{Br}$ for $\text{S}_\text{N}2$ reaction

7. Incorrect statement about alkyl halides is / are:
- (A) Tertiary alkyl halides undergo $\text{S}_\text{N}2$ substitutions
- (B) Alkyl iodides on exposure to sunlight gradually darken
- (C) Photo iodination is irreversible in presence of HIO_3
- (D) A nucleophilic substitution is most difficult in alkyl iodides

8. $\text{S}_\text{N}1$ & $\text{S}_\text{N}2$ is not favourable in

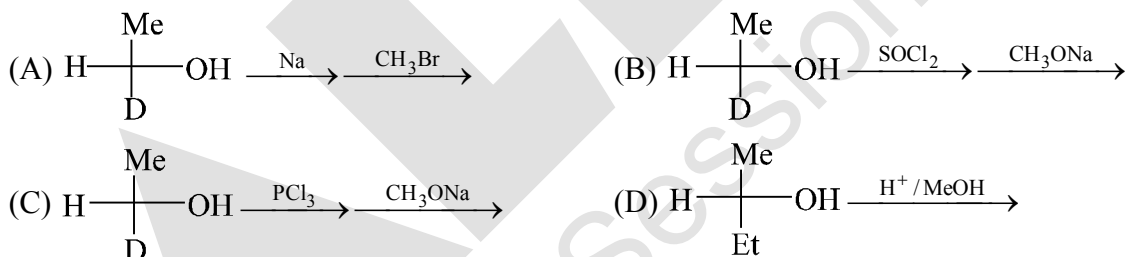


9. Correct statement(s) for the product(s) of following reaction.



- (A) Four different products are formed
- (B) Two optically active products are formed
- (C) The optically active compound formed here can also be made by the reaction of HCl
- (D) The reaction path is free radical substitution.

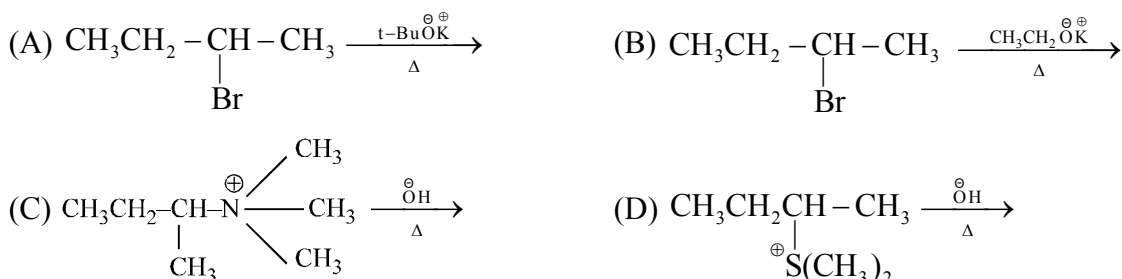
10. In which of the following reaction configuration about chiral C is retained in the final product



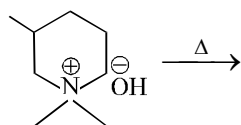
11. A gem dichloride is formed in the reaction :



12. In which product formation takes place according to Hoffmann's rule

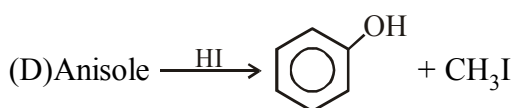
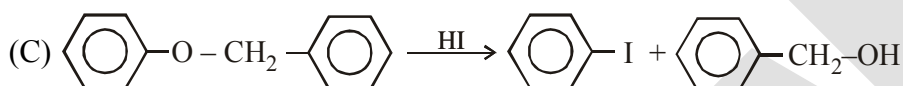


13. Which of following are correct for given reaction

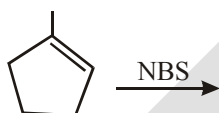


- (A) Major product of reaction is (B) Major product is
 (C) The reaction is thermal elimination reaction (D) The reaction is E₂ reaction

14. In which case incorrect products are formed :

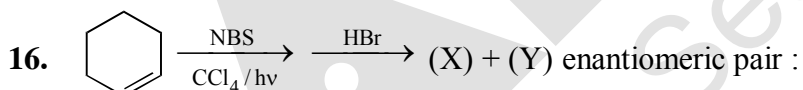


15. In the given reaction :



Find out the correct statement

- (A) It gives total 9 allylic brominated products
 (B) 6 fractions are obtained on fractional distillation of product mixture
 (C) Substrate has 7 allylic hydrogens
 (D) NBS is a brominating agent for allylic positions

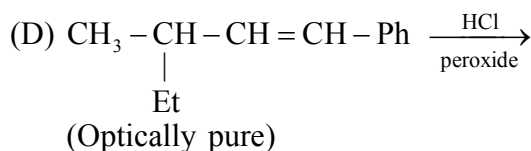
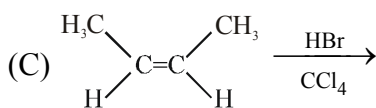
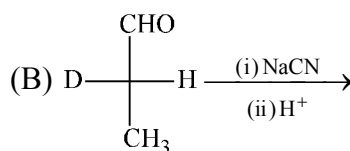
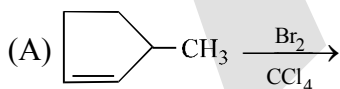


- (A) (B) (C) (D)

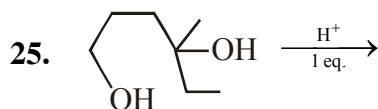
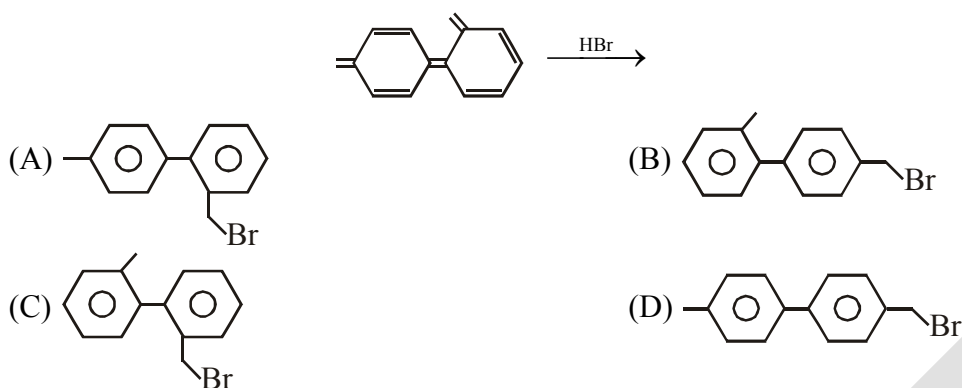
17. Which of the following can be produced by Wurtz reaction in good yield :

- (A) (B) (C) (D)

18. Products formed when HCl adds to 2,4-hexadiene is :
- (A) 4-chloro-2-hexene (B) 2-chloro-3-hexene
(C) 2-chloro-4-hexene (D) 1-chloro-2-hexene
19. Correct statement among the following is/are :
- (A) The rate of hydrolysis of tertiary butyl bromide increases by addition of Ag_2O
(B) Aqueous Ag_2O produces nucleophilic OH^-
(C) The addition of a small amount of oxygen slows down the photochemical chlorination of methane.
(D) $\text{CH}_3\text{CH}_2\text{Cl}$ is more reactive than PhCH_2Cl for bimolecular nucleophilic substitution reaction
20. Incorrect statement among the following is/are :
- (A) $\text{R}-\text{OH}$ with NaI in the presence of phosphoric acid gives $\text{R}-\text{I}$, but not in the absence of phosphoric acid
(B) 2-methyl propane on chlorination ($\text{Cl}_2, h\nu$) gives 1-chloro-2-methyl propane while bromination ($\text{Br}_2, h\nu$) gives 2-bromo-2-methyl propane
(C) Usually higher temperature prefers substitution over elimination
(D) Triphenyl chloromethane cannot be hydrolysed
21. Correct statements among the following is/are :
- (A) Dihaloalkanes having the same type of halogen atoms on same atom are named as alkylidene dihalides
(B) Dihaloalkanes having the same type of halogen atoms on adjacent atoms are named as alkylene dihalides
(C) In common name system gem-dihalides are named as alkylidene halide
(D) In common name system vic-dihalides are named as alkylene halide
22. Which of the following is correct order of nucleophilicity ?
- (A) $(\text{CH}_3\text{CH}_2)_3\text{N} > \text{pyridine}$ (B) $\text{HOO}^- > \text{HO}^-$ in DMSO
(C) $\text{H}_2\text{S} > \text{H}_2\text{O}$ (D) $\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\text{O}^- > \text{CH}_3-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{O}^-$
23. Which of following reaction products are diastereomer of each other :

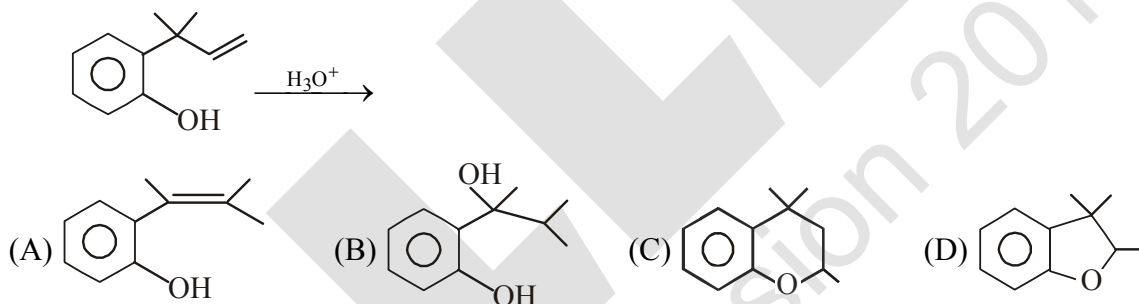


24. Product obtained in given reaction in good yield are :



Correct statements for given reaction :

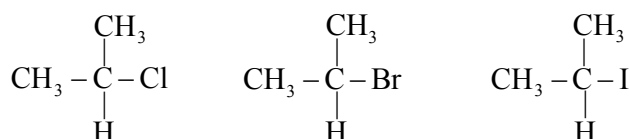
- (A) Product mixture is resolvable
 (B) Product can be separated by fractional distillation of mixture
 (C) Two products possible & both are optically active
 (D) Products are diastereomer
26. Which of the following can be formed during this reaction ?



27. Select **true** statement(s) :

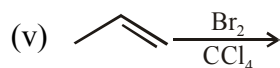
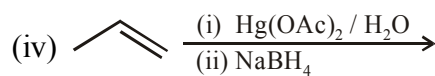
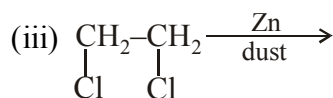
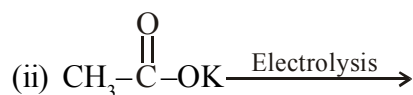
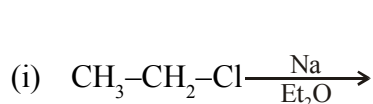
- (A) Cyclopropane decolorizes bromine water
 (B) In general, bromination is more selective than chlorination.
 (C) The 2,4,6-tri-tert, butylphenoxy radical is resistant to dimerization.
 (D) The radical-catalysed chlorination, $\text{ArCH}_3 \rightarrow \text{ArCH}_2\text{Cl}$, occurs faster when $\text{Ar} = \text{phenyl}$ than when $\text{Ar} = \text{p-nitrophenyl}$.

28. From left to right, correct statements are :



- (A) Rate of $\text{S}_{\text{N}}1$ mechanism increases in polar protic solvent
 (B) Rate of $\text{S}_{\text{N}}2$ mechanism increases in DMSO
 (C) Rate of E_2 mechanism increases
 (D) Rate of E_1 mechanism increases

29. Number of following reactions which produces hydrocarbon as major product ?

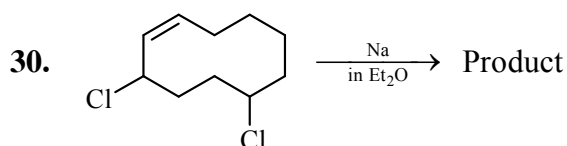


(A) 2

(B) 4

(C) 5

(D) 6



Correct statement is/are :

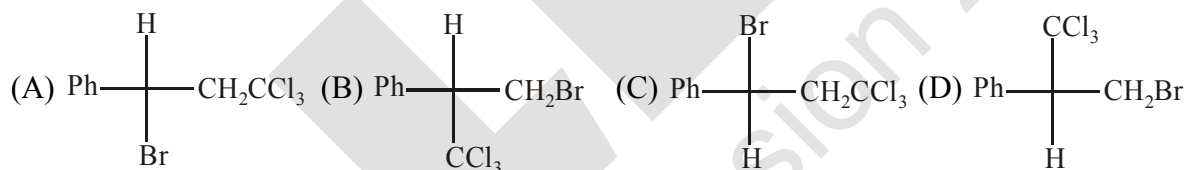
(A) odd no. of double bond equivalent in product

(B) product is bicyclic compound

(C) product can show geometrical isomerism

(D) reaction involve carbocation as intermediate

31. $\text{Ph}-\text{CH}=\text{CH}_2 + \text{BrCCl}_3 \xrightarrow{\text{Peroxide}}$ Product is :



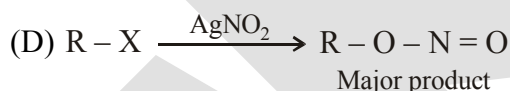
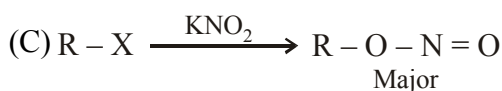
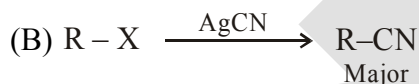
EXERCISE # II (B)

Comprehension Type :

Paragraph for Q.No. 01 to 02

Groups like CN & $[-O-\ddot{N}=O]$ possess two nucleophilic centre and are called ambident nucleophiles. Actually cyanide group is hybriide of two contributing structures and therefore can act as nucleophile in two different ways $[\overset{\ominus}{C}\equiv N \longleftrightarrow :C \equiv N^{\ominus}]$. Similarly nitrite ion also represents an ambident nucleophile with two different points of linkage $[O-\ddot{N}=O]$.

1. Correct option among the following :



2. Incorrect statement



(A) KCN is predominantly ionic in nature

(B) AgCN is mainly covalent in nature

(C) In AgCN, carbon is the donor atom

(D) In AgCN nitrogen is the donor atom

3. **Statement-1** : HBr shows antimarkownikoff's addition on propene but not HCl.

Statement-2 : H-Br is stronger acid than H-Cl.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

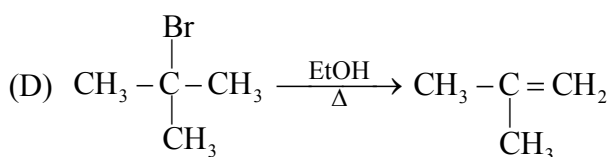
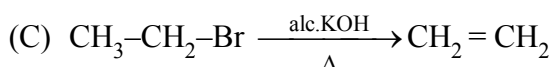
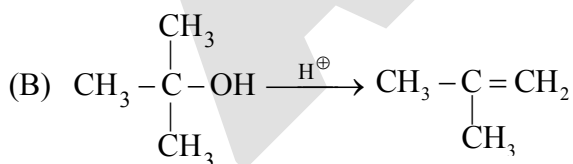
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

4. Match the List I with List II and select the correct answer using the codes given below the Lists.

List I



List II

(P) Elimination Reaction

(Q) Carbocation

(R) Carbanion

(S) Free radical

5. Match List I with List II and select the correct answer from the codes given below:

List I

(Reactions)

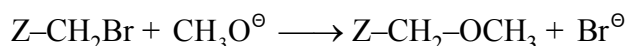
- (A) $\text{CH}_3\text{O}-\text{SO}_2\text{CH}_3 + \text{C}_2\text{H}_5\text{O}^\ominus$
 (B) $\text{CH}_3-\text{CH}_2-\text{I} + \text{PH}_3$
 (C) $\text{HC} \equiv \text{C}^\oplus\text{Na} + \text{CH}_3-\text{CH}_2-\text{Br}$
 (D) $\text{CH}_3-\text{Cl} + \text{CH}_3-\text{O}^\ominus$

List II

(Products)

- (P) $\text{CH}_3-\text{CH}_2-\text{PH}_2$
 (Q) $\text{CH}_3-\text{O}-\text{C}_2\text{H}_5$
 (R) $\text{CH}_3-\text{O}-\text{CH}_3$
 (S) $\text{CH} \equiv \text{C}-\text{CH}_2-\text{CH}_3$

6. Match List-I with List-II for given $\text{S}_\text{N}2$ reaction & select the correct answer from the codes given below



List-I (Z-)

- (A) $\text{H}-$
 (B) CH_3-
 (C) C_2H_5-
 (D) $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} \\ \diagup \\ \text{CH}_3 \end{array} -$

List-II (relative reactivity)

- (P) 0.1
 (Q) 3
 (R) 1
 (S) 100

7. Match the List I with List II and select the correct answer using the codes given below the Lists.

List I

- (A) $\text{E}_{1\text{CB}}$
 (B) Saytzeff alkene as major product
 (C) E_2
 (D) E_i

List II

- (P) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{N}^\oplus\text{O}^\ominus \\ | \\ \text{CH}_3 \end{array}$
 (Q) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{O}-\text{C}(=\text{S})-\text{S}-\text{CH}_3$
 (R) $\begin{array}{c} \text{Cl} \\ | \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_3 \end{array}$
 (S) $\begin{array}{c} \text{C}_6\text{H}_5-\text{CH}_2-\text{CH}-\text{CH}_3 \\ | \\ \text{F} \end{array}$

8. **Column - I**
(Reactions)

- (A) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 \xrightarrow{\text{HBr}}$
 (B) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 \xrightarrow{\text{HBr, Peroxide}}$
 (C) $\text{PhCH}(\text{CH}_3)\text{OH} \xrightarrow{\text{SOCl}_2}$
 (D) $\text{PhCH}(\text{CH}_3)\text{OH} \xrightarrow{\text{HBr}}$

Column - II
(Characteristics)

- (P) Bimolecular
 (Q) Carbocation intermediate
 (R) Regioselective
 (S) Racemic modification
 (T) Stereospecific reaction

9. Column - I

(Statements)

- (A) Reactions are concerted
 (B) CH_3X cannot react
 (C) $3^\circ \text{R-X} > 2^\circ \text{R-X} > 1^\circ \text{R-X}$
 (D) R-I reacts faster than R-Cl

Column - II

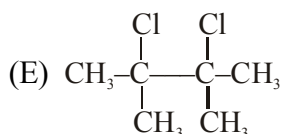
(Consistent path of reaction)

- (P) $\text{S}_{\text{N}}1$
 (Q) $\text{S}_{\text{N}}2$
 (R) $\text{E}1$
 (S) $\text{E}2$

10. Each of the compounds in column A is subjected to further chlorination. Match the following for them :

Column - A

- (A) $\text{CHCl}_2\text{-CH}_2\text{-CH}_3$
 (B) $\text{CH}_2\text{Cl-CHCl-CH}_3$
 (C) $\text{CH}_2\text{Cl-CH}_2\text{-CH}_2\text{-Cl}$
 (D) $\text{CH}_3\text{-CCl}_2\text{-CH}_3$



Column - B

- (P) Optically active original compound
 (Q) Only one trichloro product
 (R) Three trichloro product.
 (S) Four trichloro product
 (T) At least one of the trichloro product is optically active
 (U) Two trichloro products.

11. Column - I

(Intermediate)

- (A) Carbocation
 (B) Carbanion
 (C) Free radical
 (D) Octet complete in one of the intermediate

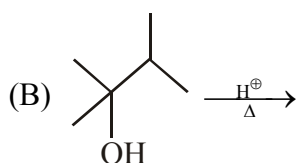
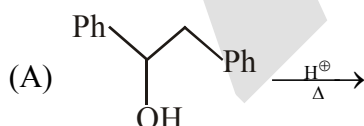
Column - II

- (P) Kolbey Electrolysis
 (Q) Wurtz reaction
 (R) Dehydration of alcohol
 (S) Monocarboxylic acid with sodalime

12. Match the column

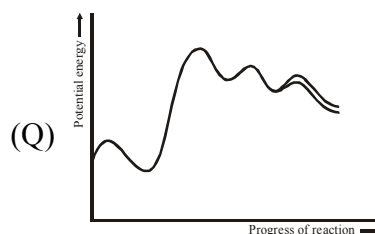
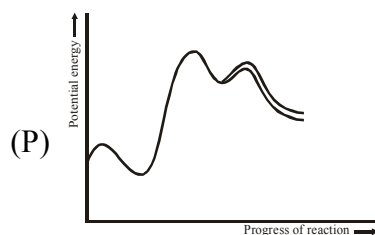
Column-I

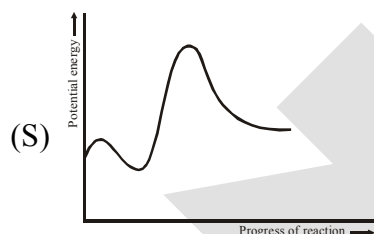
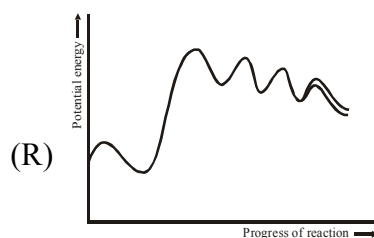
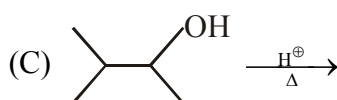
(Reaction)



Column-II

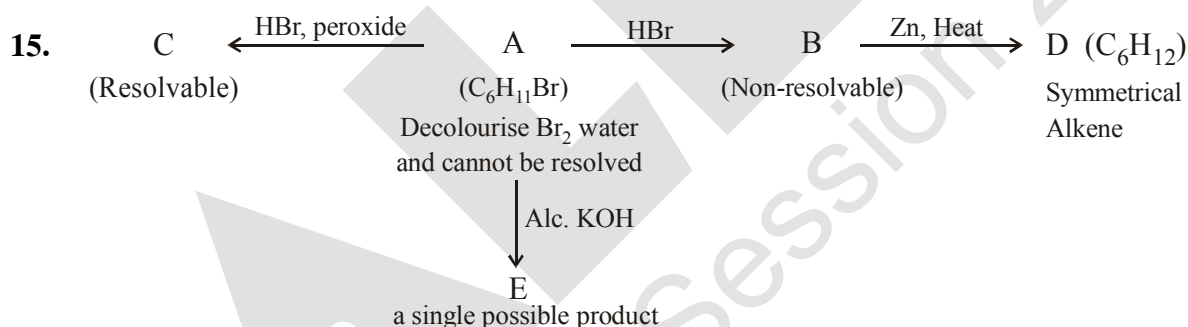
(Potential energy curve)





Subjective Type :

13. RCl is treated with Li in ether to form R – Li, R – Li reacts with water to form isopentane. R – Cl also reacts with sodium to form 2, 7-dimethyloctane. What is the structure of R – Cl.
14. A chloroderivative 'X' on reduction gave a hydrocarbon with five carbon atoms in the molecule. When X is dissolved in ether and treated with sodium, 2, 2, 5, 5-tetramethyl hexane is obtained. What is compound X.



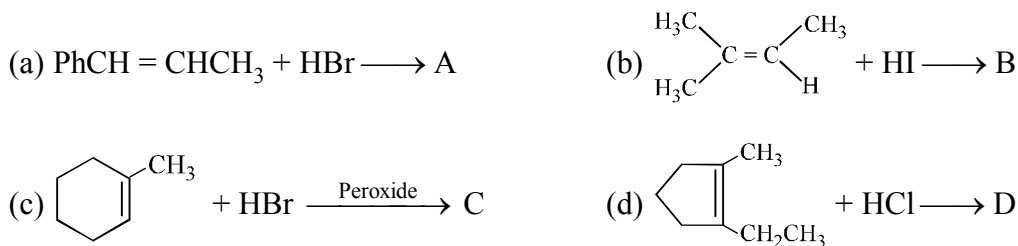
Identify A, C & E in the sequence of reaction.

16. With the help of following data show HBr exhibits the peroxide effect.

	$\Delta H_1^0/\text{kJ mol}^{-1}$	$\Delta H_2^0/\text{kJ mol}^{-1}$
H-X	$\dot{\text{X}} + \text{CH}_2 = \text{CH}_2 \rightarrow \text{XCH}_2 - \dot{\text{C}}\text{H}_2$	$\text{XCH}_2 - \dot{\text{C}}\text{H}_2 + \text{H-X} \rightarrow \text{XCH}_2\text{CH}_3 + \dot{\text{X}}$
	$\downarrow$	
HCl	-67	+ 12.6
HBr	-25.1	- 50.2
HI	+46	-117.1

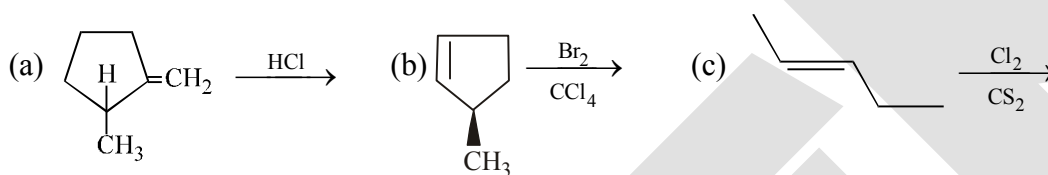
17. Write all the monochlorinated products (including stereo) of isohexane.

18. What are the products of the following reactions ?



19. It required 0.7 g of a hydrocarbon (A) to react completely with Br_2 (2.0 g) and form a non resolvable product. On treatment of (A) with HBr it yielded monobromo alkane (B). The same compound (B) was obtained when (A) was treated with HBr in presence of peroxide. Write down the structure formula of (A) and (B) and explain the reactions involved.

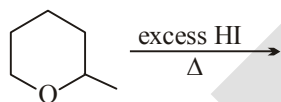
20. Complete following reaction :



21. $\text{CH}_3\text{CH}_2\text{I}$ reacts more rapidly with strong base in comparison to $\text{CD}_3\text{CH}_2\text{I}$.

22. $\text{CH}\equiv\text{C}-\text{CH}_2-\text{CH}=\text{CH}_2$, adds up HBr to give $\text{CH}\equiv\text{C}-\text{CH}_2-\text{CHBr}-\text{CH}_3$ while $\text{CH}\equiv\text{C}-\text{CH}=\text{CH}_2$ adds up HBr to give $\text{CH}_2=\underset{\text{Br}}{\text{C}}-\text{CH}=\text{CH}_2$

23. Predict the product(s) and write the mechanism of the given reaction :



24. What are the products of the following reactions?

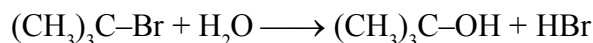


25. A primary alkyl bromide (A), $\text{C}_4\text{H}_9\text{Br}$, reacted with alcoholic KOH to give compound (B). Compound (B) reacted with HBr to give an isomer of (A). When (A) was reacted with sodium metal it gave compound (D), C_8H_{18} , which was different from the compound produced when n-butyl bromide was reacted with sodium. Draw the structure of (A) and write equations for all the reactions.

26. In study of chlorination of propane four products (A,B,C,D) of molecular formula $\text{C}_3\text{H}_7\text{Cl}_2$ were obtained. On further chlorination of the above products A gave one trichloro product, B gave two whereas C and D gave three each. When optically active C was chlorinated one of trichloro propanes was optically active and remaining two were optically inactive. Identify the structures of A, B, C and D, and explain formation of products.

EXERCISE # III (J-MAINS)

1. Following reaction :



is an example of-

[AIEEE-2002]

- (1) Elimination reaction (2) Free radical substitution
(3) Nucleophilic substitution (4) Electrophilic substitution

2. SN^1 reaction is feasible in-

[AIEEE-2002]



3. Bottles containing $\text{C}_6\text{H}_5\text{I}$ and $\text{C}_6\text{H}_5\text{CH}_2\text{I}$ lost their original labels. They were labelled A and B for testing. A and B were separately taken in a test tube and boiled with NaOH solution. The end solution in each tube was made acidic with dilute HNO_3 and then some AgNO_3 solution was added. Substance B gave a yellow precipitate. Which one of the following statements is true for this experiment.

[AIEEE-2003]

- (1) A was $\text{C}_6\text{H}_5\text{I}$ (2) A was $\text{C}_6\text{H}_5\text{CH}_2\text{I}$
(3) B was $\text{C}_6\text{H}_5\text{I}$ (4) Addition of HNO_3 was unnecessary

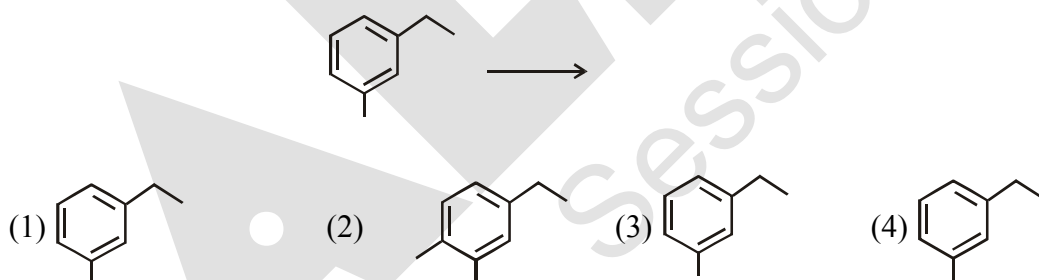
4. The compound formed on heating chlorobenzene with chloral in the presence of concentrated sulphuric acid is-

[AIEEE-2004]

- (1) Gammaxe (2) DDT (3) Freon (4) Hexa chloro ethane

5. The structure of the major product formed in the following reaction is :

[AIEEE-2006]

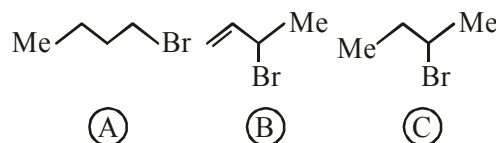


6. Which of the following on heating with aqueous KOH , produces acetaldehyde ? [AIEEE-2009]

- (1) $\text{CH}_2\text{ClCH}_2\text{Cl}$ (2) CH_3CHCl_2 (3) CH_3COCl (4) $\text{CH}_3\text{CH}_2\text{Cl}$

7. Consider the following bromides :-

[AIEEE-2010]



The correct order of S_N^1 reactivity is

- (1) $\text{A} > \text{B} > \text{C}$ (2) $\text{B} > \text{C} > \text{A}$ (3) $\text{B} > \text{A} > \text{C}$ (4) $\text{C} > \text{B} > \text{A}$

8. In S_N^2 reactions, the correct order of reactivity for the following compounds : [JEE(Main)-2014]
 CH_3Cl , $\text{CH}_3\text{CH}_2\text{Cl}$, $(\text{CH}_3)_2\text{CHCl}$ and $(\text{CH}_3)_3\text{CCl}$ is :

- (1) $\text{CH}_3\text{CH}_2\text{Cl} > \text{CH}_3\text{Cl} > (\text{CH}_3)_2\text{CHCl} > (\text{CH}_3)_3\text{CCl}$
 (2) $(\text{CH}_3)_2\text{CHCl} > \text{CH}_3\text{CH}_2\text{Cl} > \text{CH}_3\text{Cl} > (\text{CH}_3)_3\text{CCl}$
 (3) $\text{CH}_3\text{Cl} > (\text{CH}_3)_2\text{CHCl} > \text{CH}_3\text{CH}_2\text{Cl} > (\text{CH}_3)_3\text{CCl}$
 (4) $\text{CH}_3\text{Cl} > \text{CH}_3\text{CH}_2\text{Cl} > (\text{CH}_3)_2\text{CHCl} > (\text{CH}_3)_3\text{CCl}$

9. In a nucleophilic substitution reaction : [JEE(Main)-On-Line-2014]



which one of the following undergoes complete inversion of configuration?

- (1) $\text{C}_6\text{H}_5\text{CCH}_3\text{C}_6\text{H}_5\text{Br}$ (2) $\text{C}_6\text{H}_5\text{CHCH}_3\text{Br}$
 (3) $\text{C}_6\text{H}_5\text{CHC}_6\text{H}_5\text{Br}$ (4) $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$

10. The major product obtained in the photo catalysed bromination of 2-methylbutane is :-

- (1) 2-bromo-2-methylbutane
 (2) 2-bromo-3-methylbutane
 (3) 1-bromo-2-methylbutane
 (4) 1-bromo-3-methylbutane

[JEE(Main)-On-Line-2014]

11. In the presence of peroxide, HCl and HI do not give anti-Markownikoff's addition to alkenes because :- [JEE(Main)-On-Line-2014]

- (1) All the steps are exothermic in HCl and HI
 (2) One of the steps is endothermic in HCl and HI
 (3) HCl is oxidizing and the HI is reducing
 (4) Both HCl and HI are strong acids

12. The major product formed when 1,1,1 - trichloro - propane is treated with aqueous potassium hydroxide is : [JEE(Main)-On-Line-2014]

- (1) 2 - Propanol (2) Propionic acid (3) Propyne (4) 1 - Propanol

13. The synthesis of alkyl fluoride is best accomplished by : [JEE(Main)-2015]

- (1) Finkelstein reaction (2) Swarts reaction
 (3) Free radical fluorination (4) Sandmeyer's reaction

14. 2-chloro-2-methylpentane on reaction with sodium methoxide in methanol yields :

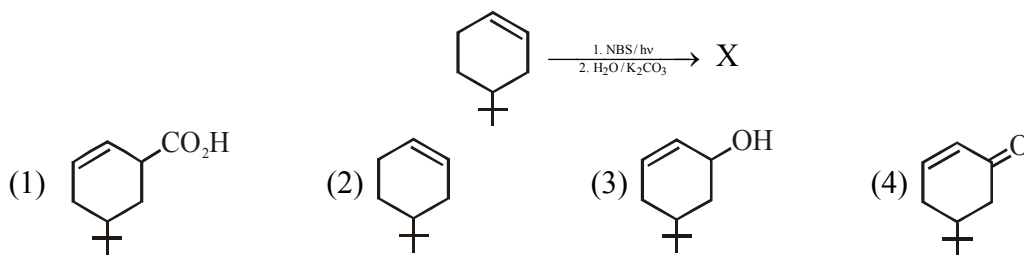
- (1) $\text{C}_2\text{H}_5\text{CH}_2\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{OCH}_3$ (2) $\text{C}_2\text{H}_5\text{CH}_2\overset{\text{CH}_3}{\text{C}}=\text{CH}_2$ (3) $\text{C}_2\text{H}_5\text{CH}_2=\overset{\text{CH}_3}{\text{C}}-\text{CH}_3$

[JEE-MAIN-2016]

- (1) (1) and (2) (2) All of these (3) (1) and (3) (4) (3) only

15. The product of the reaction given below is :

[JEE-MAIN-2016]



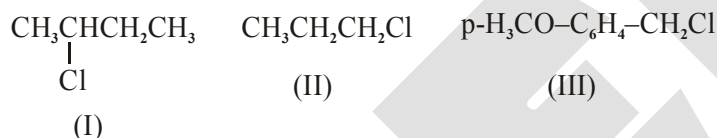
16. The reaction of propene with HOCl (Cl₂ + H₂O) proceeds through the intermediate :

[JEE-MAIN-2016]

- (1) CH₃-CHCl-CH₂⁺
- (2) CH₃-CH⁺-CH₂-OH
- (3) CH₃-CH⁺-CH₂-Cl
- (4) CH₃-CH(OH)-CH₂⁺

17. The increasing order of the reactivity of the following halides for the S_N1 reaction is :

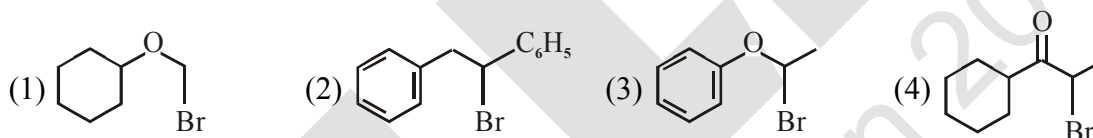
[JEE-MAIN-2017]



- (1) (III) < (II) < (I)
- (2) (II) < (I) < (III)
- (3) (I) < (III) < (II)
- (4) (II) < (III) < (I)

18. Which of the following, upon treatment with tert-BuONa followed by addition of bromine water, fails to decolourize the colour of bromine ?

[JEE-MAIN-2017]



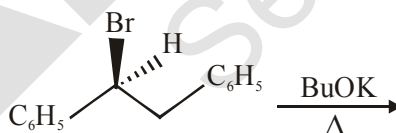
19. 3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is :

[JEE-MAIN-2017]

- (1) Six
- (2) Zero
- (3) Two
- (4) Four

20. The major product obtained in the following reaction is :

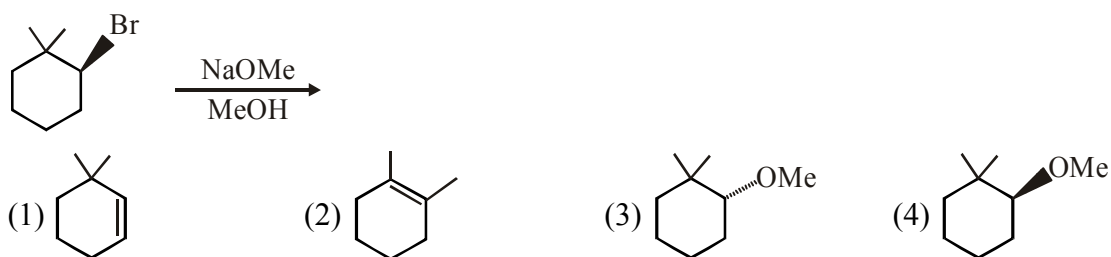
[JEE-MAIN-2017]



- (1) $(\pm)\text{C}_6\text{H}_5\text{CH}(\text{O}^t\text{Bu})\text{CH}_2\text{CH}_2\text{C}_6\text{H}_5$
- (2) $\text{C}_6\text{H}_5\text{CH}=\text{CHC}_6\text{H}_5$
- (3) $(+)\text{C}_6\text{H}_5\text{CH}(\text{O}^t\text{Bu})\text{CH}_2\text{C}_6\text{H}_5$
- (4) $(-)\text{C}_6\text{H}_5\text{CH}(\text{O}^t\text{Bu})\text{CH}_2\text{C}_6\text{H}_5$

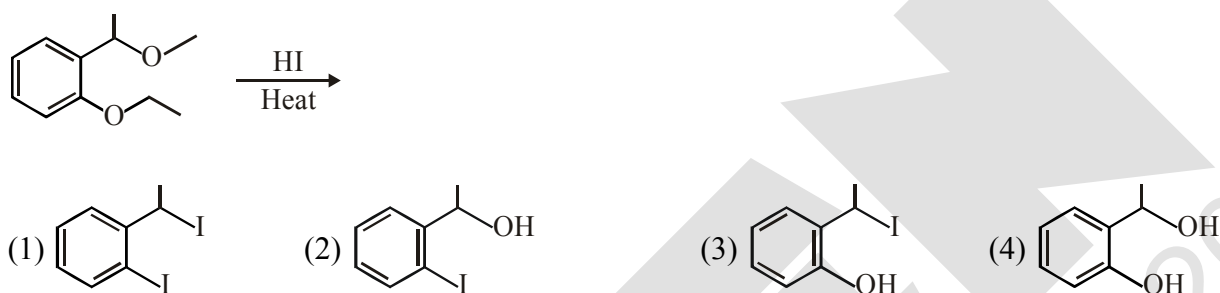
21. The major product of the following reaction is :

[JEE-MAIN-2018]



22. The major product formed in the following reaction is :

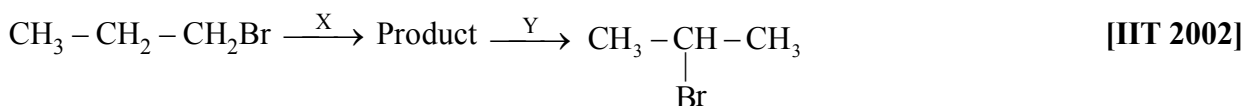
[JEE-MAIN-2018]



EXERCISE # IV (A) (J-ADVANCE OBJECTIVE)

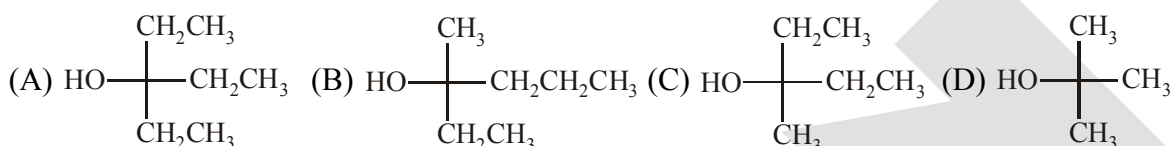
- Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives : [IIT 1990]
 (A) o-cresol (B) p-cresol
 (C) 2,4-dihydroxytoluene (D) Benzoic acid
- Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to [IIT 1990]
 (A) The formation of less stable carbonium ion
 (B) Resonance stabilization
 (C) The inductive effect
 (D) sp^2 hybridised carbon attached to the halogen
- 1-Chlorobutane on reaction with alcoholic potash gives : [IIT 1991]
 (A) 1-butene (B) 1-butanol (C) 2-butene (D) 2-butanol
- The products of reaction of alcoholic $AgNO_2$ with ethyl bromide are [IIT 1991]
 (A) Ethane (B) Ethyl nitrite (C) Nitroethane (D) Ethyl alcohol
- Arrange the following compounds in order of increasing dipole moment [IIT 1996]
 Toluene m-dichlorobenzene o-dichlorobenzene p-dichlorobenzene
 I II III IV
 (A) $I < IV < II < III$ (B) $IV < I < II < III$ (C) $IV < I < III < II$ (D) $IV < II < I < III$
- $(CH_3)_3CMgCl$ reaction with D_2O produces: [IIT 1997]
 (A) $(CH_3)_3CD$ (B) $(CH_3)_3OD$ (C) $(CD_3)_3CD$ (D) $(CH_3)_3OD$
- Benzyl chloride ($C_6H_5CH_2Cl$) can be prepared from toluene by chlorination with: [IIT 1998]
 (A) SO_2Cl_2 (B) $SOCl_2$ (C) $Cl_2, (h\nu)$ (D) $NaOCl$
- The order of reactivity of the following alkyl halides for a S_N2 reaction is: [IIT 2000]
 (A) $RF > RC > R-Br > R-I$ (B) $R-F > R-Br > R-Cl > R-I$
 (C) $R-Cl > R-Br > RF > RI$ (D) $R-I > RBr > R-Cl > R-F$
- Which of the following has the highest nucleophilicity? [IIT 2000]
 (A) F^- (B) OH^- (C) CH_3^- (D) NH_2^-
- An S_N2 reaction at an asymmetric carbon of a compound always gives. [IIT 2001]
 (A) an enantiomer of the substance (B) a product with opposite optical rotation
 (C) a mixture of diastereomers (D) a single stereoisomer
- The compound that will react most readily with NaOH to form methanol is [IIT 2001]
 (A) $(CH_3)_4N^+I^-$ (B) CH_3OCH_3 (C) $(CH_3)_3S^+I^-$ (D) $(CH_3)_3CCl$

12. Identify the set of reagents / reaction conditions 'X' and 'Y' in the following set of transformation:

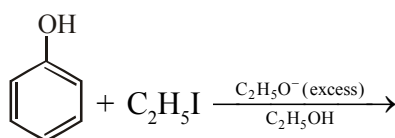


- (A) X = dilute aqueous NaOH, 20°C; Y = HBr / acetic acid, 20°C
 (B) X = concentrated alcoholic NaOH, 80°C; Y = HBr / acetic acid 20°C
 (C) X = dilute aqueous NaOH, 20°C; Y = Br₂ / CHCl₃, 0°C
 (D) X = concentrated alcoholic NaOH, 80°C; Y = Br₂/CHCl₃, 0°C

13. CH₃MgBr + Ethyl ester → which can be formed as product. (excess) [IIT 2003]

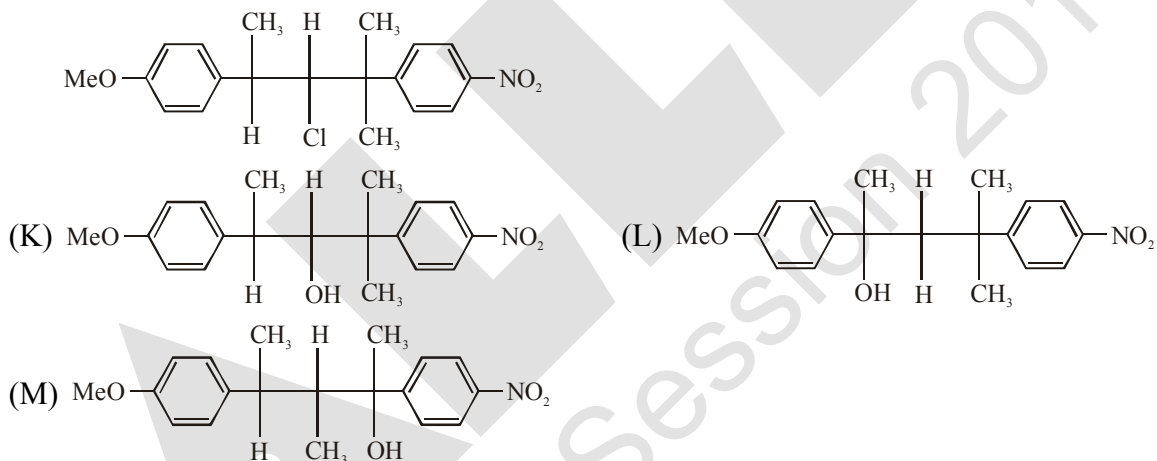


14. The product of following reaction is [IIT 2003]



- (A) C₆H₅OC₂H₅ (B) C₂H₅OC₂H₅ (C) C₆H₅OC₆H₅ (D) C₆H₅I

15. The following compound on hydrolysis in aqueous acetone will give: [IIT 2005]



It mainly gives

- (A) K and L (B) Only K (C) L and M (D) Only M

16. Match the following: [IIT 2006]

Column I

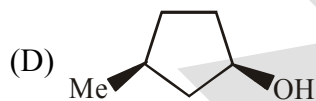
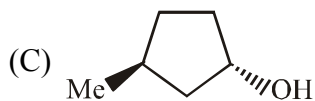
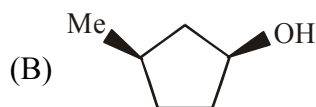
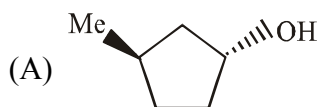
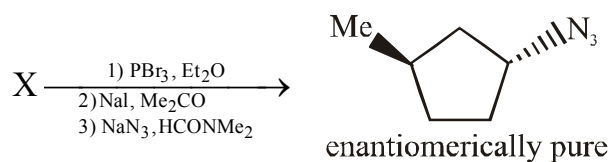
- (A) CH₃-CHBr-CD₃ on treatment with alc. KOH gives CH₂=CH-CD₃ as a major product.
 (B) Ph-CHBr-CH₃ reacts faster than Ph-CHBr-CD₃.
 (C) Ph-CD₂-CH₂Br on treatment with C₂H₅OD/C₂H₅O⁻ gives Ph-CD=CH₂ as the major product.
 (D) PhCH₂CH₂Br and PhCD₂CH₂Br react with same rate.

Column II

- (P) E1 reaction
 (Q) E2 reaction
 (R) E1cb reaction
 (S) First order reaction

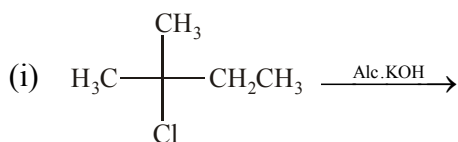
22. In the following reaction sequence, the correct structure(s) of X is (are)

[IIT-2018]

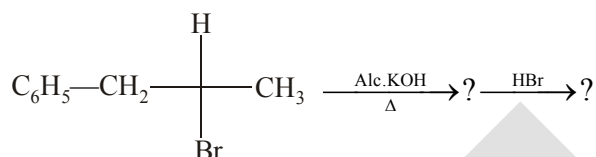


EXERCISE # IV (B) (J-ADVANCE SUBJECTIVE)

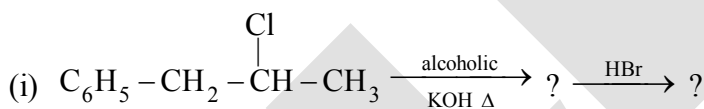
- Fill in the blanks :
 - Butane nitrile can be prepared by heating _____ with alcoholic KCN. [IIT 1992]
 - Amongst three isomers of nitrophenol, the one that is least soluble in water is _____. [IIT 1992]
- Arrange the following in order of their
 - Increasing basicity
 H_2O , OH^- , CH_3OH , CH_3O^-
 - Increasing reactivity in nucleophilic substitution reactions
 CH_3F , CH_3I , CH_3Br , CH_3Cl [IIT 1992]
- Write the structures of the major organic product expected from each of the following reactions: [IIT 1992]



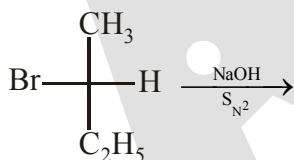
- Identify the major product in the following reaction. [IIT 1993]



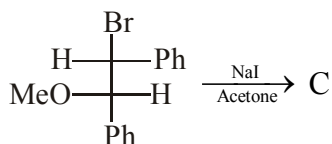
- Identify the major product in the following reactions : [IIT 1993]



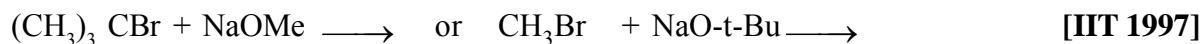
- Aryl halides are less reactive than alkyl halides towards nucleophilic reagents. Give reason. [IIT 1994]
- Draw the stereochemical structure of product in the following reaction. [IIT 1994]



- Optically active 2-iodobutane on treatment with NaI in acetone gives a product which does not show optical activity. Explain briefly. [IIT 1995]
- An alkyl halide X of formula $\text{C}_6\text{H}_{13}\text{Cl}$ on treatment with potassium tertiary butoxide gives two isomeric alkenes Y and Z (C_6H_{12}). Both alkenes on hydrogenation give 2,3-dimethylbutane. Predict the structures of X, Y and Z. [IIT 1996]
- Predict the structure of the intermediates/products in the following reaction sequence – [IIT 1996]



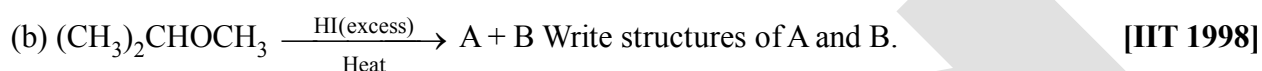
11. Which of the following is the correct method for synthesising methyl-t-butyl ether and why?



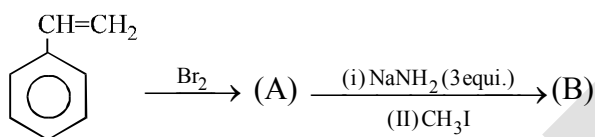
12. Write the structures of the products: [IIT 1998]



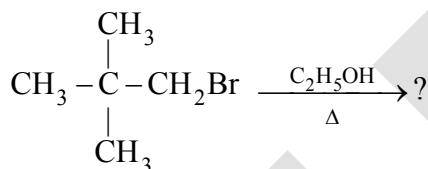
13. (a) $\text{C}_6\text{H}_5\text{CH}_2\overset{\text{C}_6\text{H}_5}{\underset{|}{\text{CH}}}\text{Cl} \xrightarrow[\text{Heat}]{\text{alc. KOH}} \text{A} + \text{B}$ Write structures of (A) and (B).



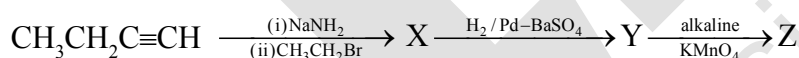
14. Complete the following reaction with appropriate structures of products/reagents. [IIT 1998]



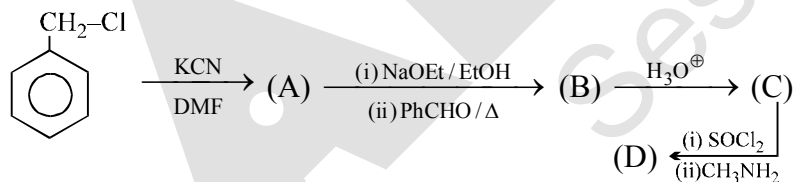
15. What would be major product? [IIT 2000]



16. Identify X, Y and Z in the following synthetic scheme and write their structures [IIT 2002]

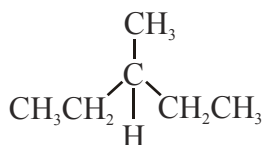


17. Give major products A, B, C and D in following reaction sequence. [IIT 2004]



18. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is [IIT 2011]

19. The maximum number of isomers (including stereoisomers) that are possible on mono-chlorination of the following compounds, is [IIT 2011]



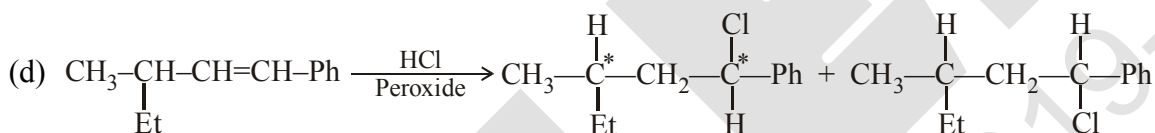
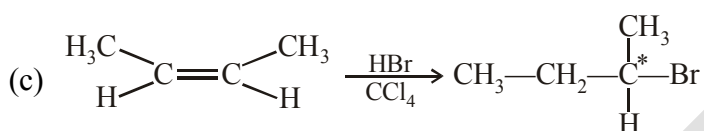
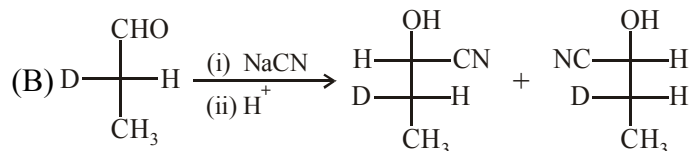
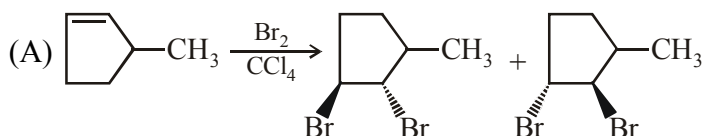
ANSWER-KEY**EXERCISE # I (MAINS ORIENTED)**

1. Ans. (A)	2. Ans. (C)	3. Ans. (D)	4. Ans. (D)	5. Ans. (C)
6. Ans. (C)	7. Ans. (B)	8. Ans. (D)	9. Ans. (B)	10. Ans. (B)
11. Ans. (C)	12. Ans. (C)	13. Ans. (C)	14. Ans. (A)	15. Ans. (D)
16. Ans. (B)	17. Ans. (A)	18. Ans. (D)	19. Ans. (C)	20. Ans. (D)
21. Ans. (D)	22. Ans. (A)	23. Ans. (D)	24. Ans. (A)	25. Ans. (B)
26. Ans. (C)	27. Ans. (B)	28. Ans. (C)	29. Ans. (D)	30. Ans. (B)
31. Ans. (D)	32. Ans. (B)	33. Ans. (B)	34. Ans. (C)	35. Ans. (A)
36. Ans. (A)	37. Ans. (C)	38. Ans. (D)	39. Ans. (A)	40. Ans. (B)
41. Ans. (B)	42. Ans. (C)	43. Ans. (A)	44. Ans. (C)	45. Ans. (A)
46. Ans. (C)	47. Ans. (D)	48. Ans. (D)	49. Ans. (B)	50. Ans. (C)
51. Ans. (B)	52. Ans. (C)	53. Ans. (A)	54. Ans. (C)	55. Ans. (A)
56. Ans. (D)	57. Ans. (C)	58. Ans. (B)	59. Ans. (B)	60. Ans. (C)
61. Ans. (C)	62. Ans. (D)	63. Ans. (A)	64. Ans. (B)	65. Ans. (A)
66. Ans. (C)	67. Ans. (D)	68. Ans. (B)	69. Ans. (B)	70. Ans. (C)
71. Ans. (B)	72. Ans. (C)	73. Ans. (C)	74. Ans. (A)	75. Ans. (B)
76. Ans. (A)	77. Ans. (A)	78. Ans. (D)	79. Ans. (C)	80. Ans. (B)
81. Ans. (C)	82. Ans. (C)	83. Ans. (C)	84. Ans. (B)	85. Ans. (D)
86. Ans. (C)	87. Ans. (B)	88. Ans. (A)	89. Ans. (C)	90. Ans. (C)
91. Ans. (A)	92. Ans. (C)	93. Ans. (B)	94. Ans. (B)	95. Ans. (A)
96. Ans. (B)	97. Ans. (D)	98. Ans. (B)	99. Ans. (B)	100. Ans. (D)

EXERCISE # II (A)

1. Ans. (A,B,C,D)	2. Ans. (A,B,C)	3. Ans. (A,B)	4. Ans. (B,D)
5. Ans. (B,D)	6. Ans. (A,C)	7. Ans. (A,C,D)	8. Ans. (A,C)
9. Ans. (A,B,D)	10. Ans. (A,C)	11. Ans. (A,B,D)	12. Ans. (A,C,D)
13. Ans. (A,CD)	14. Ans. (A,B,C)	15. Ans. (A,C,D)	16. Ans. (B,D)
17. Ans. (B,D)	18. Ans. (A,B)	19. Ans. (A,B,C)	20. Ans. (C,D)
21. Ans. (B,C)	22. Ans. (B,C,D)		

23. Ans.



24. Ans. (A,B)

25. Ans. (A,C)

26. Ans. (A,B,D)

27. Ans. (A,B,C,D)

28. Ans. (A,B,C,D)

29. Ans. (B)

30. Ans. (A,B,C)

31. Ans. (A,C)

EXERCISE # II (B)

1. Ans. (C)

2. Ans. (C)

3. Ans. (B)

4. Ans. (A) → P, R ; (B) → P, Q ; (C) → P ; (D) → P, Q

5. Ans. (A) → Q ; (B) → P ; (C) → S ; (D) → R

6. Ans. (A) → S ; (B) → Q ; (C) → R ; (D) → P

7. Ans. (A) → S ; (B) → R, S ; (C) → R ; (D) → P, Q

8. Ans. (A) → P, Q, R, S ; (B) → P, R ; (C) → P, T ; (D) → Q, S

9. Ans. (A) → Q, S ; (B) → P, R, S ; (C) → P, R, S ; (D) → P, Q, R, S

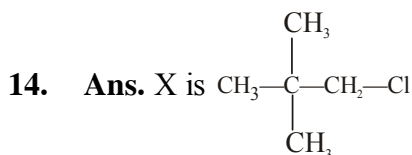
10. Ans. (A) → S, T ; (B) → P, S, T ; (C) → U ; (D) → Q ; (E) → T, U

11. Ans. (A) → R ; (B) → Q, S ; (C) → P, Q ; (D) → Q, S

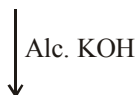
12. Ans. (A) → P ; (B) → P ; (C) → Q ; (D) → R

13. Ans.

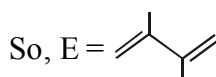
C-I bond being less stable than C-Cl bond and thus on heating heterolytic cleavage of C-I form I^- which gives yellow precipitate with $AgNO_3$.



15. Ans. Molecule A, $C_6H_{11}Br$ has 1 unsaturation

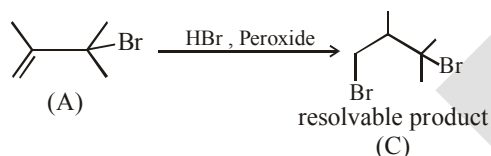


A single possible product, it suggests a symmetrical arrangement



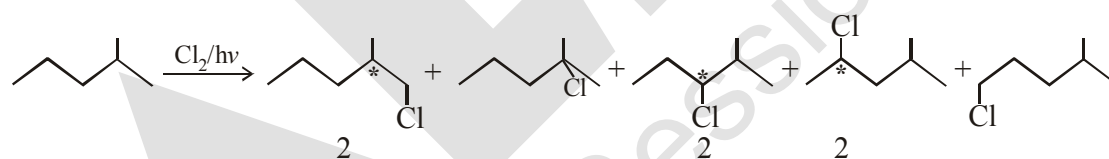
There are only two possibilities of A $\begin{array}{c} \diagup \\ \diagdown \end{array}CH_2Br$ or $\begin{array}{c} \diagup \\ \diagdown \end{array}Br$ structure I can be resolved

while structure II cannot be resolved so 'A' :

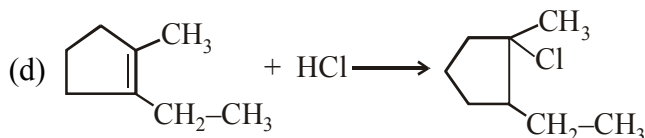
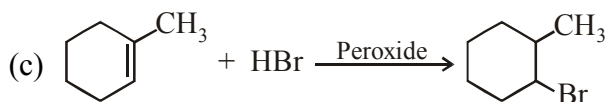
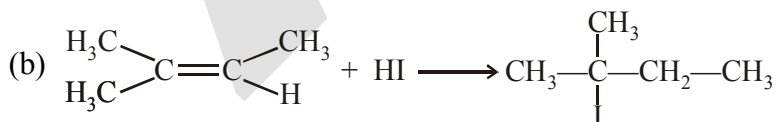


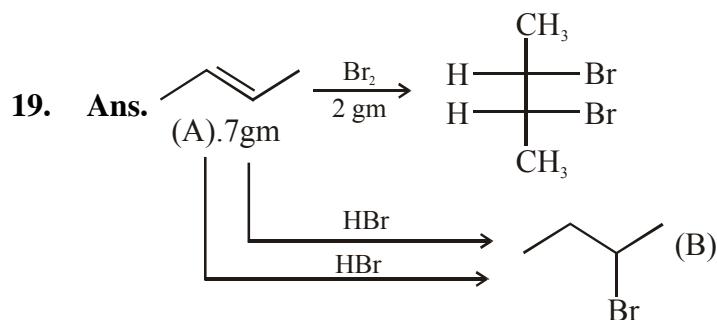
16. Ans. Both step is exothermic with HBr

17. Ans.

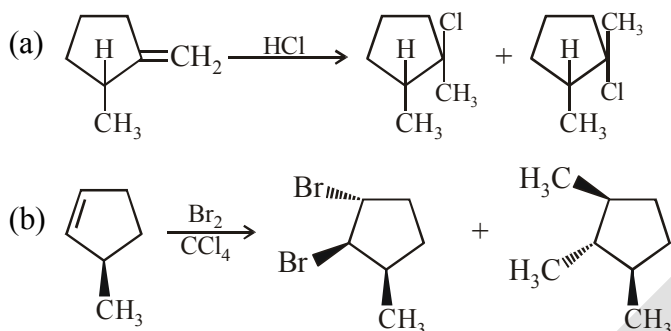


18. Ans.





20. Ans.

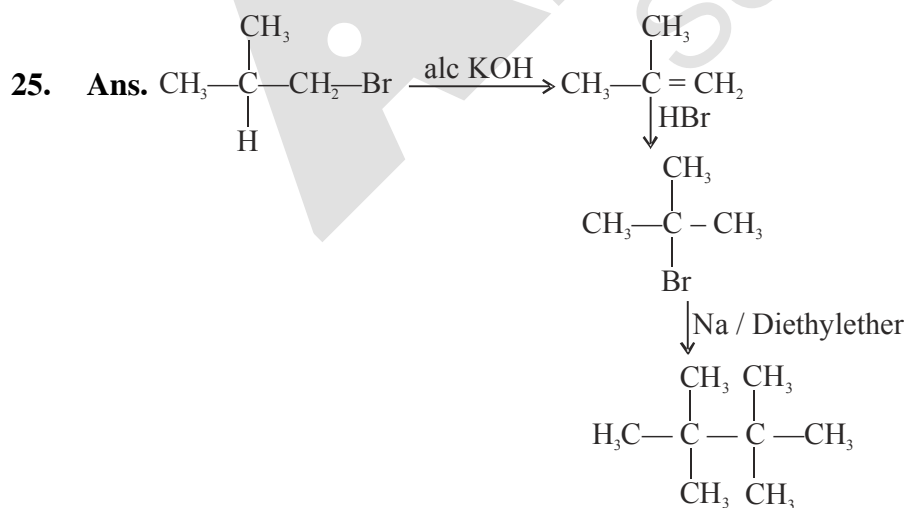
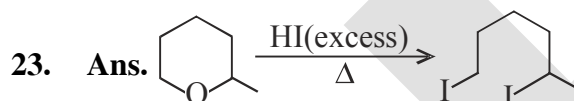


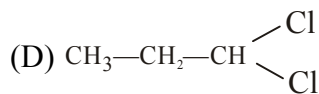
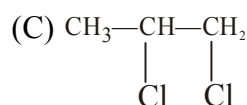
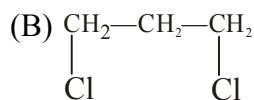
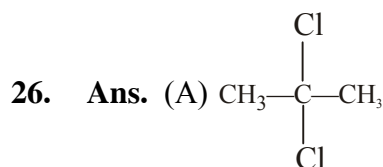
21. Ans.

The elimination of HI (or DI) in presence of strong base shows E2 elimination. The rate determining step involves breaking up of C – H (or C–D) bond. The C–D bond being stronger than C–H and thus elimination is faster in case of $\text{CH}_3 - \text{CH}_2\text{I}$.

22. Ans.

In second compound π bonds are conjugated so due to resonance given product is formed as major product.





EXERCISE # III (J-MAINS)

1. Ans. (3)

2. Ans. (1)

3. Ans. (1)

4. Ans. (2)

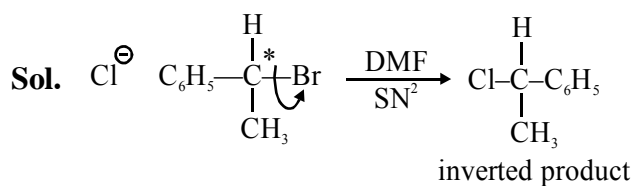
5. Ans. (4)

6. Ans. (2)

7. Ans. (2)

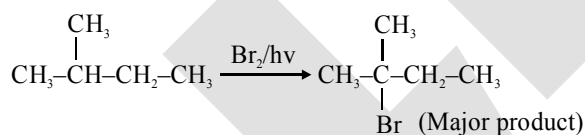
8. Ans. (4)

9. Ans. (4)

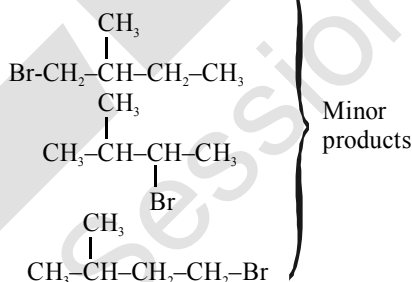


10. Ans. (1)

Sol.



2-Bromo-2-methyl butane



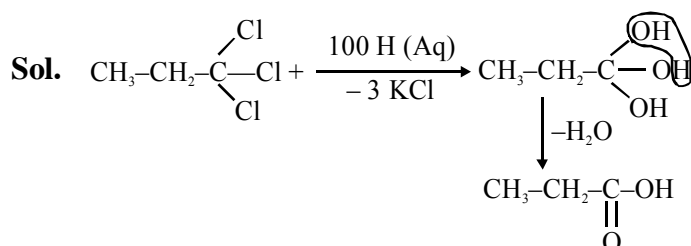
selectivity ratio for bromination is

$1^\circ : 2^\circ : 3^\circ :: 1 : 82 : 1600$

Hence 3° product will be major product.

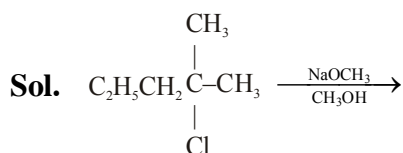
11. Ans. (2)

12. Ans. (2)

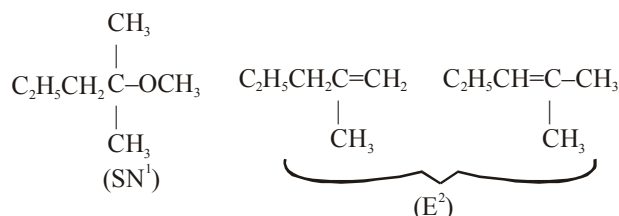


13. Ans. (2)

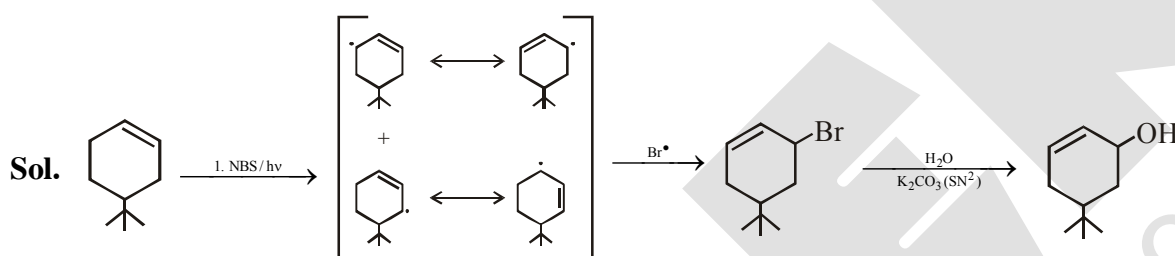
14. Ans. (2)



possible mechanism which takes place is E^2 & SN^1 mechanism. Hence possible products are.

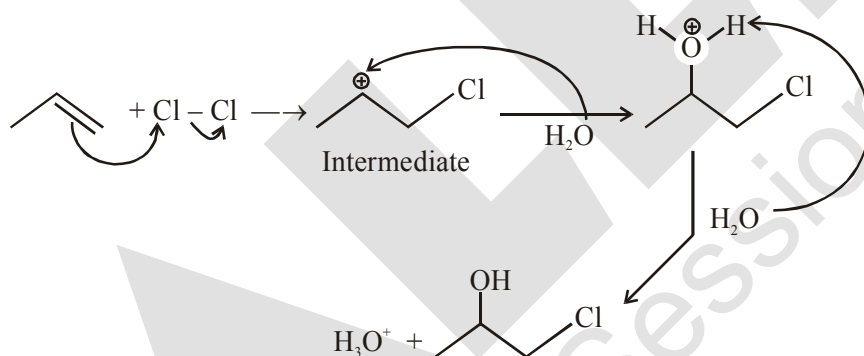


15. Ans. (3)



16. Ans. (3)

Sol.



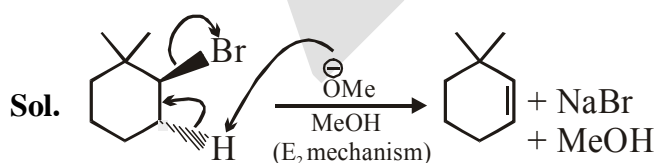
17. Ans. (2)

18. Ans. (1)

19. Ans. (4)

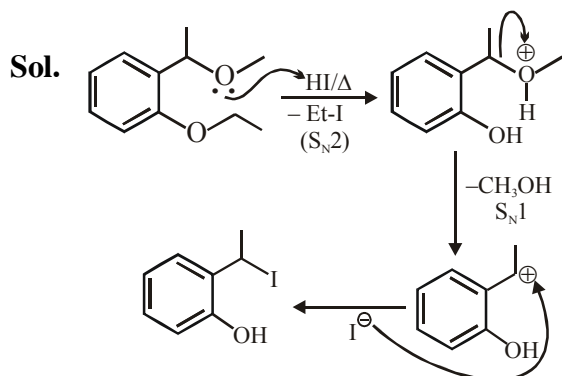
20. Ans. (2)

21. Ans. (1)



Reaction is dehydrohalogenation E^2 -elimination reaction. Elimination takes place in single step and proceed by formation of transition state from anti position.

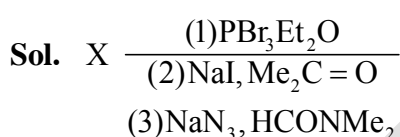
22. Ans. (3)



It is nucleophilic substitution reaction.

EXERCISE # IV (A) (J-ADVANCE OBJECTIVE)

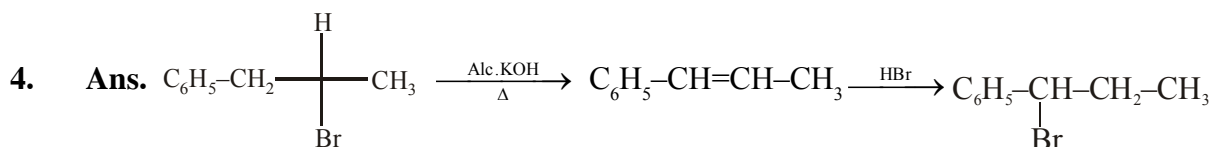
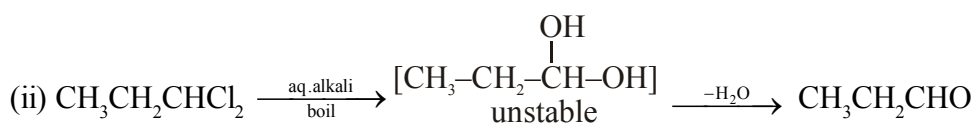
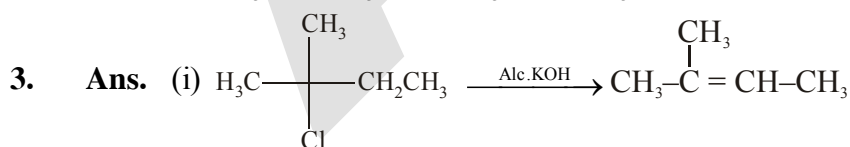
- | | | | | |
|--|---------------|--------------|--------------|--------------|
| 1. Ans. (D) | 2. Ans. (B,D) | 3. Ans. (A) | 4. Ans. (C) | 5. Ans. (B) |
| 6. Ans. (A) | 7. Ans. (A,C) | 8. Ans. (D) | 9. Ans. (C) | 10. Ans. (D) |
| 11. Ans. (A) | 12. Ans. (B) | 13. Ans. (D) | 14. Ans. (B) | 15. Ans. (A) |
| 16. Ans. (A) → Q ; (B) → Q ; (C) → R, S ; (D) → P, S | | | | |
| 17. Ans. (A) | 18. Ans. (D) | 19. Ans. (B) | 20. Ans. (D) | 21. Ans. (5) |
| 22. Ans. (B) | | | | |



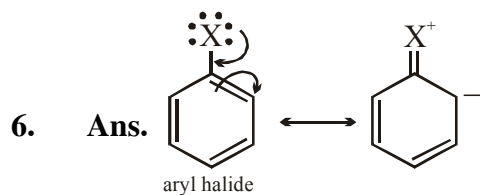
all the three reaction are S_N2 so X is

EXERCISE # IV (B) (J-ADVANCE SUBJECTIVE)

1. Ans. (a) propyl chloride, (b) ortho
 2. Ans. (i) $H_2O < CH_3OH < HO^- < CH_3O^-$
 (ii) $CH_3F < CH_3Cl < CH_3Br < CH_3I$

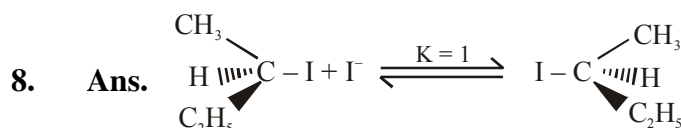
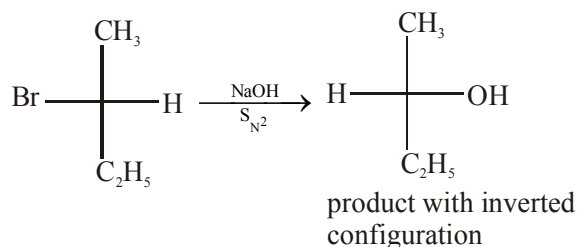


5. Ans. (i) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH} - \text{CH}_3$; $\text{C}_6\text{H}_5 - \overset{\text{Br}}{\underset{|}{\text{CH}}} - \text{CH}_2 - \text{CH}_3$ (ii) $\text{C}_6\text{H}_5\text{COOMgI} + \text{CH}_4$



Due to the above resonance phenomena, C-X bond acquire partial double bond character and becomes difficult to break in the rate determining step of $\text{S}_{\text{N}}2$ reaction.

7. Ans. $\text{S}_{\text{N}}2$ reactions leads to inversion of configuration at a-C.



Above equilibrium is established which has equilibrium constant equal to 1. Therefore, equilibrium mixture will have both the enantiomers in equal amount giving racemic mixture.

